PROOF OF THE BOUNDEDNESS CONJECTURE FOR MINIMAL LOG DISCREPANCIES OF KOLLÁR COMPONENTS

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Abstract. We prove Zhuang's boundedness conjecture for minimal log discrepancies of Kollár components, which asserts that for klt pairs with coefficients in a fixed DCC set, any point has a Kollár component whose log discrepancy is bounded in terms of the codimension of the point and the DCC set. As a corollary, we prove another conjecture of Zhuang, namely that the Weil indices of weakly exceptional Fano varieties of fixed dimension are bounded. We also construct K-semistable Fano surfaces with arbitrarily large Weil index, so the bound does not extend to K-semistable Fano varieties.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let (X, B) be a klt pair and $\eta \in X$ a point, not necessarily closed. A Kollár component of $(X \ni \eta, B)$ is a prime divisor E over X with center η that is extracted, near η, by a projective birational morphism $f : Y \to X$ such that $-E$ is ample over X and $(Y, f_*^{-1}B + E)$ is plt near E, for η of codimension at least two (see Definition 2.3 for the general case). Any klt singularity has a Kollár component [Xu14, Lemma 1], and Kollár components are a basic tool in the local stability theory of klt singularities [LX20]. Zhuang introduced the minimal log discrepancy of Kollár components $\mathrm{mld}^K(X \ni \eta, B)$, the infimum of the log discrepancies $a(E, X, B)$ of the

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Kollár components E of $(X \ni \eta, B)$ [Zhu24b, Definition 1.4], as an approach to the conjectures on the boundedness of klt singularities whose local volumes [XZ24, Definition 2.1] are bounded from below [HLQ20, Conjectures 1.6 and 8.9]. Zhuang proposed the following conjecture. For DCC sets and the notation $B \in I$, see Definition 2.2.

Conjecture 1.1 ([Zhu24b, Conjecture 1.8]). *Let d be a positive integer and $I \subset [0,1]$ a DCC set. Then there exists a positive integer N depending only on d and I satisfying the following. Let (X, B) be a klt pair such that B is a $\mathbb{Q}$-divisor and $B \in I$, and let $\eta \in X$ be a point of codimension d. Then* $\mathrm{mld}^K(X \ni \eta, B) \leq N$.

Conjecture 1.1 is the analogue for Kollár components of the boundedness conjecture for minimal log discrepancies, which follows from Shokurov's ACC conjecture for minimal log discrepancies [Sho88, Problem 5]. The boundedness conjecture for minimal log discrepancies was proved in [Liu26, Theorem 0.2], using [OAI26a, Proposition 2.1]. Since $\mathrm{mld}(X \ni \eta, B) \leq \mathrm{mld}^K(X \ni \eta, B)$, Theorem 1.2 also implies it for klt pairs with coefficients in a DCC set. Two examples of Zhuang show how mld^K differs from mld. Let $1 < a < b$ be coprime integers and let $C_{a,b} = (x^a + y^b = 0) \subset \mathbb{A}^2$. Then

$$\mathrm{mld}^K\left(\mathbb{A}^2 \ni 0, \tfrac{1}{a} C_{a,b}\right) = a$$

[Zhu24b, Example 7.1], while the exceptional divisor E_0 of the blow-up of $\mathbb{A}^2$ at the origin satisfies $a(E_0, \mathbb{A}^2, \frac{1}{a} C_{a,b}) = 2 - \frac{1}{a} \cdot a = 1$. The set $\{\frac{1}{a} \mid a \geq 2\}$ is not a DCC set, so [Zhu24b, Example 7.1] does not contradict Conjecture 1.1, but it shows that the constant N must depend on I. For $d \geq 3$, the canonical hypersurface singularity

$$0 \in (x_0^{d+1} + x_1^d + \cdots + x_d^d = 0) \subset \mathbb{A}^{d+1}$$

is resolved by the blow-up of $\mathbb{A}^{d+1}$ at the origin, the exceptional divisor E of the induced resolution is the only divisor computing its minimal log discrepancy, and E is not a Kollár component [Zhu24b, Example 7.2]. Thus an upper bound on mld does not directly imply Conjecture 1.1.

Conjecture 1.1 was proved in the following cases.

(1) When $d = 2$ and I is a DCC set whose closure is contained in $\mathbb{Q}$ [Zhu24b, Proposition 6.4].
(2) When η is a closed point, I is finite, and the local volume of $(X \ni \eta, B)$ is at least a fixed positive real number ϵ [XZ24, Theorem 4.3]. The proof of [XZ24, Theorem 4.3] degenerates $(X \ni \eta, B)$ to a K-semistable log Fano cone singularity by the stable degeneration [XZ24, Theorem 2.6 and Definitions 2.4 and 2.5] (see [Zhu24c, Theorem 2.19] for real coefficients), and then uses the boundedness of K-semistable log Fano cone singularities with local volume at least ϵ [XZ24, Theorem 1.1].
(3) When η is a closed point and the local volume of $(X \ni \eta, B)$ is at least a fixed positive real number ϵ, for an arbitrary DCC set I [HLQ24, Theorem 1.5].
(4) When η is a closed point and $(X \ni \eta, B)$ admits a δ-plt blow-up [HLS26, Definition 3.6] for a fixed positive real number δ. In this case $\mathrm{mld}^K(X \ni \eta, B)$ is bounded from above by a constant depending only on d and δ [HLS26, Proposition 4.3(1)], [Zhu24b, Lemma 4.2], and the coefficients of B play no role; see also [Mor21, Lemma 2.20] when $B = 0$ and K_X is Cartier at η.
(5) When η is a closed point and $(X \ni \eta, B)$ is an exceptional singularity [HLS26, Definition 3.9]. This case follows from [HLS26, Lemma 3.13 and Proposition 4.3(1)].
(6) When X is affine, $(X \ni \eta, B)$ is a log Fano cone singularity of complexity one [LMS26, Definition 2.7], and the coefficients of B are larger than a fixed positive real number [LMS26, Theorem 5.25]. Here complexity one means that an algebraic torus T of dimension $d-1$ acts effectively on X with full-dimensional pointed weight cone and with only constant invariant regular functions, that η is the unique T-fixed point, and that B is T-invariant.

In general dimension, without additional hypotheses such as a δ-plt blow-up, exceptionality, or complexity one, the cases of Conjecture 1.1 proved before this paper assume a lower bound on the local volume. Zhuang asks whether such a lower bound is necessary [Zhu25, Question 6.11].

We now explain a difficulty that concerns points which are not closed. After localizing at a point of codimension two, the proof of [Zhu24b, Proposition 6.4] reduces to surfaces over a field that is not algebraically closed [Zhu24b, Section 6.1]. More generally, localizing at a point of codimension d leads to a germ of dimension d over a field that is not algebraically closed, so results on closed points of varieties over an algebraically closed field cannot be used directly. When $\dim X = 2$ and $B = 0$, at any closed point $x \in X$ some Kollár component computes $\mathrm{mld}(X \ni x)$ [LX23, Theorem 1.2], and hence $\mathrm{mld}^K(X \ni x) = \mathrm{mld}(X \ni x)$ [Zhu24b, Section 7].

For example, let $V = (x^2 - ty^2 = z^3) \subset \mathbb{A}^4$ and let ξ be the generic point of the t-axis, a point of codimension two. For $t_0 \neq 0$, the surface $S_{t_0} = (x^2 - t_0y^2 = z^3) \subset \mathbb{A}^3$ has an A_2 singularity at the origin, and the exceptional curves E_1 and E_2 of the minimal resolution of S_{t_0} are Kollár components with log discrepancy 1. Going once around $t = 0$ sends $\sqrt{t}$ to $-\sqrt{t}$ and exchanges $x + y\sqrt{t}$ and $x - y\sqrt{t}$, so it exchanges E_1 and E_2. By Proposition 3.8 (take $t_1 = t$ in the proof of Proposition 3.8), the Kollár components of $(V \ni \xi)$ are the Kollár components over $k = \mathbb{C}(t)$ of the surface $S = (x^2 - ty^2 = z^3) \subset \mathbb{A}^3_k$ at the origin. Over the algebraic closure $\bar{k}$ of k, the divisors with center the origin and log discrepancy 1 correspond to E_1 and E_2, which are not defined over k, and by Lemma 3.6(1) a Kollár component of $(S \ni 0)$ over k remains a prime divisor over $\bar{k}$, invariant under $\mathrm{Gal}(\bar{k}/k)$, and its log discrepancy over $\bar{k}$ equals its log discrepancy over k, so none has log discrepancy 1. Any other divisor over $\bar{k}$ with center the origin has log discrepancy at least 2. The exceptional divisor of the weighted blow-up of S with weights $(3, 3, 2)$ on (x, y, z) is defined over k and is a Kollár component with log discrepancy 2 (its log discrepancy is $3 + 3 + 2 - 6 = 2$, and it is a Kollár component over $\bar{k}$ by [Zhu24b, Lemma 2.5], hence over k by Lemma 3.6(2)). Thus

$$\mathrm{mld}^K(V \ni \xi) = 2 > 1 = \mathrm{mld}^K(S_{t_0} \ni 0),$$

where S_{t_0} is the section $V \cap (t = t_0)$ and 0 is the closed point $(0, 0, 0, t_0)$ of the t-axis. In particular, cutting by general hyperplanes down to a closed point can strictly decrease mld^K.

We now explain a second conjecture of Zhuang, on Fano varieties, which is related to Conjecture 1.1. A Fano variety is a normal projective variety X such that K_X is $\mathbb{Q}$-Cartier and $-K_X$ is ample. A Fano variety X is weakly exceptional if (X, D) is lc for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$. The Weil index $q_W(X)$ of a Fano variety X is the supremum of the positive integers q such that $-K_X \sim_{\mathbb{Q}} qA$ for some Weil divisor A on X (see Definition 15.1 for the three notions). The term weakly exceptional is taken from [CPS10, Definition 1.1.10]. The notion of exceptional pairs goes back to [Sho00, Definition 1.5]. For weakly exceptional singularities, see [Pro00, Kud01, CS14, CPS18]. In [Zhu24b, Section 7], weakly exceptional Fano varieties are called *weakly special*.

Zhuang conjectured that there exists a positive integer N depending only on d such that $q_W(X) \leq N$ for any weakly exceptional Fano variety X of dimension d [Zhu24b, Conjecture 7.4] (see also [Zhu25, Question 6.13]). Zhuang observed that Conjecture 1.1 implies [Zhu24b, Conjecture 7.4] through the orbifold cone [Zhu24b, Section 7]. For a Fano variety X and a Weil divisor A on X with $-K_X \sim_{\mathbb{Q}} qA$, the orbifold cone is

$$\mathrm{Spec} \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mA))$$

[Kol04, Paragraph 14], [Zhu24c, Definition 3.2].

Before this paper, [Zhu24b, Conjecture 7.4] was known when the anti-canonical volume is bounded from below. Indeed, for any K-semistable Fano variety X of dimension d (see [BJ20, Theorem B] for a characterization of K-semistability),

$$q_W(X) \cdot (-K_X)^d \leq (d + 1)^{d+1}$$

[BL21, Theorem 37], and weakly exceptional Fano varieties are K-semistable [BJ20, Theorems A and B]. The proof of [BL21, Theorem 37] goes through the orbifold cone: if $-K_X \sim_{\mathbb{Q}} qA$ for a Weil divisor A on X, then the normalized volume [XZ24, Definition 2.1] of the divisorial valuation of the zero section at the vertex is $q \cdot (-K_X)^d$, and it equals the local volume of the vertex when X is K-semistable [LX20, Theorem A]. Thus the inequality bounds $q_W(X)$ only when $(-K_X)^d$ is bounded from below. Weakly exceptional Fano varieties can have arbitrarily small anti-canonical volume. For a positive integer n, a quasismooth hypersurface $X_{8n+4} \subset \mathbb{P}(2, 2n+1, 2n+1, 4n+1)$ of degree $8n+4$ is weakly exceptional [CPS10, Lemma 2.1.1 and Definition 1.1.10]. Since $8n+4-\big(2+2(2n+1)+(4n+1)\big) = -1$ and both $\mathbb{P}(2, 2n+1, 2n+1, 4n+1)$ and X_{8n+4} are well-formed (cf. [CPS10, Section 1.2]), by adjunction $\mathcal{O}_{X_{8n+4}}(-K_{X_{8n+4}}) \cong \mathcal{O}_{X_{8n+4}}(1)$, so

$$(-K_{X_{8n+4}})^2 = \frac{8n+4}{2(2n+1)^2(4n+1)} = \frac{2}{(2n+1)(4n+1)}.$$

Thus the case of [Zhu24b, Conjecture 7.4] left open was the case when $(-K_X)^d$ is small. Through the orbifold cone, the cases of Conjecture 1.1 with local volume bounded from below [XZ24, Theorem 4.3], [HLQ24, Theorem 1.5] only cover the weakly exceptional Fano varieties X for which the local volume $q_W(X) \cdot (-K_X)^d$ of the vertex is bounded from below by a fixed positive real number.

Boundedness theorems for Fano varieties also bound the Weil index on the classes of Fano varieties they cover. Indeed, let X be a Fano variety of dimension d, H a very ample divisor on X, and A a Weil divisor on X with $-K_X \sim_{\mathbb{Q}} qA$. A general complete intersection curve C of $d-1$ members of $|H|$ is contained in the smooth locus of X, so $(A \cdot C)$ is a positive integer and

$$q \leq q(A \cdot C) = (-K_X \cdot C) = (-K_X \cdot H^{d-1}).$$

For a bounded family of Fano varieties and the restrictions of a relatively very ample divisor, $(-K_X \cdot H^{d-1})$ takes finitely many values by Noetherian induction on the base of the family. Bounded families of Fano varieties of dimension d include K-semistable Fano varieties X with $(-K_X)^d$ bounded from below [Jia20, Theorem 1.1], ϵ-lc Fano varieties for a fixed positive real number ϵ [Bir21, Theorem 1.1], and exceptional Fano varieties [Bir19, Theorem 1.3], that is, Fano varieties X such that (X, D) is klt for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$ [CPS10, Definition 1.1.10]. Weakly exceptional Fano surfaces do not form a bounded family, since $(-K_{X_{8n+4}})^2 \to 0$ as $n \to +\infty$, so [Jia20, Theorem 1.1], [Bir21, Theorem 1.1], and [Bir19, Theorem 1.3] do not cover them.

After [Zhu24b, Conjecture 7.4], Zhuang suspects that the bound $q_W(X) \leq N$, with N depending only on d, holds for all K-semistable Fano varieties X of dimension d [Zhu24b, Section 7]. As evidence, the Weil index of a weighted projective space can be arbitrarily large, while the only K-semistable weighted projective spaces are projective spaces [Zhu24b, Section 7]. In the setting of orbifolds, Ross and Thomas show that a Fano orbifold whose orbifold anti-canonical class is r times an orbi-divisor $D \geq 0$ with $r > d+1$ is slope unstable, the index obstruction [RT11, Theorem 8.7], and by the index obstruction any weighted projective space other than a projective space is slope unstable [RT11, Example 8.9]. Since weakly exceptional Fano varieties are K-semistable, the K-semistable version is stronger than [Zhu24b, Conjecture 7.4]. In a later survey, Zhuang doubts the K-semistable version: he writes that there seems to be no particular reason for the Weil index of a K-semistable Fano variety to be bounded [Zhu25, Section 5].

The goal of this paper is to prove Conjecture 1.1.

Our main theorem is the following.

Theorem 1.2. *Let d be a positive integer and $I \subset [0,1]$ a DCC set. Then there exists a positive integer N depending only on d and I satisfying the following. Let (X, B) be a klt pair such that $B \in I$, and let $\eta \in X$ be a point of codimension d. Then there exists a Kollár component E of $(X \ni \eta, B)$ such that $a(E, X, B) \leq N$. In particular, $\mathrm{mld}^K(X \ni \eta, B) \leq N$.*

We denote by $N(d, I)$ the constant in Theorem 1.2 given in Section 16. Theorem 1.2 proves Conjecture 1.1.

As a consequence of Theorem 1.2, we obtain a bound on the Weil indices of weakly exceptional Fano varieties (Definition 15.1).

Corollary 1.3. *Let d be a positive integer. Then there exists a positive integer N depending only on d such that $q_W(X) \leq N$ for any weakly exceptional Fano variety X of dimension d.*

Corollary 1.3 is [Zhu24b, Conjecture 7.4]. Let X be a weakly exceptional Fano variety of dimension d, let q be a positive integer, and let A be a Weil divisor on X with $-K_X \sim_{\mathbb{Q}} qA$. By Lemma 15.3 and Construction-Proposition 15.4, the orbifold cone over X with respect to A is a klt variety of dimension $d+1$, and the zero section of the total space $\operatorname{Spec}_X \bigoplus_{m\geq 0} \mathcal{O}_X(mA)$ is a Kollár component of the vertex with log discrepancy q. Since X is weakly exceptional, the zero section is the unique Kollár component of the vertex (Lemma 15.6). Since the vertex is a point of codimension $d+1$, Theorem 1.2 with $I = \{0\}$ provides a Kollár component of the vertex with log discrepancy at most $N(d+1, \{0\})$, so $q \leq N(d+1, \{0\})$, and hence

$$q_W(X) \leq N(d+1, \{0\}).$$

We prove Corollary 1.3 in Section 16.

Let X be a klt Fano variety of dimension d. Following [BJ20, Section 4.1], the *α-invariant* $\alpha(X)$ of X, an algebraic version of the invariant introduced by Tian [Tia87], is the infimum of the log canonical thresholds of the $\mathbb{Q}$-divisors $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$ with respect to X. A weakly exceptional Fano variety is klt (Lemma 15.3), and a klt Fano variety X is weakly exceptional if and only if $\alpha(X) \geq 1$. Let $\delta(X)$ be the stability threshold of $-K_X$ [BJ20, Section 4.2]. By [BJ20, Theorem A],

$$\delta(X) \geq \frac{d+1}{d}\alpha(X).$$

By [BJ20, Theorem B], X is K-semistable if and only if $\delta(X) \geq 1$, and X is uniformly K-stable if and only if $\delta(X) > 1$. Thus any weakly exceptional Fano variety X of dimension d satisfies $\delta(X) \geq \frac{d+1}{d} > 1$, so it is uniformly K-stable. The following example of toric surfaces [CLS11, Section 3.1] shows that Corollary 1.3 does not extend to K-semistable Fano varieties, contrary to the expectation in the paragraph after [Zhu24b, Conjecture 7.4].

Example 1.4. Let t be a positive integer, let $m := 358050t$, and let $v_0, v_1, v_2, v_3 \in \mathbb{Q}^2$ be the vectors

$$v_0 = \Big(-1, \frac{19m+1}{11m-1}\Big), \quad v_1 = \Big(-1, \frac{5-m}{7m-5}\Big), \quad v_2 = \Big(-\frac{m+5}{7m+5}, -1\Big), \quad v_3 = \Big(\frac{19m-1}{11m+1}, -1\Big).$$

Let $\Lambda_t := \mathbb{Z}v_0 + \mathbb{Z}v_1 + \mathbb{Z}v_2 + \mathbb{Z}v_3$, which is a lattice of rank two, let Q_t be the convex hull of v_0, v_1, v_2, v_3, and let Σ_t be the set consisting of the zero cone and the cones over the proper faces of Q_t. Then every v_i is primitive in Λ_t, the interior of Q_t contains 0, and Σ_t is a complete fan in $\Lambda_t \otimes \mathbb{R}$. Moreover, the toric surface X_t associated to Σ_t is a $\mathbb{Q}$-factorial klt Fano surface, $\delta(X_t) = 1$, and $q_W(X_t) = 429660t$. In particular, X_t is K-semistable, and the Weil indices of K-semistable Fano surfaces are not bounded.

The polygon of $-K_{X_t}$ is the polar of Q_t, and its barycenter is 0, so $\delta(X_t) = 1$ by the formula for the stability threshold of toric Fano varieties [BJ20, Corollaries 7.16 and 7.17]. Since $v_0, \dots, v_3$ generate Λ_t, $q_W(X_t)$ is the index in Λ_t of the subgroup generated by $v_1 - v_0, v_2 - v_0, v_3 - v_0$. We prove Example 1.4 in Section 15.

Since $\delta(X_t) = 1 < \frac{3}{2}$, the surface X_t is not weakly exceptional by the paragraph before Example 1.4. Let A_t be a Weil divisor on X_t with $-K_{X_t} \sim_{\mathbb{Q}} q_W(X_t)A_t$. Theorem 1.2 with $d = 3$ and $I = \{0\}$, applied to the vertex of the orbifold cone over X_t with respect to A_t, provides a Kollár component of the vertex with log discrepancy at most $N(3, \{0\})$, while the zero section of $\operatorname{Spec}_{X_t} \bigoplus_{m\geq 0} \mathcal{O}_{X_t}(mA_t)$ is a Kollár component of the vertex with log discrepancy $q_W(X_t) = 429660t$ (Construction-Proposition 15.4). If $429660t > N(3, \{0\})$, then the zero section

is not the Kollár component provided by Theorem 1.2. Thus the argument for Corollary 1.3 fails exactly at the uniqueness in Lemma 15.6.

Sketch of the proof. Theorem 1.2 is proved by four reductions.

(1) By Proposition 3.8, η becomes a closed point x of a d-dimensional klt pair over a purely transcendental extension k of $\mathbb{C}$, whose Kollár components over k are those of $(X \ni \eta, B)$. Thus, for $d \geq 2$, it suffices to prove Theorem 14.1, uniformly in all fields k finitely generated over $\mathbb{C}$; for $d = 1$ the closure of η is, near η, a Kollár component (Section 16).

(2) Theorem 14.1 follows from Theorem 13.1, which treats rational boundaries with fixed denominators, once B is enlarged to an lc boundary Δ having x as an lc center, with coefficients in a finite set J (Corollary 7.5), K_X and the sums of the components of the equality part B^{eq} of B (the part of B along the prime divisors on which the coefficients of B and Δ are equal and positive) with a given coefficient become $\mathbb{Q}$-Cartier on torus quasi-torsors (Construction-Theorem 8.5; see (b) below), and the coefficients become rational (Theorem 4.5).

(3) Theorem 13.1 follows from the reduced case, Definition-Theorem 11.1, since Fermat suspensions of index covers absorb the klt boundary into the variety and make the lc boundary reduced (Theorems 12.12 and 12.7).

(4) Definition-Theorem 11.1 is proved by contradiction: root covers of degrees $m_i \to +\infty$ turn counterexamples into cyclic quotients whose Kollár components over the ground fields have log discrepancy at least 1 (Lemma 11.6), contradicting Lemma 11.4 and Theorem 10.2, a version of [OAI26b, Theorem G.1] over finitely generated fields.

The paragraphs (a)–(d) below explain the main points: (a) explains how Kollár components over k are obtained in all four reductions, (b) concerns reduction (2), and (c) and (d) concern reduction (4).

(a) Kollár components over the ground field. As in [Zhu24b, Section 6.1], localization leads to germs over k, and results over $\bar{k} \cong \mathbb{C}$ (Lemma 3.2) provide Kollár components over $\bar{k}$, in general not defined over k. We obtain Kollár components over k in three ways, without summing Galois conjugates.

(i) Among the quasi-monomial valuations v centered at a given closed point with $v(h) = 1$ which are lc places of an lc pair $(X, B+cH)$, where $H = \mathrm{div}(h)$ and $c > 0$, and which satisfy finitely many incidence conditions, the volume multiplied by the d-th power of a positive linear function of finitely many values $v(g)$, called the height, has a unique minimizer (Section 5, Definition-Proposition 5.2), hence the minimizer is Galois invariant, and it descends to k (Definition-Proposition 6.3). Nearby rational valuations define Kollár components over k (Theorem 6.6), since their valuation ideals are generated by monomials in finitely many elements over k (Lemma 6.5), although the monomial charts of [LX18, Lemma 2.10] are not defined over k. In this way (i) provides the Kollár component S over k (Corollary 6.7) to which Theorem 7.4 is applied in Corollary 7.5, used in reduction (2), and the Galois invariant minimizers of Proposition 10.16 used in Section 10.

(ii) Restricting valuations from suspensions and torus quasi-torsors to the base involves no choice (Proposition 13.2 and Construction-Theorem 8.5(3)). (iii) A Kollár component of a quasi-étale quotient, whose center has exactly one preimage in the cover, lifts uniquely, hence invariantly, to the cover (Lemma 9.3).

(b) From real coefficients to fixed denominators. A small $\mathbb{Q}$-factorial modification ([BCHM10, Corollary 1.4.3] and the second paragraph after it) is in general not an isomorphism over x, and a Kollár component on it does not induce a prime extraction of X. Instead, for a Weil divisor D not $\mathbb{Q}$-Cartier at x, the rank one torus quasi-torsor $\mathrm{Spec} \bigoplus_{m\in\mathbb{Z}} \Gamma(X, \mathcal{O}_X(mD))$ has a unique $\mathbb{G}_m$-fixed point over x, with residue field $\kappa(x)$, and pulls D back to a principal divisor (Construction-Proposition 8.4), and Kollár components there that are lc places of the pullback of Δ descend to X (Proposition 8.2). We only make K_X and the sums of the components of B^{eq} with a given coefficient $\mathbb{Q}$-Cartier. Since B^{eq} has coefficients in J, at most $|J\backslash\{0\}| + 1$ layers suffice,

one for K_X and one for each coefficient (Construction-Theorem 8.5), and a convex combination with Δ recovers the plt property for B (Definition-Lemma 4.4). By [Nak16, Corollary 3.9], there is a rational substitution depending only on d and J (Definition-Theorem 4.1).

(c) Reduced boundaries. The proof of [OAI26b, Theorem G.1] needs a generator of the canonical sheaf and an equation of the boundary with a common character, while on the index cover of K_X and the boundary P (Construction-Proposition 9.1), the ratio of their characters is only known to have order dividing n_0, where $n_0(K_X + P)$ is Cartier and n_0 depends only on d [Bir19, Corollary 1.9].

A product of n_0 root covers of degree m aligns the characters (Construction-Proposition 11.2), but the return to X can multiply log discrepancies by a factor of order m (Example 11.7). We therefore take counterexamples $(X_i \ni x_i, P_i)$ over k_i with $m = m_i \to +\infty$. If the Kollár components over k_i of $(X_i \ni x_i, 0)$ that are lc places of (X_i, P_i) have log discrepancy larger than m_i^2, then every Kollár component over k_i of the corresponding cyclic quotient has log discrepancy at least 1 (Lemma 11.6). This contradicts Lemma 11.4 and Theorem 10.2 for $m_i \gg 0$.

(d) Pinning over finitely generated fields. The pinning obstruction [OAI26b, Theorem G.1], which yields the bounded klt complements of [OAI26b, Theorem 2], is proved over an algebraically closed field for ϵ-lc quotients, whereas for the quotients in (c) we only bound from below the log discrepancies of Kollár components over the ground field. We change the proof at two places. First, the ϵ-lc assumption is used only in the final case of [OAI26b, Subsection G.2], where a rank one valuation with finitely generated klt graded ring yields prime divisors with log discrepancies tending to zero. By Theorem 2.10, they are Kollár components over the ground field (Lemma 10.23).

Second, the rank refinement [OAI26b, Proposition F.7] enforces the order condition [OAI26b, (C.5)] by tilting the height along a violating $\mathbb{Q}$-divisor Γ, and uniqueness makes the tilted minimizer Galois invariant only for Γ defined over the ground field. Thus Proposition 10.16 only yields the order condition (MO_k) for $\mathbb{Q}$-divisors over the ground field (Set-up 10.6), and the uses of [OAI26b, (C.5)] only need such divisors (Remarks 10.7 and 10.17).

The only new bound is the constant $N_0(d)$ of Definition-Theorem 11.1, used in dimensions bounded in terms of d and I. The minimizers in (a) and (d) only make choices over k and bound no log discrepancy, so the proof uses no lower bound on the local volume. In the other reductions, log discrepancies do not increase except by the factor 2^{c_J} of Theorem 4.5, where c_J depends only on J, and the unbounded degrees of the index covers cancel (Lemma 9.4). The DCC assumption on I is used only in Corollary 7.5. The constant $N(d, I)$ is not explicit: $N_0(d)$ is obtained by contradiction, and J and the rational substitution of the coefficients in Definition-Theorem 4.1 come from the existence results of [HLS26] used in Theorem 7.4 and from [Nak16, Corollary 3.9].

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce Kollár components and mld^K, marked graded pairs and the Xu–Zhuang criterion (Theorem 2.10), and minimizers of the normalized volume for pairs with real coefficients (Theorem 2.11).

In Section 3 we study extension of constants and descent of Kollár components, we reduce a point of codimension d to a closed point of a d-dimensional variety over a field finitely generated over $\mathbb{C}$, and we reduce further to geometrically integral varieties and rational points.

In Section 4 we replace finitely many real coefficients of a boundary by rational numbers depending only on the dimension and on the set of coefficients, at the cost of a bounded factor between log discrepancies, also over fields finitely generated over $\mathbb{C}$. We also define the equality part of a klt boundary with respect to a larger lc boundary, and we show that if the prime extraction of an lc place of the larger boundary is plt near the exceptional divisor for the equality part, then it is plt near the exceptional divisor for the klt boundary.

In Section 5 we show that, among the quasi-monomial valuations centered at a given closed point which are lc places of an lc pair whose boundary is the boundary of a klt pair plus a positive multiple of the divisor of a regular function, which take the value one at that function, and which

satisfy finitely many incidence conditions, the volume multiplied by the d-th power of a height has a unique minimizer, and that the minimizer is a limit of normalized Kollár components of the klt pair.

In Section 6 we show that the minimizer of Section 5 over an algebraic closure of the ground field, being unique, descends to the ground field together with its associated graded ring, and we construct Kollár components over the ground field whose valuations approximate the minimizer.

In Section 7 we construct decomposable complements over the ground field, which enlarge a boundary with coefficients in I to an lc boundary with coefficients in a finite set depending only on the dimension and on I such that the given closed point is an lc center.

In Section 8 we construct towers of rank one torus quasi-torsors over a klt germ, on which the canonical divisor and finitely many given Weil divisors become $\mathbb{Q}$-Cartier, and we show that the Kollár components over them which are lc places of the pullback of an lc boundary descend to Kollár components of the germ.

In Section 9 we construct index covers over the ground field which keep the residue field and have exactly one point over the given point, a version of [Zhu24b, Lemma-Definition 2.15], we lift and descend prime extractions along quasi-étale Galois quotients, and we study the restriction of divisorial valuations along finite covers and from a product with an affine space to the first factor.

In Section 10 we carry the proof of [OAI26b, Theorem G.1], which occupies [OAI26b, Sections C–G] and uses the lemmas of [OAI26b, Sections A and B], over to fields finitely generated over $\mathbb{C}$. We write out only the places that change, and the remarks on the modifications list the results of [OAI26b] that we use.

In Section 11 we construct, over the ground field, Kollár components of bounded log discrepancy which are lc places of a given lc pair with a reduced boundary, using index covers, products of root covers, and Section 10.

In Section 12 we construct the additive and the marked Fermat suspensions. The additive suspension turns an lc boundary with a fixed denominator into a reduced boundary, the marked suspension absorbs a klt boundary with a fixed denominator into a variety of higher dimension, and in both cases the Kollár components compatible with the boundary return to Kollár components of the base.

In Section 13 we combine Sections 11 and 12 to construct, over the ground field, Kollár components with bounded log discrepancy for boundaries with fixed denominators.

In Section 14 we prove a version of Theorem 1.2 for closed points of varieties over fields finitely generated over $\mathbb{C}$, with Kollár components defined over the ground field and a bound independent of the field.

In Section 15 we construct orbifold cones over Fano varieties with respect to Weil divisors, show that the vertex has a unique Kollár component when the Fano variety is weakly exceptional, and prove Example 1.4.

In Section 16 we prove Theorem 1.2 and Corollary 1.3.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and Kollár components.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is

$\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Contractions in Definition 2.1 are projective morphisms π with $\pi_* \mathcal{O}_X = \mathcal{O}_Z$ (cf. [HLS26, Definition 3.2]). Replacing $\mathbb{C}$ with any field of characteristic zero, Definition 2.1 defines the same notions for normal quasi-projective varieties over that field, and from Section 3 on we use them for such varieties.

Definition 2.2. A set $I \subset \mathbb{R}$ is a *DCC set* if any non-increasing sequence in I is eventually constant. For an $\mathbb{R}$-divisor B and a set $I \subset \mathbb{R}$, we write $B \in I$ if the coefficients of B belong to I.

For a prime divisor E over X, we denote by $\operatorname{center}_X E \in X$ the generic point of the center of E on X.

Definition 2.3. Let X be a normal variety. A prime divisor E over X is *primitive* over X if there exists a projective birational morphism $f : Y \to X$ such that Y is normal, E is a prime divisor on Y, and $-E$ is a $\mathbb{Q}$-Cartier f-ample divisor. Such a morphism f is unique, and we call it the *prime extraction* of E. Let (X, B) be a klt pair and $\eta \in X$ a point. A *Kollár component* of $(X \ni \eta, B)$ is a prime divisor E over X with $\operatorname{center}_X E = \eta$ such that for some open neighborhood U of η, E is primitive over U and $(Y, f_*^{-1} B_U + E)$ is plt near E, that is, plt (cf. [KM98, Definition 2.34]) on an open neighborhood of E in Y, where $f : Y \to U$ is the prime extraction of E and B_U is the sum of the components of $B|_U$ other than E, with their coefficients. The *minimal log discrepancy of Kollár components* of $(X \ni \eta, B)$ is

$$\operatorname{mld}^K(X \ni \eta, B) := \inf\{a(E, X, B) \mid E \text{ is a Kollár component of } (X \ni \eta, B)\}.$$

The uniqueness of the prime extraction is part of [Zhu24b, Definition 2.1(1)], and by Lemma 2.7 the prime extraction of a primitive prime divisor E is determined by the valuation ord_E. If η is a codimension one point of X, then E can be a prime divisor on X. In this case, possibly shrinking U, $E|_U$ is Cartier because X is normal, and hence the prime extraction of E is the identity morphism of U by Lemma 2.7. This is why B_U omits E.

Since the prime extraction is determined by ord_E, when we descend and lift Kollár components under extension of constants (Section 3), finite covers (Section 9), torus quasi-torsors (Section 8), and suspensions (Section 12), we work with the valuation ideals of ord_E rather than with the prime divisor E.

Part (2) of Lemma 2.4 below is used in Lemma 3.6, Theorem 4.5, and Lemma 9.3. For example, under the hypotheses of Lemmas 3.6 and 9.3, a Kollár component defined over a field k at a k-rational point remains a prime divisor after extension to an algebraic closure of k (Lemma 3.6), and a Kollár component of a quotient by a finite group lifts uniquely to an invariant Kollár component on the cover (Lemma 9.3).

Lemma 2.4. *Let $f : Y \to X$ be a projective birational morphism between normal varieties over a field of characteristic zero and $x \in X$ a closed point.*

(1) If E is an f-exceptional prime divisor such that $f(E) = \{x\}$ and $-E$ is $\mathbb{Q}$-Cartier and f-ample, then f is an isomorphism over $X \setminus \{x\}$ and $\operatorname{Supp} f^{-1}(x) = E$.

(2) *If S is a reduced divisor on Y such that* $\operatorname{Supp} S = \operatorname{Supp} f^{-1}(x)$ *and $(Y, S + B_Y)$ is plt near S for some $\mathbb{R}$-divisor $B_Y \geq 0$ on Y, then S is a prime divisor.*

Proof. (1) Let $C \subset Y$ be a curve such that $f(C)$ is a point. Since $-E$ is f-ample, $(E \cdot C) < 0$, and hence $C \subset E$. By Lemma 2.7, f is an isomorphism over $X \backslash f(E) = X \backslash \{x\}$. By [Har77, Corollary III.11.4], $f^{-1}(x)$ is connected. Since $f^{-1}(x)$ contains E and any curve in $f^{-1}(x)$ is contained in E, $\operatorname{Supp} f^{-1}(x) = E$.

(2) If $\dim Y = 1$, then f is an isomorphism, S is a point, and there is nothing to prove. Assume that $\dim Y \geq 2$. Let V be an open neighborhood of S in Y such that $(V, (S + B_Y)|_V)$ is plt. Since f is proper and $f^{-1}(x) \subset V$, $W := X \backslash f(Y \backslash V)$ is an open neighborhood of x with $f^{-1}(W) \subset V$. We use [Kol13], whose results hold for pairs over any field of characteristic zero [Kol13, Definition 1.5], and whose discrepancy is the log discrepancy minus 1. Choose a rational number r with $-1 < r \leq \operatorname{discrep}(f^{-1}(W), (S + B_Y)|_{f^{-1}(W)})$, and for each of the finitely many components P of $B_Y|_{f^{-1}(W)}$ whose coefficient c_P is irrational (we have $0 < c_P < 1$, since the coefficients of the plt pair $(f^{-1}(W), (S + B_Y)|_{f^{-1}(W)})$ are at most 1 [KM98, Corollary 2.31(1)], whose proof works for $\mathbb{R}$-divisors), choose rational numbers $c_P^-, c_P^+ \in [0, 1)$ with $c_P^- < c_P < c_P^+$. By [Kol13, Proposition 2.21] applied with r and the numbers c_P^- and c_P^+, there exists a $\mathbb{Q}$-divisor Δ' on $f^{-1}(W)$ with $\operatorname{Supp} \Delta' = \operatorname{Supp}(S + B_Y)|_{f^{-1}(W)}$ such that $K_{f^{-1}(W)} + \Delta'$ is $\mathbb{Q}$-Cartier, the coefficients of Δ' are in $[0, 1]$ by (3) and (4) there, each component of S has coefficient 1 in Δ' by (4) there, and $(f^{-1}(W), \Delta')$ is plt by (5) there. By [Kol13, Definition 2.8], $(f^{-1}(W), \Delta')$ is dlt. Suppose that two components $S_1 \neq S_2$ of S intersect. By [Kol13, Theorem 4.16(1)(2)], any irreducible component T of $S_1 \cap S_2$ is an lc center of $(f^{-1}(W), \Delta')$ of codimension 2. Thus there exists an exceptional prime divisor F over $f^{-1}(W)$ with center T and $a(F, f^{-1}(W), \Delta') = 0$, which contradicts that $(f^{-1}(W), \Delta')$ is plt. Therefore the components of S are pairwise disjoint. Since $\operatorname{Supp} S = \operatorname{Supp} f^{-1}(x)$ is connected [Har77, Corollary III.11.4], S is a prime divisor. □

2.2. **Finite morphisms.**

Lemma 2.5. *Let $f : X' \to X$ be a finite surjective morphism between normal varieties over a field of characteristic zero, and let B and B' be $\mathbb{R}$-divisors on X and X' such that $K_X + B$ is $\mathbb{R}$-Cartier and $K_{X'} + B' = f^*(K_X + B)$. Let F be a prime divisor over X'. Then the restriction of* ord_F *to the function field of X is $e \operatorname{ord}_E$ for a prime divisor E over X and a positive integer e, and $a(F, X', B') = ea(E, X, B)$. In particular, (X', B') is sub-lc (resp. sub-klt) if and only if (X, B) is sub-lc (resp. sub-klt), and if $B' \geq 0$ and $B \geq 0$, then (X', B') is plt if and only if (X, B) is plt. If f is quasi-étale, that is, étale over the generic point of any prime divisor on X, then $K_{X'} = f^*K_X$, and hence $K_{X'} + f^*B = f^*(K_X + B)$.*

Proof. By [Kol13, 2.42], the restriction of ord_F to the function field of X is $e \operatorname{ord}_E$ for a prime divisor E over X and a positive integer e, and for any prime divisor E over X there exists a prime divisor F over X' such that the restriction of ord_F to the function field of X is a positive integer multiple of ord_E. Since the characteristic is zero, the ramification along F is tame, so $R(F) = r(F) - 1$ and $r(F) = e$ [Kol13, 2.41]. Thus [Kol13, (2.42.2)] becomes $a(F, X', B') = ea(E, X, B)$. The equivalences for sub-lc, sub-klt, and plt pairs are [Kol13, Corollary 2.43(2)] (see also [KM98, Proposition 5.20] for klt and lc pairs). The results of [Kol13] cited here hold over any field of characteristic zero [Kol13, Definition 1.5], and their proofs for $\mathbb{Q}$-divisors work for $\mathbb{R}$-divisors (cf. [Kol13, 2.20]).

If f is quasi-étale, then f is étale at the generic point of any prime divisor on X', so the ramification divisor of f is zero. Hence $K_{X'} = f^*K_X$ by the Hurwitz formula. □

2.3. **Valuations.**

Definition 2.6. Let X be a normal variety over a field of characteristic zero. A *valuation* on X is a valuation v of the function field of X with values in $\mathbb{R}$, trivial on the ground field, which has a center $\operatorname{center}_X v$ on X. For a real number $t \geq 0$, set $\mathfrak{a}_t(v) := \{f \in \mathcal{O}_X \mid v(f) \geq t\}$. A

valuation v is *quasi-monomial* if there exist a proper birational morphism $Y \to X$ from a normal variety Y, a regular point $y \in Y$, a regular system of parameters $y_1, \dots, y_r$ of $\mathcal{O}_{Y,y}$, and real numbers $\alpha_1, \dots, \alpha_r \geq 0$ such that v is the monomial valuation with $v(y_i) = \alpha_i$ for any i. For a pair (X, B), the log discrepancy $a(v, X, B) \in \mathbb{R} \cup \{+\infty\}$ of a valuation v on X is defined as in [HLQ20, Definition 2.7], following [JM12]. If $v = c \operatorname{ord}_E$ for a prime divisor E over X and a real number $c > 0$, then $a(v, X, B) = ca(E, X, B)$ as in Definition 2.1.

If $x := \operatorname{center}_X v$ is a closed point and $d := \dim X$, the *volume* of v [ELS03], [HLQ20, Definition 2.10] is

$$\operatorname{vol}(v) := \lim_{t \to +\infty} \frac{\dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_t(v)}{t^d/d!},$$

and the *normalized volume* of v with respect to (X, B) [Li18], [HLQ20, Definition 2.11] is $\widehat{\operatorname{vol}}_{X,B}(v) := a(v, X, B)^d \operatorname{vol}(v)$ if $a(v, X, B) < +\infty$, and $\widehat{\operatorname{vol}}_{X,B}(v) := +\infty$ otherwise. Since x is a closed point, $\kappa(x)$ is a finite separable extension of the ground field, so by Hensel's lemma the Artinian local ring $\mathcal{O}_{X,x}/\mathfrak{a}_t(v)$ contains a unique copy of $\kappa(x)$ containing the ground field, and $\dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_t(v)$ equals the length of $\mathcal{O}_{X,x}/\mathfrak{a}_t(v)$, as in [HLQ20, Definition 2.10]. Here the ideals $\mathfrak{a}_m(v)\mathcal{O}_{X,x}$, $m \geq 1$ an integer, form a graded family of $\mathfrak{m}_x$-primary ideals of $\mathcal{O}_{X,x}$, and $\mathcal{O}_{X,x}$ is analytically unramified, that is, its completion is reduced, since it is a reduced excellent local ring [Cut14, Section 2]. Thus the limit over integers $t = m$ exists by [Cut14, Theorem 4.7]. We omit the passage to real numbers t, which follows from

$$\mathfrak{a}_{\lceil t \rceil}(v) \subset \mathfrak{a}_t(v) \subset \mathfrak{a}_{\lfloor t \rfloor}(v) \quad \text{and} \quad \lim_{t \to +\infty} \left(\frac{\lceil t \rceil}{t} \right)^d = 1.$$

Lemma 2.7. *Let X be a normal variety over a field of characteristic zero, $f : Y \to X$ a projective birational morphism from a normal variety Y, and E a prime divisor on Y such that $-E$ is $\mathbb{Q}$-Cartier and f-ample. Let r be a positive integer such that rE is Cartier. Then f is an isomorphism over $X \backslash f(E)$, $f_*\mathcal{O}_Y(-mE) = \mathfrak{a}_m(\operatorname{ord}_E)$ for any integer $m \geq 0$, the $\mathcal{O}_X$-algebra $\bigoplus_{m \geq 0} \mathfrak{a}_{mr}(\operatorname{ord}_E)$ is finitely generated, and there exists an isomorphism*

$$Y \cong \operatorname{Proj}_X \bigoplus_{m \geq 0} \mathfrak{a}_{mr}(\operatorname{ord}_E)$$

over X under which $\mathcal{O}(m)$ corresponds to $\mathcal{O}_Y(-mrE)$ for any sufficiently large integer m. In particular, f is determined by ord_E, and if $f(E)$ is a $\mathbb{Q}$-Cartier prime divisor on X, then f is an isomorphism.

Proof. Let C be a curve on Y such that $f(C)$ is a point. Since $-E$ is f-ample, $(E \cdot C) < 0$, so $C \subset E$. Thus the fibers of f over closed points of $X \backslash f(E)$ contain no curve, and hence are finite. By [Har77, Exercise II.3.22(d)], the set of points $y \in Y$ such that some irreducible component of the fiber of f containing y has positive dimension is closed in Y. Since f is proper, the image in X of the set of such points y is closed, and it contains no closed point of $X \backslash f(E)$, so it is contained in $f(E)$. Hence the fibers of f over all points of $X \backslash f(E)$ are finite, and f is finite over $X \backslash f(E)$ by [Har77, Exercise III.11.2]. Since f is birational and X is normal, f is an isomorphism over $X \backslash f(E)$.

Since X is normal and f is proper and birational, $f_*\mathcal{O}_Y = \mathcal{O}_X$. Let $m \geq 0$ be an integer and $U \subset X$ an open subset. A section of $f_*\mathcal{O}_Y(-mE)$ over U is a rational function ϕ which is regular on $f^{-1}(U)$ with $\operatorname{ord}_E(\phi) \geq m$, that is, $\phi \in \mathcal{O}_X(U)$ with $\operatorname{ord}_E(\phi) \geq m$. Thus $f_*\mathcal{O}_Y(-mE) = \mathfrak{a}_m(\operatorname{ord}_E)$.

Since rE is Cartier, $\mathcal{O}_Y(-rE)$ is an invertible f-ample sheaf and $\mathcal{O}_Y(-rE)^{\otimes m} = \mathcal{O}_Y(-mrE)$ for any $m \geq 0$. By [Sta, Tag 01VJ] ((1)⇒(5) of Lemma 29.38.4 there), the natural morphism

$$g : Y \to \operatorname{Proj}_X \bigoplus_{m \geq 0} f_*\mathcal{O}_Y(-mrE) = \operatorname{Proj}_X \bigoplus_{m \geq 0} \mathfrak{a}_{mr}(\operatorname{ord}_E)$$

over X is an open immersion. Since f is proper and the right-hand side is separated over X, $g(Y)$ is closed. Since $\bigoplus_{m\geq 0}\mathfrak{a}_{mr}(\mathrm{ord}_E)$ is a graded $\mathcal{O}_X$-subalgebra of the polynomial ring over the function field of X in one variable, the right-hand side is integral. Thus g is surjective, hence an isomorphism. The right-hand side depends only on ord_E, so f is determined by ord_E.

The remaining assertions are local on X, so we may assume that X is affine. Then $L := \mathcal{O}_Y(-rE)$ is ample [Sta, Tag 01VK], so

$$\bigoplus_{m\geq 0} H^0(Y, L^{\otimes m}) = \bigoplus_{m\geq 0} H^0(X, \mathfrak{a}_{mr}(\mathrm{ord}_E))$$

is a finitely generated $H^0(X,\mathcal{O}_X)$-algebra [Sta, Tag 0B5T], and there exists m_0 such that $L^{\otimes m}$ is generated by global sections for any $m \geq m_0$ [Sta, Tag 01Q3]. The morphism g is the morphism $r_{L,\psi}$ of [Sta, Tag 01NK] for the graded ring $\bigoplus_{m\geq 0} H^0(Y, L^{\otimes m})$ (see [Sta, Tag 01O9]), so for any $m \geq m_0$ the map $g^*\mathcal{O}(m) \to L^{\otimes m}$ of [Sta, Tag 01NK] is an isomorphism by the last statement of [Sta, Tag 01NK] and [Sta, Tag 01N8].

Assume that $P := f(E)$ is a $\mathbb{Q}$-Cartier prime divisor on X. Since E dominates P, $\mathcal{O}_{X,P} \subset \mathcal{O}_{Y,E}$ are discrete valuation rings, both with fraction field the function field of X, so $\mathcal{O}_{X,P} = \mathcal{O}_{Y,E}$ and $\mathrm{ord}_E = \mathrm{ord}_P$. Let r' be a positive integer such that $r'P$ is Cartier. Since $rr'E$ is Cartier, we may replace r with rr'. For any $m \geq 0$, a regular function ϕ on an open subset of X satisfies $\mathrm{ord}_P(\phi) \geq mr$ if and only if $\mathrm{div}(\phi) \geq mrP$ there, so

$$\mathfrak{a}_{mr}(\mathrm{ord}_E) = \mathcal{O}_X(-mrP) = \mathcal{O}_X(-rP)^{\otimes m}.$$

Since $\mathcal{O}_X(-rP)$ is invertible, $\mathrm{Proj}_X \bigoplus_{m\geq 0}\mathcal{O}_X(-rP)^{\otimes m} = X$, and f is an isomorphism. □

Definition 2.8. Let $\mathbb{K}$ be a field, v a valuation of $\mathbb{K}$ with values in $\mathbb{R}$, $\theta = (\theta_1,\dots,\theta_s) \in \mathbb{R}^s$, and $u_1,\dots,u_s$ indeterminates. The function on $\mathbb{K}[u_1,\dots,u_s]\backslash\{0\}$ given by $\sum_m g_m u^m \to \min_m(v(g_m) + \langle m,\theta\rangle)$, where $m = (m_1,\dots,m_s)$ runs over $\mathbb{N}^s$ and $u^m := u_1^{m_1}\cdots u_s^{m_s}$, extends uniquely to a valuation v_θ of $\mathbb{K}(u_1,\dots,u_s)$ [Bou89, Chapter VI, §10, no. 1, Lemma 1] (by induction on s, adjoining $u_1,\dots,u_s$ one at a time), called the *Gauss extension* of v with weights θ.

2.4. **Associated graded rings.**

Definition 2.9. Let $(X \ni x, B)$ be a klt pair over a field of characteristic zero with x a closed point, and let v be a valuation on X with $\mathrm{center}_X v = x$. The *associated graded ring* of v is

$$\mathrm{gr}_v \mathcal{O}_{X,x} := \bigoplus_{\lambda\in\mathbb{R}_{\geq 0}} \mathfrak{a}_\lambda(v)\mathcal{O}_{X,x}/\mathfrak{a}_{>\lambda}(v)\mathcal{O}_{X,x},$$

where $\mathfrak{a}_{>\lambda}(v) := \{f \in \mathcal{O}_X \mid v(f) > \lambda\}$, and for a non-zero $f \in \mathcal{O}_{X,x}$ we write $\mathrm{in}_v(f)$ for the image of f in the degree $v(f)$ part. Assume that $\mathrm{gr}_v \mathcal{O}_{X,x}$ is finitely generated, and let $X_0 := \mathrm{Spec}\,\mathrm{gr}_v \mathcal{O}_{X,x}$. For a prime divisor D on X with ideal sheaf I_D, let D_0 be the divisorial part, counted with multiplicities, of the closed subscheme of X_0 defined by the ideal generated by $\mathrm{in}_v(f)$, $f \in I_D\mathcal{O}_{X,x}\backslash\{0\}$. In particular, $D_0 = 0$ if $x \notin D$. If $B = \sum_i b_i B_i$ with prime divisors B_i, we set $B_0 := \sum_i b_i (B_i)_0$, and we call (X_0, B_0) the *marked graded pair* of v. We say that (X_0, B_0) is *klt* if X_0 is normal, $K_{X_0} + B_0$ is $\mathbb{R}$-Cartier, and $a(F, X_0, B_0) > 0$ for any prime divisor F over X_0.

Since the degree zero part of $\mathrm{gr}_v \mathcal{O}_{X,x}$ is $\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$, the ideal

$$\bigoplus_{\lambda>0} \mathfrak{a}_\lambda(v)\mathcal{O}_{X,x}/\mathfrak{a}_{>\lambda}(v)\mathcal{O}_{X,x}$$

of $\mathrm{gr}_v \mathcal{O}_{X,x}$ is maximal, and the corresponding closed point of X_0 is called the *vertex* of X_0. Since $\mathrm{in}_v(fg) = \mathrm{in}_v(f)\,\mathrm{in}_v(g)$ for any non-zero $f, g \in \mathcal{O}_{X,x}$, $\mathrm{gr}_v \mathcal{O}_{X,x}$ is an integral domain, and the function sending a non-zero element $\sum_\lambda f_\lambda$, where f_λ is homogeneous of degree λ, to $\min\{\lambda \mid f_\lambda \neq 0\}$ extends to a valuation on X_0 with center the vertex, called the *valuation given by the grading*.

The pair (X_0, B_0) is the central fiber (X_0, Δ_v) of the degeneration of (X, B) induced by v in [XZ25, Theorem 4.1]: for any prime divisor B_i, $\mathbb{Q}$-Cartier or not, $(B_i)_0$ is $\Delta_{v,i}$ defined in the paragraph after [XZ25, Theorem 4.1]. The embedded and lower-dimensional components of the closed subscheme in Definition 2.9 have codimension at least two in X_0, so they do not change the Weil divisor B_0.

The boundary B_0 keeps the coefficients of B: in Theorem 12.12, klt-ness of (X_0, B_0) is what gives Kollár components of $(X \ni x, B)$ rather than of $(X \ni x, 0)$.

Theorem 2.10 (cf. [XZ25, Theorem 4.1])**.** *Let $(X \ni x, B)$ be a klt pair over an algebraically closed field of characteristic zero such that x is a closed point and B is a $\mathbb{Q}$-divisor, and let E be a prime divisor over X with $\operatorname{center}_X E = x$. Then E is a Kollár component of $(X \ni x, B)$ if and only if $\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of ord_E is klt.*

Proof. If $\dim X = 1$, then X is regular [Har77, Theorem I.6.2A] and $E = x$ (as in the last paragraph of the proof of Lemma 2.7). In this case E is a Kollár component of $(X \ni x, B)$: the prime extraction of E is the identity (see the paragraph after Definition 2.3), and the pair in Definition 2.3 is plt near E since there is no exceptional prime divisor over X. Moreover, $\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x} = \kappa(x)[t]$, where t is the image in degree one of a uniformizer of the discrete valuation ring $\mathcal{O}_{X,x}$, is finitely generated, and the marked graded pair of ord_E is the klt pair $(\mathbb{A}^1, bo)$, where o is the origin and $b := \operatorname{mult}_x B < 1$ since $(X \ni x, B)$ is klt. Hence we may assume that $\dim X \geq 2$, so E is exceptional over X. We assemble Theorem 2.10 from several results of [XZ25]. After replacing X with an affine neighborhood of x [XZ25, Section 2], a Kollár model of $(X \ni x, B)$ is a projective birational morphism $\pi : Y \to X$, isomorphic over $X \backslash \{x\}$, together with $F := \operatorname{Exc}(\pi)$, such that $(Y, \pi_*^{-1}B + F)$ is qdlt [XZ25, Definition 2.2] and $-(K_Y + \pi_*^{-1}B + F)$ is π-ample [XZ25, Definition 3.7]. At a closed point, Definition 2.3 agrees with the definition of Kollár components in [XZ25, Section 2.4] by Lemma 2.4(1), since $-(K_Y + f_*^{-1}B + E) \equiv_f -a(E, X, B)E$ for the prime extraction $f : Y \to X$ of E. By the paragraph after Definition 2.9, the central fiber in [XZ25, Theorem 4.1] is the marked graded pair. The divisorial valuation ord_E is quasi-monomial. If $\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of ord_E is klt, then by [XZ25, Lemma 4.3], which is the implication (1)⇒(2) of [XZ25, Theorem 4.1], ord_E is a monomial lc place of a special $\mathbb{Q}$-complement Γ of $(X \ni x, B)$ with respect to a log smooth model (Y_1, F_1) [XZ25, Definition 3.1]. By [XZ25, Theorem 3.14], applied to the rational simplex $\sigma := \mathbb{R}_{\geq 0} \operatorname{ord}_E \subset \operatorname{LC}(\Gamma; Y_1, F_1)$ spanned by the single divisorial valuation ord_E, there exists a model (Y', E') of (X, B) of qdlt Fano type [XZ25, Definition 3.5] with $\operatorname{QM}(Y', E') = \sigma$ (see [XZ25, Section 2.2.2]), so E' is the prime divisor E ($E' \neq 0$ since $\operatorname{QM}(Y', E')$ contains the non-trivial valuation ord_E, each component E'_i of E' gives $\operatorname{ord}_{E'_i} \in \operatorname{QM}(Y', E') = \mathbb{R}_{\geq 0} \operatorname{ord}_E$, and $\operatorname{ord}_{E'_i} = \operatorname{ord}_E$ since both have value group $\mathbb{Z}$). Since $\operatorname{center}_X E = x$, [XZ25, Proposition 3.9] turns (Y', E) into a Kollár model of $(X \ni x, B)$ whose exceptional divisor is E, and E is a Kollár component of $(X \ni x, B)$ by [XZ25, Lemma 3.8]. Conversely, if E is a Kollár component of $(X \ni x, B)$, then the prime extraction of E (Definition 2.3) is a Kollár model by [XZ25, Lemma 3.8], and (3)⇒(1) of [XZ25, Theorem 4.1] implies that $\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of ord_E is klt. □

Theorem 2.10 turns the condition of being a Kollár component into finite generation of the graded ring and klt-ness of the marked graded pair. We use Theorem 2.10 in Sections 5, 10, 11, and 12.

In Section 5, the minimizing valuations on slices are reduced to minimizers of $\widehat{\operatorname{vol}}_{X,B}$ for klt pairs $(X \ni x, B)$ with real coefficients. Thus we state the following theorem for $\mathbb{R}$-divisors B.

Theorem 2.11 ([HLQ20, Theorems 3.3 and 3.4], [XZ25, Theorem 1.2], and [Zhu24c, Section 2.6])**.** *Let $(X \ni x, B)$ be a klt pair over $\mathbb{C}$ with x a closed point. Then there exists a quasi-monomial valuation w on X with $\operatorname{center}_X w = x$ which minimizes $\widehat{\operatorname{vol}}_{X,B}$ among the valuations v on X with $\operatorname{center}_X v = x$, and w is unique up to multiplication by a positive real number. Moreover, $\operatorname{gr}_w \mathcal{O}_{X,x}$ is finitely generated, and the marked graded pair (X_0, B_0) of w is klt.*

The existence of w, the quasi-monomiality of w, and the uniqueness up to multiplication by a positive real number are [HLQ20, Theorems 3.3 and 3.4]. The finite generation of $\mathrm{gr}_w \mathcal{O}_{X,x}$ and the klt-ness of (X_0, B_0) are [XZ25, Theorem 1.2(1)] for $\mathbb{Q}$-divisors B. They hold for $\mathbb{R}$-divisors B by [Zhu24c, Section 2.6, the sentence before Theorem 2.19], and the central fiber there is a log Fano cone singularity, hence klt by [Zhu24c, Definition 2.10] (klt-ness near the vertex suffices: the non-normal locus of X_0, the locus where $K_{X_0} + B_0$ is not $\mathbb{R}$-Cartier, and the non-klt locus of (X_0, B_0) are closed and invariant under the torus of the grading, and the closure of any orbit contains the vertex, so they are empty). Here the boundary Δ_0 in [Zhu24c, Theorem 2.19] is the degeneration of B in the sense of [XZ25, Theorem 1.2 and the paragraph after Theorem 4.1], that is, B_0 of Definition 2.9.

3. Change of the ground field

In this section, we compare pairs and Kollár components over a field k finitely generated over $\mathbb{C}$ with those over an algebraic closure $\bar{k}$ of k. A Kollár component over $\bar{k}$ is in general not defined over k. Thus we only descend a valuation that is already determined and invariant under $\mathrm{Gal}(\bar{k}/k)$. We never replace a Kollár component E' over $\bar{k}$ by the sum of the Galois conjugates of E', since at a k-rational point the base change of a Kollár component over k is a prime divisor (Lemma 3.6(1)).

Definition 3.1. Let X be a variety over a field k, L/k a field extension, and B an $\mathbb{R}$-divisor on X. We define $X_L := X \times_k \operatorname{Spec} L$, and we define B_L to be the pullback of B to X_L.

3.1. **Extension of constants.** In this subsection, we show that algebraic extensions of the ground field preserve log discrepancies and the singularities of pairs, and that at a rational point a Kollár component over $\bar{k}$ with Galois invariant valuation descends to k.

Lemma 3.2. *Let k be a field finitely generated over $\mathbb{C}$ and $\bar{k}$ an algebraic closure of k. Then $\bar{k}$ is isomorphic to $\mathbb{C}$ as a field.*

Proof. The fields $\bar{k}$ and $\mathbb{C}$ are algebraically closed of characteristic zero. Since k is generated over $\mathbb{C}$ by finitely many elements, the transcendence degree of $\bar{k}$ over $\mathbb{Q}$ is equal to the transcendence degree of $\mathbb{C}$ over $\mathbb{Q}$, which is the cardinality of $\mathbb{C}$. Thus $\bar{k} \cong \mathbb{C}$ by [Ste10]. □

Fix a field isomorphism $\tau : \bar{k} \to \mathbb{C}$. Base change along τ sends a variety, an $\mathbb{R}$-divisor, or a morphism over $\bar{k}$ to a variety, an $\mathbb{R}$-divisor, or a morphism over $\mathbb{C}$, and the scheme over $\bar{k}$ is isomorphic to the scheme over $\mathbb{C}$ as an abstract scheme. Hence the notions of Definitions 2.1 and 2.3 and the log discrepancies over $\bar{k}$ correspond to the complex ones. The isomorphism τ need not fix the subfield $\mathbb{C}$ of k. This does not matter: Definitions 2.1 and 2.3 only involve the structure of schemes, and the real coefficients of a boundary are formal coefficients. When we use a result on varieties over $\mathbb{C}$ for varieties over $\bar{k}$, we refer to Lemma 3.2.

For a normal variety X over k and an algebraic extension L/k, the scheme X_L is of finite type over L and normal by Lemma 3.3(1) below, so X_L is a finite disjoint union of normal varieties over L. When X is not geometrically integral, X_L may be disconnected. In this case, a prime divisor over X_L means a prime divisor over a connected component of X_L, and we say that (X_L, B_L) is lc (resp. klt, plt) if the restriction of (X_L, B_L) to any connected component of X_L is lc (resp. klt, plt).

Lemma 3.3. *Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k, L/k an algebraic field extension, and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Then the following hold.*

(1) X_L is normal, and for any prime divisor E on a normal variety Y over k birational to X, the pullback of E to Y_L is a reduced divisor along each component E_1 of which a local equation of E has order one and $\mathrm{ord}_{E_1}|_{k(Y)} = \mathrm{ord}_E$.

(2) *For any normal variety Y over k birational to X, the pullback to Y_L of a canonical divisor K_Y of Y is a canonical divisor of Y_L, that is, $K_{Y_L} = (K_Y)_L$. For any prime divisor E' over X_L, the restriction of $\mathrm{ord}_{E'}$ to $k(X)$ is ord_E for a prime divisor E over X, and $a(E', X_L, B_L) = a(E, X, B)$.*

(3) *(X, B) is lc (resp. klt, plt) if and only if (X_L, B_L) is lc (resp. klt, plt).*

Proof. (1) Since L/k is separable, X_L and Y_L are normal [Sta, Tag 0C3M]. Let η be the generic point of E and t a local equation of E at η, that is, a generator of the maximal ideal of the discrete valuation ring $\mathcal{O}_{Y,\eta}$. The morphism $Y_L \to Y$ is flat and integral. Thus the points of Y_L over η are the generic points of the irreducible components of the preimage of E, and they have codimension one in Y_L [Mat89, Theorem 15.1(ii)]. Let η' be a point of Y_L over η. Then $\mathcal{O}_{Y_L,\eta'}$ is a discrete valuation ring, and $\mathcal{O}_{Y_L,\eta'}/t\mathcal{O}_{Y_L,\eta'}$ is the localization of $\kappa(\eta) \otimes_k L$ at a prime ideal. Any finite subextension l/k of L/k is of the form $l \cong k[s]/(p)$ for a separable polynomial p, so $\kappa(\eta) \otimes_k l \cong \kappa(\eta)[s]/(p)$ is reduced. Hence $\kappa(\eta) \otimes_k L$, the union of the rings $\kappa(\eta) \otimes_k l$, is reduced. Moreover, $\dim(\kappa(\eta) \otimes_k L) = 0$ since $\kappa(\eta) \otimes_k L$ is integral over $\kappa(\eta)$. Thus the localizations of $\kappa(\eta) \otimes_k L$ at prime ideals are fields, so t generates the maximal ideal of $\mathcal{O}_{Y_L,\eta'}$. Let E_1 be the prime divisor on Y_L with generic point η'. Any $g \in k(Y)\backslash\{0\}$ can be written as $g = ut^m$, where $m = \mathrm{ord}_E(g)$ and u is a unit of $\mathcal{O}_{Y,\eta}$. Since $\mathcal{O}_{Y,\eta} \to \mathcal{O}_{Y_L,\eta'}$ is a local homomorphism, u is a unit of $\mathcal{O}_{Y_L,\eta'}$, so $\mathrm{ord}_{E_1}(g) = m \,\mathrm{ord}_{E_1}(t) = m$. Thus

$$\mathrm{ord}_{E_1}|_{k(Y)} = \mathrm{ord}_E .$$

This proves (1).

(2) Let $d := \dim X$. We first show that $K_{Y_L} = (K_Y)_L$ for any normal variety Y over k birational to X and any canonical divisor K_Y of Y. Let ω be a non-zero rational section of $\Omega^d_{Y/k}$ with $K_Y = \mathrm{div}(\omega)$. Since $\Omega_{Y_L/L}$ is the pullback of $\Omega_{Y/k}$, the pullback ω_L of ω is a non-zero rational section of $\Omega^d_{Y_L/L}$. Any codimension one point of Y_L maps to a codimension one point of Y [Mat89, Theorem 15.1(ii)], at which Y is smooth and $\Omega^d_{Y/k}$ is free of rank one. By (1), $\mathrm{div}(\omega_L)$ is the pullback of $\mathrm{div}(\omega)$, that is, $K_{Y_L} = (K_Y)_L$.

Let X' be the connected component of X_L over which E' is a prime divisor, K' the function field of X', and $v := \mathrm{ord}_{E'}$. The morphism $X' \to X$ is flat, so K' contains $k(X)$, and K' is generated by $k(X)$ and L. In particular, K' is algebraic over $k(X)$. Let w be the restriction of v to $k(X)$.

We show that w is a non-zero multiple of a divisorial valuation of $k(X)$. Let $g \in K'\backslash\{0\}$. There exist $b_0, \dots, b_p \in k(X)$, not all zero, such that $\sum_{i=0}^p b_i g^i = 0$. At least two of the non-zero terms $b_i g^i$ attain the minimum of $v(b_i g^i)$, so $(i-j)v(g) \in w(k(X)\backslash\{0\})$ for some $i \neq j$. Since $v(K'\backslash\{0\}) = \mathbb{Z}$, $w(k(X)\backslash\{0\})$ is a non-zero subgroup of $\mathbb{Z}$. If $v(g) = 0$, then dividing the relation by a non-zero b_j with $w(b_j)$ minimal and reducing modulo the maximal ideal of the valuation ring of v, we see that the residue class of g is algebraic over the residue field of w. The residue field of v is the function field of E', which has transcendence degree $d-1$ over L, hence over k. Thus the residue field of w has transcendence degree $d-1$ over k. Moreover, w has a center on X, namely the image of the center of v on X'. By [KM98, Lemma 2.45], after taking a normalization, there exist a normal variety Y, a projective birational morphism $h : Y \to X$, and a prime divisor E on Y with generic point η such that the valuation ring of w is $\mathcal{O}_{Y,\eta}$. Hence $w = c\,\mathrm{ord}_E$ for a positive integer c.

We show that $c = 1$. The restriction of $h_L : Y_L \to X_L$ over X' is proper and birational, so v has a center y' on Y_L, and y' maps to the center η of w on Y. Since $Y_L \to Y$ is flat with zero-dimensional fibers, y' has codimension one in Y_L [Mat89, Theorem 15.1(ii)], so y' is the generic point of a component E_1 of the pullback of E to Y_L, and $\mathcal{O}_{Y_L,y'}$ is a discrete valuation ring of K'. The valuation ring of v dominates $\mathcal{O}_{Y_L,y'}$, so $\mathcal{O}_{Y_L,y'}$ is the valuation ring of v, and $v = \mathrm{ord}_{E_1}$. By (1), $w = \mathrm{ord}_{E_1}|_{k(X)} = \mathrm{ord}_E$, that is, $c = 1$.

We show that $a(E', X_L, B_L) = a(E, X, B)$. Let ω be a non-zero rational section of $\Omega^d_{X/k}$, so that $K_X = \mathrm{div}(\omega)$, and let $K_Y := \mathrm{div}(h^*\omega)$. Applying the equality $K_{Y_L} = (K_Y)_L$ shown above

to X with ω, we obtain $K_{X_L} := \operatorname{div}(\omega_L) = (K_X)_L$. Applying it to Y with $h^*\omega$, we obtain $K_{Y_L} := \operatorname{div}(h_L^*\omega_L) = (K_Y)_L$. Write $K_Y + B_Y := h^*(K_X + B)$. Since the pullback of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors commutes with base change,

$$K_{Y_L} + (B_Y)_L = h_L^*(K_{X_L} + B_L).$$

By (1), $\operatorname{mult}_{E_1}(B_Y)_L = \operatorname{mult}_E B_Y$. Since $\operatorname{ord}_{E'} = \operatorname{ord}_{E_1}$,

$$a(E', X_L, B_L) = a(E_1, X_L, B_L) = 1 - \operatorname{mult}_{E_1}(B_Y)_L = 1 - \operatorname{mult}_E B_Y = a(E, X, B).$$

(3) Write $B = \sum_i b_i B_i$, where the B_i are distinct prime divisors. By (1), $B_L = \sum_i b_i (B_i)_L$, where each $(B_i)_L$ is reduced, and $(B_i)_L$ and $(B_j)_L$ have no common component for $i \neq j$, since the generic points of the components of $(B_i)_L$ map to the generic point of B_i. Thus the coefficient of B_L along any component of $(B_i)_L$ is b_i. In particular, $B \geq 0$ if and only if $B_L \geq 0$, and $K_{X_L} + B_L = (K_X + B)_L$ is $\mathbb{R}$-Cartier.

For any prime divisor E' over X_L, (2) provides a prime divisor E over X with $\operatorname{ord}_{E'}|_{k(X)} = \operatorname{ord}_E$ and $a(E', X_L, B_L) = a(E, X, B)$. Conversely, let E be a prime divisor on a normal variety Y with a projective birational morphism $Y \to X$, and let E_1 be a component of the pullback of E to Y_L. By (1), $\operatorname{ord}_{E_1}|_{k(X)} = \operatorname{ord}_E$, so (2) implies $a(E_1, X_L, B_L) = a(E, X, B)$, since the log discrepancy only depends on the valuation. If $\operatorname{ord}_{E'}|_{k(X)} = \operatorname{ord}_E$, then the center ζ' of $\operatorname{ord}_{E'}$ on X_L maps to the center ζ of ord_E on X, and $\dim \mathcal{O}_{X_L,\zeta'} = \dim \mathcal{O}_{X,\zeta}$ by [Mat89, Theorem 15.1(ii)], since $X_L \to X$ is flat with zero-dimensional fibers. Thus E' is exceptional over X_L if and only if E is exceptional over X. Being lc, klt, or plt is a condition on the coefficients of the boundary, on the log discrepancies of the prime divisors over the variety, and on which of the prime divisors over the variety are exceptional (Definition 2.1, and cf. [KM98, Definition 2.34] for plt). Hence (3) follows. □

Corollary 6.7, Lemma 7.3, and Theorem 7.4 use the following form of the Bertini theorem to choose general members over k.

Lemma 3.4. *Let k be an infinite field of characteristic zero, W a smooth variety over k, Σ an snc divisor on W, $\mathcal{L}$ an invertible sheaf on W, and $V \subset H^0(W, \mathcal{L})$ a k-linear subspace with $\dim_k V < +\infty$ which generates $\mathcal{L}$. Then there exists a non-empty open subset V° of the affine space V over k with $0 \notin V^\circ$ such that $V^\circ(k) \neq \emptyset$ and, for any $s \in V^\circ(k)$, the divisor of zeros $\operatorname{div}(s)$ of the section s of $\mathcal{L}$ is smooth over k, has no common component with Σ, and $\Sigma + \operatorname{div}(s)$ is an snc divisor.*

Proof. Let $\bar{k}$ be an algebraic closure of k and $r := \dim_k V$, and regard V as the affine space $\mathbb{A}_k^r$. By [Har77, Corollary III.10.9 and Remark III.10.9.2], applied over $\bar{k}$ to $W_{\bar{k}}$ and to each stratum of $\Sigma_{\bar{k}}$, the non-zero $s \in V \otimes_k \bar{k}$ such that $\operatorname{div}(s)$ is smooth, has no common component with $\Sigma_{\bar{k}}$, and $\Sigma_{\bar{k}} + \operatorname{div}(s)$ is an snc divisor contain a non-empty open subset $\bar{U}_0$ of $\mathbb{A}_{\bar{k}}^r$. Since W, Σ, $\mathcal{L}$, and V are defined over k, the set of s with the three properties above is stable under $\operatorname{Gal}(\bar{k}/k)$, so $\bar{U} := \bigcup_{\sigma \in \operatorname{Gal}(\bar{k}/k)} \sigma(\bar{U}_0)$ is an open subset of $\mathbb{A}_{\bar{k}}^r$ stable under $\operatorname{Gal}(\bar{k}/k)$ on which the three properties hold. By [Sta, Tag 038B], applied to the complement of $\bar{U}$, $\bar{U}$ is the preimage of a non-empty open subset V° of $\mathbb{A}_k^r$, and $0 \notin V^\circ$. Since k is infinite, $V^\circ(k) \neq \emptyset$. For $s \in V^\circ(k)$, the base change of $\operatorname{div}(s)$ to $\bar{k}$ is $\operatorname{div}(s \otimes 1)$, which is smooth, so $\operatorname{div}(s)$ is smooth over k, and it has no common component with Σ. The components of Σ and of $\operatorname{div}(s)$ are smooth over k, and they meet transversally since they do so after the base change to $\bar{k}$. Thus $\Sigma + \operatorname{div}(s)$ is an snc divisor. □

Lemma 3.5. *Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k, and k_0 the algebraic closure of k in the function field of X. Then the following hold.*

(1) k_0 is a finite extension of k contained in $\Gamma(U, \mathcal{O}_X)$ for any non-empty open affine subset $U \subset X$, and X is geometrically integral as a variety over k_0. The canonical divisors of X over k and over k_0 coincide. For any $\mathbb{R}$-divisor B on X, (X, B) is a klt pair over k

if and only if it is a klt pair over k_0, and in this case, for any point $x \in X$, the Kollár components of $(X \ni x, B)$ over k and over k_0 coincide.

(2) If X has a k-rational point, then $k_0 = k$ and X is geometrically integral over k.

Proof. (1) Let $t_1, \dots, t_d$ be a transcendence basis of $k(X)$ over k. Since $t_1, \dots, t_d$ are algebraically independent over any algebraic extension of k, we have

$$[l : k] = [l(t_1, \dots, t_d) : k(t_1, \dots, t_d)] \leq [k(X) : k(t_1, \dots, t_d)]$$

for any finite subextension l/k of k_0/k. Hence k_0/k is finite. Any element of k_0 is integral over $k \subset \Gamma(U, \mathcal{O}_X)$, and $\Gamma(U, \mathcal{O}_X)$ is a normal domain with fraction field $k(X)$, so $k_0 \subset \Gamma(U, \mathcal{O}_X)$. Thus X is a variety over k_0. Since k_0 is algebraically closed in $k(X)$ and has characteristic zero, X is geometrically irreducible over k_0 [Gro65, Proposition 4.5.9]. Moreover, X is geometrically normal over k_0 [Sta, Tag 038O]. Hence X is geometrically integral over k_0 (see [Har77, Exercise II.3.15(c)] for the definition).

Since k_0/k is finite and separable, $\Omega_{k_0/k} = 0$ [Har77, Theorem II.8.6A], so the natural map $\Omega_{X/k} \to \Omega_{X/k_0}$ is an isomorphism [Har77, Proposition II.8.11]. Thus the canonical divisors of X over k and over k_0 coincide.

Let Y be a normal variety over k with a projective birational morphism $h : Y \to X$. Applying the argument above to Y, whose function field is $k(X)$, we obtain $k_0 \subset \Gamma(V, \mathcal{O}_Y)$ for any non-empty open affine subset $V \subset Y$. Since h induces the identity of $k(X)$, h is a morphism over k_0. Conversely, a variety over k_0 is a variety over k, since k_0/k is finite. Thus the set of prime divisors over X over k coincides with the set of prime divisors over X over k_0, and the canonical divisors of Y over k and over k_0 coincide, so for any prime divisor E over X, the log discrepancy $a(E, X, B)$ computed over k equals $a(E, X, B)$ computed over k_0. Normality, the condition $B \geq 0$, and the condition that $K_X + B$ is $\mathbb{R}$-Cartier do not depend on the base field. Hence (X, B) is a klt pair over k if and only if it is a klt pair over k_0. In this case, the center of a prime divisor over X, primitivity, the prime extraction, and the plt condition near the prime divisor do not depend on whether we work over k or over k_0, so the Kollár components of $(X \ni x, B)$ over k and over k_0 coincide.

(2) Let $x \in X$ be a k-rational point and U a non-empty open affine subset of X containing x. By (1), $k \subset k_0 \subset \Gamma(U, \mathcal{O}_X)$. The composite

$$k_0 \to \Gamma(U, \mathcal{O}_X) \to \kappa(x) = k$$

is a homomorphism of fields over k, hence injective. Thus $k_0 = k$, and X is geometrically integral over k by (1). □

When $X_{\bar{k}}$ is integral, $\mathrm{Gal}(\bar{k}/k)$ acts on $X_{\bar{k}}$ and on $\mathrm{Frac}(k(X) \otimes_k \bar{k})$ through the second factor, and for a valuation w of $\mathrm{Frac}(k(X) \otimes_k \bar{k})$ and $\sigma \in \mathrm{Gal}(\bar{k}/k)$, we denote by $w \circ \sigma$ the composite of σ and w.

Lemma 3.6(2) answers the following question: when does a Kollár component over $\bar{k}$ whose valuation is invariant under $\mathrm{Gal}(\bar{k}/k)$ come from a prime extraction over k? Later, the valuation over $\bar{k}$ is invariant under $\mathrm{Gal}(\bar{k}/k)$ for one of three reasons: its valuation ideals are the base changes of the valuation ideals of a valuation over k (Lemma 10.23), it is the restriction, along a projection defined over k, of the base change of a Kollár component over k (Lemma 11.5), or it is attached to the base change of a Kollár component over k without any choice (Proposition 13.2). We need the point x to be k-rational, and a general closed point is first reduced to a rational point by Corollary 3.10.

Lemma 3.6. *Let k be a field finitely generated over $\mathbb{C}$, and $(X \ni x, B)$ a klt pair over k such that x is a k-rational point. Let $\bar{x} \in X_{\bar{k}}$ be the point over x.*

(1) Let E be a Kollár component of $(X \ni x, B)$ with prime extraction f. Then $E_{\bar{k}}$ is a prime divisor, $E_{\bar{k}}$ is a Kollár component of $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$ with prime extraction $f_{\bar{k}}$, $\mathrm{ord}_{E_{\bar{k}}}|_{k(X)} = \mathrm{ord}_E$, $a(E_{\bar{k}}, X_{\bar{k}}, B_{\bar{k}}) = a(E, X, B)$, and $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x} \otimes_k \bar{k} \cong \mathrm{gr}_{\mathrm{ord}_{E_{\bar{k}}}} \mathcal{O}_{X_{\bar{k}},\bar{x}}$.

(2) *Let E' be a Kollár component of $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$ such that $\mathrm{ord}_{E'} \circ \sigma = \mathrm{ord}_{E'}$ for any $\sigma \in \mathrm{Gal}(\bar{k}/k)$. Then $\mathrm{ord}_{E'}|_{k(X)} = \mathrm{ord}_E$ for a Kollár component E of $(X \ni x, B)$ with $E_{\bar{k}} = E'$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix notation and prove the lemma when $\dim X = 1$.

Since x is a k-rational point, X is geometrically integral by Lemma 3.5(2), and $X_{\bar{k}}$ is a normal variety by Lemma 3.3(1), with function field $\mathrm{Frac}(k(X) \otimes_k \bar{k})$. The ring $k(X) \otimes_k \bar{k}$ is a domain, but in general not a field. The fiber of $X_{\bar{k}} \to X$ over x is $\mathrm{Spec}(\kappa(x) \otimes_k \bar{k}) = \mathrm{Spec}\, \bar{k}$, so $\bar{x}$ is the only point of $X_{\bar{k}}$ over x, and $\bar{x}$ is a closed point. Since $X_{\bar{k}} \to X$ is integral, hence closed, any open neighborhood of $\bar{x}$ contains the preimage of an open neighborhood of x. Since the statements are local around x, possibly shrinking X around x, we may assume that X is affine and that (X, B) is a klt pair. Then $(X_{\bar{k}}, B_{\bar{k}})$ is a klt pair by Lemma 3.3(3).

Let $R := \mathcal{O}_{X,x}$ with maximal ideal $\mathfrak{m}_x$. The ring $R \otimes_k \bar{k}$ is integral over R, so the maximal ideals of $R \otimes_k \bar{k}$ contain $\mathfrak{m}_x(R \otimes_k \bar{k})$. Since $(R \otimes_k \bar{k})/\mathfrak{m}_x(R \otimes_k \bar{k}) \cong \kappa(x) \otimes_k \bar{k} = \bar{k}$, the ring $R \otimes_k \bar{k}$ is local with maximal ideal $\mathfrak{m}_x(R \otimes_k \bar{k})$. Thus

$$\bar{R} := \mathcal{O}_{X_{\bar{k}},\bar{x}} = R \otimes_k \bar{k}.$$

Assume that $\dim X = 1$. Then R is a discrete valuation ring, and a projective birational morphism from a normal curve to an open subset of X is finite [Har77, Exercise III.11.2], hence an isomorphism. Thus the only prime divisor over X with center x is $\{x\}$, and the identity morphism is the prime extraction of $\{x\}$. Since a projective birational morphism from a normal curve to an open subset of $X_{\bar{k}}$ is also an isomorphism [Har77, Exercise III.11.2], the only prime divisor over $X_{\bar{k}}$ with center $\bar{x}$ is $\{\bar{x}\}$. By Lemma 3.3(1), the pullback of $\{x\}$ to $X_{\bar{k}}$ is the reduced divisor $\{\bar{x}\}$, and a generator t of $\mathfrak{m}_x$ generates the maximal ideal of $\bar{R}$. Let $B^\circ := B - (\mathrm{mult}_x B)\{x\}$. Then $\{x\}$ is a Kollár component of $(X \ni x, B)$ if and only if $(X, B^\circ + \{x\})$ is plt near x, and $\{\bar{x}\}$ is a Kollár component of $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$ if and only if $(X_{\bar{k}}, B^\circ_{\bar{k}} + \{\bar{x}\})$ is plt near $\bar{x}$. By Lemma 3.3(3) and the choice of neighborhoods above, $(X, B^\circ + \{x\})$ is plt near x if and only if $(X_{\bar{k}}, B^\circ_{\bar{k}} + \{\bar{x}\})$ is plt near $\bar{x}$. Hence $\{x\}$ is a Kollár component of $(X \ni x, B)$ if and only if $\{x\}_{\bar{k}} = \{\bar{x}\}$ is a Kollár component of $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$. Moreover, $\mathrm{ord}_{\bar{x}}|_{k(X)} = \mathrm{ord}_x$ since t generates both maximal ideals, $a(\{\bar{x}\}, X_{\bar{k}}, B_{\bar{k}}) = 1 - \mathrm{mult}_x B = a(\{x\}, X, B)$ by Lemma 3.3(1), and

$$\mathrm{gr}_{\mathrm{ord}_x} R \otimes_k \bar{k} \cong k[T] \otimes_k \bar{k} \cong \bar{k}[T] \cong \mathrm{gr}_{\mathrm{ord}_{\bar{x}}} \bar{R},$$

where T is the class of t. This proves (1) and (2) when $\dim X = 1$. From now on, we assume that $d := \dim X \geq 2$.

Step 2. In this step we prove (1).

Let $f : Y \to U$ be the prime extraction of E, where U is an open neighborhood of x. Since the statements are local around x, possibly replacing X with an affine open neighborhood $X^\circ \subset U$ of x and Y with $f^{-1}(X^\circ)$, we may assume that $U = X$. Since x is a closed point and $d \geq 2$, E is f-exceptional, and $(Y, f_*^{-1}B + E)$ is plt near E. By Lemma 2.4(1), f is an isomorphism over $X \backslash \{x\}$ and $\mathrm{Supp}\, f^{-1}(x) = E$. Since $k(Y) = k(X)$, Lemma 3.5(2) shows that k is algebraically closed in $k(Y)$, so Y is geometrically integral by Lemma 3.5(1), and $Y_{\bar{k}}$ is normal by Lemma 3.3(1). Thus $f_{\bar{k}} : Y_{\bar{k}} \to X_{\bar{k}}$ is a projective birational morphism between normal varieties. Since $\bar{x}$ is the only point over x, $f_{\bar{k}}^{-1}(\bar{x}) = f^{-1}(x) \times_k \bar{k}$, so $\mathrm{Supp}\, f_{\bar{k}}^{-1}(\bar{x}) = \mathrm{Supp}\, E_{\bar{k}}$. By Lemma 3.3(1), $E_{\bar{k}}$ is a reduced divisor. Let r be a positive integer such that rE is Cartier. Then $rE_{\bar{k}}$ is Cartier, and $-rE_{\bar{k}}$ is $f_{\bar{k}}$-ample since relative ampleness is preserved under base change.

Since E is the only f-exceptional prime divisor, we have

$$K_Y + f_*^{-1}B + E = f^*(K_X + B) + a(E, X, B)E,$$

which is $\mathbb{R}$-Cartier. Let $W \subset Y$ be an open neighborhood of E such that $(W, (f_*^{-1}B + E)|_W)$ is plt. The $\mathbb{R}$-divisor $(f_*^{-1}B)_{\bar{k}}$ has no component contained in $E_{\bar{k}}$ and coincides with $B_{\bar{k}}$ over $X_{\bar{k}} \backslash \{\bar{x}\}$, so $(f_*^{-1}B)_{\bar{k}} = (f_{\bar{k}})_*^{-1}B_{\bar{k}}$. By Lemma 3.3(3) applied to W, the pair $(Y_{\bar{k}}, (f_{\bar{k}})_*^{-1}B_{\bar{k}} + E_{\bar{k}})$

is plt on the open neighborhood $W_{\bar{k}}$ of $E_{\bar{k}}$. By Lemma 2.4(2), applied to $f_{\bar{k}}$, $S = E_{\bar{k}}$, and $B_Y = (f_{\bar{k}})_*^{-1}B_{\bar{k}}$, $E_{\bar{k}}$ is a prime divisor. Thus $E_{\bar{k}}$ is primitive over $X_{\bar{k}}$ with prime extraction $f_{\bar{k}}$, $\mathrm{center}_{X_{\bar{k}}}(E_{\bar{k}}) = \bar{x}$, and $E_{\bar{k}}$ is a Kollár component of the klt pair $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$.

By Lemma 3.3(1), $\mathrm{ord}_{E_{\bar{k}}}|_{k(X)} = \mathrm{ord}_E$. By Lemma 3.3(2), $a(E_{\bar{k}}, X_{\bar{k}}, B_{\bar{k}}) = a(E_2, X, B)$ for a prime divisor E_2 over X with $\mathrm{ord}_{E_2} = \mathrm{ord}_{E_{\bar{k}}}|_{k(X)} = \mathrm{ord}_E$, so $a(E_{\bar{k}}, X_{\bar{k}}, B_{\bar{k}}) = a(E, X, B)$.

It remains to compare the graded rings. Since k has characteristic zero, E is geometrically reduced [Gro65, Proposition 4.6.1], and $\mathrm{Supp}(E \times_k \bar{k}) = \mathrm{Supp}\, E_{\bar{k}}$ is irreducible. Thus $E \times_k \bar{k}$ is integral, and $\kappa(E) \otimes_k \bar{k}$ is a domain contained in the function field of $E_{\bar{k}}$, which is the residue field of $\mathrm{ord}_{E_{\bar{k}}}$. For $j \geq 0$, let

$$\mathfrak{a}_j := \mathfrak{a}_j(\mathrm{ord}_E)R \quad \text{and} \quad \bar{\mathfrak{a}}_j := \mathfrak{a}_j(\mathrm{ord}_{E_{\bar{k}}})\bar{R}.$$

We claim that $\bar{\mathfrak{a}}_j = \mathfrak{a}_j \otimes_k \bar{k}$ in $\bar{R} = R \otimes_k \bar{k}$. Since $\mathrm{ord}_{E_{\bar{k}}}|_{k(X)} = \mathrm{ord}_E$, we have $\mathfrak{a}_j \otimes_k \bar{k} \subset \bar{\mathfrak{a}}_j$. Conversely, let $h \in \bar{R}$ be a non-zero element, and write $h = \sum_{i=1}^p g_i \otimes c_i$, where $g_1, \dots, g_p \in R \backslash \{0\}$ and $c_1, \dots, c_p \in \bar{k}$ are linearly independent over k. Let $m := \min_i \mathrm{ord}_E(g_i)$, let t be a local equation of E at the generic point of E, and write $g_i = t^m u_i$ with $u_i \in \mathcal{O}_{Y,E}$, where at least one u_i is a unit of $\mathcal{O}_{Y,E}$. The image of $\sum_i u_i \otimes c_i$ in the residue field of $\mathrm{ord}_{E_{\bar{k}}}$ is the image of $\sum_i \bar{u}_i \otimes c_i \in \kappa(E) \otimes_k \bar{k}$, where $\bar{u}_i$ is the residue class of u_i. Since $1 \otimes c_1, \dots, 1 \otimes c_p$ are linearly independent over $\kappa(E)$ and some $\bar{u}_i \neq 0$, the image of $\sum_i u_i \otimes c_i$ is non-zero. Hence

$$\mathrm{ord}_{E_{\bar{k}}}(h) = m\, \mathrm{ord}_{E_{\bar{k}}}(t) = \min_i \mathrm{ord}_E(g_i).$$

Thus $h \in \bar{\mathfrak{a}}_j$ implies $g_i \in \mathfrak{a}_j$ for any i, so $h \in \mathfrak{a}_j \otimes_k \bar{k}$. This proves the claim. Since ord_E and $\mathrm{ord}_{E_{\bar{k}}}$ have value group $\mathbb{Z}$ and $- \otimes_k \bar{k}$ is exact, we obtain

$$\mathrm{gr}_{\mathrm{ord}_E} R \otimes_k \bar{k} = \bigoplus_{j \geq 0} (\mathfrak{a}_j/\mathfrak{a}_{j+1}) \otimes_k \bar{k} \cong \bigoplus_{j \geq 0} \bar{\mathfrak{a}}_j/\bar{\mathfrak{a}}_{j+1} = \mathrm{gr}_{\mathrm{ord}_{E_{\bar{k}}}} \bar{R},$$

and the isomorphism respects multiplication. This proves (1).

Step 3. In this step we descend the valuation ideals of $w := \mathrm{ord}_{E'}$, where E' is the Kollár component in (2).

Let $\mathfrak{m}_x \subset \mathcal{O}(X)$ also denote the maximal ideal of x. Since $\mathcal{O}(X_{\bar{k}})/\mathfrak{m}_x\mathcal{O}(X_{\bar{k}}) \cong \kappa(x) \otimes_k \bar{k} = \bar{k}$, the ideal $\mathfrak{m}_x\mathcal{O}(X_{\bar{k}})$ is the maximal ideal of $\bar{x}$. For $j \geq 0$, let

$$\mathfrak{a}_j(w) := \{h \in \mathcal{O}(X_{\bar{k}}) \mid w(h) \geq j\}.$$

Since $\mathrm{center}_{X_{\bar{k}}}(w) = \bar{x}$ and w is integer-valued, $w(h) \geq 1$ for any $h \in \mathfrak{m}_x\mathcal{O}(X_{\bar{k}})$, so

$$\mathfrak{m}_x^j \mathcal{O}(X_{\bar{k}}) \subset \mathfrak{a}_j(w).$$

For $\sigma \in \mathrm{Gal}(\bar{k}/k)$ and $h \in \mathfrak{a}_j(w)$, we have $w(\sigma(h)) = (w \circ \sigma)(h) = w(h) \geq j$, so $\sigma(\mathfrak{a}_j(w)) = \mathfrak{a}_j(w)$.

Fix $j \geq 1$. Since x is a k-rational point, $\dim_k \mathcal{O}(X)/\mathfrak{m}_x^j < +\infty$. Let $e_1, \dots, e_s$ be a basis of $\mathcal{O}(X)/\mathfrak{m}_x^j$ over k. We identify

$$\mathcal{O}(X_{\bar{k}})/\mathfrak{m}_x^j\mathcal{O}(X_{\bar{k}}) \cong (\mathcal{O}(X)/\mathfrak{m}_x^j) \otimes_k \bar{k}$$

with $\bar{k}^s$ through $e_1, \dots, e_s$, so that $\mathrm{Gal}(\bar{k}/k)$ acts on $\bar{k}^s$ coordinatewise. Let $\Lambda \subset \bar{k}^s$ be the image of $\mathfrak{a}_j(w)$, a $\bar{k}$-linear subspace stable under $\mathrm{Gal}(\bar{k}/k)$. The subspace Λ has a unique basis $\lambda_1, \dots, \lambda_q$ in reduced row echelon form. For $\sigma \in \mathrm{Gal}(\bar{k}/k)$, $\sigma(\lambda_1), \dots, \sigma(\lambda_q)$ is a basis of $\sigma(\Lambda) = \Lambda$ in reduced row echelon form, so $\sigma(\lambda_i) = \lambda_i$ for any i. Since $\bar{k}/k$ is a Galois extension, the coordinates of $\lambda_1, \dots, \lambda_q$ belong to k. Let $\Lambda_0 \subset \mathcal{O}(X)/\mathfrak{m}_x^j$ be the k-linear span of $\lambda_1, \dots, \lambda_q$, so that $\Lambda = \Lambda_0 \otimes_k \bar{k}$, and let $I_j \subset \mathcal{O}(X)$ be the preimage of Λ_0. Since $- \otimes_k \bar{k}$ is exact, the preimage of Λ in $\mathcal{O}(X_{\bar{k}})$ is $I_j \otimes_k \bar{k}$, that is,

$$\mathfrak{a}_j(w) = I_j \otimes_k \bar{k} = I_j \mathcal{O}(X_{\bar{k}}).$$

Writing $\bar{k} = k \oplus C'$ for a k-linear complement C' of k in $\bar{k}$, we obtain $I_j = \mathfrak{a}_j(w) \cap \mathcal{O}(X)$. We also set $I_0 := \mathcal{O}(X)$. Then for any $i, j \geq 0$,

$$I_i I_j \subset \mathfrak{a}_i(w)\mathfrak{a}_j(w) \cap \mathcal{O}(X) \subset \mathfrak{a}_{i+j}(w) \cap \mathcal{O}(X) = I_{i+j}.$$

Step 4. In this step we prove (2).

By Lemma 2.4(1), the prime extraction of E' over an open neighborhood U' of $\bar{x}$ is an isomorphism over $U'\backslash\{\bar{x}\}$. Gluing it with the identity morphism of $X_{\bar{k}}\backslash\{\bar{x}\}$, we obtain a proper birational morphism $f' : Y' \to X_{\bar{k}}$ from a normal variety Y' such that E' is a prime divisor on Y', $-E'$ is $\mathbb{Q}$-Cartier and f'-ample, and $(Y', (f')_*^{-1}B_{\bar{k}} + E')$ is plt near E'. Let r be a positive integer such that rE' is Cartier. Since f' is proper, $\mathcal{O}_{Y'}(-rE')$ is f'-ample, and $X_{\bar{k}}$ is affine, f' is projective [Sta, Tag 01VT]. By Lemma 2.7, $f'_*\mathcal{O}_{Y'}(-mE') = \mathfrak{a}_m(w)$ for any integer $m \geq 0$, and

$$Y' \cong \operatorname{Proj}_{X_{\bar{k}}} \mathcal{A}', \quad \text{where } \mathcal{A}' := \bigoplus_{m \geq 0} \mathfrak{a}_{mr}(w).$$

By Lemma 2.7, $\mathcal{A}'$ is a finitely generated $\mathcal{O}(X_{\bar{k}})$-algebra, since $X_{\bar{k}}$ is affine.

Let $\mathcal{A} := \bigoplus_{m \geq 0} I_{mr}$, a graded $\mathcal{O}(X)$-algebra by **Step 3**. By **Step 3**, $\mathcal{A} \otimes_k \bar{k} = \mathcal{A}'$. We show that $\mathcal{A}$ is finitely generated over $\mathcal{O}(X)$. Let $h_1, \ldots, h_p$ be homogeneous generators of $\mathcal{A}'$ over $\mathcal{O}(X_{\bar{k}})$. Each h_i is a finite sum $\sum_l g_{il} \otimes c_{il}$, where $c_{il} \in \bar{k}$ and the $g_{il} \in \mathcal{A}$ are homogeneous of degree $\deg h_i$. Let $\mathcal{A}_1 \subset \mathcal{A}$ be the $\mathcal{O}(X)$-subalgebra generated by the finitely many g_{il}. Then $\mathcal{A}_1 \otimes_k \bar{k} \to \mathcal{A} \otimes_k \bar{k}$ is injective, and it is surjective since the image contains $h_1, \ldots, h_p$. Thus $(\mathcal{A}/\mathcal{A}_1) \otimes_k \bar{k} = 0$, so $\mathcal{A} = \mathcal{A}_1$ is finitely generated.

Let $Y := \operatorname{Proj}_X \mathcal{A}$ and $f : Y \to X$ the natural projective morphism. Since the formation of Proj commutes with base change, $Y_{\bar{k}} \cong \operatorname{Proj}_{X_{\bar{k}}} \mathcal{A}' \cong Y'$ over $X_{\bar{k}}$, and $f_{\bar{k}}$ is identified with f'. Since $\mathcal{A}$ is contained in the polynomial ring $\mathcal{O}(X)[s]$, with I_{mr} in degree m, $\mathcal{A}$ is a domain, so Y is integral. For any $y \in Y$ and a point $y' \in Y_{\bar{k}}$ over y, the local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{Y_{\bar{k}},y'}$ is flat and $\mathcal{O}_{Y_{\bar{k}},y'}$ is normal, so $\mathcal{O}_{Y,y}$ is normal [Mat89, Theorem 23.9, Corollary (i)]. Thus Y is a normal variety. Since $\mathfrak{m}_x^j \subset I_j$, the stalk of I_j at any point of $X\backslash\{x\}$ is the local ring of X there, so f is an isomorphism over $X\backslash\{x\}$. In particular, f is birational.

Let E be the reduced closed subscheme of Y with support $f^{-1}(x)$. Since $\bar{x}$ is the only point over x, $f^{-1}(x) \times_k \bar{k} = f'^{-1}(\bar{x})$, and $\operatorname{Supp} f'^{-1}(\bar{x}) = E'$ by Lemma 2.4(1). Since $Y_{\bar{k}} \to Y$ is surjective and integral, E is the image of E', so E is irreducible and $\dim E = \dim E' = d - 1$. Thus E is a prime divisor on Y, and $\operatorname{center}_X E = x$. By Lemma 3.3(1), $E_{\bar{k}}$ is a reduced divisor with support equal to the preimage of E, which is E'. Hence $E_{\bar{k}} = E'$. By Lemma 3.3(1), $\operatorname{ord}_{E'}|_{k(X)} = \operatorname{ord}_E$.

The pullback of the divisorial sheaf $\mathcal{O}_Y(-rE)$ to $Y_{\bar{k}}$ is reflexive of rank one, since the formation of duals of coherent sheaves commutes with flat base change, and it coincides with $\mathcal{O}_{Y_{\bar{k}}}(-rE')$ over the preimage of the smooth locus of Y, whose complement has codimension at least two in $Y_{\bar{k}}$ [Mat89, Theorem 15.1(ii)]. Thus the pullback of $\mathcal{O}_Y(-rE)$ is the invertible sheaf $\mathcal{O}_{Y_{\bar{k}}}(-rE')$, so $\mathcal{O}_Y(-rE)$ is invertible [Gro65, Proposition 2.5.2]. Moreover, $\mathcal{O}_Y(-rE)$ is f-ample since the pullback $\mathcal{O}_{Y_{\bar{k}}}(-rE')$ is f'-ample [Gro65, Corollaire 2.7.2]. Hence E is primitive over X with prime extraction f.

Finally, we show that $(Y, f_*^{-1}B + E)$ is plt near E. Since E is the only f-exceptional prime divisor, $K_Y + f_*^{-1}B + E = f^*(K_X + B) + a(E, X, B)E$ is $\mathbb{R}$-Cartier, and, since $K_{Y_{\bar{k}}} = (K_Y)_{\bar{k}}$ by Lemma 3.3(2) and $(f_*^{-1}B)_{\bar{k}} = (f')_*^{-1}B_{\bar{k}}$ by the argument of **Step 2**,

$$(K_Y + f_*^{-1}B + E)_{\bar{k}} = K_{Y'} + (f')_*^{-1}B_{\bar{k}} + E'.$$

Let $W' \subset Y'$ be an open neighborhood of E' on which $(Y', (f')_*^{-1}B_{\bar{k}} + E')$ is plt. Since f' is proper and $f'^{-1}(\bar{x}) \subset W'$, the closed subset $f'(Y'\backslash W')$ does not contain $\bar{x}$. By **Step 1**, there exists an open neighborhood N of x such that $f'^{-1}(N_{\bar{k}}) \subset W'$. By Lemma 3.3(3) applied to $f^{-1}(N)$, whose base change is $f'^{-1}(N_{\bar{k}})$, the pair $(Y, f_*^{-1}B + E)$ is plt on the open neighborhood $f^{-1}(N)$

of E. Therefore E is a Kollár component of $(X \ni x, B)$ with $E_{\bar{k}} = E'$ and $\mathrm{ord}_{E'}|_{k(X)} = \mathrm{ord}_E$. This proves (2). □

3.2. **Points of higher codimension.** In this subsection, we show that a point η of codimension d of a klt pair (X, B) becomes a closed point x of a klt pair (V, B_V) of dimension d over a purely transcendental extension of $\mathbb{C}$, and that the equality $\mathbb{C}(X) = k(V)$ induces a bijection between the Kollár components of $(X \ni \eta, B)$ and of $(V \ni x, B_V)$ which preserves log discrepancies (Proposition 3.8).

Cutting X by general hyperplanes through a general closed point of the closure of η also yields a germ of dimension d, but monodromy may interchange the Kollár components of the section as the closed point moves, so they need not come from Kollár components of $(X \ni \eta, B)$, and mld^K of the section may be strictly smaller, as in the A_2 example of Section 1. We use the purely transcendental field k of Proposition 3.8 rather than the residue field $\kappa(\eta)$, because lifts of a transcendence basis of $\kappa(\eta)$ over $\mathbb{C}$ generate a subfield of $\mathcal{O}_{X,\eta}$, whereas $\kappa(\eta)$ itself in general does not embed into $\mathcal{O}_{X,\eta}$ over $\mathbb{C}$. For instance, if X is an affine space and the closure of η is an affine elliptic curve, then an embedding over $\mathbb{C}$ of the function field of the curve into the function field of X would induce a dominant rational map from X to the smooth projective model of the closure of η, whose restriction to a general line in X is a finite morphism from $\mathbb{P}^1$ to an elliptic curve, which is impossible by [Har77, Example IV.2.5.4].

The following lemma spreads a prime extraction over a localization of an affine variety to an open subset of the variety. We use it in the proofs of Propositions 3.8 and 3.9.

Lemma 3.7. *Let $U = \mathrm{Spec}\, R$ be a normal affine variety over a field of characteristic zero, $\Sigma \subset R$ a multiplicative subset with $0 \notin \Sigma$, $U_\Sigma := \mathrm{Spec}\, \Sigma^{-1}R$, regarded as the subset of U consisting of the points whose prime ideals do not intersect Σ, and $U_s := \mathrm{Spec}\, R[s^{-1}]$ for $s \in \Sigma$. Then the following hold.*

(1) For any closed subset $T \subset U$ such that $T \cap U_\Sigma = \emptyset$, there exists $s \in \Sigma$ such that $T \cap U_s = \emptyset$.

(2) Let $g : W \to U_\Sigma$ be a projective birational morphism from a normal integral scheme W, D a prime divisor on W, and r a positive integer such that rD is Cartier and $-rD$ is g-ample. Then there exist $s \in \Sigma$, a projective birational morphism $f : Y \to U_s$ from a normal variety Y, and a prime divisor E on Y, such that rE is Cartier, $-rE$ is f-ample, and there exists an isomorphism $Y \times_{U_s} U_\Sigma \cong W$ over U_Σ which maps $E \times_{U_s} U_\Sigma$ onto D.

Proof. By [AM69, Proposition 3.11(iv)], the morphism $U_\Sigma \to U$ is injective with image consisting of the points whose prime ideals do not intersect Σ. Moreover, $\Sigma^{-1}\mathfrak{p} \cap R = \mathfrak{p}$ for any prime ideal $\mathfrak{p} \subset R$ with $\mathfrak{p} \cap \Sigma = \emptyset$.

(1) Let $\mathfrak{p}_1, \dots, \mathfrak{p}_m \subset R$ be the prime ideals of the generic points of the irreducible components of T. Since none of the generic points is in U_Σ, for any i there exists $s_i \in \mathfrak{p}_i \cap \Sigma$. Then $s := s_1 \cdots s_m \in \Sigma$ vanishes on T, so $T \cap U_s = \emptyset$.

(2) Since g is projective and W is integral, there exist a positive integer N and a homogeneous prime ideal $J \subset \Sigma^{-1}R[z_0, \dots, z_N]$ such that W is the closed subscheme of $\mathbb{P}^N_{U_\Sigma}$ defined by J. Let $J_0 := J \cap R[z_0, \dots, z_N]$. Then J_0 is a homogeneous prime ideal with $\Sigma^{-1}J_0 = J$ [AM69, Proposition 3.11(i)], so the closed subscheme $W_0 \subset \mathbb{P}^N_U$ defined by J_0 is integral and $W_0 \times_U U_\Sigma = W$. Let $f_1 : Y_1 \to U$ be the composite of the normalization $Y_1 \to W_0$ and the projection $W_0 \to U$. Since W_0 is of finite type over the ground field, the normalization is finite, so Y_1 is a normal variety and f_1 is projective. Since normalization commutes with localization [AM69, Proposition 5.12] and W is normal, $Y_1 \times_U U_\Sigma \cong W$ over U_Σ. In particular, f_1 is birational. Let E_1 be the closure of D in Y_1. The local ring of Y_1 at the generic point of E_1 is the local ring of W at the generic point of D, so E_1 is a prime divisor on Y_1 and $E_1 \times_U U_\Sigma = D$.

The set $N_1 \subset Y_1$ of points at which $\mathcal{O}_{Y_1}(rE_1)$ is not invertible is closed, and it is disjoint from $f_1^{-1}(U_\Sigma)$ since rD is Cartier. Since f_1 is proper, $f_1(N_1)$ is a closed subset of U disjoint from U_Σ. By (1), there exists $s_1 \in \Sigma$ such that $f_1(N_1) \cap U_{s_1} = \emptyset$, so rE_1 is Cartier on $f_1^{-1}(U_{s_1})$. The open subsets $f_1^{-1}(U_{s_1 s})$, $s \in \Sigma$, form an inverse system with affine transition maps, and the limit

is $Y_1 \times_U U_\Sigma \cong W$. The pullback of $\mathcal{O}_{Y_1}(-rE_1)$ to W is $\mathcal{O}_W(-rD)$, which is ample since U_Σ is affine. By [Sta, Tag 09MT], there exists $s_2 \in \Sigma$ such that $\mathcal{O}_{Y_1}(-rE_1)$ is ample on $f_1^{-1}(U_{s_1 s_2})$. Thus $s := s_1 s_2$, $Y := f_1^{-1}(U_s)$, $f := f_1|_Y$, and $E := E_1 \cap Y$ satisfy (2). □

Proposition 3.8. *Let (X, B) be a klt pair over $\mathbb{C}$, $\eta \in X$ a point of codimension $d \geq 1$, Z the closure of η, and $e := \dim Z$. Then there exist an affine open neighborhood U of η and a dominant morphism $p : U \to \mathbb{A}^e$ satisfying the following. Let $k := \mathbb{C}(t_1, \ldots, t_e)$ be the function field of $\mathbb{A}^e$, $V := U \times_{\mathbb{A}^e} \operatorname{Spec} k$, and B_V the restriction of B to V.*

(1) V is a normal variety over k of dimension d, η defines a closed point $x \in V$ with $\kappa(x) = \mathbb{C}(Z)$, and $\mathcal{O}_{U,\eta} = \mathcal{O}_{V,x}$.
(2) (V, B_V) is a klt pair over k, and the coefficients of B_V are coefficients of B.
(3) The equality $\mathbb{C}(X) = k(V)$ induces a bijection between the Kollár components E of $(X \ni \eta, B)$ and the Kollár components E_V of $(V \ni x, B_V)$ over k, such that $a(E, X, B) = a(E_V, V, B_V)$. Moreover, after shrinking U around η, the prime extraction of E restricts to the prime extraction of E_V.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct U and p, and prove (1) and the statement on coefficients in (2).

Fix a transcendence basis $\bar{t}_1, \ldots, \bar{t}_e$ of $\mathbb{C}(Z)$ over $\mathbb{C}$, and elements $t_1, \ldots, t_e \in \mathcal{O}_{X,\eta}$ whose images in $\kappa(\eta) = \mathbb{C}(Z)$ are $\bar{t}_1, \ldots, \bar{t}_e$. Let $U = \operatorname{Spec} R$ be an affine open neighborhood of η such that $t_1, \ldots, t_e \in R$ and any component of $B|_U$ contains η, let $\mathfrak{q} \subset R$ be the prime ideal of η, and let $p : U \to \mathbb{A}^e$ be the morphism defined by $t_1, \ldots, t_e$. For any non-zero polynomial h in e variables, the image of $h(t_1, \ldots, t_e)$ in $\mathbb{C}(Z)$ is $h(\bar{t}_1, \ldots, \bar{t}_e) \neq 0$. Thus

$$\Sigma := \mathbb{C}[t_1, \ldots, t_e] \backslash \{0\}$$

is a multiplicative subset of R with $\Sigma \cap \mathfrak{q} = \emptyset$. In particular, p is dominant and $V = \operatorname{Spec} \Sigma^{-1} R$.

The scheme V is of finite type over k, and it is a normal variety since $\Sigma^{-1} R$ is a localization of a normal domain [AM69, Proposition 5.12]. Moreover,

$$\dim V = \operatorname{trdeg}_k \mathbb{C}(X) = \dim X - e = d.$$

Since $\Sigma \cap \mathfrak{q} = \emptyset$, the prime ideal $\Sigma^{-1}\mathfrak{q}$ defines a point $x \in V$ with

$$\mathcal{O}_{V,x} = (\Sigma^{-1} R)_{\Sigma^{-1}\mathfrak{q}} = R_{\mathfrak{q}} = \mathcal{O}_{U,\eta},$$

so $\kappa(x) = \kappa(\eta) = \mathbb{C}(Z)$. Since $\mathbb{C}(Z)$ is a finite extension of k, x is a closed point of V. This proves (1).

Let P be a component of $B|_U$ and $\mathfrak{p} \subset R$ the prime ideal of P. Since $\eta \in P$, we have $\mathfrak{p} \subset \mathfrak{q}$, so $\mathfrak{p} \cap \Sigma = \emptyset$. By [AM69, Proposition 3.11(iv)], $\Sigma^{-1}\mathfrak{p}$ is a prime ideal of height one with $\Sigma^{-1}\mathfrak{p} \cap R = \mathfrak{p}$. Thus $P \cap V$ is a prime divisor on V, and distinct components of $B|_U$ restrict to distinct prime divisors on V. Hence the coefficients of B_V are coefficients of B.

Step 2. In this step we compare canonical divisors and log discrepancies over U and over V, and prove (2).

Let $f : Y \to U$ be a projective birational morphism from a normal variety Y, $Y_V := Y \times_U V$, and $f_V : Y_V \to V$ the induced morphism. Since $Y_V \to Y$ is a localization, Y_V is a normal variety over k. For a prime divisor F on Y with $F \cap Y_V \neq \emptyset$, $F_V := F \cap Y_V$ is a prime divisor on Y_V, and the local rings of Y and of Y_V at the generic point of F coincide. Hence $\operatorname{ord}_F = \operatorname{ord}_{F_V}$ under the equality $\mathbb{C}(X) = k(V)$.

Let $Y^0 \subset Y$ be the open subset on which $p \circ f$ is smooth. Since the local ring of $\mathbb{A}^e$ at the generic point is k, a point of Y_V is in Y^0 if and only if Y_V is smooth over k at the point. By [Sta, Tag 02K4], the sequence

$$0 \to (p \circ f)^* \Omega_{\mathbb{A}^e/\mathbb{C}} \to \Omega_{Y^0/\mathbb{C}} \to \Omega_{Y^0/\mathbb{A}^e} \to 0$$

is exact. Since $\det \Omega_{\mathbb{A}^e/\mathbb{C}}$ is trivial, taking determinants implies $\det \Omega_{Y^0/\mathbb{C}} \cong \det \Omega_{Y^0/\mathbb{A}^e}$, and the restriction of $\det \Omega_{Y^0/\mathbb{A}^e}$ to $Y^0 \cap Y_V$ is $\det \Omega_{Y_V/k}$. Since Y_V is normal, it is regular at any point of codimension one, hence smooth over k at such a point [Liu02, Corollary 4.3.33]. Thus the restriction of K_Y to Y_V is a canonical divisor K_{Y_V} of Y_V over k. In particular, for $f = \mathrm{id}_U$, $K_V := K_U|_V$ is a canonical divisor of V, and $K_V + B_V = (K_U + B)|_V$ is $\mathbb{R}$-Cartier.

Write

$$K_Y + f_*^{-1}B = f^*(K_U + B) + \sum_F (a(F, X, B) - 1)F,$$

where F runs through the f-exceptional prime divisors. Since pullback commutes with the flat morphism $Y_V \to Y$, restricting to Y_V implies

$$K_{Y_V} + (f_V)_*^{-1}B_V = f_V^*(K_V + B_V) + \sum_{F \cap Y_V \neq \emptyset} (a(F, X, B) - 1)F_V.$$

Hence

$$a(F_V, V, B_V) = a(F, X, B) \text{ for any prime divisor } F \text{ on } Y \text{ with } F \cap Y_V \neq \emptyset. \tag{3.1}$$

Let $h : T \to U$ be a log resolution of (U, B) and $T_V := T \times_U V$. The strata of $\mathrm{Exc}(h) \cup \operatorname{Supp} h_*^{-1}B$ are smooth over $\mathbb{C}$, so their intersections with T_V are regular, hence smooth over k [Liu02, Corollary 4.3.33]. Thus $h_V : T_V \to V$ is a log resolution of (V, B_V).

The argument above proves the following for any normal variety Y' with a morphism $Y' \to U$ such that $Y'_V := Y' \times_U V$ is non-empty: for an $\mathbb{R}$-divisor Δ on Y' such that $K_{Y'} + \Delta$ is $\mathbb{R}$-Cartier and a log resolution $h : T \to Y'$ of (Y', Δ), with $T_V := T \times_U V$, the restriction of $K_{Y'}$ to Y'_V is a canonical divisor of Y'_V over k, h restricts to a log resolution $T_V \to Y'_V$ of $(Y'_V, \Delta|_{Y'_V})$, and the coefficient of the log pullback of $K_{Y'_V} + \Delta|_{Y'_V}$ along $G \cap T_V$ equals the coefficient of $h^*(K_{Y'} + \Delta) - K_T$ along G for any prime divisor G on T with $G \cap T_V \neq \emptyset$. Write $K_T + B_T = h^*(K_U + B)$ and $K_{T_V} + B_{T_V} = h_V^*(K_V + B_V)$. Taking $Y' = U$ and $\Delta = B$, the coefficients of B_{T_V} are coefficients of B_T, which are smaller than one since (X, B) is klt. Hence (V, B_V) is klt [KM98, Lemma 2.30 and Corollary 2.31(3)], whose proofs work for $\mathbb{R}$-divisors. With **Step 1**, this proves (2).

Step 3. In this step we prove (3).

If $d = 1$, then $\mathcal{O}_{U,\eta} = \mathcal{O}_{V,x}$ is a discrete valuation ring, so ord_Z is the only divisorial valuation with center η on U and with center x on V. After shrinking U so that U and $Z \cap U$ are smooth, the identity morphisms of U and of V are the prime extractions. By the choice of U in **Step 1**, any component of $B|_U$ contains η, hence equals $Z \cap U$, so the $\mathbb{R}$-divisor B_U in Definition 2.3 is zero, the pairs $(U, Z \cap U)$ and (V, x) are plt, and both log discrepancies are equal to $1 - \mathrm{mult}_Z B$. Thus we may assume that $d \geq 2$. Then any prime divisor over X with center η is exceptional over X, any prime divisor over V with center x is exceptional over V, and the $\mathbb{R}$-divisor B_U in Definition 2.3 is $B|_U$.

For $h \in R \backslash \mathfrak{q}$, replacing U with $U_h := \operatorname{Spec} R[h^{-1}]$ replaces V with the open neighborhood $V \cap U_h$ of x in V. Since Definition 2.3 only requires a prime extraction over some open neighborhood of the center, such a replacement does not change the Kollár components in (3), and the prime extraction of E_V in (3) is understood over $V \cap U_h$. Replacing U with $p^{-1}(D(s)) = U_s$ for $s \in \Sigma$ does not change V.

We first restrict Kollár components to V. Let E be a Kollár component of $(X \ni \eta, B)$. Possibly replacing U with U_h for some $h \in R \backslash \mathfrak{q}$, we may assume that E is primitive over U, with prime extraction $f : Y \to U$, and that $(Y_1, f_*^{-1}B + E)$ is plt for an open neighborhood $Y_1 \subset Y$ of E. Let r be a positive integer such that rE is Cartier. Since the generic point of E maps to $\eta \in V$, $E_V := E \cap Y_V$ is a prime divisor on Y_V with $\mathrm{ord}_{E_V} = \mathrm{ord}_E$ and $\mathrm{center}_V E_V = x$. Moreover, rE_V is Cartier and $-rE_V$ is f_V-ample, so E_V is primitive over V with prime extraction f_V. By the last paragraph of **Step 2** applied to $Y' = Y_1$ and $\Delta = f_*^{-1}B + E$, a log resolution of $(Y_1, f_*^{-1}B + E)$ restricts to a log resolution of $(Y_1 \cap Y_V, (f_V)_*^{-1}B_V + E_V)$, and the coefficients of the log pullback on the restriction are coefficients of the log pullback on the log resolution. Thus

$(Y_1 \cap Y_V, (f_V)_*^{-1}B_V + E_V)$ is plt [KM98, Lemma 2.30, Corollary 2.31(3) and Definition 2.34], whose proofs work for $\mathbb{R}$-divisors, and E_V is a Kollár component of $(V \ni x, B_V)$. By (3.1), $a(E_V, V, B_V) = a(E, X, B)$.

We now spread Kollár components from V to U. Let E_V be a Kollár component of $(V \ni x, B_V)$, and $g : W \to V_0$ the prime extraction of E_V over an open neighborhood V_0 of x such that $(W, g_*^{-1}B_V + E_V)$ is plt near E_V. Let r be a positive integer such that rE_V is Cartier. The open subset V_0 contains $V \cap U_h$ for some $h \in R\backslash\mathfrak{q}$, so possibly replacing U with U_h, we may assume that $V_0 = V$. By Lemma 2.4(1), g is an isomorphism over $V\backslash\{x\}$ and $\operatorname{Supp} g^{-1}(x) = E_V$. Since (V, B_V) is klt by **Step 2**, $(W\backslash E_V, g_*^{-1}B_V)$ is klt, so $(W, g_*^{-1}B_V + E_V)$ is plt. By Lemma 3.7(2), possibly replacing U with U_s for some $s \in \Sigma$, there exist a projective birational morphism $f : Y \to U$ from a normal variety Y and a prime divisor E on Y, such that rE is Cartier, $-rE$ is f-ample, $Y_V \cong W$ over V, and $E \cap Y_V$ corresponds to E_V. Thus E is primitive over U with prime extraction f, $\operatorname{ord}_E = \operatorname{ord}_{E_V}$, and $\operatorname{center}_X E = \eta$ since the generic point of E maps to x.

Let $F \neq E$ be an f-exceptional prime divisor. If $F \cap Y_V \neq \emptyset$, then F_V is a g-exceptional prime divisor different from E_V, which is not possible since g is an isomorphism over $V\backslash\{x\}$ and $\operatorname{Supp} g^{-1}(x) = E_V$. Thus $f(F) \cap V = f(F \cap Y_V) = \emptyset$. By Lemma 3.7(1), possibly replacing U with U_s for some $s \in \Sigma$, we may assume that E is the only f-exceptional prime divisor. Then by (3.1),

$$K_Y + f_*^{-1}B + E = f^*(K_U + B) + aE, \quad \text{where } a := a(E_V, V, B_V) = a(E, X, B).$$

In particular, $K_Y + f_*^{-1}B + E$ is $\mathbb{R}$-Cartier.

Let $h : T \to Y$ be a log resolution of $(Y, f_*^{-1}B + E)$, let $T_V := T \times_U V$, and let

$$K_T + \Delta_T := h^*(K_Y + f_*^{-1}B + E).$$

By the last paragraph of **Step 2** applied to $Y' = Y$ and $\Delta = f_*^{-1}B + E$, h restricts to a log resolution $T_V \to W$ of $(W, g_*^{-1}B_V + E_V)$, and $\operatorname{mult}_{G_V} \Delta_{T_V} = \operatorname{mult}_G \Delta_T$ for any prime divisor G on T with $G \cap T_V \neq \emptyset$, where $K_{T_V} + \Delta_{T_V}$ is the log pullback of $K_W + g_*^{-1}B_V + E_V$. Since $(W, g_*^{-1}B_V + E_V)$ is plt, any prime divisor $G \neq h_*^{-1}E$ on T with $\operatorname{mult}_G \Delta_T \geq 1$ satisfies $G \cap T_V = \emptyset$, hence $f(h(G)) \cap V = \emptyset$. There exist finitely many such G, so by Lemma 3.7(1), possibly replacing U with U_s for some $s \in \Sigma$, we may assume that there exists no such G. After replacing U with U_s, $\operatorname{mult}_G \Delta_T < 1$ for any prime divisor $G \neq h_*^{-1}E$ on T, and $\operatorname{mult}_{h_*^{-1}E} \Delta_T = 1$. Hence $(Y, f_*^{-1}B + E)$ is plt [KM98, Lemma 2.30, Corollary 2.31(3) and Definition 2.34], whose proofs work for $\mathbb{R}$-divisors, and E is a Kollár component of $(X \ni \eta, B)$ whose prime extraction f restricts to g.

Restriction and spreading preserve the valuations $\operatorname{ord}_E = \operatorname{ord}_{E_V}$, so they are inverse to each other. This proves the bijection in (3), and the equality of log discrepancies follows from (3.1). The statement on prime extractions in (3) holds by the constructions. □

3.3. **Rational centers.** In this subsection, we replace a closed point with a rational point of an étale neighborhood, without changing Kollár components and log discrepancies (Corollary 3.10). Rational centers are used in two places: at a k-rational point, a Kollár component over k is still a prime divisor after base change to $\bar{k}$ (Lemma 3.6), and linearized invertible sheaves have local generators which are eigenvectors (Construction-Proposition 9.1), which the character computations in Section 11 use. The diagonal point of Corollary 3.10 provides a rational center without embedding $\kappa(x)$ into the local ring $\mathcal{O}_{X,x}$.

Proposition 3.9. *Let $p : X' \to X$ be an étale morphism between normal varieties over a field of characteristic zero, $x' \in X'$ a point, and $x := p(x')$, such that $\dim \mathcal{O}_{X,x} \geq 2$ and p induces an isomorphism $\kappa(x) \to \kappa(x')$. Let $B \geq 0$ be an $\mathbb{R}$-divisor on X such that $(X \ni x, B)$ is klt, and let $B' := p^*B$. Then restriction of valuations induces a bijection between the Kollár components E' of $(X' \ni x', B')$ and the Kollár components E of $(X \ni x, B)$. For corresponding E' and E, the restriction of $\operatorname{ord}_{E'}$ to the function field of X is ord_E, $a(E', X', B') = a(E, X, B)$,*

$$\mathfrak{a}_j(\operatorname{ord}_E)\mathcal{O}_{X',x'} = \mathfrak{a}_j(\operatorname{ord}_{E'})\mathcal{O}_{X',x'} \quad \textit{for any integer } j \geq 0,$$

and, if x is a closed point,

$$\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x} \cong \mathrm{gr}_{\mathrm{ord}_{E'}} \mathcal{O}_{X',x'}.$$

Moreover, possibly shrinking X around x and X' around x', the prime extraction of E' is the base change of the prime extraction of E, and $a(E', X', p^*\Theta) = a(E, X, \Theta)$ for any $\mathbb{R}$-divisor Θ on X such that $K_X + \Theta$ is $\mathbb{R}$-Cartier.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case where p is surjective and $p^{-1}(x) = \{x'\}$.

Since p is étale, it is quasi-finite, so $p^{-1}(x)$ is a finite set of points, each of which is closed in $p^{-1}(x)$. In particular, x' is not contained in the closure of any point of $p^{-1}(x)\backslash\{x'\}$. Removing the closures of the finitely many points of $p^{-1}(x)\backslash\{x'\}$ from X', and then replacing X with the open subset $p(X')$, we may assume that

$(*)$ p is surjective and $p^{-1}(x) = \{x'\}$.

Condition $(*)$ is preserved if we replace X' with an open neighborhood X'_1 of x' and X with $p(X'_1)$, or if we replace X with an open neighborhood X_1 of x and X' with $p^{-1}(X_1)$. Since Definition 2.3 only requires a prime extraction over some open neighborhood of the center, such replacements do not change the Kollár components of $(X \ni x, B)$ and of $(X' \ni x', B')$. Possibly shrinking X, (X, B) is klt. Since p is étale, $K_{X'} = p^*K_X$, so $K_{X'} + B' = p^*(K_X + B)$ is $\mathbb{R}$-Cartier, and (X', B') is klt by [Kol13, Proposition 2.15] (see [Kol13, 2.20] for $\mathbb{R}$-divisors).

Let $R := \mathcal{O}_{X,x}$ and $R' := \mathcal{O}_{X',x'}$, with maximal ideals $\mathfrak{m}$ and $\mathfrak{m}'$, and let $\kappa := \kappa(x) = \kappa(x')$, where $\kappa(x)$ and $\kappa(x')$ are identified by p. The local homomorphism $R \to R'$ is flat, hence faithfully flat, and $\mathfrak{m}R' = \mathfrak{m}'$ since p is unramified.

Step 2. In this step we show that $R/\mathfrak{m}^N \to R'/\mathfrak{m}'^N$ is an isomorphism for any positive integer N, and that $\mathfrak{b} = (\mathfrak{b} \cap R)R'$ for any $\mathfrak{m}'$-primary ideal $\mathfrak{b} \subset R'$.

Let $\hat{R}$ and $\hat{R}'$ be the completions. By [Sta, Tag 039M], $\hat{R}' \cong \hat{R}^{\oplus n}$ as $\hat{R}$-modules for some positive integer n. Since $\mathfrak{m}R' = \mathfrak{m}'$ and $R'/\mathfrak{m}' = \kappa$, we have $\hat{R}'/\mathfrak{m}\hat{R}' = \kappa$, so $n = \dim_\kappa(\hat{R}'/\mathfrak{m}\hat{R}') = 1$. Thus $\hat{R}'$ is a free $\hat{R}$-module of rank one, so $\hat{R} \to \hat{R}'$ is injective, and it is surjective by Nakayama's lemma since $\hat{R}'/\mathfrak{m}\hat{R}' = \kappa$ is spanned by 1. Hence $\hat{R} \to \hat{R}'$ is an isomorphism, and since $\mathfrak{m}^N R' = \mathfrak{m}'^N$,

$$R/\mathfrak{m}^N = \hat{R}/\mathfrak{m}^N\hat{R} \to \hat{R}'/\mathfrak{m}^N\hat{R}' = R'/\mathfrak{m}'^N$$

is an isomorphism for any positive integer N.

Let $\mathfrak{b} \subset R'$ be an $\mathfrak{m}'$-primary ideal, and N a positive integer such that $\mathfrak{m}'^N \subset \mathfrak{b}$. Let $\mathfrak{a} \subset R$ be the inverse image of $\mathfrak{b}/\mathfrak{m}'^N$ under $R \to R/\mathfrak{m}^N \cong R'/\mathfrak{m}'^N$. Then $\mathfrak{a} = \mathfrak{b} \cap R$ and $\mathfrak{a}R' + \mathfrak{m}'^N = \mathfrak{b}$. Since $\mathfrak{m}'^N = \mathfrak{m}^N R' \subset \mathfrak{a}R'$, we have $\mathfrak{a}R' = \mathfrak{b}$.

Step 3. In this step we show that the prime extraction of a primitive divisor over X with center x base changes to a prime extraction over X'.

Let E be a prime divisor over X with $\mathrm{center}_X E = x$, and assume that E is primitive over X, with prime extraction $f : Y \to X$. Let $Y' := Y \times_X X'$, $q : Y' \to Y$ the projection, and $f' : Y' \to X'$ the induced morphism. We show the following.

(a) $E' := q^*E$ is a prime divisor on Y', f' is the prime extraction of E', $\mathrm{center}_{X'} E' = x'$, and the restriction of $\mathrm{ord}_{E'}$ to $k(X)$ is ord_E.
(b) $a(E', X', p^*\Theta) = a(E, X, \Theta)$ for any $\mathbb{R}$-divisor Θ on X such that $K_X + \Theta$ is $\mathbb{R}$-Cartier.
(c) $(Y, f_*^{-1}B + E)$ is plt near E if and only if $(Y', f_*'^{-1}B' + E')$ is plt near E'.

The morphism q is étale and surjective, so Y' is normal, and f' is projective and birational (any irreducible component of Y' dominates Y since q is flat, and f' is an isomorphism over the complement of the inverse image under p of the closure of x by Lemma 2.7). Since $\kappa(x) \to \kappa(x')$ is an isomorphism,

$$f'^{-1}(x') = Y \times_X \mathrm{Spec}\,\kappa(x') \cong Y \times_X \mathrm{Spec}\,\kappa(x) = f^{-1}(x),$$

and the isomorphism is induced by q. Let ξ be the generic point of E, so $f(\xi) = x$. Since q is flat, the generic point of any irreducible component of $q^{-1}(E)$ maps to ξ, so by $(*)$ it is contained in $f'^{-1}(p^{-1}(x)) = f'^{-1}(x')$. Since q is injective on $f'^{-1}(x')$, $q^{-1}(E)$ has only one irreducible component, and it is reduced since q is étale. Thus E' is a prime divisor whose generic point ξ' satisfies $f'(\xi') = x'$. Let r be a positive integer such that rE is Cartier. Then $-rE' = q^*(-rE)$ is Cartier and f'-ample, so f' is the prime extraction of E' and $\operatorname{center}_{X'} E' = x'$. The local homomorphism $\mathcal{O}_{Y,\xi} \to \mathcal{O}_{Y',\xi'}$ of discrete valuation rings is unramified, so a uniformizer of $\mathcal{O}_{Y,\xi}$ is a uniformizer of $\mathcal{O}_{Y',\xi'}$. Hence the restriction of $\operatorname{ord}_{E'}$ to $k(X)$ is ord_E. This proves (a).

Since $\dim \mathcal{O}_{X,x} \geq 2$, E is f-exceptional. Write

$$K_Y + f_*^{-1}\Theta = f^*(K_X + \Theta) + \sum_F (a(F, X, \Theta) - 1)F,$$

where F runs through the f-exceptional prime divisors. Since p and q are étale, $K_{X'} = p^*K_X$, $K_{Y'} = q^*K_Y$, and $q^*f_*^{-1}\Theta = f_*'^{-1}p^*\Theta$. Pulling back the equality by q, which is $\mathbb{R}$-linear, implies

$$K_{Y'} + f_*'^{-1}p^*\Theta = f'^*(K_{X'} + p^*\Theta) + \sum_F (a(F, X, \Theta) - 1)q^*F.$$

For $F \neq E$, E' is not a component of q^*F, since $q(E') = E$. Taking the coefficient of E' implies $a(E', X', p^*\Theta) = a(E, X, \Theta)$. This proves (b).

Let $\Delta := f_*^{-1}B + E$, so $q^*\Delta = f_*'^{-1}B' + E'$. Here $K_Y + \Delta = f^*(K_X + B) + a(E, X, B)E$ is $\mathbb{R}$-Cartier since E is the only f-exceptional prime divisor (cf. Lemma 2.7). If $(Y_1, \Delta|_{Y_1})$ is plt for an open neighborhood Y_1 of E, then $q^{-1}(Y_1)$ is an open neighborhood of E', and $(q^{-1}(Y_1), q^*\Delta|_{q^{-1}(Y_1)})$ is plt by [Kol13, Proposition 2.15] applied to the étale surjective morphism $q^{-1}(Y_1) \to Y_1$. Conversely, if $(Y_1', q^*\Delta|_{Y_1'})$ is plt for an open neighborhood Y_1' of E', then $q(Y_1')$ is open since q is étale, it contains $q(E') = E$, and $(q(Y_1'), \Delta|_{q(Y_1')})$ is plt by [Kol13, Proposition 2.15] applied to the étale surjective morphism $Y_1' \to q(Y_1')$. This proves (c).

Step 4. In this step we descend a Kollár component of $(X' \ni x', B')$ to a primitive divisor over X with center x.

Let E' be a Kollár component of $(X' \ni x', B')$, $u := \operatorname{ord}_{E'}$, and v the restriction of u to $k(X)$. Since $u \geq 0$ on R' and $u > 0$ on $\mathfrak{m}'$, v is a valuation with center x on X. For any integer $j \geq 0$, let

$$\mathfrak{b}_j' := \mathfrak{a}_j(u)R' \quad \text{and} \quad \mathfrak{b}_j := \mathfrak{a}_j(v)R = \mathfrak{b}_j' \cap R.$$

Since u is integer-valued and $u > 0$ on $\mathfrak{m}'$, we have $\mathfrak{m}'^j \subset \mathfrak{b}_j'$, so by **Step 2**,

$$\mathfrak{b}_j R' = \mathfrak{b}_j' \text{ for any integer } j \geq 0. \tag{3.2}$$

In particular, $\mathfrak{m}^j \subset \mathfrak{m}'^j \cap R \subset \mathfrak{b}_j$.

If x is a closed point, then $\operatorname{gr}_v R$ and $\operatorname{gr}_u R'$ are the associated graded rings of Definition 2.9, which defines them only at closed points, and the following argument shows that the map $\operatorname{gr}_v R \to \operatorname{gr}_u R'$ induced by $R \subset R'$ is an isomorphism. The argument itself does not use that x is closed. Since v is integer-valued, the degree λ part of $\operatorname{gr}_v R$ is zero for $\lambda \notin \mathbb{Z}$, and it is $\mathfrak{b}_j/\mathfrak{b}_{j+1}$ for $\lambda = j \in \mathbb{Z}_{\geq 0}$. Similarly, the degree λ part of $\operatorname{gr}_u R'$ is zero for $\lambda \notin \mathbb{Z}$, and it is $\mathfrak{b}_j'/\mathfrak{b}_{j+1}'$ for $\lambda = j \in \mathbb{Z}_{\geq 0}$. Since $\mathfrak{m}\mathfrak{b}_j \subset \mathfrak{b}_{j+1}$, $\mathfrak{b}_j/\mathfrak{b}_{j+1}$ is a κ-vector space, so $(\mathfrak{b}_j/\mathfrak{b}_{j+1}) \otimes_R R' = \mathfrak{b}_j/\mathfrak{b}_{j+1}$ because $R' \otimes_R \kappa = R'/\mathfrak{m}' = \kappa$. By flatness and (3.2), $(\mathfrak{b}_j/\mathfrak{b}_{j+1}) \otimes_R R' = \mathfrak{b}_j'/\mathfrak{b}_{j+1}'$. Hence the natural map $\mathfrak{b}_j/\mathfrak{b}_{j+1} \to \mathfrak{b}_j'/\mathfrak{b}_{j+1}'$ is an isomorphism for any j, and the map $\operatorname{gr}_v R \to \operatorname{gr}_u R'$ induced by $R \subset R'$ is an isomorphism of graded rings.

For an arbitrary point x, possibly replacing X' and X as in **Step 1**, we may assume that E' is primitive over X', with prime extraction $f_1' : Y_1' \to X'$, and that $(Y_1', (f_1')_*^{-1}B' + E')$ is plt near E'. Let r be a positive integer such that rE' is Cartier, and let

$$\mathfrak{A}' := \bigoplus_{m \geq 0} \mathfrak{b}_{mr}' \quad \text{and} \quad \mathfrak{A} := \bigoplus_{m \geq 0} \mathfrak{b}_{mr}.$$

By Lemma 2.7 and flat base change, $Y'_{x'} := Y'_1 \times_{X'} \operatorname{Spec} R'$ is isomorphic to $\operatorname{Proj}_{R'} \mathfrak{A}'$ over R', and the degree m part of $\mathfrak{A}'$ is $\Gamma(Y'_{x'}, \mathcal{O}_{Y'_{x'}}(-mrE'))$. Replacing r by a multiple, $\mathcal{O}_{Y'_{x'}}(-rE')$ is very ample over $\operatorname{Spec} R'$. By Serre's theorem [Har77, Theorem III.5.2(a)], any degree m part of $\mathfrak{A}'$ is a finite R'-module. By [Har77, Theorem III.5.2(b)] for the ideal sheaf of $Y'_{x'}$ under a closed embedding $Y'_{x'} \subset \mathbb{P}^N_{R'}$ given by $\mathcal{O}_{Y'_{x'}}(-rE')$, the map

$$R'[z_0, \dots, z_N]_m \to \Gamma(Y'_{x'}, \mathcal{O}_{Y'_{x'}}(-mrE'))$$

is surjective for $m \gg 0$, so for $m \gg 0$ the degree m part of $\mathfrak{A}'$ is spanned by products of elements of degree one. Thus $\mathfrak{A}'$ is a finitely generated R'-algebra. By flatness and (3.2), $\mathfrak{A} \otimes_R R' = \mathfrak{A}'$. Fix finitely many homogeneous elements of $\mathfrak{A}$ whose images generate $\mathfrak{A}'$ over R', and let $\mathfrak{A}_0 \subset \mathfrak{A}$ be the R-subalgebra generated by them. Then $\mathfrak{A}_0 \otimes_R R' \to \mathfrak{A}'$ is surjective, so $(\mathfrak{A}/\mathfrak{A}_0) \otimes_R R' = 0$, and $\mathfrak{A} = \mathfrak{A}_0$ by faithful flatness. Hence $\mathfrak{A}$ is a finitely generated R-algebra, and replacing r by a multiple, we may assume that $\mathfrak{A}$ is generated by $\mathfrak{b}_r$ over R [Sta, Tag 0EGH].

Let $Y_x := \operatorname{Proj}_R \mathfrak{A}$, which is projective over $\operatorname{Spec} R$. Since $\mathfrak{A} \otimes_R R' = \mathfrak{A}'$, there exists an isomorphism $Y_x \times_R R' \cong Y'_{x'}$ over R', and we let $q_x : Y'_{x'} \to Y_x$ be the induced morphism. It is faithfully flat, and it is a localization of an étale morphism. We show that Y_x satisfies the following.

- Y_x is a normal integral scheme. Since $Y'_{x'}$ is normal and, for any point $y' \in Y'_{x'}$, $\mathcal{O}_{Y_x, q_x(y')} \to \mathcal{O}_{Y'_{x'}, y'}$ is a flat local homomorphism, Y_x is normal [Mat89, Theorem 23.9, Corollary (i)]. Since $Y_x = q_x(Y'_{x'})$ is irreducible, Y_x is integral.
- $Y_x \to \operatorname{Spec} R$ is an isomorphism over $\operatorname{Spec} R \backslash \{\mathfrak{m}\}$, in particular birational. Since $\mathfrak{m}^j \subset \mathfrak{b}_j$ for any j, for any prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of R, we have $\mathfrak{b}_{mr} R_\mathfrak{p} = R_\mathfrak{p}$ for any m, and $\mathfrak{A} \otimes_R R_\mathfrak{p}$ is a polynomial ring over $R_\mathfrak{p}$ in one variable of degree one (with R_g for $g \in \mathfrak{m} \backslash \mathfrak{p}$ in place of $R_\mathfrak{p}$, the computation shows that $Y_x \to \operatorname{Spec} R$ is an isomorphism over the open set $\operatorname{Spec} R \backslash \{\mathfrak{m}\}$).
- There exists a prime divisor E_x on Y_x such that $q_x^* E_x = E'_{x'} := E' \times_{X'} \operatorname{Spec} R'$, E_x is contained in the fiber of Y_x over $\mathfrak{m}$, and $\operatorname{ord}_{E_x} = v$. Since $R' \otimes_R \kappa = \kappa$, q_x maps the fiber of $Y'_{x'}$ over $\mathfrak{m}'$ isomorphically onto the fiber of Y_x over $\mathfrak{m}$, and q_x^{-1} of the second fiber is the first one. Since $\operatorname{center}_{X'} E' = x'$ and the points of $\operatorname{Spec} R'$ are the generizations of x', $E'_{x'}$ is contained in the fiber over $\mathfrak{m}'$. Thus $E_x := q_x(E'_{x'})$ is closed and irreducible, and $q_x^{-1}(E_x) = E'_{x'}$. Let ξ' and ξ be the generic points of $E'_{x'}$ and E_x. The local homomorphism $\mathcal{O}_{Y_x, \xi} \to \mathcal{O}_{Y'_{x'}, \xi'}$ is flat and unramified, so $\dim \mathcal{O}_{Y_x, \xi} = \dim \mathcal{O}_{Y'_{x'}, \xi'} = 1$, a uniformizer of $\mathcal{O}_{Y_x, \xi}$ is a uniformizer of $\mathcal{O}_{Y'_{x'}, \xi'}$, and $q_x^* E_x = E'_{x'}$. Hence ord_{E_x} is the restriction of u to $k(X)$, that is, $\operatorname{ord}_{E_x} = v$.
- rE_x is Cartier and $-rE_x$ is ample over $\operatorname{Spec} R$. The sheaf $q_x^* \mathcal{O}_{Y_x}(-rE_x)$ is reflexive since q_x is flat, and it coincides with $\mathcal{O}_{Y'_{x'}}(-rE'_{x'})$ outside a closed subset of codimension at least two, so $q_x^* \mathcal{O}_{Y_x}(-rE_x) = \mathcal{O}_{Y'_{x'}}(-rE'_{x'})$, which is invertible and ample over $\operatorname{Spec} R'$. By [Gro65, Proposition 2.5.2] and [Gro65, Corollaire 2.7.2], $\mathcal{O}_{Y_x}(-rE_x)$ is invertible and ample over $\operatorname{Spec} R$.

Let $X_1 = \operatorname{Spec} R_1$ be an affine open neighborhood of x and $\Sigma := R_1 \backslash \mathfrak{p}_x$, where $\mathfrak{p}_x$ is the prime ideal of x. Then $\operatorname{Spec} \Sigma^{-1} R_1 = \operatorname{Spec} R$. By Lemma 3.7(2), there exist an open neighborhood X_0 of x, a projective birational morphism $f : Y \to X_0$ from a normal variety Y, and a prime divisor E on Y, such that rE is Cartier, $-rE$ is f-ample, $Y \times_{X_0} \operatorname{Spec} R \cong Y_x$ over R, and $E \times_{X_0} \operatorname{Spec} R$ corresponds to E_x. Thus E is primitive over X_0 with prime extraction f, $\operatorname{ord}_E = \operatorname{ord}_{E_x} = v$, and $\operatorname{center}_X E = x$ since E_x is contained in the fiber over $\mathfrak{m}$.

Step 5. In this step we conclude the proof.

Let E' be a Kollár component of $(X' \ni x', B')$, and let f'_1, $Y'_{x'}$, $f : Y \to X_0$, and E be as constructed in **Step 4**. Replacing X with X_0 and X' with $p^{-1}(X_0)$, let $Y'' := Y \times_X X'$ and $q : Y'' \to Y$ the projection. By **Step 3**(a), $E'' := q^* E$ is a prime divisor and the projection

$Y'' \to X'$ is the prime extraction of E''. Since $Y'' \times_{X'} \operatorname{Spec} R' \cong Y_x \times_R R' \cong Y'_{x'}$ over R', and $E'' \times_{X'} \operatorname{Spec} R'$ corresponds to $q_x^{-1}(E_x) = E'_{x'}$, the local ring of Y'' at the generic point of E'' equals the local ring of $Y'_{x'}$ at the generic point of $E'_{x'}$, which is the valuation ring of $\operatorname{ord}_{E'}$ in $k(X')$. Hence $E'' = E'$ as prime divisors over X'. By the uniqueness of the prime extraction in Definition 2.3, $Y'' \to X'$ coincides with f'_1 over an open neighborhood X'_1 of x'. Replacing X' with X'_1 and X with $p(X'_1)$, **Step 3**(c) implies that $(Y, f_*^{-1}B + E)$ is plt near E. Thus E is a Kollár component of $(X \ni x, B)$, the restriction of $\operatorname{ord}_{E'}$ to $k(X)$ is ord_E, and the prime extraction of E' is the base change of the prime extraction of E.

Conversely, let E be a Kollár component of $(X \ni x, B)$. Possibly replacing X and X' as in **Step 1**, E is primitive over X, with prime extraction $f : Y \to X$, and $(Y, f_*^{-1}B + E)$ is plt near E. By **Step 3**(a)(c), q^*E is a Kollár component of $(X' \ni x', B')$ whose prime extraction is the base change of the prime extraction of E, and the restriction of $\operatorname{ord}_{q^*E}$ to $k(X)$ is ord_E.

Hence $E' \to E$ and $E \to q^*E$ are inverse to each other, and restriction of valuations induces a bijection between the Kollár components of $(X' \ni x', B')$ and of $(X \ni x, B)$. For corresponding E' and E, we have $v = \operatorname{ord}_E$ in **Step 4**, so (3.2) implies

$$\mathfrak{a}_j(\operatorname{ord}_E)\mathcal{O}_{X',x'} = \mathfrak{a}_j(\operatorname{ord}_{E'})\mathcal{O}_{X',x'} \quad \text{for any integer } j \geq 0,$$

and, if x is a closed point, **Step 4** implies

$$\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x} \cong \operatorname{gr}_{\operatorname{ord}_{E'}} \mathcal{O}_{X',x'}.$$

By **Step 3**(b), $a(E', X', p^*\Theta) = a(E, X, \Theta)$ for any $\mathbb{R}$-divisor Θ on X such that $K_X + \Theta$ is $\mathbb{R}$-Cartier. For $\Theta = B$, we obtain $a(E', X', B') = a(E, X, B)$. □

Corollary 3.10. *Let k be a field finitely generated over $\mathbb{C}$, $(X \ni x, B)$ a klt pair over k of dimension $d \geq 2$ with x a closed point, $\kappa := \kappa(x)$, $x_\kappa \in X_\kappa$ the κ-rational point induced by x, X' the irreducible component of X_κ through x_κ, $p : X' \to X$ the projection, and $B' := p^*B$. Then X' is geometrically integral over κ, $(X' \ni x_\kappa, B')$ is klt, and restriction of valuations induces a bijection between the Kollár components E' of $(X' \ni x_\kappa, B')$ over κ and the Kollár components E of $(X \ni x, B)$ over k. For corresponding E' and E, $a(E', X', B') = a(E, X, B)$, and $a(E', X', p^*\Theta) = a(E, X, \Theta)$ for any $\mathbb{R}$-divisor Θ on X such that $K_X + \Theta$ is $\mathbb{R}$-Cartier.*

Proof. Since κ is a finite separable extension of k, the projection $X_\kappa \to X$ is finite étale, and X_κ is normal by Lemma 3.3(1). Thus the irreducible components of X_κ are pairwise disjoint, X' is open and closed in X_κ, and p is étale. The composite of $x_\kappa : \operatorname{Spec} \kappa \to X_\kappa$ and the projection $X_\kappa \to X$ is x, so $p(x_\kappa) = x$ and p induces the identity $\kappa(x) = \kappa \to \kappa(x_\kappa) = \kappa$. By Lemma 3.3(3), possibly shrinking X around x, (X_κ, B_κ) is klt, so (X', B') is klt as a pair over k. Since x is a closed point and $d \geq 2$, $\dim \mathcal{O}_{X,x} = d \geq 2$. By Proposition 3.9 applied to the étale morphism p of varieties over k, restriction of valuations induces a bijection between the Kollár components of $(X' \ni x_\kappa, B')$ over k and the Kollár components of $(X \ni x, B)$ over k, which satisfies $a(E', X', B') = a(E, X, B)$ and $a(E', X', p^*\Theta) = a(E, X, \Theta)$ for corresponding E' and E, where the canonical divisor of X' is taken over k.

The scheme X' is a normal variety over κ with the κ-rational point x_κ, and κ is finitely generated over $\mathbb{C}$. By Lemma 3.5(2), X' is geometrically integral over κ, so κ is algebraically closed in the function field of X'. Since $\kappa \subset \Gamma(X', \mathcal{O}_{X'})$ and κ is a finite extension of k, the algebraic closure of k in the function field of X' is κ. By Lemma 3.5(1) applied to X' over k, the canonical divisors of X' over k and over κ coincide, (X', B') is klt over κ, and the Kollár components of $(X' \ni x_\kappa, B')$ over k and over κ coincide. With the first paragraph, this proves the corollary. □

At a closed point which is not k-rational, X need not be geometrically integral over k, and this is why Corollary 3.10 passes to X', which satisfies the hypotheses of Lemma 3.5(2).

4. Rational approximation of coefficients

In this section, we replace the real coefficients of a boundary by rational numbers depending only on the dimension d and a finite set J of coefficients, since the choices of Section 13 work only for rational boundaries with fixed denominators. The boundary Δ of Corollary 7.5 in Section 7 has coefficients in the finite set $J(d, I) \subset [0, 1]$ depending only on d and the DCC set I of Corollary 7.5, but for a single pair the size of an admissible rational perturbation of the coefficients depends on the pair. A substitution depending only on d and J needs a fixed neighborhood of each irrational coefficient containing no lc threshold, which is provided by [Nak16, Corollary 3.9], and the cost is a factor 2^{c_J} between log discrepancies (Definition-Theorem 4.1).

4.1. A uniform rational substitution.

In this subsection, varieties are defined over $\mathbb{C}$. Theorem 4.5 extends Definition-Theorem 4.1 to varieties over a field finitely generated over $\mathbb{C}$.

Definition-Theorem 4.1. Let $d \geq 2$ be an integer, $J \subset [0, 1]$ a finite set, $J^+ := J \cup \{0, 1\}$, and $c_J := \dim_{\mathbb{Q}} \operatorname{Span}_{\mathbb{Q}} J^+ - 1$. Then there exist a finite set $J_{\mathbb{Q}} \subset [0, 1] \cap \mathbb{Q}$ and a strictly increasing map $\rho : J^+ \to J_{\mathbb{Q}}$ depending only on d and J satisfying the following. The map ρ fixes any element of $J^+ \cap \mathbb{Q}$ and preserves any $\mathbb{Q}$-affine relation among elements of J^+, that is, if $\lambda_0, \dots, \lambda_m \in \mathbb{Q}$ and $b_1, \dots, b_m \in J^+$ satisfy $\lambda_0 + \sum_{i=1}^m \lambda_i b_i = 0$, then $\lambda_0 + \sum_{i=1}^m \lambda_i \rho(b_i) = 0$. Moreover, let X be a normal variety of dimension d and $B \leq \Delta$ two $\mathbb{R}$-divisors on X with coefficients in J^+ such that (X, B) is klt and (X, Δ) is lc, and let B^ρ and Δ^ρ be the $\mathbb{Q}$-divisors obtained by applying ρ to the coefficients of B and Δ. Then:

(1) $B^\rho \leq \Delta^\rho$, both $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier, (X, B^ρ) is klt, and (X, Δ^ρ) is lc.
(2) For $D \in \{B, \Delta\}$ and any prime divisor F over X,

$$2^{-c_J} a(F, X, D^\rho) \leq a(F, X, D) \leq 2^{c_J} a(F, X, D^\rho).$$

In particular, the lc places of (X, Δ) and (X, Δ^ρ) coincide.
(3) For any projective birational morphism $f : Y \to X$ from a normal variety Y with exactly one exceptional prime divisor E such that $-E$ is $\mathbb{Q}$-Cartier and f-ample, $(Y, f_*^{-1}B + E)$ is plt near E if and only if $(Y, f_*^{-1}B^\rho + E)$ is plt near E.

Proof of Definition-Theorem 4.1. The proof is long, so we divide it into several steps.

Step 1. In this step we construct ρ and compare log discrepancies on $\mathbb{Q}$-factorial varieties.

Since $1 \in J^+$, there exist real numbers $r_1, \dots, r_{c_J}$ such that $1, r_1, \dots, r_{c_J}$ is a basis of the $\mathbb{Q}$-vector space $\operatorname{Span}_{\mathbb{Q}} J^+$. For any $b \in J^+$, there exist unique rational numbers $\sigma_{b,0}, \dots, \sigma_{b,c_J}$ such that $b = \sigma_{b,0} + \sum_{j=1}^{c_J} \sigma_{b,j} r_j$, and we let

$$\sigma_b(x) := \sigma_{b,0} + \sum_{j=1}^{c_J} \sigma_{b,j} x_j \quad \text{for } x = (x_1, \dots, x_{c_J}) \in \mathbb{R}^{c_J}.$$

If $b \in \mathbb{Q}$, then σ_b is the constant function b.

Let $\mathcal{T}$ be the set of tuples $(W, S, (T_b)_{b \in J^+})$ such that W is a $\mathbb{Q}$-factorial normal variety with $\dim W \leq d$, and S and T_b $(b \in J^+)$ are reduced divisors on W, no two of which have a common component. For such a tuple and any $x \in \mathbb{R}^{c_J}$, let

$$\Theta_x := S + \sum_{b \in J^+} \sigma_b(x) T_b.$$

Claim 4.2. *Let $1 \leq l \leq c_J$ be an integer and $q_{l+1}, \dots, q_{c_J}$ rational numbers. Assume that the map $J^+ \to \mathbb{R}$, $b \to \sigma_b(r_1, \dots, r_l, q_{l+1}, \dots, q_{c_J})$, is strictly increasing with values in $[0, 1]$. Then there exists a rational number q_l, depending only on d, J, and $q_{l+1}, \dots, q_{c_J}$, such that:*

(1) the map $J^+ \to \mathbb{R}$, $b \to \sigma_b(r_1, \dots, r_{l-1}, q_l, \dots, q_{c_J})$, is strictly increasing with values in $[0, 1]$, and

(2) for any $(W, S, (T_b)_{b\in J^+}) \in \mathcal{T}$*, letting*

$$\Theta := \Theta_{(r_1,\dots,r_l,q_{l+1},\dots,q_{c_J})} \quad \text{and} \quad \Theta' := \Theta_{(r_1,\dots,r_{l-1},q_l,\dots,q_{c_J})},$$

(W, Θ) *is lc if and only if* (W, Θ') *is lc, and in this case*

$$\tfrac{1}{2}a(F, W, \Theta') \le a(F, W, \Theta) \le 2a(F, W, \Theta')$$

for any prime divisor F *over* W*.*

Proof. Let $r := r_l$ and $V_0 := \mathrm{Span}_{\mathbb{Q}}\{1, r_1, \dots, r_{l-1}\}$. Since $1, r_1, \dots, r_l$ are linearly independent over $\mathbb{Q}$, $r \notin V_0$. For any $b \in J^+$ and $t \in \mathbb{R}$, let

$$s_b(t) := \sigma_b(r_1, \dots, r_{l-1}, t, q_{l+1}, \dots, q_{c_J}).$$

Then $s_b(t) = e_b + u_b t$ for some $e_b \in V_0$ and $u_b \in \mathbb{Q}$. If $u_b \ne 0$, then $s_b(r) \notin \mathbb{Q}$, as otherwise $r = u_b^{-1}(s_b(r) - e_b) \in V_0$. Since $s_b(r) \in [0, 1]$ by assumption, $0 < s_b(r) < 1$ for any $b \in J^+$ with $u_b \ne 0$. Since J^+ is finite and the map $b \to s_b(r)$ is strictly increasing, there exist rational numbers $t_- < r < t_+$ such that for any $t \in [t_-, t_+]$, $0 < s_b(t) < 1$ for any $b \in J^+$ with $u_b \ne 0$, and $s_b(t) < s_{b'}(t)$ for any $b < b'$ in J^+. If $u_b = 0$, then $s_b(t) = s_b(r) \in [0, 1]$ for any t. Thus, for any $t \in [t_-, t_+]$,

$$\text{the map } J^+ \to \mathbb{R},\ b \to s_b(t), \text{ is strictly increasing with values in } [0, 1]. \tag{4.1}$$

Let N be a positive integer such that $Nu_b \in \mathbb{Z}$ for any $b \in J^+$. For any $b \in J^+$, let $i_b := s_b(t_-)$ and $k_b := Nu_b \in \mathbb{Z}$, so that

$$s_b(t_- + Ny) = i_b + k_b y \quad \text{for any } y \in \mathbb{R}.$$

Let $I_0 := \{i_b \mid b \in J^+\} \cup \{1\}$, a finite subset of $[0, 1]$ by (4.1). Since $t_- \in \mathbb{Q}$, $I_0 \subset V_0$, hence $\mathrm{Span}_{\mathbb{Q}}(I_0 \cup \{1\}) \subset V_0$. For any $t \in \mathbb{R}$, let $y_t := (t - t_-)/N$. Since $r \notin V_0$ and $t_-, N \in \mathbb{Q}$, $y_r \notin V_0$.

By [Nak16, Definition 3.1], the coefficients allowed at a parameter $c \in \mathbb{R}_{\ge 0}$ are the affine functions of $y \in \mathbb{R}$ which are either 1 or of the form $(m - 1 + f + ky)/m$ with $m \in \mathbb{N}^+$, $k \in \mathbb{Z}$, and $f + ky = \sum_j (f_j + k_j y)$ for some $f_j \in I_0 \cup \{0\}$ and $k_j \in \mathbb{Z}$ with $f_j + k_j c \ge 0$. The set $L_d(I_0) \subset \mathbb{R}_{\ge 0}$ [Nak16, Definition 3.3] consists of the real numbers $c \ge 0$ for which there exist a $\mathbb{Q}$-Gorenstein normal variety Z with $\dim Z \le d$ and an $\mathbb{R}$-divisor $\Gamma(y)$ on Z whose coefficients are allowed at c, such that $\Gamma(a)$ is $\mathbb{R}$-Cartier for any $a \in \mathbb{R}$, $(Z, \Gamma(c))$ is lc, and either $(Z, \Gamma(c + \delta))$ is not lc for any $\delta > 0$ or $(Z, \Gamma(c - \delta))$ is not lc for any $\delta > 0$. Here a pair $(Z, \Gamma(a))$ with $\Gamma(a) \not\ge 0$ is not lc [Nak16, Remark 3.4]. By [Nak16, Corollary 3.9], the accumulation points of $L_d(I_0)$ belong to $\mathrm{Span}_{\mathbb{Q}}(I_0 \cup \{1\}) \subset V_0$. Hence y_r is not an accumulation point of $L_d(I_0)$, and there exists a real number $\epsilon > 0$, depending only on d, I_0, N, t_-, t_+, and r, such that $[r - \epsilon, r + \epsilon] \subset (t_-, t_+)$ and

$$L_d(I_0) \cap [y_{r-\epsilon}, y_{r+\epsilon}] \subset \{y_r\}. \tag{4.2}$$

Let q_l be a rational number such that $|q_l - r| < \epsilon/2$. Then q_l depends only on d, J, and $q_{l+1}, \dots, q_{c_J}$. Since $q_l \in (t_-, t_+)$ and $\sigma_b(r_1, \dots, r_{l-1}, q_l, \dots, q_{c_J}) = s_b(q_l)$, (1) follows from (4.1).

We prove (2). Fix $(W, S, (T_b)_{b\in J^+}) \in \mathcal{T}$. For any $t \in \mathbb{R}$, let $\Theta(t) := S + \sum_{b\in J^+} s_b(t) T_b$. Then $\Theta(r) = \Theta$ and $\Theta(q_l) = \Theta'$. Let

$$\Lambda := \{t \in \mathbb{R} \mid \Theta(t) \ge 0 \text{ and } (W, \Theta(t)) \text{ is lc}\}.$$

Since W is $\mathbb{Q}$-factorial, K_W, S, and T_b are $\mathbb{Q}$-Cartier. For any prime divisor F over W and $t \in \mathbb{R}$, let $\varphi_F(t) := a(F, W, \Theta(t))$. Then

$$\varphi_F(t) = a(F, W, S) - \sum_{b\in J^+} s_b(t) \operatorname{mult}_F T_b,$$

where $a(F, W, S) \in \mathbb{Q}$ and $\operatorname{mult}_F T_b \in \mathbb{Q}$. Thus $\varphi_F(t) = \alpha_F + \beta_F t$ for some $\alpha_F \in V_0$ and $\beta_F \in \mathbb{Q}$. Let $h : W' \to W$ be a fixed log resolution of $(W, S + \sum_{b\in J^+} T_b)$, and let $\mathcal{P}$ be the finite set of the h-exceptional prime divisors and the strict transforms of the components of $S + \sum_{b\in J^+} T_b$. Any other prime divisor P on W' satisfies $a(P, W, S) = 1$ and $\operatorname{mult}_P T_b = 0$ for any $b \in J^+$, so $\varphi_P \equiv 1$. Since $\operatorname{Supp} \Theta(t) \subset \operatorname{Supp}(S + \sum_{b\in J^+} T_b)$ for any t, by [KM98, Lemma 2.30 and Corollary 2.31(3)]

(whose proofs work for $\mathbb{R}$-divisors), if $\Theta(t) \geq 0$, then $(W, \Theta(t))$ is lc if and only if $\varphi_P(t) \geq 0$ for any $P \in \mathcal{P}$. Thus Λ is the set of $t \in \mathbb{R}$ satisfying the finitely many inequalities

$$s_b(t) \geq 0 \quad (b \in J^+,\ T_b \neq 0) \quad \text{and} \quad \varphi_P(t) \geq 0 \quad (P \in \mathcal{P}), \tag{4.3}$$

each of the form $\alpha + \beta t \geq 0$ with $\alpha \in V_0$ and $\beta \in \mathbb{Q}$. In particular, Λ is a closed convex subset of $\mathbb{R}$, possibly empty, unbounded, or a single point.

First, let $\tau \in \{\min \Lambda, \max \Lambda\}$ such that $\tau \in (t_-, t_+)$. We show that $y_\tau \in L_d(I_0)$. Consider the $\mathbb{R}$-divisor $\Theta(t_- + Ny) = S + \sum_{b \in J^+}(i_b + k_b y)T_b$ on W with parameter y. Since W is $\mathbb{Q}$-factorial, W is $\mathbb{Q}$-Gorenstein and $\Theta(t_- + Ny)$ is $\mathbb{R}$-Cartier for any $y \in \mathbb{R}$. Moreover, $\dim W \leq d$. Any coefficient of $\Theta(t_- + Ny)$ is either 1 or of the form $i_b + k_b y$ with $i_b \in I_0$ and $k_b \in \mathbb{Z}$, and $i_b + k_b y_\tau = s_b(\tau) \geq 0$ by (4.1). Thus $\Theta(t_- + Ny)$ satisfies the conditions of [Nak16, Definition 3.1] at the parameter y_τ, with denominator $m = 1$. Since Λ is closed, $(W, \Theta(\tau))$ is lc. If $\tau = \max \Lambda$ (resp. $\tau = \min \Lambda$), then for any $\delta > 0$, $\tau + \delta \notin \Lambda$ (resp. $\tau - \delta \notin \Lambda$), that is, either $\Theta(\tau \pm \delta) \geq 0$ fails or $(W, \Theta(\tau \pm \delta))$ is not lc. In both cases the pair at the parameter $y_\tau \pm \delta/N$ is not lc in the sense of [Nak16, Definition 3.3]. Since $\tau > t_-$, $y_\tau > 0$. Thus [Nak16, Definition 3.3] implies that $y_\tau \in L_d(I_0)$.

Next, we show that $r \notin \{\min \Lambda, \max \Lambda\}$. Suppose that $r \in \{\min \Lambda, \max \Lambda\}$. Then $r \in \Lambda$, and there exist real numbers $t \notin \Lambda$ arbitrarily close to r. The inequalities of (4.3) which are strict at r hold on a neighborhood of r. Thus one of the inequalities, say $\alpha + \beta t \geq 0$, satisfies $\alpha + \beta r = 0$ and fails at some t near r, so $\beta \neq 0$. Hence $r = -\alpha/\beta \in V_0$, a contradiction.

Now we show that $\{\min \Lambda, \max \Lambda\} \cap [r - \epsilon, r + \epsilon] = \emptyset$. If $\tau \in \{\min \Lambda, \max \Lambda\} \cap [r - \epsilon, r + \epsilon]$, then $\tau \in (t_-, t_+)$, so $y_\tau \in L_d(I_0) \cap [y_{r-\epsilon}, y_{r+\epsilon}]$. By (4.2), $y_\tau = y_r$, hence $\tau = r$, contradicting the previous paragraph. Since Λ is a closed convex subset of $\mathbb{R}$, either $\Lambda \cap [r - \epsilon, r + \epsilon] = \emptyset$ or $[r - \epsilon, r + \epsilon] \subset \Lambda$.

Since $r, q_l \in [r - \epsilon, r + \epsilon]$ and $\Theta(r), \Theta(q_l) \geq 0$ by (4.1), (W, Θ) is lc if and only if $r \in \Lambda$, if and only if $q_l \in \Lambda$, if and only if (W, Θ') is lc. Assume that (W, Θ) is lc. Then $[r - \epsilon, r + \epsilon] \subset \Lambda$, so for any prime divisor F over W, $\varphi_F(t) \geq 0$ for any $t \in [r - \epsilon, r + \epsilon]$. Thus

$$\varphi_F(r) - |\beta_F|\epsilon = \min\{\varphi_F(r - \epsilon), \varphi_F(r + \epsilon)\} \geq 0,$$

and

$$|\varphi_F(q_l) - \varphi_F(r)| = |\beta_F| \cdot |q_l - r| \leq \tfrac{1}{2}|\beta_F|\epsilon \leq \tfrac{1}{2}\varphi_F(r).$$

Hence $\frac{1}{2}\varphi_F(r) \leq \varphi_F(q_l) \leq \frac{3}{2}\varphi_F(r)$, which implies that

$$\tfrac{1}{2}\varphi_F(q_l) \leq \varphi_F(r) \leq 2\varphi_F(q_l).$$

Since $\varphi_F(r) = a(F, W, \Theta)$ and $\varphi_F(q_l) = a(F, W, \Theta')$, we obtain (2). □

Proof of Definition-Theorem 4.1 continued. Since $\sigma_b(r_1, \dots, r_{c_J}) = b$ for any $b \in J^+$, the assumption of Claim 4.2 holds for $l = c_J$. Applying Claim 4.2 to $l = c_J$, then to $l = c_J - 1$ with the rational number q_{c_J}, and so on until $l = 1$, where the assumption for l is Claim 4.2(1) for $l + 1$, we obtain rational numbers $q_{c_J}, \dots, q_1$ depending only on d and J. Let

$$\rho(b) := \sigma_b(q_1, \dots, q_{c_J}) \quad \text{for any } b \in J^+,$$

and $J_{\mathbb{Q}} := \rho(J^+)$. If $c_J = 0$, then $J^+ \subset \mathbb{Q}$ and ρ is the identity map of J^+. Since $\sigma_{b,j} \in \mathbb{Q}$, $J_{\mathbb{Q}} \subset \mathbb{Q}$, and by Claim 4.2(1) for $l = 1$, $\rho : J^+ \to J_{\mathbb{Q}}$ is strictly increasing and $J_{\mathbb{Q}} \subset [0, 1]$. If $b \in J^+ \cap \mathbb{Q}$, then σ_b is constant, so $\rho(b) = b$. Let $\lambda_0, \lambda_1, \dots, \lambda_m \in \mathbb{Q}$ and $b_1, \dots, b_m \in J^+$ such that $\lambda_0 + \sum_{i=1}^m \lambda_i b_i = 0$. Then

$$\lambda_0 + \sum_{i=1}^m \lambda_i \sigma_{b_i}(x) = \Big(\lambda_0 + \sum_{i=1}^m \lambda_i \sigma_{b_i,0}\Big) + \sum_{j=1}^{c_J} \Big(\sum_{i=1}^m \lambda_i \sigma_{b_i,j}\Big) x_j$$

has rational coefficients and vanishes at $x = (r_1, \dots, r_{c_J})$. Since $1, r_1, \dots, r_{c_J}$ are linearly independent over $\mathbb{Q}$, all of the coefficients are zero. Taking $x = (q_1, \dots, q_{c_J})$, we obtain $\lambda_0 + \sum_{i=1}^m \lambda_i \rho(b_i) = 0$. Thus ρ preserves any $\mathbb{Q}$-affine relation among elements of J^+.

Claim 4.3. *Let $(W, S, (T_b)_{b\in J^+}) \in \mathcal{T}$, $\Theta := S + \sum_{b\in J^+} bT_b$, and $\Theta^\rho := S + \sum_{b\in J^+} \rho(b)T_b$. Then (W, Θ) is lc if and only if (W, Θ^ρ) is lc, and in this case*

$$2^{-c_J} a(F, W, \Theta^\rho) \le a(F, W, \Theta) \le 2^{c_J} a(F, W, \Theta^\rho)$$

for any prime divisor F over W. In particular, (W, Θ) is klt (resp. plt) if and only if (W, Θ^ρ) is klt (resp. plt).

Proof. For $0 \le j \le c_J$, let $\Theta^{(j)} := \Theta_{(r_1,\dots,r_j,q_{j+1},\dots,q_{c_J})}$. Then $\Theta^{(c_J)} = \Theta$ and $\Theta^{(0)} = \Theta^\rho$. By Claim 4.2(2) for $l = c_J, \dots, 1$, $(W, \Theta^{(l)})$ is lc if and only if $(W, \Theta^{(l-1)})$ is lc, and in this case

$$\frac{1}{2} a(F, W, \Theta^{(l-1)}) \le a(F, W, \Theta^{(l)}) \le 2a(F, W, \Theta^{(l-1)})$$

for any prime divisor F over W. Multiplying the inequalities for $l = c_J, \dots, 1$, we obtain the first statement. The coefficients of Θ and Θ^ρ belong to $[0, 1]$. Thus (W, Θ) is klt if and only if $a(F, W, \Theta) > 0$ for any prime divisor F over W, and (W, Θ) is plt if and only if (W, Θ) is lc and $a(F, W, \Theta) > 0$ for any prime divisor F exceptional over W (cf. [KM98, Definition 2.34]). Both criteria also apply to (W, Θ^ρ). Hence the last statement follows from the first. □

Step 2. In this step we show that $B^\rho \le \Delta^\rho$ and that $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier.

Let $D \in \{B, \Delta\}$. Write $D = \sum_{b\in J^+} bD_b$, where D_b ($b \in J^+$) are reduced divisors without common components, and let

$$L_0 := K_X + \sum_{b\in J^+} \sigma_{b,0} D_b, \qquad L_j := \sum_{b\in J^+} \sigma_{b,j} D_b \quad (1 \le j \le c_J).$$

Then $L_0, \dots, L_{c_J}$ are $\mathbb{Q}$-divisors, and

$$K_X + D = L_0 + \sum_{j=1}^{c_J} r_j L_j, \qquad K_X + D^\rho = L_0 + \sum_{j=1}^{c_J} q_j L_j.$$

Since $K_X + D$ is $\mathbb{R}$-Cartier, $K_X + D \sim_{\mathbb{R},X} 0$. Since $1, r_1, \dots, r_{c_J}$ are linearly independent over $\mathbb{Q}$, by [HLS26, Lemma 5.3] applied with $f = \mathrm{id}_X$, $L_0, \dots, L_{c_J}$ are $\mathbb{Q}$-Cartier. Hence $K_X + D^\rho$ is $\mathbb{Q}$-Cartier.

For any prime divisor P on X, $\mathrm{mult}_P B \le \mathrm{mult}_P \Delta$ and both belong to J^+. Since ρ is increasing,

$$\mathrm{mult}_P B^\rho = \rho(\mathrm{mult}_P B) \le \rho(\mathrm{mult}_P \Delta) = \mathrm{mult}_P \Delta^\rho.$$

Since $\rho(0) = 0$, $B^\rho \ge 0$. Thus $0 \le B^\rho \le \Delta^\rho$.

Step 3. In this step we prove (1) and (2).

The klt and lc conditions in (1) and the inequalities in (2) are local on X, since $a(F, X, D)$ only depends on a neighborhood of $\mathrm{center}_X F$. Possibly replacing X with an affine open subset, we may assume that X is affine. Since (X, B) is klt, by [BCHM10, Corollary 1.4.3] and the second paragraph after it, there exists a small birational morphism $g : X' \to X$ such that X' is $\mathbb{Q}$-factorial. For $D \in \{B, \Delta\}$, let $D_{X'} := g_*^{-1} D$ and $D^\rho_{X'} := g_*^{-1} D^\rho$. Then $D^\rho_{X'}$ is obtained by applying ρ to the coefficients of $D_{X'}$. Since $K_X + D$ and $K_X + D^\rho$ are $\mathbb{R}$-Cartier (**Step 2**) and g is small,

$$K_{X'} + D_{X'} = g^*(K_X + D), \qquad K_{X'} + D^\rho_{X'} = g^*(K_X + D^\rho).$$

Indeed, in both equalities the difference between the left hand side and the right hand side is supported on the g-exceptional divisors, and g has no exceptional divisor. Thus (cf. [KM98, Lemma 2.30]),

$$a(F, X, D) = a(F, X', D_{X'}) \quad \text{and} \quad a(F, X, D^\rho) = a(F, X', D^\rho_{X'}) \tag{4.4}$$

for any prime divisor F over X. In particular, $(X', B_{X'})$ is klt and $(X', \Delta_{X'})$ is lc.

Recall that $\mathcal{T}$ is the set of tuples $(W, S, (T_b)_{b\in J^+})$ of **Step 1**. For $D \in \{B, \Delta\}$ and $b \in J^+$, let T^D_b be the sum of the components of $D_{X'}$ with coefficient b. Then $(X', 0, (T^D_b)_{b\in J^+}) \in \mathcal{T}$,

$D_{X'} = \sum_{b\in J^+} bT_b^D$, and $D^\rho_{X'} = \sum_{b\in J^+} \rho(b)T_b^D$. By Claim 4.3, $(X', B^\rho_{X'})$ is klt, $(X', \Delta^\rho_{X'})$ is lc, and

$$2^{-c_J} a(F, X', D^\rho_{X'}) \le a(F, X', D_{X'}) \le 2^{c_J} a(F, X', D^\rho_{X'})$$

for any $D \in \{B, \Delta\}$ and any prime divisor F over X'. Since $B^\rho, \Delta^\rho \ge 0$ and $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier (**Step 2**), by (4.4), (X, B^ρ) is klt, (X, Δ^ρ) is lc, and the inequalities of (2) hold. Together with **Step 2**, this proves (1) and (2). By (2) for $D = \Delta$, $a(F, X, \Delta) = 0$ if and only if $a(F, X, \Delta^\rho) = 0$, so the lc places of (X, Δ) and (X, Δ^ρ) coincide.

Step 4. In this step we prove (3).

Let $f : Y \to X$ and E be as in (3). For $D \in \{B, B^\rho\}$ and an open cover $\{U_i\}$ of X, the pair $(Y, f_*^{-1}D + E)$ is plt near E if and only if each $(f^{-1}(U_i), f_*^{-1}D|_{f^{-1}(U_i)} + E|_{f^{-1}(U_i)})$ is plt near $E \cap f^{-1}(U_i)$. Possibly replacing X with an affine open subset, we may assume that X is affine. Then Y is quasi-projective. For $D \in \{B, B^\rho\}$, let $D_Y := f_*^{-1}D$. Since E is the only exceptional prime divisor of f,

$$K_Y + D_Y + E = f^*(K_X + D) + a(E, X, D)E. \tag{4.5}$$

Since $-E$ is $\mathbb{Q}$-Cartier and $K_X + B$ and $K_X + B^\rho$ are $\mathbb{R}$-Cartier (**Step 2**), $K_Y + B_Y + E$ and $K_Y + B^\rho_Y + E$ are $\mathbb{R}$-Cartier.

Let $\{D, D'\} = \{B, B^\rho\}$ and assume that $(Y, D_Y + E)$ is plt near E. We show that $(Y, D'_Y + E)$ is plt near E. Let $V \subset Y$ be an open neighborhood of E such that $(V, D_Y + E)$ is plt, where B_Y, B^ρ_Y, and E also denote their restrictions to V. Then V is quasi-projective. Fix a real number $0 < \epsilon' < 1$. Since E is $\mathbb{Q}$-Cartier, for any prime divisor F over V,

$$a(F, V, D_Y + (1-\epsilon')E) = a(F, V, D_Y + E) + \epsilon' \operatorname{mult}_F E \ge a(F, V, D_Y + E).$$

If F is exceptional over V, then $a(F, V, D_Y + E) > 0$. If F is a prime divisor on V, then

$$a(F, V, D_Y + (1-\epsilon')E) = \begin{cases} \epsilon' > 0 & \text{if } F = E, \\ 1 - \operatorname{mult}_F D_Y > 0 & \text{if } F \ne E, \end{cases}$$

since (X, B) and (X, B^ρ) are klt by (1). Thus $(V, D_Y + (1-\epsilon')E)$ is klt. By [BCHM10, Corollary 1.4.3] and the second paragraph after it, there exists a small birational morphism $h : Y' \to V$ such that Y' is $\mathbb{Q}$-factorial. Let $E' := h_*^{-1}E$, $B_{Y'} := h_*^{-1}B_Y$, and $B^\rho_{Y'} := h_*^{-1}B^\rho_Y$, and let $D_{Y'} := h_*^{-1}D_Y$ and $D'_{Y'} := h_*^{-1}D'_Y$. Since h is small and $K_V + B_Y + E$ and $K_V + B^\rho_Y + E$ are $\mathbb{R}$-Cartier,

$$K_{Y'} + B_{Y'} + E' = h^*(K_V + B_Y + E), \qquad K_{Y'} + B^\rho_{Y'} + E' = h^*(K_V + B^\rho_Y + E).$$

Since h is small, a prime divisor over V is exceptional over V if and only if it is exceptional over Y'. Thus (cf. [KM98, Lemma 2.30]), for $D'' \in \{D, D'\}$, $(V, D''_Y + E)$ is plt if and only if $(Y', D''_{Y'} + E')$ is plt. In particular, $(Y', D_{Y'} + E')$ is plt.

For $b \in J^+$, let T_b be the sum of the components of $B_{Y'}$ with coefficient b. Since E is f-exceptional, E' is not a component of $B_{Y'}$. Thus $(Y', E', (T_b)_{b\in J^+}) \in \mathcal{T}$, where $\mathcal{T}$ is the set of tuples of **Step 1**, and $\Theta = E' + B_{Y'}$ and $\Theta^\rho = E' + B^\rho_{Y'}$ in the notation of Claim 4.3. By Claim 4.3, $(Y', B_{Y'} + E')$ is plt if and only if $(Y', B^\rho_{Y'} + E')$ is plt. Hence $(Y', D'_{Y'} + E')$ is plt, so $(V, D'_Y + E)$ is plt. Thus $(Y, D'_Y + E)$ is plt near E. Since (D, D') is either (B, B^ρ) or (B^ρ, B), we obtain (3). □

The exponent c_J in Definition-Theorem 4.1(2) is the number of substitution steps: since [Nak16, Corollary 3.9] controls lc thresholds only in one-parameter families, the irrational basis elements $r_1, \dots, r_{c_J}$ are replaced by rational numbers q_l one at a time (Claim 4.2). The base 2 reflects the choice of q_l in the middle half of the interval $[r_l - \epsilon, r_l + \epsilon]$, on which each pair $(W, \Theta(t))$ is either lc for any $t \in [r_l - \epsilon, r_l + \epsilon]$ or lc for no $t \in [r_l - \epsilon, r_l + \epsilon]$ (by (4.2), which allows $y_{r_l} \in L_d(I_0)$, and since $r_l \notin V_0$ implies $r_l \notin \{\min \Lambda, \max \Lambda\}$), and, in the first case, log discrepancies are non-negative affine functions of t on it. Taking q_l closer to r_l makes the factor

of each step arbitrarily close to 1, at the cost of larger denominators of $J_{\mathbb{Q}}$, which enter the constant of Theorem 14.1.

Definition-Theorem 4.1 is related to the uniform lc rational polytopes of [HLS26, Theorem 5.6]: there, for a fixed coefficient vector v, an open neighborhood of v in the rational envelope of v, depending only on d, the number of coefficients, and v, preserves the lc, klt, and plt properties when one moves from v to a nearby point. Here we need two-sided bounds between log discrepancies and the equivalence of the plt property for a given prime extraction, with a substitution which only depends on d and J.

4.2. **Equality parts.** In this subsection, we define the equality part of a klt boundary with respect to a larger lc boundary, and we show that if the prime extraction of an lc place of the larger boundary is plt near the exceptional divisor for the equality part, then it is plt near the exceptional divisor for the klt boundary. The equality part is used in Construction-Theorem 8.5, and both conclusions of Definition-Lemma 4.4, the plt statement and the inequality of log discrepancies, are used in the proof of Construction-Theorem 8.5(3).

In the notation of Definition-Lemma 4.4 below, to deduce that $(Y, f_*^{-1}B + E)$ is plt near E from a pair $(Y, f_*^{-1}B' + E)$ that is plt near E with $B' \leq B$ and from the lc pair $(Y, f_*^{-1}\Delta + E)$, we use $B \leq (1-\theta)B' + \theta\Delta$ for some $\theta < 1$. On a prime divisor D with $\operatorname{mult}_D B = \operatorname{mult}_D \Delta > 0$ the inequalities $B \leq (1-\theta)B' + \theta\Delta$ and $B' \leq B$ force $\operatorname{mult}_D B' = \operatorname{mult}_D B$, so $B' = 0$ does not work, while $B' = \Delta$ is not a klt boundary, and the equality part below keeps exactly these D. On the other prime divisors we take θ to be the maximum of $\operatorname{mult}_D B / \operatorname{mult}_D \Delta$ over the prime divisors D with $\operatorname{mult}_D B < \operatorname{mult}_D \Delta$, so $\theta < 1$, and θ depends on (X, B) and Δ but enters no bound, so there is no quantitative loss when the gaps $\operatorname{mult}_D \Delta - \operatorname{mult}_D B$ tend to zero.

Definition-Lemma 4.4. Let (X, B) be a klt pair over a field of characteristic zero and $\Delta \geq B$ an $\mathbb{R}$-divisor on X such that (X, Δ) is lc. Let

$$B^{\mathrm{eq}} := \sum_D (\operatorname{mult}_D B) D,$$

where D runs over the prime divisors on X such that $\operatorname{mult}_D B = \operatorname{mult}_D \Delta > 0$. We call B^{eq} the *equality part* of B with respect to Δ. Assume that $K_X + B^{\mathrm{eq}}$ is $\mathbb{R}$-Cartier. Let E be a prime divisor exceptional over X with prime extraction $f : Y \to X$, and assume that $a(E, X, \Delta) = 0$ and $(Y, f_*^{-1}B^{\mathrm{eq}} + E)$ is plt near E. Then $(Y, f_*^{-1}B + E)$ is plt near E and $a(E, X, B) \leq a(E, X, B^{\mathrm{eq}})$.

Proof of Definition-Lemma 4.4. Since $B \geq 0$, $B^{\mathrm{eq}} \leq B$. Thus

$$B - B^{\mathrm{eq}} = (K_X + B) - (K_X + B^{\mathrm{eq}}) \geq 0$$

is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor, and for any prime divisor F over X,

$$a(F, X, B) = a(F, X, B^{\mathrm{eq}}) - \operatorname{mult}_F(B - B^{\mathrm{eq}}) \leq a(F, X, B^{\mathrm{eq}}),$$

since the pullback of $B - B^{\mathrm{eq}} \geq 0$ is effective (cf. [KM98, Lemma 3.39], whose proof works for $\mathbb{R}$-divisors). In particular, $a(E, X, B) \leq a(E, X, B^{\mathrm{eq}})$.

Let θ be the maximum of the ratios $\operatorname{mult}_D B / \operatorname{mult}_D \Delta$, where D runs over the prime divisors on X such that $\operatorname{mult}_D B < \operatorname{mult}_D \Delta$, and let $\theta := 0$ if there exists no such D. There exist only finitely many such D, and $0 \leq \operatorname{mult}_D B / \operatorname{mult}_D \Delta < 1$ for any such D, so $0 \leq \theta < 1$. Since $B \leq \Delta$, any prime divisor D on X satisfies either $\operatorname{mult}_D B = \operatorname{mult}_D \Delta > 0$, or $\operatorname{mult}_D B < \operatorname{mult}_D \Delta$, or $\operatorname{mult}_D B = \operatorname{mult}_D \Delta = 0$. In the first case $\operatorname{mult}_D((1-\theta)B^{\mathrm{eq}} + \theta\Delta) = \operatorname{mult}_D B$, and in the second case $\operatorname{mult}_D((1-\theta)B^{\mathrm{eq}} + \theta\Delta) = \theta \operatorname{mult}_D \Delta \geq \operatorname{mult}_D B$. Thus

$$B \leq (1-\theta)B^{\mathrm{eq}} + \theta\Delta. \tag{4.6}$$

Since $-E$ is f-ample, any curve $C \subset Y$ contracted by f satisfies $(E \cdot C) < 0$, hence $C \subset E$. Thus E is the only f-exceptional prime divisor. By the definition of log discrepancies and since $a(E, X, \Delta) = 0$,

$$K_Y + f_*^{-1}B^{\mathrm{eq}} + E = f^*(K_X + B^{\mathrm{eq}}) + a(E, X, B^{\mathrm{eq}})E,$$

$$K_Y + f_*^{-1}B + E = f^*(K_X + B) + a(E, X, B)E, \quad K_Y + f_*^{-1}\Delta + E = f^*(K_X + \Delta),$$

and the three log divisors are $\mathbb{R}$-Cartier since E is $\mathbb{Q}$-Cartier. Let $V \subset Y$ be an open neighborhood of E such that $(V, (f_*^{-1}B^{\mathrm{eq}} + E)|_V)$ is plt (cf. [KM98, Definition 2.34]). Let

$$B_Y := (1-\theta)f_*^{-1}B^{\mathrm{eq}} + \theta f_*^{-1}\Delta.$$

Then $B_Y \geq 0$ and

$$K_Y + B_Y + E = (1-\theta)(K_Y + f_*^{-1}B^{\mathrm{eq}} + E) + \theta(K_Y + f_*^{-1}\Delta + E)$$

is $\mathbb{R}$-Cartier. For any prime divisor F exceptional over Y with $\mathrm{center}_Y F \in V$,

$$\begin{aligned} a(F, Y, B_Y + E) &= (1-\theta)a(F, Y, f_*^{-1}B^{\mathrm{eq}} + E) + \theta a(F, Y, f_*^{-1}\Delta + E) \\ &= (1-\theta)a(F, Y, f_*^{-1}B^{\mathrm{eq}} + E) + \theta a(F, X, \Delta) > 0, \end{aligned}$$

where the second equality follows from $K_Y + f_*^{-1}\Delta + E = f^*(K_X + \Delta)$, and the inequality holds since $1-\theta > 0$, $a(F, Y, f_*^{-1}B^{\mathrm{eq}} + E) > 0$, and $a(F, X, \Delta) \geq 0$. Thus $(V, (B_Y + E)|_V)$ is plt.

Let

$$G := (1-\theta)f_*^{-1}B^{\mathrm{eq}} + \theta f_*^{-1}\Delta - f_*^{-1}B = (K_Y + B_Y + E) - (K_Y + f_*^{-1}B + E).$$

By (4.6), $G \geq 0$, and G is $\mathbb{R}$-Cartier. Thus, for any prime divisor F over Y,

$$a(F, Y, B_Y + E - G) = a(F, Y, B_Y + E) + \mathrm{mult}_F G \geq a(F, Y, B_Y + E),$$

so $(V, (f_*^{-1}B + E)|_V) = (V, (B_Y + E - G)|_V)$ is plt (cf. [KM98, Corollary 2.35(1)], whose proof works for $\mathbb{R}$-divisors). Therefore $(Y, f_*^{-1}B + E)$ is plt near E. □

4.3. **Substitution over the ground field.** In this subsection, we show that the rational substitution ρ of Definition-Theorem 4.1 works over any field k finitely generated over $\mathbb{C}$, for prime extractions defined over k.

Theorem 4.5. *Let $d \geq 2$ be an integer and $J \subset [0,1]$ a finite set, and let J^+, c_J, and ρ be as in Definition-Theorem 4.1. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension d, and $B \leq \Delta$ two $\mathbb{R}$-divisors on X with coefficients in J^+ such that (X, B) is klt and (X, Δ) is lc, and let B^ρ and Δ^ρ be the $\mathbb{Q}$-divisors obtained by applying ρ to the coefficients of B and Δ. Then the following hold.*

(1) $B^\rho \leq \Delta^\rho$, both $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier, (X, B^ρ) is klt, and (X, Δ^ρ) is lc.

(2) For $D \in \{B, \Delta\}$ and any prime divisor F over X,

$$2^{-c_J}a(F, X, D^\rho) \leq a(F, X, D) \leq 2^{c_J}a(F, X, D^\rho),$$

and for any prime divisor F' over $X_{\bar{k}}$,

$$2^{-c_J}a(F', X_{\bar{k}}, (D^\rho)_{\bar{k}}) \leq a(F', X_{\bar{k}}, D_{\bar{k}}) \leq 2^{c_J}a(F', X_{\bar{k}}, (D^\rho)_{\bar{k}}).$$

In particular, the lc places of (X, Δ) and (X, Δ^ρ) coincide, and the lc places of $(X_{\bar{k}}, \Delta_{\bar{k}})$ and $(X_{\bar{k}}, (\Delta^\rho)_{\bar{k}})$ coincide.

(3) For any closed point $x \in X$ and any projective birational morphism $f : Y \to X$ from a normal variety Y over k with exactly one exceptional prime divisor E such that $\mathrm{center}_X E = x$ and $-E$ is $\mathbb{Q}$-Cartier and f-ample, $(Y, f_*^{-1}B + E)$ is plt near E if and only if $(Y, f_*^{-1}B^\rho + E)$ is plt near E.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that Definition-Theorem 4.1 holds over $\bar{k}$, and we decompose $X_{\bar{k}}$.

Fix an isomorphism of fields $\sigma : \bar{k} \to \mathbb{C}$ (Lemma 3.2). For a variety V over $\bar{k}$, the scheme V with the structure morphism $V \to \mathrm{Spec}\,\bar{k} \to \mathrm{Spec}\,\mathbb{C}$ induced by σ is a variety over $\mathbb{C}$ of dimension $\dim V$, and normality, prime divisors over V, the coefficients of $\mathbb{R}$-divisors, log discrepancies, the klt, lc, and plt conditions, the $\mathbb{Q}$-Cartier condition, and relative ampleness depend only on the scheme V and the morphisms of schemes involved. Thus Definition-Theorem 4.1 holds, with ρ as

defined there for d and J, for normal varieties of dimension d over $\bar{k}$. The isomorphism σ need not restrict to the identity on $\mathbb{C} \subset k$, and this does not matter since ρ depends only on d and J.

By Lemma 3.3(1), $X_{\bar{k}}$ is normal. Thus $X_{\bar{k}}$ is a disjoint union of finitely many normal varieties $X_1, \dots, X_m$ over $\bar{k}$. The projection $X_{\bar{k}} \to X$ is the base change of $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$, hence is flat and integral (cf. [Sta, Tag 01U9] and [Sta, Tag 01WL]). Since generalizations lift along flat morphisms [Sta, Tag 03HV], the generic point of any X_i maps to the generic point of X, and since integral morphisms are closed [Sta, Tag 01WM], $X_i \to X$ is surjective, so $\dim X_i = d$ by [Sta, Tag 0ECG]. Let P be a prime divisor on X. By Lemma 3.3(1), $P_{\bar{k}}$ is a reduced divisor, and any component of $P_{\bar{k}}$ maps onto P, by the argument just given for $X_i \to X$, since $P_{\bar{k}} \to P$ is again the base change of $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$. Thus, for any $\mathbb{R}$-divisor D on X, the coefficient in $D_{\bar{k}}$ of any component of $P_{\bar{k}}$ is $\operatorname{mult}_P D$. In particular, for $D \in \{B, \Delta\}$, $D_{\bar{k}}$ has coefficients in J^+ and $(D_{\bar{k}})^\rho = (D^\rho)_{\bar{k}}$. By Lemma 3.3(3), $(X_{\bar{k}}, B_{\bar{k}})$ is klt and $(X_{\bar{k}}, \Delta_{\bar{k}})$ is lc. Thus Definition-Theorem 4.1 holds for any X_i and the restrictions of $B_{\bar{k}} \le \Delta_{\bar{k}}$ to X_i.

Step 2. In this step we prove (1).

Since ρ is strictly increasing and fixes 0, $0 \le B^\rho \le \Delta^\rho$.

We show on X that $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier. Let $r_0 := 1, r_1, \dots, r_{c_J}$ be a basis of the $\mathbb{Q}$-vector space $\operatorname{Span}_{\mathbb{Q}} J^+$. If $\sum_b q_b b = q_0$ with $b \in J^+$ and $q_b, q_0 \in \mathbb{Q}$, then $\sum_b q_b \rho(b) = q_0$, since ρ preserves $\mathbb{Q}$-affine relations among elements of J^+ (Definition-Theorem 4.1). Thus

$$\phi : \operatorname{Span}_{\mathbb{Q}} J^+ \to \mathbb{Q}, \quad \sum_b q_b b \to \sum_b q_b \rho(b)$$

is a well-defined $\mathbb{Q}$-linear map with $\phi(1) = 1$ and $\phi|_{J^+} = \rho$. Let $D \in \{B, \Delta\}$. Writing the coefficients of D in the basis $r_0, \dots, r_{c_J}$, we obtain $\mathbb{Q}$-divisors $L_0, \dots, L_{c_J}$ on X such that

$$K_X + D = L_0 + \sum_{j=1}^{c_J} r_j L_j.$$

Since $K_X + D$ is $\mathbb{R}$-Cartier and $1, r_1, \dots, r_{c_J}$ are linearly independent over $\mathbb{Q}$, any L_j is $\mathbb{Q}$-Cartier by [HLS26, Lemma 5.3] with $f = \mathrm{id}_X$, whose proof works over k. Applying ϕ to the coefficients, we obtain

$$K_X + D^\rho = L_0 + \sum_{j=1}^{c_J} \phi(r_j) L_j, \quad \phi(r_j) \in \mathbb{Q}.$$

Thus $K_X + D^\rho$ is $\mathbb{Q}$-Cartier.

By **Step 1** and Definition-Theorem 4.1(1) on $X_1, \dots, X_m$, $(X_{\bar{k}}, (B^\rho)_{\bar{k}})$ is klt and $(X_{\bar{k}}, (\Delta^\rho)_{\bar{k}})$ is lc. Since $K_X + B^\rho$ and $K_X + \Delta^\rho$ are $\mathbb{Q}$-Cartier, Lemma 3.3(3) implies that (X, B^ρ) is klt and (X, Δ^ρ) is lc.

Step 3. In this step we prove (2).

Let $D \in \{B, \Delta\}$. Let F' be a prime divisor over $X_{\bar{k}}$. Then F' is a prime divisor over X_i for some i, and $a(F', X_{\bar{k}}, D_{\bar{k}}) = a(F', X_i, D_{\bar{k}}|_{X_i})$, and likewise for D^ρ. Since $(D_{\bar{k}}|_{X_i})^\rho = (D^\rho)_{\bar{k}}|_{X_i}$ (**Step 1**), Definition-Theorem 4.1(2) on X_i implies

$$2^{-c_J} a(F', X_{\bar{k}}, (D^\rho)_{\bar{k}}) \le a(F', X_{\bar{k}}, D_{\bar{k}}) \le 2^{c_J} a(F', X_{\bar{k}}, (D^\rho)_{\bar{k}}).$$

Let F be a prime divisor over X, and let $h : X' \to X$ be a projective birational morphism from a normal variety X' such that F is a prime divisor on X'. By Lemma 3.3(1), $X'_{\bar{k}}$ is normal and $F_{\bar{k}}$ is a reduced divisor. Let F' be a component of $F_{\bar{k}}$. Since $h_{\bar{k}} : X'_{\bar{k}} \to X_{\bar{k}}$ is projective and birational, F' is a prime divisor over $X_{\bar{k}}$. Since $F_{\bar{k}} \to F$ is flat, F' dominates F [Sta, Tag 03HV], so the restriction of $\operatorname{ord}_{F'}$ to $k(X)$ is non-negative on $\mathcal{O}_{X',F}$ and positive on the maximal ideal of $\mathcal{O}_{X',F}$. Since a local equation of F has order one along F' (Lemma 3.3(1)), the restriction of $\operatorname{ord}_{F'}$ to $k(X)$ is ord_F. By Lemma 3.3(2) and (1),

$$a(F', X_{\bar{k}}, D_{\bar{k}}) = a(F, X, D), \quad a(F', X_{\bar{k}}, (D^\rho)_{\bar{k}}) = a(F, X, D^\rho).$$

Thus the inequalities over $\bar{k}$ imply

$$2^{-c_J} a(F, X, D^\rho) \leq a(F, X, D) \leq 2^{c_J} a(F, X, D^\rho).$$

In particular, $a(F, X, \Delta) = 0$ if and only if $a(F, X, \Delta^\rho) = 0$, and $a(F', X_{\bar{k}}, \Delta_{\bar{k}}) = 0$ if and only if $a(F', X_{\bar{k}}, (\Delta^\rho)_{\bar{k}}) = 0$. This proves (2).

Step 4. In this step we prove (3).

Fix x, $f : Y \to X$, and E as in (3). For $D \in \{B, B^\rho\}$, the definition of log discrepancies implies

$$K_Y + f_*^{-1} D + E = f^*(K_X + D) + a(E, X, D)E,$$

which is $\mathbb{R}$-Cartier by (1).

Since x is a closed point and $\operatorname{center}_X E = x$, $f(E) = \{x\}$. By Lemma 2.4(1), f is an isomorphism over $X \backslash \{x\}$ and $\operatorname{Supp} f^{-1}(x) = E$. Since the residue field $\kappa(x)$ is a finite extension of k and k is of characteristic zero, the preimage of x in $X_{\bar{k}}$ is a finite set of closed points $\bar{x}_1, \dots, \bar{x}_s$. Thus $f_{\bar{k}} : Y_{\bar{k}} \to X_{\bar{k}}$ is an isomorphism over $X_{\bar{k}} \backslash \{\bar{x}_1, \dots, \bar{x}_s\}$, and

$$\operatorname{Supp} f_{\bar{k}}^{-1}(\{\bar{x}_1, \dots, \bar{x}_s\}) = \operatorname{Supp} E_{\bar{k}}.$$

By Lemma 3.3(1), $Y_{\bar{k}}$ is normal and $E_{\bar{k}}$ is a reduced divisor. The image of any component of $E_{\bar{k}}$ is irreducible, hence is one of the points $\bar{x}_j$. Thus $E_{\bar{k}} = S_1 + \cdots + S_s$, where S_j is the sum of the components of $E_{\bar{k}}$ that map to $\bar{x}_j$, and $S_j \neq 0$ since $f_{\bar{k}}$ is surjective. For any j, let U_j be the connected component of $X_{\bar{k}} \backslash \{\bar{x}_l \mid l \neq j\}$ containing $\bar{x}_j$, $Y_j := f_{\bar{k}}^{-1}(U_j)$, and $f_j := f_{\bar{k}}|_{Y_j} : Y_j \to U_j$. By **Step 1**, U_j is a normal variety over $\bar{k}$ with $\dim U_j = d$. The morphism f_j is an isomorphism over $U_j \backslash \{\bar{x}_j\}$ and $\operatorname{Supp} f_j^{-1}(\bar{x}_j) = \operatorname{Supp} S_j$. The open subset $Y_j \backslash \operatorname{Supp} S_j \cong U_j \backslash \{\bar{x}_j\}$ of Y_j is irreducible, and it is dense in Y_j since $\dim Y' = d$ for any irreducible component Y' of $Y_{\bar{k}}$ by the argument of **Step 1**. Thus Y_j is a normal variety, and f_j is a projective birational morphism whose exceptional prime divisors are the components of S_j. For $D \in \{B, B^\rho, \Delta\}$, let $D_{U_j} := D_{\bar{k}}|_{U_j}$.

Let $\{D, D'\} = \{B, B^\rho\}$ and assume that $(Y, f_*^{-1} D + E)$ is plt near E. Let $V \subset Y$ be an open neighborhood of E such that $(V, (f_*^{-1} D + E)|_V)$ is plt. By Lemma 3.3(1), the pullback to $Y_{\bar{k}}$ of any component P of $f_*^{-1} D$ is a reduced divisor whose components dominate P, hence are not contained in $\operatorname{Supp} E_{\bar{k}}$. Since $f_{\bar{k}}$ is an isomorphism over the complement of $\{\bar{x}_1, \dots, \bar{x}_s\}$, $(f_*^{-1} D)_{\bar{k}} = (f_{\bar{k}})_*^{-1} D_{\bar{k}}$. By Lemma 3.3(3), $(V_{\bar{k}}, ((f_{\bar{k}})_*^{-1} D_{\bar{k}} + E_{\bar{k}})|_{V_{\bar{k}}})$ is plt. Restricting to Y_j, the pair $(Y_j, (f_j)_*^{-1} D_{U_j} + S_j)$ is plt near S_j. Since $\operatorname{Supp} S_j = \operatorname{Supp} f_j^{-1}(\bar{x}_j)$ and $(f_j)_*^{-1} D_{U_j} \geq 0$, Lemma 2.4(2) implies that S_j is a prime divisor. Thus S_j is the only exceptional prime divisor of f_j. Moreover, $S_j = E_{\bar{k}}|_{Y_j}$ is $\mathbb{Q}$-Cartier, and $-S_j$ is f_j-ample since relative ampleness is preserved under base change, and restriction to an open subset of the base is a base change ([Sta, Tag 0893], applied to $\mathcal{O}_Y(-mE)$ for m such that mE is Cartier). Since $B_{U_j} \leq \Delta_{U_j}$ have coefficients in J^+, (U_j, B_{U_j}) is klt, (U_j, Δ_{U_j}) is lc, and $(B_{U_j})^\rho = B^\rho_{U_j}$ (**Step 1**), Definition-Theorem 4.1(3) over $\bar{k}$ (**Step 1**) implies that $(Y_j, (f_j)_*^{-1} D'_{U_j} + S_j)$ is plt near S_j.

For any j, let $W_j \subset Y_j$ be an open neighborhood of S_j such that $(W_j, ((f_j)_*^{-1} D'_{U_j} + S_j)|_{W_j})$ is plt, and let $W := W_1 \cup \cdots \cup W_s$. Since Y_j is open in $Y_{\bar{k}}$ and $((f_{\bar{k}})_*^{-1} D'_{\bar{k}} + E_{\bar{k}})|_{Y_j} = (f_j)_*^{-1} D'_{U_j} + S_j$, and since the plt condition is local on W, the pair $(W, ((f_{\bar{k}})_*^{-1} D'_{\bar{k}} + E_{\bar{k}})|_W)$ is plt. Since $Y_{\bar{k}} \to Y$ is integral, hence closed ([Sta, Tag 01WL] and [Sta, Tag 01WM]), the image T of $Y_{\bar{k}} \backslash W$ in Y is closed. Since the preimage of E in $Y_{\bar{k}}$ is $\operatorname{Supp} E_{\bar{k}} \subset W$, $T \cap E = \emptyset$. Let $V' := Y \backslash T$. Then $V'_{\bar{k}} \subset W$, so $(V'_{\bar{k}}, ((f_{\bar{k}})_*^{-1} D'_{\bar{k}} + E_{\bar{k}})|_{V'_{\bar{k}}})$ is plt. Since $(f_*^{-1} D')_{\bar{k}} = (f_{\bar{k}})_*^{-1} D'_{\bar{k}}$, Lemma 3.3(3) implies that $(V', (f_*^{-1} D' + E)|_{V'})$ is plt. Thus $(Y, f_*^{-1} D' + E)$ is plt near E. Since (D, D') is either (B, B^ρ) or (B^ρ, B), this proves (3). □

5. Constrained log canonical slices

In this section, we prove that the volume multiplied by the d-th power of a height has a unique minimizer among the quasi-monomial valuations centered at a closed point x which are lc places of an lc pair $(X, B + cH)$, which take the value one at h, and which satisfy finitely many incidence conditions, where (X, B) is a klt pair, $c > 0$ is a real number, and $H = \mathrm{div}(h)$ is the divisor of a non-zero regular function h. We also study the associated graded ring of the minimizer. As in Subsection 4.1 and Section 12, the varieties in this section are varieties over $\mathbb{C}$, and Section 6 uses the results over $\bar{k}$ through Lemma 3.2.

The valuations used in Lemma 11.4, Theorem 10.2, and Corollary 6.7 minimize the volume multiplied by the d-th power of a height, with heights that change in Section 10, among the lc places of a given lc pair that take the value one at h and satisfy the incidence conditions of Set-up 5.1, so the global minimizers of Theorem 2.11 do not suffice. The exact penalization of Definition-Proposition 5.2 reduces the constrained problem to the global problem for a klt perturbation of B, as [OAI26b, Theorem E.8] does for a $\mathbb{Q}$-Gorenstein klt germ without boundary with the coefficient c equal to one and a rational height, and [OAI26a, Proposition 4.2] for a klt germ without boundary, a rational coefficient, and a height defined by the log discrepancy and the order of one ideal, while here B, c, and the height are real and X need not be $\mathbb{Q}$-Gorenstein.

5.1. **Exact penalization.**

In this subsection, we construct the minimizer on a constrained slice and realize it as the minimizer of the normalized volume for a klt perturbation of the boundary.

Set-up 5.1. Let d, $(X \ni x, B)$, h, H, c, $f_1, \dots, f_s$, $m_1, \dots, m_s$, $\mathcal{Q}$, $\mathcal{Q}_x$, $g_1, \dots, g_t$, b_0, $\beta_1, \dots, \beta_t$, and b be as follows.

(1) $d \geq 2$ is an integer and $(X \ni x, B)$ is a klt pair of dimension d, where X is an affine variety and x is a closed point.
(2) h is a non-zero regular function on X with $h(x) = 0$, $H := \mathrm{div}(h)$, and $c > 0$ is a real number such that $(X, B + cH)$ is lc and any lc center of $(X, B + cH)$ contains x.
(3) $f_1, \dots, f_s$ are non-zero regular functions on X and $m_1, \dots, m_s$ are positive integers, where $s \geq 0$.
(4) $\mathcal{Q}$ is the set of non-trivial quasi-monomial valuations v on X with $v \geq 0$ on $\mathcal{O}_{X,x}$ such that $a(v, X, B) = cv(h)$, $v(h) = 1$, and $v(f_j) \geq m_j$ for any j.
(5) $\mathcal{Q}_x \subset \mathcal{Q}$ is the subset of the valuations v with $\mathrm{center}_X v = x$, and $\mathcal{Q}_x \neq \emptyset$.
(6) $g_1, \dots, g_t$ are non-zero regular functions on X, b_0 is a real number, and $\beta_1, \dots, \beta_t \geq 0$ are real numbers, where $t \geq 0$, such that the function $b(v) := b_0 v(h) - \sum_l \beta_l v(g_l)$ satisfies $b(v) > 0$ for any $v \in \mathcal{Q}$.

We call $\mathcal{Q}$ the *constrained slice*, b the *height*, and the conditions $v(f_j) \geq m_j$ in (4) the *incidence conditions*.

If h, $f_1, \dots, f_s$, and $g_1, \dots, g_t$ are only given as elements of $\mathcal{O}_{X,x}$, then, replacing X with an affine neighborhood of x on which they are regular and which does not meet the finitely many lc centers of $(X, B + cH)$ not containing x, we reach Set-up 5.1. The sets $\mathcal{Q}$ and $\mathcal{Q}_x$ and the function b only depend on $\mathcal{O}_{X,x}$, so the replacement does not change them. We ask the height to be positive on $\mathcal{Q}$, that is, also on the valuations in $\mathcal{Q}$ that are not centered at x, so that the perturbation Ψ_λ of Definition-Proposition 5.2 is klt. The heights used in Section 10 are positive on $\mathcal{Q}$ by [OAI26b, (G.4), Subsection G.6, and Lemma E.11].

Definition-Proposition 5.2. Notation and conditions as in Set-up 5.1. Then the following hold.

(1) There exists a unique valuation $w \in \mathcal{Q}_x$ at which the restriction of $b^d \cdot \mathrm{vol}$ to $\mathcal{Q}_x$ attains the minimum.
(2) There exist an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$, which is a non-negative real linear combination of principal divisors, and a real number $\lambda_0 > 0$, such that for any real number $0 < \lambda < \lambda_0$,

the $\mathbb{R}$-divisor

$$\Psi_\lambda := B + D - \lambda b_0 H + \lambda \sum_l \beta_l \operatorname{div}(g_l)$$

satisfies $\Psi_\lambda \geq B$, (X, Ψ_λ) is klt, w minimizes $\widehat{\operatorname{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x, and $a(v, X, \Psi_\lambda) = \lambda b(v)$ for any $v \in \mathcal{Q}$.

(3) For any positive rational numbers $\tau_1, \dots, \tau_s$ with $\tau := \sum_j \tau_j < 1$ ($\tau := 0$ if $s = 0$), (2) holds for some D and λ_0 such that $\lambda_0 b_0 \leq c(1-\tau)$ and

$$D = c(1-\tau)H + \frac{c}{qr} \sum_{p=1}^{r} \operatorname{div}(P_p),$$

where q and r are positive integers with $q\tau_j/m_j \in \mathbb{Z}$ for any j, and $P_1, \dots, P_r$ are regular functions on X such that each P_p is a linear combination of products having, for any j, exactly $q\tau_j/m_j$ factors in $\{h^{m_j}, f_j\}$, and $v(P_p) = q\tau$ for any $v \in \mathcal{Q}$ and any p.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we encode the conditions defining $\mathcal{Q}$ by an ideal.

Fix positive rational numbers $\tau_1, \dots, \tau_s$ with $\tau := \sum_j \tau_j < 1$, where $\tau := 0$ if $s = 0$, and let q be a positive integer such that $q\tau_j/m_j \in \mathbb{Z}$ for any j. Then $q\tau = \sum_j m_j \cdot q\tau_j/m_j$ is an integer, and $q(1-\tau)$ is a positive integer. Let

$$\mathfrak{b} := (h)^{q(1-\tau)} \prod_{j=1}^{s} (h^{m_j}, f_j)^{q\tau_j/m_j} \subset \mathcal{O}_X.$$

For a valuation v on X and a non-zero ideal $\mathfrak{a} \subset \mathcal{O}_X$, we write $v(\mathfrak{a}) := \min\{v(f) \mid f \in \mathfrak{a}\}$. Since $v((h^{m_j}, f_j)) = \min\{m_j v(h), v(f_j)\}$ and v is additive on products of ideals,

$$v(\mathfrak{b}) = q(1-\tau)v(h) + q \sum_j \tau_j \min\Big\{v(h), \frac{v(f_j)}{m_j}\Big\}.$$

Since $a(v, X, B + cH) = a(v, X, B) - cv(h)$, for any valuation v on X with $a(v, X, B) < +\infty$ we obtain

$$a(v, X, B) - \frac{c}{q} v(\mathfrak{b}) = a(v, X, B + cH) + c \sum_j \tau_j \Big(v(h) - \min\Big\{v(h), \frac{v(f_j)}{m_j}\Big\}\Big). \tag{5.1}$$

If v is quasi-monomial, then $a(v, X, B + cH) \geq 0$ since $(X, B + cH)$ is lc [HLQ20, Lemma 2.8]. Since $\tau_j > 0$ for any j, the right hand side of (5.1) is non-negative for any quasi-monomial valuation v on X, and it is zero if and only if $a(v, X, B) = cv(h)$ and $v(f_j) \geq m_j v(h)$ for any j.

Step 2. In this step we construct D and describe the lc places of $(X, B + D)$.

Let V be the $\mathbb{C}$-linear span of the products

$$h^{q(1-\tau)} \prod_{j=1}^{s} \prod_{i=1}^{q\tau_j/m_j} y_{j,i}, \qquad y_{j,i} \in \{h^{m_j}, f_j\}.$$

Then $\dim V < +\infty$ and V generates $\mathfrak{b}$. Let $\mathfrak{m}_x \subset \mathcal{O}_X$ be the ideal of x. Let $\pi : Y \to X$ be a log resolution of (X, B) such that $\mathfrak{b}\mathcal{O}_Y$, $\mathfrak{m}_x\mathcal{O}_Y$, $h\mathcal{O}_Y$, $f_j\mathcal{O}_Y$, and $g_l\mathcal{O}_Y$ are invertible for any j and l, and such that the union Φ_0 of $\operatorname{Exc}(\pi)$, the strict transform of $\operatorname{Supp} B$, and the supports of the divisors on Y defined by $\mathfrak{b}\mathcal{O}_Y$, $\mathfrak{m}_x\mathcal{O}_Y$, $h\mathcal{O}_Y$, $f_j\mathcal{O}_Y$, and $g_l\mathcal{O}_Y$ is an snc divisor. Write $\mathfrak{b}\mathcal{O}_Y = \mathcal{O}_Y(-F)$, and fix an integer $r > c/q$. By [Laz04b, Lemma 9.1.9] and the Bertini theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied to the base point free linear system defined by V on Y after removing F and to the strata of Φ_0, general elements $\sigma_1, \dots, \sigma_r \in V$

satisfy $\pi^* \operatorname{div}(\sigma_p) = F + M_p$, where $M_1, \dots, M_r$ are reduced divisors without common components with each other and with Φ_0, and $\Phi := \Phi_0 + \sum_p M_p$ is a reduced snc divisor. Let

$$D := \frac{c}{qr} \sum_{p=1}^{r} \operatorname{div}(\sigma_p).$$

Any element of V is divisible by $h^{q(1-\tau)}$, so $\sigma_p = h^{q(1-\tau)} P_p$, where P_p is a non-zero linear combination of products having, for any j, exactly $q\tau_j/m_j$ factors in $\{h^{m_j}, f_j\}$. Thus

$$D = c(1-\tau)H + \frac{c}{qr} \sum_{p=1}^{r} \operatorname{div}(P_p). \tag{5.2}$$

In particular, $D \geq 0$ is a non-negative real linear combination of principal divisors, and $K_X + B + D$ is $\mathbb{R}$-Cartier.

For a valuation v on X and a prime divisor E on Y, we write $v(E)$ for the value of v at a local equation of E at the center of v on Y. For any component E of Φ, let

$$\alpha_E := a(E, X, B), \qquad \gamma_E := a(E, X, B + D), \qquad e_E := b(\operatorname{ord}_E).$$

Let $K_Y + B_Y := \pi^*(K_X + B)$. By [HLQ20, Lemma 2.9], applied to π and to id_Y, for any quasi-monomial valuation v on X, $a(v, X, B) = a(v, Y, 0) - v(B_Y)$ and $a(v, Y, \Phi) = a(v, Y, 0) - v(\Phi)$. Since $B_Y = \sum_E (1 - \alpha_E)E$, where E runs over the components of Φ, we obtain

$$a(v, X, B) = a(v, Y, 0) - v(B_Y) = a(v, Y, \Phi) + \sum_E \alpha_E v(E).$$

Since $\pi^* D$, $\pi^* H$, and $\pi^* \operatorname{div}(g_l)$ are supported on Φ, the computation of $a(v, X, B)$ above with B replaced by $B + D$ and the definition of b imply that, for any quasi-monomial valuation v on X,

$$\begin{aligned} a(v, X, B) &= a(v, Y, \Phi) + \sum_E \alpha_E v(E), \\ a(v, X, B + D) &= a(v, Y, \Phi) + \sum_E \gamma_E v(E), \\ b(v) &= \sum_E e_E v(E). \end{aligned} \tag{5.3}$$

Since (X, B) is klt, $\alpha_E > 0$ for any E. If E is a component of some M_p, then the coefficient of E in $B_Y + \pi^* D$ is $c/(qr)$, so $\gamma_E = 1 - c/(qr) > 0$. If E is a component of Φ_0, then E is not a component of any M_p, so $\operatorname{ord}_E(\sigma_p) = \operatorname{mult}_E F = \operatorname{mult}_E \mathfrak{b}$ for any p, and

$$\gamma_E = \alpha_E - \frac{c}{q} \operatorname{mult}_E \mathfrak{b} \geq 0$$

by (5.1) for $v = \operatorname{ord}_E$. Thus $\gamma_E \geq 0$ for any E, and $(X, B + D)$ is lc by [KM98, Lemma 2.30 and Corollary 2.31(3)], whose proofs work for $\mathbb{R}$-divisors. By [JM12, Corollary 5.4], $a(v, Y, \Phi) \geq 0$ for any quasi-monomial valuation v on X, with equality if and only if v is a monomial valuation in the components of Φ. Hence, by (5.3), the quasi-monomial valuations v on X with $a(v, X, B + D) = 0$ are exactly the monomial valuations in the components E of Φ with $\gamma_E = 0$, and no such component is a component of some M_p.

We show that, for any non-trivial quasi-monomial valuation v on X, the following hold.

(i) $a(v, X, B) > 0$.
(ii) $a(v, X, B + D) = 0$ if and only if $a(v, X, B) = cv(h)$ and $v(f_j) \geq m_j v(h)$ for any j.
(iii) If $a(v, X, B + D) = 0$, then $v(h) > 0$ and $v(\sigma_p) = v(\mathfrak{b})$ for any p.

If $a(v, Y, \Phi) > 0$, then (i) follows from (5.3). Otherwise v is a non-trivial monomial valuation in the components of Φ, so $v(E) > 0$ for some E, and (i) follows from $\alpha_E > 0$. Assume that

$a(v,X,B)=cv(h)$ and $v(f_j)\geq m_jv(h)$ for any j. By **Step 1**, $a(v,X,B)=\frac{c}{q}v(\mathfrak{b})$. Since $\sigma_p\in\mathfrak{b}$, $v(\sigma_p)\geq v(\mathfrak{b})$ for any p, so

$$0\leq a(v,X,B+D)=a(v,X,B)-\frac{c}{qr}\sum_p v(\sigma_p)\leq a(v,X,B)-\frac{c}{q}v(\mathfrak{b})=0.$$

Thus $a(v,X,B+D)=0$ and $v(\sigma_p)=v(\mathfrak{b})$ for any p. Conversely, assume that $a(v,X,B+D)=0$. Since $\gamma_E>0$ for the components E of M_p, (5.3) implies $v(M_p)=0$ for any p, so $v(\sigma_p)=v(F)+v(M_p)=v(\mathfrak{b})$, and

$$a(v,X,B)-\frac{c}{q}v(\mathfrak{b})=a(v,X,B)-\frac{c}{qr}\sum_p v(\sigma_p)=a(v,X,B+D)=0.$$

By **Step 1**, $a(v,X,B)=cv(h)$ and $v(f_j)\geq m_jv(h)$ for any j. This proves (ii). If $a(v,X,B+D)=0$, then $v(h)=a(v,X,B)/c>0$ by (ii) and (i), and $v(\sigma_p)=v(\mathfrak{b})$ for any p was shown above. This proves (iii).

Let $v\in\mathcal{Q}$. By (ii) and (iii), $a(v,X,B+D)=0$ and $v(\sigma_p)=v(\mathfrak{b})$ for any p. Since $v(h)=1$ and $v(f_j)\geq m_j$ for any j, **Step 1** implies $v(\mathfrak{b})=q(1-\tau)+q\tau=q$. Since $v(\sigma_p)=q(1-\tau)v(h)+v(P_p)$, we obtain $v(P_p)=q\tau$ for any p.

Step 3. In this step we construct Ψ_λ and show that (X,Ψ_λ) is klt for small λ.

Let Σ be the reduced divisor whose components are the components E of Φ with $\gamma_E=0$. Let E be a component of Σ. By **Step 2**(ii), $a(E,X,B+cH)=0$ and $\mathrm{ord}_E(f_j)\geq m_j\,\mathrm{ord}_E(h)$ for any j. Thus $\pi(E)$ is an lc center of $(X,B+cH)$, and $x\in\pi(E)$ by Set-up 5.1(2). Hence $\mathrm{ord}_E\geq 0$ on $\mathcal{O}_{X,x}$, $\mathrm{ord}_E(h)>0$ by **Step 2**(iii), and $\mathrm{ord}_E/\mathrm{ord}_E(h)\in\mathcal{Q}$. By Set-up 5.1(6),

$$e_E=\mathrm{ord}_E(h)\cdot b\Big(\frac{\mathrm{ord}_E}{\mathrm{ord}_E(h)}\Big)>0.$$

Fix a positive integer n such that $e_E+n\gamma_E>0$ for any component E of Φ. For a quasi-monomial valuation v on X, let $\phi(v):=b(v)+na(v,X,B+D)$. By (5.3),

$$\phi(v)=na(v,Y,\Phi)+\sum_E(e_E+n\gamma_E)v(E).$$

Since $n>0$, $\alpha_E>0$, and $e_E+n\gamma_E>0$ for the finitely many components E of Φ, (5.3) implies that there exists a real number $C_1\geq 1$ such that

$$C_1^{-1}a(v,X,B)\leq\phi(v)\leq C_1a(v,X,B) \tag{5.4}$$

for any quasi-monomial valuation v on X.

Let $v\in\mathcal{Q}_x$. Since $v\geq 0$ on $\mathcal{O}_{X,x}$, $v(g_l)\geq 0$ for any l. Since $\beta_l\geq 0$ for any l, $b_0=b(v)+\sum_l\beta_lv(g_l)\geq b(v)>0$. For a real number $\epsilon>0$, let $\lambda:=\epsilon/(1+\epsilon n)$. Since $\Psi_\lambda-(B+D)=-\lambda b_0H+\lambda\sum_l\beta_l\,\mathrm{div}(g_l)$, for any quasi-monomial valuation v on X,

$$a(v,X,\Psi_\lambda)=a(v,X,B+D)+\lambda b(v)=\frac{a(v,X,B+D)+\epsilon\phi(v)}{1+\epsilon n}. \tag{5.5}$$

Assume that $\lambda b_0\leq c(1-\tau)$. By (5.2),

$$\Psi_\lambda-B=\big(c(1-\tau)-\lambda b_0\big)H+\frac{c}{qr}\sum_p\mathrm{div}(P_p)+\lambda\sum_l\beta_l\,\mathrm{div}(g_l)\geq 0,$$

since h, P_p, and g_l are regular functions on X. Moreover, $K_X+\Psi_\lambda$ is $\mathbb{R}$-Cartier, and for any prime divisor E over X, (5.5), (5.4), and $a(E,X,B+D)\geq 0$ imply

$$a(E,X,\Psi_\lambda)\geq\frac{\epsilon a(E,X,B)}{C_1(1+\epsilon n)}>0.$$

Thus (X,Ψ_λ) is klt. By **Step 2**, $a(v,X,B+D)=0$ for any $v\in\mathcal{Q}$, so $a(v,X,\Psi_\lambda)=\lambda b(v)$ for any $v\in\mathcal{Q}$.

Step 4. In this step we bound a minimizer of $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$.

Let $\epsilon > 0$ be a real number with $\lambda b_0 \leq c(1-\tau)$. By Theorem 2.11, there exists a quasi-monomial valuation u on X centered at x which minimizes $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x. Since the normalized volume does not change when the valuation is multiplied by a positive real number, we may assume that $u(\mathfrak{m}_x) = 1$. Fix $v_* \in \mathcal{Q}_x$. By **Step 2**, $a(v_*, X, B+D) = 0$, so $\phi(v_*) = b(v_*)$, and (5.5) and the minimality of u imply

$$\big(a(u, X, B+D) + \epsilon\phi(u)\big)^d \,\mathrm{vol}(u) \leq \epsilon^d b(v_*)^d \,\mathrm{vol}(v_*). \tag{5.6}$$

In particular, by (5.4), $\widehat{\mathrm{vol}}_{X,B}(u) \leq C_1^d b(v_*)^d \,\mathrm{vol}(v_*)$. By [HLQ20, Lemma 2.14] (see [Li18, Theorem 3.1] for the Izumi type estimate when $B = 0$ and X is $\mathbb{Q}$-Gorenstein), there exists a real number $C_2 > 0$ such that

$$\widehat{\mathrm{vol}}_{X,B}(v) \geq C_2^{-1} \frac{a(v, X, B)}{v(\mathfrak{m}_x)} \quad \text{and} \quad v(f) \leq C_2 a(v, X, B) \,\mathrm{ord}_x(f)$$

for any quasi-monomial valuation v on X centered at x and any non-zero $f \in \mathcal{O}_{X,x}$, where $\mathrm{ord}_x(f) := \max\{m \in \mathbb{Z}_{\geq 0} \mid f \in \mathfrak{m}_x^m \mathcal{O}_{X,x}\}$. Let $C_3 := C_2 C_1^d b(v_*)^d \,\mathrm{vol}(v_*)$. Then $a(u, X, B) \leq C_3$ and $u(f) \leq C_2 C_3 \,\mathrm{ord}_x(f)$ for any non-zero $f \in \mathcal{O}_{X,x}$. Thus $\mathfrak{a}_t(u) \subset \mathfrak{m}_x^{\lceil t/(C_2C_3)\rceil} \mathcal{O}_{X,x}$ for any $t > 0$, and

$$\mathrm{vol}(u) \geq \frac{e(\mathfrak{m}_x)}{(C_2 C_3)^d} > 0,$$

where $e(\mathfrak{m}_x)$ is the Hilbert–Samuel multiplicity of $\mathfrak{m}_x \mathcal{O}_{X,x}$. By (5.6) and (5.4), there exists a real number $C_4 > 0$, independent of ϵ, such that

$$a(u, X, B+D) \leq C_4 \epsilon \quad \text{and} \quad \phi(u) \leq C_4. \tag{5.7}$$

Step 5. In this step we compare u with the monomial retraction of u on Σ.

Let ξ be the center of u on Y, and let $E_1, \ldots, E_m$ be the components of Σ containing ξ. The *monomial retraction* u_Σ of u on Σ is the monomial valuation, at the generic point of the component of $E_1 \cap \cdots \cap E_m$ containing ξ, in local equations of $E_1, \ldots, E_m$ with weights $u(E_1), \ldots, u(E_m)$ [JM12, Section 4.3]. By [JM12, Lemma 4.7 and Corollary 4.8], $u_\Sigma(f) \leq u(f)$ for any $f \in \mathcal{O}_{Y,\xi}$ and $u_\Sigma(E) = u(E)$ for any component E of Σ, and $a(u_\Sigma, Y, \Sigma) = 0$ by [JM12, (5.1)]. Since Φ is snc, a component E of Φ which is not a component of Σ does not contain the center of u_Σ, so $u_\Sigma(E) = 0$. Thus $a(u_\Sigma, Y, \Phi) = a(u_\Sigma, Y, \Sigma) = 0$, and (5.3) implies

$$a(u_\Sigma, X, B+D) = 0 \quad \text{and} \quad \phi(u_\Sigma) = \sum_{E \subset \Sigma} e_E u(E) \leq \phi(u), \tag{5.8}$$

where the inequality holds because

$$\phi(u) - \phi(u_\Sigma) = n a(u, Y, \Phi) + \sum_{E \not\subset \Sigma} (e_E + n\gamma_E) u(E) \geq 0.$$

Since $\gamma_E > 0$ for the finitely many components E of Φ which are not components of Σ, (5.3) implies $a(u, Y, \Phi) \leq a(u, X, B+D)$ and $\gamma_E u(E) \leq a(u, X, B+D)$ for any such E. We have $a(u, Y, \Sigma) = a(u, Y, \Phi) + \sum_{E \not\subset \Sigma} u(E)$. Moreover, $\mathfrak{m}_x \mathcal{O}_Y$ is the ideal of the divisor $\sum_E (\mathrm{mult}_E \mathfrak{m}_x) E$, so

$$u(\mathfrak{m}_x) - u_\Sigma(\mathfrak{m}_x) = \sum_{E \not\subset \Sigma} (\mathrm{mult}_E \mathfrak{m}_x) u(E).$$

Hence there exists a real number $C_5 > 0$, independent of ϵ, such that

$$a(u, Y, \Sigma) \leq C_5 a(u, X, B+D) \quad \text{and} \quad u_\Sigma(\mathfrak{m}_x) \geq 1 - C_5 a(u, X, B+D). \tag{5.9}$$

Let $z_1, \ldots, z_N$ be regular functions on X generating $\mathfrak{m}_x$. By [OAI26b, Lemma E.1], applied to Y, Σ, and $z_1, \ldots, z_N$, there exists a real number $C_6 > 0$ such that, for any positive integer L, any polynomial G in $z_1, \ldots, z_N$ of degree less than L which is non-zero as a regular function on X, and any quasi-monomial valuation v on X centered at x,

$$v_\Sigma(G) \leq v(G) \leq v_\Sigma(G) + C_6 L a(v, Y, \Sigma), \tag{5.10}$$

where v_Σ is the monomial retraction of v on Σ. The standing assumption of [OAI26b, Section E] is that the base is a $\mathbb{Q}$-Gorenstein klt germ, but [OAI26b, Lemma E.1] is an estimate on the smooth variety Y for the reduced snc divisor Σ and finitely many fixed functions, and the argument there only uses étale coordinates on Y, the snc structure of Σ, the equations satisfied by the fixed functions, and the estimate $v(\mathcal{L}f) \geq v(f) - a(v, Y, \Sigma)$ for logarithmic vector fields $\mathcal{L}$ along Σ. The canonical divisor of the base does not enter, so X need not be $\mathbb{Q}$-Gorenstein.

Fix a real number $\epsilon_0 > 0$ such that $2C_4C_5\epsilon_0 \leq 1$ and $7C_4C_5C_6\epsilon_0 \leq 1$, and assume from now on that $0 < \epsilon < \epsilon_0$. By (5.7) and (5.9), we have $u_\Sigma(\mathfrak{m}_x) \geq 1/2$ and $7C_5C_6a(u, X, B + D) < 1$. Since $u_\Sigma \geq 0$ on $\mathcal{O}_X$ and $u_\Sigma > 0$ on $\mathfrak{m}_x$, u_Σ is a non-trivial quasi-monomial valuation on X centered at x. We show that

$$u(f) \leq \big(1 + 7C_5C_6a(u, X, B + D)\big)u_\Sigma(f) \tag{5.11}$$

for any non-zero $f \in \mathcal{O}_{X,x}$. If f is a unit, then $u(f) = u_\Sigma(f) = 0$. Assume that $f \in \mathfrak{m}_x\mathcal{O}_{X,x}$. Then $u_\Sigma(f) \geq u_\Sigma(\mathfrak{m}_x) \geq 1/2$. Let $L_f := \lceil 3u_\Sigma(f)\rceil + 1$. Since $\mathcal{O}_{X,x}/\mathfrak{m}_x^{L_f}\mathcal{O}_{X,x} \cong \mathcal{O}_X(X)/\mathfrak{m}_x^{L_f}$ and $\mathfrak{m}_x$ is generated by $z_1, \dots, z_N$, there exists a polynomial G in $z_1, \dots, z_N$ of degree less than L_f with $f - G \in \mathfrak{m}_x^{L_f}\mathcal{O}_{X,x}$. Since $u(\mathfrak{m}_x) = 1$ and $u_\Sigma(\mathfrak{m}_x) \geq 1/2$,

$$u(f - G) \geq L_f \quad \text{and} \quad u_\Sigma(f - G) \geq \frac{L_f}{2} > u_\Sigma(f).$$

Hence $u_\Sigma(G) = u_\Sigma(f)$, and in particular $G \neq 0$. Since $u_\Sigma(f) \geq 1/2$, $L_f \leq 3u_\Sigma(f) + 2 \leq 7u_\Sigma(f)$. By (5.10) and (5.9),

$$u(G) \leq u_\Sigma(f) + 7C_6u_\Sigma(f)a(u, Y, \Sigma) \leq \big(1 + 7C_5C_6a(u, X, B + D)\big)u_\Sigma(f) < 2u_\Sigma(f) < L_f.$$

Since $u(f - G) \geq L_f > u(G)$, $u(f) = u(G)$, which proves (5.11).

Step 6. In this step we show that $a(u, X, B + D) = 0$ and conclude the proof.

Let $\omega := 1 + 7C_5C_6a(u, X, B + D)$. By (5.11), $\mathfrak{a}_{\omega t}(u) \subset \mathfrak{a}_t(u_\Sigma)$ for any $t > 0$, so $\mathrm{vol}(u) \geq \omega^{-d}\,\mathrm{vol}(u_\Sigma)$. Suppose that $a(u, X, B + D) > 0$. By (5.7) and $7C_4C_5C_6\epsilon < 1$,

$$(\omega - 1)\epsilon\phi(u) = 7C_5C_6a(u, X, B + D)\epsilon\phi(u) \leq 7C_4C_5C_6\epsilon a(u, X, B + D) < a(u, X, B + D).$$

Thus, by (5.5) and (5.8),

$$\begin{aligned}(1 + \epsilon n)^d\widehat{\mathrm{vol}}_{X,\Psi_\lambda}(u_\Sigma) &= \big(\epsilon\phi(u_\Sigma)\big)^d\,\mathrm{vol}(u_\Sigma) \leq \big(\omega\epsilon\phi(u)\big)^d\,\mathrm{vol}(u)\\ &< \big(a(u, X, B + D) + \epsilon\phi(u)\big)^d\,\mathrm{vol}(u) = (1 + \epsilon n)^d\widehat{\mathrm{vol}}_{X,\Psi_\lambda}(u).\end{aligned}$$

Since u_Σ is centered at x, this contradicts the minimality of u. Hence $a(u, X, B + D) = 0$.

By **Step 2**(iii), $u(h) > 0$. Let $w := u/u(h)$. Since u is centered at x, **Step 2**(ii) implies $w \in \mathcal{Q}_x$. By **Step 3**, $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}(v) = \lambda^d b(v)^d\,\mathrm{vol}(v)$ for any $v \in \mathcal{Q}_x$. Since u minimizes $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ and $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}(w) = \widehat{\mathrm{vol}}_{X,\Psi_\lambda}(u)$, $b(w)^d\,\mathrm{vol}(w) \leq b(v)^d\,\mathrm{vol}(v)$ for any $v \in \mathcal{Q}_x$. If $v \in \mathcal{Q}_x$ satisfies $b(v)^d\,\mathrm{vol}(v) = b(w)^d\,\mathrm{vol}(w)$, then v minimizes $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x, so v is a positive real multiple of w by Theorem 2.11, and $v = w$ since $v(h) = w(h) = 1$. This proves (1). Since w is determined by (1), for any $0 < \epsilon < \epsilon_0$ with $\lambda b_0 \leq c(1 - \tau)$, the minimizer u of $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ constructed in **Step 4** is a positive real multiple of w, so w minimizes $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x.

Let

$$\lambda_0 := \min\left\{\frac{\epsilon_0}{1 + \epsilon_0 n}, \frac{c(1 - \tau)}{b_0}\right\}.$$

Then $\lambda_0 b_0 \leq c(1-\tau)$. Since $\epsilon \to \epsilon/(1+\epsilon n)$ is an increasing bijection from $(0, \epsilon_0)$ to $(0, \epsilon_0/(1+\epsilon_0 n))$, any real number $0 < \lambda < \lambda_0$ is equal to $\epsilon/(1 + \epsilon n)$ for some $0 < \epsilon < \epsilon_0$, and $\lambda b_0 < c(1 - \tau)$. By **Step 3** and the previous paragraph, $\Psi_\lambda \geq B$, (X, Ψ_λ) is klt, w minimizes $\widehat{\mathrm{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x, and $a(v, X, \Psi_\lambda) = \lambda b(v)$ for any $v \in \mathcal{Q}$. By **Step 2**, $D \geq 0$ is a non-negative real linear combination of principal divisors, D satisfies (5.2), and $v(P_p) = q\tau$ for

any $v \in \mathcal{Q}$ and any p. Thus (2) and (3) hold for D and λ_0. Since $\tau_1, \dots, \tau_s$ are arbitrary positive rational numbers with $\tau < 1$, this proves (3), and (2) follows from (3). □

5.2. **Kollár components on the slice.** In this subsection, we prove that the marked graded pair of the valuation w of Definition-Proposition 5.2 is klt, and that w is a pointwise limit of normalized Kollár components of $(X \ni x, B)$ on the constrained slice.

Remark 5.3. [OAI26a, Lemma 5.4] is formulated for boundaries $B \geq 0$ with rational coefficients, in the setting of [OAI26a, Lemmas 5.2 and 5.3], where the germ is an affine klt germ with a canonical generator. The proofs of [OAI26a, Lemmas 5.2 and 5.4] only use that w is quasi-monomial, that $\operatorname{gr}_w \mathcal{O}_{X,x}$ is finitely generated, an Izumi-type estimate for w, the one-parameter family of weights of [OAI26a, Lemma 5.3] but not the part of [OAI26a, Lemma 5.3] on canonical generators, and the normality of X_0. They do not use that X is klt or has a canonical generator. For Lemma 5.4, the Izumi-type estimate is [HLQ20, Lemma 2.14(2)] for the klt pair (X, Ψ_λ), and the normality of X_0 is part (1) of Lemma 5.4, which is proved before part (4).

Lemma 5.4. *Notation and conditions as in Set-up 5.1. Let w be as in Definition-Proposition 5.2, let $\tau_1, \dots, \tau_s$, τ, q, r, $P_1, \dots, P_r$, D, and λ_0 be as in Definition-Proposition 5.2(3), and let $0 < \lambda < \lambda_0$ be a real number. Let $h_0 := \operatorname{in}_w(h)$, $H_0 := \operatorname{div}(h_0)$, and $g_{l,0} := \operatorname{in}_w(g_l)$ for any l. For an $\mathbb{R}$-divisor $\Delta = \sum_k \delta_k \Delta_k$ on X with prime divisors Δ_k, let $\Delta_0 := \sum_k \delta_k (\Delta_k)_0$ as in Definition 2.9. Then the following hold.*

(1) $\operatorname{gr}_w \mathcal{O}_{X,x}$ is finitely generated, and $X_0 := \operatorname{Spec} \operatorname{gr}_w \mathcal{O}_{X,x}$ is a normal affine variety with $\dim X_0 = d$.
(2) The marked graded pair $(X_0, (\Psi_\lambda)_0)$ of w with respect to $(X \ni x, \Psi_\lambda)$ is klt.
(3) $c(1-\tau) - \lambda b_0 > 0$.
(4) We have

$$D_0 = c(1-\tau) H_0 + \frac{c}{qr} \sum_{p=1}^{r} \operatorname{div}(\operatorname{in}_w P_p)$$

and

$$(\Psi_\lambda)_0 = B_0 + D_0 - \lambda b_0 H_0 + \lambda \sum_l \beta_l \operatorname{div}(g_{l,0}). \tag{5.12}$$

Proof. By Definition-Proposition 5.2(2), $\Psi_\lambda \geq B$, (X, Ψ_λ) is klt, and w minimizes $\widehat{\operatorname{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x. By Theorem 2.11 applied to $(X \ni x, \Psi_\lambda)$, there exists a minimizer w' of $\widehat{\operatorname{vol}}_{X,\Psi_\lambda}$ such that $\operatorname{gr}_{w'} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of w' with respect to $(X \ni x, \Psi_\lambda)$ is klt, and $w = \mu w'$ for some real number $\mu > 0$. Since $\mathfrak{a}_\gamma(w) = \mathfrak{a}_{\gamma/\mu}(w')$ for any $\gamma \geq 0$, the rings $\operatorname{gr}_w \mathcal{O}_{X,x}$ and $\operatorname{gr}_{w'} \mathcal{O}_{X,x}$ coincide after multiplying the grading by μ, and $\operatorname{in}_w(\varphi) = \operatorname{in}_{w'}(\varphi)$ for any non-zero $\varphi \in \mathcal{O}_{X,x}$. Thus $\operatorname{gr}_w \mathcal{O}_{X,x}$ is finitely generated and $(X_0, (\Psi_\lambda)_0)$ is klt. In particular, X_0 is normal. Moreover, X_0 is the central fiber of the special degeneration of X induced by the minimizer w [Zhu24c, Section 2.6], so $\dim X_0 = d$.

By [OAI26a, Lemma 5.4], for any non-zero regular function φ on X,

$$(\operatorname{div}(\varphi))_0 = \operatorname{div}(\operatorname{in}_w \varphi). \tag{5.13}$$

For the setting of [OAI26a, Lemma 5.4], see the remark before Lemma 5.4. We use [OAI26a, Lemma 5.4] for one principal divisor at a time, and the map $\Delta \to \Delta_0$ is $\mathbb{R}$-linear because it is defined prime divisor by prime divisor. Thus (5.13) extends to any $\mathbb{R}$-linear combination of divisors of non-zero regular functions on X. Since $\mathcal{Q}_x \neq \emptyset$ and $v(P_p) = q\tau < +\infty$ for $v \in \mathcal{Q}$, each P_p is non-zero. Hence the formulas for D_0 and $(\Psi_\lambda)_0$ follow from Definition-Proposition 5.2(3) and from

$$\Psi_\lambda = B + D - \lambda b_0 H + \lambda \sum_l \beta_l \operatorname{div}(g_l).$$

Finally, pick $v \in \mathcal{Q}_x$. Since $v(h) = 1$, $\beta_l \geq 0$, and $v(g_l) \geq 0$ for any l, we have $0 < b(v) \leq b_0$. Thus $\lambda b_0 < \lambda_0 b_0 \leq c(1-\tau)$. □

The difficulty in Proposition 5.5 is that the theory of Kollár models and special complements in [XZ25] concerns $\mathbb{Q}$-divisors, so we rationalize B and B_0 simultaneously by one map ρ of Definition-Theorem 4.1, which depends on the germ and preserves any $\mathbb{Q}$-affine relation among the coefficients of B and of B_0, in particular the relations expressing each coefficient of B_0 as a non-negative integral combination of coefficients of B (so that the rationalization of B_0 is the degeneration of the rationalization of B), and we return to the $\mathbb{R}$-divisor B by Definition-Theorem 4.1(3).

Proposition 5.5. *Notation and conditions as in Set-up 5.1, and let w be as in Definition-Proposition 5.2. Then $\mathrm{gr}_w \mathcal{O}_{X,x}$ is finitely generated, and the marked graded pair (X_0, B_0) of w is klt. Moreover, there exist Kollár components $E_1, E_2, \dots$ of $(X \ni x, B)$ such that $a(E_i, X, B) = c\,\mathrm{ord}_{E_i}(h)$ and $\mathrm{ord}_{E_i}(f_j) \geq m_j \mathrm{ord}_{E_i}(h)$ for any i and j, and $\mathrm{ord}_{E_i} / \mathrm{ord}_{E_i}(h)$ converges to w pointwise on $\mathcal{O}_{X,x}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that $\mathrm{gr}_w \mathcal{O}_{X,x}$ is finitely generated and that (X_0, B_0) is klt.

Fix $\tau_1, \dots, \tau_s$, τ, q, r, $P_1, \dots, P_r$, D, and λ_0 as in Definition-Proposition 5.2(3), and fix a real number $0 < \lambda < \lambda_0$. By Lemma 5.4, $\mathrm{gr}_w \mathcal{O}_{X,x}$ is finitely generated, $(X_0, (\Psi_\lambda)_0)$ is klt, and

$$\Gamma_0 := (\Psi_\lambda)_0 - B_0 = (c(1-\tau) - \lambda b_0)H_0 + \frac{c}{qr}\sum_{p=1}^{r} \mathrm{div}(\mathrm{in}_w P_p) + \lambda \sum_l \beta_l \,\mathrm{div}(g_{l,0})$$

is a non-negative real linear combination of divisors of non-zero regular functions on X_0. Thus $K_{X_0} + B_0 = (K_{X_0} + (\Psi_\lambda)_0) - \Gamma_0$ is $\mathbb{R}$-Cartier, and for any prime divisor F over X_0,

$$a(F, X_0, B_0) = a(F, X_0, (\Psi_\lambda)_0) + \mathrm{mult}_F \Gamma_0 > 0.$$

Hence (X_0, B_0) is klt.

Step 2. In this step we rationalize B and B_0 simultaneously.

Write $B = \sum_i e_i B_i$ with distinct prime divisors B_i, and write $(B_i)_0 = \sum_\alpha n_{i\alpha} B_{0,\alpha}$, where $B_{0,\alpha}$ are the distinct prime components of B_0 and $n_{i\alpha}$ are non-negative integers. Then

$$B_0 = \sum_\alpha e^0_\alpha B_{0,\alpha}, \qquad e^0_\alpha = \sum_i n_{i\alpha} e_i.$$

Since (X, B) is klt, $0 \leq e_i < 1$ for any i. Since (X_0, B_0) is klt by **Step 1**, $a(B_{0,\alpha}, X_0, B_0) = 1 - e^0_\alpha > 0$, so $0 \leq e^0_\alpha < 1$ for any α. Let $J := \{e_i\}_i \cup \{e^0_\alpha\}_\alpha \subset [0,1]$, and let $\rho : J^+ \to J_{\mathbb{Q}}$ be the map of Definition-Theorem 4.1 for d and J. Since $0 \in J^+$ and $\rho(0) = 0$, each equality $e^0_\alpha = \sum_i n_{i\alpha} e_i$ is a $\mathbb{Q}$-affine relation among elements of J^+, so $\rho(e^0_\alpha) = \sum_i n_{i\alpha}\rho(e_i)$. Thus

$$(B^\rho)_0 = \sum_i \rho(e_i)(B_i)_0 = \sum_\alpha \rho(e^0_\alpha) B_{0,\alpha} = (B_0)^\rho.$$

By Lemma 5.4, X_0 is a normal variety with $\dim X_0 = d$. By Definition-Theorem 4.1(1) applied on X with $\Delta := B$, and on X_0 with B_0 in place of both B and Δ (since $e_i, e^0_\alpha \in J \subset J^+$), the pairs (X, B^ρ) and $(X_0, (B_0)^\rho)$ are klt, and $K_X + B^\rho$ and $K_{X_0} + (B_0)^\rho$ are $\mathbb{Q}$-Cartier. Hence the marked graded pair of w with respect to $(X \ni x, B^\rho)$, which is $(X_0, (B^\rho)_0) = (X_0, (B_0)^\rho)$, is klt.

Step 3. In this step we construct a rational polyhedral cone of valuations containing w on which $a(\cdot, X, B)$, h, and the f_j are linear.

The valuation $w \in \mathcal{Q}_x$ is quasi-monomial. By **Step 2** and [XZ25, Theorem 4.1] applied to $(X \ni x, B^\rho)$ and w, there exists a Kollár model $\pi : Y \to X$ of $(X \ni x, B^\rho)$ with $w \in \mathrm{QM}(Y, E)$, where $E = \mathrm{Exc}(\pi)$. Recall from [XZ25, Definition 3.7] that π is an isomorphism over $X \backslash \{x\}$, $(Y, \pi_*^{-1}B^\rho + E)$ is qdlt, and $-(K_Y + \pi_*^{-1}B^\rho + E)$ is π-ample, and recall from [XZ25, Section 2.2.2] that $\mathrm{QM}(Y, E)$ is the union of the cones $\mathrm{QM}_\eta(Y, F)$ of valuations monomial at η with respect to F, over the points $\eta \in Y$ and the reduced divisors $F \leq E$ such that (Y, F) is toroidal at η. Choose η and F with $w \in \mathrm{QM}_\eta(Y, F)$. A toroidal log resolution of (Y, F) near η subdivides $\mathrm{QM}_\eta(Y, F)$

into finitely many rational simplicial cones, each consisting of the valuations monomial at a regular point of the resolution with respect to an snc divisor through that point (cf. [KKMS73, Chapter II, §2, Theorems 4*, 6*, and 11*]). Thus there exist a projective birational morphism $\pi' : Y' \to Y$, a point $\eta' \in Y'$ such that $\mathcal{O}_{Y',\eta'}$ is regular, and a regular system of parameters $z_1, \dots, z_N$ of $\mathcal{O}_{Y',\eta'}$ with the following properties. Near η', z_k defines a prime divisor F'_k with $\pi'(F'_k) \subset E$ for any k. For $\alpha \in \mathbb{R}^N_{\geq 0}$, let v_α be the monomial valuation at η' with $v_\alpha(z_k) = \alpha_k$ for any k. Then $v_\alpha \in \mathrm{QM}(Y, E)$ for any α, and $w = v_{\alpha^w}$ for some $\alpha^w \in \mathbb{R}^N_{\geq 0}$. Let $\sigma := \mathbb{R}^N_{\geq 0}$. For any $0 \neq \alpha \in \sigma$, the center of v_α on Y' is contained in some F'_k, and $\pi(E) = \{x\}$, so $\mathrm{center}_X(v_\alpha) = x$.

Let $0 \neq \varphi \in \mathcal{O}_{Y',\eta'}$, for example the pullback of a non-zero element of $\mathcal{O}_{X,x}$. Since $\widehat{\mathcal{O}}_{Y',\eta'} \cong \kappa(\eta')[[z_1, \dots, z_N]]$, we write $\varphi = \sum_{\mathbf{n}} c_{\mathbf{n}} z^{\mathbf{n}}$ with $\mathbf{n} \in \mathbb{Z}^N_{\geq 0}$ and $c_{\mathbf{n}} \in \kappa(\eta')$. Then $v_\alpha(\varphi) = \min\{\langle \alpha, \mathbf{n} \rangle \mid c_{\mathbf{n}} \neq 0\}$. By Dickson's lemma [CLO15, Chapter 2, §4, Theorem 5], the set $\{\mathbf{n} \mid c_{\mathbf{n}} \neq 0\}$ has finitely many minimal elements $\mathbf{n}^1, \dots, \mathbf{n}^{N_\varphi}$ with respect to the componentwise order, so

$$v_\alpha(\varphi) = \min\{\langle \alpha, \mathbf{n}^1 \rangle, \dots, \langle \alpha, \mathbf{n}^{N_\varphi} \rangle\} \quad \text{for any } \alpha \in \sigma.$$

In particular, $\alpha \to v_\alpha(\varphi)$ is continuous and piecewise linear with rational coefficients. Let $B_{Y'}$ be the $\mathbb{R}$-divisor with $K_{Y'} + B_{Y'} = (\pi \circ \pi')^*(K_X + B)$ near η', and write $B_{Y'} = \sum_a \gamma'_a \operatorname{div}(y_a)$ near η' with $\gamma'_a \in \mathbb{R}$ and non-zero $y_a \in \mathcal{O}_{Y',\eta'}$, which is possible since $\mathcal{O}_{Y',\eta'}$ is regular. By [HLQ20, Lemma 2.9] applied to $\pi \circ \pi'$ and the pairs (X, B) and $(Y', 0)$ near η', where $K_{Y'} - (\pi \circ \pi')^*(K_X + B) = -B_{Y'}$, we have $a(v_\alpha, X, B) = a(v_\alpha, Y', 0) - \sum_a \gamma'_a v_\alpha(y_a)$. Since $(Y', \sum_k F'_k)$ is log smooth near η' and $a(F'_k, Y', 0) = 1$ for any k, [HLQ20, Definition 2.7(2)] implies $a(v_\alpha, Y', 0) = \sum_k \alpha_k$. Thus

$$a(v_\alpha, X, B) = \sum_{k=1}^N \alpha_k - \sum_a \gamma'_a v_\alpha(y_a).$$

Hence there exists a subdivision of σ into finitely many rational polyhedral cones such that on each cone, the function $\alpha \to a(v_\alpha, X, B)$ is linear, and the functions $\alpha \to v_\alpha(h)$ and $\alpha \to v_\alpha(f_j)$, $1 \leq j \leq s$, are linear with rational coefficients. Let σ_w be a cone of the subdivision with $\alpha^w \in \sigma_w$.

Step 4. In this step we construct the Kollár components E_i.

Let $\ell(\alpha) := a(v_\alpha, X, B) - c v_\alpha(h)$ for $\alpha \in \sigma_w$. Then ℓ is linear. For any $0 \neq \alpha \in \sigma_w \cap \mathbb{Q}^N$, v_α is an Abhyankar valuation of rational rank one, so $v_\alpha = \mu \operatorname{ord}_{E'}$ for a prime divisor E' over X and a rational number $\mu > 0$ [KM98, Lemma 2.45], and

$$\ell(\alpha) = \mu\big(a(E', X, B) - c \operatorname{ord}_{E'}(h)\big) = \mu a(E', X, B + cH) \geq 0$$

because $(X, B + cH)$ is lc. By continuity, $\ell \geq 0$ on σ_w. Since σ_w is a rational polyhedral cone, the zero set of ℓ on σ_w is a face of σ_w [Ful93, Section 1.2], hence a rational polyhedral cone. Thus

$$\sigma^0_w := \{\alpha \in \sigma_w \mid \ell(\alpha) = 0,\ v_\alpha(f_j) \geq m_j v_\alpha(h) \text{ for any } j\}$$

is a rational polyhedral cone. Since $w \in \mathcal{Q}$, we have $a(w, X, B) = cw(h)$ and $w(f_j) \geq m_j = m_j w(h)$, so $\alpha^w \in \sigma^0_w$. The rational points of σ^0_w are dense in σ^0_w, so there exist non-zero $\alpha^1, \alpha^2, \dots \in \sigma^0_w \cap \mathbb{Q}^N$ converging to α^w. By [KM98, Lemma 2.45], $v_{\alpha^i} = \mu_i \operatorname{ord}_{E_i}$ for a prime divisor E_i over X and a rational number $\mu_i > 0$, and $\mathrm{center}_X E_i = x$ by **Step 3**.

We show that E_i is a Kollár component of $(X \ni x, B)$. Since $v_{\alpha^i} \in \mathrm{QM}(Y, E)$, [XZ25, Theorem 4.1] implies that $\mathrm{gr}_{v_{\alpha^i}} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of v_{α^i} with respect to $(X \ni x, B^\rho)$ is klt. Since $\operatorname{ord}_{E_i} = \mu_i^{-1} v_{\alpha^i}$, $\mathrm{gr}_{\operatorname{ord}_{E_i}} \mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of ord_{E_i} with respect to $(X \ni x, B^\rho)$ is klt. By Theorem 2.10, E_i is a Kollár component of $(X \ni x, B^\rho)$. Thus there exists a projective birational morphism $\pi_i : Y_i \to X$ with exceptional locus E_i such that $(Y_i, \pi_{i*}^{-1} B^\rho + E_i)$ is plt and $-(K_{Y_i} + \pi_{i*}^{-1} B^\rho + E_i)$ is π_i-ample. By Definition 2.3, Lemma 2.4(1), and Lemma 2.7, π_i is the prime extraction of E_i, which is defined over X and is an isomorphism over $X \backslash \{x\}$, and $-E_i$ is $\mathbb{Q}$-Cartier and π_i-ample, which also follows from

$$K_{Y_i} + \pi_{i*}^{-1} B^\rho + E_i = \pi_i^*(K_X + B^\rho) + a(E_i, X, B^\rho) E_i$$

since $a(E_i, X, B^\rho)$ is a positive rational number. By Definition-Theorem 4.1(3), $(Y_i, \pi_{i*}^{-1}B + E_i)$ is plt near E_i. Since π_i is an isomorphism over $X\backslash\{x\}$ and (X, B) is klt, $(Y_i, \pi_{i*}^{-1}B + E_i)$ is plt. Moreover,

$$K_{Y_i} + \pi_{i*}^{-1}B + E_i = \pi_i^*(K_X + B) + a(E_i, X, B)E_i$$

with $a(E_i, X, B) > 0$, so $-(K_{Y_i} + \pi_{i*}^{-1}B + E_i)$ is π_i-ample. Hence E_i is a Kollár component of $(X \ni x, B)$.

Since $\alpha^i \in \sigma_w^0$, we have $a(v_{\alpha^i}, X, B) = cv_{\alpha^i}(h)$ and $v_{\alpha^i}(f_j) \geq m_j v_{\alpha^i}(h)$. Dividing by μ_i, we obtain $a(E_i, X, B) = c\operatorname{ord}_{E_i}(h)$ and $\operatorname{ord}_{E_i}(f_j) \geq m_j \operatorname{ord}_{E_i}(h)$ for any j. In particular, $\operatorname{ord}_{E_i}(h) = a(E_i, X, B)/c > 0$ and $\operatorname{ord}_{E_i}/\operatorname{ord}_{E_i}(h) = v_{\alpha^i}/v_{\alpha^i}(h)$. For any non-zero $\varphi \in \mathcal{O}_{X,x}$, $\alpha \to v_\alpha(\varphi)$ is continuous on σ by **Step 3**, so $v_{\alpha^i}(\varphi) \to w(\varphi)$ and $v_{\alpha^i}(h) \to w(h) = 1$. Thus $\operatorname{ord}_{E_i}/\operatorname{ord}_{E_i}(h)$ converges to w pointwise on $\mathcal{O}_{X,x}$. □

5.3. **The central slice.** In this subsection, we prove that the valuation given by the grading of X_0 is the unique minimizer on the constrained slice of the marked graded pair, that the minimum value on the constrained slice of the marked graded pair equals the minimum value on $\mathcal{Q}_x$, and that the valuations on both constrained slices are bounded in terms of w and w_0.

Lemma 5.6. *Let $(X \ni x, B)$ be a klt pair of dimension d over $\mathbb{C}$ with x a closed point, and let w be a valuation on X with $\operatorname{center}_X w = x$ which minimizes $\widehat{\operatorname{vol}}_{X,B}$ among the valuations on X centered at x. Then for any valuation v on X with $\operatorname{center}_X v = x$ and $a(v, X, B) < +\infty$ and any non-zero $f \in \mathcal{O}_{X,x}$,*

$$v(f) \leq 2^{d+1}\frac{a(v, X, B)}{a(w, X, B)}w(f).$$

Proof. If B is a $\mathbb{Q}$-divisor and v is quasi-monomial, an inequality $v(f) \leq C\frac{a(v,X,B)}{a(w,X,B)}w(f)$ with some constant C is [OAI26a, Lemma 4.1]. We prove it for any v with $a(v, X, B) < +\infty$ and with the constant 2^{d+1}. For an integer $m \geq 1$ and a valuation u on X with $\operatorname{center}_X u = x$, let

$$T_m(u) := \sum_{i\geq 1} \dim_{\kappa(x)} \frac{\mathfrak{a}_i(u)\mathcal{O}_{X,x}}{\mathfrak{a}_i(u)\mathcal{O}_{X,x} \cap \mathfrak{a}_m(w)\mathcal{O}_{X,x}}.$$

By [XZ21, Theorem 3.10], w is K-semistable in the sense of [XZ21, Definition 3.4], and by [XZ21, Lemma 3.3] this means

$$a(v, X, B) \geq a(w, X, B) \lim_{m\to+\infty} \frac{T_m(v)}{T_m(w)}$$

for any valuation v on X with $\operatorname{center}_X v = x$ and $a(v, X, B) < +\infty$. For $\mathbb{R}$-divisors B, [XZ21, Theorem 3.10] holds as well, since $T_m(u)$ does not depend on B and the summation formula for multiplier ideals holds for $\mathbb{R}$-divisors, see the paragraph before [HLQ20, Theorem 3.4] and [Zhu24c, Section 2.6].

Since (X, B) is klt and w is a minimizer, $0 < a(w, X, B) < +\infty$ and $\operatorname{vol}(w) > 0$ by [HLQ20, Lemma 2.14]. Let v be a valuation on X with $\operatorname{center}_X v = x$ and $a(v, X, B) < +\infty$, and let $0 \neq f \in \mathcal{O}_{X,x}$. If f is a unit, then $v(f) = w(f) = 0$. Assume that $f \in \mathfrak{m}_x$. Then $w(f) > 0$. For an integer $m \geq 1$, let $k_m := \lfloor m/(2w(f))\rfloor$ and $t_m := m - k_m w(f)$, so $t_m \geq m/2$. If $g \in \mathcal{O}_{X,x}$ and $f^{k_m}g \in \mathfrak{a}_m(w)\mathcal{O}_{X,x}$, then $w(g) \geq t_m$. Thus multiplication by f^{k_m} induces an injective $\kappa(x)$-linear map

$$\iota_m : \mathcal{O}_{X,x}/\mathfrak{a}_{t_m}(w)\mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{a}_m(w)\mathcal{O}_{X,x}.$$

Since $v(f^{k_m}g) \geq k_m v(f)$, the image of ι_m is contained in the image of $\mathfrak{a}_i(v)\mathcal{O}_{X,x}$ for any integer $1 \leq i \leq k_m v(f)$. Since $\mathfrak{a}_{t_m}(w) \subset \mathfrak{a}_{m/2}(w)$, we obtain

$$T_m(v) \geq \lfloor k_m v(f)\rfloor \dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_{m/2}(w)\mathcal{O}_{X,x}.$$

Since $\mathfrak{a}_i(w) \subset \mathfrak{a}_m(w)$ for $i \geq m$,

$$T_m(w) \leq m \dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_m(w)\mathcal{O}_{X,x}.$$

By Definition 2.6 and $\mathrm{vol}(w) > 0$,

$$\frac{\dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_{m/2}(w)\mathcal{O}_{X,x}}{\dim_{\kappa(x)} \mathcal{O}_{X,x}/\mathfrak{a}_m(w)\mathcal{O}_{X,x}} \to 2^{-d} \quad (m \to +\infty),$$

and $\lfloor k_m v(f) \rfloor / m$ converges to $v(f)/(2w(f))$. Hence

$$\frac{a(v,X,B)}{a(w,X,B)} \geq \lim_{m\to+\infty} \frac{T_m(v)}{T_m(w)} \geq 2^{-d-1} \frac{v(f)}{w(f)}.$$

Thus $v(f) \leq 2^{d+1} \frac{a(v,X,B)}{a(w,X,B)} w(f)$. □

Remark 5.7. [LX18, Lemma 2.58] concerns $\mathbb{Q}$-divisors. The proof of [LX18, Lemma 2.58] uses the boundary only through the following fact, which does not involve the coefficients: for the valuation w of [LX18, Lemma 2.58] and a non-zero regular function φ, the degeneration of $\mathrm{div}(\varphi)$ is $\mathrm{div}(\mathrm{in}_w \varphi)$. Thus [LX18, Lemma 2.58] holds for $\mathbb{R}$-divisors as well (cf. [Zhu24c, Section 2.6]).

Proposition 5.8. *Notation and conditions as in Set-up 5.1, let w be as in Definition-Proposition 5.2, and let (X_0, B_0) be the marked graded pair of w (Proposition 5.5). Let $o \in X_0$ be the vertex, w_0 the valuation on X_0 given by the grading, $h_0 := \mathrm{in}_w(h)$, $f_{j,0} := \mathrm{in}_w(f_j)$, and $g_{l,0} := \mathrm{in}_w(g_l)$. Define $\mathcal{Q}^0$, $\mathcal{Q}^0_o$, and b^0 by the conditions of Set-up 5.1 for $(X_0 \ni o, B_0)$, h_0, c, $f_{j,0}$, m_j, $g_{l,0}$, b_0, and β_l. Then $(X_0, B_0 + c\,\mathrm{div}(h_0))$ is lc, $b^0 > 0$ on $\mathcal{Q}^0$, $w_0 \in \mathcal{Q}^0_o$, and the minimum of $(b^0)^d \cdot \mathrm{vol}$ on $\mathcal{Q}^0_o$ is attained only at w_0 and is equal to the minimum of $b^d \cdot \mathrm{vol}$ on $\mathcal{Q}_x$. Moreover, for any $v \in \mathcal{Q}_x$, any $u \in \mathcal{Q}^0_o$, any non-zero $f \in \mathcal{O}_{X,x}$, and any non-zero $f_0 \in \mathcal{O}_{X_0,o}$,*

$$v(f) \leq 2^{d+1} \frac{b(v)}{b(w)} w(f) \quad \text{and} \quad u(f_0) \leq 2^{d+1} \frac{b^0(u)}{b^0(w_0)} w_0(f_0).$$

Proof. The proof is long, so we divide it into several steps. Here b_0 and β_l are the real numbers of Set-up 5.1(6), not degenerations. Let $H_0 := \mathrm{div}(h_0)$. For an $\mathbb{R}$-linear combination $\Gamma = \sum_k \gamma_k \mathrm{div}(\varphi_k)$ of divisors of non-zero regular functions φ_k on X_0 and a valuation u on X_0 which is non-negative on regular functions, let $u(\Gamma) := \sum_k \gamma_k u(\varphi_k)$. Then $a(u, X_0, \Delta) = a(u, X_0, \Delta + \Gamma) + u(\Gamma)$ for any $\mathbb{R}$-divisor Δ on X_0 with $K_{X_0} + \Delta$ $\mathbb{R}$-Cartier and any $\Gamma \geq 0$ [HLQ20, Lemma 2.9], applied to the identity morphism of X_0. We use it only for $\Gamma = (\Psi_\lambda)_0 - B_0 \geq 0$ and $\Gamma = D_0 \geq 0$. Moreover, a non-trivial quasi-monomial valuation u on X_0 satisfies $a(u, X_0, \Delta) \geq 0$ if (X_0, Δ) is lc, and $a(u, X_0, \Delta) > 0$ if (X_0, Δ) is klt [HLQ20, Lemma 2.8].

Step 1. In this step we prove that $(X_0, B_0 + cH_0)$ is lc.

Let $\tau_1, \dots, \tau_s$, τ, q, r, $P_1, \dots, P_r$, D, and λ_0 be as in Definition-Proposition 5.2(3), and let $0 < \lambda < \lambda_0$. By Lemma 5.4, $(X_0, (\Psi_\lambda)_0)$ is klt and (5.12) holds. Let F be a prime divisor over X_0. Since $\mathrm{mult}_F \mathrm{div}(g_{l,0}) \geq 0$ for any l, (5.12) implies

$$0 < a(F, X_0, (\Psi_\lambda)_0) \leq a(F, X_0, B_0 + D_0) + \lambda b_0 \,\mathrm{mult}_F H_0.$$

Letting $\lambda \to 0$, we obtain $a(F, X_0, B_0 + D_0) \geq 0$. Since (X_0, B_0) is klt by Proposition 5.5, $K_{X_0} + B_0$ is $\mathbb{R}$-Cartier (Definition 2.9), so $(X_0, B_0 + D_0)$ is lc. By Lemma 5.4,

$$D_0 - c(1-\tau)H_0 = \frac{c}{qr} \sum_p \mathrm{div}(\mathrm{in}_w P_p) \geq 0,$$

so

$$a(F, X_0, B_0) - c(1-\tau)\,\mathrm{mult}_F H_0 \geq a(F, X_0, B_0 + D_0) \geq 0.$$

The valuation w, and hence X_0, B_0, and H_0, do not depend on $\tau_1, \dots, \tau_s$. If $s = 0$, then $\tau = 0$. If $s \geq 1$, then τ can be any positive rational number less than 1. In both cases, $a(F, X_0, B_0) \geq c\,\mathrm{mult}_F H_0$. Hence $(X_0, B_0 + cH_0)$ is lc.

From now on, we fix $\tau_1, \dots, \tau_s$, τ, q, r, $P_1, \dots, P_r$, D, λ_0, and $0 < \lambda < \lambda_0$ as in **Step 1**. In particular, $(X_0, B_0 + D_0)$ is lc.

Step 2. In this step we prove that $w_0 \in \mathcal{Q}^0_o$.

We use four properties of w_0. First, let ξ_w be the vector given by the grading of $\operatorname{gr}_w \mathcal{O}_{X,x}$, a Reeb vector for the torus action on X_0 induced by the grading [LX18, Definitions 2.13 and 2.56]. The valuation $\operatorname{wt}_{\xi_w}$ of ξ_w [LX18, the paragraph before Lemma 2.16] is the valuation given by the grading of Definition 2.9, so $w_0 = \operatorname{wt}_{\xi_w}$. Since $\operatorname{gr}_{w_0} \mathcal{O}_{X_0,o} \cong \operatorname{gr}_w \mathcal{O}_{X,x}$ and $\dim \operatorname{gr}_w \mathcal{O}_{X,x} = \dim X_0 = d$, w_0 is quasi-monomial [LX18, Lemma 2.11(2)], and $\operatorname{center}_{X_0} w_0 = o$ since $w_0 > 0$ exactly on the ideal of elements of positive degree. Since the grading is by non-negative degrees and $\mathcal{O}_{X,x} \neq \kappa(x)$, w_0 is non-trivial and $w_0 \geq 0$ on $\mathcal{O}_{X_0,o}$. Second, for any non-zero $\varphi \in \mathcal{O}_{X,x}$, $\operatorname{in}_w(\varphi)$ is homogeneous of degree $w(\varphi)$, so $w_0(\operatorname{in}_w \varphi) = w(\varphi)$. Thus

$$w_0(h_0) = 1, \quad w_0(f_{j,0}) = w(f_j) \geq m_j, \quad w_0(g_{l,0}) = w(g_l), \quad w_0(\operatorname{in}_w P_p) = w(P_p) = q\tau,$$

and $b^0(w_0) = b(w)$. Third, $\operatorname{vol}(w_0) = \operatorname{vol}(w)$ and $a(w_0, X_0, (\Psi_\lambda)_0) = a(w, X, \Psi_\lambda)$ by [LX18, Lemma 2.58]. For real coefficients, see the remark before Proposition 5.8. Fourth, $a(w, X, \Psi_\lambda) = \lambda b(w)$ by Definition-Proposition 5.2(2).

By (5.12) and the formula for D_0 in Lemma 5.4,

$$w_0\big((\Psi_\lambda)_0 - B_0\big) = c(1-\tau) - \lambda b_0 + \frac{c}{qr} \cdot r \cdot q\tau + \lambda \sum_l \beta_l w(g_l) = c - \lambda b(w).$$

Hence

$$a(w_0, X_0, B_0) = a(w_0, X_0, (\Psi_\lambda)_0) + w_0\big((\Psi_\lambda)_0 - B_0\big) = \lambda b(w) + c - \lambda b(w) = c = c w_0(h_0).$$

Thus $w_0 \in \mathcal{Q}^0_o$.

Step 3. In this step we prove that $a(u, X_0, (\Psi_\lambda)_0) = \lambda b^0(u)$ and $b^0(u) > 0$ for any $u \in \mathcal{Q}^0$.

Let $u \in \mathcal{Q}^0$, and let $1 \leq p \leq r$. By Definition-Proposition 5.2(3), $P_p = \sum_k c_k \Pi_k$ with $c_k \in \mathbb{C}$, where each Π_k is a product having, for any j, exactly $q\tau_j/m_j$ factors in $\{h^{m_j}, f_j\}$. Since $w(h) = 1$ and $w(f_j) \geq m_j$, we have $w(\Pi_k) \geq \sum_j (q\tau_j/m_j) m_j = q\tau = w(P_p)$ for any k. Thus $\operatorname{in}_w(P_p)$, the image of P_p in degree $q\tau$, is equal to $\sum_{w(\Pi_k) = q\tau} c_k \operatorname{in}_w(\Pi_k)$, and $\operatorname{in}_w(\Pi_k)$ is the corresponding product of factors in $\{h_0^{m_j}, f_{j,0}\}$. Since $u(h_0) = 1$ and $u(f_{j,0}) \geq m_j$, we have $u(\operatorname{in}_w \Pi_k) \geq q\tau$ for any k, so $u(\operatorname{in}_w P_p) \geq q\tau$. By the formula for D_0 in Lemma 5.4,

$$u(D_0) \geq c(1-\tau) + \frac{c}{qr} \cdot r \cdot q\tau = c.$$

On the other hand, $(X_0, B_0 + D_0)$ is lc by **Step 1**, so

$$c = c u(h_0) = a(u, X_0, B_0) = a(u, X_0, B_0 + D_0) + u(D_0) \geq u(D_0).$$

Thus $u(D_0) = c$, and (5.12) implies

$$a(u, X_0, (\Psi_\lambda)_0) = a(u, X_0, B_0) - u(D_0) + \lambda b_0 u(h_0) - \lambda \sum_l \beta_l u(g_{l,0}) = \lambda b^0(u).$$

Since $(X_0, (\Psi_\lambda)_0)$ is klt and u is a non-trivial quasi-monomial valuation, $a(u, X_0, (\Psi_\lambda)_0) > 0$. Hence $b^0(u) > 0$.

Step 4. In this step we prove that w_0 is the unique minimizer of $(b^0)^d \cdot \operatorname{vol}$ on $\mathcal{Q}^0_o$.

By [XZ25, Theorem 1.2(1)] applied to $(X \ni x, \Psi_\lambda)$ and the minimizer w, the log Fano cone singularity $(X_0, (\Psi_\lambda)_0; \xi_w)$ [LX18, Definition 2.23] is K-semistable. Since $\operatorname{gr}_{w_0} \operatorname{gr}_w \mathcal{O}_{X,x} = \operatorname{gr}_w \mathcal{O}_{X,x}$, the degeneration of $(X_0 \ni o, (\Psi_\lambda)_0)$ induced by w_0 is trivial (the prime components of $(\Psi_\lambda)_0$ are defined by homogeneous prime ideals, which are their own initial ideals with respect to w_0, and $\xi_{w_0} = \xi_w$). Thus [XZ25, Theorem 1.2(2)] applied to $(X_0 \ni o, (\Psi_\lambda)_0)$ and w_0 implies that w_0 minimizes $\widehat{\operatorname{vol}}_{X_0,(\Psi_\lambda)_0}$ among the valuations on X_0 centered at o. As in the paragraph after Theorem 2.11, parts (1) and (2) of [XZ25, Theorem 1.2] hold for pairs with real coefficients by [Zhu24c, Section 2.6, the sentence before Theorem 2.19].

By **Step 3**, $\widehat{\operatorname{vol}}_{X_0,(\Psi_\lambda)_0}(u) = \lambda^d b^0(u)^d \operatorname{vol}(u)$ for any $u \in \mathcal{Q}^0_o$. Since $w_0 \in \mathcal{Q}^0_o$ by **Step 2**, w_0 minimizes $(b^0)^d \cdot \operatorname{vol}$ on $\mathcal{Q}^0_o$. If $u \in \mathcal{Q}^0_o$ also attains the minimum, then u minimizes $\widehat{\operatorname{vol}}_{X_0,(\Psi_\lambda)_0}$ among the valuations on X_0 centered at o, so $u = \mu w_0$ for some real number $\mu > 0$ by Theorem 2.11

applied to $(X_0 \ni o, (\Psi_\lambda)_0)$. Since $u(h_0) = w_0(h_0) = 1$, we have $\mu = 1$. Finally, by **Step 2** and Definition-Proposition 5.2(1),

$$b^0(w_0)^d \operatorname{vol}(w_0) = b(w)^d \operatorname{vol}(w) = \min_{v \in \mathcal{Q}_x} b(v)^d \operatorname{vol}(v).$$

Step 5. In this step we prove the inequalities for v and u.

Let $v \in \mathcal{Q}_x$ and $0 \neq f \in \mathcal{O}_{X,x}$. By Definition-Proposition 5.2(2), (X, Ψ_λ) is klt, w minimizes $\widehat{\operatorname{vol}}_{X,\Psi_\lambda}$ among the valuations on X centered at x, and $a(v, X, \Psi_\lambda) = \lambda b(v) < +\infty$. Lemma 5.6 applied to $(X \ni x, \Psi_\lambda)$ implies

$$v(f) \leq 2^{d+1} \frac{\lambda b(v)}{\lambda b(w)} w(f) = 2^{d+1} \frac{b(v)}{b(w)} w(f).$$

Let $u \in \mathcal{Q}^0_o$ and $0 \neq f_0 \in \mathcal{O}_{X_0,o}$. By Lemma 5.4, $(X_0 \ni o, (\Psi_\lambda)_0)$ is a klt pair with $\dim X_0 = d$. By **Step 4**, w_0 minimizes $\widehat{\operatorname{vol}}_{X_0,(\Psi_\lambda)_0}$ among the valuations on X_0 centered at o, and by **Step 3**, $a(u, X_0, (\Psi_\lambda)_0) = \lambda b^0(u)$ and $a(w_0, X_0, (\Psi_\lambda)_0) = \lambda b^0(w_0)$. Lemma 5.6 applied to $(X_0 \ni o, (\Psi_\lambda)_0)$ implies

$$u(f_0) \leq 2^{d+1} \frac{\lambda b^0(u)}{\lambda b^0(w_0)} w_0(f_0) = 2^{d+1} \frac{b^0(u)}{b^0(w_0)} w_0(f_0).$$

□

6. Kollár components over the ground field

In this section, we descend the minimizer of Definition-Proposition 5.2 and the Kollár components near it from $\bar{k}$ to k. Kollár components of a klt germ over $\bar{k}$ are in general not defined over k, but the minimizer on a constrained slice is unique, hence Galois invariant, and it descends to k together with the associated graded ring (Definition-Proposition 6.3). The rational valuations near the minimizer are reconstructed from monomials in elements defined over k (Lemma 6.5), so they are Galois invariant as well, although the monomial charts defining them are not defined over k. The uniqueness of the minimizer also implies invariance under the groups acting on the covers considered in Sections 10 and 11.

Set-up 6.1. Let k, d, $(X \ni x, B)$, h, H, c, $f_1, \dots, f_s$, $m_1, \dots, m_s$, $g_1, \dots, g_t$, b_0, $\beta_1, \dots, \beta_t$, and $\bar{x}$ be as follows.

(1) k is a field finitely generated over $\mathbb{C}$, and $d \geq 2$ is an integer.
(2) $(X \ni x, B)$ is a klt pair over k of dimension d, where X is affine and x is a closed point.
(3) h is a non-zero regular function on X with $h(x) = 0$, $H := \operatorname{div}(h)$, and $c > 0$ is a real number such that $(X, B + cH)$ is lc.
(4) $f_1, \dots, f_s$ are non-zero regular functions on X and $m_1, \dots, m_s$ are positive integers, where $s \geq 0$.
(5) $g_1, \dots, g_t$ are non-zero regular functions on X, and $b_0, \beta_1, \dots, \beta_t$ are real numbers with $\beta_l \geq 0$ for any l, where $t \geq 0$.
(6) $\bar{x} \in X_{\bar{k}}$ is a point over x.

Let X' be the connected component of $X_{\bar{k}}$ containing $\bar{x}$. Fix an affine open neighborhood of $\bar{x}$ in X' disjoint from the lc centers of $(X_{\bar{k}}, B_{\bar{k}} + cH_{\bar{k}})$ not containing $\bar{x}$, which exists by Lemma 6.2(2), and call it the *germ of $X_{\bar{k}}$ at $\bar{x}$*. Let $\mathcal{Q}$, $\mathcal{Q}_{\bar{x}}$, and b be defined by (4)–(6) of Set-up 5.1 for the germ of $X_{\bar{k}}$ at $\bar{x}$, the pullbacks of B, h, f_j, and g_l, the integers m_j, and the real numbers c, b_0, and β_l. By Lemma 6.2(3), they do not depend on the choice of the germ. Assume that $\mathcal{Q}_{\bar{x}} \neq \emptyset$ and $b > 0$ on $\mathcal{Q}$. Then, by Lemma 6.2(3), Set-up 5.1 holds for the germ of $X_{\bar{k}}$ at $\bar{x}$, so Definition-Proposition 5.2 and Proposition 5.5 apply to it.

Lemma 6.2. *Let k, d, $(X \ni x, B)$, h, H, c, $f_1, \dots, f_s$, $m_1, \dots, m_s$, $g_1, \dots, g_t$, b_0, $\beta_1, \dots, \beta_t$, and $\bar{x}$ be as in (1)–(6) of Set-up 6.1, and let X' be the connected component of $X_{\bar{k}}$ containing $\bar{x}$. Then the following hold.*

(1) *X' is an integral affine variety over $\bar{k}$, the restriction of $(X_{\bar{k}}, B_{\bar{k}})$ to X' is klt, the restriction of $(X_{\bar{k}}, B_{\bar{k}} + cH_{\bar{k}})$ to X' is lc, and the pullbacks of h, f_j, and g_l to X' are non-zero.*
(2) *There exists an affine open neighborhood of $\bar{x}$ in X' disjoint from the lc centers of $(X_{\bar{k}}, B_{\bar{k}} + cH_{\bar{k}})$ not containing $\bar{x}$.*
(3) *$\bar{k} \cong \mathbb{C}$ as fields. For any affine open neighborhood X° of $\bar{x}$ as in (2), conditions (1)–(3) of Set-up 5.1 hold for $(X^\circ \ni \bar{x}, B_{\bar{k}}|_{X^\circ})$, the pullbacks of h and f_j, the real number c, and the integers m_j. The sets $\mathcal{Q}$ and $\mathcal{Q}_{\bar{x}}$ and the function b defined by (4)–(6) of Set-up 5.1 for X°, the pullbacks of g_l, and the real numbers b_0 and β_l only depend on $\mathcal{O}_{X_{\bar{k}},\bar{x}}$.*

Proof. By Lemma 3.3(1), $X_{\bar{k}}$ is normal, so X' is an integral affine variety over $\bar{k}$, although X may be geometrically non-integral. By Lemma 3.3(3), the restriction of $(X_{\bar{k}}, B_{\bar{k}})$ to X' is klt and the restriction of $(X_{\bar{k}}, B_{\bar{k}} + cH_{\bar{k}})$ to X' is lc. Since $X' \to X$ is dominant, the pullbacks of h, f_j, and g_l to X' are non-zero. This proves (1). An lc pair has only finitely many lc centers, namely the images of the strata of the divisors with log discrepancy zero on a log resolution, which implies (2). By Lemma 3.2, $\bar{k} \cong \mathbb{C}$. The point $\bar{x}$ is closed since x is, so conditions (1)–(3) of Set-up 5.1 hold for X° by (1), the choice of X° in (2), $h(\bar{x}) = 0$, and $\dim X^\circ = d$ [Har77, Exercise II.3.20(e),(f)]. The conditions defining $\mathcal{Q}$, $\mathcal{Q}_{\bar{x}}$, and b only involve $\mathcal{O}_{X_{\bar{k}},\bar{x}}$. □

6.1. **Descent of the minimizer.** In this subsection, we show that the minimizer of Definition-Proposition 5.2 on the germ of $X_{\bar{k}}$ at $\bar{x}$ restricts to a valuation of $k(X)$ centered at x, and that the associated graded ring descends to $\kappa(x)$.

Definition-Proposition 6.3. Notation and conditions as in Set-up 6.1, and let w be the valuation in Definition-Proposition 5.2 for the germ of $X_{\bar{k}}$ at $\bar{x}$. Let v be the restriction of w to $k(X)$, and let $\iota : \kappa(x) \to \bar{k}$ be the embedding determined by $\bar{x}$. Then the following hold.

(1) v does not depend on the choice of $\bar{x}$, and $\operatorname{center}_X v = x$.
(2) The subgroup $M \subset \mathbb{R}$ generated by $v(\mathcal{O}_{X,x}\backslash\{0\})$ is a free abelian group of finite rank.
(3) $R_0 := \operatorname{gr}_v \mathcal{O}_{X,x}$ is a finitely generated $\kappa(x)$-algebra, and
$$R_0 \otimes_{\kappa(x),\iota} \bar{k} \cong \operatorname{gr}_w \mathcal{O}_{X_{\bar{k}},\bar{x}}$$
as graded rings, where the isomorphism sends $\operatorname{in}_v(f) \otimes 1$ to $\operatorname{in}_w(f)$ for any non-zero $f \in \mathcal{O}_{X,x}$.
(4) The marked graded pair of w, defined by Definition 2.9 for the germ of $X_{\bar{k}}$ at $\bar{x}$ with the pullback of B, is klt.

Proof. The proof is long, so we divide it into several steps. Let $R := \mathcal{O}_{X,x}$, let $\mathfrak{m}$ be the maximal ideal of R, and let $R' := R \otimes_k \bar{k}$. The ring R' is a localization of $\mathcal{O}(X) \otimes_k \bar{k}$, hence Noetherian.

Step 1. In this step we describe the points of $X_{\bar{k}}$ over x and the action of $\operatorname{Gal}(\bar{k}/k)$ on them.

Since $\kappa(x)$ is a finite separable extension of k, the map $a \otimes e \to (\iota(a)e)_\iota$ is an isomorphism
$$\kappa(x) \otimes_k \bar{k} \to \prod_\iota \bar{k},$$
where ι runs over the k-embeddings $\kappa(x) \to \bar{k}$. The fiber of $X_{\bar{k}} \to X$ over x is $\operatorname{Spec}(\kappa(x) \otimes_k \bar{k})$. Thus the points of $X_{\bar{k}}$ over x are the points $\bar{x}_\iota$, one for any k-embedding ι, and the residue map $\kappa(x) \to \kappa(\bar{x}_\iota) = \bar{k}$ is ι. In particular, $\bar{x} = \bar{x}_\iota$ for the embedding ι determined by $\bar{x}$. The ring R' is integral over R, so by [AM69, Corollary 5.8, Theorem 5.10] the maximal ideals of R' are the prime ideals over $\mathfrak{m}$. Hence the maximal ideals of R' are the ideals $\mathfrak{n}_\iota$ corresponding to the points $\bar{x}_\iota$, and $R'_{\mathfrak{n}_\iota} = \mathcal{O}_{X_{\bar{k}},\bar{x}_\iota}$.

For $\sigma \in \operatorname{Gal}(\bar{k}/k)$, let $\sigma_{R'} := \mathrm{id}_R \otimes \sigma$, a ring automorphism of R', and let $\sigma_X := \mathrm{id}_X \times \operatorname{Spec} \sigma$, an automorphism of the scheme $X_{\bar{k}}$. The prime ideal $\sigma_{R'}^{-1}(\mathfrak{n}_\iota)$ is the kernel of the map
$$R' \to \bar{k}, \quad r \otimes e \to \iota(\bar{r})\sigma(e) = \sigma\big(\sigma^{-1}(\iota(\bar{r}))e\big),$$

where $\bar{r}$ is the image of r in $\kappa(x)$. Hence

$$\sigma_X(\bar{x}_\iota) = \bar{x}_{\sigma^{-1}\circ\iota}.$$

Since any k-embedding $\kappa(x) \to \bar{k}$ extends to an automorphism of $\bar{k}$, the group $\mathrm{Gal}(\bar{k}/k)$ acts transitively on the points of $X_{\bar{k}}$ over x.

Step 2. In this step we show that v does not depend on the choice of $\bar{x}$ and that $\mathrm{center}_X v = x$.

Let $\sigma \in \mathrm{Gal}(\bar{k}/k)$ and let $\bar{y} \in X_{\bar{k}}$ be a point over x. The automorphism σ_X induces an isomorphism of local rings

$$\sigma_X^\sharp : \mathcal{O}_{X_{\bar{k}},\sigma_X(\bar{y})} \to \mathcal{O}_{X_{\bar{k}},\bar{y}},$$

which is the localization of $\sigma_{R'}$. For a valuation u on $X_{\bar{k}}$ with $u \geq 0$ on $\mathcal{O}_{X_{\bar{k}},\bar{y}}$, let $\sigma_* u := u \circ \sigma_X^\sharp$. Then $\sigma_* u \geq 0$ on $\mathcal{O}_{X_{\bar{k}},\sigma_X(\bar{y})}$, and $\sigma_* u(f) = u(f)$ for any $f \in R$, since $\sigma_{R'}$ is the identity on $R \otimes 1$. The square formed by σ_X, $\mathrm{Spec}\,\sigma$, and the projection $X_{\bar{k}} \to \mathrm{Spec}\,\bar{k}$ is cartesian, and σ_X fixes the pullbacks of B, h, f_j, and g_l. Hence σ_X carries log resolutions of $X_{\bar{k}}$ and their canonical divisors to log resolutions and their canonical divisors, and $\sigma_X^\sharp$ maps $\mathfrak{a}_t(\sigma_* u)$ isomorphically onto $\mathfrak{a}_t(u)$ for any t. Thus $u \to \sigma_* u$ preserves quasi-monomiality, log discrepancies with respect to $B_{\bar{k}}$, volumes, and the values at h, f_j, and g_l. Therefore σ_* maps the sets $\mathcal{Q}$ and $\mathcal{Q}_{\bar{y}}$ of the germ at $\bar{y}$ bijectively onto the corresponding sets of the germ at $\sigma_X(\bar{y})$, and $b(\sigma_* u) = b(u)$ and $\mathrm{vol}(\sigma_* u) = \mathrm{vol}(u)$ for any u in $\mathcal{Q}$.

By **Step 1**, for any ι there exists $\sigma \in \mathrm{Gal}(\bar{k}/k)$ with $\sigma_X(\bar{x}) = \bar{x}_\iota$. Hence the conditions of Set-up 6.1 hold with $\bar{x}$ replaced by $\bar{x}_\iota$. Let w_ι be the valuation in Definition-Proposition 5.2 for the germ of $X_{\bar{k}}$ at $\bar{x}_\iota$. By the uniqueness in Definition-Proposition 5.2(1), $w_\iota = \sigma_* w$ for any σ with $\sigma_X(\bar{x}) = \bar{x}_\iota$, and more generally $\sigma_* w_\iota = w_{\sigma^{-1}\circ\iota}$ for any σ and ι. By the definition of σ_*, the last equality means

$$w_{\sigma^{-1}\circ\iota}(f) = w_\iota(\sigma_{R'}(f)) \quad \text{for any } f \in R',$$

where $w_\iota(f)$ denotes the value of the image of f in $R'_{\mathfrak{n}_\iota}$, with $w_\iota(0) = +\infty$. Moreover, $w_\iota(f) = w(f)$ for any $f \in R$. Since any point of $X_{\bar{k}}$ over x is of the form $\bar{x}_\iota$, the restriction v of w to $k(X) = \mathrm{Frac}\,R$ does not depend on the choice of $\bar{x}$.

Since $R \subset \mathcal{O}_{X_{\bar{k}},\bar{x}}$ and $\mathfrak{m} \subset \mathfrak{m}_{\bar{x}}$, where $\mathfrak{m}_{\bar{x}}$ is the maximal ideal of $\mathcal{O}_{X_{\bar{k}},\bar{x}}$, and since $\mathrm{center}_{X_{\bar{k}}}(w) = \bar{x}$, we have $v \geq 0$ on R and $v > 0$ on $\mathfrak{m}$. Thus $\mathrm{center}_X v = x$. In particular, $\mathfrak{a}_{>0}(v)\mathcal{O}_{X,x} = \mathfrak{m}$, and the degree zero part of R_0 is $R/\mathfrak{m} = \kappa(x)$.

Step 3. In this step we show that the valuation ideals of v in R generate the corresponding ideals of the family $(w_\iota)_\iota$ in R'.

For a real number $t \geq 0$, let

$$J_t := \{f \in R' \mid w_\iota(f) \geq t \text{ for any } \iota\}, \quad J_{>t} := \{f \in R' \mid w_\iota(f) > t \text{ for any } \iota\}.$$

Both J_t and $J_{>t}$ are ideals of R'. By the formula in **Step 2**, $\sigma_{R'}(J_t) = J_t$ and $\sigma_{R'}(J_{>t}) = J_{>t}$ for any $\sigma \in \mathrm{Gal}(\bar{k}/k)$. Since $w_\iota = v$ on R for any ι, we have $J_t \cap R = \mathfrak{a}_t(v)\mathcal{O}_{X,x} = \{f \in R \mid v(f) \geq t\}$ and $J_{>t} \cap R = \mathfrak{a}_{>t}(v)\mathcal{O}_{X,x}$.

Let J be one of the ideals J_t and $J_{>t}$. Since R' is Noetherian, J is generated by finitely many elements. Hence there exists a finite Galois extension k'/k in $\bar{k}$ such that J is generated by elements of $R \otimes_k k'$. Let $J' := J \cap (R \otimes_k k')$. The ideal J is generated by elements of J', and $\bar{k}$ is flat over k', so $J' \otimes_{k'} \bar{k} = J$. Since any element of $\mathrm{Gal}(k'/k)$ extends to an element of $\mathrm{Gal}(\bar{k}/k)$, which preserves J and $R \otimes_k k'$, the ideal J' is stable under $\mathrm{Gal}(k'/k)$. By [Sta, Tag 0CDR], J' descends to an ideal I of R with $I \otimes_k k' = J'$. Thus $I \otimes_k \bar{k} = J$. Write $\bar{k} = k \oplus W$ as k-vector spaces. Then $R' = R \oplus (R \otimes_k W)$ and $I \otimes_k \bar{k} = I \oplus (I \otimes_k W)$, so $(I \otimes_k \bar{k}) \cap R = I$. Hence $I = J \cap R$, and we obtain

$$\mathfrak{a}_t(v)\mathcal{O}_{X,x} \otimes_k \bar{k} = J_t, \quad \mathfrak{a}_{>t}(v)\mathcal{O}_{X,x} \otimes_k \bar{k} = J_{>t}.$$

Step 4. In this step we compare the graded rings by the Chinese remainder theorem.

Write $\mathcal{O}_\iota := \mathcal{O}_{X_{\bar{k}},\bar{x}_\iota} = R'_{\mathfrak{n}_\iota}$. For a real number $t \geq 0$ and any ι, let $J_{t,\iota}$ and $J_{>t,\iota}$ be the preimages of $\mathfrak{a}_t(w_\iota)\mathcal{O}_\iota$ and $\mathfrak{a}_{>t}(w_\iota)\mathcal{O}_\iota$ under $R' \to \mathcal{O}_\iota$. Then $J_t = \bigcap_\iota J_{t,\iota}$ and $J_{>t} = \bigcap_\iota J_{>t,\iota}$. The maximal ideal of $\mathcal{O}_\iota$ is finitely generated and w_ι is positive on it, so a power of the maximal ideal is contained in $\mathfrak{a}_{>t}(w_\iota)\mathcal{O}_\iota$. Hence, if $J_{t,\iota} \neq R'$, then the radical of $J_{t,\iota}$ is $\mathfrak{n}_\iota$, and similarly for $J_{>t,\iota}$. In this case $R'/J_{t,\iota}$ is a local ring with maximal ideal $\mathfrak{n}_\iota/J_{t,\iota}$, so the elements of $R'\backslash\mathfrak{n}_\iota$ are units in $R'/J_{t,\iota}$, and the natural map

$$R'/J_{t,\iota} \to \mathcal{O}_\iota/\mathfrak{a}_t(w_\iota)\mathcal{O}_\iota$$

is an isomorphism. Likewise, the natural map

$$R'/J_{>t,\iota} \to \mathcal{O}_\iota/\mathfrak{a}_{>t}(w_\iota)\mathcal{O}_\iota$$

is an isomorphism. Since the ideals $\mathfrak{n}_\iota$ are pairwise distinct maximal ideals, the ideals $J_{t,\iota}$ are pairwise coprime, and so are the ideals $J_{>t,\iota}$. By the Chinese remainder theorem [AM69, Proposition 1.10],

$$R'/J_t \cong \prod_\iota \mathcal{O}_\iota/\mathfrak{a}_t(w_\iota)\mathcal{O}_\iota, \quad R'/J_{>t} \cong \prod_\iota \mathcal{O}_\iota/\mathfrak{a}_{>t}(w_\iota)\mathcal{O}_\iota.$$

Taking the kernels of $R'/J_{>t} \to R'/J_t$ and using **Step 3** and the exactness of $- \otimes_k \bar{k}$, we obtain for any real number $t \geq 0$

$$\left(\mathfrak{a}_t(v)\mathcal{O}_{X,x}/\mathfrak{a}_{>t}(v)\mathcal{O}_{X,x}\right) \otimes_k \bar{k} \cong J_t/J_{>t} \cong \prod_\iota \mathfrak{a}_t(w_\iota)\mathcal{O}_\iota/\mathfrak{a}_{>t}(w_\iota)\mathcal{O}_\iota. \tag{6.1}$$

The isomorphisms in (6.1) are induced by the ring homomorphisms $R' \to \mathcal{O}_\iota$, so the direct sum over t is an isomorphism of graded rings

$$\Phi : R_0 \otimes_k \bar{k} \to \prod_\iota \operatorname{gr}_{w_\iota} \mathcal{O}_\iota.$$

By **Step 2**, the degree zero part of R_0 is $\kappa(x)$, so R_0 is a $\kappa(x)$-algebra and

$$R_0 \otimes_k \bar{k} = R_0 \otimes_{\kappa(x)} (\kappa(x) \otimes_k \bar{k}) \cong \prod_\iota R_0 \otimes_{\kappa(x),\iota} \bar{k}$$

by **Step 1**. In degree zero, Φ is the isomorphism $\kappa(x) \otimes_k \bar{k} \to \prod_\iota \bar{k}$ of **Step 1**, because the residue map of $\bar{x}_\iota$ is ι. Since Φ is a ring homomorphism, it maps the idempotent of $\kappa(x) \otimes_k \bar{k}$ corresponding to ι to the idempotent of the factor indexed by ι. Hence Φ maps $R_0 \otimes_{\kappa(x),\iota} \bar{k}$ isomorphically onto $\operatorname{gr}_{w_\iota} \mathcal{O}_\iota$. For the embedding ι determined by $\bar{x}$, we have $w_\iota = w$, and we obtain

$$R_0 \otimes_{\kappa(x),\iota} \bar{k} \cong \operatorname{gr}_w \mathcal{O}_{X_{\bar{k}},\bar{x}}.$$

Since Φ is induced by the homomorphisms $R' \to \mathcal{O}_\iota$ and $v(f) = w(f)$ for any non-zero $f \in R$, the isomorphism in the last display sends $\operatorname{in}_v(f) \otimes 1$ to $\operatorname{in}_w(f)$.

Step 5. In this step we prove the remaining properties of M, R_0, and w.

Recall that $R = \mathcal{O}_{X,x}$ and $\iota : \kappa(x) \to \bar{k}$ is the embedding determined by $\bar{x}$. For $t \geq 0$, we have $t \in v(R\backslash\{0\})$ if and only if the degree t part of R_0 is non-zero, and $t \in w(\mathcal{O}_{X_{\bar{k}},\bar{x}}\backslash\{0\})$ if and only if the degree t part of $\operatorname{gr}_w \mathcal{O}_{X_{\bar{k}},\bar{x}}$ is non-zero. Since $\bar{k}$ is faithfully flat over $\kappa(x)$ via ι, **Step 4** implies

$$v(R\backslash\{0\}) = w(\mathcal{O}_{X_{\bar{k}},\bar{x}}\backslash\{0\}).$$

Since w is quasi-monomial, there exist real numbers $\alpha_1, \ldots, \alpha_r$ such that

$$w(\mathcal{O}_{X_{\bar{k}},\bar{x}}\backslash\{0\}) \subset \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_r.$$

Thus M is a subgroup of the finitely generated torsion-free abelian group $\mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_r \subset \mathbb{R}$, so M is a free abelian group of finite rank.

By Proposition 5.5, $\operatorname{gr}_w \mathcal{O}_{X_{\bar{k}},\bar{x}}$ is a finitely generated $\bar{k}$-algebra. Choose finitely many homogeneous generators. Under the isomorphism of **Step 4**, each of them can be written as $\sum_q z_q \otimes e_q$, where the sum is finite, $e_q \in \bar{k}$, and $z_q \in R_0$ is homogeneous of the degree of that

generator. Let $S \subset R_0$ be the $\kappa(x)$-subalgebra generated by the finitely many elements z_q so obtained. Then the map

$$S \otimes_{\kappa(x),\iota} \bar{k} \to R_0 \otimes_{\kappa(x),\iota} \bar{k}$$

is surjective, and it is injective since $\bar{k}$ is flat over $\kappa(x)$. By faithful flatness, $S = R_0$. Thus R_0 is a finitely generated $\kappa(x)$-algebra.

Finally, by Proposition 5.5 applied to the germ of $X_{\bar{k}}$ at $\bar{x}$, the marked graded pair of w is klt. □

6.2. **Kollár components near the minimizer.** In this subsection, we prove Theorem 6.6, which constructs Kollár components over k near the valuation v of Definition-Proposition 6.3. The rational valuations near the valuation w of Definition-Proposition 6.3 (the minimizer of Definition-Proposition 5.2 on the germ of $X_{\bar{k}}$ at $\bar{x}$) given by [LX18, Lemma 2.10] are defined on monomial charts over $\bar{k}$, and Lemma 6.5 writes their valuation ideals using only finitely many elements of $\mathcal{O}_{X,x}$ and numerical weights, provided that an Izumi type upper bound holds and the graded ring is generated by the images of the elements used.

Definition 6.4. Let $(R, \mathfrak{m})$ be a Noetherian local domain and u a valuation of $\operatorname{Frac} R$ with $u \geq 0$ on R and $u > 0$ on $\mathfrak{m}$. For $t \in \mathbb{R}$, let $\mathfrak{a}_t(u) := \{f \in R \mid u(f) \geq t\}$. We define $\operatorname{gr}_u R$ and in_u by the formulas of Definition 2.9 with $\mathcal{O}_{X,x}$ replaced by R, and we regard $\operatorname{gr}_u R$ as an $R/\mathfrak{m}$-algebra, where $R/\mathfrak{m}$ is the degree zero part of $\operatorname{gr}_u R$.

Lemma 6.5. *Let $(R, \mathfrak{m})$ be a Noetherian local domain and u a valuation of $\operatorname{Frac} R$ with $u \geq 0$ on R and $u > 0$ on $\mathfrak{m}$, such that $u(f) \leq C \operatorname{ord}_{\mathfrak{m}}(f)$ for some real number $C > 0$ and any non-zero $f \in R$, where $\operatorname{ord}_{\mathfrak{m}}(f) := \max\{j \mid f \in \mathfrak{m}^j\}$. Let $s_1, \dots, s_n \in \mathfrak{m}$ be elements which generate the ideal $\mathfrak{m}$ such that $\operatorname{in}_u(s_1), \dots, \operatorname{in}_u(s_n)$ generate $\operatorname{gr}_u R$ over $R/\mathfrak{m}$. Then for any real number $t > 0$, $\mathfrak{a}_t(u)$ is the ideal of R generated by the monomials $s_1^{\beta_1} \cdots s_n^{\beta_n}$ with $\beta \in \mathbb{Z}^n_{\geq 0}$ and $\sum_i \beta_i u(s_i) \geq t$.*

Proof. Fix $t > 0$, and let I be the ideal of R generated by the monomials $s^\beta := s_1^{\beta_1} \cdots s_n^{\beta_n}$ with $\sum_i \beta_i u(s_i) \geq t$. Since $u(s^\beta) = \sum_i \beta_i u(s_i)$, we have $I \subset \mathfrak{a}_t(u)$. We prove the reverse inclusion.

Let $\epsilon := \min_i u(s_i)$, which is positive since $s_i \in \mathfrak{m}$, and let N be a positive integer with $N\epsilon \geq t$. Since $s_1, \dots, s_n$ generate $\mathfrak{m}$, the ideal $\mathfrak{m}^N$ is generated by the monomials s^β with $\sum_i \beta_i = N$, and $u(s^\beta) \geq N\epsilon \geq t$ for any such β. Hence $\mathfrak{m}^N \subset I$. Let $\Gamma \subset \mathbb{R}$ be the semigroup generated by $u(s_1), \dots, u(s_n)$. An element $\sum_i \beta_i u(s_i)$ of Γ with $\sum_i \beta_i u(s_i) \leq CN$ satisfies $\sum_i \beta_i \leq CN/\epsilon$, so $\Gamma \cap [0, CN]$ is a finite set.

Let $g \in R$ be non-zero and $\lambda := u(g)$. Since $\operatorname{in}_u(s_i)$ is homogeneous of degree $u(s_i)$ for any i and $\operatorname{in}_u(s_1), \dots, \operatorname{in}_u(s_n)$ generate $\operatorname{gr}_u R$ over $R/\mathfrak{m}$, there exist finitely many $\beta \in \mathbb{Z}^n_{\geq 0}$ with $\sum_i \beta_i u(s_i) = \lambda$ and elements $e_\beta \in R$ such that

$$\operatorname{in}_u(g) = \sum_\beta \bar{e}_\beta \operatorname{in}_u(s_1)^{\beta_1} \cdots \operatorname{in}_u(s_n)^{\beta_n},$$

where $\bar{e}_\beta$ is the image of e_β in $R/\mathfrak{m}$. Since $\operatorname{in}_u(g) \neq 0$, the sum is non-empty, so $\lambda \in \Gamma$. The right hand side is the image of $g' := \sum_\beta e_\beta s^\beta \in \mathfrak{a}_\lambda(u)$ in $\mathfrak{a}_\lambda(u)/\mathfrak{a}_{>\lambda}(u)$. Hence $g - g' = 0$ or $u(g - g') > \lambda$. Moreover, if $\lambda \geq t$, then $g' \in I$.

Now let $f \in \mathfrak{a}_t(u)$ be non-zero. Let $f_0 := f$, and as long as $f_p \neq 0$ and $u(f_p) \leq CN$, let $f_{p+1} := f_p - f_p'$, where f_p' is the element g' above for $g = f_p$. Then $t \leq u(f_0) < u(f_1) < \cdots$ are elements of Γ, and $f_p - f_{p+1} = f_p' \in I$ for any p. Since $\Gamma \cap [0, CN]$ is finite, after finitely many steps we obtain f_q with $f_q = 0$ or $u(f_q) > CN$. If $u(f_q) > CN$, then $\operatorname{ord}_{\mathfrak{m}}(f_q) \geq u(f_q)/C > N$, so $f_q \in \mathfrak{m}^N \subset I$. Thus

$$f = f_q + \sum_{p=0}^{q-1} (f_p - f_{p+1}) \in I.$$

□

Theorem 6.6. *Notation and conditions as in Set-up 6.1, and let v and M be as in Definition-Proposition 6.3. Then the following hold.*

(1) There exist elements $y_1, \dots, y_n \in \mathfrak{m}_x$ and an open rational polyhedral cone $\mathcal{C} \subset \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{R})$ containing the inclusion $M \subset \mathbb{R}$ satisfying the following. For any $\ell \in \mathcal{C}$ with $\ell(M) \subset \mathbb{Q}$, there exists a valuation v_ℓ on X with $\mathrm{center}_X(v_\ell) = x$ such that for any real number $\lambda > 0$, $\mathfrak{a}_\lambda(v_\ell)\mathcal{O}_{X,x}$ is generated by the monomials $y^\gamma := y_1^{\gamma_1} \cdots y_n^{\gamma_n}$ with $\gamma \in \mathbb{Z}^n_{\geq 0}$ and $\sum_i \gamma_i \ell(v(y_i)) \geq \lambda$. Moreover, $\ell(M) = \delta\mathbb{Z}$ for a rational number $\delta > 0$, and $v_\ell = \delta\,\mathrm{ord}_{F_\ell}$ for a Kollár component F_ℓ of $(X \ni x, B)$ over k which is geometrically integral over $\kappa(x)$ and satisfies $a(F_\ell, X, B + cH) = 0$, $a(F_\ell, X, B) = c\,\mathrm{ord}_{F_\ell}(h) = c\ell(v(h))/\delta$, and $\mathrm{ord}_{F_\ell}(f_j) \geq m_j\,\mathrm{ord}_{F_\ell}(h)$ for any j.

(2) If ℓ runs over the elements of $\mathcal{C}$ as in (1) converging to the inclusion $M \subset \mathbb{R}$, then $\mathrm{ord}_{F_\ell} / \mathrm{ord}_{F_\ell}(h)$ converges to v pointwise on $\mathcal{O}_{X,x}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a rational substitution of the coefficients, a special complement, and a chart on which the geometric minimizer is monomial with $\mathbb{Q}$-linearly independent weights.

Let $\bar{X}$ be the germ of $X_{\bar{k}}$ at $\bar{x}$, let $\bar{B}$ and $\bar{H}$ be the pullbacks of B and H to $\bar{X}$, and denote the pullbacks of h and f_j to $\bar{X}$ again by h and f_j. Let w be the valuation of Definition-Proposition 5.2 for $\bar{X}$. Then v is the restriction of w to $k(X)$ (Definition-Proposition 6.3), and $w(h) = 1$ since $w \in \mathcal{Q}_{\bar{x}}$. By Lemma 3.2, $\bar{k} \cong \mathbb{C}$, so the results of [LX18], [XZ25], and [HLQ20] hold over $\bar{k}$. By Lemma 3.3(3), $(\bar{X}, \bar{B})$ is klt and $(\bar{X}, \bar{B} + c\bar{H})$ is lc. The local homomorphism $\mathcal{O}_{X,x} \to \mathcal{O}_{\bar{X},\bar{x}}$ is flat, hence faithfully flat, so

$$I\mathcal{O}_{\bar{X},\bar{x}} \cap \mathcal{O}_{X,x} = I \quad \text{for any ideal } I \subset \mathcal{O}_{X,x}. \tag{6.2}$$

Since $\kappa(x) \otimes_k \bar{k}$ is reduced, $\mathfrak{m}_x\mathcal{O}_{\bar{X},\bar{x}} = \mathfrak{m}_{\bar{x}}$.

Since $R_0 \otimes_{\kappa(x),\iota} \bar{k} \cong \mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ as graded rings (Definition-Proposition 6.3), the degrees of the non-zero homogeneous elements of R_0 and of $\mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ coincide. Thus M is the subgroup of $\mathbb{R}$ generated by $w(\mathcal{O}_{\bar{X},\bar{x}}\backslash\{0\})$, that is, the value group of w. Let $r := \mathrm{rank}\, M$. Since $w \in \mathcal{Q}_{\bar{x}}$ is quasi-monomial and its value group M has rank r, w is monomial with $\mathbb{Q}$-linearly independent weights on local equations of the r components of the boundary through the generic point of a codimension r stratum of some log smooth model of $\bar{X}$ ([LX18, Definition 2.8], see also the remark after [XZ25, Definition 2.8]). Thus the residue field of w is the residue field of that generic point, which has transcendence degree $d - r$ over $\bar{k}$.

Let (X_0, B_0) be the marked graded pair of w. By Definition-Proposition 6.3, (X_0, B_0) is klt. Moreover, $\dim X_0 = d$, because the homogeneous fractions of degree zero of $\mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ form a field isomorphic to the residue field of w and the degrees of the homogeneous elements generate M, so that the fraction field of $\mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ has transcendence degree $(d - r) + r = d$ over $\bar{k}$.

Write $\bar{B} = \sum_i b_i \bar{B}_i$, where the $\bar{B}_i$ are distinct prime divisors and $b_i \in (0, 1)$, and write $(\bar{B}_i)_0 = \sum_\mu n_{i\mu} C_\mu$, where the C_μ are distinct prime divisors on X_0 and $n_{i\mu} \in \mathbb{Z}_{\geq 0}$. Then

$$B_0 = \sum_\mu \Big(\sum_i n_{i\mu} b_i \Big) C_\mu.$$

Let $J \subset [0, 1]$ be the finite set of the numbers b_i and $\sum_i n_{i\mu} b_i$ (the numbers $\sum_i n_{i\mu} b_i$ are the coefficients of B_0, hence less than 1 since (X_0, B_0) is klt), let ρ be the map of Definition-Theorem 4.1 for d and J, and let $\bar{B}^\rho = \sum_i \rho(b_i)\bar{B}_i$. For any μ, the numbers $e_\mu := \sum_i n_{i\mu} b_i$, b_i, and 0 in J^+ satisfy the $\mathbb{Q}$-affine relation $e_\mu - \sum_i n_{i\mu} b_i + (\sum_i n_{i\mu} - 1) \cdot 0 = 0$. Since ρ preserves $\mathbb{Q}$-affine relations among elements of J^+ and $\rho(0) = 0$,

$$\rho\Big(\sum_i n_{i\mu} b_i \Big) = \sum_i n_{i\mu} \rho(b_i)$$

for any μ. Thus the marked graded pair of w with respect to $\bar{B}^\rho$ is (X_0, B_0^ρ). By Definition-Theorem 4.1(1), applied to $\bar{X}$ with $B = \Delta := \bar{B}$ and to X_0 with $B = \Delta := B_0$ (both of dimension

d, with coefficients in J), the pairs $(\bar{X}, \bar{B}^\rho)$ and (X_0, B_0^ρ) are klt and $K_{\bar{X}} + \bar{B}^\rho$ and $K_{X_0} + B_0^\rho$ are $\mathbb{Q}$-Cartier. Since ρ is strictly increasing and $\rho(0) = 0$, $\operatorname{Supp} \bar{B}^\rho = \operatorname{Supp} \bar{B}$.

By [XZ25, Lemma 4.3] (the implication (1)⇒(2) of [XZ25, Theorem 4.1]), applied to $(\bar{X} \ni \bar{x}, \bar{B}^\rho)$ and w, there exist a log smooth model (Y, E) of $(\bar{X}, \bar{B}^\rho)$ with $w \in \mathrm{QM}(Y, E)$ (see [XZ25, Section 2.2.2]) and a special $\mathbb{Q}$-complement Γ of $(\bar{X} \ni \bar{x}, \bar{B}^\rho)$ with respect to (Y, E) such that w is a monomial lc place of Γ. That is, $\Gamma \geq 0$ is a $\mathbb{Q}$-divisor such that $(\bar{X}, \bar{B}^\rho + \Gamma)$ is lc, $K_{\bar{X}} + \bar{B}^\rho + \Gamma \sim_{\mathbb{Q}} 0$ near $\bar{x}$, the strict transform of Γ to Y satisfies the positivity condition of [XZ25, Definition 3.1], and $a(w, \bar{X}, \bar{B}^\rho + \Gamma) = 0$. Since $\operatorname{Supp} \bar{B}^\rho = \operatorname{Supp} \bar{B}$, (Y, E) is also a log smooth model of $(\bar{X}, \bar{B})$.

A projective toroidal modification of (Y, E) given by a subdivision of the conical complex of (Y, E) [KKMS73, Chapter II, §2, Theorem 6*] yields a log smooth model (Y', E') of $(\bar{X}, \bar{B})$, the generic point η' of a codimension r stratum of E', local equations $z_1, \dots, z_r$ at η' of the components $E'_1, \dots, E'_r$ of E' through η', and $\alpha \in \mathbb{R}^r_{>0}$ with $\mathbb{Q}$-linearly independent coordinates such that $w = w_\alpha$ ([LX18, Definition 2.8], see also the remark after [XZ25, Definition 2.8]). Here, for $\beta \in \mathbb{R}^r_{>0}$, w_β is the monomial valuation with weights β_i on z_i: for a non-zero $f \in \mathcal{O}_{Y',\eta'}$ with expansion $f = \sum_\gamma c_\gamma z^\gamma$ in $\widehat{\mathcal{O}}_{Y',\eta'} \cong \kappa(\eta')[[z_1, \dots, z_r]]$,

$$w_\beta(f) = \min\{\langle \beta, \gamma \rangle \mid c_\gamma \neq 0\}, \quad \text{where } \langle \beta, \gamma \rangle := \sum_i \beta_i \gamma_i.$$

To obtain (Y', E'), let ξ be the center of w on Y, σ_ξ the cone of (Y, E) at ξ, and ω the weight vector of w, so that $\omega \in \operatorname{relint} \sigma_\xi$. We first take a projective regular subdivision of the conical complex of (Y, E), of all its cones and not only of σ_ξ, with $\omega \in \operatorname{relint} \vartheta$ for a cone ϑ of dimension r, and the toroidal modification given by it (cf. [KKMS73, Chapter II, §2, Theorems 4*, 6*, 10*, and 11*]). Since $\omega \in \operatorname{relint} \sigma_\xi$, $\operatorname{relint} \vartheta \subset \operatorname{relint} \sigma_\xi$. Next we blow up the closure of the stratum corresponding to ϑ. Then, as long as the cone ϑ' with $\omega \in \operatorname{relint} \vartheta'$ has a primitive ray generator $e \notin \operatorname{relint} \vartheta$, we take a primitive ray generator $u \in \operatorname{relint} \vartheta$ of ϑ' and blow up the closure of the codimension 2 stratum corresponding to the face of ϑ' spanned by u and e. Since the coefficients of u and e in ω have irrational ratio, after finitely many such blow-ups, as in the subtractive Euclidean algorithm, e is no longer a ray generator of ϑ', and the new ray generators $u + je$ belong to $\operatorname{relint} \vartheta$. We let (Y', E') be the model obtained in the end. Since $Y \to \bar{X}$ is projective [XZ25, Section 2.2.2] and $Y' \to Y$ is the composition of a projective toroidal modification and blow-ups, $Y' \to \bar{X}$ is projective [Har77, Exercise II.4.9]. All primitive ray generators of the cone of (Y', E') at η' belong to $\operatorname{relint} \vartheta \subset \operatorname{relint} \sigma_\xi$, so the center of each E'_i on Y is the closure of ξ, and the center of each E'_i on $\bar{X}$ is $\bar{x}$. Thus w satisfies [LX18, Definition 2.8] literally on (Y', E') at η'. Since each blow-up above is the toroidal modification given by a star subdivision, $Y' \to Y$ is the toroidal modification given by a subdivision of the conical complex of (Y, E), so $\mathrm{QM}_{\eta'}(Y', E') \subset \mathrm{QM}(Y, E)$, and a rational weight vector β corresponds to a rational weight vector on (Y, E). For any $\beta \in \mathbb{R}^r_{>0}$, the center of w_β on Y' is η', so $\operatorname{center}_{\bar{X}}(w_\beta) = \operatorname{center}_{\bar{X}} w = \bar{x}$.

For a non-zero $f \in \mathcal{O}_{Y',\eta'}$, the set of the minimal elements, for the coordinatewise order, of $\{\gamma \mid c_\gamma \neq 0\}$ is finite, and for $\beta \in \mathbb{R}^r_{>0}$ the minimum defining $w_\beta(f)$ is attained on it. Since the coordinates of α are $\mathbb{Q}$-linearly independent, $\langle \alpha, \gamma \rangle \neq \langle \alpha, \gamma' \rangle$ for $\gamma \neq \gamma'$, so exactly one exponent γ_f attains $w(f)$. Hence, writing an element of $k(\bar{X})^*$ as a quotient of two elements of $\mathcal{O}_{Y',\eta'}$, we obtain for any $f \in k(\bar{X})^*$ a vector $\gamma_f \in \mathbb{Z}^r$ such that

$$w_\beta(f) = \langle \beta, \gamma_f \rangle \quad \text{for any } \beta \text{ in a neighborhood of } \alpha \text{ in } \mathbb{R}^r_{>0}. \tag{6.3}$$

The map $\mathbb{Z}^r \to \mathbb{R}$, $\gamma \to \langle \alpha, \gamma \rangle$, is injective, and by (6.3) the image $\{\langle \alpha, \gamma \rangle \mid \gamma \in \mathbb{Z}^r\}$ contains M. For $m \in M$ let $\gamma(m) \in \mathbb{Z}^r$ be the vector with $\langle \alpha, \gamma(m) \rangle = m$. Then $m \to \gamma(m)$ is an injective homomorphism whose image has finite index in $\mathbb{Z}^r$, since $\operatorname{rank} M = r$. For $\beta \in \mathbb{R}^r$ define $\ell_\beta \in \operatorname{Hom}_{\mathbb{Z}}(M, \mathbb{R})$ by $\ell_\beta(m) := \langle \beta, \gamma(m) \rangle$. Then $\beta \to \ell_\beta$ is an $\mathbb{R}$-linear isomorphism $\mathbb{R}^r \to \operatorname{Hom}_{\mathbb{Z}}(M, \mathbb{R})$, ℓ_α is the inclusion $M \subset \mathbb{R}$, and $\ell_\beta(M) \subset \mathbb{Q}$ if and only if $\beta \in \mathbb{Q}^r$. By (6.3),

for any non-zero $f \in \mathcal{O}_{\bar{X},\bar{x}}$,

$$w_\beta(f) = \ell_\beta(w(f)) \quad \text{for any } \beta \text{ in a neighborhood of } \alpha. \tag{6.4}$$

Step 2. In this step we show that the monomial valuations near w are lc places of $(\bar{X}, \bar{B} + c\bar{H})$ satisfying the incidence conditions, and that the rational ones are multiples of Kollár components of $(\bar{X} \ni \bar{x}, \bar{B})$.

Since $K_{\bar{X}} + \bar{B}^\rho$ is $\mathbb{Q}$-Cartier and $K_{\bar{X}} + \bar{B}^\rho + \Gamma \sim_{\mathbb{Q}} 0$ near $\bar{x}$, there exist an integer $m_\Gamma > 0$ and $\psi \in k(\bar{X})^*$ with $m_\Gamma \Gamma = \operatorname{div}(\psi)$ near $\bar{x}$. For any quasi-monomial valuation u on $\bar{X}$ with $\operatorname{center}_{\bar{X}} u = \bar{x}$, since $(\bar{X}, \bar{B}^\rho + \Gamma)$ and $(\bar{X}, \bar{B} + c\bar{H})$ are lc,

$$a(u, \bar{X}, \bar{B}^\rho) - \frac{u(\psi)}{m_\Gamma} = a(u, \bar{X}, \bar{B}^\rho + \Gamma) \geq 0, \qquad a(u, \bar{X}, \bar{B}) - cu(h) = a(u, \bar{X}, \bar{B} + c\bar{H}) \geq 0.$$

Since (Y', E') is a log smooth model of $(\bar{X}, \bar{B}^\rho)$ and of $(\bar{X}, \bar{B})$ and the components of E' through η' are $E'_1, \dots, E'_r$, the functions $\beta \to a(w_\beta, \bar{X}, \bar{B}^\rho)$ and $\beta \to a(w_\beta, \bar{X}, \bar{B})$ are linear on $\mathbb{R}^r_{>0}$ ([XZ25, Definition 2.12(2)] and [HLQ20, Definition 2.7(2)], respectively). By (6.3) for ψ, h, and f_j, there exists a convex open neighborhood Ω of α in $\mathbb{R}^r_{>0}$ on which

$$L_1(\beta) := a(w_\beta, \bar{X}, \bar{B}^\rho) - \frac{w_\beta(\psi)}{m_\Gamma} \quad \text{and} \quad L_2(\beta) := a(w_\beta, \bar{X}, \bar{B}) - cw_\beta(h)$$

are restrictions of linear functions and $w_\beta(f_j) - m_j w_\beta(h) = \langle \beta, \gamma_{f_j} - m_j \gamma_h \rangle$ for any j. Now $L_1 \geq 0$ and $L_2 \geq 0$ on Ω, $L_1(\alpha) = 0$ since w is an lc place of Γ, and $L_2(\alpha) = 0$ since $w \in \mathcal{Q}_{\bar{x}}$. A linear function which is non-negative on Ω and vanishes at $\alpha \in \Omega$ vanishes on Ω, because $L(\alpha + \epsilon\zeta) + L(\alpha - \epsilon\zeta) = 2L(\alpha) = 0$ for any $\zeta \in \mathbb{R}^r$ and small $\epsilon > 0$. Thus, for any $\beta \in \Omega$,

$$a(w_\beta, \bar{X}, \bar{B}^\rho + \Gamma) = 0 \quad \text{and} \quad a(w_\beta, \bar{X}, \bar{B}) = cw_\beta(h). \tag{6.5}$$

For any j, $\langle \alpha, \gamma_{f_j} - m_j \gamma_h \rangle = w(f_j) - m_j w(h) \geq 0$ since $w \in \mathcal{Q}_{\bar{x}}$. If $\langle \alpha, \gamma_{f_j} - m_j \gamma_h \rangle = 0$, then $\gamma_{f_j} - m_j \gamma_h = 0$ since $\gamma_{f_j} - m_j \gamma_h \in \mathbb{Z}^r$ and the coordinates of α are $\mathbb{Q}$-linearly independent. If $\langle \alpha, \gamma_{f_j} - m_j \gamma_h \rangle > 0$, then $\langle \beta, \gamma_{f_j} - m_j \gamma_h \rangle > 0$ for any β near α. Possibly shrinking Ω, we obtain

$$w_\beta(f_j) \geq m_j w_\beta(h) \quad \text{for any } \beta \in \Omega \text{ and any } j. \tag{6.6}$$

Let $\beta \in \Omega \cap \mathbb{Q}^r$. By (6.5), w_β is a monomial lc place of the special $\mathbb{Q}$-complement Γ with respect to (Y, E), and w_β is a divisorial valuation since β is rational (see the remark after [XZ25, Definition 2.8]). By [XZ25, Theorem 3.14] applied to the rational simplex $\mathbb{R}_{\geq 0} w_\beta \subset \mathrm{LC}(\Gamma; Y, E)$ spanned by the single divisorial valuation w_β, together with [XZ25, Proposition 3.9 and Lemma 3.8], w_β is a positive multiple of $\operatorname{ord}_{\bar{F}}$ for a Kollár component $\bar{F}$ of $(\bar{X} \ni \bar{x}, \bar{B}^\rho)$. Let $\bar{g} : \bar{Y} \to \bar{U}$ be the prime extraction of $\bar{F}$ over an affine neighborhood $\bar{U}$ of $\bar{x}$. By Definition-Theorem 4.1(3) applied to $\bar{g}$, $(\bar{Y}, \bar{g}_*^{-1}\bar{B} + \bar{F})$ is plt near $\bar{F}$. Hence $\bar{F}$ is a Kollár component of $(\bar{X} \ni \bar{x}, \bar{B})$.

Step 3. In this step we fix $y_1, \dots, y_n$ and show that for rational weights near α, the valuation ideals of w_β are generated by monomials in the y_i.

Since $w(h) = 1$ and $w(f_j) \geq m_j w(h) > 0$, we have $h, f_j \in \mathfrak{m}_{\bar{x}} \cap \mathcal{O}_{X,x} = \mathfrak{m}_x$. The degree zero part of R_0 is $\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$ and R_0 is a finitely generated $\kappa(x)$-algebra. Thus there exist $y_1, \dots, y_n \in \mathfrak{m}_x$ such that $\operatorname{in}_v(y_1), \dots, \operatorname{in}_v(y_n)$ generate R_0 over $\kappa(x)$, $y_1, \dots, y_n$ generate $\mathfrak{m}_x$, and $h, f_1, \dots, f_s$ are among the y_i. Since the isomorphism $R_0 \otimes_{\kappa(x),\iota} \bar{k} \cong \operatorname{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ sends $\operatorname{in}_v(f) \otimes 1$ to $\operatorname{in}_w(f)$ (Definition-Proposition 6.3(3)), the elements $\operatorname{in}_w(y_i)$ generate $\operatorname{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ over $\bar{k}$, and the y_i generate $\mathfrak{m}_{\bar{x}}$. Moreover, any degree of a non-zero homogeneous element of R_0 is a sum of the $v(y_i)$, so the $v(y_i)$ generate M.

Since $w = w_\alpha$ on (Y', E') with $\alpha \in \mathbb{R}^r_{>0}$ of $\mathbb{Q}$-linearly independent coordinates, that is, w is a quasi-monomial valuation of rational rank r in the sense of [LX18, Definition 2.8] at η', and $\operatorname{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$ is finitely generated, by [LX18, Lemma 2.10] there exist $p_1, \dots, p_q \in \mathcal{O}_{\bar{X},\bar{x}}$ such that

$\mathrm{in}_w(p_1),\dots,\mathrm{in}_w(p_q)$ are homogeneous generators of $\mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}}$, and an open neighborhood $\Omega' \subset \Omega$ of α such that for any $\beta \in \Omega' \cap \mathbb{Q}^r$ there exists an isomorphism of $\bar{k}$-algebras

$$\phi_\beta : \mathrm{gr}_w \mathcal{O}_{\bar{X},\bar{x}} \to \mathrm{gr}_{w_\beta} \mathcal{O}_{\bar{X},\bar{x}} \quad \text{with} \quad \phi_\beta(\mathrm{in}_w(p_l)) = \mathrm{in}_{w_\beta}(p_l) \text{ for any } l.$$

For any i, fix a polynomial Q_i with coefficients in $\bar{k}$ such that $\sum_l \mu_l w(p_l) = w(y_i)$ for any monomial p^μ occurring in Q_i and $\mathrm{in}_w(y_i) = Q_i(\mathrm{in}_w(p_1),\dots,\mathrm{in}_w(p_q))$. Then $y_i = Q_i(p)$ or $w(y_i - Q_i(p)) > w(y_i)$, where $Q_i(p) := Q_i(p_1,\dots,p_q)$. By (6.4) for the finitely many elements p_l, y_i, and $y_i - Q_i(p)$, and since $w(y_i - Q_i(p)) > w(y_i)$, ℓ_α is the inclusion $M \subset \mathbb{R}$, and $\beta \to \ell_\beta(m)$ is continuous for each $m \in M$, possibly shrinking Ω', for any $\beta \in \Omega' \cap \mathbb{Q}^r$ and any i and l,

$$w_\beta(p_l) = \ell_\beta(w(p_l)), \qquad w_\beta(y_i - Q_i(p)) > \ell_\beta(w(y_i)) \text{ if } y_i \neq Q_i(p).$$

Fix $\beta \in \Omega' \cap \mathbb{Q}^r$. By the first equality, ϕ_β maps the degree λ part to the degree $\ell_\beta(\lambda)$ part for any $\lambda \in M$. Any monomial p^μ in Q_i satisfies $w_\beta(p^\mu) = \ell_\beta(w(y_i))$, and the sum of the classes of the monomials of $Q_i(p)$ in degree $\ell_\beta(w(y_i))$ is

$$Q_i(\mathrm{in}_{w_\beta}(p_1),\dots,\mathrm{in}_{w_\beta}(p_q)) = \phi_\beta(\mathrm{in}_w(y_i)) \neq 0.$$

Thus $w_\beta(Q_i(p)) = \ell_\beta(w(y_i))$ and $\mathrm{in}_{w_\beta}(Q_i(p)) = \phi_\beta(\mathrm{in}_w(y_i))$. Since $w_\beta(y_i - Q_i(p)) > \ell_\beta(w(y_i))$, we get

$$w_\beta(y_i) = \ell_\beta(v(y_i)) \quad \text{and} \quad \mathrm{in}_{w_\beta}(y_i) = \phi_\beta(\mathrm{in}_w(y_i)).$$

Since ϕ_β is an isomorphism, the elements $\mathrm{in}_{w_\beta}(y_i)$ generate $\mathrm{gr}_{w_\beta} \mathcal{O}_{\bar{X},\bar{x}}$ over $\bar{k}$.

By [HLQ20, Lemma 2.14], applied to the klt pair $(\bar{X} \ni \bar{x}, \bar{B})$, there exists a real number $C > 0$ such that $u(f) \le Ca(u,\bar{X},\bar{B})\,\mathrm{ord}_{\mathfrak{m}_{\bar{x}}}(f)$ for any valuation u on $\bar{X}$ with $\mathrm{center}_{\bar{X}} u = \bar{x}$ and any non-zero $f \in \mathcal{O}_{\bar{X},\bar{x}}$. Since $a(w_\beta,\bar{X},\bar{B}) < +\infty$, Lemma 6.5, applied to $R = \mathcal{O}_{\bar{X},\bar{x}}$, $u = w_\beta$, $s_i := y_i$, and the constant $Ca(w_\beta,\bar{X},\bar{B})$, implies that for any $\beta \in \Omega' \cap \mathbb{Q}^r$ and any real number $\lambda > 0$,

$$\mathfrak{a}_\lambda(w_\beta)\mathcal{O}_{\bar{X},\bar{x}} = \Big(y^\gamma \;\Big|\; \gamma \in \mathbb{Z}^n_{\ge 0},\ \sum_i \gamma_i \ell_\beta(v(y_i)) \ge \lambda\Big)\mathcal{O}_{\bar{X},\bar{x}}. \tag{6.7}$$

Step 4. In this step we construct $\mathcal{C}$ and v_ℓ, and we prove the statement on $\mathfrak{a}_\lambda(v_\ell)$ in (1).

For $\tau > 0$, $w_{\tau\beta} = \tau w_\beta$, $\ell_{\tau\beta} = \tau\ell_\beta$, and $\mathfrak{a}_\lambda(w_{\tau\beta}) = \mathfrak{a}_{\lambda/\tau}(w_\beta)$. Let $\beta \in \mathbb{Q}^r \cap \mathbb{R}_{>0}\Omega'$. The set of real numbers $\tau > 0$ with $\beta/\tau \in \Omega'$ is open and non-empty, so it contains a rational number τ, and we write $\beta = \tau\beta_1$ with $\beta_1 \in \Omega' \cap \mathbb{Q}^r$. Since (6.5), (6.6), and (6.7) hold for β_1, they hold for β, and $w_\beta = \tau w_{\beta_1}$ is a positive multiple of $\mathrm{ord}_{\bar{F}}$ for the Kollár component $\bar{F}$ of $(\bar{X} \ni \bar{x}, \bar{B})$ given by **Step 2** for β_1.

The set $\{\ell_\beta \mid \beta \in \mathbb{R}_{>0}\Omega'\}$ is an open cone in $\mathrm{Hom}_{\mathbb{Z}}(M,\mathbb{R})$ containing ℓ_α. Since $\mathrm{Hom}_{\mathbb{Z}}(M,\mathbb{Q})$ is dense in $\mathrm{Hom}_{\mathbb{Z}}(M,\mathbb{R})$, there exists an open rational polyhedral cone $\mathcal{C}$ with $\ell_\alpha \in \mathcal{C} \subset \{\ell_\beta \mid \beta \in \mathbb{R}_{>0}\Omega'\}$, for example the cone over a small rational cube around ℓ_α. Let $\ell \in \mathcal{C}$ with $\ell(M) \subset \mathbb{Q}$. Then $\ell = \ell_\beta$ for some $\beta \in \mathbb{Q}^r \cap \mathbb{R}_{>0}\Omega'$, and we let v_ℓ be the restriction of w_β to $k(X)$. Since $w_\beta \ge 0$ on $\mathcal{O}_{\bar{X},\bar{x}}$ and $w_\beta > 0$ on $\mathfrak{m}_{\bar{x}}$, $\mathrm{center}_X(v_\ell) = x$. For $\lambda > 0$, let $I_\lambda \subset \mathcal{O}_{X,x}$ be the ideal generated by the monomials y^γ with $\sum_i \gamma_i \ell(v(y_i)) \ge \lambda$. By (6.7) and (6.2),

$$\mathfrak{a}_\lambda(v_\ell)\mathcal{O}_{X,x} = \mathfrak{a}_\lambda(w_\beta)\mathcal{O}_{\bar{X},\bar{x}} \cap \mathcal{O}_{X,x} = I_\lambda \mathcal{O}_{\bar{X},\bar{x}} \cap \mathcal{O}_{X,x} = I_\lambda.$$

This proves the statement on $\mathfrak{a}_\lambda(v_\ell)$ in (1).

Step 5. In this step we prove the statements on F_ℓ in (1).

Let $\ell = \ell_\beta$ and v_ℓ be as in **Step 4**, write $\beta = \tau\beta_1$ as there, and let $\bar{F}$ and its prime extraction $\bar{g} : \bar{Y} \to \bar{U}$ be as in **Step 2** for β_1, so that w_β is a positive multiple of $\mathrm{ord}_{\bar{F}}$ and $\bar{F}$ is a Kollár component of $(\bar{X} \ni \bar{x}, \bar{B})$. The group $\ell(M)$ is a finitely generated subgroup of $\mathbb{Q}$, and it is non-zero since $\ell(v(y_i)) = w_\beta(y_i) > 0$. Hence $\ell(M) = \delta\mathbb{Z}$ for some $\delta > 0$, and δ is rational since $\delta \in \ell(M) \subset \mathbb{Q}$. Let $S \subset \mathbb{R}_{\ge 0}$ be the set of the numbers $\sum_i \gamma_i \ell(v(y_i))$ with $\gamma \in \mathbb{Z}^n_{\ge 0}$. Then $S \subset \ell(M)$ and S is discrete. For a non-zero $f \in \mathcal{O}_{\bar{X},\bar{x}}$, let $\lambda := w_\beta(f)$. If $\lambda \notin S$, then $\lambda' := \min(S \cap [\lambda, +\infty)) > \lambda$ and $\mathfrak{a}_\lambda(w_\beta) = \mathfrak{a}_{\lambda'}(w_\beta)$ by (6.7), which contradicts $w_\beta(f) = \lambda$. Thus

$w_\beta(\mathcal{O}_{\bar X,\bar x}\backslash\{0\}) \subset S$. Since the $w_\beta(y_i) = \ell(v(y_i))$ generate $\ell(M)$, the value group of w_β is $\delta\mathbb{Z}$. Since w_β is a positive multiple of $\operatorname{ord}_{\bar F}$, $w_\beta = \delta \operatorname{ord}_{\bar F}$. By Lemma 3.3(2), the restriction of $\operatorname{ord}_{\bar F}$ to $k(X)$ is $\operatorname{ord}_{F_\ell}$ for a prime divisor F_ℓ over X, and $a(\bar F, X_{\bar k}, B_{\bar k}) = a(F_\ell, X, B)$. Thus $v_\ell = \delta \operatorname{ord}_{F_\ell}$, and $\operatorname{center}_X(F_\ell) = x$.

Since h is among the y_i, $w_\beta(h) = \ell(v(h))$. By (6.5),

$$a(F_\ell, X, B) = a(\bar F, \bar X, \bar B) = \frac{a(w_\beta, \bar X, \bar B)}{\delta} = \frac{c w_\beta(h)}{\delta} = c \operatorname{ord}_{F_\ell}(h) = \frac{c\ell(v(h))}{\delta},$$

so $a(F_\ell, X, B + cH) = a(F_\ell, X, B) - c \operatorname{ord}_{F_\ell}(h) = 0$. By (6.6), $\operatorname{ord}_{F_\ell}(f_j) \ge m_j \operatorname{ord}_{F_\ell}(h)$ for any j.

It remains to show that F_ℓ is a Kollár component of $(X \ni x, B)$ over k and that F_ℓ is geometrically integral over $\kappa(x)$. Let $d_i := \ell(v(y_i))/\delta$, which are positive integers. Let $U \subset X$ be an affine open neighborhood of x on which the y_i are regular and generate the ideal of x. For an integer $j \ge 0$, let $\mathfrak{b}_j \subset \mathcal{O}(U)$ be the ideal generated by the monomials y^γ with $\sum_i \gamma_i d_i \ge j$. By **Step 4**, $\mathfrak{b}_j \mathcal{O}_{X,x} = \mathfrak{a}_j(\operatorname{ord}_{F_\ell})\mathcal{O}_{X,x}$. At any point of U other than x, some y_i is a unit, and $\operatorname{center}_X(F_\ell) = x$, so both ideals are the unit ideal there. Thus $\mathfrak{b}_j = \mathfrak{a}_j(\operatorname{ord}_{F_\ell})|_U$. The Rees algebra

$$\mathcal{A} := \bigoplus_{j\ge 0} \mathfrak{b}_j T^j \subset \mathcal{O}(U)[T]$$

is generated over $\mathcal{O}(U)$ by the elements $y_i T^e$ with $1 \le e \le d_i$, since for $y^\gamma T^j$ with $\sum_i \gamma_i d_i \ge j$ we can write $j = \sum_i \sum_{p=1}^{\gamma_i} e_{ip}$ with integers $0 \le e_{ip} \le d_i$, and then $y^\gamma T^j = \prod_i \prod_{p=1}^{\gamma_i} y_i T^{e_{ip}}$, where $y_i T^0 = y_i \in \mathcal{O}(U)$. Let $g : Y_\ell := \operatorname{Proj}_U \mathcal{A} \to U$. Then g is projective, Y_ℓ is integral since $\mathcal{A}$ is a domain, and g is an isomorphism over $U\backslash\{x\}$ since $\mathfrak{b}_j$ is the unit ideal there.

For $\sigma \in \operatorname{Gal}(\bar k/k)$, let σ_X be the automorphism $\mathrm{id}_X \times \operatorname{Spec}\sigma$ of the scheme $X_{\bar k} = X \times_k \operatorname{Spec}\bar k$. The automorphisms σ_X fix the pullbacks of regular functions on X and of B, and they act transitively on the finitely many points of $X_{\bar k}$ over x. Possibly shrinking $\bar U$, we may assume that $\bar U \subset U_{\bar k}$ and that $\bar U$ contains no point over x other than $\bar x$. For a point $\bar x'$ over x, fix σ with $\sigma_X(\bar x) = \bar x'$, and let $\bar F_{\bar x'}$ be the prime divisor over $X_{\bar k}$ whose prime extraction is the conjugate $\bar g^\sigma : \bar Y^\sigma \to \sigma_X(\bar U)$ of $\bar g$ by σ. Since σ_X fixes $B_{\bar k}$, $\bar F_{\bar x'}$ is a Kollár component of $(X_{\bar k} \ni \bar x', B_{\bar k})$. By (6.7),

$$\mathfrak{a}_j(\operatorname{ord}_{\bar F})\mathcal{O}_{\bar X,\bar x} = \mathfrak{a}_{j\delta}(w_\beta)\mathcal{O}_{\bar X,\bar x} = \mathfrak{b}_j \mathcal{O}_{\bar X,\bar x}.$$

Since σ_X fixes the y_i, and since both ideals are the unit ideal at the points of $\sigma_X(\bar U)$ other than $\bar x'$, we obtain

$$\mathfrak{a}_j(\operatorname{ord}_{\bar F_{\bar x'}})|_{\sigma_X(\bar U)} = \mathfrak{b}_j \mathcal{O}_{\sigma_X(\bar U)}.$$

In particular, $\operatorname{ord}_{\bar F_{\bar x'}}(f) = \max\{j \mid f \in \mathfrak{b}_j \mathcal{O}_{X_{\bar k},\bar x'}\}$ for any non-zero $f \in \mathcal{O}_{X_{\bar k},\bar x'}$, so $\bar F_{\bar x'}$ does not depend on the choice of σ. Let r_0 be a positive integer such that $r_0 \bar F$ is Cartier on $\bar Y$. Then $r_0 \bar F_{\bar x'}$ is Cartier on $\bar Y^\sigma$. The base change $Y_{\ell,\bar k} \to U_{\bar k}$ of g is $\operatorname{Proj}_{U_{\bar k}}(\mathcal{A} \otimes_k \bar k)$, and Proj does not change when we pass to the Veronese subalgebra $\bigoplus_m \mathfrak{b}_{mr_0} T^{mr_0}$ [Sta, Tag 0B5J]. Thus Lemma 2.7 implies the following.

(6.8) The morphism $g_{\bar k} : Y_{\ell,\bar k} \to U_{\bar k}$ is an isomorphism over the complement of the points over x, and over $\sigma_X(\bar U)$ it is the prime extraction $\bar g^\sigma$ of $\bar F_{\bar x'}$, for any point $\bar x' = \sigma_X(\bar x)$ over x.

We show that Y_ℓ is normal. By (6.8) and Lemma 3.3(1), $Y_{\ell,\bar k}$ is normal. Let $\operatorname{Spec} A \subset Y_\ell$ be an affine open subset, and let $a, b \in A$ with $b \ne 0$ such that a/b is integral over A. Since $A \otimes_k \bar k$ is normal and flat over A, $a \in b(A \otimes_k \bar k) \cap A = bA$. Thus A is normal.

By (6.8), Lemma 2.7, and Lemma 2.4(1), applied to $\bar g^\sigma$, $\bar x'$, and $\bar F_{\bar x'}$, $g_{\bar k}^{-1}$ of the points over x is the disjoint union of the $\bar F_{\bar x'}$, and $\operatorname{Gal}(\bar k/k)$ permutes them transitively. The preimage in $Y_{\ell,\bar k}$ of any irreducible component of $g^{-1}(x)$ is a $\operatorname{Gal}(\bar k/k)$-stable union of some of the $\bar F_{\bar x'}$. Thus $g^{-1}(x)$ is irreducible of dimension $d-1$, and g has exactly one exceptional prime divisor F'. By Lemma 3.3(1), the pullback of F' to $Y_{\ell,\bar k}$ is $\sum_{\bar x'} \bar F_{\bar x'}$, and a local equation of F' has order one

along $\bar{F}$. Since $\bar{F}$ dominates F', the restriction of $\mathrm{ord}_{\bar{F}}$ to $k(X)$ is $\mathrm{ord}_{F'}$. Hence $F' = F_\ell$, that is, F_ℓ is the exceptional prime divisor of g.

We show that $-F_\ell$ is $\mathbb{Q}$-Cartier and g-ample. Let $\mathcal{L} := \mathcal{O}_{Y_\ell}(-r_0 F_\ell)$. Since Y_ℓ is normal, $\mathcal{L}$ is reflexive. Since $Y_{\ell,\bar{k}} \to Y_\ell$ is flat and $Y_{\ell,\bar{k}}$ is normal, the pullback of $\mathcal{L}$ to each connected component of $Y_{\ell,\bar{k}}$ is reflexive [Sta, Tag 0EBF]. Let $\Sigma \subset Y_\ell$ be the complement of the regular locus of Y_ℓ. On $Y_\ell \backslash \Sigma$, $\mathcal{L}$ is invertible, and the pullback of $\mathcal{L}|_{Y_\ell \backslash \Sigma}$ is $\mathcal{O}(-r_0 \sum_{\bar{x}'} \bar{F}_{\bar{x}'})$, since the pullback of F_ℓ to $Y_{\ell,\bar{k}}$ is $\sum_{\bar{x}'} \bar{F}_{\bar{x}'}$. Since Y_ℓ is normal, Σ has codimension at least two in Y_ℓ, and since $Y_{\ell,\bar{k}} \to Y_\ell$ is flat with zero-dimensional fibers, the preimage $\Sigma_{\bar{k}}$ of Σ has codimension at least two in $Y_{\ell,\bar{k}}$. A normal scheme has depth at least two at the points of codimension at least two, so two reflexive sheaves on $Y_{\ell,\bar{k}}$ whose restrictions to $Y_{\ell,\bar{k}} \backslash \Sigma_{\bar{k}}$ are isomorphic are isomorphic [Sta, Tag 0EBJ]. Thus the pullback of $\mathcal{L}$ to $Y_{\ell,\bar{k}}$ is $\mathcal{O}_{Y_{\ell,\bar{k}}}(-r_0 \sum_{\bar{x}'} \bar{F}_{\bar{x}'})$, which is invertible by (6.8). Hence $\mathcal{L}$ is invertible [Sta, Tag 05B2]. By (6.8), the pullback of $\mathcal{L}$ is $g_{\bar{k}}$-ample. Since $U_{\bar{k}} \to U$ is flat and surjective, $\{U_{\bar{k}} \to U\}$ is an fpqc covering. Since $g_{\bar{k}}$ is the base change of g and the pullback of $\mathcal{L}$ is $g_{\bar{k}}$-ample, $\mathcal{L}$ is g-ample [Sta, Tag 0D2P]. Hence F_ℓ is primitive over U, and g is the prime extraction of F_ℓ.

We show that $(Y_\ell, g_*^{-1} B_U + F_\ell)$ is plt near F_ℓ. Since F_ℓ is exceptional, $B_U = B|_U$. Since F_ℓ is $\mathbb{Q}$-Cartier,

$$K_{Y_\ell} + g_*^{-1} B_U + F_\ell = g^*(K_U + B_U) + a(F_\ell, X, B) F_\ell$$

is $\mathbb{R}$-Cartier. By (6.8), the pullback of $(Y_\ell, g_*^{-1} B_U + F_\ell)$ to $Y_{\ell,\bar{k}}$ is plt on an open neighborhood of $\bigcup_{\bar{x}'} \bar{F}_{\bar{x}'}$, whose complement Z is closed and $\mathrm{Gal}(\bar{k}/k)$-stable. The image of Z in Y_ℓ is closed, since $Y_{\ell,\bar{k}} \to Y_\ell$ is integral, and it is disjoint from F_ℓ. Let $Y_\ell^\circ \subset Y_\ell$ be the complement of the image of Z. By Lemma 3.3(3), $(Y_\ell^\circ, g_*^{-1} B_U|_{Y_\ell^\circ} + F_\ell)$ is plt. Thus F_ℓ is a Kollár component of $(X \ni x, B)$ over k.

Finally,

$$F_\ell \times_k \bar{k} = \bigsqcup_\iota F_\ell \times_{\kappa(x),\iota} \bar{k},$$

where ι runs over the k-embeddings $\kappa(x) \to \bar{k}$, which correspond to the points over x. By Lemma 3.3(1), $F_\ell \times_k \bar{k}$ is the divisor $\sum_{\bar{x}'} \bar{F}_{\bar{x}'}$, and $F_\ell \times_k \bar{k}$ is reduced since F_ℓ is reduced and $\mathrm{char}\, k = 0$ [Har77, Exercise II.3.15(b)], and $\bar{F}_{\bar{x}'}$ maps to $\bar{x}'$. Thus $F_\ell \times_{\kappa(x),\iota} \bar{k} \cong \bar{F}_{\bar{x}'}$ for the point $\bar{x}'$ corresponding to ι, which is integral. Hence F_ℓ is geometrically integral over $\kappa(x)$ [Har77, Exercise II.3.15(c)]. This proves (1).

Step 6. In this step we prove (2).

Let $\ell = \ell_\beta$ be as in (1). Then $\mathrm{ord}_{F_\ell} / \mathrm{ord}_{F_\ell}(h) = v_\ell / v_\ell(h)$ is the restriction of $w_\beta / w_\beta(h)$ to $k(X)$, and ℓ converges to the inclusion $M \subset \mathbb{R}$ if and only if β converges to α. Let $f \in \mathcal{O}_{X,x}$ be non-zero. By (6.3), for β close to α,

$$\frac{w_\beta(f)}{w_\beta(h)} = \frac{\langle \beta, \gamma_f \rangle}{\langle \beta, \gamma_h \rangle} \to \frac{\langle \alpha, \gamma_f \rangle}{\langle \alpha, \gamma_h \rangle} = \frac{w(f)}{w(h)} = v(f),$$

since $w(h) = 1$. This proves (2). □

Corollary 7.5 uses only the existence of a Kollár component over k given by the following corollary, and not its log discrepancy.

Corollary 6.7. *Let k be a field finitely generated over $\mathbb{C}$ and $(X \ni x, B)$ a klt pair over k of dimension $d \geq 2$ with X affine and x a closed point. Then there exist a non-zero regular function h on X with $h(x) = 0$ and a real number $0 < c < 1$ such that the conditions of Set-up 6.1 hold with $s = t = 0$ and $b_0 = 1$, so that $b(v) = v(h)$. In particular, $(X \ni x, B)$ has a Kollár component over k.*

Proof. Let $\mathfrak{m} \subset \mathcal{O}(X)$ be the ideal of x. Let $f : W \to X$ be a log resolution of (X, B) and $\mathfrak{m}$ over k [Hir64, Main Theorems I and II], that is, W is smooth over k, $\mathfrak{m}\mathcal{O}_W = \mathcal{O}_W(-D)$ for a divisor $D \geq 0$, and $\mathrm{Exc}(f) \cup \mathrm{Supp}\, f_*^{-1} B \cup \mathrm{Supp}\, D$ is an snc divisor. Write $D = \sum_q e_q D_q$, where the

D_q are the prime divisors in $\mathrm{Exc}(f)\cup\mathrm{Supp}\,D$ and $e_q\geq 0$. Since x is a closed point and $d\geq 2$, $D\neq 0$, and $f(D_q)=x$ whenever $e_q>0$. Let

$$\lambda:=\min\left\{\frac{a(D_q,X,B)}{e_q}\;\middle|\;e_q>0\right\},$$

which is a positive real number since (X,B) is klt. Fix q_0 with $e_{q_0}>0$ and $a(D_{q_0},X,B)=\lambda e_{q_0}$, and fix an integer $N>\lambda$.

Let $V\subset\mathfrak{m}^N$ be a k-subspace with $\dim_k V<+\infty$ which generates $\mathfrak{m}^N$. Then $\mathfrak{m}^N\mathcal{O}_W=\mathcal{O}_W(-ND)$, and for $\varphi\in V\backslash\{0\}$ we write $\mathrm{div}(f^*\varphi)=ND+D_\varphi$ with $D_\varphi\geq 0$. Since $\mathfrak{m}^N\mathcal{O}_W=\mathcal{O}_W(-ND)$, the sections $f^*\varphi$, $\varphi\in V$, of the invertible sheaf $\mathcal{O}_W(-ND)$ generate it, and D_φ is the divisor of $f^*\varphi$ as a section of $\mathcal{O}_W(-ND)$. By Lemma 3.4, applied to W, the snc divisor $\mathrm{Exc}(f)\cup\mathrm{Supp}\,f_*^{-1}B\cup\mathrm{Supp}\,D$, $\mathcal{O}_W(-ND)$, and f^*V, there exists a non-zero $h\in V$ such that D_h is smooth, has no common component with $\mathrm{Exc}(f)\cup\mathrm{Supp}\,f_*^{-1}B\cup\mathrm{Supp}\,D$, and $\mathrm{Exc}(f)\cup\mathrm{Supp}\,f_*^{-1}B\cup\mathrm{Supp}\,D\cup D_h$ is an snc divisor. Let $H:=\mathrm{div}(h)$ and $c:=\lambda/N\in(0,1)$. Then $h\neq 0$ and $h(x)=0$.

Let $K_W+B_W:=f^*(K_X+B)$. Then $f^*(K_X+B+cH)=K_W+B_W+cND+cD_h$. The coefficient of D_q in B_W+cND is $1-a(D_q,X,B)+\lambda e_q\leq 1$, with equality for $q=q_0$, the coefficients of the components of $f_*^{-1}B$ are less than 1, and the coefficient of D_h is $c<1$. By Lemma 3.3(3) and [KM98, Lemma 2.30 and Corollary 2.31(3)] (the proofs there, written for $\mathbb{Q}$-divisors, apply to $\mathbb{R}$-divisors), applied on $W_{\bar{k}}$, $(X,B+cH)$ is lc, and any lc center of $(X_{\bar{k}},B_{\bar{k}}+cH_{\bar{k}})$ is the image of a stratum of the components of $(B_W+cND)_{\bar{k}}$ with coefficient one [Kol13, Definition 4.15 and the paragraphs after it], hence a point over x. Moreover $a(D_{q_0},X,B+cH)=a(D_{q_0},X,B)-cNe_{q_0}=0$.

The morphism $(D_{q_0})_{\bar{k}}\to\{x\}\times_k\bar{k}$ is surjective, so some component $\bar{D}$ of the pullback of D_{q_0} to $W_{\bar{k}}$ has center $\bar{x}$ on $X_{\bar{k}}$. By Lemma 3.3(1)(2), the restriction of $\mathrm{ord}_{\bar{D}}$ to $k(X)$ is $\mathrm{ord}_{D_{q_0}}$ and $a(\bar{D},X_{\bar{k}},B_{\bar{k}}+cH_{\bar{k}})=0$. Thus $u:=\mathrm{ord}_{\bar{D}}/\mathrm{ord}_{\bar{D}}(h)$, where $\mathrm{ord}_{\bar{D}}(h)=Ne_{q_0}>0$, is a non-trivial quasi-monomial valuation with $\mathrm{center}_{X_{\bar{k}}}(u)=\bar{x}$, $u(h)=1$, and $a(u,X_{\bar{k}},B_{\bar{k}})=cu(h)$. Hence $u\in\mathcal{Q}_{\bar{x}}$ for $s=t=0$, and with $b_0=1$ we have $b(v)=v(h)=1>0$ for any $v\in\mathcal{Q}$. Thus the conditions of Set-up 6.1 hold with $s=t=0$ and $b_0=1$.

By Theorem 6.6(1), the open rational polyhedral cone $\mathcal{C}$ contains an element ℓ with $\ell(M)\subset\mathbb{Q}$, and F_ℓ is a Kollár component of $(X\ni x,B)$ over k. □

7. Decomposable complements over the ground field

In this section, we prove Corollary 7.5, which enlarges a boundary with DCC coefficients to an lc boundary with coefficients in a finite set, by the decomposable $\mathbb{R}$-complements of [HLS26]. Over k, the sum of a geometric complement over a Galois orbit has denominators depending on the size of the orbit, and Construction-Theorem 8.5 needs x to be an lc center of (X,Δ). Lemma 7.3 removes the first difficulty, and we remove the second by constructing complements on the prime extraction of a Kollár component S over k, with the coefficient of S equal to 1.

We recall the following notions from [HLS26, Definitions 3.2 and 3.11].

Definition 7.1. Let $(X/Z\ni z,B)$ be a pair over a field of characteristic zero. We say that X is *of Fano type* over Z if there exists an $\mathbb{R}$-divisor $\Delta\geq 0$ on X such that (X,Δ) is klt and $-(K_X+\Delta)$ is big and nef over Z. An $\mathbb{R}$-*complement* of $(X/Z\ni z,B)$ is a pair $(X/Z\ni z,B^+)$ such that $B^+\geq B$, (X,B^+) is lc over a neighborhood of z, and $K_X+B^+\sim_{\mathbb{R}}0$ over a neighborhood of z. We say that $(X/Z\ni z,B)$ is $\mathbb{R}$-*complementary* if it has an $\mathbb{R}$-complement. For a positive integer n, a *monotonic n-complement* of $(X/Z\ni z,B)$ is an $\mathbb{R}$-complement $(X/Z\ni z,B^+)$ of $(X/Z\ni z,B)$ such that nB^+ is a Weil divisor and $n(K_X+B^+)\sim 0$ over a neighborhood of z.

Lemma 7.2. *Let k be a field finitely generated over $\mathbb{C}$ and (X,B) a klt pair over k. Then there exists a small projective birational morphism $g:X'\to X$ over k from a normal variety X' such that any Weil divisor on X' is $\mathbb{Q}$-Cartier.*

Proof. Write $B = \sum_i b_i B_i$, where the B_i are the distinct prime components of B. Fix real numbers $r_1 = 1, r_2, \dots, r_m$ that are linearly independent over $\mathbb{Q}$ and whose $\mathbb{Q}$-span contains all b_i. Then

$$K_X + B = \sum_{i=1}^{m} r_i D_i$$

for some $\mathbb{Q}$-divisors $D_1, \dots, D_m$ on X. Since $K_X + B$ is $\mathbb{R}$-Cartier, $K_X + B \sim_{\mathbb{R}} 0$ over X. Hence each D_i is $\mathbb{Q}$-Cartier by [HLS26, Lemma 5.3] applied with $f = \mathrm{id}_X$, whose proof (given in version 2 of arXiv:1903.04338 and omitted in the published version) works over k since it only writes $K_X + B$ as an $\mathbb{R}$-linear combination of Cartier divisors and compares coefficients in a basis of a $\mathbb{Q}$-vector space of real numbers.

Let $\mu : X'' \to X$ be a log resolution of (X, B) [Hir64, Main Theorems I and II], and write $K_{X''} + B'' := \mu^*(K_X + B)$. Since (X, B) is klt, all coefficients of B'' are less than 1. By [Gro66, Sections 8 and 9], there exist a smooth integral variety T over $\mathbb{C}$ with function field k and a projective birational morphism $\mu_T : \mathcal{X}'' \to \mathcal{X}$ of normal quasi-projective varieties over T whose generic fiber over T is μ, satisfying the following: $\mathcal{X}''$ is smooth, the closure $\mathcal{D}_i$ of D_i in $\mathcal{X}$ is $\mathbb{Q}$-Cartier for any i, and, if $\mathcal{B}$ and $\mathcal{B}''$ are the closures of B and B'' and $K_{\mathcal{X}}$ is a canonical divisor of $\mathcal{X}$ whose restriction to X is K_X, then

$$K_{\mathcal{X}} + \mathcal{B} = \sum_{i=1}^{m} r_i \mathcal{D}_i, \quad K_{\mathcal{X}''} + \mathcal{B}'' = \mu_T^*(K_{\mathcal{X}} + \mathcal{B}),$$

$\mathrm{Exc}(\mu_T)$ is a divisor, and $\mathrm{Exc}(\mu_T) \cup \operatorname{Supp} \mathcal{B}''$ is an snc divisor. Each of the conditions above holds over a non-empty open subset of T because it holds on the generic fiber, and since T is smooth, $K_{\mathcal{X}}$ differs from the relative canonical divisor $K_{\mathcal{X}/T}$ by the pullback of the Cartier divisor K_T. Since the coefficients of $\mathcal{B}''$ are the coefficients of B'', which are less than 1, the pair $(\mathcal{X}, \mathcal{B})$ is klt by [Kol13, Corollary 2.13(2)], applied to the log resolution μ_T. This corollary holds for the $\mathbb{R}$-divisor $\mathcal{B}$ by [Kol13, 2.20], since $K_{\mathcal{X}} + \mathcal{B} = \sum_{i=1}^{m} r_i \mathcal{D}_i$ is a linear combination of Cartier divisors with real coefficients, each $\mathcal{D}_i$ being $\mathbb{Q}$-Cartier.

By [BCHM10, Corollary 1.4.3] and the second paragraph after it, there exists a small projective birational morphism $g_T : \mathcal{X}' \to \mathcal{X}$ such that $\mathcal{X}'$ is $\mathbb{Q}$-factorial. Let $g : X' \to X$ be the generic fiber of g_T over T. Then g is a small projective birational morphism over k, and X' is normal since X' is a localization of $\mathcal{X}'$. Let P be a prime divisor on X' and $\mathcal{P}$ the closure of P in $\mathcal{X}'$. Then $\mathcal{P}$ is a prime divisor, $m\mathcal{P}$ is Cartier for some positive integer m, and $mP = m\mathcal{P}|_{X'}$ is Cartier. Thus any Weil divisor on X' is $\mathbb{Q}$-Cartier. □

Lemma 7.3 descends the existence of geometric monotonic n-complements to k without enlarging n: the existence of the geometric complements implies that the fixed part of the linear system $|-nK_W - \lceil nD_0 \rceil|$ defines a sub-lc sub-boundary near the fiber over x, and since k is infinite, Bertini's theorem provides a general member of $|-nK_W - \lceil nD_0 \rceil|$ over k.

Lemma 7.3. *Let k be a field finitely generated over $\mathbb{C}$, $h : W \to X$ a projective birational morphism between normal varieties over k with X affine, $x \in X$ a closed point, n a positive integer, and $D_0 \geq 0$ a $\mathbb{Q}$-divisor on W such that $K_W + D_0$ is $\mathbb{Q}$-Cartier. Assume that for any point $\bar{x} \in X_{\bar{k}}$ over x, if X° is the connected component of $X_{\bar{k}}$ containing $\bar{x}$ and $W^\circ := h_{\bar{k}}^{-1}(X^\circ)$, then $(W^\circ / X^\circ \ni \bar{x}, (D_0)_{\bar{k}}|_{W^\circ})$ has a monotonic n-complement. Then, possibly shrinking X around x, there exists a $\mathbb{Q}$-divisor $\Theta \geq D_0$ on W such that (W, Θ) is lc, $n\Theta$ is a Weil divisor, and $n(K_W + \Theta) = \mathrm{div}(\phi)$ for some $\phi \in k(W)^*$.*

Proof. By Lemma 3.3(1), $X_{\bar{k}}$ and $W_{\bar{k}}$ are normal, so the connected components of $X_{\bar{k}}$ are irreducible. For any connected component X° of $X_{\bar{k}}$, the scheme $W^\circ := h_{\bar{k}}^{-1}(X^\circ)$ is a normal variety, $W^\circ \to X^\circ$ is projective and birational, and since X° is normal, $(h_{\bar{k}}|_{W^\circ})_* \mathcal{O}_{W^\circ}$ is a finite $\mathcal{O}_{X^\circ}$-algebra contained in the function field of X°, hence equal to $\mathcal{O}_{X^\circ}$. Thus $W^\circ \to X^\circ$ is a contraction.

Fix a canonical divisor K_W. Since differentials commute with base change, the pullback of K_W to $W_{\bar{k}}$ is a canonical divisor $K_{W_{\bar{k}}}$ of $W_{\bar{k}}$. For any non-zero $s \in H^0(W, \mathcal{O}_W(-nK_W - \lceil nD_0 \rceil)) \subset k(W)$, we have $\operatorname{div}_W(s) \geq nK_W + \lceil nD_0 \rceil$, so

$$\Theta_s := -K_W + \frac{1}{n} \operatorname{div}_W(s)$$

satisfies $\Theta_s \geq \frac{1}{n}\lceil nD_0 \rceil \geq D_0$, $n\Theta_s$ is a Weil divisor, and $n(K_W + \Theta_s) = \operatorname{div}_W(s)$. It suffices to find s such that (W, Θ_s) is lc after shrinking X around x.

Since h is projective, $h_*\mathcal{O}_W(-nK_W - \lceil nD_0 \rceil)$ is coherent [Har77, Theorem III.8.8(b)]. Since X is affine, it is generated by finitely many global sections $s_1, \dots, s_t \in H^0(W, \mathcal{O}_W(-nK_W - \lceil nD_0 \rceil))$. Since W is normal, $\mathcal{O}_W(-nK_W - \lceil nD_0 \rceil)$ is reflexive [Sta, Tag 0AY6]. Since $W_{\bar{k}} \to W$ is flat and $W_{\bar{k}}$ is normal, the pullback of $\mathcal{O}_W(-nK_W - \lceil nD_0 \rceil)$ to each connected component of $W_{\bar{k}}$ is reflexive [Sta, Tag 0EBF]. Since $(\lceil nD_0 \rceil)_{\bar{k}} = \lceil n(D_0)_{\bar{k}} \rceil$ by Lemma 3.3(1), the pullback coincides with

$$\mathcal{O}_{W_{\bar{k}}}(-nK_{W_{\bar{k}}} - \lceil n(D_0)_{\bar{k}} \rceil)$$

over the inverse image in $W_{\bar{k}}$ of the regular locus of W. Since W is normal, the complement of the regular locus of W has codimension at least 2 in W, and since field extension preserves dimensions, the complement in $W_{\bar{k}}$ of the inverse image of the regular locus of W has codimension at least 2 in each connected component of $W_{\bar{k}}$, which is integral since $W_{\bar{k}}$ is normal, so the local rings of $W_{\bar{k}}$ at the points outside the inverse image of the regular locus of W have depth at least 2 by Serre's criterion [Sta, Tag 031S]. Since both sheaves are reflexive, they coincide on $W_{\bar{k}}$ [Sta, Tag 0EBJ]. By [Har77, Proposition III.9.3] applied to the flat morphism $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$, $s_1, \dots, s_t$ generate

$$(h_{\bar{k}})_*\mathcal{O}_{W_{\bar{k}}}(-nK_{W_{\bar{k}}} - \lceil n(D_0)_{\bar{k}} \rceil).$$

Let $\mu : W' \to W$ be a log resolution over k of $(W, \operatorname{Supp} K_W)$ that also resolves the base locus of the linear system spanned by $s_1, \dots, s_t$ [Hir64, Main Theorems I and II]. Namely, W' is smooth, and if F is the divisor on W' with $\operatorname{mult}_P F = \min_i \operatorname{ord}_P(s_i)$ for any prime divisor P on W', then for any non-zero $s = \sum_i c_i s_i$ with $c_i \in k$,

$$\operatorname{div}_{W'}(s) = F + H_s,$$

where $H_s \geq 0$ belongs to a base point free linear system, and $\operatorname{Exc}(\mu) \cup \mu_*^{-1} \operatorname{Supp} K_W \cup \operatorname{Supp} F$ is an snc divisor. The divisor F may have negative coefficients. Let $K_{W'}$ be the canonical divisor of W' with $\mu_* K_{W'} = K_W$, and let

$$B_{W'} := -K_{W'} + \frac{1}{n} F.$$

Then $\operatorname{Supp} B_{W'} \cup \operatorname{Exc}(\mu)$ is an snc divisor, and for any non-zero $s = \sum_i c_i s_i$ with $c_i \in k$,

$$K_{W'} + B_{W'} + \frac{1}{n} H_s = \frac{1}{n} \operatorname{div}_{W'}(s) = \mu^*(K_W + \Theta_s). \tag{7.1}$$

We show that $\operatorname{mult}_P B_{W'} \leq 1$ for any prime divisor P on W' such that $x \in h(\mu(P))$. Since $\operatorname{Spec} \bar{k} \to \operatorname{Spec} k$ is surjective, there exist a point $\bar{x} \in X_{\bar{k}}$ over x and a prime component Q of $P_{\bar{k}}$ such that $\bar{x} \in h_{\bar{k}}(\mu_{\bar{k}}(Q))$. Let X° be the connected component of $X_{\bar{k}}$ containing $\bar{x}$, let $W^\circ := h_{\bar{k}}^{-1}(X^\circ)$, and let $(W^\circ / X^\circ \ni \bar{x}, \Theta^+)$ be a monotonic n-complement of $(W^\circ / X^\circ \ni \bar{x}, (D_0)_{\bar{k}}|_{W^\circ})$. Then there exist an open neighborhood V of $\bar{x}$ in X° and $\bar{s} \in \bar{k}(W^\circ)^*$ such that (W°, Θ^+) is lc over V and

$$n(K_{W_{\bar{k}}} + \Theta^+) = \operatorname{div}(\bar{s}) \text{ over } V.$$

Since $n\Theta^+$ is a Weil divisor and $\Theta^+ \geq (D_0)_{\bar{k}}$, we have $n\Theta^+ \geq \lceil n(D_0)_{\bar{k}} \rceil$. Thus $\bar{s}$ is a section of $\mathcal{O}_{W_{\bar{k}}}(-nK_{W_{\bar{k}}} - \lceil n(D_0)_{\bar{k}} \rceil)$ over $h_{\bar{k}}^{-1}(V)$, and possibly shrinking V, $\bar{s} = \sum_i \bar{a}_i s_i$ for some regular functions $\bar{a}_i$ on V. By Lemma 3.3(1), $\operatorname{ord}_Q(s_i) = \operatorname{ord}_P(s_i)$ for any i and $\operatorname{mult}_Q K_{W'_{\bar{k}}} = \operatorname{mult}_P K_{W'}$, where $K_{W'_{\bar{k}}}$ is the pullback of $K_{W'}$. Hence

$$\operatorname{ord}_Q(\bar{s}) \geq \min_i \operatorname{ord}_P(s_i) = \operatorname{mult}_P F.$$

On the other hand,

$$(\mu_{\bar{k}})^*(K_{W_{\bar{k}}}+\Theta^+)=\frac{1}{n}\operatorname{div}_{W'_{\bar{k}}}(\bar{s})=K_{W'_{\bar{k}}}+\Big(-K_{W'_{\bar{k}}}+\frac{1}{n}\operatorname{div}_{W'_{\bar{k}}}(\bar{s})\Big)\text{ over }V.$$

Since (W°,Θ^+) is lc over V and Q intersects the inverse image of V, we have

$$\operatorname{mult}_Q\Big(-K_{W'_{\bar{k}}}+\frac{1}{n}\operatorname{div}_{W'_{\bar{k}}}(\bar{s})\Big)\le 1.$$

Therefore,

$$\operatorname{mult}_P B_{W'}=-\operatorname{mult}_P K_{W'}+\frac{1}{n}\operatorname{mult}_P F\le-\operatorname{mult}_Q K_{W'_{\bar{k}}}+\frac{1}{n}\operatorname{ord}_Q(\bar{s})\le 1.$$

Since $h\circ\mu$ is proper, possibly shrinking X around x, we may assume that $\operatorname{mult}_P B_{W'}\le 1$ for any prime divisor P on W'. Let $\mathcal{M}:=\mathcal{O}_{W'}(-F)$, an invertible sheaf since W' is smooth, and let $\mathcal{S}\subset H^0(W',\mathcal{M})$ be the k-subspace spanned by $\mu^*s_1,\dots,\mu^*s_t$. By the choice of μ, $\mathcal{S}$ generates $\mathcal{M}$, so $\mathcal{M}$ is base point free. For any non-zero $s=\sum_i c_is_i$ with $c_i\in k$, H_s is the divisor of zeros of s as a section of $\mathcal{M}$. By Lemma 3.4, applied to W', the snc divisor $\operatorname{Supp}B_{W'}\cup\operatorname{Exc}(\mu)$, $\mathcal{M}$, and $\mathcal{S}$, there exist $c_1,\dots,c_t\in k$ such that $s:=\sum_i c_is_i\neq 0$, H_s is reduced, H_s and $B_{W'}$ have no common components, and $\operatorname{Supp}B_{W'}\cup\operatorname{Exc}(\mu)\cup\operatorname{Supp}H_s$ is an snc divisor. All coefficients of $B_{W'}+\frac{1}{n}H_s$ are at most 1, so $(W',B_{W'}+\frac{1}{n}H_s)$ is sub-lc [KM98, Corollary 2.31(3)]. By (7.1) and [KM98, Lemma 2.30], (W,Θ_s) is lc. We let $\Theta:=\Theta_s$ and $\phi:=s$. □

Theorem 7.4. *Let $d\ge 2$ be an integer and $I\subset[0,1]$ a DCC set. Then there exist a positive integer n, a finite set $\mathcal{A}\subset(0,1]$, and a finite set $J\subset[0,1]$ depending only on d and I satisfying the following. Let k be a field finitely generated over $\mathbb{C}$, and $(X\ni x,B)$ a klt pair over k of dimension d such that x is a closed point and $B\in I$. Let S be a Kollár component of $(X\ni x,B)$ over k. Then, possibly shrinking X around x, there exist $\lambda_1,\dots,\lambda_r\in\mathcal{A}$ with $\sum_j\lambda_j=1$ and $\mathbb{Q}$-divisors $\Delta_1,\dots,\Delta_r$ on X such that:*

(1) for any j, (X,Δ_j) is lc, $n(K_X+\Delta_j)=\operatorname{div}(\phi_j)$ for some $\phi_j\in k(X)^*$, and S is an lc place of (X,Δ_j),
(2) $\Delta:=\sum_j\lambda_j\Delta_j$ satisfies $\Delta\ge B$, $\Delta\in J$, (X,Δ) is lc, S is an lc place of (X,Δ), and $K_X+\Delta\sim_{\mathbb{R}}0$, and
(3) $\Delta-B$ is $\mathbb{R}$-Cartier and x is an lc center of (X,Δ).

One may take J to be the set of the numbers $\sum_{j=1}^r\lambda_jm_j/n$ with $\lambda_j\in\mathcal{A}$, $\sum_j\lambda_j=1$, $r\le\lfloor 1/\min\mathcal{A}\rfloor$, and integers $0\le m_j\le n$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix n, $\mathcal{A}$, and J.

Let $\Lambda:=I\cup\{1\}$, a DCC set, and let $\bar{\Lambda}$ be the closure of Λ. By [HLS26, Theorem 5.20(1) and (6)] (in the numbering of version 2 of arXiv:1903.04338, whose published version states (6) as [HLS26, Theorem 5.12] but does not state (1)), there exist a finite set $\Lambda'\subset\bar{\Lambda}$ and a map $g:\bar{\Lambda}\to\Lambda'$ with $g(\gamma)\ge\gamma$ for any $\gamma\in\Lambda$, depending only on d and I, satisfying the following: if $(W/Z\ni z,\sum_i b_iP_i)$ is an $\mathbb{R}$-complementary pair of dimension d such that W is of Fano type over Z, each $P_i\ge 0$ is a $\mathbb{Q}$-Cartier Weil divisor, and $b_i\in\Lambda$ for any i, then $(W/Z\ni z,\sum_i g(b_i)P_i)$ is $\mathbb{R}$-complementary. Since $g(1)\ge 1$ and $\Lambda'\subset[0,1]$, we have $g(1)=1$, so $1\in\Lambda'$.

Let $\mathbf{v}:=(b)_{b\in\Lambda'}\in\mathbb{R}^{\Lambda'}$, and let $\mathcal{L}\subset\mathbb{R}^{\Lambda'}$ be the smallest affine subspace defined over $\mathbb{Q}$ containing $\mathbf{v}$. By [HLS26, Theorem 5.9], there exists an open neighborhood $\mathcal{N}$ of $\mathbf{v}$ in $\mathcal{L}$, depending only on d and $\mathbf{v}$, satisfying the following: if $(W/Z\ni z,\sum_{b\in\Lambda'}bP_b)$ is an $\mathbb{R}$-complementary pair of dimension d such that W is of Fano type over Z and each $P_b\ge 0$ is a Weil divisor, then $(W/Z\ni z,\sum_{b\in\Lambda'}v'_bP_b)$ is $\mathbb{R}$-complementary for any $\mathbf{v}'=(v'_b)_{b\in\Lambda'}\in\mathcal{N}$. For $b\in\Lambda'\cap\{0,1\}$, the hyperplane $\{v'_b=b\}$ is defined over $\mathbb{Q}$ and contains $\mathbf{v}$, so $v'_b=b$ for any $\mathbf{v}'\in\mathcal{L}$. Possibly shrinking $\mathcal{N}$, we may assume that $v'_b\in(0,1)$ for any $\mathbf{v}'\in\mathcal{N}$ and any $b\in\Lambda'\backslash\{0,1\}$. Since $\mathcal{L}$

is defined over $\mathbb{Q}$, rational points are dense in $\mathcal{L}$, so there exist a simplex in $\mathcal{N}$ with vertices $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathcal{N} \cap \mathbb{Q}^{\Lambda'}$ and real numbers $\lambda_1, \dots, \lambda_r > 0$ such that

$$\sum_{j=1}^{r} \lambda_j = 1 \text{ and } \mathbf{v} = \sum_{j=1}^{r} \lambda_j \mathbf{v}_j.$$

Write $\mathbf{v}_j = (v_{j,b})_{b \in \Lambda'}$. Then $v_{j,b} \in [0,1] \cap \mathbb{Q}$ and $v_{j,1} = 1$ for any j. Let $\mathcal{A} := \{\lambda_1, \dots, \lambda_r\}$ and $\Lambda_1 := \{v_{j,b} \mid 1 \le j \le r, b \in \Lambda'\}$, finite sets depending only on d and I, with $r \le \lfloor 1/\min \mathcal{A} \rfloor$.

By [HLS26, Theorem 1.10] (in the numbering of version 2 of arXiv:1903.04338) applied to the finite set $\Lambda_1 \subset \mathbb{Q}$, there exists a positive integer n depending only on d and Λ_1 such that any $\mathbb{R}$-complementary pair $(W/Z \ni z, D)$ of dimension d such that W is of Fano type over Z and $D \in \Lambda_1$ has a monotonic n-complement. Let J be the set of the numbers $\sum_{j=1}^{r} \lambda_j m_j/n$ as in the statement. Then n, $\mathcal{A}$, and J depend only on d and I, and $J \subset [0,1]$ is finite.

Step 2. In this step we construct a projective birational morphism $h : W \to X$ such that W is $\mathbb{Q}$-factorial and of Fano type over X, S is a prime divisor on W, and the strict transform of $B + S$ is $\mathbb{R}$-complementary.

Since $d \ge 2$ and $\operatorname{center}_X S = x$, S is exceptional over X. By Definition 2.3, possibly shrinking X around x, we may assume that X is affine and that S has a prime extraction $f : Y \to X$ such that $(Y, B_Y + S)$ is plt near S, where $B_Y := f_*^{-1}B$. Since $K_X + B$ is an $\mathbb{R}$-linear combination of Cartier divisors and Cartier divisors are principal near x, possibly shrinking X further, we may assume that $K_X + B \sim_{\mathbb{R}} 0$. By Lemma 2.4(1), f is an isomorphism over $X \backslash \{x\}$ and $\operatorname{Supp} f^{-1}(x) = S$, so $(Y, B_Y + S)$ is plt since (X, B) is klt. By the definition of the log discrepancy,

$$K_Y + B_Y + S = f^*(K_X + B) + a(S, X, B)S. \tag{7.2}$$

We show that Y is of Fano type over X, which is the reason to work on the prime extraction. Fix a real number $0 < \epsilon < \min\{1, a(S, X, B)\}$. Since $(Y, B_Y + S)$ is plt and the coefficients of B_Y are less than 1, $(Y, B_Y + (1-\epsilon)S)$ is klt. By (7.2),

$$-(K_Y + B_Y + (1-\epsilon)S) \sim_{\mathbb{R}} -(a(S, X, B) - \epsilon)S,$$

which is f-ample. Thus Y is of Fano type over X.

Since $-S$ is f-ample and X is affine, there exists an integer $l > a(S, X, B)$ such that lS is Cartier and $\mathcal{O}_Y(-lS)$ is generated by global sections [Har77, Theorem II.7.6]. Let $\psi : Y'' \to Y$ be a log resolution of $(Y, B_Y + S)$ over k and write $K_{Y''} + B_{Y''} := \psi^*(K_Y + B_Y + S)$. Since $(Y, B_Y + S)$ is plt, all coefficients of $B_{Y''}$ are at most 1. Let $V \subset H^0(Y, \mathcal{O}_Y(-lS))$ be the k-linear subspace spanned by finitely many global sections generating $\mathcal{O}_Y(-lS)$. Then $\dim_k V < +\infty$ and V generates $\mathcal{O}_Y(-lS)$. By Lemma 3.4, applied to Y'', the snc divisor $\operatorname{Supp} B_{Y''} \cup \operatorname{Exc}(\psi)$, $\psi^*\mathcal{O}_Y(-lS)$, and ψ^*V, there exists a member H of $|-lS|$ over k such that ψ^*H is reduced, ψ^*H and $B_{Y''}$ have no common components, and $\operatorname{Supp} B_{Y''} \cup \operatorname{Exc}(\psi) \cup \operatorname{Supp} \psi^*H$ is an snc divisor. Since $a(S, X, B)/l < 1$, $(Y'', B_{Y''} + \frac{a(S,X,B)}{l}\psi^*H)$ is sub-lc [KM98, Corollary 2.31(3)], so $(Y, B_Y + S + \frac{a(S,X,B)}{l}H)$ is lc [KM98, Lemma 2.30]. By (7.2) and since $H \sim -lS$,

$$K_Y + B_Y + S + \frac{a(S, X, B)}{l}H \sim_{\mathbb{R}} f^*(K_X + B) \sim_{\mathbb{R}} 0.$$

By Lemma 7.2 applied to the klt pair $(Y, B_Y + (1-\epsilon)S)$, there exists a small projective birational morphism $g_W : W \to Y$ over k from a normal variety W such that any Weil divisor on W is $\mathbb{Q}$-Cartier. Let $h := f \circ g_W$, and let B_W, S_W, and H_W be the strict transforms of B_Y, S, and H on W. Since g_W is small, $g_W^*(K_Y + \Delta_Y) = K_W + (g_W)_*^{-1}\Delta_Y$ for any $\mathbb{R}$-divisor Δ_Y on Y such that $K_Y + \Delta_Y$ is $\mathbb{R}$-Cartier. Thus $(W, B_W + (1-\epsilon)S_W)$ is klt and $-(K_W + B_W + (1-\epsilon)S_W)$ is the pullback of an f-ample $\mathbb{R}$-divisor, hence nef and big over X. Moreover, $(W, B_W + S_W + \frac{a(S,X,B)}{l}H_W)$ is lc and $K_W + B_W + S_W + \frac{a(S,X,B)}{l}H_W \sim_{\mathbb{R}} 0$. Since h is projective and birational and X is normal, $h_*\mathcal{O}_W$ is a finite $\mathcal{O}_X$-algebra contained in the

function field, hence $h_*\mathcal{O}_W = \mathcal{O}_X$. Thus W is of Fano type over X, and $(W/X \ni x, B_W + S_W)$ is $\mathbb{R}$-complementary.

Step 3. In this step we construct $\mathbb{Q}$-divisors $D_1, \dots, D_r$ on W whose geometric pairs are $\mathbb{R}$-complementary.

Write $B_W + S_W = \sum_i b_i P_i$, where the P_i are distinct prime divisors on W and $b_i \in \Lambda$. Let $\bar{x} \in X_{\bar{k}}$ be a point over x, X° the connected component of $X_{\bar{k}}$ containing $\bar{x}$, and $W^\circ := h_{\bar{k}}^{-1}(X^\circ)$. For an $\mathbb{R}$-divisor P on W, we let $P^\circ := P_{\bar{k}}|_{W^\circ}$. By Lemma 3.2, $\bar{k} \cong \mathbb{C}$, so the results of [HLS26] hold for varieties over $\bar{k}$. Since X° is normal by Lemma 3.3(1) and $W^\circ \to X^\circ$ is projective and birational, $W^\circ \to X^\circ$ is a contraction as in **Step 2**. By Lemma 3.3(3) and since the pullback of an f-ample $\mathbb{R}$-divisor to $Y_{\bar{k}}$ is $f_{\bar{k}}$-ample, W° is of Fano type over X° and $(W^\circ/X^\circ \ni \bar{x}, (B_W + S_W)^\circ)$ is $\mathbb{R}$-complementary. By Lemma 3.3(1), each P_i° is a reduced $\mathbb{Q}$-Cartier Weil divisor. A prime divisor on W may split into several prime divisors on $W_{\bar{k}}$, but P_i° is still a $\mathbb{Q}$-Cartier Weil divisor, and this is why we use [HLS26, Theorem 5.12] for Weil divisors that are not necessarily prime. By **Step 1**,

$$\Big(W^\circ/X^\circ \ni \bar{x}, \sum_i g(b_i) P_i^\circ\Big)$$

is $\mathbb{R}$-complementary.

Let $D' := \sum_i g(b_i) P_i$. Since $g(\gamma) \geq \gamma$ for any $\gamma \in \Lambda$ and $b_i \in \Lambda$, $D' \geq B_W + S_W$. For $b \in \Lambda'$, let $P_b := \sum_{i:g(b_i)=b} P_i$, a $\mathbb{Q}$-Cartier Weil divisor on W, possibly zero. For $1 \leq j \leq r$, let

$$D_j := \sum_{b \in \Lambda'} v_{j,b} P_b.$$

Then D_j is a $\mathbb{Q}$-divisor on W with $D_j \in \Lambda_1$, $K_W + D_j$ is $\mathbb{Q}$-Cartier, and

$$D' = \sum_{b \in \Lambda'} b P_b = \sum_{j=1}^{r} \lambda_j D_j. \tag{7.3}$$

Since $g(1) = 1$, S_W is a component of P_1, so $\operatorname{mult}_{S_W} D_j = v_{j,1} = 1$. Since $(W^\circ/X^\circ \ni \bar{x}, \sum_b b P_b^\circ)$ is $\mathbb{R}$-complementary and $\mathbf{v}_j \in \mathcal{N}$, by **Step 1**, $(W^\circ/X^\circ \ni \bar{x}, D_j^\circ)$ is $\mathbb{R}$-complementary. By Lemma 3.3(1), $D_j^\circ \in \Lambda_1$. By **Step 1**, $(W^\circ/X^\circ \ni \bar{x}, D_j^\circ)$ has a monotonic n-complement for any $\bar{x}$ over x and any j.

Step 4. In this step we construct complements of $D_1, \dots, D_r$ over k.

By **Step 3** and Lemma 7.3 applied to h and $D_0 := D_j$ for $j = 1, \dots, r$, possibly shrinking X around x finitely many times, there exist $\mathbb{Q}$-divisors $\Theta_j \geq D_j$ on W such that (W, Θ_j) is lc, $n\Theta_j$ is a Weil divisor, and $n(K_W + \Theta_j) = \operatorname{div}_W(\phi_j)$ for some $\phi_j \in k(W)^* = k(X)^*$. Since $\operatorname{mult}_{S_W} D_j = 1$ and (W, Θ_j) is lc, $\operatorname{mult}_{S_W} \Theta_j = 1$.

Step 5. In this step we push $\Theta_1, \dots, \Theta_r$ down to X and conclude the proof.

Let $K_X := h_* K_W$ and $\Delta_j := h_* \Theta_j$. Then

$$n(K_X + \Delta_j) = h_* \operatorname{div}_W(\phi_j) = \operatorname{div}_X(\phi_j),$$

so $h^*(K_X + \Delta_j) = \frac{1}{n} \operatorname{div}_W(\phi_j) = K_W + \Theta_j$. Thus $a(E, X, \Delta_j) = a(E, W, \Theta_j)$ for any prime divisor E over X, so (X, Δ_j) is lc, and $a(S, X, \Delta_j) = 1 - \operatorname{mult}_{S_W} \Theta_j = 0$. This implies (1). Moreover, $\Delta_j \geq 0$, $n\Delta_j$ is a Weil divisor, and the coefficients of Δ_j are at most 1, so $\Delta_j \in \{0, \frac{1}{n}, \dots, 1\}$.

Let $\Delta := \sum_j \lambda_j \Delta_j$. By (7.3),

$$\Delta = h_* \sum_j \lambda_j \Theta_j \geq h_* \sum_j \lambda_j D_j = h_* D' \geq h_*(B_W + S_W) = B.$$

Since $\sum_j \lambda_j = 1$,

$$K_X + \Delta = \sum_j \lambda_j (K_X + \Delta_j) = \frac{1}{n} \sum_j \lambda_j \operatorname{div}_X(\phi_j) \sim_{\mathbb{R}} 0$$

and $a(E,X,\Delta)=\sum_j \lambda_j a(E,X,\Delta_j)$ for any prime divisor E over X. Hence (X,Δ) is lc and $a(S,X,\Delta)=0$. Since $\lambda_j\in\mathcal{A}$, $\sum_j\lambda_j=1$, $r\le\lfloor 1/\min\mathcal{A}\rfloor$, and $\Delta_j\in\{0,\frac{1}{n},\dots,1\}$, we have $\Delta\in J$. This implies (2). Finally, $\Delta-B=(K_X+\Delta)-(K_X+B)$ is $\mathbb{R}$-Cartier, and since $\operatorname{center}_X S=x$ and S is an lc place of (X,Δ), x is an lc center of (X,Δ). This implies (3). □

Corollary 7.5. *Let $d\ge 2$ be an integer and $I\subset[0,1]$ a DCC set. Then there exists a finite set $J=J(d,I)\subset[0,1]$ depending only on d and I satisfying the following. Let k be a field finitely generated over $\mathbb{C}$ and $(X\ni x,B)$ a klt pair over k of dimension d such that x is a closed point and $B\in I$. Then, possibly shrinking X around x, there exists an $\mathbb{R}$-divisor $\Delta\ge B$ on X such that $\Delta\in J$, (X,Δ) is lc, and x is an lc center of (X,Δ).*

Proof. Theorem 7.4 provides a finite set $J\subset[0,1]$ depending only on d and I, and we let $J(d,I):=J$. Possibly shrinking X around x, we may assume that X is affine. By Corollary 6.7, $(X\ni x,B)$ has a Kollár component S over k. By Theorem 7.4(2) and (3), possibly shrinking X around x, there exists an $\mathbb{R}$-divisor $\Delta\ge B$ on X such that $\Delta\in J(d,I)$, (X,Δ) is lc, and x is an lc center of (X,Δ). □

8. Torus quasi-torsors

In this section, for a klt germ $(X\ni x,B)$ and an $\mathbb{R}$-divisor $\Delta\ge B$ on X such that (X,Δ) is lc and x is an lc center of (X,Δ), we construct a tower of rank one torus quasi-torsors over X on whose top layer K_X and the sums of the components of the equality part B^{eq} of B with respect to Δ (Definition-Lemma 4.4) with a given coefficient become $\mathbb{Q}$-Cartier near a closed point over x, and we show that the Kollár components over the quasi-torsors which are lc places of the pullback of Δ descend to Kollár components of $(X\ni x,B)$ (Construction-Theorem 8.5). As explained in the introduction, we use rank one torus quasi-torsors instead of a small $\mathbb{Q}$-factorial modification of X.

8.1. **Invariant valuations.** In this subsection, we show that the log discrepancies of $\mathbb{G}_m$-invariant prime divisors over a rank one torus quasi-torsor are positive integer multiples of log discrepancies over the base, and that $\mathbb{G}_m$-invariant Kollár components descend to the base.

Definition 8.1. Let k be a field of characteristic zero and $X=\operatorname{Spec}R$ a normal affine variety over k. A *rank one torus quasi-torsor* over X is a morphism $p:Y=\operatorname{Spec}S\to X$ over k, where S is a $\mathbb{Z}$-graded k-algebra of finite type which is a normal domain with $S_0=R$, such that for the induced action of $\mathbb{G}_m$ on Y, there exists an open subset $U\subset X$ with $\operatorname{codim}_X(X\backslash U)\ge 2$ and $\operatorname{codim}_Y(Y\backslash p^{-1}(U))\ge 2$ such that $p^{-1}(U)\to U$ is a $\mathbb{G}_m$-torsor. For an $\mathbb{R}$-divisor B on X, we let B_Y be the closure in Y of the pullback of $B|_U$ to $p^{-1}(U)$. A point $y\in Y$ is *$\mathbb{G}_m$-fixed* if the prime ideal of S corresponding to y contains S_m for any $m\ne 0$. A valuation w on Y is *$\mathbb{G}_m$-invariant* if $\{f\in S\mid w(f)\ge t\}$ is a homogeneous ideal of S for any real number $t>0$, and a prime divisor F over Y is *$\mathbb{G}_m$-invariant* if ord_F is $\mathbb{G}_m$-invariant.

In Definition 8.1, any prime divisor on X intersects U and any prime divisor on Y intersects $p^{-1}(U)$, so pullbacks of divisors and of canonical divisors are determined on the $\mathbb{G}_m$-torsor $p^{-1}(U)\to U$. This is used in the proofs of Proposition 8.2(1) and (2).

Proposition 8.2. *Let k be a field finitely generated over $\mathbb{C}$, X a normal affine variety over k, $p:Y\to X$ a rank one torus quasi-torsor over X, and $B\ge 0$ an $\mathbb{R}$-divisor on X such that K_X+B is $\mathbb{R}$-Cartier. Then the following hold.*

1. *For any canonical divisor K_X of X, $(K_X)_Y$ is a canonical divisor of Y, and for any prime divisor P on X, P_Y is a prime divisor on Y, so that $B_Y=\sum_P(\operatorname{mult}_P B)P_Y$. Moreover, K_Y+B_Y is $\mathbb{R}$-Cartier, and (Y,B_Y) is klt (resp. lc) if and only if (X,B) is klt (resp. lc).*
2. *For any $\mathbb{G}_m$-invariant prime divisor F over Y with a center on Y, the restriction of ord_F to $k(X)$ is $c\operatorname{ord}_E$ for a positive integer c and a prime divisor E over X, and $a(F,Y,B_Y)=ca(E,X,B)$.*

(3) *If (X, B) is lc, then any lc place of (Y, B_Y) is $\mathbb{G}_m$-invariant.*
(4) *Let $x \in X$ be a closed point and $y \in Y$ a $\mathbb{G}_m$-fixed closed point with $p(y) = x$, and assume that (X, B) is klt. Let F be a $\mathbb{G}_m$-invariant Kollár component of $(Y \ni y, B_Y)$, and let c be the positive integer and E the prime divisor over X such that $\operatorname{ord}_F|_{k(X)} = c \operatorname{ord}_E$ (see (2)). Then E is a Kollár component of $(X \ni x, B)$ and $a(E, X, B) = a(F, Y, B_Y)/c$.*
(5) *Let $x \in X$ be a closed point and $y \in Y$ a $\mathbb{G}_m$-fixed closed point with $p(y) = x$, assume that (X, B) is klt, and let $\Delta \geq B$ be an $\mathbb{R}$-divisor on X such that (X, Δ) is lc and x is an lc center of (X, Δ). Then (Y, Δ_Y) is lc, y is an lc center of (Y, Δ_Y), and any Kollár component F of $(Y \ni y, B_Y)$ with $a(F, Y, \Delta_Y) = 0$ is $\mathbb{G}_m$-invariant. Moreover, for such F, the prime divisor E over X with $\operatorname{ord}_F|_{k(X)} = c \operatorname{ord}_E$ for a positive integer c satisfies $a(E, X, \Delta) = 0$.*

Proof. The proof is long, so we divide it into several steps. Write $X = \operatorname{Spec} R$ and $Y = \operatorname{Spec} S$ as in Definition 8.1, and write $\psi = \sum_m \psi_m$ for the decomposition of $\psi \in S$ into homogeneous components $\psi_m \in S_m$. For $\lambda \in k^*$, the action of $\lambda \in \mathbb{G}_m(k)$ on Y is given by the automorphism λ^* of S with $\lambda^*\psi = \sum_m \lambda^m \psi_m$.

Step 1. In this step we fix coordinates on Y and prove the first two sentences of (1).

Since $p^{-1}(U) \to U$ is a $\mathbb{G}_m$-torsor and $\mathbb{G}_m$-torsors are Zariski locally trivial, there exists a non-empty open subset $V_0 \subset U$ such that $p^{-1}(V_0) \cong V_0 \times \mathbb{G}_m$ $\mathbb{G}_m$-equivariantly over V_0. Thus the degree one part $(S \otimes_R k(X))_1$ is a one-dimensional $k(X)$-vector space, and for a basis u of it, $S \otimes_R k(X) = k(X)[u, u^{-1}]$. In particular, $k(Y) = k(X)(u)$, $S \subset k(X)[u, u^{-1}]$, $S_m \subset k(X)u^m$ for any $m \in \mathbb{Z}$, and $\lambda^* u = \lambda u$ for any $\lambda \in k^*$.

Let ω be a non-zero rational differential form of top degree on X with $K_X = \operatorname{div}_X(\omega)$, and let $\Omega := p^*\omega \wedge du/u$. Since $\mathbb{G}_m$-torsors are Zariski locally trivial, $p^{-1}(U)$ is covered by open subsets $p^{-1}(V) \cong V \times \mathbb{G}_m$ with a fiber coordinate τ of degree one. On such $p^{-1}(V)$, $u = \alpha\tau$ for some $\alpha \in k(X)^*$, so $\Omega = p^*\omega \wedge d\tau/\tau$ and $\operatorname{div}(\Omega) = p^* \operatorname{div}_V(\omega)$. Since $\operatorname{codim}_Y(Y \backslash p^{-1}(U)) \geq 2$, $\operatorname{div}_Y(\Omega) = (K_X)_Y$, so $(K_X)_Y$ is a canonical divisor of Y. We take $K_Y := \operatorname{div}_Y(\Omega)$. For a prime divisor P on X, $P \cap U \neq \emptyset$ since $\operatorname{codim}_X(X \backslash U) \geq 2$, $p^{-1}(P \cap U)$ is integral as a $\mathbb{G}_m$-torsor over the integral scheme $P \cap U$, and p^*P is reduced since p is smooth over U. Thus P_Y is a prime divisor and $B_Y = \sum_P (\operatorname{mult}_P B) P_Y$.

Let $V \subset X$ be an open subset on which $K_X + B = \sum_j b_j \operatorname{div}_V(h_j)$ for finitely many $b_j \in \mathbb{R}$ and $h_j \in k(X)^*$. Then

$$K_Y + B_Y = \sum_j b_j \operatorname{div}_{p^{-1}(V)}(h_j) \quad \text{on } p^{-1}(V), \tag{8.1}$$

since both sides are the pullbacks of $K_X + B$ on $p^{-1}(V \cap U)$ and the complement of $p^{-1}(V \cap U)$ in $p^{-1}(V)$ has codimension at least two. Hence $K_Y + B_Y$ is $\mathbb{R}$-Cartier. By Definition 2.1,

$$\begin{aligned} a(D, X, B) &= 1 + \operatorname{ord}_D(\omega) - \sum_j b_j \operatorname{ord}_D(h_j), \\ a(D', Y, B_Y) &= 1 + \operatorname{ord}_{D'}(\Omega) - \sum_j b_j \operatorname{ord}_{D'}(h_j) \end{aligned} \tag{8.2}$$

for any prime divisor D over X with $\operatorname{center}_X D \in V$ and any prime divisor D' over Y with $\operatorname{center}_Y D' \in p^{-1}(V)$, where $\operatorname{ord}_D(\omega)$ is the order of ω along D and $\operatorname{ord}_{D'}(\Omega)$ is the order of Ω along D'.

Step 2. In this step we show that $\mathbb{G}_m$-invariant valuations are Gauss extensions, and we prove a criterion for $\mathbb{G}_m$-invariance.

Let w be a $\mathbb{G}_m$-invariant valuation on Y and $\theta := w(u)$. We show that

$$w\Big(\sum_m f_m u^m\Big) = \min_m \big(w(f_m) + m\theta\big) \quad \text{for any } \sum_m f_m u^m \in k(X)[u, u^{-1}], \tag{8.3}$$

that is, $w = (w|_{k(X)})_\theta$ in the notation of Definition 2.8. Let $\psi \in S$ and $t := w(\psi)$. Since w has a center on Y, we have $t \geq 0$ and $w(\psi_m) \geq 0$ for any m. If $t > 0$, then $w(\psi_m) \geq t$ for any m since $\{f \in S \mid w(f) \geq t\}$ is homogeneous. Thus $\min_m w(\psi_m) \geq w(\psi) \geq \min_m w(\psi_m)$ in both cases. For $\sum_m f_m u^m \in k(X)[u, u^{-1}]$, there exists $0 \neq \beta \in R$ with $\beta f_m u^m \in S$ for any m, and applying the previous equality to $\psi = \sum_m \beta f_m u^m$ implies (8.3).

Conversely, let w be a valuation on Y such that $w(\lambda^*\psi) = w(\psi)$ for any $\psi \in S$ and $\lambda \in k^*$. We show that w is $\mathbb{G}_m$-invariant. Let $t > 0$, $\psi \in S$ with $w(\psi) \geq t$, and M the set of m with $\psi_m \neq 0$. Then $(2^i)^*\psi = \sum_{m\in M}(2^m)^i\psi_m$ satisfies $w((2^i)^*\psi) \geq t$ for $0 \leq i < |M|$. The matrix $((2^m)^i)_{0\leq i<|M|,m\in M}$ is a Vandermonde matrix in the distinct elements 2^m of k, so any ψ_m is a k-linear combination of the $(2^i)^*\psi$ and $w(\psi_m) \geq t$.

Step 3. In this step we prove (2).

Let F be a $\mathbb{G}_m$-invariant prime divisor over Y, $w := \operatorname{ord}_F$, and $\theta := w(u) \in \mathbb{Z}$. If w is trivial on $k(X)$, then $w \geq 0$ on $S \otimes_R k(X) = k(X)[u, u^{-1}]$, so $w(u) \geq 0$ and $w(u^{-1}) \geq 0$, hence $\theta = 0$ and w is trivial by (8.3), a contradiction. Thus the value group of $w|_{k(X)}$ is $c\mathbb{Z}$ for a positive integer c. The residue field of w has transcendence degree $\dim Y - 1 = \dim X$ over k. By [Bou89, Chapter VI, §10, no. 3, Corollary 1] applied to $k(X)(u)/k(X)$, the residue field of $w|_{k(X)}$ has transcendence degree at least $\dim X - 1$ over k, and by [Abh56, Theorem 1] it has transcendence degree at most $\dim X - 1$. Since $w \geq 0$ on R, [KM98, Lemma 2.45] implies that $c^{-1}w|_{k(X)} = \operatorname{ord}_E$ for a prime divisor E over X. Moreover, p maps $\operatorname{center}_Y F$ to $\operatorname{center}_X E$.

Let V be as in **Step 1** with $\operatorname{center}_X E \in V$, and $d := \dim X$. Let $X'' \to X$ be a projective birational morphism from a normal variety X'' on which E is a prime divisor. Near the generic point of E, X'' is smooth, and we choose a local equation z of E and regular functions $x_2, \dots, x_d$ such that $dz \wedge dx_2 \wedge \cdots \wedge dx_d$ generates the differential forms of top degree. Then $\omega = ez^\ell dz \wedge dx_2 \wedge \cdots \wedge dx_d$, where $\ell := \operatorname{ord}_E(\omega)$ and e is a unit at the generic point of E. By (8.3), $\operatorname{ord}_F = (c \operatorname{ord}_E)_\theta$, and the value group $c\mathbb{Z} + \theta\mathbb{Z}$ of ord_F is $\mathbb{Z}$. Thus ord_F is the monomial valuation in z, u with weights c, θ at the generic point of $E \times \{0\} \subset X'' \times \mathbb{P}^1$ if $\theta \geq 0$, and the monomial valuation in z, u^{-1} with weights $c, -\theta$ at the generic point of $E \times \{\infty\}$ if $\theta < 0$. If $\theta \geq 0$, then on a smooth open neighborhood of the generic point of $E \times \{0\}$ in $X'' \times \mathbb{P}^1$ on which $dz \wedge dx_2 \wedge \cdots \wedge dx_d \wedge du$ generates the differential forms of top degree, the log discrepancy of the monomial valuation ord_F in z, u with weights c, θ is $c + \theta$ by [JM12, Proposition 5.1] (as in **Step 6**), so $\operatorname{ord}_F(dz \wedge dx_2 \wedge \cdots \wedge dx_d \wedge du) = c + \theta - 1$. Since $\operatorname{ord}_F(u) = \theta$,

$$\operatorname{ord}_F(dz \wedge dx_2 \wedge \cdots \wedge dx_d \wedge du/u) = c - 1.$$

If $\theta < 0$, the equality $\operatorname{ord}_F(dz \wedge dx_2 \wedge \cdots \wedge dx_d \wedge du/u) = c - 1$ follows from the argument above with u^{-1} and weights $c, -\theta$ in place of u and c, θ, since $du/u = -d(u^{-1})/u^{-1}$. Since $\operatorname{ord}_F(e) = c \operatorname{ord}_E(e) = 0$ and $\operatorname{ord}_F(z) = c$, $\operatorname{ord}_F(\Omega) = c\ell + c - 1$. Hence by (8.2),

$$a(F, Y, B_Y) = c\ell + c - \sum_j b_j c \operatorname{ord}_E(h_j) = c\Big(1 + \ell - \sum_j b_j \operatorname{ord}_E(h_j)\Big) = ca(E, X, B).$$

Step 4. In this step we lift prime divisors from the base to $\mathbb{G}_m$-invariant prime divisors.

Let $S' \subset k(X)[u, u^{-1}]$ be a k-subalgebra of finite type with $\operatorname{Frac}(S') = k(Y)$ which is generated by elements of the form fu^m with $f \in k(X)$ and $m \in \mathbb{Z}$, and let $R' := S' \cap k(X)$. Assume that R' is a normal k-algebra of finite type with $\operatorname{Frac}(R') = k(X)$. We regard S' as a $\mathbb{Z}$-graded algebra with $S'_m := S' \cap k(X)u^m$, and define $\mathbb{G}_m$-fixed points and $\mathbb{G}_m$-invariant valuations on $\operatorname{Spec} S'$ as in Definition 8.1. Let P be a prime divisor over $\operatorname{Spec} R'$ and $v := \operatorname{ord}_P$. We show the following.

(i) There exist a positive integer c' and a $\mathbb{G}_m$-invariant prime divisor P' over $\operatorname{Spec} S'$ such that $\operatorname{ord}_{P'}|_{k(X)} = c' \operatorname{ord}_P$.

(ii) If $y' \in \operatorname{Spec} S'$ is a $\mathbb{G}_m$-fixed closed point whose image $x' \in \operatorname{Spec} R'$ is $\operatorname{center}_{\operatorname{Spec} R'} P$, then P' in (i) can be taken with $\operatorname{center}_{\operatorname{Spec} S'} P' = y'$.

Let $f_1u^{m_1},\dots,f_nu^{m_n}$ with $f_i\in k(X)$ and $m_i\neq 0$ be generators of the R'-algebra S'. For $m_i>0>m_j$, the element $(f_iu^{m_i})^{-m_j}(f_ju^{m_j})^{m_i}=f_i^{-m_j}f_j^{m_i}$ belongs to R', on which $v\geq 0$, so

$$-\frac{v(f_i)}{m_i}\leq\frac{v(f_j)}{-m_j}. \tag{8.4}$$

Thus there exists $\theta\in\mathbb{Q}$ with $v(f_i)+m_i\theta\geq 0$ for any i, so $v_\theta\geq 0$ on S' by Definition 2.8. Write $\theta=a/b$ with coprime integers a and $b>0$, and let $\pi\in k(X)$ with $v(\pi)=1$. Then the value group of v_θ is $b^{-1}\mathbb{Z}$. The element $\pi^{-a}u^b$ satisfies $v_\theta(\sum_i g_i(\pi^{-a}u^b)^i)=\min_i v(g_i)$ for $g_i\in k(X)$, so the residue of $\pi^{-a}u^b$ is transcendental over the residue field of v, and the residue field of v_θ has transcendence degree at least $\dim X=\dim\operatorname{Spec}S'-1$ over k, and at most $\dim X$ by [Abh56, Theorem 1], as in **Step 3**. By [KM98, Lemma 2.45], $bv_\theta=\operatorname{ord}_{P'}$ for a prime divisor P' over $\operatorname{Spec}S'$, and $\operatorname{ord}_{P'}|_{k(X)}=bv$. By Definition 2.8, $v_\theta(\psi)=\min_m v_\theta(\psi_m)$ for $\psi\in S'$, so the valuation ideals of $\operatorname{ord}_{P'}$ in S' are homogeneous. This proves (i) with $c':=b$.

For (ii), let $\mathfrak{m}_{y'}$ and $\mathfrak{m}_{x'}$ be the maximal ideals of y' and x'. Since $\mathfrak{m}_{y'}\supset S'_m$ for $m\neq 0$ and $\mathfrak{m}_{y'}\cap R'=\mathfrak{m}_{x'}$, we have

$$\mathfrak{m}_{y'}=\mathfrak{m}_{x'}\oplus\bigoplus_{m\neq 0}S'_m=\mathfrak{m}_{x'}S'+\sum_i f_iu^{m_i}S'.$$

The elements $f_i^{-m_j}f_j^{m_i}$ with $m_i>0>m_j$ belong to $\mathfrak{m}_{y'}\cap R'=\mathfrak{m}_{x'}$, and $v>0$ on $\mathfrak{m}_{x'}$, so the inequalities (8.4) are strict. Thus we may take $\theta\in\mathbb{Q}$ with $v(f_i)+m_i\theta>0$ for any i. Then $v_\theta>0$ on $\mathfrak{m}_{y'}$ and $v_\theta\geq 0$ on S', so $\operatorname{center}_{\operatorname{Spec}S'}P'=y'$.

Step 5. In this step we prove the last sentence of (1).

Assume that (X,B) is lc (resp. klt). Let $h:W\to Y$ be a $\mathbb{G}_m$-equivariant log resolution of (Y,B_Y) [BM08, Theorems 1.1 and 1.3], where the $\mathbb{G}_m$-equivariance follows from the functoriality with respect to the smooth morphisms $\mathbb{G}_m\times Y\to Y$ given by the action and the projection. The set of prime divisors in $\operatorname{Exc}(h)\cup\operatorname{Supp}h_*^{-1}B_Y$ is finite and $\mathbb{G}_m$-stable, as any P_Y is $\mathbb{G}_m$-stable. Since $\mathbb{G}_m$ is connected, $\mathbb{G}_m$ maps any such prime divisor D to itself, so $\operatorname{ord}_D(\lambda^*\psi)=\operatorname{ord}_D(\psi)$ for $\psi\in S$ and $\lambda\in k^*$, and D is $\mathbb{G}_m$-invariant by **Step 2**. By (2), $a(D,Y,B_Y)=c_Da(E_D,X,B)\geq 0$ (resp. >0) for a positive integer c_D and a prime divisor E_D over X. Hence (Y,B_Y) is lc (resp. klt) by [KM98, Lemma 2.30 and Corollary 2.31(3)].

Conversely, assume that (Y,B_Y) is lc (resp. klt), and let E be a prime divisor over X. By **Step 4**(i) with $S'=S$, there exist a positive integer c and a $\mathbb{G}_m$-invariant prime divisor F over Y with $\operatorname{ord}_F|_{k(X)}=c\operatorname{ord}_E$. By (2), $a(E,X,B)=a(F,Y,B_Y)/c\geq 0$ (resp. >0). Thus (X,B) is lc (resp. klt).

Step 6. In this step we prove (3).

Assume that (X,B) is lc, so (Y,B_Y) is lc by (1). Let F be an lc place of (Y,B_Y), and $h:W\to Y$ as in **Step 5**. Let $D_1,\dots,D_N$ be the prime divisors in $\operatorname{Exc}(h)\cup\operatorname{Supp}h_*^{-1}B_Y$, so $D:=\sum_i D_i$ is a reduced snc divisor. Write $K_W+\Gamma=h^*(K_Y+B_Y)$ and $\Gamma=\sum_i\gamma_iD_i$. Since (Y,B_Y) is lc, $\gamma_i\leq 1$ for any i. Let $w:=\operatorname{ord}_F$ and let $r_{W,D}$ be the retraction onto the valuations which are quasi-monomial along (W,D). We take D rather than $\operatorname{Supp}\Gamma$, since the D_i with $\gamma_i=0$ are not contained in $\operatorname{Supp}\Gamma$. Since W is a smooth variety over a field of characteristic zero, it is a separated, regular, connected, and excellent scheme over $\mathbb{Q}$, which is the setting of [JM12] (see [JM12, Section 1.2]). By [JM12, Corollary 5.4], $a(r_{W,D}(w),W,0)\leq a(w,W,0)$, with equality if and only if w is quasi-monomial along (W,D). By [JM12, Proposition 5.1], a valuation v which is quasi-monomial along (W,D) satisfies $a(v,W,0)=\sum_i v(D_i)$, so $a(v,W,\Gamma)=\sum_i(1-\gamma_i)v(D_i)$ since $v(\Gamma)=\sum_i\gamma_iv(D_i)$. Taking $v=r_{W,D}(w)$ and using $r_{W,D}(w)(D_i)=w(D_i)$ for any i, we have

$$0=a(w,W,\Gamma)\geq a(r_{W,D}(w),W,\Gamma)=\sum_i(1-\gamma_i)w(D_i)\geq 0.$$

Thus $w=r_{W,D}(w)$ and $w(D_i)=0$ whenever $\gamma_i<1$. So w is the monomial valuation in the local equations of the D_i with $w(D_i)>0$, all of which satisfy $\gamma_i=1$, at the generic point η

of an irreducible component of $\bigcap_{w(D_i)>0} D_i$. Since $\mathbb{G}_m$ is connected, it maps any D_i and η to themselves, so for $\lambda \in k^*$, λ^* multiplies the local equations of the D_i at η by units. A monomial valuation in a regular system of parameters does not change when the parameters are multiplied by units, so $\mathrm{ord}_F(\lambda^*\psi) = \mathrm{ord}_F(\psi)$ for $\psi \in S$ and $\lambda \in k^*$. By **Step 2**, F is $\mathbb{G}_m$-invariant.

Step 7. In this step we prove (4).

By (1), (Y, B_Y) is klt. By (2), $a(E, X, B) = a(F, Y, B_Y)/c$, and $\mathrm{center}_X E = p(y) = x$ by **Step 3**. It remains to show that E is a Kollár component of $(X \ni x, B)$.

Let Y° be an open neighborhood of y over which F is primitive with prime extraction $Y'^\circ \to Y^\circ$ such that $(Y'^\circ, B_{Y'^\circ} + F)$ is plt near F, where $B_{Y'^\circ}$ is the strict transform of $B_Y|_{Y^\circ}$ (Definition 2.3), and let r be a positive integer such that rF is Cartier on Y'°. Since $\mathrm{center}_Y F = y$ is a closed point, the ideals $\mathfrak{a}_j(\mathrm{ord}_F)$ are $\mathfrak{m}_y$-primary for $j > 0$. Thus the S-algebra $\bigoplus_{j\geq 0} \mathfrak{a}_{rj}(\mathrm{ord}_F)$ is finitely generated, since it is so over Y° by Lemma 2.7 and equal to $\bigoplus_{j\geq 0} \mathcal{O}_Y$ over $Y\backslash\{y\}$. By Lemma 2.7, the morphism

$$h_Y : Y' := \mathrm{Proj}_Y \bigoplus_{j\geq 0} \mathfrak{a}_{rj}(\mathrm{ord}_F) \to Y$$

is $Y'^\circ \to Y^\circ$ over Y° and an isomorphism over $Y\backslash\{y\}$. Hence F is primitive over Y with prime extraction h_Y, and $(Y', B_{Y'} + F)$ is plt near F, where $B_{Y'} := (h_Y)_*^{-1} B_Y$.

Let $A_Y := \bigoplus_{j\geq 0} \mathfrak{a}_{crj}(\mathrm{ord}_F)$, which is a finitely generated R-algebra graded by j and, since F is $\mathbb{G}_m$-invariant, by the weight of $\mathbb{G}_m$. Since $\mathrm{ord}_F(f) = c\,\mathrm{ord}_E(f)$ for $f \in R$, $\mathfrak{a}_{crj}(\mathrm{ord}_F) \cap R = \mathfrak{a}_{rj}(\mathrm{ord}_E)$, so the weight-zero part of A_Y is $A_X := \bigoplus_{j\geq 0} \mathfrak{a}_{rj}(\mathrm{ord}_E)$. Let $\psi_1, \dots, \psi_n$ be generators of the R-algebra A_Y which are homogeneous for both gradings, with weights $m_1, \dots, m_n$. Then A_X is spanned over R by the monomials $\psi_1^{\beta_1} \cdots \psi_n^{\beta_n}$ with $\beta \in \mathbb{N}^n$ and $\sum_i \beta_i m_i = 0$, and the set of such β is a finitely generated monoid by Gordan's lemma [Ful93, Section 1.2, Proposition 1]. Thus A_X is a finitely generated R-algebra.

The algebra $\bigoplus_{j\in\mathbb{Z}} \mathfrak{a}_j(\mathrm{ord}_E)s^{-j}$, where $\mathfrak{a}_j(\mathrm{ord}_E) = R$ for $j \leq 0$, is the intersection of $R[s, s^{-1}]$ with the valuation ring of the Gauss extension of ord_E with weight 1 on s (Definition 2.8), hence normal, and

$$\Big(\bigoplus_{j\in\mathbb{Z}} \mathfrak{a}_j(\mathrm{ord}_E)s^{-j}\Big)\Big/(s) = \bigoplus_{j\geq 0} \mathfrak{a}_j(\mathrm{ord}_E)/\mathfrak{a}_{j+1}(\mathrm{ord}_E),$$

which is $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x}$ of Definition 2.9 since the ideals $\mathfrak{a}_j(\mathrm{ord}_E)$ are $\mathfrak{m}_x$-primary for $j > 0$, and which is a domain since ord_E is a valuation. Let $\mathcal{R} := \bigoplus_{j\in\mathbb{Z}} \mathfrak{a}_j(\mathrm{ord}_E)s^{-j}$. Since $\mathrm{Proj}_X A_X \cong \mathrm{Proj}_X \bigoplus_{j\geq 0} \mathfrak{a}_j(\mathrm{ord}_E)s^{-j}$ [Sta, Tag 0B5J], $X' := \mathrm{Proj}_X A_X$ is covered by the spectra of the degree zero parts of the rings $\mathcal{R}[(\varphi s^{-j})^{-1}]$ with $j > 0$ and $0 \neq \varphi \in \mathfrak{a}_j(\mathrm{ord}_E)$. The degree zero part of each $\mathcal{R}[(\varphi s^{-j})^{-1}]$ is normal, and on its spectrum the exceptional locus of $h_X : X' \to X$ is defined by the degree zero part of $s\mathcal{R}[(\varphi s^{-j})^{-1}]$, whose quotient is the degree zero part of the localization of $\bigoplus_{j\geq 0} \mathfrak{a}_j(\mathrm{ord}_E)/\mathfrak{a}_{j+1}(\mathrm{ord}_E)$ at the image of φs^{-j}, hence a domain if $\varphi \notin \mathfrak{a}_{j+1}(\mathrm{ord}_E)$ and zero otherwise, in which case the chart does not meet the exceptional locus. Here the exceptional locus is $h_X^{-1}(x)$, since h_X is an isomorphism over $X\backslash\{x\}$, see below, and $\dim X \geq 2$ as Y has a $\mathbb{G}_m$-fixed point. On the spectrum of the degree zero part of $\mathcal{R}[(\varphi s^{-j})^{-1}]$, the radical of the ideal of $h_X^{-1}(x)$ equals the radical of the degree zero part of $s\mathcal{R}[(\varphi s^{-j})^{-1}]$, since $\mathfrak{m}_x \subset \mathfrak{a}_1(\mathrm{ord}_E)$ and $\varphi \in \mathfrak{a}_j(\mathrm{ord}_E) \subset \mathfrak{m}_x$, the j-th power of any degree zero element of $s\mathcal{R}[(\varphi s^{-j})^{-1}]$ being φ times a degree zero element of $\mathcal{R}[(\varphi s^{-j})^{-1}]$. Thus X' is normal and the exceptional locus of h_X is one prime divisor, which induces ord_E and which we denote by E. Since $\mathfrak{a}_j(\mathrm{ord}_E)$ is $\mathfrak{m}_x$-primary for $j > 0$, h_X is an isomorphism over $X\backslash\{x\}$. Fix a positive integer j_0 such that $\bigoplus_{m\geq 0} \mathfrak{a}_{rj_0m}(\mathrm{ord}_E)$ is generated by $I := \mathfrak{a}_{rj_0}(\mathrm{ord}_E)$ over R [Sta, Tag 0EGH] and $\mathrm{ord}_E(\varphi_0) = rj_0$ for some $\varphi_0 \in R$. Such j_0 exists, since the first condition is preserved when j_0 is replaced by a multiple, and the values of ord_E on $R\backslash\{0\}$ form a semigroup generating $\mathbb{Z}$, hence contain rj_0 for any sufficiently large j_0. Then $X' \cong \mathrm{Proj}_X \bigoplus_{m\geq 0} I^m$ [Sta, Tag 0B5J] is the blow-up of I, so $I\mathcal{O}_{X'}$ is invertible [Har77, Proposition II.7.13(a)]. Since $I\mathcal{O}_{X'}$ is trivial over $X\backslash\{x\}$, its support

is E, and $\min\{\mathrm{ord}_E(\varphi) \mid \varphi \in I\} = rj_0$, we have $I\mathcal{O}_{X'} = \mathcal{O}_{X'}(-rj_0E)$. The natural surjection $h_X^* \bigoplus_{m\geq 0} I^m \to \bigoplus_{m\geq 0}(I\mathcal{O}_{X'})^{\otimes m}$ induces the identity of X', so $\mathcal{O}_{X'}(-rj_0E)$ is h_X-ample by [Sta, Tag 01VJ] ((4) implies (1)). Hence $-E$ is $\mathbb{Q}$-Cartier and h_X-ample, and E is primitive over X with prime extraction h_X. Moreover, A_Y is the c-th Veronese subalgebra of $\bigoplus_{j\geq 0}\mathfrak{a}_{rj}(\mathrm{ord}_F)$, so $Y' = \mathrm{Proj}_Y A_Y$ [Sta, Tag 0B5J], and $\mathcal{O}_{Y'}(j) = \mathcal{O}_{Y'}(-crjF)$ for any sufficiently large j by Lemma 2.7 applied to h_Y.

For a sufficiently large multiple j of j_0 and $0 \neq \varphi \in \mathfrak{a}_{rj}(\mathrm{ord}_E)$, regarded as an element of degree j of A_X and of A_Y, let $D_{X',\varphi} := \mathrm{Spec}(A_X)_{(\varphi)} \subset X'$ and $D_{Y',\varphi} := \mathrm{Spec}(A_Y)_{(\varphi)} \subset Y'$ be the open subsets where φ does not vanish as a section of $\mathcal{O}(j)$. Since φ has weight zero, $(A_X)_{(\varphi)}$ is the weight-zero part of $(A_Y)_{(\varphi)}$. The open subsets $D_{X',\varphi}$ cover X', and

$$\mathrm{div}_{X'}(\varphi)|_{D_{X',\varphi}} = rjE|_{D_{X',\varphi}}, \qquad \mathrm{div}_{Y'}(\varphi)|_{D_{Y',\varphi}} = crjF|_{D_{Y',\varphi}}. \tag{8.5}$$

Here the first equality holds since $\mathfrak{a}_{rj}(\mathrm{ord}_E) = I^{j/j_0}$ and φ generates $I^{j/j_0}\mathcal{O}_{X'} = \mathcal{O}_{X'}(-rjE)$ on $D_{X',\varphi}$. Moreover, $S' := (A_Y)_{(\varphi)}$ and $R' := (A_X)_{(\varphi)}$ satisfy the assumptions of **Step 4**. The algebra S' is of finite type over k since A_Y is a finitely generated R-algebra, and it is generated by the elements ψ/φ^l with $\psi \in A_Y$ of degree lj and homogeneous of weight m, each of which belongs to $k(X)u^m$ since $\psi \in S_m$ and $\varphi \in R$. We have $\mathrm{Frac}(S') = k(Y)$ since $D_{Y',\varphi}$ is a non-empty open subset of Y'. Finally, $S' \cap k(X)$ is the weight-zero part $(A_X)_{(\varphi)}$ of S', which is normal of finite type over k with fraction field $k(X)$ since $D_{X',\varphi}$ is a non-empty affine open subset of the normal variety X'.

Let $B_{X'}$ be the strict transform on X' of the sum of the components of B other than E, with their coefficients. Then

$$K_{X'}+B_{X'}+E = h_X^*(K_X+B)+a(E,X,B)E, \qquad K_{Y'}+B_{Y'}+F = h_Y^*(K_Y+B_Y)+a(F,Y,B_Y)F,$$

so $K_{X'} + B_{X'} + E$ is $\mathbb{R}$-Cartier, and the coefficients of $B_{X'}$ are less than one. Let P be a prime divisor over X' which is exceptional over X'. If $\mathrm{center}_{X'} P \notin E$, then $a(P, X', B_{X'} + E) = a(P, X, B) > 0$ since h_X is an isomorphism over $X\backslash\{x\}$. Assume that $\mathrm{center}_{X'} P \in E$, and choose φ with $\mathrm{center}_{X'} P \in D_{X',\varphi}$. By **Step 4**(i), there exist a positive integer c' and a $\mathbb{G}_m$-invariant prime divisor P' over Y with a center on $D_{Y',\varphi}$ such that $\mathrm{ord}_{P'}|_{k(X)} = c'\,\mathrm{ord}_P$. By (8.5),

$$c\,\mathrm{mult}_{P'} F = \frac{\mathrm{ord}_{P'}(\varphi)}{rj} = \frac{c'\,\mathrm{ord}_P(\varphi)}{rj} = c'\,\mathrm{mult}_P E > 0.$$

Thus $\mathrm{center}_{Y'} P' \in F$. Moreover, $P' \neq F$, since the center of $\mathrm{ord}_F|_{k(X)} = c\,\mathrm{ord}_E$ on X' is the generic point of E, while $\mathrm{center}_{X'} P$ has codimension at least two in X'. So P' is exceptional over Y'. By (2) applied to P' and to F,

$$\begin{aligned} a(P', Y', B_{Y'} + F) &= a(P', Y, B_Y) - a(F, Y, B_Y)\,\mathrm{mult}_{P'} F \\ &= c'a(P, X, B) - a(E, X, B)c\,\mathrm{mult}_{P'} F \\ &= c'\big(a(P, X, B) - a(E, X, B)\,\mathrm{mult}_P E\big) = c'a(P, X', B_{X'} + E). \end{aligned}$$

Since $(Y', B_{Y'} + F)$ is plt near F, $a(P', Y', B_{Y'} + F) > 0$, hence $a(P, X', B_{X'} + E) > 0$. Thus $(X', B_{X'} + E)$ is plt [KM98, Definition 2.34], and E is a Kollár component of $(X \ni x, B)$.

Step 8. In this step we prove (5).

The pair (X, Δ) satisfies the assumptions on (X, B) in (1), (2), and (3). By (1) applied to Δ, (Y, Δ_Y) is lc. Let E_0 be an lc place of (X, Δ) with $\mathrm{center}_X E_0 = x$. By **Step 4**(ii) with $S' = S$, there exist a positive integer c_0 and a $\mathbb{G}_m$-invariant prime divisor F_0 over Y with $\mathrm{center}_Y F_0 = y$ and $\mathrm{ord}_{F_0}|_{k(X)} = c_0\,\mathrm{ord}_{E_0}$. By (2) applied to Δ, $a(F_0, Y, \Delta_Y) = c_0a(E_0, X, \Delta) = 0$, so y is an lc center of (Y, Δ_Y). Let F be a Kollár component of $(Y \ni y, B_Y)$ with $a(F, Y, \Delta_Y) = 0$. Then F is an lc place of (Y, Δ_Y), so F is $\mathbb{G}_m$-invariant by (3) applied to Δ. By (2) applied to Δ, $a(E, X, \Delta) = a(F, Y, \Delta_Y)/c = 0$. □

8.2. **The tower.** In this subsection, we prove Construction-Proposition 8.4, which constructs one layer of the tower, and Construction-Theorem 8.5, which constructs the tower and descends Kollár components to the base.

In Construction-Proposition 8.4, if D is not $\mathbb{Q}$-Cartier at x, then there is a $\mathbb{G}_m$-fixed point y over x with $\kappa(y) = \kappa(x)$, and the pullback of D is the principal divisor $\operatorname{div}(t)$.

Lemma 8.3. *Let X be a normal variety over a field, $x \in X$ a point, and D a Weil divisor on X which is not Cartier at x. Then $\mathcal{O}_X(D)_x \cdot \mathcal{O}_X(-D)_x \subset \mathfrak{m}_x$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$.*

Proof. For $f \in \mathcal{O}_X(D)_x$ and $g \in \mathcal{O}_X(-D)_x$, $\operatorname{div}(fg) \geq 0$ near x, so $fg \in \mathcal{O}_{X,x}$ since X is normal. Suppose that $\mathcal{O}_X(D)_x \cdot \mathcal{O}_X(-D)_x \not\subset \mathfrak{m}_x$. Since $\mathcal{O}_{X,x}$ is local and the product is generated by the elements fg, there exist $f \in \mathcal{O}_X(D)_x$ and $g \in \mathcal{O}_X(-D)_x$ such that fg is a unit. For any $f' \in \mathcal{O}_X(D)_x$, $f' = f \cdot gf' \cdot (fg)^{-1}$ with $gf' \in \mathcal{O}_{X,x}$. Thus $\mathcal{O}_X(D)_x = f\mathcal{O}_{X,x}$ is free of rank one, and D is Cartier at x, a contradiction. □

Construction-Proposition 8.4. Let k be a field finitely generated over $\mathbb{C}$, and let X, B, x, and D be as follows.

(i) X is a geometrically integral normal affine variety over k, and $B \geq 0$ is an $\mathbb{R}$-divisor on X such that (X, B) is klt.
(ii) $x \in X$ is a closed point, and D is a Weil divisor on X.

Let $S_D := \bigoplus_{m \in \mathbb{Z}} \Gamma(X, \mathcal{O}_X(mD))t^m \subset k(X)[t, t^{-1}]$, where t is an indeterminate of degree one, and $Y := \operatorname{Spec} S_D$. Then the following hold.

(1) S_D is a k-algebra of finite type, Y is geometrically integral over k, and $p : Y \to X$ is a rank one torus quasi-torsor.
(2) If D is not $\mathbb{Q}$-Cartier at x, then there exists a unique $\mathbb{G}_m$-fixed point $y \in Y$ with $p(y) = x$; moreover $\kappa(y) = \kappa(x)$, $\operatorname{div}(t) = D_Y$, and for any Weil divisor D' on X which is $\mathbb{Q}$-Cartier at x, D'_Y is $\mathbb{Q}$-Cartier at y.

Proof. Let $R := \Gamma(X, \mathcal{O}_X)$, so that $X = \operatorname{Spec} R$, and let $U \subset X$ be the open subset on which D is Cartier. Since X is normal, $\operatorname{codim}_X(X \backslash U) \geq 2$. If $D = 0$, then $S_D = R[t, t^{-1}]$ and (1) holds with $U = X$, so for (1) we may assume that $D \neq 0$ (this is used below for [ADHL10, Chapter I, Construction 3.2.4 and Proposition 3.2.8]: the subgroup of Weil divisors generated by $D_{\bar{k}}$ is then free of rank one, so that its sheaf of divisorial algebras is $\mathbb{Z}$-graded as $S_D \otimes_k \bar{k}$).

We first show that $X_{\bar{k}}$ is of klt type. There exists a $\mathbb{Q}$-divisor $B' \geq 0$ on X with $\operatorname{Supp} B' = \operatorname{Supp} B$ such that (X, B') is klt. The perturbation takes place over k: write the coefficients of B as rational linear combinations of real numbers $1, r_1, \ldots, r_e$ which are linearly independent over $\mathbb{Q}$. $K_X + B'$ is $\mathbb{Q}$-Cartier for any $\mathbb{Q}$-divisor B' obtained by replacing $r_1, \ldots, r_e$ with rational numbers (cf. [HLS26, Lemma 5.4(1)]). Write $K_X + B = M_0 + \sum_{i=1}^{e} r_i M_i$ with $\mathbb{Q}$-divisors $M_0, \ldots, M_e$, and $K_X + B = \sum_q a_q H_q$ with $a_q \in \mathbb{R}$ and Cartier divisors H_q. Expanding each a_q in a basis of $\mathbb{R}$ over $\mathbb{Q}$ which begins with $1, r_1, \ldots, r_e$ and comparing the coefficients of each prime divisor, we see that each M_i is a $\mathbb{Q}$-linear combination of the H_q, hence $\mathbb{Q}$-Cartier, so $K_X + B'$, which is M_0 plus a rational linear combination of $M_1, \ldots, M_e$, is $\mathbb{Q}$-Cartier. Moreover, (X, B') is klt if and only if finitely many strict inequalities hold on a log resolution of $(X, \operatorname{Supp} B)$ [KM98, Lemma 2.30 and Corollary 2.31(3)]. Thus any such B' close enough to B is as desired. Since X is geometrically integral, $X_{\bar{k}}$ is a normal variety (Lemma 3.3(1)) over $\bar{k} \cong \mathbb{C}$ (Lemma 3.2), and $(X_{\bar{k}}, B'_{\bar{k}})$ is klt by Lemma 3.3(3). Hence $X_{\bar{k}}$ is of klt type.

Next we show that

$$S_D \otimes_k \bar{k} = \bigoplus_{m \in \mathbb{Z}} \Gamma(X_{\bar{k}}, \mathcal{O}_{X_{\bar{k}}}(mD_{\bar{k}}))t^m. \tag{8.6}$$

Fix $m \in \mathbb{Z}$. Since $\mathcal{O}_X(mD)$ is reflexive and double duals of coherent sheaves commute with the flat base change $X_{\bar{k}} \to X$ [Mat89, Theorem 7.11], $\mathcal{O}_X(mD) \otimes_k \bar{k}$ is a reflexive sheaf on $X_{\bar{k}}$. On an open subset of U on which $D = \operatorname{div}(a)$ for a rational function a, both $\mathcal{O}_X(mD) \otimes_k \bar{k}$ and

$\mathcal{O}_{X_{\bar{k}}}(mD_{\bar{k}})$ are equal to $a^{-m}\mathcal{O}_{X_{\bar{k}}}$. Since $R \to R\otimes_k \bar{k}$ is flat and integral, heights of prime ideals do not change under contraction [Mat89, Theorem 15.1(ii)], so $\operatorname{codim}_{X_{\bar{k}}}(X_{\bar{k}}\backslash U_{\bar{k}}) \geq 2$. Two reflexive subsheaves of the constant sheaf $k(X_{\bar{k}})$ on the normal variety $X_{\bar{k}}$ which coincide outside a closed subset of codimension at least two coincide [Sta, Tag 0EBJ]. Thus $\mathcal{O}_X(mD)\otimes_k \bar{k} = \mathcal{O}_{X_{\bar{k}}}(mD_{\bar{k}})$, and (8.6) follows by taking global sections on the affine variety $X_{\bar{k}}$.

We prove (1). By [Zhu24a, Lemma 2.5] applied to $D_{\bar{k}}$ and $-D_{\bar{k}}$ on $X_{\bar{k}}$, the sheaves of $\mathcal{O}_{X_{\bar{k}}}$-algebras $\bigoplus_{m\geq 0}\mathcal{O}_{X_{\bar{k}}}(mD_{\bar{k}})$ and $\bigoplus_{m\geq 0}\mathcal{O}_{X_{\bar{k}}}(-mD_{\bar{k}})$ are finitely generated. Since $X_{\bar{k}}$ is affine, the algebras of their global sections are finitely generated over $R\otimes_k \bar{k}$. The algebras of global sections are generated as follows: on a finite affine cover, the sheaves of algebras are generated in degrees at most some integer N, and global generators of the coherent modules of degree at most N generate the algebra of global sections, since equality of quasi-coherent subsheaves can be checked on the cover. The generators for $D_{\bar{k}}$ and for $-D_{\bar{k}}$ together generate the right-hand side of (8.6). Write each of the finitely many generators as a finite sum of elements $s\otimes\lambda$ with $s\in S_D$ and $\lambda\in\bar{k}$. Let $S'\subset S_D$ be the R-subalgebra generated by the finitely many elements s. Since $k\to\bar{k}$ is flat, $S'\otimes_k\bar{k}$ is a subring of $S_D\otimes_k\bar{k}$ which contains $R\otimes_k\bar{k}$ and the generators, so $S'\otimes_k\bar{k} = S_D\otimes_k\bar{k}$. Since $\bar{k}$ is a non-zero free k-module, $S' = S_D$. Since R is of finite type over k, S_D is a k-algebra of finite type. It is a $\mathbb{Z}$-graded domain with $(S_D)_0 = R$, since $S_D\subset k(X)[t,t^{-1}]$. Since X is geometrically integral, $X_{\bar{k}}$ is integral, so by (8.6), $S_D\otimes_k\bar{k}$ is a subring of $k(X_{\bar{k}})[t,t^{-1}]$, hence a domain. Thus Y is geometrically integral over k.

Let $V\subset U$ be an affine open subset on which $D = \operatorname{div}(a)$ for a rational function a. Since $\Gamma(V,\mathcal{O}_X(mD)) = a^{-m}\Gamma(V,\mathcal{O}_X)$ for any m, we have

$$p^{-1}(V) = \operatorname{Spec}\bigoplus_{m\in\mathbb{Z}}\Gamma(V,\mathcal{O}_X(mD))t^m = \operatorname{Spec}\Gamma(V,\mathcal{O}_X)\big[(a^{-1}t),(a^{-1}t)^{-1}\big],$$

where $a^{-1}t$ has degree one. Thus $p^{-1}(V)\to V$ is a trivial $\mathbb{G}_m$-torsor, and $p^{-1}(U)\to U$ is a $\mathbb{G}_m$-torsor (see also [ADHL10, Chapter I, Remark 3.2.7]). By (8.6), $Y_{\bar{k}} = \operatorname{Spec}(S_D\otimes_k\bar{k})$ is the relative spectrum over $X_{\bar{k}}$ of the sheaf of divisorial algebras associated with the subgroup of Weil divisors generated by $D_{\bar{k}}$ [ADHL10, Chapter I, Construction 3.2.4]. By [ADHL10, Chapter I, Proposition 3.2.8], $Y_{\bar{k}}$ is normal, and the preimage of $X_{\bar{k}}\backslash U_{\bar{k}}$ in $Y_{\bar{k}}$ has codimension at least two. Since $S_D\to S_D\otimes_k\bar{k}$ is faithfully flat, S_D is normal [Mat89, Theorem 23.9, Corollary (i)]. Since $S_D\to S_D\otimes_k\bar{k}$ is moreover integral, any prime ideal $\mathfrak{q}$ of S_D is the contraction of a prime ideal of $S_D\otimes_k\bar{k}$ of height $\operatorname{ht}\mathfrak{q}$ [Mat89, Theorem 15.1(ii)], so $\operatorname{codim}_Y(Y\backslash p^{-1}(U))\geq 2$. Thus p is a rank one torus quasi-torsor (Definition 8.1).

We prove (2). Assume that D is not $\mathbb{Q}$-Cartier at x, and let $\mathfrak{m}_x\subset\mathcal{O}_{X,x}$ be the maximal ideal. We claim that for any integer $m\neq 0$,

$$\mathcal{O}_X(mD)_x\cdot\mathcal{O}_X(-mD)_x\subset\mathfrak{m}_x. \tag{8.7}$$

Since D is not $\mathbb{Q}$-Cartier at x, mD is not Cartier at x for any integer $m\neq 0$, so (8.7) holds by Lemma 8.3.

Let $\mathfrak{p}_x := \mathfrak{m}_x\cap R$ and

$$\mathfrak{n} := \mathfrak{p}_x\oplus\bigoplus_{m\neq 0}(S_D)_m\subset S_D.$$

By (8.7), $(S_D)_m\cdot(S_D)_{-m}\subset\mathfrak{p}_x$ for any $m\neq 0$, so $\mathfrak{n}$ is an ideal of S_D, and $S_D/\mathfrak{n} = R/\mathfrak{p}_x = \kappa(x)$. Thus $\mathfrak{n}$ is a maximal ideal. Let $y\in Y$ be the corresponding point. Then y is $\mathbb{G}_m$-fixed, $p(y) = x$, and $\kappa(y) = \kappa(x)$. Since x is a closed point of a variety over k, $\kappa(x)$ is finite over k, so y is a closed point. If $y'\in Y$ is a $\mathbb{G}_m$-fixed point with $p(y') = x$, then the prime ideal of y' contains $\mathfrak{p}_x$ and $(S_D)_m$ for any $m\neq 0$, hence contains $\mathfrak{n}$, so $y' = y$ since $\mathfrak{n}$ is maximal.

Let $V\subset U$ be any affine open subset on which $D|_V = \operatorname{div}(a)|_V$ for a rational function a. Since D is Cartier on U, such V cover U. Since $p^{-1}(V) = \operatorname{Spec}\Gamma(V,\mathcal{O}_X)\big[(a^{-1}t),(a^{-1}t)^{-1}\big]$, $t = a\cdot(a^{-1}t)$ on $p^{-1}(V)$ with $a^{-1}t$ a unit, so $\operatorname{div}(t)|_{p^{-1}(V)}$ is the pullback of $D|_V$. Hence $\operatorname{div}(t)$ and D_Y coincide over $p^{-1}(U)$. Since $\operatorname{codim}_Y(Y\backslash p^{-1}(U))\geq 2$, $\operatorname{div}(t) = D_Y$. Finally, let D' be a

Weil divisor on X which is $\mathbb{Q}$-Cartier at x. There exist an integer $r > 0$, an open neighborhood V' of x, and a rational function g on X such that $rD'|_{V'} = \mathrm{div}(g)|_{V'}$. Then $\mathrm{div}(g)|_{p^{-1}(V')}$ and $rD'_Y|_{p^{-1}(V')}$ coincide over $p^{-1}(V' \cap U)$, whose complement in $p^{-1}(V')$ has codimension at least two. Thus $rD'_Y = \mathrm{div}(g)$ on the open neighborhood $p^{-1}(V')$ of y, and D'_Y is $\mathbb{Q}$-Cartier at y. □

The number of layers in Construction-Theorem 8.5 depends only on the number $|J\backslash\{0\}|$ of possible coefficients of the equality part, plus one layer for K_X, and not on the number of components of the boundary or on the rank of the class group, so the dimension increases by a bounded amount.

Construction-Theorem 8.5. Let k be a field finitely generated over $\mathbb{C}$ and $J \subset [0,1]$ a finite set, and let X, x, B, Δ, B^{eq}, and D_b be as follows.

(i) X is a geometrically integral normal variety over k of dimension $d \geq 2$, and $x \in X$ is a closed point.
(ii) $B \leq \Delta$ are $\mathbb{R}$-divisors on X such that (X, B) is klt, (X, Δ) is lc, x is an lc center of (X, Δ), and $\Delta \in J$.
(iii) $B^{\mathrm{eq}} := \sum_P (\mathrm{mult}_P B)P$, where P runs over the prime divisors on X with $\mathrm{mult}_P B = \mathrm{mult}_P \Delta > 0$; that is, B^{eq} is the equality part of B with respect to Δ (Definition-Lemma 4.4).
(iv) For $b \in J\backslash\{0\}$, D_b is the sum of the prime divisors P on X with $\mathrm{mult}_P B^{\mathrm{eq}} = b$.

Then, possibly shrinking X around x, there exist an integer $0 \leq l \leq |J\backslash\{0\}| + 1$, rank one torus quasi-torsors $Y_i \to Y_{i-1}$ over k for $1 \leq i \leq l$ with $Y_0 = X$, and closed points $y_i \in Y_i$ with $y_0 = x$ such that y_i is a $\mathbb{G}_m$-fixed point over y_{i-1} and $\kappa(y_i) = \kappa(x)$ for any i, satisfying the following, where B_i, Δ_i, and B_i^{eq} are the pullbacks of B, Δ, and B^{eq} to Y_i.

(1) For any i, Y_i is a geometrically integral normal variety over k of dimension $d + i$, (Y_i, B_i) is klt, (Y_i, Δ_i) is lc, y_i is an lc center of (Y_i, Δ_i), $B_i \leq \Delta_i$, and the coefficients of B_i, Δ_i, and B_i^{eq} belong to the sets of coefficients of B, Δ, and B^{eq}, respectively.
(2) K_{Y_l} is $\mathbb{Q}$-Cartier near y_l, the pullback of D_b to Y_l is $\mathbb{Q}$-Cartier near y_l for any b, and (Y_l, B_l^{eq}) is klt near y_l.
(3) After shrinking Y_l around y_l so that (Y_l, B_l^{eq}) is klt, for any Kollár component F of $(Y_l \ni y_l, B_l^{\mathrm{eq}})$ over k with $a(F, Y_l, \Delta_l) = 0$, there exists a Kollár component E of $(X \ni x, B)$ over k such that $a(E, X, \Delta) = 0$ and $a(E, X, B) \leq a(F, Y_l, B_l^{\mathrm{eq}})$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the divisors which become $\mathbb{Q}$-Cartier on the tower.

Possibly shrinking X around x, we may assume that X is affine. Let $h := |J\backslash\{0\}|$ and write $J\backslash\{0\} = \{b_1, \dots, b_h\}$. Let $G_0 := K_X$ be a canonical divisor of X, and let $G_j := D_{b_j}$ for $1 \leq j \leq h$. Since the coefficients of B^{eq} are coefficients of Δ and are positive, they belong to $J\backslash\{0\}$, so

$$B^{\mathrm{eq}} = \sum_{j=1}^{h} b_j G_j. \tag{8.8}$$

For a quasi-torsor $Y_i \to Y_{i-1}$ in the tower below and a Weil divisor G on X, the pullback of G to Y_i is defined inductively as the pullback to Y_i, in the sense of Definition 8.1, of the pullback of G to Y_{i-1}. We denote by $(G_j)_i$ the pullback of G_j to Y_i.

Step 2. In this step we construct one layer of the tower and show that it preserves the properties in (1).

Let $i \geq 1$, and let Y_{i-1} be an affine variety over k with a closed point y_{i-1} such that $\kappa(y_{i-1}) = \kappa(x)$ and such that Y_{i-1}, y_{i-1}, and the pullbacks B_{i-1}, Δ_{i-1}, and B_{i-1}^{eq} satisfy the properties in (1) with $i-1$ in place of i. Assume moreover that $K_{Y_{i-1}} + B_{i-1}$ and $K_{Y_{i-1}} + \Delta_{i-1}$ are $\mathbb{R}$-Cartier, and that B_{i-1}^{eq} is the equality part of B_{i-1} with respect to Δ_{i-1} (Definition-Lemma 4.4). Let D be a Weil divisor on Y_{i-1} which is not $\mathbb{Q}$-Cartier at y_{i-1}.

By Construction-Proposition 8.4 applied to Y_{i-1}, B_{i-1}, y_{i-1}, and D, the variety $Y_i := \operatorname{Spec} S_D$ with the projection $p_i : Y_i \to Y_{i-1}$ is a rank one torus quasi-torsor, and there exists a unique $\mathbb{G}_m$-fixed point $y_i \in Y_i$ with $p_i(y_i) = y_{i-1}$. Moreover $\kappa(y_i) = \kappa(y_{i-1}) = \kappa(x)$, the pullback of D to Y_i is $\operatorname{div}(t)$, and the pullback to Y_i of any Weil divisor on Y_{i-1} which is $\mathbb{Q}$-Cartier at y_{i-1} is $\mathbb{Q}$-Cartier at y_i. Since $\kappa(x)$ is finite over k, y_i is a closed point of Y_i.

The variety Y_i is normal and affine. By Construction-Proposition 8.4(1), Y_i is geometrically integral over k. The ring $S_D \subset k(Y_{i-1})[t, t^{-1}]$ contains $\Gamma(Y_{i-1}, \mathcal{O}_{Y_{i-1}})$ and ft for a non-zero $f \in \Gamma(Y_{i-1}, \mathcal{O}_{Y_{i-1}}(D))$, so $k(Y_i) = k(Y_{i-1})(t)$ and $\dim Y_i = \dim Y_{i-1} + 1 = d + i$.

By Proposition 8.2(1), $K_{Y_i} := (K_{Y_{i-1}})_{Y_i}$ is a canonical divisor of Y_i, both $K_{Y_i} + B_i$ and $K_{Y_i} + \Delta_i$ are $\mathbb{R}$-Cartier, and (Y_i, B_i) is klt. By Proposition 8.2(5) applied to p_i, y_{i-1}, y_i, B_{i-1}, and Δ_{i-1}, (Y_i, Δ_i) is lc and y_i is an lc center of (Y_i, Δ_i). By Proposition 8.2(1), for any prime divisor P on Y_{i-1}, P_{Y_i} is a prime divisor on Y_i, and

$$B_i = \sum_P (\operatorname{mult}_P B_{i-1}) P_{Y_i}, \quad \Delta_i = \sum_P (\operatorname{mult}_P \Delta_{i-1}) P_{Y_i}, \quad B_i^{\mathrm{eq}} = \sum_P (\operatorname{mult}_P B_{i-1}^{\mathrm{eq}}) P_{Y_i}.$$

Distinct prime divisors P have distinct pullbacks P_{Y_i}, since $p_i(P_{Y_i})$ is dense in P. Thus the coefficients of B_i, Δ_i, and B_i^{eq} belong to the sets of coefficients of B_{i-1}, Δ_{i-1}, and B_{i-1}^{eq} respectively, $B_i \le \Delta_i$, and B_i^{eq} is the equality part of B_i with respect to Δ_i. Moreover, if $1 \le j \le h$ and $(G_j)_{i-1}$ is the sum of the prime divisors P on Y_{i-1} with $\operatorname{mult}_P B_{i-1}^{\mathrm{eq}} = b_j$, then $(G_j)_i$ is the sum of the prime divisors Q on Y_i with $\operatorname{mult}_Q B_i^{\mathrm{eq}} = b_j$.

Step 3. In this step we construct the tower and prove (1) and (2).

Let $Y_0 := X$ and $y_0 := x$. The assumptions of **Step 2** hold for Y_0 and y_0. We treat $G_0, G_1, \dots, G_h$ one by one, in the order of their indices. Suppose that $G_0, \dots, G_{j-1}$ have been treated and that $Y_0, \dots, Y_i$ and $y_0, \dots, y_i$ have been constructed so that Y_i, y_i satisfy the assumptions of **Step 2** with $i+1$ in place of i, and $(G_{j'})_i$ is $\mathbb{Q}$-Cartier at y_i for any $j' < j$. If $(G_j)_i$ is $\mathbb{Q}$-Cartier at y_i, we construct nothing. Otherwise, we apply **Step 2** to Y_i, y_i, and $D := (G_j)_i$, and obtain $p_{i+1} : Y_{i+1} \to Y_i$ and y_{i+1}. Then $(G_j)_{i+1} = \operatorname{div}(t)$ is Cartier, and $(G_{j'})_{i+1}$ is $\mathbb{Q}$-Cartier at y_{i+1} for any $j' < j$ by **Step 2**. After G_h has been treated, we obtain the tower $Y_l \to \cdots \to Y_0 = X$ with $l \le h + 1 = |J \backslash \{0\}| + 1$, and $(G_j)_l$ is $\mathbb{Q}$-Cartier at y_l, hence near y_l, for any $0 \le j \le h$.

By **Step 2** and induction on i, (1) holds, y_i is a $\mathbb{G}_m$-fixed closed point over y_{i-1}, and $\kappa(y_i) = \kappa(x)$ for any i. We prove (2). By **Step 2** and induction on i, $K_{Y_l} = (G_0)_l$, so K_{Y_l} is $\mathbb{Q}$-Cartier near y_l, and the pullback $(G_j)_l$ of D_{b_j} is $\mathbb{Q}$-Cartier near y_l for any $1 \le j \le h$. Pulling back (8.8), we obtain $B_l^{\mathrm{eq}} = \sum_{j=1}^h b_j (G_j)_l$, so $K_{Y_l} + B_l^{\mathrm{eq}}$ is $\mathbb{R}$-Cartier near y_l. Since $K_{Y_l} + B_l$ is $\mathbb{R}$-Cartier, $B_l - B_l^{\mathrm{eq}} \ge 0$ is $\mathbb{R}$-Cartier near y_l. Thus, near y_l, for any prime divisor E over Y_l,

$$a(E, Y_l, B_l^{\mathrm{eq}}) = a(E, Y_l, B_l) + \operatorname{mult}_E(B_l - B_l^{\mathrm{eq}}) \ge a(E, Y_l, B_l) > 0,$$

and (Y_l, B_l^{eq}) is klt near y_l.

Step 4. In this step we prove (3): we first pass from B_l^{eq} to B_l on Y_l, and then descend through the layers one by one.

Let $V \subset Y_l$ be an open neighborhood of y_l on which $(V, B_l^{\mathrm{eq}}|_V)$ is klt, and let F be a Kollár component of $(V \ni y_l, B_l^{\mathrm{eq}}|_V)$ over k with $a(F, Y_l, \Delta_l) = 0$. Since $\dim Y_l = d + l \ge 2$ and the center of F on Y_l is the closed point y_l, F is exceptional over Y_l. Possibly shrinking V around y_l, F is primitive over V with prime extraction $f : W \to V$, and $(W, f_*^{-1}(B_l^{\mathrm{eq}}|_V) + F)$ is plt near F. By **Step 2**, $B_l^{\mathrm{eq}}|_V$ is the equality part of $B_l|_V$ with respect to $\Delta_l|_V$. By Definition-Lemma 4.4 applied to $(V, B_l|_V)$, $\Delta_l|_V$, and F, the pair $(W, f_*^{-1}(B_l|_V) + F)$ is plt near F, and

$$a(F, Y_l, B_l) \le a(F, Y_l, B_l^{\mathrm{eq}}). \tag{8.9}$$

Since (Y_l, B_l) is klt, F is a Kollár component of $(Y_l \ni y_l, B_l)$.

Let $F_l := F$. Suppose that $1 \le i \le l$ and that F_i is a Kollár component of $(Y_i \ni y_i, B_i)$ with $a(F_i, Y_i, \Delta_i) = 0$. Recall that $p_i : Y_i \to Y_{i-1}$ is a rank one torus quasi-torsor over the affine

variety Y_{i-1} and y_i is a $\mathbb{G}_m$-fixed closed point with $p_i(y_i) = y_{i-1}$. By Proposition 8.2(5) applied to p_i, y_{i-1}, y_i, B_{i-1}, and Δ_{i-1}, F_i is $\mathbb{G}_m$-invariant, and there exist a positive integer c_i and a prime divisor F_{i-1} over Y_{i-1} such that $\operatorname{ord}_{F_i}|_{k(Y_{i-1})} = c_i \operatorname{ord}_{F_{i-1}}$ and $a(F_{i-1}, Y_{i-1}, \Delta_{i-1}) = 0$. By Proposition 8.2(4), F_{i-1} is a Kollár component of $(Y_{i-1} \ni y_{i-1}, B_{i-1})$ and

$$a(F_{i-1}, Y_{i-1}, B_{i-1}) = \frac{a(F_i, Y_i, B_i)}{c_i}.$$

By descending induction on i, we obtain a Kollár component $E := F_0$ of $(X \ni x, B)$ over k with $a(E, X, \Delta) = 0$. Since $a(F, Y_l, B_l) > 0$ and $c_1, \dots, c_l$ are positive integers, (8.9) implies that

$$a(E, X, B) = \frac{a(F, Y_l, B_l)}{c_1 \cdots c_l} \le a(F, Y_l, B_l) \le a(F, Y_l, B_l^{\mathrm{eq}}),$$

where the product $c_1 \cdots c_l$ is 1 if $l = 0$. This proves (3). □

9. Finite covers

In this section, we prove the results on finite covers used in Sections 11, 12, and 13: the index cover, quotients and lifts of invariant prime extractions, the divisibility of restriction indices, and the restriction of divisorial valuations from $X \times \mathbb{A}^s$ to X.

Construction-Proposition 9.1 differs from [Zhu24b, Lemma-Definition 2.15] in that it works over k with explicit Weil divisors D'_t and rational functions ϕ_t, and keeps the residue field, that is, the point x' over x has residue field $\kappa(x)$. As in [Zhu24b, Lemma-Definition 2.15], it has exactly one point over x. In Section 11, the uniqueness of x', the equality of the residue fields of x' and x, and the trivial action of $H^\vee$ on $\kappa(x')$ show that a power of the ratio of the characters of a local generator of the canonical sheaf and of a local equation of the boundary is 1. In Sections 11 and 13, the last statement of Construction-Proposition 9.1 provides local generators and local equations at x' which are eigenvectors of $H^\vee$.

Construction-Proposition 9.1. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k, $x \in X$ a closed point, and $D_1, \dots, D_l$ Weil divisors on X which are $\mathbb{Q}$-Cartier at x. Let H be the subgroup of $\mathrm{Cl}(\mathcal{O}_{X,x})$ generated by the classes of $D_1, \dots, D_l$, which is finite. Choose Weil divisors $D'_1, \dots, D'_n$ whose classes generate H as a direct sum of cyclic groups of orders $e_1, \dots, e_n \ge 2$ (so $n = 0$ if $H = 0$), and rational functions $\phi_1, \dots, \phi_n$ such that $\operatorname{div}(\phi_t) = e_t D'_t$ near x for any t. Possibly shrinking X around x, the *index cover* of $D_1, \dots, D_l$ is

$$f : X' := \operatorname{Spec}_X \bigoplus_{0 \le m_t < e_t} \mathcal{O}_X\Big(\sum_{t=1}^n m_t D'_t\Big) \to X,$$

where the product of a local section s of $\mathcal{O}_X(\sum_t m_t D'_t)$ and a local section s' of $\mathcal{O}_X(\sum_t m'_t D'_t)$ is the local section

$$ss' \prod_{t : m_t + m'_t \ge e_t} \phi_t \quad \text{of} \quad \mathcal{O}_X\Big(\sum_t \bar{m}_t D'_t\Big),$$

with $\bar{m}_t$ the remainder of $m_t + m'_t$ modulo e_t. Then X' is normal, f is a finite quasi-étale Galois morphism with group $H^\vee := \operatorname{Hom}(H, k^*)$, there exists a unique point $x' \in X'$ with $f(x') = x$, $\kappa(x') = \kappa(x)$, $H^\vee$ acts trivially on $\kappa(x')$, and $f^* D_j$ is Cartier near x' for any j. If moreover x is a k-rational point, then any $H^\vee$-linearized invertible sheaf on an $H^\vee$-invariant neighborhood of x' has a local generator at x' which is an eigenvector of $H^\vee$.

Proof. Since any D_j is $\mathbb{Q}$-Cartier at x, the class of D_j in $\mathrm{Cl}(\mathcal{O}_{X,x})$ is torsion, so H is finite. Possibly shrinking X around x, we may assume that $\operatorname{div}(\phi_t) = e_t D'_t$ on X for any t, and that for any j there exist integers $c_{j1}, \dots, c_{jn}$ and $\rho_j \in k(X)$ such that

$$D_j = \sum_{t=1}^n c_{jt} D'_t + \operatorname{div}(\rho_j) \quad \text{on } X.$$

For $h = \sum_t m_t[D'_t] \in H$ with $0 \le m_t < e_t$, let $\mathcal{A}_h := \mathcal{O}_X(\sum_t m_t D'_t)$, and let $\mathcal{A} := \bigoplus_{h\in H} \mathcal{A}_h$. Then $\mathcal{A}$ is an H-graded commutative $\mathcal{O}_X$-algebra (the product of three sections carries the total factor $\prod_t \phi_t^{\lfloor (m_t+m'_t+m''_t)/e_t \rfloor}$, which does not depend on the order of multiplication) with $\mathcal{A}_0 = \mathcal{O}_X$ and $X' = \operatorname{Spec}_X \mathcal{A}$. Let τ_t be the element 1 of $\mathcal{A}_{[D'_t]} \otimes_{\mathcal{O}_X} k(X) = k(X)$. Since $e_t \ge 2$, the e_t-th factor in the product $\tau_t^{e_t}$ is the first one with $m_t + m'_t \ge e_t$, so by the definition of the multiplication, $\tau_t^{e_t} = \phi_t$, and the local sections of $\mathcal{A}_h$ for $h = \sum_t m_t[D'_t]$ are the elements $s\prod_t \tau_t^{m_t}$ with s a local section of $\mathcal{O}_X(\sum_t m_t D'_t)$.

We show that f is quasi-étale. Let P be a prime divisor on X, π a uniformizer of $\mathcal{O}_{X,P}$, and $b_t := \pi^{\operatorname{mult}_P D'_t}$. Then $\phi_t = u_t b_t^{e_t}$ for a unit $u_t \in \mathcal{O}_{X,P}$, and

$$\mathcal{A}_P = \mathcal{O}_{X,P}[y_1, \dots, y_n]/(y_1^{e_1} - u_1, \dots, y_n^{e_n} - u_n), \qquad y_t = \tau_t/b_t.$$

Since e_t is a unit in k and u_t is a unit in $\mathcal{O}_{X,P}$, f is étale over the generic point of P.

We show that X' is normal. Any $\mathcal{A}_h$ is a reflexive $\mathcal{O}_X$-module of rank one on the normal variety X, hence satisfies (S2) [Sta, Tag 0AVB]. Since $\mathcal{A}$ is finite and torsion free over the normal scheme X, going-down holds for $\mathcal{O}_X \subset \mathcal{A}$, so the heights and depths of primes of $\mathcal{A}$ can be computed over $\mathcal{O}_X$. Thus the finite $\mathcal{O}_X$-algebra $\mathcal{A}$ satisfies (S2). Since $\mathcal{A}$ is torsion free, any irreducible component of X' dominates X, so any point of codimension one of X' maps to a point of codimension one of X, over which f is étale. Thus X' satisfies (R1). By Serre's criterion [Har77, Theorem II.8.22A], which holds for any Noetherian ring and does not assume that the ring is a domain, X' is normal.

We show that x has exactly one preimage, whose residue field is $\kappa(x)$. Let $A := \mathcal{A}_x = \bigoplus_{h\in H} A_h$, so $A_0 = \mathcal{O}_{X,x}$. We claim that $A_h A_{-h} \subset \mathfrak{m}_x$ for any $0 \ne h \in H$. Write $h = \sum_t m_t[D'_t]$ with $0 \le m_t < e_t$, and let $D := \sum_t m_t D'_t$. By the definition of the multiplication, the product of A_h and A_{-h} in $A_0 = \mathcal{O}_{X,x}$ is the product of rational functions multiplied by $\prod_{t:m_t\ne 0} \phi_t$, and $\prod_{t:m_t\ne 0} \phi_t \cdot A_{-h} = \mathcal{O}_X(-D)_x$. Hence $A_h A_{-h} = \mathcal{O}_X(D)_x \cdot \mathcal{O}_X(-D)_x$. Since $h \ne 0$ in $\operatorname{Cl}(\mathcal{O}_{X,x})$, D is not Cartier at x, so $A_h A_{-h} \subset \mathfrak{m}_x$ by Lemma 8.3. Let $\bar{A} := A/\mathfrak{m}_x A = \bigoplus_{h\in H} \bar{A}_h$, so $\bar{A}_0 = \kappa(x)$. If $b \in \bar{A}_h$ with $h \ne 0$ and r is the order of h, then $b^r = b \cdot b^{r-1} \in \bar{A}_h \bar{A}_{-h} = 0$. Let $I \subset \bar{A}$ be the ideal generated by the homogeneous elements of non-zero degree. Since A is a finite $\mathcal{O}_{X,x}$-module, $\dim_{\kappa(x)} \bar{A} < +\infty$, so I is generated by finitely many homogeneous elements of non-zero degree, which are nilpotent. Since $\bar{A}$ is commutative, I is nilpotent. Since I is homogeneous and the degree zero part of I is contained in $\sum_{h\ne 0} \bar{A}_h \bar{A}_{-h} = 0$, we have $\bar{A}/I = \bar{A}_0 = \kappa(x)$. Thus $\bar{A}_{\mathrm{red}} = \kappa(x)$, that is, there exists a unique point $x' \in X'$ with $f(x') = x$, and $\kappa(x') = \kappa(x)$. Since any irreducible component of X' dominates X and f is finite, any irreducible component of X' contains x'. Since the irreducible components of the normal scheme X' are disjoint, X' is integral.

We show that f is Galois with group $H^\vee$. Let $\chi \in H^\vee$ act on $\mathcal{A}_h$ by multiplication by $\chi(h)$. Since χ is a homomorphism, χ acts on $\mathcal{A}$ by $\mathcal{O}_X$-algebra automorphisms. Since k contains all roots of unity, $|H^\vee| = |H|$, and for any $0 \ne h \in H$ there exists $\chi \in H^\vee$ with $\chi(h) \ne 1$. Thus $H^\vee$ acts faithfully on X', and the invariant subalgebra of $\mathcal{A}$ is $\mathcal{A}_0 = \mathcal{O}_X$. Since $[k(X') : k(X)] = \operatorname{rank} \mathcal{A} = |H| = |H^\vee|$, f is Galois with group $H^\vee$. Moreover, $H^\vee$ acts trivially on $\bar{A}_0 = \kappa(x) = \kappa(x')$.

We show that $f^* D_j$ is Cartier. Since $\tau_t^{e_t} = \phi_t$, we have $e_t \operatorname{div}_{X'}(\tau_t) = f^* \operatorname{div}(\phi_t) = e_t f^* D'_t$, so $f^* D'_t = \operatorname{div}_{X'}(\tau_t)$. Hence

$$f^* D_j = \operatorname{div}_{X'} \left(\rho_j \prod_{t=1}^{n} \tau_t^{c_{jt}} \right)$$

is Cartier for any j.

Finally, assume that x is a k-rational point, and let $\mathcal{L}$ be an $H^\vee$-linearized invertible sheaf on an $H^\vee$-invariant neighborhood of x'. Since $\kappa(x') = \kappa(x) = k$ and $H^\vee$ acts trivially on $\kappa(x')$, $H^\vee$ acts k-linearly on the one-dimensional k-vector space $\mathcal{L}_{x'} \otimes \kappa(x')$, through a character $\psi : H^\vee \to k^*$.

Let s be a generator of $\mathcal{L}_{x'}$, and let

$$s_\psi := \frac{1}{|H^\vee|} \sum_{\chi \in H^\vee} \psi(\chi)^{-1} \chi \cdot s.$$

Then $\chi \cdot s_\psi = \psi(\chi) s_\psi$ for any $\chi \in H^\vee$, and the image of s_ψ in $\mathcal{L}_{x'} \otimes \kappa(x')$ is equal to the image of s, which is non-zero. By Nakayama's lemma, s_ψ is a generator of $\mathcal{L}_{x'}$. □

Lemma 9.2. *Let k be a field finitely generated over $\mathbb{C}$, V a normal variety of dimension at least 2 over k with an action of a finite group G, $q : V \to W := V/G$ the quotient morphism, and $v \in V$ a closed point with $q^{-1}(q(v)) = \{v\}$. Let $\Delta_W \geq 0$ and $\Delta_V \geq 0$ be $\mathbb{R}$-divisors on W and V such that $K_W + \Delta_W$ is $\mathbb{R}$-Cartier and $K_V + \Delta_V = q^*(K_W + \Delta_W)$. Let F be a G-invariant prime divisor over V, primitive over V, with $\operatorname{center}_V F = v$ and prime extraction $f_V : Y_V \to V$, and let L be the prime divisor over W with $\operatorname{ord}_F |_{k(W)} = e \operatorname{ord}_L$ for a positive integer e. Then G acts on Y_V so that f_V is G-equivariant, the induced morphism $f_W : Y_W := Y_V/G \to W$ is the prime extraction of L, and*

$$K_{Y_V} + (f_V)_*^{-1}\Delta_V + F = q_Y^*\big(K_{Y_W} + (f_W)_*^{-1}\Delta_W + L\big)$$

for the induced morphism $q_Y : Y_V \to Y_W$. In particular, $(Y_V, (f_V)_*^{-1}\Delta_V + F)$ is plt near F if and only if $(Y_W, (f_W)_*^{-1}\Delta_W + L)$ is plt near L, and $a(F, V, \Delta_V) = e a(L, W, \Delta_W)$.

Proof. Let $w := q(v)$, let r be a positive integer such that rF is Cartier, and let $\mathcal{S} := \bigoplus_{m \geq 0} \mathfrak{a}_{mr}(\operatorname{ord}_F)$. By Lemma 2.7, $Y_V \cong \operatorname{Proj}_V \mathcal{S}$ over V, and f_V is an isomorphism over $V \backslash \{v\}$. Since F is G-invariant, the ideals $\mathfrak{a}_{mr}(\operatorname{ord}_F)$ are G-invariant, so G acts on $\mathcal{S}$ by graded automorphisms. Thus G acts on Y_V and f_V is G-equivariant. Since taking G-invariants commutes with the grading, $Y_W = \operatorname{Proj}_W (q_* \mathcal{S})^G$, so f_W is projective. Since $q^{-1}(W \backslash \{w\}) = V \backslash \{v\}$, f_W is an isomorphism over $W \backslash \{w\}$. In particular, f_W is birational, and Y_W is normal as the quotient of a normal variety by a finite group.

Let $L' := q_Y(F)$. By Lemma 2.4(1), $\operatorname{Supp} f_V^{-1}(v) = F$. Since $q^{-1}(w) = \{v\}$, we have $f_W^{-1}(w) = q_Y(f_V^{-1}(v))$, so $\operatorname{Supp} f_W^{-1}(w) = L'$. Since the fibers of q_Y are G-orbits and F is G-invariant, $q_Y^{-1}(L') = F$. The valuation ring of $\operatorname{ord}_F |_{k(W)}$ dominates the discrete valuation ring $\mathcal{O}_{Y_W, L'}$, so it is equal to $\mathcal{O}_{Y_W, L'}$. Hence $\operatorname{ord}_F |_{k(W)} = e' \operatorname{ord}_{L'}$, where e' is the ramification index of q_Y along F. Thus $L' = L$, $e' = e$, and

$$q_Y^* L = eF.$$

We show that L is $\mathbb{Q}$-Cartier. Let $y \in L$. Since $q_Y^{-1}(y)$ is a G-orbit and $\mathcal{O}_{Y_V}(-rF)$ is invertible, rF is principal on the semi-local ring of Y_V at $q_Y^{-1}(y)$ [Bou89, Chapter II, §5, no. 3, Proposition 5] (a finitely generated projective module of constant rank over a semi-local ring is free), so there exist an affine open neighborhood U of y and a regular function ϕ on $q_Y^{-1}(U)$ with $rF = \operatorname{div}(\phi)$ on $q_Y^{-1}(U)$. Since F is G-invariant, $\prod_{g \in G} g^* \phi$ is a G-invariant regular function on $q_Y^{-1}(U)$, hence a regular function on U, and

$$q_Y^* \operatorname{div}\Big(\prod_{g \in G} g^* \phi\Big) = |G| rF \quad \text{on } q_Y^{-1}(U).$$

Since $q_Y^* L = eF$, we have $e \operatorname{div}(\prod_{g \in G} g^* \phi) = |G| rL$ on U. Thus L is $\mathbb{Q}$-Cartier. Since $q_Y^*(-L) = -eF$ is f_V-ample and q_Y is finite and surjective, $-L$ is f_W-ample [Laz04a, Corollary 1.2.28 and Theorem 1.7.8], whose proofs work over any field. Hence L is primitive over W and f_W is the prime extraction of L.

Let $\Theta_V := (f_V)_*^{-1}\Delta_V + F$ and $\Theta_W := (f_W)_*^{-1}\Delta_W + L$. Since

$$K_{Y_W} + \Theta_W = f_W^*(K_W + \Delta_W) + a(L, W, \Delta_W) L$$

and L is $\mathbb{Q}$-Cartier, $K_{Y_W} + \Theta_W$ is $\mathbb{R}$-Cartier. Let Θ' be the $\mathbb{R}$-divisor on Y_V such that $K_{Y_V} + \Theta' = q_Y^*(K_{Y_W} + \Theta_W)$. For a prime divisor P on Y_V, let e_P be the ramification index of q_Y along P.

By the Hurwitz formula

$$K_{Y_V} = q_Y^* K_{Y_W} + \sum_P (e_P - 1)P,$$

we have

$$\operatorname{mult}_P \Theta' = 1 - e_P\big(1 - \operatorname{mult}_{q_Y(P)} \Theta_W\big).$$

For $P = F$, we have $\operatorname{mult}_F \Theta' = 1 = \operatorname{mult}_F \Theta_V$. For $P \neq F$, the morphisms f_V and f_W are isomorphisms near the generic points of P and $q_Y(P)$, and the Hurwitz formula for q together with $K_V + \Delta_V = q^*(K_W + \Delta_W)$ implies

$$\operatorname{mult}_{f_V(P)} \Delta_V = 1 - e_P\big(1 - \operatorname{mult}_{q(f_V(P))} \Delta_W\big) = \operatorname{mult}_P \Theta'.$$

Thus $\Theta' = \Theta_V$, which proves the equality of the lemma.

For any open neighborhood U' of F, the intersection $\bigcap_{g\in G} g(U')$ is a G-invariant open neighborhood of F, which is the preimage of the open neighborhood $q_Y(\bigcap_{g\in G} g(U'))$ of L. Thus, by Lemma 2.5 for q_Y over open neighborhoods of L, (Y_V, Θ_V) is plt near F if and only if (Y_W, Θ_W) is plt near L. By Lemma 2.5 for q, we have $a(F, V, \Delta_V) = ea(L, W, \Delta_W)$. □

Kollár components are taken on the quotient and lifted to the cover: by Lemma 9.3 the lift is unique, hence invariant under the group, whereas a Kollár component of the cover is in general not invariant under the group and does not descend to the quotient.

Lemma 9.3. *Let k be a field finitely generated over $\mathbb{C}$, and V a normal variety of dimension at least 2 over k with an action of a finite group G such that the quotient morphism $q : V \to W := V/G$ is quasi-étale. Let $v \in V$ be a closed point with $q^{-1}(q(v)) = \{v\}$. Let $\Delta_W \geq 0$ be an $\mathbb{R}$-divisor on W such that (W, Δ_W) is klt, and let L be a Kollár component of $(W \ni q(v), \Delta_W)$ over k. Then there exists a unique prime divisor F over V such that $\operatorname{ord}_F|_{k(W)} = e \operatorname{ord}_L$ for some positive integer e. Moreover, F is G-invariant, F is a Kollár component of $(V \ni v, q^*\Delta_W)$ over k, and $a(F, V, q^*\Theta) = ea(L, W, \Theta)$ for any $\mathbb{R}$-divisor Θ on W such that $K_W + \Theta$ is $\mathbb{R}$-Cartier.*

Proof. Let $w := q(v)$. Since the conclusion only concerns divisors centered over w and $q^{-1}(w) = \{v\}$, possibly shrinking W around w and replacing V with the preimage, by Definition 2.3 we may assume that L is primitive over W with prime extraction $f_W : Y_W \to W$, and that (Y_W, Θ_W) is plt near L, where $\Theta_W := (f_W)_*^{-1}\Delta_W + L$. By Lemma 2.4(1), f_W is an isomorphism over $W\backslash\{w\}$ and $\operatorname{Supp} f_W^{-1}(w) = L$.

Let Y_V be the normalization of Y_W in $k(V)$. Since V is the normalization of W in $k(V)$, the universal property of normalization implies that G acts on Y_V, and that there exist a finite morphism $q_Y : Y_V \to Y_W$ and a projective birational morphism $f_V : Y_V \to V$ such that $q \circ f_V = f_W \circ q_Y$. Since Y_V/G is normal, finite, and birational over Y_W, we have $Y_V/G = Y_W$. Since $q^{-1}(w) = \{v\}$, f_V is an isomorphism over $V\backslash\{v\}$ and $f_V^{-1}(v) = q_Y^{-1}(f_W^{-1}(w))$. Thus $\operatorname{Supp} f_V^{-1}(v) = F_1 \cup \cdots \cup F_l$, where $F_1, \ldots, F_l$ are the prime divisors on Y_V over L.

Since L is $\mathbb{Q}$-Cartier and $K_W + \Delta_W$ is $\mathbb{R}$-Cartier, $K_{Y_W} + \Theta_W$ is $\mathbb{R}$-Cartier. Let Θ_V be the $\mathbb{R}$-divisor on Y_V such that $K_{Y_V} + \Theta_V = q_Y^*(K_{Y_W} + \Theta_W)$. For a prime divisor P on Y_V, let e_P be the ramification index of q_Y along P. The Hurwitz formula implies

$$\operatorname{mult}_P \Theta_V = 1 - e_P\big(1 - \operatorname{mult}_{q_Y(P)} \Theta_W\big).$$

Thus $\operatorname{mult}_{F_i} \Theta_V = 1$ for any i. If $P \neq F_i$ for any i, then f_V and f_W are isomorphisms near the generic points of P and $q_Y(P)$, and q is étale over the generic point of $q(f_V(P))$, so $e_P = 1$ and $\operatorname{mult}_P \Theta_V = \operatorname{mult}_P \big((f_V)_*^{-1} q^* \Delta_W\big)$. Hence

$$K_{Y_V} + (f_V)_*^{-1} q^* \Delta_W + \sum_{i=1}^{l} F_i = q_Y^*(K_{Y_W} + \Theta_W).$$

Any open neighborhood of L has preimage an open neighborhood of $\sum_i F_i$, and any G-invariant open neighborhood of $\sum_i F_i$ is such a preimage. Thus, by Lemma 2.5 for q_Y over an open neighborhood of L on which (Y_W, Θ_W) is plt, $(Y_V, (f_V)_*^{-1} q^* \Delta_W + \sum_i F_i)$ is plt near $\sum_i F_i$. By

Lemma 2.4(2), applied to f_V, $S = \sum_i F_i$, and $B_Y = (f_V)_*^{-1} q^* \Delta_W$, we have $l = 1$. Let $F := F_1$. Since G permutes the prime divisors on Y_V over L, F is G-invariant.

Let e be the ramification index of q_Y along F. Then $q_Y^* L = eF$ and $\operatorname{ord}_F|_{k(W)} = e \operatorname{ord}_L$. Since L is $\mathbb{Q}$-Cartier, F is $\mathbb{Q}$-Cartier. Since $q_Y^*(-L) = -eF$ and $-L$ is f_W-ample, $-F$ is f_V-ample [Laz04a, Proposition 1.2.13 and Theorem 1.7.8], whose proofs work over any field. Hence F is primitive over V with prime extraction f_V, and $\operatorname{center}_V F = v$. By Lemma 2.5, $K_V + q^* \Delta_W = q^*(K_W + \Delta_W)$ and $(V, q^* \Delta_W)$ is klt. Since $(Y_V, (f_V)_*^{-1} q^* \Delta_W + F)$ is plt near F, F is a Kollár component of $(V \ni v, q^* \Delta_W)$, and F and f_V are defined over k.

We show the uniqueness. Let F' be a prime divisor over V such that $\operatorname{ord}_{F'}|_{k(W)} = e'' \operatorname{ord}_L$ for a positive integer e''. Since q_Y is finite, $\operatorname{ord}_{F'}$ has a center on Y_V, which maps to the generic point of L. Thus the center of $\operatorname{ord}_{F'}$ on Y_V is the generic point of F, and the valuation ring of $\operatorname{ord}_{F'}$ dominates the discrete valuation ring $\mathcal{O}_{Y_V, F}$. Hence the valuation ring of $\operatorname{ord}_{F'}$ is equal to $\mathcal{O}_{Y_V, F}$, so $F' = F$.

Finally, let Θ be an $\mathbb{R}$-divisor on W such that $K_W + \Theta$ is $\mathbb{R}$-Cartier. Since q is quasi-étale, $K_V + q^* \Theta = q^*(K_W + \Theta)$, and Lemma 2.5 implies $a(F, V, q^* \Theta) = e a(L, W, \Theta)$. □

Lemma 9.4 is used in Sections 11 and 13, where X is the given germ, S is a variety built from a finite cover of X (a product of root covers of the index cover with $\mathbb{A}^1$ in Section 11, a Fermat suspension of the index cover in Section 13), W is the quotient of S by a finite group, and F is the prime divisor over S to which a prime divisor L over W is lifted. There the log discrepancy of the prime divisor E over X below F is bounded through S and W: an inequality bounds $ca(E, X, \cdot)$ from above by a positive multiple of $a(F, S, \cdot)$, where c is the ramification index of F over E, and $a(F, S, \cdot) = e a(L, W, \cdot)$ for the ramification index e of F over L. The degrees of the covers, on which c and e depend, are not bounded, but c/e is a positive integer, so c and e do not enter the bound on $a(E, X, \cdot)$.

Lemma 9.4. *Let k be a field finitely generated over $\mathbb{C}$ and X, W, and S varieties over k such that $k(X) \subset k(W) \subset k(S)$. Let F be a prime divisor over S, and let L and E be prime divisors over W and X such that $\operatorname{ord}_F|_{k(W)} = e \operatorname{ord}_L$ and $\operatorname{ord}_F|_{k(X)} = c \operatorname{ord}_E$ for positive integers e and c. Then c/e is a positive integer and $\operatorname{ord}_L|_{k(X)} = (c/e) \operatorname{ord}_E$.*

Proof. For any $\varphi \in k(X)$, we have $e \operatorname{ord}_L(\varphi) = \operatorname{ord}_F(\varphi) = c \operatorname{ord}_E(\varphi)$, so $\operatorname{ord}_L|_{k(X)} = (c/e) \operatorname{ord}_E$. Choosing φ with $\operatorname{ord}_E(\varphi) = 1$, we obtain $c/e = \operatorname{ord}_L(\varphi) \in \mathbb{Z}$, and $c/e > 0$ since $c, e > 0$. □

Lemma 9.5. *Let X be a normal variety over a field k_0 of characteristic zero, s a positive integer, and F a prime divisor over $X \times \mathbb{A}^s$. Then the restriction of ord_F to the function field of X is either trivial or $c \operatorname{ord}_E$ for a positive integer c and a prime divisor E over X. If $\dim X \geq 1$ and $\operatorname{center}_{X \times \mathbb{A}^s} F = (x, 0)$ for a closed point $x \in X$, then the second case occurs and $\operatorname{center}_X E = x$.*

Proof. Let $v := \operatorname{ord}_F|_{k_0(X)}$. For a valuation u trivial on k_0, let $\kappa(u)$ and Γ_u be the residue field and the value group of u. Let $z := \operatorname{center}_{X \times \mathbb{A}^s} F$ and let $x_z \in X$ be the image of z. Since ord_F is non-negative on $\mathcal{O}_{X \times \mathbb{A}^s, z}$ and positive on the maximal ideal of $\mathcal{O}_{X \times \mathbb{A}^s, z}$, v is non-negative on $\mathcal{O}_{X, x_z}$ and positive on $\mathfrak{m}_{x_z}$, that is, v is centered at x_z.

Assume that v is non-trivial. Then $\Gamma_v = c\mathbb{Z} \subset \Gamma_{\operatorname{ord}_F} = \mathbb{Z}$ for a positive integer c, so $\dim_{\mathbb{Q}} \left((\Gamma_{\operatorname{ord}_F} / \Gamma_v) \otimes \mathbb{Q} \right) = 0$. By the relative Abhyankar inequality [Bou89, Chapter VI, §10, no. 3, Corollary 1] for the extension $k_0(X \times \mathbb{A}^s)/k_0(X)$,

$$\operatorname{trdeg}_{k_0} \kappa(\operatorname{ord}_F) - \operatorname{trdeg}_{k_0} \kappa(v) \leq \operatorname{trdeg}_{k_0(X)} k_0(X \times \mathbb{A}^s) = s.$$

Since F is a prime divisor over $X \times \mathbb{A}^s$, $\operatorname{trdeg}_{k_0} \kappa(\operatorname{ord}_F) = \dim X + s - 1$, so $\operatorname{trdeg}_{k_0} \kappa(v) \geq \dim X - 1$. By the Abhyankar inequality for $k_0(X)/k_0$ [Abh56, Theorem 1],

$$\operatorname{trdeg}_{k_0} \kappa(v) + \dim_{\mathbb{Q}}(\Gamma_v \otimes \mathbb{Q}) \leq \dim X,$$

so $\operatorname{trdeg}_{k_0} \kappa(v) = \dim X - 1$. Thus v/c is a valuation of $k_0(X)$ centered at x_z with value group $\mathbb{Z}$ and residue transcendence degree $\dim X - 1$, so $v/c = \operatorname{ord}_E$ for a prime divisor E over X with $\operatorname{center}_X E = x_z$ [KM98, Lemma 2.45].

If $z = (x, 0)$ for a closed point $x \in X$, then $x_z = x$, $v(\varphi) > 0$ for any non-zero $\varphi \in \mathfrak{m}_x$, and $\mathfrak{m}_x \neq 0$ since $\dim X \geq 1$. Hence v is non-trivial and $\operatorname{center}_X E = x$. □

10. Arithmetic pinning

In this section, we prove Theorem 10.2 below.

Set-up 10.1. Let d, k_i, S_i, x_i, G_i, p_i, X_i, o_i, ω_i, χ_i, h_i, and T_i be as follows, where i runs through the positive integers.

(1) $d \geq 2$ is an integer.
(2) k_i is a field finitely generated over $\mathbb{C}$, S_i is a geometrically integral normal affine variety over k_i with $\dim S_i = d$, and $x_i \in S_i$ is a k_i-rational point.
(3) G_i is a finite cyclic group of automorphisms of S_i over k_i fixing x_i.
(4) S_i is klt, the quotient morphism $p_i : S_i \to X_i := S_i/G_i$ is quasi-étale, and $o_i := p_i(x_i)$.
(5) ω_i is a generator of the reflexive canonical sheaf of S_i at x_i which is an eigenvector of G_i with character χ_i.
(6) $h_i \in \mathcal{O}_{S_i,x_i}$ is a non-zero eigenvector of G_i with character χ_i such that $(S_i, \operatorname{div}(h_i))$ is lc.
(7) T_i is a quasi-monomial valuation on $S_{i,\bar{k}_i}$ centered at the point over x_i such that $a(T_i, S_{i,\bar{k}_i}, 0) = T_i(h_i) = 1$.

Theorem 10.2. *Notation and conditions as in Set-up 10.1. Let $\epsilon > 0$ be a real number. Assume that $a(F, X_i, 0) \geq \epsilon$ for any i and any Kollár component F of $(X_i \ni o_i, 0)$ over k_i. Then there exist a positive integer m and infinitely many i such that $T_i(\sigma) < m$ for some non-zero $\sigma \in \mathcal{O}_{S_i,x_i}$ satisfying $g^*\sigma = \chi_i(g)^m\sigma$ for any $g \in G_i$.*

Theorem 10.2 is a version of [OAI26b, Theorem G.1] over the fields k_i, with two changes. First, [OAI26b, Theorem G.1] assumes that the quotient germs are ϵ-lc. The ϵ-lc assumption is used only in the final case of [OAI26b, Subsection G.2], for a rank one valuation whose associated graded ring is finitely generated and klt, and the valuation there is a Kollár component over k_i (Lemma 10.23). Thus a lower bound on the log discrepancies of Kollár components over k_i is enough. Second, the valuations in the argument are over k_i, and the order condition [OAI26b, (C.5)] is only tested by $\mathbb{Q}$-divisors over k_i, which is condition (MO_k) of Set-up 10.6. Theorem 10.2 is the only quantitative input of the argument by contradiction in Section 11. We prove Theorem 10.2 in Subsection 10.4.

10.1. Scaled phases over the ground field.

In this subsection, we prove versions over finitely generated fields of [OAI26b, Lemmas C.1 and C.2], namely Lemmas 10.10 and 10.11.

Definition 10.3. Let V be a normal variety. A *higher model* of V is a normal variety V' with a projective birational morphism $V' \to V$. A ***b**-divisor* $\mathbf{N}$ over V assigns to any higher model $V' \to V$ a $\mathbb{Q}$-divisor $\mathbf{N}_{V'}$ on V', called the *trace* of $\mathbf{N}$ on V', such that $g_*\mathbf{N}_{V''} = \mathbf{N}_{V'}$ for any morphism $g : V'' \to V'$ of higher models of V. For a prime divisor P over V, $\mathbf{N}(P)$ is the coefficient of P in $\mathbf{N}_{V'}$ for a higher model V' on which P is a prime divisor. A ***b***-divisor $\mathbf{N}$ is *Cartier* if there exists a higher model V' such that $\mathbf{N}_{V'}$ is $\mathbb{Q}$-Cartier and $\mathbf{N}_{V''}$ is the pullback of $\mathbf{N}_{V'}$ for any higher model $V'' \to V'$, and it is *nef* if moreover $\mathbf{N}_{V'}$ is nef. The *Cartier* ***b**-divisor* of a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on V is the ***b***-divisor whose trace on any higher model is the pullback of D.

Definition 10.4. A *generalized pair* $(V, B, \mathbf{M})$ consists of a normal variety V, a $\mathbb{Q}$-divisor B on V, and a nef Cartier ***b***-divisor $\mathbf{M}$ over V, called the *nef part*, such that $K_V + B + \mathbf{M}_V$ is $\mathbb{Q}$-Cartier. For a prime divisor P on a higher model $g : V' \to V$, the *log discrepancy* of P with respect to $(V, B, \mathbf{M})$ is $1 - \operatorname{mult}_P B'$, where $K_{V'} + B' + \mathbf{M}_{V'} = g^*(K_V + B + \mathbf{M}_V)$. The pair $(V, B, \mathbf{M})$ is *generalized lc* (resp. *generalized klt*) if the log discrepancy of any prime divisor over V is ≥ 0 (resp. > 0). [OAI26b, Subsection C.1] calls the function sending a prime divisor P over V to the log discrepancy of P the discrepancy function, and we call it the *log discrepancy function* of $(V, B, \mathbf{M})$.

Definition 10.5. Following [OAI26b, Subsection C.1], *rational generalized lc Calabi–Yau data* $(V, B, \mathbf{M})$ over a field k is a generalized lc pair over k such that $B \geq 0$ and $K_V + B + \mathbf{M}_V \sim_{\mathbb{Q}} 0$ over k. Let a be the log discrepancy function of such data, $D \geq 0$ a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on V, and $\mathbf{D}$ the Cartier $\boldsymbol{b}$-divisor of D. We say that $a + \mathbf{D}$ *is generalized klt with trace* ≥ 0 *on* V if the generalized pair $(V, B - D, \mathbf{M} + \mathbf{D})$ is generalized klt and $B - D \geq 0$. The log discrepancy function of $(V, B - D, \mathbf{M} + \mathbf{D})$ is $a + \mathbf{D}$, so the condition means that $a + \mathbf{D} > 0$ and $a(Q) + \mathbf{D}(Q) \leq 1$ for any prime divisor Q on V.

We keep the letters a_i and $\mathbf{D}_i$ of [OAI26b], so that the lemmas below can be compared with [OAI26b] one by one.

Set-up 10.6. Let d, k_i, V_i, B_i, $\mathbf{M}_i$, a_i, D_i, $\mathbf{D}_i$, L_i, and η_i be as follows, where i runs through the positive integers.

(1) d is a positive integer, and k_i is a field finitely generated over $\mathbb{C}$.
(2) V_i is a geometrically integral normal projective variety over k_i with $\dim V_i \leq d$ such that $V_{i,\bar{k}_i}$ is of Fano type.
(3) For a prime divisor P over $V_{i,\bar{k}_i}$ and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor Γ on V_i, $\mathrm{mult}_P \Gamma$ is the coefficient of P in the pullback of Γ to a normal variety on which P is a prime divisor.
(4) $(V_i, B_i, \mathbf{M}_i)$ is rational generalized lc Calabi–Yau data over k_i, and a_i is the log discrepancy function of $(V_i, B_i, \mathbf{M}_i)$ on prime divisors over $V_{i,\bar{k}_i}$.
(5) $a_i(P) \in \mathbb{Z}$ for any prime divisor P over $V_{i,\bar{k}_i}$.
(6) $D_i \geq 0$ is a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on V_i, $\mathbf{D}_i$ is the Cartier $\boldsymbol{b}$-divisor of D_i, so that $\mathbf{D}_i(P) = \mathrm{mult}_P D_i$, and $a_i + \mathbf{D}_i$ is generalized klt with trace ≥ 0 on V_i.
(7) $L_i \geq 1$ and $\eta_i \geq 0$ are real numbers with $\lim_i \eta_i = 0$.
(8) We say that (MO_k) *holds at* a prime divisor P over $V_{i,\bar{k}_i}$ if

$$\mathrm{mult}_P \Gamma \leq (1 + \eta_i)\mathbf{D}_i(P) \tag{10.1}$$

for any $\mathbb{Q}$-divisor $\Gamma \geq 0$ on V_i with $\Gamma \sim_{\mathbb{Q}} D_i$ over k_i.
(9) For any subsequence and any prime divisors P_i over $V_{i,\bar{k}_i}$ with $a_i(P_i) = 0$ and $\lim_i L_i \mathbf{D}_i(P_i) = 0$, (MO_k) holds at P_i for any sufficiently large i.

Condition (5) is the integrality assumption of [OAI26b, Subsection C.1], and in the application it holds by Definition-Proposition 10.14. We allow the factor $1+\eta_i$ in (8) since the rank refinement implies the order condition only asymptotically: the excess in Proposition 10.16(2) tends to zero but need not vanish. Condition (9) is the form in which [OAI26b, Lemma C.2] uses [OAI26b, (C.5)].

Condition (MO_k) tests only $\mathbb{Q}$-divisors over k_i for the following reason. In the rank refinement of Proposition 10.16, a $\mathbb{Q}$-divisor Γ violating (10.1) is used to tilt the height, and the valuation minimizing the tilted height is Galois invariant by uniqueness only when Γ is over k_i. On the other hand, the arguments below and in Lemma 10.22 use (10.1) only for $\mathbb{Q}$-divisors over k_i: for $D_i + (\beta/2C)\Xi$ in the proof of Lemma 10.10, for $\Gamma_i/(c_i L_i)$ and for $f_*(\Sigma + t\widetilde{E})$ on V_i at the first threshold in the proof of Lemma 10.11, and in Lemma 10.22.

Remark 10.7. The items below refer to Construction 10.9 and to the proofs of Lemmas 10.10 and 10.11, where the notation is introduced. The proofs of [OAI26b, Lemmas C.1 and C.2] change over k_i at the following places.

(1) In the proof of [OAI26b, Lemma C.1], the extractions and MMPs are over $\bar{k}_i$, and [OAI26b, (C.5)] is applied to a test divisor that need not be over k_i. Construction 10.9 runs them over k_i, and **Step 3** of the proof of Lemma 10.10 tests only $D_i + (\beta/2C)\Xi$, which is over k_i. The small lifts [OAI26b, (C.7)] are taken over $\bar{k}_i$ (**Step 1** of the proof of Lemma 10.10).
(2) At [OAI26b, (C.14)], the test divisor $G/(c_g L)$ becomes $\Gamma_i/(c_i L_i)$, which is over k_i (**Step 1** of the proof of Lemma 10.11).

(3) At [OAI26b, (C.19)], the threshold prime divisor is over $\bar{k}_i$, and the effective representative is taken over $\bar{k}_i$. **Step 3** of the proof of Lemma 10.11 uses the prime divisor E over k_i, whose base change may be reducible, and a representative over k_i.
(4) In the paragraph "A bounded principalization multiple" of the proof of [OAI26b, Lemma C.2], the multiple is bounded on a geometric generic fiber and the trivialization descends to the field of definition of the generic fiber by flat base change. **Step 4** of the proof of Lemma 10.11 bounds the multiple q_0 over an algebraic closure of the function field $\mathcal{K}_{\tau_i}$ of the cutoff over k_i by [OAI26b, Lemma B.5], and descends the trivialization to $\mathcal{K}_{\tau_i}$ by flat base change.
(5) At the definition of h before [OAI26b, (C.33)], which enters [OAI26b, (C.35)], the principal divisor subtracted from the bounded multiple of ℓ is over $\bar{k}_i$. **Step 4** of the proof of Lemma 10.11 subtracts a principal divisor over k_i from $q_0\ell$, and **Step 5** uses that the values of ℓ° at a prime divisor over $V_{i,\bar{k}_i}$ are those at the restricted prime divisor over V_i.
(6) In the proof of [OAI26b, Lemma C.2], [OAI26b, (C.25)] uses the proportionality on a Mori contraction of relative Picard number one, and the third paragraph of [OAI26b, Subsection C.1] restricts it to a geometric generic fiber. Over k_i the Picard number may increase after base change. **Step 3** of the proof of Lemma 10.11 checks the proportionality on curves by Lemma 10.8(2) and passes to $\mathbb{Q}$-linear equivalence by Lemma 10.8(5).

Lemma 10.8. *Let k be a field finitely generated over $\mathbb{C}$ and Y a geometrically integral normal projective variety over k. Then the following hold.*

(1) For any coherent sheaf $\mathcal{F}$ on Y, $H^0(Y_{\bar{k}}, \mathcal{F}_{\bar{k}}) = H^0(Y, \mathcal{F}) \otimes_k \bar{k}$. In particular, if N is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y such that $N_{\bar{k}}$ is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-divisor, then $N \sim_{\mathbb{Q}} N^+$ over k for some $\mathbb{Q}$-divisor $N^+ \geq 0$ on Y.
(2) Let $g : Y \to Z$ be a contraction over k and N a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y such that $(N \cdot C) = 0$ for any curve C on Y with $\dim g(C) = 0$. Then $(N_{\bar{k}} \cdot C') = 0$ for any curve C' on $Y_{\bar{k}}$ with $\dim g_{\bar{k}}(C') = 0$.
(3) If $Y_{\bar{k}}$ is of Fano type, then there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ on Y such that (Y, Δ) is klt and $-(K_Y + \Delta)$ is ample.
(4) If $Y_{\bar{k}}$ is of Fano type and any prime divisor on Y is $\mathbb{Q}$-Cartier, then for any $\mathbb{Q}$-divisor N on Y, an N-MMP over k exists and terminates, either with a variety on which the strict transform of N is semi-ample, or with a Mori fiber space over k.
(5) If $Y_{\bar{k}}$ is of Fano type and N is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y with $N_{\bar{k}} \equiv 0$, then $N \sim_{\mathbb{Q}} 0$.

Proof. (1) The first statement is flat base change [Har77, Proposition III.9.3]. For the second, pick $r > 0$ such that rN is Cartier and $H^0(Y_{\bar{k}}, \mathcal{O}_{Y_{\bar{k}}}(rN_{\bar{k}})) \neq 0$. By the first statement, $H^0(Y, \mathcal{O}_Y(rN)) \neq 0$.

(2) Let C' be such a curve. There exists a finite Galois extension k'/k such that C' is the base change of a curve $C'_{k'}$ on $Y_{k'}$. Let C be the image of $C'_{k'}$ in Y. Then $\dim g(C) = 0$, and since $\operatorname{char} k = 0$, $C_{k'}$ is the reduced sum of the r distinct $\mathrm{Gal}(k'/k)$-conjugates of $C'_{k'}$. Since N is over k, $(N_{k'} \cdot C'') = (N_{k'} \cdot C'_{k'})$ for any conjugate C'' of $C'_{k'}$. Thus

$$0 = (N \cdot C) = (N_{k'} \cdot C_{k'}) = r(N_{k'} \cdot C'_{k'}) = r(N_{\bar{k}} \cdot C').$$

(3) Since $Y_{\bar{k}}$ is of Fano type, there exists a $\mathbb{Q}$-divisor $\bar{\Delta} \geq 0$ on $Y_{\bar{k}}$ such that $(Y_{\bar{k}}, \bar{\Delta})$ is klt and $-(K_{Y_{\bar{k}}} + \bar{\Delta})$ is ample (by [PS09, Lemma-Definition 2.6], in the definition of Fano type we may require $-(K_{Y_{\bar{k}}} + \bar{\Delta})$ to be ample, and then $\bar{\Delta}$ to have rational coefficients since klt and ampleness are open conditions). Since $\bar{\Delta}$ has finitely many components and coefficients, there exist a finite Galois extension k'/k in $\bar{k}$ and a $\mathbb{Q}$-divisor $\Delta' \geq 0$ on $Y_{k'}$ with $\Delta'_{\bar{k}} = \bar{\Delta}$, and $m(K_{Y_{k'}} + \Delta')$ is Cartier for some positive integer m, since invertibility descends along the faithfully flat morphism $Y_{\bar{k}} \to Y_{k'}$. Then $(Y_{k'}, \Delta')$ is klt by Lemma 3.3(3) applied to $\bar{k}/k'$, and $-(K_{Y_{k'}} + \Delta')$ is ample by Serre's criterion [Har77, Proposition III.5.3] and flat base change [Har77, Proposition III.9.3]. The sum of the $\mathrm{Gal}(k'/k)$-conjugates of Δ' divided by $[k' : k]$ is

Galois invariant, hence equal to $\Delta_{k'}$ for a $\mathbb{Q}$-divisor $\Delta \geq 0$ on Y. Since $K_{Y_{k'}}$ is Galois invariant, $-(K_{Y_{k'}} + \Delta_{k'})$ is a mean of ample $\mathbb{Q}$-divisors, and $a(P, Y_{k'}, \Delta_{k'})$ is the mean of the positive numbers $a(P, Y_{k'}, \Delta'')$ over the conjugates Δ'' of Δ', for any prime divisor P over $Y_{k'}$. Thus $(Y_{k'}, \Delta_{k'})$ is klt with ample $-(K_{Y_{k'}} + \Delta_{k'})$. By Lemma 3.3(3), (Y, Δ) is klt, and $-(K_Y + \Delta)$ is ample by [Har77, Exercise III.5.7(d)], applied to a Cartier multiple of $-(K_Y + \Delta)$ and the finite surjective morphism $Y_{k'} \to Y$.

(4) Let Δ be as in (3). For $0 < \epsilon \ll 1$, $\epsilon N - (K_Y + \Delta)$ is ample, and we let $H \geq 0$ be a general $\mathbb{Q}$-divisor over k with $H \sim_{\mathbb{Q}} \epsilon N - (K_Y + \Delta)$. Then $(Y, \Delta + H)$ is klt, Y is $\mathbb{Q}$-factorial over k, and $K_Y + \Delta + H \sim_{\mathbb{Q}} \epsilon N$, so an N-MMP over k is a $(K_Y + \Delta + H)$-MMP over k, and the cone and contraction theorems over k for it are [BP24, Theorems 3.1 and 3.3]. A contraction over k given by [BP24, Theorems 3.1 and 3.3] contracts, after base change, a $\mathrm{Gal}(\bar{k}/k)$-invariant extremal face of $\overline{\mathrm{NE}}(Y_{\bar{k}})$. The relevant rings of sections are finitely generated over $\bar{k}$ by [BCHM10, Corollary 1.1.9] and (3), hence over k by (1), so the flips exist.

Since $Y_{\bar{k}}$ is of Fano type, there is a small $\mathbb{Q}$-factorial modification $Y^q \to Y_{\bar{k}}$ [BCHM10, Corollary 1.4.3] and the second paragraph after it. Then Y^q is of Fano type, since $Y_{\bar{k}}$ is of Fano type and $Y^q \to Y_{\bar{k}}$ is small: the strict transform of a klt boundary $\bar{\Delta}$ of $Y_{\bar{k}}$ is a klt boundary Δ^q with $K_{Y^q} + \Delta^q$ the pullback of $K_{Y_{\bar{k}}} + \bar{\Delta}$, so $-(K_{Y^q} + \Delta^q)$ is big and nef, and Y^q carries a klt boundary with ample anti-log canonical divisor [PS09, Lemma-Definition 2.6]. Hence Y^q is a Mori dream space [BCHM10, Corollary 1.3.2]. Let $Y = Y_0 \dashrightarrow Y_1 \dashrightarrow \cdots$ be an N-MMP over k. For each j, the birational map $Y^q \dashrightarrow Y_{j,\bar{k}}$ is a rational contraction of Y^q, and Y^q has only finitely many rational contractions up to isomorphism [HK00, Proposition 1.11], see also [BCHM10, Corollary 1.1.5(2)]. Let Δ_j and H_j be the strict transforms of Δ and H on Y_j. By [KM98, Lemma 3.38], the log discrepancies of $(Y_j, \Delta_j + H_j)$ do not decrease along the MMP, and each step strictly increases some of them, so the models $Y_{j,\bar{k}}$ are pairwise distinct. Thus the MMP terminates.

It ends either with a Mori fiber space over k, or with a variety Y_m on which the strict transform N_m of N is nef. If the MMP ends with Y_m, then $Y_{m,\bar{k}}$ is of Fano type, since $Y_{\bar{k}} \dashrightarrow Y_{m,\bar{k}}$ does not extract divisors and a normal projective variety is of Fano type if and only if it carries a big klt boundary Θ with $\mathbb{Q}$-linearly trivial log canonical divisor (by [PS09, Lemma-Definition 2.6(i)] and Kodaira's lemma [KM98, Lemma 2.60]), whose pushforward is again such a boundary by the negativity lemma. Thus (3) gives a $\mathbb{Q}$-divisor $\Delta'_m \geq 0$ on Y_m such that (Y_m, Δ'_m) is klt and $-(K_{Y_m} + \Delta'_m)$ is ample. Thus $N_{m,\bar{k}}$ is semi-ample by [KM98, Theorem 3.3], and N_m is semi-ample by (1).

(5) Pick $r > 0$ such that rN is Cartier, and let Δ be as in (3). Then $(Y_{\bar{k}}, \Delta_{\bar{k}})$ is a proper klt pair with $\Delta_{\bar{k}} \geq 0$, $\pm rN_{\bar{k}}$ is nef, and

$$t(\pm rN_{\bar{k}}) - (K_{Y_{\bar{k}}} + \Delta_{\bar{k}}) \equiv -(K_{Y_{\bar{k}}} + \Delta_{\bar{k}})$$

is ample for any real number $t > 0$. By [KM98, Theorem 3.3], there is a positive integer b such that $|\pm brN_{\bar{k}}|$ is basepoint free, so $H^0(Y_{\bar{k}}, \mathcal{O}(\pm brN_{\bar{k}})) \neq 0$. By (1) there exist non-zero $s \in H^0(Y, \mathcal{O}_Y(brN))$ and $s' \in H^0(Y, \mathcal{O}_Y(-brN))$. Since Y is geometrically integral and projective, $ss' \in H^0(Y, \mathcal{O}_Y) = k$ is a non-zero constant. Thus s vanishes nowhere, and $brN \sim 0$. □

Construction 10.9. Notation and conditions as in Set-up 10.6. Fix i and a rational number $0 < \beta \leq 1$. Let $\mathcal{P}$ be the set of exceptional prime divisors P over $V_{i,\bar{k}_i}$ with $a_i(P) = 0$ and $\mathbf{D}_i(P) \leq 1$. Since a_i takes values in $\mathbb{Z}_{\geq 0}$, $\mathcal{P}$ is the set of exceptional prime divisors at which the log discrepancy function $a_i + \mathbf{D}_i/2$ of the generalized klt pair $(V_i, B_i - D_i/2, \mathbf{M}_i + \mathbf{D}_i/2)$ is $\leq 1/2$. Thus $\mathcal{P}$ is finite, and it is invariant under $\mathrm{Gal}(\bar{k}_i/k_i)$ by Lemma 3.3(2). There exists a projective birational morphism $f : U \to V_i$ over k_i such that U is normal, any prime divisor on U is $\mathbb{Q}$-Cartier, $U_{\bar{k}_i}$ is of Fano type, and the exceptional prime divisors of $f_{\bar{k}_i}$ are exactly the elements of $\mathcal{P}$. To construct f, take a log resolution over k_i and a boundary over k_i as in [OAI26b, Subsection A.3], which is possible since $\mathcal{P}$ is Galois invariant. The extraction of $\mathcal{P}$ over V_i [BCHM10, Corollary 1.4.3] is obtained by a relative MMP with scaling over V_i [BCHM10,

Theorem 1.2 and Corollary 1.4.2], which we run over k_i, with the cone and contraction theorems over k_i from [BP24, Theorems 3.1 and 3.3] (the flips exist since finite generation descends by Lemma 10.8(1), and the MMP with scaling terminates: after base change, small $\mathbb{Q}$-factorial modifications of the models of the MMP are weak log canonical models over $V_{i,\bar{k}_i}$ of the boundary plus s times the scaling divisor for s in a fixed segment, so there are finitely many of them [BCHM10, Theorem E], while two models admitting a common small $\mathbb{Q}$-factorial modification have equal log discrepancies, and the log discrepancies do not decrease along the MMP, each step strictly increasing some of them [KM98, Lemma 3.38]). After base change, each step contracts a Galois-invariant extremal face. Fix such an f, and let $D_U := f^*D_i$.

Let I_β be the set of prime divisors P over $V_{i,\bar{k}_i}$ with $a_i(P) = 0$ and $\mathbf{D}_i(P) \le \beta$. Any element of I_β is either in $\mathcal{P}$ or a component of $B_{i,\bar{k}_i}$ with coefficient one, so I_β is a finite set of prime divisors on $U_{\bar{k}_i}$. Let

$$\Upsilon_\beta := D_U - \sum_{P \in I_\beta} \mathbf{D}_i(P)P.$$

By Lemma 3.3(2), the coefficients are constant on Galois orbits, so Υ_β is a $\mathbb{Q}$-divisor on U, and $\Upsilon_\beta \ge 0$. Run a Υ_β-MMP $U \dashrightarrow U'$ over k_i (Lemma 10.8(4)). Since $\Upsilon_\beta \ge 0$, the strict transform Υ'_β is semi-ample on U'. Let $p : U' \to W_\beta$ be the contraction defined by Υ'_β, and let $D_W \ge 0$ be the $\mathbb{Q}$-divisor on W_β with $p^*D_W = \Upsilon'_\beta$. It exists because, if $\Upsilon'_\beta \sim_{\mathbb{Q}} p^*D^0$ for an ample $\mathbb{Q}$-divisor D^0, then the principal $\mathbb{Q}$-divisor $\Upsilon'_\beta - p^*D^0$ is zero on the generic fiber of p, hence is the pullback of a principal $\mathbb{Q}$-divisor on W_β. Let $\boldsymbol{\Upsilon}$ be the Cartier $\boldsymbol{b}$-divisor of Υ'_β, and let $\mathcal{K}_\beta := k_i(W_\beta) \subset k_i(U') = k_i(V_i)$. Since W_β is the ample model of Υ_β, neither W_β nor $\mathcal{K}_\beta$ depends on the MMP. We call W_β the *cutoff model* and $\mathcal{K}_\beta$ the *cutoff field*.

By the negativity lemma, $0 \le \boldsymbol{\Upsilon} \le \mathbf{D}_i$, and $\boldsymbol{\Upsilon}(P) \le \operatorname{mult}_P \Upsilon_\beta$ for any prime divisor P on $U_{\bar{k}_i}$.

Lemma 10.10. *Notation and conditions as in Set-up 10.6, and for any rational number $0 < \beta \le 1$ let I_β, W_β, D_W, and $\mathcal{K}_\beta$ be as in Construction 10.9. Then the following hold.*

1. *For any rational number $0 < \beta \le 1$, the set I_β is finite, and the cutoff field $\mathcal{K}_\beta \subset k_i(V_i)$ is algebraically closed in $k_i(V_i)$.*
2. *Let $0 < \beta_i \le 1$ be rational numbers such that $\dim W_{\beta_i} > 0$ for any i. Then, possibly passing to a subsequence, there exist real numbers $\delta, C > 0$, which depend only on the sequence of (V_i, a_i, D_i, β_i), such that the following hold for any i, where $\beta = \beta_i$ and $e := \dim W_\beta$.*
   - (a) *There exists a projective birational morphism $\widehat{W} \to W_\beta$ over k_i such that any prime divisor on $\widehat{W}$ is $\mathbb{Q}$-Cartier, $\widehat{W}_{\bar{k}_i}$ is of Fano type and δ-lc, and any positive coefficient of the pullback of D_W/β to $\widehat{W}$ is $\ge \delta$.*
   - (b) *The k_i-vector space*

     $$\mathcal{V} := \{\phi \in \mathcal{K}_\beta^* \mid \operatorname{div}(\phi) + CD_W/\beta \ge 0\} \cup \{0\}$$

     generates $\mathcal{K}_\beta$ over k_i and contains elements $\phi_1, \dots, \phi_e$ algebraically independent over k_i.
   - (c) *There exists a nef Cartier $\boldsymbol{b}$-divisor $\mathbf{H}$ over W_β, over k_i, whose coefficients at prime divisors have denominators bounded by C, such that for any prime divisor T over $V_{i,\bar{k}_i}$:*
     - (i) *$\mathbf{H}(T) \le C\mathbf{D}_i(T)/\beta$ if $a_i(T) = 0$, and $\mathbf{H}(T) \le -1$ if moreover T is non-trivial on $\mathcal{K}_\beta$ and $\operatorname{ord}_T \phi_j \ge 0$ for any j.*
     - (ii) *$\mathbf{H}(T) \le 0$ if $\mathbf{D}_i(T) = 0$ and $a_i(T) \le 1$.*
     - (iii) *$\mathbf{H}(T) \ge -C\eta_i\mathbf{D}_i(T)/\beta$ if $a_i(T) = 0$ and (MO_k) holds at T.*

   Here $\mathbf{H}(T)$ is the coefficient at T of the pullback of $\mathbf{H}$ along the dominant rational map $V_{i,\bar{k}_i} \dashrightarrow W_{\beta,\bar{k}_i}$.

(3) *If $0 < \beta_i \le 1$ are rational numbers, which may depend on i, such that* (MO_k) *holds at any element of I_{β_i} for any i, then any element of I_{β_i} is trivial on $\mathcal{K}_{\beta_i}$ for any sufficiently large i.*

Lemma 10.10(2) is [OAI26b, Lemma C.1] over k_i, with [OAI26b, (C.5)] replaced by (MO_k).

Proof of Lemma 10.10. We use Construction 10.9. The set I_β is finite by construction, and $\mathcal{K}_\beta$ is algebraically closed in $k_i(V_i)$ since p is a contraction and $\operatorname{char} k_i = 0$. This proves (1). The rest of the proof is long, so we divide it into several steps. In **Steps 1–3**, the rational numbers β_i are as in (2), $\beta = \beta_i$, and $e = \dim W_\beta$. Remark 10.7 lists the places where the argument differs from [OAI26b].

Step 1. In this step we prove (2)(a) and (2)(b). By Lemma 3.2, the results of [OAI26b] hold over $\bar{k}_i$. Let

$$\mathbf{d} := \frac{\mathbf{D}_i - \boldsymbol{\Upsilon}}{\beta} + \boldsymbol{\Upsilon},$$

which is [OAI26b, (C.6)]. Then $\mathbf{d}$ is a $\boldsymbol{b}$-divisor over k_i, $\mathbf{d} \ge 0$, $\mathbf{d} > 0$ at the prime divisors with $a_i = 0$, and $a_i + t\mathbf{d}$ has trace ≥ 0 for $0 \le t \le 1$. By Lemma 3.3(2), the values of a_i and $\mathbf{d}$ at a prime divisor over $V_{i,\bar{k}_i}$ are their values at the restricted prime divisor over V_i. By [OAI26b, Lemma A.3] for $p_{\bar{k}_i}$, the function a_i, and the direction $\mathbf{d}$, there exists a fixed $t_1 > 0$ such that, for $0 < t \le t_1$, the discriminant of $a_i + t\mathbf{d}$ on $W_{\beta,\bar{k}_i}$, that is, the log discrepancy function of the induced generalized pair, is $a' + t\mathbf{d}'$, where the denominators of the values of a' are bounded, and $a' + t\mathbf{d}'$ is generalized klt for $t > 0$.

By [OAI26b, (C.7)], if prime divisors E_i over $W_{\beta,\bar{k}_i}$ satisfy $a'(E_i) = 0$ and $\mathbf{d}'(E_i) \to 0$, and positive numbers κ_i tend to 0, then, possibly passing to a subsequence, there exist prime divisors T_i over $V_{i,\bar{k}_i}$ such that ord_{T_i} restricts to $e_{T_i} \operatorname{ord}_{E_i}$ on $\bar{k}_i(W_\beta)$ for a positive integer e_{T_i}, $a_i(T_i) = 0$, $\mathbf{d}(T_i) \to 0$, and $e_{T_i}\kappa_i \to 0$. By Lemma 3.3(2), the restriction of T_i is a prime divisor over V_i.

Fix a sufficiently small rational number $0 < t \le t_1$. We show that there is no sequence of prime divisors E over $W_{\beta,\bar{k}_i}$ with $(a' + t\mathbf{d}')(E) \to 0$ and $0 < \operatorname{mult}_E D_W/\beta \to 0$ [OAI26b, (C.8)]. Suppose there is such a sequence. Since the denominators of the values of a' are bounded, $a'(E) = 0$ for $i \gg 0$. By [OAI26b, (C.7)] with $\kappa_i = \operatorname{mult}_E D_W/\beta$, there exist prime divisors T over $V_{i,\bar{k}_i}$ with $a_i(T) = 0$ and

$$\frac{\mathbf{D}_i(T)}{\beta} \le \mathbf{d}(T) + \frac{e_T \operatorname{mult}_E D_W}{\beta} \to 0,$$

so $T \in I_\beta$ for $i \gg 0$. However, $\boldsymbol{\Upsilon}(T) = e_T \operatorname{mult}_E D_W > 0$, which contradicts $\boldsymbol{\Upsilon}(P) \le \operatorname{mult}_P \Upsilon_\beta = 0$ for $P \in I_\beta$.

Thus, extracting the exceptional prime divisors over $W_{\beta,\bar{k}_i}$ at which $a' + t\mathbf{d}'$ is less than a fixed $\delta_* > 0$, we obtain $\widehat{W}$ such that, for a fixed $\delta > 0$, $\widehat{W}_{\bar{k}_i}$ is δ-lc and any positive coefficient of the pullback of D_W/β to $\widehat{W}$ is $\ge \delta$ ($\widehat{W}_{\bar{k}_i}$ differs from the extraction in [OAI26b] by a small birational map, so its δ-lc property and Fano type follow from the comparison of the log discrepancies of $\widehat{W}_{\bar{k}_i}$ with $a' + t\mathbf{d}'$ in [OAI26b, Lemma A.3], the trace being over k_i. The gap and the lower bound on the coefficients come from [OAI26b, (C.7)–(C.8)]). The set of extracted prime divisors is determined by a' and $\mathbf{d}'$, which are over k_i, so it is Galois invariant, and $\widehat{W}$ exists over k_i as in Construction 10.9. This proves (2)(a).

For (2)(b), let C be the constant of [OAI26b, Lemma C.1] for the base change to $\bar{k}_i$, so that the space $\mathcal{V}_{\bar{k}_i}$ of $\phi \in \bar{k}_i(W_\beta)^*$ with $\operatorname{div}(\phi) + CD_{W,\bar{k}_i}/\beta \ge 0$, together with 0, generates $\bar{k}_i(W_\beta)$. Since the pullback of a prime divisor on W_β to $W_{\beta,\bar{k}_i}$ is reduced (Lemma 3.3(1)), the round-down of CD_W/β base changes to the round-down of $CD_{W,\bar{k}_i}/\beta$, so the pullback of the reflexive sheaf $\mathcal{O}_{W_\beta}(\lfloor CD_W/\beta \rfloor)$ to $W_{\beta,\bar{k}_i}$ is $\mathcal{O}_{W_{\beta,\bar{k}_i}}(\lfloor CD_{W,\bar{k}_i}/\beta \rfloor)$ (both sheaves are reflexive [Mat89, Theorem 7.11], they are equal outside the preimage of the singular locus of W_β, which has codimension at least two since W_β is normal [Mat89, Theorem 15.1(ii)], so they are equal

[Sta, Tag 0EBJ]), and Lemma 10.8(1) applied to $\mathcal{O}_{W_\beta}(\lfloor CD_W/\beta \rfloor)$ shows that $\mathcal{V} \otimes_{k_i} \bar{k}_i = \mathcal{V}_{\bar{k}_i}$. Let $\mathcal{K}' \subset \mathcal{K}_\beta$ be the subfield generated by $\mathcal{V}$ over k_i. Since V_i is geometrically integral, $\mathcal{K}_\beta \otimes_{k_i} \bar{k}_i$ is a domain, and its fraction field $\bar{k}_i(W_\beta)$ is generated by $\mathcal{V} \otimes_{k_i} \bar{k}_i$. Thus $\mathcal{K}_\beta$ is algebraic over $\mathcal{K}'$, and $[\mathcal{K}_\beta : \mathcal{K}'] = 1$ since the degree does not change after $\otimes_{k_i} \bar{k}_i$. Hence $\mathcal{V}$ generates $\mathcal{K}_\beta$, and we pick $\phi_1, \dots, \phi_e \in \mathcal{V}$ algebraically independent over k_i.

Step 2. In this step we construct $\mathbf{H}$ and prove (2)(c)(i) and (2)(c)(ii). Let $\mathbf{H}$ be the $\boldsymbol{b}$-divisor of [OAI26b, Subsection C.1] attached to $\phi_1, \dots, \phi_e$: for a prime divisor E over $W_{\beta,\bar{k}_i}$,

$$\mathbf{H}(E) = a'(E) - 1 - \operatorname{ord}_E(d\phi_1 \wedge \cdots \wedge d\phi_e),$$

where a' is the log discrepancy function of the generalized pair induced on W_β by $(V_i, B_i, \mathbf{M}_i)$ and p [OAI26b, (C.11)]. [OAI26b, (C.11)] defines $\mathbf{H}$ for any algebraically independent elements of $\mathcal{V} \otimes_{k_i} \bar{k}_i$, in particular for $\phi_1, \dots, \phi_e$. Since a_i and the ϕ_j are over k_i, $\mathbf{H}$ is over k_i, nef, and Cartier with bounded denominators.

We prove (2)(c)(i) and (2)(c)(ii). Let $\omega := d\phi_1 \wedge \cdots \wedge d\phi_e$, and let T be a prime divisor over $V_{i,\bar{k}_i}$. If T is trivial on $\mathcal{K}_\beta$, then $\mathbf{H}(T) = 0$, and (2)(c)(i) and (2)(c)(ii) hold at T. Otherwise, the restriction of ord_T to $\mathcal{K}_\beta$ is $e_T \operatorname{ord}_E$ for a prime divisor E over $W_{\beta,\bar{k}_i}$ and a positive integer e_T, so

$$\mathbf{H}(T) = e_T \mathbf{H}(E), \qquad \boldsymbol{\Upsilon}(T) = e_T \operatorname{mult}_E D_W.$$

By the definition of the discriminant of the induced generalized pair, $e_T a'(E) \le a_i(T)$, so $a'(E) = 0$ if $a_i(T) = 0$. Suppose $a_i(T) = 0$. Since $\phi_j \in \mathcal{V}$, $\operatorname{ord}_E \phi_j \ge -C \operatorname{mult}_E D_W/\beta$, so, after enlarging C, $1 + \operatorname{ord}_E \omega \ge -C \operatorname{mult}_E D_W/\beta$. Thus

$$\mathbf{H}(T) = -e_T(1 + \operatorname{ord}_E \omega) \le \frac{C\boldsymbol{\Upsilon}(T)}{\beta} \le \frac{C\mathbf{D}_i(T)}{\beta}.$$

If moreover $\operatorname{ord}_T \phi_j \ge 0$ for any j, then ω is regular on the discrete valuation ring of E, so $1 + \operatorname{ord}_E \omega \ge 1$ and $\mathbf{H}(T) \le -e_T \le -1$. This proves (2)(c)(i). For (2)(c)(ii), suppose $\mathbf{D}_i(T) = 0$ and $a_i(T) \le 1$. Since $\boldsymbol{\Upsilon} \le \mathbf{D}_i$, $\operatorname{mult}_E D_W = 0$, so $\operatorname{ord}_E \phi_j \ge 0$ for any j and $1 + \operatorname{ord}_E \omega \ge 1$. Thus

$$\mathbf{H}(T) = e_T a'(E) - e_T(1 + \operatorname{ord}_E \omega) \le a_i(T) - e_T \le 0.$$

Step 3. In this step we prove (2)(c)(iii). Let λ be a rational number with $0 < \lambda \le C$. Let $W' \to W_\beta$ be a higher model over k_i on which $\mathbf{H}$ descends, and let $D_{W'}$ be the pullback of D_W. Since D_W is ample, $\mathbf{H}_{W'} + \lambda D_{W'}/\beta$ is nef and big. By Lemma 10.8(1), there exists a $\mathbb{Q}$-divisor $\Sigma \ge 0$ on W' such that $\Sigma \sim_{\mathbb{Q}} \mathbf{H}_{W'} + \lambda D_{W'}/\beta$ over k_i. Let Ξ be the pullback to V_i of the principal $\mathbb{Q}$-divisor $\Sigma - \mathbf{H}_{W'} - \lambda D_{W'}/\beta$. Then $\Xi = \frac{1}{r}\operatorname{div}(\psi)$ for some $r > 0$ and $\psi \in \mathcal{K}_\beta^*$, and since the pullback of D_W to V_i is $\boldsymbol{\Upsilon}$,

$$\operatorname{mult}_T \Xi \ge -\mathbf{H}(T) - \lambda \boldsymbol{\Upsilon}(T)/\beta$$

for any prime divisor T over $V_{i,\bar{k}_i}$. Let

$$\Gamma := D_i + \frac{\beta}{2C}\Xi.$$

Then $\Gamma \sim_{\mathbb{Q}} D_i$ over k_i. We show $\Gamma \ge 0$. Let Q be a prime divisor on $V_{i,\bar{k}_i}$. Since $a_i + \mathbf{D}_i$ has trace ≥ 0 on V_i, $a_i(Q) \in \{0, 1\}$, and $\mathbf{D}_i(Q) = 0$ if $a_i(Q) = 1$. If $a_i(Q) = 0$, then (2)(c)(i) and $\boldsymbol{\Upsilon} \le \mathbf{D}_i$ imply

$$\operatorname{mult}_Q \Gamma \ge \mathbf{D}_i(Q) - \frac{\beta}{2C}\Big(\frac{C\mathbf{D}_i(Q)}{\beta} + \frac{\lambda \mathbf{D}_i(Q)}{\beta}\Big) \ge 0.$$

If $a_i(Q) = 1$, then $\boldsymbol{\Upsilon}(Q) = 0$ and $\mathbf{H}(Q) \le 0$ by (2)(c)(ii), so $\operatorname{mult}_Q \Gamma \ge 0$. Now let T be a prime divisor over $V_{i,\bar{k}_i}$ with $a_i(T) = 0$ at which (MO_k) holds. Since Ξ is principal,

$$\operatorname{mult}_T \Gamma = \mathbf{D}_i(T) + \frac{\beta}{2C} \operatorname{mult}_T \Xi,$$

so (10.1) implies

$$-\mathbf{H}(T) - \frac{\lambda \boldsymbol{\Upsilon}(T)}{\beta} \le \operatorname{mult}_T \Xi \le \frac{2C\eta_i \mathbf{D}_i(T)}{\beta}.$$

Letting λ tend to 0, $\mathbf{H}(T) \ge -2C\eta_i \mathbf{D}_i(T)/\beta$. Replacing C with $2C$, (2)(c)(iii) holds.

Step 4. In this step we prove (3). If $\dim W_{\beta_i} = 0$, then $\mathcal{K}_{\beta_i}$ is algebraic over k_i and algebraically closed in $k_i(V_i)$, so $\mathcal{K}_{\beta_i} = k_i$ since V_i is geometrically integral, and any prime divisor is trivial on $\mathcal{K}_{\beta_i}$. Suppose that, for infinitely many i, $\dim W_{\beta_i} > 0$ and some $P_i \in I_{\beta_i}$ is non-trivial on $\mathcal{K}_{\beta_i}$. We apply (2) to the sequence β_i along the subsequence of such i. Since $\boldsymbol{\Upsilon}(P_i) = 0$ and $\operatorname{div}(\phi_j) + CD_W/\beta_i \ge 0$, we have $\operatorname{ord}_{P_i} \phi_j \ge -C\boldsymbol{\Upsilon}(P_i)/\beta_i = 0$ for any j. Thus (2)(c)(i) implies $\mathbf{H}(P_i) \le -1$. Since (MO_k) holds at P_i and $\mathbf{D}_i(P_i) \le \beta_i$, (2)(c)(iii) implies $-1 \ge \mathbf{H}(P_i) \ge -C\eta_i$, which contradicts $\lim_i \eta_i = 0$. □

Lemma 10.11. *Notation and conditions as in Set-up 10.6. For any i, let $\ell_i := \frac{1}{r_i}\operatorname{div}(\varphi_i)$ for a positive integer r_i and $\varphi_i \in k_i(V_i)^*$, and let $\Gamma_i \ge 0$ be a $\mathbb{Q}$-divisor on V_i over k_i with $\Gamma_i \sim_{\mathbb{Q}} c_i L_i D_i$ over k_i for a real number $0 \le c_i \le C_0$ such that $c_i L_i \in \mathbb{Q}$, where C_0 does not depend on i. Assume that for any prime divisor P on $V_{i,\bar{k}_i}$ with $a_i(P) > 0$ there exists an integer $p_{i,P}$ such that $|p_{i,P} - \operatorname{mult}_P \ell_i| \le \operatorname{mult}_P \Gamma_i$. Then there exist a positive integer q and a real number $C_1 > 0$ such that, for any subsequence and any prime divisors P_i over $V_{i,\bar{k}_i}$ with $a_i(P_i) = 0$ and $\lim_i L_i \mathbf{D}_i(P_i) = 0$,*

$$\operatorname{dist}\left(q \operatorname{mult}_{P_i} \ell_i, \mathbb{Z}\right) \le C_1 L_i \mathbf{D}_i(P_i)$$

for any sufficiently large i.

Proof. To simplify notation, we write a, D, $\mathbf{D}$, L, η, ℓ, Γ, and c for a_i, D_i, $\mathbf{D}_i$, L_i, η_i, ℓ_i, Γ_i, and c_i, and C denotes positive constants that depend only on the sequence. For $\mathbb{Q}$-divisors N_1, N_2 we write $N_1 \le_{\mathbb{Q}} N_2$ if $N_2 - N_1$ is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-divisor. The proof is long, so we divide it into several steps. Remark 10.7 lists the places where the argument differs from [OAI26b].

Step 1. In this step we bound Γ at small prime divisors and fix conventions on Galois orbits. By Lemma 3.3(2), for a prime divisor P over $V_{i,\bar{k}_i}$, ord_P restricts to ord_{P_0} for a prime divisor P_0 over V_i, with ramification index one, and $a(P) = a(P_0)$. Thus a, $\mathbf{D}$, $\operatorname{mult}\ell$, and $\operatorname{mult}\Gamma$ are constant on Galois orbits of prime divisors over $V_{i,\bar{k}_i}$, and the integers $p_{i,P}$ and any rounding below can be fixed once for each prime divisor over V_i. Lemma 3.3(2) also holds with k_i replaced by the function field of the base of any contraction over k_i, which is finitely generated over $\mathbb{C}$, so the integers and roundings can be fixed on Galois orbits there as well. If $c > 0$, then $\Gamma/(cL) \ge 0$ and $\Gamma/(cL) \sim_{\mathbb{Q}} D$ over k_i. If $c = 0$, then $\Gamma \ge 0$ is a $\mathbb{Q}$-linearly trivial $\mathbb{Q}$-divisor on the geometrically integral projective variety V_i, so $\Gamma = 0$. By Set-up 10.6(9), for any subsequence and any prime divisors P_i with $a(P_i) = 0$ and $\lim_i L\mathbf{D}(P_i) = 0$,

$$\operatorname{mult}_{P_i} \Gamma \le C_0 L(1+\eta)\mathbf{D}(P_i) \tag{10.2}$$

for any sufficiently large i.

Step 2. In this step we assume that the lemma fails and fix a threshold τ_i. Suppose the lemma fails. Since $\operatorname{dist}(q'x, \mathbb{Z}) \le (q'/q)\operatorname{dist}(qx, \mathbb{Z})$ when q divides q', a diagonal argument over $q = j!$ yields a subsequence and prime divisors P_i over $V_{i,\bar{k}_i}$ with $a(P_i) = 0$, $\beta_i := \mathbf{D}(P_i)$, which is rational since D_i is a $\mathbb{Q}$-divisor, $\lim_i L\beta_i = 0$, and

$$\lim_i \frac{\operatorname{dist}(q \operatorname{mult}_{P_i} \ell, \mathbb{Z})}{L\beta_i} = +\infty \quad \text{for any positive integer } q. \tag{10.3}$$

Among such subsequences, pick one along which $\operatorname{tr.deg}_{k_i} \mathcal{K}_{\beta_i}$ is constant and maximal. If, for infinitely many i, (MO_k) failed at some $Q_i \in I_{\beta_i}$, then $a(Q_i) = 0$ and $L\mathbf{D}(Q_i) \le L\beta_i \to 0$ would contradict Set-up 10.6(9). Thus, by Lemma 10.10(3), P_i is trivial on $\mathcal{K}_{\beta_i}$ for $i \gg 0$. Since P_i is non-trivial on $k_i(V_i)$, $\mathcal{K}_{\beta_i} \ne k_i(V_i)$. The fields $\mathcal{K}_\beta$ decrease as β increases and change only at the values $\mathbf{D}(Q)$, $Q \in I_1$. Let τ_i be the smallest number in $(0, \beta_i]$ with $\mathcal{K}_{\tau_i} = \mathcal{K}_{\beta_i}$. Then $\lim_i L\tau_i = 0$.

Replacing ℓ, Γ, c, C_0 and the integers $p_{i,P}$ by their products with a fixed positive integer M keeps the hypotheses of the lemma and (10.3), since (10.3) holds for any positive integer q. We show that, after replacing ℓ, Γ, c, and C_0 by a fixed multiple, for any subsequence and any prime divisors Q_i over $V_{i,\bar{k}_i}$ with $a(Q_i)=0$ and $\mathbf{D}(Q_i)<\tau_i$,

$$\operatorname{dist}(\operatorname{mult}_{Q_i}\ell,\mathbb{Z})\le CL\mathbf{D}(Q_i) \tag{10.4}$$

for $i\gg 0$. Otherwise, the diagonal argument above yields such Q_i satisfying (10.3) with β_i replaced by $\mathbf{D}(Q_i)$. Since $\mathbf{D}(Q_i)<\tau_i$, $\mathcal{K}_{\mathbf{D}(Q_i)}\supsetneq\mathcal{K}_{\tau_i}=\mathcal{K}_{\beta_i}$, and both fields are algebraically closed in $k_i(V_i)$, so

$$\operatorname{tr.deg}_{k_i}\mathcal{K}_{\mathbf{D}(Q_i)}>\operatorname{tr.deg}_{k_i}\mathcal{K}_{\beta_i}.$$

This contradicts the maximality.

Step 3. In this step we show that, after replacing ℓ, Γ, c, and C_0 by a further fixed multiple, (10.4) also holds when $\mathbf{D}(Q_i)=\tau_i$. Suppose not. Possibly replacing $\mathbf{D}(Q_i)<\tau_i$ with $\mathbf{D}(Q_i)=\tau_i$, the argument of **Step 2** yields prime divisors P_i' with $a(P_i')=0$, $\mathbf{D}(P_i')=\tau_i$, and (10.3) with P_i and β_i replaced by P_i' and τ_i. Since $\mathbf{D}(P_i')\le 1$, P_i' is a prime divisor on $U_{\bar{k}_i}$, where U is as in Construction 10.9. Let E be the prime divisor on U over k_i whose base change contains P_i'. Then $E_{\bar{k}_i}$ is the reduced sum of the Galois conjugates of P_i'. Fix a rational number $0<\gamma<1/4$, and let

$$\widetilde{E}:=E+\sum_Q\gamma\min\left\{1,\frac{\mathbf{D}(Q)}{\tau_i}\right\}Q,$$

where Q runs through the prime divisors on U over k_i other than E with $a(Q)=0$. Let t be the largest number such that $D_U-t\widetilde{E}$ is pseudo-effective. Since $U_{\bar{k}_i}$ is of Fano type, there is a small $\mathbb{Q}$-factorial modification $U^q\to U_{\bar{k}_i}$ [BCHM10, Corollary 1.4.3] and the second paragraph after it. Then U^q is of Fano type, since $U_{\bar{k}_i}$ is of Fano type and $U^q\to U_{\bar{k}_i}$ is small, so that the strict transform of a klt boundary of $U_{\bar{k}_i}$ is a klt boundary of U^q with crepant pullback, and U^q carries a klt boundary with ample anti-log canonical divisor [PS09, Lemma-Definition 2.6]. Hence U^q is a Mori dream space [BCHM10, Corollary 1.3.2], so its pseudo-effective cone is rational polyhedral [HK00, Proposition 1.11]. Since $U^q\to U_{\bar{k}_i}$ is small, $D_U-s\widetilde{E}$ is pseudo-effective if and only if its pullback to U^q is, for any s, so t is rational. Since $D_U-\tau_i\widetilde{E}\ge 0$, $t\ge\tau_i$. The pullback of $D_U-t\widetilde{E}$ to the Mori dream space U^q is pseudo-effective, hence $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-divisor [HK00, Proposition 1.11], whose pushforward to $U_{\bar{k}_i}$ is effective. By Lemma 10.8(1), there exists $\Sigma\ge 0$ on U with $\Sigma\sim_{\mathbb{Q}}D_U-t\widetilde{E}$ over k_i. Then $f_*(\Sigma+t\widetilde{E})\ge 0$ is $\mathbb{Q}$-linearly equivalent to D over k_i, and the pullback of $f_*(\Sigma+t\widetilde{E})$ to U is $\Sigma+t\widetilde{E}$, whose multiplicity along P_i' is $\ge t$. Since $a(P_i')=0$ and $L\mathbf{D}(P_i')=L\tau_i\to 0$, Set-up 10.6(9) implies

$$\tau_i\le t\le(1+\eta)\tau_i\quad\text{for } i\gg 0.$$

Run a $(D_U-t\widetilde{E})$-MMP over k_i (Lemma 10.8(4)). It ends with U'' on which the strict transform of $D_U-t\widetilde{E}$ is semi-ample and not big, defining a contraction $U''\to Z$ with $\dim Z<\dim U$. Let F_0 be the generic fiber over $k_i(Z)$, and for a $\mathbb{Q}$-divisor G on U let G'' be the strict transform of G on U''. Since the strict transform of $D_U-t\widetilde{E}$ on U'' is $\mathbb{Q}$-linearly equivalent to the pullback of an ample $\mathbb{Q}$-divisor on Z, we have $D''-t\widetilde{E}''\sim_{\mathbb{Q}}0$ on F_0. The coefficient of E'' in $D''-t\widetilde{E}''$ is $\tau_i-t\ge-\eta\tau_i$. The exceptional prime divisors of f satisfy $a=0$, and a prime divisor Q on $V_{i,\bar{k}_i}$ with $\mathbf{D}(Q)>0$ satisfies $a(Q)=0$ since $a+\mathbf{D}$ has trace ≥ 0 on V_i. Thus D_U is supported on prime divisors Q with $a(Q)=0$, and any such $Q\ne E$ has coefficient $\gamma\min\{1,\mathbf{D}(Q)/\tau_i\}\le\gamma\mathbf{D}(Q)/\tau_i$ in $\widetilde{E}$. Since $t\le(1+\eta)\tau_i\le 2\tau_i$, the coefficient of Q'' in $D''-t\widetilde{E}''$ is $\ge\mathbf{D}(Q)-2\gamma\mathbf{D}(Q)\ge\mathbf{D}(Q)/2$. Thus, on F_0,

$$\sum_{Q\ne E}\mathbf{D}(Q)Q''\le_{\mathbb{Q}}2\eta\tau_iE''. \tag{10.5}$$

Here E'' is the strict transform of the prime divisor E over k_i, so after base change it contains all Galois conjugates of P_i'. Since D'' is big on F_0, so is E''.

By (10.3) for P_i', there exist non-zero integers $m = m_i$ such that $\sigma := \{m \operatorname{mult}_{P_i'} \ell\}$ satisfies $\sigma \to 0^+$ and $\eta + |m|L\tau_i = o(\sigma)$ (the choice of m is made as for [OAI26b, (C.22)], using (10.3) for every positive integer q). Let $N := m\ell + n$ be a Weil divisor over k_i on U, where $\operatorname{mult}_P n = -\sigma$ along the components of $E_{\bar{k}_i}$, $|\operatorname{mult}_Q n| \le C|m|L\mathbf{D}(Q)$ for $a(Q) = 0$ and $\mathbf{D}(Q) < \tau_i$ by (10.4), $0 \le \operatorname{mult}_Q n < 1$ for the other Q with $a(Q) = 0$, and $|\operatorname{mult}_Q n| \le |m| \operatorname{mult}_Q \Gamma$ for $a(Q) > 0$ by assumption. By **Step 1**, the rounding is constant on Galois orbits. Write $n = -\sigma E + n_1 + n_2 + n_3$, where n_1 is supported on the prime divisors $Q \ne E$ with $a(Q) = 0$ and $\mathbf{D}(Q) < \tau_i$, n_2 on those with $a(Q) = 0$ and $\mathbf{D}(Q) \ge \tau_i$, and n_3 on those with $a(Q) > 0$. By the three bounds above, and since $\operatorname{mult}_Q n < 1 \le \mathbf{D}(Q)/\tau_i$ for $\mathbf{D}(Q) \ge \tau_i$,

$$\pm n_1 \le C|m|L \sum_{Q \ne E} \mathbf{D}(Q)Q, \qquad 0 \le n_2 \le \frac{1}{\tau_i} \sum_{Q \ne E} \mathbf{D}(Q)Q, \qquad \pm n_3 \le |m|\Gamma.$$

On F_0, since $m\ell$ is a principal $\mathbb{Q}$-divisor, $N'' \sim_{\mathbb{Q}} n''$. By (10.5),

$$D'' = \tau_i E'' + \sum_{Q \ne E} \mathbf{D}(Q)Q'' \le_{\mathbb{Q}} (1 + 2\eta)\tau_i E''$$

on F_0, and $|m|\Gamma'' \sim_{\mathbb{Q}} |m|cLD'' \le_{\mathbb{Q}} C|m|L\tau_i E''$. Thus, on F_0,

$$\pm n_1'' \le_{\mathbb{Q}} C|m|L\eta\tau_i E'', \qquad 0 \le n_2'' \le_{\mathbb{Q}} 2\eta E'', \qquad \pm n_3'' \le_{\mathbb{Q}} C|m|L\tau_i E''.$$

Since $\eta + |m|L\tau_i = o(\sigma)$,

$$(\sigma - o(\sigma))E'' \le_{\mathbb{Q}} -N'' \le_{\mathbb{Q}} (\sigma + o(\sigma))E'' \quad \text{on } F_0.$$

By the left inequality, N'' is not pseudo-effective on F_0, and an N''-MMP over $k_i(Z)$ ends with a Mori fiber space $Y \to Z'$ (Lemma 10.8(4)). Let F be the geometric generic fiber of $Y \to Z'$, over an algebraic closure of $k_i(Z')$, which is isomorphic to $\mathbb{C}$ by Lemma 3.2. For a $\mathbb{Q}$-divisor G on U, let G_Y be its strict transform on Y and G_F the restriction of G_Y to F. Let $A := -N_F$. Then A is ample. Each step of the N''-MMP is a birational map that does not extract divisors, so the pushforward preserves $\le_{\mathbb{Q}}$, and so does the restriction to the geometric generic fiber. Thus the right inequality implies $A \le_{\mathbb{Q}} (\sigma + o(\sigma))E_F$. Since the relative Picard number of $Y \to Z'$ over $k_i(Z')$ is one, $-K_Y$ is numerically a positive multiple of $-N_Y$ over Z' on curves over $k_i(Z')$. By Lemma 10.8(2) and (5), $-K_F \sim_{\mathbb{Q}} I_i A$ for a positive rational number I_i. Since $a(E) = 0$, E has coefficient one in the boundary of the generalized pair induced by $(V_i, B_i, \mathbf{M}_i)$, so $-K_F - E_F$ is pseudo-effective. Thus $(I_i(\sigma + o(\sigma)) - 1)E_F$ is pseudo-effective, and since E_F is big, $I_i \ge 1/(\sigma + o(\sigma)) \to +\infty$.

Let $w = e \operatorname{ord}_T$, where T is a prime divisor over F and e is a positive integer, with $a(w, F, 0) \to 0$ (all estimates for w in **Step 3** are homogeneous in w, since the distance from ex to $\mathbb{Z}$ is at most e times the distance from x to $\mathbb{Z}$ for $x \in \mathbb{R}$). For a $\mathbb{Q}$-divisor G on U, write G^* and $(G_Y)^*$ for the pullbacks of G and G_Y to a common higher model, and $\operatorname{mult}_w G_F := w((G_Y)^*)$. Then $a(w) = 0$ and $\operatorname{mult}_w \widetilde{E}_F \le a(w, F, 0)$. By negativity for the $(D_U - t\widetilde{E})$-MMP, crepancy of $D'' - t\widetilde{E}''$ over Z, and nefness of $\mathbf{D}$, we have

$$0 \le e_D := (D_Y)^* - \mathbf{D} \le t\big((\widetilde{E}_Y)^* - \widetilde{E}^*\big)$$

[OAI26b, (C.21)], so $w(\widetilde{E}^*) \le \operatorname{mult}_w \widetilde{E}_F$. Since $D_U \le \gamma^{-1}\widetilde{E}$, $\mathbf{D}(w) \le \gamma^{-1}a(w, F, 0) \to 0$, so the restriction of ord_T given by Lemma 3.3(2) is ord_{T_0} for an element T_0 of I_1, hence a prime divisor on U, and the coefficient of T_0 in $\widetilde{E}$ is $w(\widetilde{E}^*)/e \le a(w, F, 0)$. Thus T_0 is neither E nor a prime divisor with $\mathbf{D} \ge \tau_i$, whose coefficients in $\widetilde{E}$ are 1 and γ, and the coefficient of T_0 in $\widetilde{E}$ is $\gamma\mathbf{D}(T_0)/\tau_i$. Hence $\mathbf{D}(w) = e\mathbf{D}(T_0) \le C\tau_i a(w, F, 0)$. Since $\mathbf{D}(T_0) \le \mathbf{D}(w) < \tau_i$ for $i \gg 0$, (10.4) at T_0, $\operatorname{mult}_w \ell = e \operatorname{mult}_{T_0} \ell$, and $|m|L\tau_i \to 0$ imply $m \operatorname{mult}_w \ell = o(a(w, F, 0))$ modulo $\mathbb{Z}$. Since $\Gamma - cLD$ is principal, $w((\Gamma_Y)^*) = \operatorname{mult}_w \Gamma + cLe_D(w)$, and $e_D(w) \le t \operatorname{mult}_w \widetilde{E}_F = O(\tau_i a(w, F, 0))$, so (10.2) gives $|m|w((\Gamma_Y)^*) = o(a(w, F, 0))$ [OAI26b, (C.28)]. The negative part of n is at most

$\sigma E + C|m|LD_U + |m|\Gamma$, and its positive part is at most the reduced sum of the prime divisors with $a = 0$ plus $C|m|LD_U + |m|\Gamma$, so

$$-o(a(w, F, 0)) \le w((n_Y)^*) \le Ca(w, F, 0)$$

[OAI26b, (C.29)] (the multiplicities at w are those of the pullbacks of the strict transforms on the models after the MMPs, and the pullback errors are bounded as in [OAI26b, (C.21), (C.28), and (C.29)]).

Since A is a divisor with integer coefficients, the $\mathbb{Q}$-divisor $R(A)$ of [OAI26b, Section B] is zero. Since F is klt of Fano type, A is ample, and $-K_F \sim_{\mathbb{Q}} I_i A$ with $I_i \to +\infty$, the pair (F, A) satisfies the hypotheses of [OAI26b, Lemma B.4]. Let $\mathcal{C}_F$ be the compactified Seifert cone of [OAI26b, Lemma B.4], that is, the contraction of the zero section E_0 of the compactified Seifert bundle of [OAI26b, Lemma B.2] for $H = A$, and let D_∞ be the infinity divisor. Let $w_{\mathcal{C}}$ be a primitive divisorial valuation of the function field of $\mathcal{C}_F$ with $w_{\mathcal{C}}(D_\infty) > 0$ and $a(w_{\mathcal{C}}, \mathcal{C}_F, 0) \to 0$. Since D_∞ is disjoint from E_0, $w_{\mathcal{C}}(E_0) = 0$. Since $\mathcal{C}_F$ is klt [OAI26b, Lemma B.4], the pullback of $K_{\mathcal{C}_F}$ to the compactified Seifert bundle is the canonical divisor of the bundle plus a rational multiple of E_0, so $a(w_{\mathcal{C}}, \mathcal{C}_F, 0)$ is the log discrepancy of $w_{\mathcal{C}}$ with respect to the bundle with boundary zero. By [OAI26b, Lemma B.2] with $B = 0$ and nef part zero, $w_{\mathcal{C}}$ restricts to a non-trivial valuation w of the function field of F (otherwise $a(w_{\mathcal{C}}, \mathcal{C}_F, 0) = 1$), and

$$a(w_{\mathcal{C}}, \mathcal{C}_F, 0) \ge a(w, F, 0) + w_{\mathcal{C}}(D_\infty), \qquad w_{\mathcal{C}}(D_\infty) \equiv \operatorname{mult}_w A = -m \operatorname{mult}_w \ell - w((n_Y)^*) \pmod{\mathbb{Z}}$$

[OAI26b, Lemma B.2]. Since $w_{\mathcal{C}}$ takes values in $\mathbb{Z}$ and the residue field of $w_{\mathcal{C}}$ has transcendence degree $\dim F$ over the ground field of F, $w = e \operatorname{ord}_T$ for a prime divisor T over F and a positive integer e by [Bou89, Chapter VI, §10, no. 3, Corollary 1] and [KM98, Lemma 2.45]. Since $a(w, F, 0) \le a(w_{\mathcal{C}}, \mathcal{C}_F, 0) - w_{\mathcal{C}}(D_\infty) < a(w_{\mathcal{C}}, \mathcal{C}_F, 0)$, $a(w, F, 0) \to 0$, so $m \operatorname{mult}_w \ell = o(a(w, F, 0))$ modulo $\mathbb{Z}$ and $-o(a(w, F, 0)) \le w((n_Y)^*) \le Ca(w, F, 0)$. Thus $w_{\mathcal{C}}(D_\infty)$ is congruent modulo $\mathbb{Z}$ to a number in $[-Ca(w, F, 0), o(a(w, F, 0))]$. Since $0 < w_{\mathcal{C}}(D_\infty) \le a(w_{\mathcal{C}}, \mathcal{C}_F, 0) \to 0$, we obtain $w_{\mathcal{C}}(D_\infty) = o(a(w, F, 0)) = o(a(w_{\mathcal{C}}, \mathcal{C}_F, 0))$. The estimate $w_{\mathcal{C}}(D_\infty) = o(a(w_{\mathcal{C}}, \mathcal{C}_F, 0))$ also holds if $w_{\mathcal{C}}(D_\infty) = 0$. Thus $w_{\mathcal{C}}(D_\infty) = o(a(w_{\mathcal{C}}, \mathcal{C}_F, 0))$ for any sequence of primitive divisorial valuations $w_{\mathcal{C}}$ of the function field of $\mathcal{C}_F$ with $a(w_{\mathcal{C}}, \mathcal{C}_F, 0) \to 0$, which contradicts [OAI26b, Lemma B.4].

Step 4. In this step we principalize ℓ on the generic fiber of the cutoff model. By **Steps 2** and **3**, after replacing ℓ, Γ, c, and C_0 by a fixed multiple, (10.4) holds when $\mathbf{D}(Q_i) \le \tau_i$. Let $p : U' \to W_{\tau_i}$ be as in Construction 10.9 with $\beta = \tau_i$, and let F_τ be the generic fiber of p, over $\mathcal{K}_{\tau_i}$. Then $D' = \sum_{Q \in I_{\tau_i}} \mathbf{D}(Q)Q'$ on F_τ. Let $N := \ell' + n$ be a divisor with integer coefficients over $\mathcal{K}_{\tau_i}$ on F_τ, where, for a prime divisor Q on $U'_{\bar{k}_i}$ horizontal over W_{τ_i}, $\operatorname{mult}_Q n := p_{i,Q} - \operatorname{mult}_Q \ell$ if $a(Q) > 0$, and $|\operatorname{mult}_Q n| \le CL\mathbf{D}(Q)$ if $a(Q) = 0$, by (10.4). Here a prime divisor Q with $a(Q) > 0$ is the strict transform of a prime divisor on $V_{i,\bar{k}_i}$, since the exceptional prime divisors of f satisfy $a = 0$, and $|\operatorname{mult}_Q n| \le \operatorname{mult}_Q \Gamma$ by the assumption of the lemma. By **Step 1**, n is constant on Galois orbits. If N or $-N$ is not pseudo-effective on F_τ, then, replacing E'' with $R := \sum_{Q \in I_{\tau_i}} Q'$ and σ with $CL\tau_i$, the argument of **Step 3** contradicts [OAI26b, Lemma B.4], using $\mathbf{D} \le (D'')^* \le \tau_i (R'')^*$ for the strict transforms D'' and R'' on the models after the MMP. Thus N and $-N$ are pseudo-effective. After base change to an algebraic closure of $\mathcal{K}_{\tau_i}$, F_τ is of Fano type (as $U'_{\bar{k}_i}$ is), so a small $\mathbb{Q}$-factorial modification of it is of Fano type [PS09, Lemma-Definition 2.6], hence a Mori dream space [BCHM10, Corollary 1.3.2], on which the pullbacks of $\pm N$ are $\mathbb{Q}$-linearly equivalent to effective $\mathbb{Q}$-divisors [HK00, Proposition 1.11], whose pushforwards to the base change of F_τ are effective. By Lemma 10.8(1), $N \sim_{\mathbb{Q}} N^+ \ge 0$ and $-N \sim_{\mathbb{Q}} N^- \ge 0$ over $\mathcal{K}_{\tau_i}$, and $N^+ + N^- \ge 0$ is $\mathbb{Q}$-linearly trivial, hence zero. So $N \sim_{\mathbb{Q}} 0$ on F_τ.

Run an R-MMP $F_\tau \dashrightarrow F''_\tau$ over $\mathcal{K}_{\tau_i}$ (Lemma 10.8(4)). Since $R \ge 0$, it ends with F''_τ on which the strict transform R'' of R is nef. Let N'' be the strict transform of N on F''_τ. Since $N \sim_{\mathbb{Q}} 0$, the pullbacks of N and N'' to a common higher model coincide. Fix a rational number $0 < \epsilon < 1$, let $\bar{F}$ be the base change of F''_τ to an algebraic closure of $\mathcal{K}_{\tau_i}$, and let $a_* := a + \epsilon \mathbf{R}''$ on $\bar{F}$, where

$\mathbf{R}''$ is the Cartier $\boldsymbol{b}$-divisor of R''. We verify the hypotheses of [OAI26b, Lemma B.5] for $\bar{F}$, N'', and a_*.

First, $\bar{F}$ is of Fano type, since $U'_{\bar{k}_i}$ is of Fano type and $\bar{F}$ is obtained by an MMP from a geometric generic fiber of $p_{\bar{k}_i}$, and $N'' \sim_{\mathbb{Q}} 0$ is $\mathbb{Q}$-Cartier. Second, a_* is the log discrepancy function of effective generalized klt Calabi–Yau data. Its boundary is ≥ 0 since R is reduced and supported on prime divisors with $a = 0$, and the added part $\epsilon\mathbf{R}''$ is nef. By the negativity lemma, since $\mathbf{D}$ is nef, and since $D'' = \sum_{Q \in I_{\tau_i}} \mathbf{D}(Q)Q'' \leq \tau_i R''$, we have $\mathbf{D} \leq (D'')^* \leq \tau_i \mathbf{R}''$. Since $\mathbf{D} > 0$ where $a = 0$, it follows that $a_* > 0$, so a_* is generalized klt.

Third, let w be prime divisors over $\bar{F}$ with $a_*(w) \to 0$. Since a takes integer values, $a(w) = 0$, and

$$\mathbf{D}(w) \leq \tau_i \mathbf{R}''(w) \leq \frac{\tau_i a_*(w)}{\epsilon} < \tau_i.$$

By Lemma 3.3(2), applied over $\mathcal{K}_{\tau_i}$, w restricts to a prime divisor over V_i, so $\operatorname{dist}(\operatorname{mult}_w \ell, \mathbb{Z}) \leq CL\mathbf{D}(w)$ by (10.4). Let Γ' and Γ'' be the strict transforms of Γ on F_τ and F''_τ. Since $|n| \leq CL\tau_i R + \Gamma'$ coefficientwise on F_τ, we have $\pm n'' \leq CL\tau_i R'' + \Gamma''$ on F''_τ. Since $\Gamma - cLD$ is principal and $D'' \leq \tau_i R''$, $w((\Gamma'')^*) \leq \operatorname{mult}_w \Gamma + cL\tau_i \mathbf{R}''(w)$. Together with (10.2),

$$\operatorname{dist}\big(w((N'')^*), \mathbb{Z}\big) \leq CL\mathbf{D}(w) + CL\tau_i \mathbf{R}''(w) + \operatorname{mult}_w \Gamma \leq \frac{CL\tau_i a_*(w)}{\epsilon} = o(a_*(w)),$$

since $L\tau_i \to 0$. Fourth, N'' has integer coefficients, so $R(N'') = 0$ in the notation of [OAI26b, (B.1)], and the condition of [OAI26b, Lemma B.5] on the boundary holds since the boundary of a_* is ≥ 0. By the third point, the signed phase $\theta(w)$ of N'' at prime divisors w with $a_*(w) \to 0$ satisfies $|\theta(w)| = o(a_*(w))$.

By [OAI26b, Lemma B.5], the least positive integer q_0 such that $q_0 N''$, equivalently $q_0 N$, is principal over an algebraic closure of $\mathcal{K}_{\tau_i}$ is bounded. Then $q_0 N$ is Cartier over $\mathcal{K}_{\tau_i}$ by faithfully flat descent, and $\mathcal{O}_{F_\tau}(q_0 N)$ is trivial after base change. By Lemma 10.8(1), $\mathcal{O}_{F_\tau}(\pm q_0 N)$ have non-zero sections over $\mathcal{K}_{\tau_i}$, and their product is a non-zero constant. Thus $q_0 N = \operatorname{div}(\psi)$ on F_τ for some $\psi \in \mathcal{K}_{\tau_i}(F_\tau)^* = k_i(V_i)^*$, with q_0 the integer bounded over the algebraic closure of $\mathcal{K}_{\tau_i}$. Let

$$\ell^\circ := q_0 \ell - \operatorname{div}(\psi).$$

Then ℓ° is over k_i, and $\operatorname{mult}_Q \ell^\circ = -q_0 \operatorname{mult}_Q n$ for any prime divisor Q on $U'_{\bar{k}_i}$ horizontal over W_{τ_i}, so

$$|\operatorname{mult}_Q \ell^\circ| \leq q_0 CL\mathbf{D}(Q) \ \text{ if } a(Q) = 0, \qquad |\operatorname{mult}_Q \ell^\circ| \leq q_0 \operatorname{mult}_Q \Gamma \ \text{ if } a(Q) > 0.$$

Since $\operatorname{mult}_{P_i} \operatorname{div}(\psi) \in \mathbb{Z}$, (10.3) implies $\lim_i |\operatorname{mult}_{P_i} \ell^\circ| / (L\mathbf{D}(P_i)) = +\infty$.

Step 5. In this step we conclude the proof. Fix a rational number $0 < b < 1$, and let $J = J_i \to +\infty$ be real numbers such that

$$J = o\Big(\frac{|\operatorname{mult}_{P_i} \ell^\circ|}{L\mathbf{D}(P_i)}\Big),$$

and let $R_i \to +\infty$ be integers with $R_i / J \to 0$, for example $R_i = \lfloor J^{1/2} \rfloor$. On the geometric generic fiber of p, for an integer s let

$$a^{(0)} := a + \frac{b\mathbf{D}}{\tau_i}, \qquad a^{(s)} := a^{(0)} + \frac{s\ell^\circ - q_0 |s| \Gamma}{JL\tau_i}.$$

Let $|s| \leq R_i$. Since ℓ° is principal and $\Gamma \sim_{\mathbb{Q}} cLD$, the nef part of $a^{(s)}$ is $\mathbf{M}_i + (b - q_0|s|c/J)\mathbf{D}/\tau_i$, and

$$b - \frac{q_0 |s| c}{J} \geq b - \frac{q_0 R_i C_0}{J} > 0 \quad \text{for } i \gg 0.$$

We check the trace of $a^{(s)}$ by the bound on ℓ° in **Step 4**. At a prime divisor Q on the geometric generic fiber with $a(Q) > 0$, we have $a(Q) = 1$, $\mathbf{D}(Q) = 0$, and $s \operatorname{mult}_Q \ell^\circ \leq q_0 |s| \operatorname{mult}_Q \Gamma$, so

$a^{(s)}(Q) \le 1$. At a prime divisor Q on the geometric generic fiber with $a(Q) = 0$, we have $Q \in I_{\tau_i}$, so $\mathbf{D}(Q) \le \tau_i$ and

$$a^{(s)}(Q) \le b + \frac{|s| q_0 CL\mathbf{D}(Q)}{JL\tau_i} \le b + \frac{q_0 CR_i}{J} < 1 \quad \text{for } i \gg 0.$$

Thus $a^{(s)}$ is the log discrepancy function of generalized Calabi–Yau data whose boundary is ≥ 0, and $a^{(0)}$ is generalized klt. At a prime divisor over the geometric generic fiber with $a^{(0)} < \min\{1, b\}/2$, integrality of a forces $a = 0$ and $\mathbf{D} < \tau_i/2$, so its restriction is an element of I_{τ_i}, a prime divisor on U' horizontal over W_{τ_i} (it has coefficient zero in Υ_{τ_i}, so it is not contracted by the Υ_{τ_i}-MMP of Construction 10.9, which contracts only components of the strict transforms of Υ_{τ_i}, and it is horizontal since it is the restriction of a prime divisor over the geometric generic fiber of p), and $|\operatorname{mult} \ell^\circ| + q_0 \operatorname{mult} \Gamma \le CL\mathbf{D}$ there by the bound on ℓ° in **Step 4** and (10.2). Thus, for any fixed s, $(a^{(s)} - a^{(0)})/a^{(0)} = O(|s|/J) \to 0$ there. The family $s \to a^{(s)}$, $|s| \le R_i$, is piecewise affine and concave in s at each prime divisor, since $\Gamma \ge 0$, and $R_i \to +\infty$. Thus [OAI26b, Lemma A.1], in its two-sided case with $\delta = \min\{1, b\}/2$, implies that $a^{(s)}$ is generalized klt for any fixed s and $i \gg 0$. By Lemma 3.3(2) applied over $\mathcal{K}_{\tau_i}$, P_i restricts to a prime divisor over the geometric generic fiber at which a, $\mathbf{D}$, $\operatorname{mult} \ell^\circ$, and $\operatorname{mult} \Gamma$ take their values at P_i. Let $s = \pm 1$ have sign opposite to $\operatorname{mult}_{P_i} \ell^\circ$. Since $a(P_i) = 0$,

$$a^{(s)}(P_i) \le \frac{\mathbf{D}(P_i)}{\tau_i}\left(b - \frac{|\operatorname{mult}_{P_i} \ell^\circ|}{JL\mathbf{D}(P_i)}\right) < 0$$

for $i \gg 0$, which contradicts the generalized klt property of $a^{(s)}$ for the fixed $s = \pm 1$. □

10.2. **Rank refinement over the ground field.** In this subsection, we prove Proposition 10.16, the version over the fields k_i of [OAI26b, Proposition F.7].

Set-up 10.12. Let d_0 and, for any positive integer i, k_i, S_i, x_i, $\bar{x}_i$, G_i, ω_i, χ_i, h_i, n_i, F_{ij}, m_{ij}, f_{ij}, $\mathcal{Q}_i$, w_i, w_i^0, $\mathcal{Q}_i^0$, and C be as follows.

(1) $d_0 \ge 2$ is an integer.
(2) k_i is a field finitely generated over $\mathbb{C}$, S_i is a geometrically integral normal affine variety over k_i of dimension between 2 and d_0, $x_i \in S_i$ is a k_i-rational point, and $\bar{x}_i$ is the point of $S_{i,\bar{k}_i}$ over x_i.
(3) G_i is a finite cyclic group of automorphisms of S_i over k_i fixing x_i such that S_i is klt and $S_i \to S_i/G_i$ is quasi-étale.
(4) The reflexive canonical sheaf of S_i has a generator ω_i at x_i which is an eigenvector of G_i with character χ_i.
(5) $h_i \in \mathcal{O}_{S_i,x_i}$ is a non-zero eigenvector of G_i with character χ_i such that $(S_i, \operatorname{div}(h_i))$ is lc.
(6) $n_i \ge 0$ is an integer, and $F_{i1}, \dots, F_{in_i} \in \mathcal{O}_{S_i,x_i}$ are eigenvectors of G_i with characters $\chi_i^{m_{ij}}$ for positive integers m_{ij}, and $f_{ij} := F_{ij}/h_i^{m_{ij}}$.
(7) $\mathcal{Q}_i$ is the set of quasi-monomial valuations ς on $S_{i,\bar{k}_i}$ which are non-negative on $\mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ and satisfy $a(\varsigma, S_{i,\bar{k}_i}, 0) = \varsigma(h_i) = 1$ and $\varsigma(f_{ij}) \ge 0$ for any j.
(8) $\mathcal{Q}_i$ has an element centered at $\bar{x}_i$, and $\varsigma(f_{ij}) = 0$ for any $\varsigma \in \mathcal{Q}_i$ and any j.
(9) $w_i \in \mathcal{Q}_i$ is a valuation centered at $\bar{x}_i$ which is invariant under G_i and $\operatorname{Gal}(\bar{k}_i/k_i)$ such that $\operatorname{gr}_{w_i} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is finitely generated, normal, and klt, and w_i^0 is the valuation given by the grading.
(10) $\mathcal{Q}_i^0$ is the set defined by the conditions of (7) on $\operatorname{Spec} \operatorname{gr}_{w_i} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ for $\operatorname{in}_{w_i}(h_i)$ and $\operatorname{in}_{w_i}(F_{ij})$.
(11) $C > 0$ is a real number such that $\varsigma^0(\sigma) \le C w_i^0(\sigma)$ for any i, any $\varsigma^0 \in \mathcal{Q}_i^0$, and any non-zero homogeneous element σ of $\operatorname{gr}_{w_i} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$.

Condition (8) is condition (I1) of [OAI26b, Subsection G.1] for the tests f_{ij}, and it is used in **Step 1** of the proof of Proposition 10.16 together with [OAI26b, Proposition F.4 and Corollary F.5]. Condition (11) is the hypothesis of [OAI26b, Proposition F.7] in (I5) of [OAI26b,

Subsection G.1], and it is used in **Steps 4 and 5** of the proof of Proposition 10.16. In the first use of Proposition 10.16 in the proof of Theorem 10.2, condition (11) follows from [OAI26b, Corollary E.10] and [OAI26b, (G.4)].

Lemma 10.13. *Let k be a field finitely generated over $\mathbb{C}$, X a geometrically integral normal variety over k, $x \in X$ a k-rational point, $\bar{x} \in X_{\bar{k}}$ the point over x, w a valuation on $X_{\bar{k}}$ with $\operatorname{center}_{X_{\bar{k}}}(w) = \bar{x}$ which is invariant under $\operatorname{Gal}(\bar{k}/k)$, and v the restriction of w to $k(X)$. Then $\mathcal{O}_{X,x} \otimes_k \bar{k} = \mathcal{O}_{X_{\bar{k}},\bar{x}}$, and for any real number $t \geq 0$,*

$$\mathfrak{a}_t(w)\mathcal{O}_{X_{\bar{k}},\bar{x}} = (\mathfrak{a}_t(v)\mathcal{O}_{X,x}) \otimes_k \bar{k}.$$

In particular, the natural homomorphism

$$\operatorname{gr}_v \mathcal{O}_{X,x} \otimes_k \bar{k} \to \operatorname{gr}_w \mathcal{O}_{X_{\bar{k}},\bar{x}}$$

is an isomorphism of graded rings.

Proof. Let $\operatorname{Spec} A \subset X$ be an affine open neighborhood of x and $\mathfrak{p} \subset A$ the prime ideal of x. Then $\mathcal{O}_{X,x} \otimes_k \bar{k}$ is the localization of $A \otimes_k \bar{k}$ at the image of $A \setminus \mathfrak{p}$, and the prime ideals of $\mathcal{O}_{X,x} \otimes_k \bar{k}$ correspond to the prime ideals $\mathfrak{q} \subset A \otimes_k \bar{k}$ with $\mathfrak{q} \cap A \subset \mathfrak{p}$. Since $A \otimes_k \bar{k}$ is integral over A, by going-up any maximal such $\mathfrak{q}$ satisfies $\mathfrak{q} \cap A = \mathfrak{p}$. Since x is k-rational, the fiber of $X_{\bar{k}} \to X$ over x is $\operatorname{Spec}(\kappa(x) \otimes_k \bar{k}) = \operatorname{Spec} \bar{k}$, so $\bar{x}$ is the only point of $X_{\bar{k}}$ over x. Thus $\mathcal{O}_{X,x} \otimes_k \bar{k}$ is a local ring whose maximal ideal is the ideal of $\bar{x}$, hence $\mathcal{O}_{X,x} \otimes_k \bar{k} = \mathcal{O}_{X_{\bar{k}},\bar{x}}$. Since X is geometrically integral, $k(X) \subset \bar{k}(X_{\bar{k}})$, and $v(f) = w(f \otimes 1)$ for any non-zero $f \in \mathcal{O}_{X,x}$.

Fix a real number $t \geq 0$ and let $\mathfrak{b} := \mathfrak{a}_t(w)\mathcal{O}_{X_{\bar{k}},\bar{x}}$. The group $\operatorname{Gal}(\bar{k}/k)$ acts on $\mathcal{O}_{X_{\bar{k}},\bar{x}} = \mathcal{O}_{X,x} \otimes_k \bar{k}$ through the second factor, and $\sigma(\mathfrak{b}) = \mathfrak{b}$ for any $\sigma \in \operatorname{Gal}(\bar{k}/k)$ since w is invariant. Since $\mathcal{O}_{X_{\bar{k}},\bar{x}}$ is Noetherian, $\mathfrak{b}$ is generated by finitely many elements, and their coefficients in $\bar{k}$ are contained in a finite Galois extension k' of k. Let $\mathfrak{b}' := \mathfrak{b} \cap (\mathcal{O}_{X,x} \otimes_k k')$. Then $\mathfrak{b}'$ generates $\mathfrak{b}$, and since $\bar{k}$ is flat over k', the natural map $\mathfrak{b}' \otimes_{k'} \bar{k} \to \mathfrak{b}$ is an isomorphism. Any element of $\operatorname{Gal}(k'/k)$ extends to an element of $\operatorname{Gal}(\bar{k}/k)$, so $\mathfrak{b}'$ is stable under $\operatorname{Gal}(k'/k)$. By Galois descent [Sta, Tag 0CDR] for $\operatorname{Spec} \mathcal{O}_{X,x}$ and k'/k, $\mathfrak{b}' = \mathfrak{b}_0 \otimes_k k'$ for the ideal $\mathfrak{b}_0 := \mathfrak{b}' \cap \mathcal{O}_{X,x}$. By the first paragraph,

$$\mathfrak{b}_0 = \{f \in \mathcal{O}_{X,x} \mid w(f) \geq t\} = \mathfrak{a}_t(v)\mathcal{O}_{X,x},$$

hence $\mathfrak{b} = \mathfrak{b}_0 \otimes_k \bar{k}$, which is the claimed equality.

We have

$$\mathfrak{a}_{>t}(w)\mathcal{O}_{X_{\bar{k}},\bar{x}} = \bigcup_{t'>t} \mathfrak{a}_{t'}(w)\mathcal{O}_{X_{\bar{k}},\bar{x}} \quad \text{and} \quad \mathfrak{a}_{>t}(v)\mathcal{O}_{X,x} = \bigcup_{t'>t} \mathfrak{a}_{t'}(v)\mathcal{O}_{X,x}.$$

Since tensor products commute with unions, the claimed equality also holds with $\mathfrak{a}_{>t}$ in place of $\mathfrak{a}_t$. Since $\bar{k}$ is flat over k, tensoring the exact sequence

$$0 \to \mathfrak{a}_{>t}(v)\mathcal{O}_{X,x} \to \mathfrak{a}_t(v)\mathcal{O}_{X,x} \to \mathfrak{a}_t(v)\mathcal{O}_{X,x}/\mathfrak{a}_{>t}(v)\mathcal{O}_{X,x} \to 0$$

with $\bar{k}$ implies that the natural homomorphism is bijective in degree t. It is a homomorphism of graded rings since it is induced by the inclusions $\mathfrak{a}_t(v)\mathcal{O}_{X,x} \subset \mathfrak{a}_t(w)\mathcal{O}_{X_{\bar{k}},\bar{x}}$. □

Definition-Proposition 10.14. Notation and conditions as in Set-up 10.12, and fix i. Let w be a valuation in $\mathcal{Q}_i$ centered at $\bar{x}_i$ which is invariant under G_i and $\operatorname{Gal}(\bar{k}_i/k_i)$, and assume that $\operatorname{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is finitely generated, normal, and klt, that the induced action of G_i on $\operatorname{Spec} \operatorname{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is quasi-étale, and that the generator of the canonical sheaf there has character χ_i. Let v be the restriction of w to $k_i(S_i)$, $R := (\operatorname{gr}_v \mathcal{O}_{S_i,x_i})^{G_i}$, graded by the group of values of v, M the group generated by the degrees of the homogeneous elements of $\operatorname{Frac}(R)$, $\mathcal{K} \subset \operatorname{Frac}(R)$ the subfield of homogeneous elements of degree 0, $u \in M \otimes \mathbb{Q}$ the degree of $\operatorname{in}_v(h_i)$, and

$$V_u := \operatorname{Proj} \bigoplus_n R_{nu},$$

where n runs over the non-negative integers with $nu \in M$. Choose a basis $\mu_1, \dots, \mu_{\operatorname{rank} M}$ of M and homogeneous elements $z_j \in \operatorname{Frac}(R)$ of degree μ_j, write $z^\mu := \prod_j z_j^{c_j}$ for $\mu = \sum_j c_j \mu_j \in M$, let $r := |G_i|$, and write $\operatorname{in}_v(h_i)^r = f_h z^{ru}$ with $f_h \in \mathcal{K}$. For a prime divisor P over $V_{u,\bar{k}_i}$, let Π_P be the set of $y \in \operatorname{Hom}(M, \mathbb{R})$ such that $\operatorname{ord}_P(\phi) + \langle y, \mu \rangle \geq 0$ for any non-zero homogeneous element ϕz^μ of R with $\phi \in \mathcal{K}$, let

$$D(P) := \min_{y \in \Pi_P} \big(\operatorname{ord}_P(f_h)/r + \langle y, u \rangle \big),$$

and let $a(P) := 1 + \operatorname{ord}_P(\theta_0)$, where, for a homogeneous generator ω^0 of the canonical module of $\operatorname{gr}_v \mathcal{O}_{S_i,x_i}$, the G_i-invariant form $\omega^0 / \operatorname{in}_v(h_i)$ is written as $\theta_0 \wedge d\log z_1 \wedge \cdots \wedge d\log z_{\operatorname{rank} M}$. Then the following hold.

(1) $R \otimes_{k_i} \bar{k}_i = (\operatorname{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i})^{G_i}$, which is a domain, and u is contained in the interior of the cone in $M \otimes \mathbb{R}$ generated by the degrees of the non-zero homogeneous elements of R.
(2) V_u is a geometrically integral normal projective variety over k_i with function field $\mathcal{K}$, and $V_{u,\bar{k}_i}$ is the ray quotient of [OAI26b, Proposition D.1] for $R \otimes_{k_i} \bar{k}_i$.
(3) The canonical module of $\operatorname{gr}_v \mathcal{O}_{S_i,x_i}$ is generated by a homogeneous element ω^0, which is an eigenvector of G_i with character χ_i and is unique up to multiplication by an element of k_i^*. In particular, $a(P)$ does not depend on the choice of ω^0.
(4) For any prime divisor P over $V_{u,\bar{k}_i}$, the minimum defining $D(P)$ is attained at some $y \in \Pi_P \cap \operatorname{Hom}(M, \mathbb{Q})$.
(5) There exists an ample $\mathbb{Q}$-divisor $D \geq 0$ on V_u over k_i such that $\operatorname{mult}_P D = D(P)$ for any prime divisor P over $V_{u,\bar{k}_i}$, with mult_P as in Set-up 10.6.

We call (R, M, u, V_u, D, a) the *angular data* of v.

Proof. In this proof we drop the index i, and we let $A := \operatorname{gr}_v \mathcal{O}_{S,x}$, $A_{\bar{k}} := A \otimes_k \bar{k}$, and $h_0 := \operatorname{in}_v(h)$. Since $w \in \mathcal{Q}_i$, the degree u of h_0 is $v(h) = w(h) = 1$ under $M \otimes \mathbb{Q} \subset \mathbb{R}$. We keep writing u for it.

We prove (1). By Lemma 10.13, $A_{\bar{k}} = \operatorname{gr}_w \mathcal{O}_{S_{\bar{k}},\bar{x}}$ as graded rings. Since $A \to A_{\bar{k}}$ is faithfully flat, A is a domain. Finitely many elements of A whose images generate $A_{\bar{k}}$ generate a subalgebra A' with $A' \otimes_k \bar{k} = A_{\bar{k}}$, hence $A' = A$ by faithful flatness, so A is finitely generated. If $\xi \in \operatorname{Frac}(A)$ is integral over A, then $\xi \in A_{\bar{k}}$, and writing $\xi = a/b$ with $a, b \in A$ we obtain $a \in bA_{\bar{k}} \cap A = bA$, so A is normal. Since w is invariant under G and G acts over k, v is invariant under G, and G acts on A by graded automorphisms. The G-invariant part is the image of the projector $r^{-1} \sum_{g \in G} g$, which commutes with $\otimes_k \bar{k}$, so $R_{\bar{k}} := R \otimes_k \bar{k} = (A_{\bar{k}})^G$. Thus $R_{\bar{k}}$ is contained in the domain $A_{\bar{k}} = \operatorname{gr}_w \mathcal{O}_{S_{\bar{k}},\bar{x}}$, so it is a domain. In particular, the set of degrees of non-zero homogeneous elements of R equals that of $R_{\bar{k}}$. By Lemma 3.2, $\bar{k} \cong \mathbb{C}$, and the hypotheses of [OAI26b, Section F] hold for $S_{\bar{k}}$ by Lemma 3.3(2) and (3), since G-invariants and the branch locus of $S \to S/G$ commute with the base change $\bar{k}/k$. For a non-zero $y \in \operatorname{Hom}(M, \mathbb{Q})$ which is non-negative on the degrees of the non-zero homogeneous elements of R, let ς_y be the valuation of $\operatorname{Frac}(A_{\bar{k}})$ such that $\varsigma_y(\sigma) = \min_\nu \langle y, \nu \rangle$ for any non-zero $\sigma \in A_{\bar{k}}$, where ν runs over the degrees of the non-zero homogeneous components of σ and y is extended to $M \otimes \mathbb{Q}$, which contains any such ν since $r\nu \in M$. The valuations ς_y are the pure weights of [OAI26b, Section D]. By [OAI26b, Corollary F.5] for the germ $S_{\bar{k}} \ni \bar{x}$, ω, and w, $a(\varsigma_y, \operatorname{Spec} A_{\bar{k}}, 0) = \varsigma_y(h_0)$ for any such y. Thus, by [OAI26b, Proposition D.1(i)], u is contained in the interior of the cone of $R_{\bar{k}}$, which is the cone of R.

We prove (2). The base change $V_{u,\bar{k}} = \operatorname{Proj} \bigoplus_n (R_{\bar{k}})_{nu}$ is the ray quotient of [OAI26b, Proposition D.1] for $R_{\bar{k}}$, which is a normal projective variety over $\bar{k}$ whose function field is the subfield of degree 0 elements of $\operatorname{Frac}(R_{\bar{k}})$ [OAI26b, Proposition D.1(ii)]. Thus V_u is a geometrically integral normal projective variety over k, where normality descends as for A in the proof of (1). The function field of V_u is the set of $\operatorname{Gal}(\bar{k}/k)$-invariant elements of the function field of $V_{u,\bar{k}}$. Since $\mathcal{K}$ is $\operatorname{Gal}(\bar{k}/k)$-invariant, contained in the function field of $V_{u,\bar{k}}$, and contains the function field of V_u, the function field of V_u is $\mathcal{K}$.

We prove (3). By [OAI26b, Proposition F.2 and Corollary F.3] for the germ $S_{\bar{k}} \ni \bar{x}$, ω, and w, the canonical module $\omega_{A_{\bar{k}}}$ of $A_{\bar{k}}$ is a free module generated by a homogeneous element $\bar{\omega}^0$ with character χ, and $\deg \bar{\omega}^0 = \deg h_0$. The ring A is graded by the non-negative elements of the group of values of v, and $A_0 = k$ since x is k-rational and v is centered at x. The canonical module ω_A of A is a graded reflexive A-module of rank one. Since $A_{\bar{k}} = A \otimes_k \bar{k}$ (Lemma 10.13), top degree differentials on the smooth locus commute with the base change $\bar{k}/k$ in characteristic zero [Har77, Proposition II.8.10], and pushforward commutes with flat base change [Har77, Proposition III.9.3], we have $\omega_A \otimes_k \bar{k} = \omega_{A_{\bar{k}}}$. Since $\omega_{A_{\bar{k}}}$ is free, ω_A is a finitely generated projective A-module of rank one by faithfully flat descent. Since $A_0 = k$, the graded Nakayama lemma implies that ω_A is generated by a homogeneous element ω^0. Homogeneous elements of positive degree of $A_{\bar{k}}$ are not invertible, so the homogeneous units of $A_{\bar{k}}$ are elements of $\bar{k}^*$. Thus $\omega^0 \otimes 1 = c\bar{\omega}^0$ for some $c \in \bar{k}^*$, so $\deg \omega^0 = \deg \bar{\omega}^0 = \deg h_0$ and ω^0 has character χ. Since homogeneous elements of positive degree of A are not invertible, the homogeneous units of A are the elements of $A_0^* = k^*$, so any homogeneous generator of ω_A is $c'\omega^0$ for some $c' \in k^*$. Since ω^0 and h_0 both have character χ, ω^0/h_0 is G-invariant.

Since the degrees μ_j are linearly independent and any homogeneous element of $\mathrm{Frac}(R)$ of degree μ belongs to $\mathcal{K}z^\mu$, we have $\mathrm{Frac}(R) = \mathcal{K}(z_1, \dots, z_{\operatorname{rank} M})$ with $z_1, \dots, z_{\operatorname{rank} M}$ algebraically independent over $\mathcal{K}$. Since ω^0/h_0 is G-invariant and homogeneous of degree 0, and $\mathrm{Frac}(A)^G = \mathrm{Frac}(R)$, ω^0/h_0 is a differential form of top degree over $\mathrm{Frac}(R)$, so it is written uniquely as $\theta_0 \wedge d\log z_1 \wedge \cdots \wedge d\log z_{\operatorname{rank} M}$ with θ_0 in the top exterior power of $\Omega_{\mathcal{K}/k}$. Replacing ω^0 with $c'\omega^0$ replaces θ_0 with $c'\theta_0$, which does not change $\operatorname{ord}_P(\theta_0)$. Thus $a(P)$ does not depend on the choice of ω^0.

We prove (4). Since A is finitely generated and G is finite, $R = A^G$ is finitely generated. Since ord_P of a product is the sum of the orders and ord_P of a sum is at least the minimum of the orders of the terms, Π_P is defined by the inequalities for the finitely many homogeneous generators ϕz^μ of R. Π_P is non-empty, since t times the inclusion $M \subset \mathbb{R}$ is in Π_P for $t \gg 0$ (the homogeneous generators of R of degree zero are constants, as $R_0 = k$, and the others have positive degree), and $\langle y, u\rangle$ is bounded below on Π_P, since by (1) u is a non-negative combination of the degrees of the generators (cf. [OAI26b, Subsection D.1]). Since Π_P is defined by finitely many linear inequalities with rational coefficients, the minimum defining $D(P)$ is attained at some $y \in \Pi_P \cap \mathrm{Hom}(M, \mathbb{Q})$.

We prove (5). By [OAI26b, Proposition D.1(ii)] for $R_{\bar{k}}$, there exists an ample $\mathbb{Q}$-divisor $\bar{D} \geq 0$ on $V_{u,\bar{k}}$ with $\operatorname{mult}_P \bar{D} = D(P)$ for any prime divisor P over $V_{u,\bar{k}}$. Here Π_P defined with ϕz^μ running over R equals Π_P defined with ϕz^μ running over $R_{\bar{k}}$, since any homogeneous element of $R_{\bar{k}}$ of degree μ is a $\bar{k}$-linear combination of elements $\phi z^\mu \in R$ with $\phi \in \mathcal{K}$, and ord_P of such a combination is at least the minimum of the orders of the terms. For $\sigma \in \mathrm{Gal}(\bar{k}/k)$ acting on $V_{u,\bar{k}}$, we have $\operatorname{ord}_{\sigma(P)}(\phi) = \operatorname{ord}_P(\phi)$ for any $\phi \in \mathcal{K}$. Since $f_h \in \mathcal{K}$, we obtain $\Pi_{\sigma(P)} = \Pi_P$ and $D(\sigma(P)) = D(P)$. Hence the coefficients of $\bar{D}$ are constant on the $\mathrm{Gal}(\bar{k}/k)$-orbits of prime divisors on $V_{u,\bar{k}}$. By Lemma 3.3(1), the pullback of a prime divisor E on V_u is reduced, and $\mathrm{Gal}(\bar{k}/k)$ acts transitively on the points of $V_{u,\bar{k}}$ over the generic point η_E of E [Sta, Tag 04KP] (applied to $\kappa(\eta_E) \otimes_k \bar{k}$), so the pullback of E is the reduced sum of one $\mathrm{Gal}(\bar{k}/k)$-orbit of prime divisors. Thus $\bar{D}$ is the pullback of a $\mathbb{Q}$-divisor $D \geq 0$ on V_u over k, and $\operatorname{mult}_P D = \operatorname{mult}_P \bar{D} = D(P)$. For a positive integer q such that $q\bar{D}$ is Cartier, $\mathcal{O}_{V_u}(qD) \otimes_k \bar{k} = \mathcal{O}_{V_{u,\bar{k}}}(q\bar{D})$ is invertible, so qD is Cartier [Sta, Tag 05B2]. For any coherent sheaf $\mathcal{F}$ on V_u and any $j > 0$, we have

$$H^j(V_u, \mathcal{F} \otimes \mathcal{O}_{V_u}(nqD)) \otimes_k \bar{k} = H^j(V_{u,\bar{k}}, \mathcal{F}_{\bar{k}} \otimes \mathcal{O}_{V_{u,\bar{k}}}(nq\bar{D}))$$

[Har77, Proposition III.9.3], which vanishes for $n \gg 0$ since $q\bar{D}$ is ample. Since V_u is projective over k, qD is ample by Serre's criterion [Har77, Proposition III.5.3]. □

Proposition 10.16 replaces w_i by valuations w_i' over k_i which are close to w_i and whose angular data satisfy the inequality in (MO_k) up to factors $1 + \eta_i$ with $\lim_i \eta_i = 0$.

Remark 10.15. We list the changes to the proof of [OAI26b, Proposition F.7] made in the proof of Proposition 10.16.

(1) The order inequality is tested only by $\mathbb{Q}$-divisors $\Gamma \sim_{\mathbb{Q}} D$ over k_i. Thus a failure of the inequality implies the existence of $\phi \in \mathcal{K}$ and $\Phi_0 \in R$ over k_i, and no product of Galois conjugates of ϕ is taken.
(2) The valuation $\hat{q}$ of **Step 2**, a rescaled extension of the Gauss extension $q_{P,y}$, is defined only over $\bar{k}_i$. It is used only as an element of $\mathcal{Q}_i^0(w)$, in [OAI26b, Lemma E.11] and in **Step 5**, and is not descended to k_i.
(3) The minimizer v_λ of the tilted height is defined over k_i by the uniqueness in Definition-Proposition 5.2, since b_λ and $\mathcal{Q}_i^0(w)$ are defined over k_i, and Definition-Proposition 6.3 implies the descent of gr_{v_λ}.
(4) In the proof of [OAI26b, Lemma F.6], the lifts of the generators of gr_w and gr_{v_λ} may be any G_i-eigenvectors s whose images $\mathrm{in}_w(s)$ are the generators, as long as h_i and the F_{ij} are among them. We take the lifts s_l over k_i of **Step 3**, together with G_i-eigenvectors generating $\mathfrak{m}_{x_i}$. With the lifts s_l, the proof of [OAI26b, Lemma F.6] shows that $w_\delta(s_l) = \alpha_l + \delta\alpha_l'$ and that the isomorphism of graded rings sends $\mathrm{in}_{w_\delta}(s_l)$ to $\mathrm{in}_{v_\lambda}(\mathrm{in}_w(s_l))$.
(5) The invariance of w_δ under $\mathrm{Gal}(\bar{k}_i/k_i)$ and G_i is proved by Lemma 6.5 and [Li18, Theorem 3.1] in **Step 3**; no such argument is used in [OAI26b].
(6) **Steps 2–4** are applied at most $d_0 - 1$ times by [Abh56, Theorem 1], and the fields k_i and the groups G_i are not changed.

Proposition 10.16. *Notation and conditions as in Set-up 10.12. Then, possibly passing to a subsequence, there exist valuations $w_i' \in \mathcal{Q}_i$ centered at $\bar{x}_i$ which are invariant under G_i and $\mathrm{Gal}(\bar{k}_i/k_i)$, and real numbers $e_i \geq 0$ and $\eta_i \geq 0$ with $\lim_i e_i = \lim_i \eta_i = 0$, satisfying the following.*

(1) The ring $\mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is finitely generated, normal, and klt, $w_i'(f_{ij}) = 0$ for any j, the induced action of G_i on $\mathrm{Spec}\,\mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is quasi-étale and the generator of the canonical sheaf there has character χ_i, condition (11) of Set-up 10.12 holds for w_i' in place of w_i with a constant independent of i, and $(1-e_i)w_i(\phi) \leq w_i'(\phi) \leq (1+e_i)w_i(\phi)$ for any non-zero $\phi \in \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$.

(2) Let v_i' be the restriction of w_i' to $k_i(S_i)$, let $(R_i, M_i, u_i, V_{u_i}, D_i, a_i)$ be the angular data of v_i', and let $\bar{f}_{ij} := \mathrm{in}_{v_i'}(F_{ij})/\mathrm{in}_{v_i'}(h_i)^{m_{ij}}$, an element of the function field of V_{u_i}. Then $\mathrm{mult}_P \Gamma \leq (1+\eta_i)\,\mathrm{mult}_P D_i$ for any prime divisor P over $V_{u_i,\bar{k}_i}$ with $a_i(P) = 0$ and $\mathrm{ord}_P(\bar{f}_{ij}) \geq 0$ for any j, and any $\mathbb{Q}$-divisor $\Gamma \geq 0$ on V_{u_i} over k_i with $\Gamma \sim_{\mathbb{Q}} D_i$ over k_i.

Proof. The proof is long, so we divide it into several steps. For a valuation w satisfying condition (9) of Set-up 10.12 in place of w_i, let w^0 be the valuation given by the grading of $\mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$, and let $\mathcal{Q}_i^0(w)$ be the set defined by condition (10) with w in place of w_i. By Lemma 3.2, $\bar{k}_i \cong \mathbb{C}$, so the results of [OAI26b] on complex germs apply to $S_{i,\bar{k}_i}$ at $\bar{x}_i$.

Step 1. In this step we show that any valuation w satisfying condition (9) of Set-up 10.12 in place of w_i satisfies the assumptions of Definition-Proposition 10.14, that $\varsigma^0(\mathrm{in}_w(F_{ij})) = m_{ij}$ for any $\varsigma^0 \in \mathcal{Q}_i^0(w)$ and any j, and that $\bar{f}_{ij} := \mathrm{in}_v(F_{ij})/\mathrm{in}_v(h_i)^{m_{ij}} \in \mathcal{K}$, where v is the restriction of w to $k_i(S_i)$ and $\mathcal{K}$ is as in Definition-Proposition 10.14.

By [OAI26b, Proposition F.2], where the hypotheses of [OAI26b, Section F] hold for $S_{i,\bar{k}_i}$ by Lemma 3.3(2) and (3), since G_i-invariants and the branch locus of $S_i \to S_i/G_i$ commute with the base change $\bar{k}_i/k_i$, the induced action of G_i on $\mathrm{Spec}\,\mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is quasi-étale, and the canonical sheaf there has a generator with the character χ_i of ω_i. Thus w satisfies the assumptions of Definition-Proposition 10.14. By condition (8), $\varsigma(f_{ij}) = 0$ for any $\varsigma \in \mathcal{Q}_i$, so [OAI26b, Proposition F.4 and Corollary F.5] imply

$$\varsigma^0(\mathrm{in}_w(F_{ij})) = m_{ij}\varsigma^0(\mathrm{in}_w(h_i)) = m_{ij}$$

for any $\varsigma^0 \in \mathcal{Q}_i^0(w)$. Since $w(h_i) = 1$ and $w(f_{ij}) = 0$, the elements $\mathrm{in}_v(F_{ij})$ and $\mathrm{in}_v(h_i)^{m_{ij}}$ are homogeneous of degree $m_{ij}u$. Let $r := |G_i|$. Then

$$\bar{f}_{ij} = \frac{\mathrm{in}_v(F_{ij})\,\mathrm{in}_v(h_i)^{m_{ij}(r-1)}}{\mathrm{in}_v(h_i)^{m_{ij}r}},$$

where the numerator and the denominator are homogeneous of degree $m_{ij}ru$ and G_i-invariant since $\chi_i^r = 1$. Thus both belong to R, and $\bar{f}_{ij} \in \mathcal{K}$.

Step 2. In this step, for a pair in $\mathcal{A}_i$ we construct a homogeneous element Φ_0 and a valuation $\hat{q}$ on the cone of w_i, and, assuming that the order inequality fails by a fixed proportion, a valuation v_λ over k_i on the cone of w_i which is positive on an element of degree 0.

For any i, let v_i be the restriction of w_i to $k_i(S_i)$ and $(R_i, M_i, u_i, V_{u_i}, D_i, a_i)$ the angular data of v_i, which exist by **Step 1**. Let $\mathcal{A}_i$ be the set of pairs (P, Γ) such that P is a prime divisor over $V_{u_i,\bar{k}_i}$ with $a_i(P) = 0$ and $\mathrm{ord}_P(\bar{f}_{ij}) \geq 0$ for any j, and $\Gamma \geq 0$ is a $\mathbb{Q}$-divisor on V_{u_i} over k_i with $\Gamma \sim_{\mathbb{Q}} D_i$ over k_i. Fix i and $(P, \Gamma) \in \mathcal{A}_i$, and drop the index i from k_i, S_i, x_i, $\bar{x}_i$, G_i, χ_i, h_i, n_i, F_{ij}, m_{ij}, f_{ij}, $\bar{f}_{ij}$, $\mathcal{Q}_i$, w_i, v_i, and the angular data. Let $A := \mathrm{gr}_v \mathcal{O}_{S,x}$, $X_0 := \mathrm{Spec}\, A$ with vertex o, $h_0 := \mathrm{in}_v(h)$, and $F_{j,0} := \mathrm{in}_v(F_j)$. By Lemma 10.13, $X_{0,\bar{k}} = \mathrm{Spec}\,\mathrm{gr}_w \mathcal{O}_{S_{\bar{k}},\bar{x}}$, and $\mathcal{Q}^0 := \mathcal{Q}_i^0(w)$ is a set of valuations on $X_{0,\bar{k}}$.

By [OAI26b, Proposition D.1(iii)], $a + D$ is the log discrepancy function (called the discrepancy function there) of an effective generalized klt Calabi–Yau pair, so $a(P) = 0$ implies $D(P) > 0$. Since $\Gamma \sim_{\mathbb{Q}} D$ over k, there exist a positive integer m divisible by r and $\phi \in \mathcal{K}\backslash\{0\}$ such that $m\Gamma = mD + \mathrm{div}(\phi)$. Let

$$\Phi_0 := h_0^m \phi = f_h^{m/r} \phi z^{mu} \in \mathrm{Frac}(R).$$

Since $\mathrm{div}(\phi) + mD = m\Gamma \geq 0$, we have $\Phi_0 = h_0^m \phi \in A_{\bar{k}}$ by [OAI26b, Lemma D.3] (formula (D.13)) applied to $A_{\bar{k}}$ with the action of G, h_0, and $b = m$. Since r divides m, h_0^m and hence Φ_0 are G-invariant, so $\Phi_0 \in R_{\bar{k}}$. Since Φ_0 is $\mathrm{Gal}(\bar{k}/k)$-invariant, $\Phi_0 \in R$. In particular, $\Phi_0 \in A$ is a G-invariant homogeneous element of degree mu.

By Definition-Proposition 10.14(4), the minimum defining $D(P)$ is attained at some $y \in \Pi_P \cap \mathrm{Hom}(M, \mathbb{Q})$. Let $q_{P,y}$ be the Gauss extension of ord_P with weight y, that is,

$$q_{P,y}\Big(\sum_\mu \psi_\mu z^\mu\Big) = \min_\mu \big(\mathrm{ord}_P(\psi_\mu) + \langle y, \mu\rangle\big)$$

for ψ_μ in the function field of $V_{u,\bar{k}}$, by the Gauss rule of [OAI26b, Subsection D.1], and let q be a valuation of $\mathrm{Frac}(A_{\bar{k}})$ extending $q_{P,y}$. Since $y \in \Pi_P$, $q \geq 0$ on $R_{\bar{k}}$, hence on the integral extension $A_{\bar{k}}$. Since y is rational, $q_{P,y}$ is a positive multiple of a divisorial valuation [OAI26b, Subsection D.1]. Since $\mathrm{Frac}(A_{\bar{k}})/\mathrm{Frac}(R_{\bar{k}})$ is finite and $q \geq 0$ on $A_{\bar{k}}$, q is also a positive multiple of a divisorial valuation [KM98, Lemma 2.45], hence quasi-monomial. Moreover, $q \geq 0$ on $\mathcal{O}_{X_{0,\bar{k}},o}$ since any non-empty closed subset of $X_{0,\bar{k}}$ invariant under the torus contains o (the center of q maps to that of $q_{P,y}$, and o is the only point over the vertex of $\mathrm{Spec}\, R_{\bar{k}}$ [OAI26b, Lemma D.2]). We have

$$q(h_0) = \frac{q(h_0^r)}{r} = \frac{\mathrm{ord}_P(f_h)}{r} + \langle y, u\rangle = D(P), \quad q(F_{j,0}) - m_j q(h_0) = \mathrm{ord}_P(\bar{f}_j) \geq 0,$$

$$q(\phi) = \mathrm{ord}_P(\phi) = m(\mathrm{mult}_P \Gamma - D(P)), \quad q(\Phi_0) = mq(h_0) + q(\phi) = m\,\mathrm{mult}_P \Gamma.$$

By [OAI26b, Corollary F.3 and Lemma D.2],

$$a(q, X_{0,\bar{k}}, \mathrm{div}(h_0)) = a(P) = 0, \quad \text{that is,} \quad a(q, X_{0,\bar{k}}, 0) = q(h_0).$$

Let $\hat{q} := q/D(P)$. Then $\hat{q} \in \mathcal{Q}^0$, and for the function $g(\varsigma) := \varsigma(\Phi_0)/m - \varsigma(h_0)$ on $\mathcal{Q}^0$,

$$g(\hat{q}) = \frac{\mathrm{mult}_P \Gamma}{D(P)} - 1. \tag{10.6}$$

In **Steps 2–4**, assume that there exist a rational number $\tau > 0$ and an infinite set I of positive integers such that for any $i \in I$ some $(P, \Gamma) \in \mathcal{A}_i$ satisfies $\operatorname{mult}_P \Gamma - D_i(P) \geq 2\tau D_i(P)$, and that $i \in I$ and (P, Γ) is such a pair. Then $g(\hat{q}) \geq 2\tau$ by (10.6).

For a rational number $\lambda \geq 0$, let

$$b_\lambda(\varsigma) := \varsigma(h_0) + \lambda\big(\tau\varsigma(h_0) - g(\varsigma)\big) = (1 + \lambda + \lambda\tau)\varsigma(h_0) - \frac{\lambda}{m}\varsigma(\Phi_0).$$

By [OAI26b, Lemma E.11] for $X_{0,\bar{k}}$, h_0, Φ_0, m, g, and the constant τ, the function b_λ is positive on $\mathcal{Q}^0$ if $0 \leq \lambda < \lambda^* := (\max_{\mathcal{Q}^0} g - \tau)^{-1}$.

For any rational number $\lambda \in [0, \lambda^*)$, we apply Set-up 6.1 with k, $d := \dim S$, $(X_0 \ni o, 0)$, h_0, $c = 1$, $F_{j,0}$, m_j, $g_1 = \Phi_0$, $b_0 = 1 + \lambda(1 + \tau)$, $\beta_1 = \lambda/m$, and the vertex of $X_{0,\bar{k}}$. Here $\dim X_0 = \dim S$ since $\dim \operatorname{Spec} \operatorname{gr}_w \mathcal{O}_{S_{\bar{k}},\bar{x}} = \dim S$ for the quasi-monomial valuation w (cf. [OAI26b, Subsection F.1]), K_{X_0} is Cartier since ω^0 is a generator of the canonical sheaf (Definition-Proposition 10.14(3)), and $X_{0,\bar{k}}$ is klt by condition (9), so X_0 is klt (Lemma 3.3). The pair $(X_{0,\bar{k}}, \operatorname{div}(h_0))$ is lc by [OAI26b, Corollary F.3], hence so is $(X_0, \operatorname{div}(h_0))$ (Lemma 3.3). The torus acting on $X_{0,\bar{k}}$ preserves $\operatorname{div}(h_0)$, so any lc center of $(X_{0,\bar{k}}, \operatorname{div}(h_0))$ is invariant under the torus (a connected group fixes each of the finitely many lc centers) and contains the vertex. With $c = 1$, the slice of Set-up 5.1 for the data above is $\mathcal{Q}^0$, the height is b_λ, which is positive on $\mathcal{Q}^0$, and $w^0 \in \mathcal{Q}^0$ is centered at the vertex since $a(w^0, X_{0,\bar{k}}, 0) = w^0(h_0) = 1$ [OAI26b, Proposition F.2] and $w^0(F_{j,0}) = m_j$. Let v_λ be the unique minimizer of $b_\lambda^d \cdot \operatorname{vol}$ on the valuations in $\mathcal{Q}^0$ centered at the vertex (Definition-Proposition 5.2(1)). Let v_λ^0 be the valuation given by the grading of $\operatorname{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$, let the central slice of v_λ be the set defined by the conditions of Set-up 10.12(7) on $\operatorname{Spec} \operatorname{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$ for $\operatorname{in}_{v_\lambda}(h_0)$ and $\operatorname{in}_{v_\lambda}(F_{j,0})$, and let b_λ^0 be the height b_λ with $\operatorname{in}_{v_\lambda}(h_0)$ and $\operatorname{in}_{v_\lambda}(\Phi_0)$ in place of h_0 and Φ_0.

By [OAI26b, Lemma E.11], there exists a rational number $\lambda \in [0, \lambda^*)$ such that $g(v_\lambda) > 0$, $b_\lambda(v_\lambda) \geq 1$, $b_\lambda \leq 2 + 1/\tau$ on $\mathcal{Q}^0$, and $b_\lambda^0 \leq 2 + 1/\tau$ on the central slice of v_λ. We fix such a λ.

The data h_0, $F_{j,0}$, Φ_0, m, τ, λ are defined over k. Since Φ_0 and h_0^m are G-invariant and homogeneous, $\mathcal{Q}^0$ and b_λ are preserved by $\operatorname{Gal}(\bar{k}/k)$, G, and the torus with character group the value group of w. By uniqueness (Galois conjugation preserves volumes and log discrepancies), v_λ is invariant under $\operatorname{Gal}(\bar{k}/k)$, G, and the torus. By Definition-Proposition 6.3, $\operatorname{gr}_{v_\lambda} \mathcal{O}_{X_0,o}$ is a finitely generated k-algebra whose base change to $\bar{k}$ is $\operatorname{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$, and the marked graded pair of v_λ is klt, where we write v_λ also for the restriction of v_λ to $k(X_0)$. Finally,

$$v_\lambda(\phi) = v_\lambda(\Phi_0) - m v_\lambda(h_0) = m g(v_\lambda) > 0,$$

while ϕ is homogeneous of degree 0.

Step 3. In this step we construct the compressed valuation w_δ of [OAI26b, Lemma F.6] and show that it is invariant under $\operatorname{Gal}(\bar{k}/k)$ and G.

Since v_λ is invariant under the torus, the filtration of v_λ on A splits into the homogeneous parts of A. Since localization at o does not change the graded ring (cf. [OAI26b, Subsection F.5]), $\operatorname{gr}_{v_\lambda} \mathcal{O}_{X_0,o}$ is bigraded by pairs (a, b), where a is the degree in A and b is the value of v_λ. Since $k \supset \mathbb{C}$ contains the r-th roots of unity, the action of G is diagonalizable over k. Thus there exist homogeneous G-eigenvectors $t_1, \dots, t_p \in A$ of positive degree such that $t_1, \dots, t_p$ generate A, $\operatorname{in}_{v_\lambda}(t_1), \dots, \operatorname{in}_{v_\lambda}(t_p)$ generate $\operatorname{gr}_{v_\lambda} \mathcal{O}_{X_0,o}$, and h_0 and $F_{1,0}, \dots, F_{n,0}$ are among the t_l. For any l, choose $s_l \in \mathcal{O}_{S,x}$ with $\operatorname{in}_v(s_l) = t_l$ which is a G-eigenvector with the character of t_l, by applying the projector to the character of t_l to any $s \in \mathcal{O}_{S,x}$ with $\operatorname{in}_v(s) = t_l$. We take $s_l = h$ if $t_l = h_0$, and $s_l = F_j$ if $t_l = F_{j,0}$. Add to the s_l finitely many G-eigenvectors generating $\mathfrak{m}_x$, which also generate $\mathfrak{m}_{\bar{x}}$ since x is k-rational. Let

$$\alpha_l := w(s_l) > 0 \quad \text{and} \quad \alpha_l' := v_\lambda(\operatorname{in}_w(s_l)) > 0.$$

By [OAI26b, Lemma F.6] for w, v_λ, and the lifts s_l, with the changes in Remark 10.15, there exists $\delta_0 > 0$ such that for any $\delta \in (0, \delta_0)$ outside a countable set, there exists a quasi-monomial valuation w_δ on $S_{\bar{k}}$ centered at $\bar{x}$ satisfying the following: $\mathrm{gr}_{w_\delta} \mathcal{O}_{S_{\bar{k}},\bar{x}}$ is isomorphic to $\mathrm{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$, where an element of bidegree (a,b) has degree $a+\delta b$, the isomorphism sends $\mathrm{in}_{w_\delta}(s_l)$ to $\mathrm{in}_{v_\lambda}(\mathrm{in}_w(s_l))$, $w_\delta(s_l) = \alpha_l + \delta\alpha'_l$ for any l, and $a(w_\delta, S_{\bar{k}}, 0) = w_\delta(h)$. In particular, $\mathrm{gr}_{w_\delta} \mathcal{O}_{S_{\bar{k}},\bar{x}}$ is finitely generated, normal, and klt, since the marked graded pair of v_λ is klt.

The variety $S_{\bar{k}}$ is klt at $\bar{x}$ (Lemma 3.3) and has the generator ω of the canonical sheaf. By [Li18, Theorem 3.1], there exists a constant $c_1 > 0$ with $w_\delta \le c_1 a(w_\delta, S_{\bar{k}}, 0)\,\mathrm{ord}_{\bar{x}}$, where the right hand side is finite since w_δ is quasi-monomial. Since the elements $\mathrm{in}_{w_\delta}(s_l)$ generate $\mathrm{gr}_{w_\delta} \mathcal{O}_{S_{\bar{k}},\bar{x}}$ and the s_l generate $\mathfrak{m}_{\bar{x}}$, Lemma 6.5 applied to $(\mathcal{O}_{S_{\bar{k}},\bar{x}}, \mathfrak{m}_{\bar{x}})$, w_δ and the constant $c_1 a(w_\delta, S_{\bar{k}}, 0)$ implies that for any $t > 0$, $\mathfrak{a}_t(w_\delta)$ is generated by the monomials $s^\nu := \prod_l s_l^{\nu_l}$ with $\sum_l \nu_l(\alpha_l + \delta\alpha'_l) \ge t$. Each s^ν belongs to $\mathcal{O}_{S,x}$ and is a G-eigenvector. Thus $\mathfrak{a}_t(w_\delta)$ is stable under $\mathrm{Gal}(\bar{k}/k)$ and G for any $t > 0$, so w_δ is invariant under $\mathrm{Gal}(\bar{k}/k)$ and G.

Let $N \subset \mathbb{R}^2$ be the group generated by the bidegrees of the non-zero bihomogeneous elements of $\mathrm{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$. The first projection maps N onto the value group of w. Writing $\phi = \phi_1/\phi_2$ with $\phi_1, \phi_2 \in A$ homogeneous and $\deg \phi_1 = \deg \phi_2$, the element $\mathrm{in}_{v_\lambda}(\phi_1)/\mathrm{in}_{v_\lambda}(\phi_2)$ shows that $(0, v_\lambda(\phi)) \in N$, and $v_\lambda(\phi) > 0$ by **Step 2**. Hence the rational rank of N is larger than that of the value group of w. Each non-zero $(a,b) \in N$ satisfies $a + \delta b = 0$ for at most one δ, and N is countable. Choosing δ outside the countable set of such δ as well, the map $(a,b) \to a + \delta b$ is injective on N, so the value group of w_δ is isomorphic to N, and the bihomogeneous elements of $\mathrm{gr}_{v_\lambda} \mathcal{O}_{X_{0,\bar{k}},o}$ are exactly the homogeneous elements of $\mathrm{gr}_{w_\delta} \mathcal{O}_{S_{\bar{k}},\bar{x}}$. Thus the rational rank of the value group of w_δ is larger than that of the value group of w.

Step 4. In this step we show that $w^+ := w_\delta/(1+\delta)$ satisfies conditions (9) and (11) of Set-up 10.12 with the constant $C^+ := 2^{d_0+1}(2 + 1/\tau)C$, and that $(1 - 1/i)w \le w^+ \le (1 + 1/i)w$ for δ small enough.

Since $w(h) = 1$ and $v_\lambda(h_0) = 1$, we have $w_\delta(h) = 1 + \delta$, so $a(w^+, S_{\bar{k}}, 0) = w^+(h) = 1$. Since $w(F_j) = m_j$ and $v_\lambda(F_{j,0}) = m_j$ by **Step 1**, $w_\delta(F_j) = m_j(1+\delta)$, so $w^+(f_j) = 0$. Thus $w^+ \in \mathcal{Q}_i$, and by **Step 3**, w^+ satisfies condition (9).

Choose $\delta \le \min\{1, 1/i\}$ such that $\delta\alpha'_l \le \alpha_l/i$ for any l. By [Li18, Theorem 3.1] and Lemma 6.5 for w and the s_l, as for w_δ, $\mathfrak{a}_t(w)$ is generated by the monomials s^ν with $\sum_l \nu_l\alpha_l \ge t$. Since $\alpha_l \le \alpha_l + \delta\alpha'_l \le (1 + 1/i)\alpha_l$, we obtain

$$\mathfrak{a}_t(w) \subset \mathfrak{a}_t(w_\delta) \subset \mathfrak{a}_{t/(1+1/i)}(w)$$

for any $t > 0$. Hence $w \le w_\delta \le (1 + 1/i)w$ on the non-zero elements of $\mathcal{O}_{S_{\bar{k}},\bar{x}}$. Since $1/(1+\delta) \ge 1 - \delta \ge 1 - 1/i$, we obtain $(1 - 1/i)w \le w^+ \le (1 + 1/i)w$.

Let σ be a non-zero homogeneous element of $\mathrm{gr}_{w^+} \mathcal{O}_{S_{\bar{k}},\bar{x}}$. By **Step 3**, σ corresponds to a bihomogeneous element $\mathrm{in}_{v_\lambda}(\tilde{\sigma})$ of bidegree (a,b), where $\tilde{\sigma} \in \mathrm{gr}_w \mathcal{O}_{S_{\bar{k}},\bar{x}}$ is homogeneous with $w^0(\tilde{\sigma}) = a$ and $v_\lambda(\tilde{\sigma}) = b$, and $(w^+)^0(\sigma) = (a + \delta b)/(1+\delta)$. The minimizer v_λ belongs to $\mathcal{Q}^0$, the central slice of w, so condition (11) for w implies

$$b = v_\lambda(\tilde{\sigma}) \le C w^0(\tilde{\sigma}) = Ca \le C(a + \delta b) = C(1+\delta)(w^+)^0(\sigma) \le 2C(w^+)^0(\sigma).$$

The central slice $\mathcal{Q}_i^0(w^+)$ is the central slice of v_λ for the data of **Step 2**, since the isomorphism of **Step 3** sends $\mathrm{in}_{w^+}(h)$ and $\mathrm{in}_{w^+}(F_j)$ to $\mathrm{in}_{v_\lambda}(h_0)$ and $\mathrm{in}_{v_\lambda}(F_{j,0})$. Then $v_\lambda^0(\mathrm{in}_{v_\lambda}(\tilde{\sigma})) = b$ and $b_\lambda^0(v_\lambda^0) = b_\lambda(v_\lambda) \ge 1$. For any $\varsigma^0 \in \mathcal{Q}_i^0(w^+)$, [OAI26b, Corollary E.10] for the height b_λ^0 and the bound $b_\lambda^0 \le 2 + 1/\tau$ on the central slice of v_λ of **Step 2** imply

$$\varsigma^0(\sigma) \le 2^{d_0} \frac{b_\lambda^0(\varsigma^0)}{b_\lambda^0(v_\lambda^0)} v_\lambda^0(\sigma) \le 2^{d_0}\left(2 + \frac{1}{\tau}\right) b \le C^+(w^+)^0(\sigma).$$

Thus w^+ satisfies condition (11) with the constant C^+, which depends only on d_0, τ, and C.

Step 5. In this step we iterate **Steps 2–4** and conclude the proof.

For any i, let ϵ_i be the supremum of $\{0\} \cup \{\mathrm{mult}_P \Gamma/D_i(P) - 1\}$, where (P, Γ) runs over $\mathcal{A}_i$. For $(P, \Gamma) \in \mathcal{A}_i$, **Step 2** up to (10.6), which uses no assumption on τ, gives $\hat{q} \in \mathcal{Q}_i^0$ with $\hat{q}(\Phi_0)/m = \mathrm{mult}_P \Gamma/D_i(P)$, where Φ_0 is homogeneous with $w_i^0(\Phi_0) = m$. By condition (11), $\hat{q}(\Phi_0) \leq Cm$. Hence $0 \leq \epsilon_i \leq C$.

Suppose that $\lim_i \epsilon_i = 0$ does not hold. Then there exist a rational number $\tau > 0$ and an infinite set I such that $\epsilon_i > 2\tau$ for any $i \in I$, so the assumption of **Steps 2–4** holds. Possibly passing to the subsequence I, **Steps 2–4** imply that w_i^+ satisfies conditions (7)–(11) of Set-up 10.12 in place of w_i with the constant C^+ in place of C, that $(1 - 1/i)w_i \leq w_i^+ \leq (1 + 1/i)w_i$, and that the rational rank of the value group of w_i^+ is larger than that of the value group of w_i. Possibly replacing w_i with w_i^+ and C with C^+, we may repeat the argument. By the Abhyankar inequality [Abh56, Theorem 1], the rational rank of the value group of a valuation on $S_{i,\bar{k}_i}$ centered at $\bar{x}_i$ is at most $\dim S_i \leq d_0$, and the rational rank of the value group of w_i is at least 1. Thus after at most $d_0 - 1$ replacements, we obtain a subsequence and valuations w_i' satisfying conditions (7)–(11) with a constant independent of i, such that $\lim_i \epsilon_i = 0$ for the sets $\mathcal{A}_i$ defined with w_i' in place of w_i, and

$$\left(1 - \frac{1}{i}\right)^{d_0} w_i \leq w_i' \leq \left(1 + \frac{1}{i}\right)^{d_0} w_i.$$

Let $e_i := (1+1/i)^{d_0} - 1$ and $\eta_i := \epsilon_i$ for w_i'. Then $\lim_i e_i = \lim_i \eta_i = 0$, and $1 - e_i \leq (1 - 1/i)^{d_0}$, which implies the inequalities in (1). By **Step 1**, the induced action of G_i on $\mathrm{Spec}\, \mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i}, \bar{x}_i}$ is quasi-étale and the generator of the canonical sheaf there has character χ_i. Since $w_i' \in \mathcal{Q}_i$, condition (8) implies $w_i'(f_{ij}) = 0$ for any j. Thus (1) holds. For any $(P, \Gamma) \in \mathcal{A}_i$ defined with w_i', $\mathrm{mult}_P \Gamma \leq (1 + \epsilon_i) D_i(P) = (1 + \eta_i) \mathrm{mult}_P D_i$ by Definition-Proposition 10.14. If $\mathcal{A}_i = \emptyset$, then $\epsilon_i = 0$. Thus (2) holds. □

10.3. **Section counts, characters, and termination.** In this subsection, we prove the versions over k_i of the inputs of [OAI26b, Theorem G.1] which depend on the ground field, and Lemma 10.23. The changes to the proofs in [OAI26b] are listed in the following remark.

Remark 10.17. The proofs of [OAI26b, Lemma C.3, Corollary C.7, Proposition D.4, Corollaries D.5, D.6, D.7, and Theorem G.1] are written over an algebraically closed field, and the proof of [OAI26b, Theorem G.1] uses that the quotient is ϵ-lc. Below we list the places where the proofs break over k_i, the reason, and the replacement.

(1) In the proofs of [OAI26b, Lemma C.3 and Corollary C.7], the order condition [OAI26b, (C.5)] is used through [OAI26b, Lemma C.1] at two places: the triviality on a cutoff field of the prime divisors P with $a(P) = 0$ and small $\mathbf{D}(P)$, and the lower bound of the nef Cartier $\boldsymbol{b}$-divisor $\mathbf{H}$ at the elements of the test set $\mathcal{L}$ in the application of [OAI26b, Lemma A.3]. Over k_i, the condition [OAI26b, (C.5)] is not available, since Proposition 10.16 only implies (MO_k), in which the $\mathbb{Q}$-divisors Γ are defined over k_i. Both places follow from (MO_k) on $\mathcal{L}$, by Lemma 10.10(3) and (2)(c)(iii), respectively. The jet body estimates are proved over $\bar{k}_i$, and the numerical bounds hold over k_i, since dimensions of spaces of sections and ranks of multiplication maps do not change under the extension $\bar{k}_i/k_i$. Lemma 10.19 replaces [OAI26b, Lemma C.3], applied at the beginning of [OAI26b, Subsection G.3], and [OAI26b, Corollary C.7]. Lemmas 10.10 and 10.11 replace [OAI26b, Lemmas C.1 and C.2]. The last sentence of [OAI26b, Corollary C.7] is applied to prime divisors over $V_{i,\bar{k}_i}$ and subfields of $k_i(V_i)$, so it is used for the compositum with $\bar{k}_i$.

(2) [OAI26b, Proposition D.4 and Corollary D.7] do not assume [OAI26b, (C.5)]. Over k_i, they are applied over $\bar{k}_i$ to data defined over k_i, and the non-zero sections are taken over k_i. The proof of [OAI26b, Corollary D.5] uses [OAI26b, (C.5)] at the $\mathbb{Q}$-divisor $\Theta/(2m_1)$, where $\Theta := 2m_1 D + \mathrm{div}(g_+ g_-)$, and uses [OAI26b, Lemma C.2]. Over k_i, the functions g_+ and g_- are taken over k_i, so that (MO_k) can be applied to $\Theta/(2m_1)$, and [OAI26b, Lemma C.2] is replaced by Lemma 10.11. The proof of [OAI26b, Corollary D.6]

uses [OAI26b, Lemma C.3], which is replaced by Lemma 10.19. Lemma 10.22 replaces [OAI26b, Corollary D.5] in [OAI26b, Subsections G.3 and G.8], [OAI26b, Corollary D.6] in the count [OAI26b, (G.11)] of [OAI26b, Subsection G.4], which is used again in [OAI26b, Subsections G.6 and G.8], and [OAI26b, Corollary D.7] in [OAI26b, Subsection G.6].

(3) In (I5) of [OAI26b, Subsection G.1], the rank refinement [OAI26b, Proposition F.7] implies [OAI26b, (C.5)], which is not available over k_i for the reason in (1). It is replaced by Proposition 10.16. The exact minimizers of the induction are defined over k_i by Definition-Proposition 6.3.

(4) In [OAI26b, Sections A–G], the ϵ-lc condition is used only in the statement of [OAI26b, Theorem G.1] and in the final case, treated in the first paragraph of [OAI26b, Subsection G.2], where the log discrepancy $1/\gamma$ of the divisor induced on the quotient by the rank one working valuation tends to zero; [OAI26b, Subsection G.9] refers back to this case. The use in the final case is replaced by Lemma 10.23.

(5) The remaining choices in [OAI26b, Section G], namely character projections, finite linear combinations, and Bertini parameters in non-empty Zariski open subsets of coefficient spaces of finite dimension, can be made over k_i, since k_i is infinite and contains all roots of unity. The dimension inequalities of [OAI26b, (G.11) and Lemma G.3] concern spaces over k_i and multiplication maps between them, and hold over k_i for the reason in (1). [OAI26b, Lemma G.2] holds over k_i, since its proof uses, besides the inclusions $S(C_1L_iD_i) \subset \mathcal{K}_*$ and the triviality on $\mathcal{K}_*$ given by Lemma 10.19(1) and (2), only [OAI26b, Proposition F.4] over $\bar{k}_i$ and the triviality on algebraic extensions in **Step 3** of the proof of Lemma 10.19. With the replacements (1)–(4), the induction in the proof of [OAI26b, Theorem G.1] runs over k_i, provided that condition [OAI26b, (G.1)] holds for T_i and the semi-invariants in $\mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$, namely, for any positive integer m and any sufficiently large i, $T_i(\sigma) \geq m$ for any non-zero $\sigma \in \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ with $g^*\sigma = \chi_i(g)^m\sigma$ for any $g \in G_i$, that the working valuations are defined over k_i and invariant under G_i, and that Proposition 10.16, Lemma 10.19, and Lemma 10.22 can be applied at each stage. We verify the three conditions in Subsection 10.4.

Definition 10.18. Notation and conditions as in Set-up 10.6. We use the notation of [OAI26b, Lemma C.3]. For $f_1, \dots, f_s \in k_i(V_i)^*$, where s may depend on i, and a prime divisor P over $V_{i,\bar{k}_i}$, let $J(P) := \max\{0, \max_j(-\operatorname{ord}_P f_j)\}$, a non-negative integer. The cost scale of [OAI26b, Lemma C.3], denoted L_0 there, is denoted by real numbers $\Lambda_i \geq 1$ here. The test set is $\mathcal{L} := \{P \mid a_i(P) = J(P) = 0\}$. For a real number $t \geq 0$, let

$$S(tD_i) := \{0\} \cup \{f \in k_i(V_i)^* \mid \operatorname{div}(f) + tD_i \geq 0\}$$

as in [OAI26b, (C.1)], which is also the space of sections of the $\boldsymbol{b}$-divisor $t\mathbf{D}_i$.

Lemma 10.19. *Notation and conditions as in Set-up 10.6, and notation as in Definition 10.18 for $f_1, \dots, f_s \in k_i(V_i)^*$, where s may depend on i, and for real numbers Λ_i. Assume that there exist real numbers $C, C' > 0$ independent of i such that $0 \leq J \leq C\Lambda_i\mathbf{D}_i$ and $1 \leq \Lambda_i \leq C'L_i$ for any i, which is* [OAI26b, (C.36)]*, that the other hypotheses of* [OAI26b, Lemma C.3] *except* [OAI26b, (C.5)] *hold, and that* (MO_k) *holds at any element of $\mathcal{L}$ for any i. Then, possibly passing to a subsequence, there exist subfields $\mathcal{K}_* \subset k_i(V_i)$ containing k_i whose transcendence degree e_* over k_i is independent of i, a real number $B_* > 0$, and real numbers $v_i > 0$ such that the following hold.*

(1) For any real number $C_1 > 0$, $S(C_1L_iD_i) \subset \mathcal{K}_*$ for any sufficiently large i.

(2) For any sufficiently large i, $S(B_*L_iD_i)$ generates $\mathcal{K}_*$ as a field over k_i, and any $P \in \mathcal{L}$ with $\operatorname{ord}_P(g) \geq 0$ for any $g \in S(B_*L_iD_i)$ is trivial on $\mathcal{K}_*$.

(3) There exists a real number $C_* > 0$ such that for any positive integer n and any sufficiently large i, $\dim_{k_i} S(nL_iD_i) \leq C_*v_in^{e_*}$, and the dimension over k_i of the k_i-vector space spanned by the products $g_1 \cdots g_n$ with $g_1, \dots, g_n \in S(B_*L_iD_i)$ is at least $v_in^{e_*}$.

(4) *For any real number $C_1 > 0$ and any finite subsets $\{g_{i,1}, \dots, g_{i,r_i}\} \subset S(C_1 L_i D_i)$, the algebraic closure in $k_i(V_i)$ of the subfield $k_i(g_{i,1}, \dots, g_{i,r_i})$ is contained in $\mathcal{K}_*$ for any sufficiently large i.*

The other assertions of [OAI26b, Lemma C.3 and Corollary C.7] *also hold over k_i.*

Proof. By Remark 10.17(1), it suffices to prove the following four statements.

Step 1. In this step we show that there exists a real number $c > 0$ independent of i such that $I_\beta \subset \mathcal{L}$ for any i and any rational number $0 < \beta \le c/L_i$. In particular, if $0 < \beta_i \le c/L_i$ are rational numbers, then any element of I_{β_i} is trivial on $\mathcal{K}_{\beta_i}$ for any sufficiently large i.

Let $c := \min\{1, 1/(2CC')\}$ for the real numbers C and C' of the statement. Since $L_i \ge 1$, any rational number $0 < \beta \le c/L_i$ satisfies $\beta \le 1$, so I_β is as in Construction 10.9. Let $P \in I_\beta$. Then $a_i(P) = 0$ and

$$J(P) \le C\Lambda_i \mathbf{D}_i(P) \le CC' L_i \beta \le \frac{1}{2}.$$

Since $J(P)$ is a non-negative integer, $J(P) = 0$. Thus $P \in \mathcal{L}$. Hence (MO_k) holds at any element of I_{β_i}, and the second statement follows from Lemma 10.10(3).

Step 2. In this step we show that the cutoff fields are subfields of $k_i(V_i)$ and that, for any $P \in \mathcal{L}$, the nef Cartier $\boldsymbol{b}$-divisor $\mathbf{H}$ satisfies at P the lower bound of [OAI26b, Lemma C.1].

The cutoff fields are the fields $\mathcal{K}_\beta$ of Construction 10.9, which are algebraically closed in $k_i(V_i)$ by Lemma 10.10(1). The nef Cartier $\boldsymbol{b}$-divisors $\mathbf{H}$ of Lemma 10.10(2)(c) are defined over k_i, and (MO_k) holds at any $P \in \mathcal{L}$ by assumption. Thus the lower bound at P is Lemma 10.10(2)(c)(iii).

Step 3. In this step we show the following. Let $K \subset k_i(V_i)$ be a subfield containing k_i, and let P be a prime divisor over $V_{i,\bar{k}_i}$ such that $\mathrm{ord}_P(y) = 0$ for any $y \in K^*$. Then $\mathrm{ord}_P(y) = 0$ for any non-zero $y \in \bar{k}_i(V_i)$ which is algebraic over K. In particular, ord_P is trivial on the compositum $K\bar{k}_i$.

This step replaces the last sentence of [OAI26b, Corollary C.7], since here P is a prime divisor over $V_{i,\bar{k}_i}$ while K is a subfield of $k_i(V_i)$. Let $\sum_{j=0}^n c_j y^{n-j} = 0$, where $c_0 := 1$, be the minimal polynomial of y over K. Then $c_n \ne 0$. If $\mathrm{ord}_P(y) < 0$, then $\mathrm{ord}_P(y^n) < \mathrm{ord}_P(c_j y^{n-j})$ for any $j \ge 1$ with $c_j \ne 0$, which is impossible since the sum is zero. If $\mathrm{ord}_P(y) > 0$, then $\mathrm{ord}_P(c_n) = 0 < \mathrm{ord}_P(c_j y^{n-j})$ for any $j < n$ with $c_j \ne 0$, which is again impossible. Since $\bar{k}_i$ is algebraic over $k_i \subset K$, the compositum $K\bar{k}_i$ is algebraic over K.

Step 4. In this step we show that the dimensions and the field generation in [OAI26b, Lemma C.3] do not change under the base change from k_i to $\bar{k}_i$.

For any real number $t \ge 0$, the natural homomorphism $S(tD_i) \otimes_{k_i} \bar{k}_i \to S(tD_{i,\bar{k}_i})$ is an isomorphism by Lemma 10.8(1), applied to the reflexive sheaf $\mathcal{O}_{V_i}(\lfloor tD_i \rfloor)$ (the round-down commutes with the base change, since the pullback of a prime divisor on V_i to $V_{i,\bar{k}_i}$ is reduced by Lemma 3.3(1)). For a subspace $W \subset k_i(V_i)$ with $\dim_{k_i} W < +\infty$ and a positive integer N, the span of products of N elements of W is the image of the multiplication map $W^{\otimes N} \to k_i(V_i)$, and the image of the base change of the multiplication map to $\bar{k}_i$ is the base change of the image. Thus the dimensions in (3) computed over k_i coincide with the dimensions over $\bar{k}_i$, and the bounds proved over $\bar{k}_i$ with jet bodies hold over k_i. Let $K' \subset \mathcal{K}_*$ be the subfield generated over k_i by $S(B_* L_i D_i)$. The subfield of $\bar{k}_i(V_i)$ generated over $\bar{k}_i$ by $S(B_* L_i D_i) \otimes_{k_i} \bar{k}_i$ is $K' \otimes_{k_i} \bar{k}_i$. If $K' \otimes_{k_i} \bar{k}_i = \mathcal{K}_* \otimes_{k_i} \bar{k}_i$, then $(\mathcal{K}_*/K') \otimes_{k_i} \bar{k}_i = 0$, and hence $K' = \mathcal{K}_*$. □

Lemma 10.21 below verifies the conditions of Set-up 10.6 for the angular data of Definition-Proposition 10.14, the last one under the cost bound of Lemma 10.21(3).

Set-up 10.20. Notation and conditions as in Set-up 10.12. Let $w_i' \in \mathcal{Q}_i$ be valuations centered at $\bar{x}_i$ which are invariant under G_i and $\mathrm{Gal}(\bar{k}_i/k_i)$, and let $e_i \ge 0$ and $\eta_i \ge 0$ be real numbers with $\lim_i e_i = \lim_i \eta_i = 0$, such that conditions (1) and (2) of Proposition 10.16

hold. Let v_i' be the restriction of w_i' to $k_i(S_i)$, $(R_i, M_i, u_i, V_{u_i}, D_i, a_i)$ the angular data of v_i', $\bar{f}_{ij} := \mathrm{in}_{v_i'}(F_{ij})/\mathrm{in}_{v_i'}(h_i)^{m_{ij}}$, and $\mathbf{D}_i$ the Cartier $\boldsymbol{b}$-divisor of D_i.

Lemma 10.21. *Notation and conditions as in Set-up 10.20, and let $L_i \geq 1$ be real numbers. Then the following hold.*

(1) $\mathrm{ord}_P(\bar{f}_{ij}) \geq -m_{ij}\mathbf{D}_i(P)$ for any i, any j, and any prime divisor P over $V_{u_i,\bar{k}_i}$.
(2) V_{u_i}, a_i, D_i, L_i, and η_i satisfy the conditions of Set-up 10.6 with d_0 in place of d, except possibly the condition in the last sentence of Set-up 10.6.
(3) If moreover there exists a real number $C' > 0$ such that $m_{ij} \leq C'L_i$ for any i and j, then the condition in the last sentence of Set-up 10.6 also holds.

Proof. Fix i and j. Let $r := |G_i|$ and $m := m_{ij}$, and let $\mathcal{K}_i$ be the function field of V_{u_i}.

(1) Since $w_i'(f_{ij}) = 0$ by Proposition 10.16(1), $\mathrm{in}_{v_i'}(F_{ij})$ and $\mathrm{in}_{v_i'}(h_i)^m$ are homogeneous of degree mu_i, and both are eigenvectors of G_i with character χ_i^m. Thus $\bar{f}_{ij} \in \mathcal{K}_i$, and $\mathrm{in}_{v_i'}(F_{ij})^r \in R_i$ is homogeneous of degree rmu_i. Write $\mathrm{in}_{v_i'}(F_{ij})^r = \phi z^{rmu_i}$ with $\phi \in \mathcal{K}_i$. Since $\mathrm{in}_{v_i'}(h_i)^r = f_h z^{ru_i}$, we have $\bar{f}_{ij}^r = \phi f_h^{-m}$. Let P be a prime divisor over $V_{u_i,\bar{k}_i}$. By Definition-Proposition 10.14(4), there exists $y \in \Pi_P$ such that $\mathbf{D}_i(P) = \mathrm{ord}_P(f_h)/r + \langle y, u_i\rangle$. Then $\mathrm{ord}_P(\phi) + rm\langle y, u_i\rangle \geq 0$, and hence

$$r\,\mathrm{ord}_P(\bar{f}_{ij}) = \mathrm{ord}_P(\phi) - m\,\mathrm{ord}_P(f_h) \geq -rm\Big(\frac{\mathrm{ord}_P(f_h)}{r} + \langle y, u_i\rangle\Big) = -rm\mathbf{D}_i(P).$$

(2) By Definition-Proposition 10.14(1)(2)(5), V_{u_i} is a geometrically integral normal projective variety over k_i, $R_i \otimes_{k_i} \bar{k}_i = (\mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i})^{G_i}$, and $D_i \geq 0$ is an ample $\mathbb{Q}$-divisor over k_i. Let $\Omega_i := \mathrm{Frac}(\mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i})$. The field Ω_i is generated over the subfield of homogeneous fractions of degree zero, which embeds into the residue field of w_i', by homogeneous elements whose degrees generate a group of rank at most the rational rank of w_i'. By [Bou89, Chapter VI, §10, no. 3, Corollary 1], applied to the trivial valuation on $\bar{k}_i$ and to w_i', the transcendence degree of Ω_i over $\bar{k}_i$ is at most $\dim S_i \leq d_0$. Since $\mathcal{K}_i \otimes_{k_i} \bar{k}_i$ embeds into Ω_i, $\dim V_{u_i} \leq d_0$.

By [OAI26b, Proposition D.1(i)(iii)], $V_{u_i,\bar{k}_i}$ is of Fano type, a_i is the log discrepancy function (called the discrepancy function there) of an lc Calabi–Yau pair on $V_{u_i,\bar{k}_i}$, and $a_i + \mathbf{D}_i$ is the log discrepancy function of a generalized klt Calabi–Yau pair $(V_{u_i,\bar{k}_i}, B_i', \mathbf{D}_i)$ with $B_i' \geq 0$ and nef part $\mathbf{D}_i$. We take $B_i := -\mathrm{div}(\theta_0)$ and $\mathbf{M}_i := 0$ in Set-up 10.6, where θ_0 is the form of Definition-Proposition 10.14, which is defined over k_i. Then B_i is defined over k_i, $K_{V_{u_i}} + B_i \sim 0$, and $a(P, V_{u_i}, B_i) = 1 + \mathrm{ord}_P(\theta_0) = a_i(P)$ for any prime divisor P over $V_{u_i,\bar{k}_i}$. Since $a_i(P) = 1 + \mathrm{ord}_P(\theta_0)$, $a_i(P) \in \mathbb{Z}$ for any prime divisor P over $V_{u_i,\bar{k}_i}$. Since $a(P, V_{u_i}, B_i) = a_i(P)$, we have $B_i' = B_i - D_i$, which is the trace of $a_i + \mathbf{D}_i$ on V_{u_i}, and $B_i' \geq 0$ as in condition (6) of Set-up 10.6. Moreover, $B_i \geq 0$ by [OAI26b, Proposition D.1(iii)]. Finally, $\lim_i \eta_i = 0$ by Set-up 10.20.

(3) Let P_i be prime divisors over $V_{u_i,\bar{k}_i}$ with $a_i(P_i) = 0$ and $\lim_i L_i\mathbf{D}_i(P_i) = 0$ along a subsequence. By (1),

$$-\mathrm{ord}_{P_i}(\bar{f}_{ij}) \leq C'L_i\mathbf{D}_i(P_i) < 1$$

for any j and any sufficiently large i. Since $\mathrm{ord}_{P_i}(\bar{f}_{ij})$ is an integer, $\mathrm{ord}_{P_i}(\bar{f}_{ij}) \geq 0$ for any j. Thus (MO_k) holds at P_i by Proposition 10.16(2). □

Lemma 10.22. *Notation and conditions as in Set-up 10.20. Assume the hypotheses of* [OAI26b, Proposition D.4 and Corollaries D.5, D.6, D.7] *for $(R_i, M_i, u_i, V_{u_i}, D_i, a_i)$, with the order condition* [OAI26b, (C.5)]*, wherever it is assumed there, replaced by the inequality of Proposition 10.16(2). Then the conclusions of* [OAI26b, Proposition D.4 and Corollaries D.5, D.6, D.7] *hold, the rational functions, sections, and eigenvectors in the conclusions are defined over k_i, and the dimensions of the spaces of sections are computed over k_i.*

Proof. Write $V := V_{u_i}$ and $v := v_i'$, let $\mathcal{K}$ be the function field of V, and let L_i be the scale of [OAI26b, Corollary D.5]. In [OAI26b, Corollary D.5], [OAI26b, (C.5)] is assumed at the

prime divisors P_i with $a_i(P_i)=0$ and $\lim_i L_i\mathbf{D}_i(P_i)=0$. The inequality of Proposition 10.16(2) is (10.1), so the replaced hypothesis is condition (9) of Set-up 10.6. The other conditions of Set-up 10.6 hold for V, a_i, D_i, L_i, and η_i by Lemma 10.21(2). When the lemma is applied in the proof of Theorem 10.2, condition (9) of Set-up 10.6 is verified by Lemma 10.21(3). By Remark 10.17(2), it suffices to prove the following statements.

Step 1. In this step we show that the spaces $S(qJ_m)$ of [OAI26b, Proposition D.4] are defined over k_i and commute with the base change to $\bar{k}_i$.

Let $\xi\in\mathrm{Frac}(R_i)$ be homogeneous of degree $\mu\in M_i$, and let m be an integer such that $\mu+mu_i$ is contained in the interior of the cone generated by the degrees of the non-zero homogeneous elements of R_i. By [OAI26b, Lemma D.3], for a positive integer q (this follows from [OAI26b, Lemma D.3] over $\bar{k}_i$ and $(A\otimes_{k_i}\bar{k}_i)\cap\mathrm{Frac}(A)=A$ for $A=\mathrm{gr}_v\mathcal{O}_{S_i,x_i}$ by faithful flatness),

$$S(qJ_m)=\{g\in\mathcal{K}\mid g\xi^q\,\mathrm{in}_v(h_i)^{qm}\in\mathrm{gr}_v\mathcal{O}_{S_i,x_i}\},$$

where the ring $\breve{R}$ of [OAI26b, Section D] is $\mathrm{gr}_v\mathcal{O}_{S_i,x_i}$ here, and the element h there is $\mathrm{in}_v(h_i)$. The $\boldsymbol{b}$-divisor J_m is determined by the Gauss inequalities of the homogeneous elements of R_i, so J_m is defined over k_i. Since $\mathrm{gr}_v\mathcal{O}_{S_i,x_i}\otimes_{k_i}\bar{k}_i\cong\mathrm{gr}_{w_i'}\mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ by Lemma 10.13, and since $\bar{k}_i$ is a free k_i-module,

$$S(qJ_m)\otimes_{k_i}\bar{k}_i=\{g\in\bar{k}_i(V)\mid g\xi^q\,\mathrm{in}_{w_i'}(h_i)^{qm}\in\mathrm{gr}_{w_i'}\mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}\},$$

which is the space $S(qJ_m)$ over $\bar{k}_i$.

Step 2. In this step we show the conclusion of [OAI26b, Proposition D.4] over k_i.

[OAI26b, Proposition D.4] does not assume [OAI26b, (C.5)], and the sandwich inequalities in the hypothesis concern the $\boldsymbol{b}$-divisors J_{m_0} and J_m, which are defined over k_i. By [OAI26b, Proposition D.4] over $\bar{k}_i$, there exists a bounded positive integer q such that $S(qJ_m)\otimes_{k_i}\bar{k}_i\neq 0$. Thus $S(qJ_m)\neq 0$ by **Step 1**.

Step 3. In this step we show the conclusion of [OAI26b, Corollary D.5] over k_i.

Let $L_i'\geq L_i$ be real numbers. The conditions of Set-up 10.6 hold with L_i' in place of L_i. Indeed, among these conditions only (7) and (9) involve L_i, and (7) holds since $L_i'\geq L_i\geq 1$. For (9), let P_i be prime divisors over $V_{\bar{k}_i}$, along a subsequence, such that $a_i(P_i)=0$ and $\lim_i L_i'\mathbf{D}_i(P_i)=0$. Since $D_i\geq 0$, we have $0\leq L_i\mathbf{D}_i(P_i)\leq L_i'\mathbf{D}_i(P_i)$, so $\lim_i L_i\mathbf{D}_i(P_i)=0$, and (MO_k) holds at P_i for any sufficiently large i by condition (9) of Set-up 10.6.

Let $\mu\in M_i$ satisfy the width condition of [OAI26b, Corollary D.5] at the scale L_i', that is, $|\langle y,\mu\rangle|\leq C_0L_i'\langle y,u_i\rangle$ for any y in the dual cone of the cone of **Step 1**, where $C_0>0$ is a fixed real number, and let $m_1=O(L_i')$ be a positive integer such that $\mu+m_1u_i$ and $-\mu+m_1u_i$ are contained in the interior of the cone of **Step 1**. For $\xi=z^{\pm\mu}$, the $\boldsymbol{b}$-divisor J_{m_1} of **Step 1** is, up to a principal $\mathbb{Q}$-divisor, the $\boldsymbol{b}$-pullback of an ample $\mathbb{Q}$-divisor on the ray quotient of $\pm\mu+m_1u_i$ [OAI26b, Proposition D.1(ii)]. Hence $S(nJ_{m_1})\otimes_{k_i}\bar{k}_i\neq 0$ for some positive integer n, and so $S(nJ_{m_1})\neq 0$ by **Step 1**. Take non-zero $\phi_\pm\in S(nJ_{m_1})$ for $\xi=z^{\pm\mu}$, with a common n, and set $g_\pm:=\phi_\pm^{1/n}$. The functions $g_\pm$ are used only through the $\mathbb{Q}$-divisors $\mathrm{div}(g_\pm):=\frac{1}{n}\mathrm{div}(\phi_\pm)$ on V and the numbers $\mathrm{ord}_P(g_\pm):=\frac{1}{n}\mathrm{ord}_P(\phi_\pm)$ for prime divisors P over $V_{\bar{k}_i}$. In particular, ℓ_i and Θ below are $\mathbb{Q}$-divisors over k_i.

For a prime divisor P over $V_{\bar{k}_i}$ and $y\in\Pi_P$, with Π_P, f_h, and r as in Definition-Proposition 10.14, let $\lambda_P(y):=\mathrm{ord}_P(f_h)/r+\langle y,u_i\rangle$. Then $\mathbf{D}_i(P)=\min_{y\in\Pi_P}\lambda_P(y)$, and the minimum is attained, by Definition-Proposition 10.14(4) and (5). For $\xi=z^\nu$ with $\nu\in M_i$ and an integer m such that $\nu+mu_i$ is contained in the interior of the cone of **Step 1**, we have $J_m(P)=\min_{y\in\Pi_P}\big(\langle y,\nu\rangle+m\lambda_P(y)\big)$ by [OAI26b, (D.3) and (D.11)]. Since $\mathrm{div}(\phi_\pm)+nJ_{m_1}\geq 0$ for $\xi=z^{\pm\mu}$, we obtain

$$\pm\langle y,\mu\rangle+m_1\lambda_P(y)+\mathrm{ord}_P(g_\pm)\geq 0$$

for any prime divisor P over $V_{\bar{k}_i}$ and any $y\in\Pi_P$. Let $\ell_i:=-\mathrm{div}(g_+)$ and $\Theta:=2m_1D_i+\mathrm{div}(g_+g_-)$. Adding the two inequalities and taking the minimum over $y\in\Pi_P$, we obtain

$\operatorname{mult}_P \Theta = 2m_1\mathbf{D}_i(P) + \operatorname{ord}_P(g_+) + \operatorname{ord}_P(g_-) \geq 0$ for any prime divisor P over $V_{\bar{k}_i}$. Thus $\Theta \geq 0$ and $\Theta \sim_{\mathbb{Q}} 2m_1 D_i$ over k_i. Let P be a prime divisor on $V_{\bar{k}_i}$ with $a_i(P) > 0$. By [OAI26b, Proposition D.1(iii), (D.8)], there exists $y \in \Pi_P \cap \operatorname{Hom}(M_i, \mathbb{Z})$ with $\lambda_P(y) = 0$, and hence $\mathbf{D}_i(P) = 0$ since $D_i \geq 0$. The two inequalities at this y give

$$0 \leq \langle y, \mu\rangle - \operatorname{mult}_P \ell_i \leq \operatorname{ord}_P(g_+) + \operatorname{ord}_P(g_-) = \operatorname{mult}_P \Theta,$$

so the integer $p_P := \langle y, \mu\rangle$ satisfies $|p_P - \operatorname{mult}_P \ell_i| \leq \operatorname{mult}_P \Theta$. Since $2m_1 = c_i L_i'$ with c_i bounded, Lemma 10.11, applied with L_i' in place of L_i and with $\Gamma_i = \Theta$, gives a positive integer q' and a real number $C_1 > 0$ such that, for any subsequence and any prime divisors P_i over $V_{\bar{k}_i}$ with $a_i(P_i) = 0$ and $\lim_i L_i'\mathbf{D}_i(P_i) = 0$, $\operatorname{dist}(q' \operatorname{mult}_{P_i} \ell_i, \mathbb{Z}) \leq C_1 L_i'\mathbf{D}_i(P_i)$ for any sufficiently large i. Moreover, for such P_i, (MO_k) holds at P_i for any sufficiently large i, as shown at the beginning of this step, and (MO_k), applied to $\Gamma = \Theta/(2m_1)$, gives

$$\operatorname{mult}_{P_i} \Theta \leq 2m_1(1 + \eta_i)\mathbf{D}_i(P_i).$$

Let $m_0 := q'm_1 + \lceil C_1 L_i' \rceil$ and $m := 2m_0$. Then $m_0 = O(L_i')$ and $m - m_0 = m_0 \geq C_1 L_i'$. Since u_i is contained in the interior of the cone of **Step 1** by Definition-Proposition 10.14(1), and $q'\mu + m_0 u_i = q'(\mu + m_1 u_i) + (m_0 - q'm_1)u_i$, the weights $q'\mu + m_0 u_i$ and $q'\mu + mu_i$ are contained in this interior. Let J_{m_0} and J_m be the $\boldsymbol{b}$-divisors of **Step 1** for $\xi = z^{q'\mu}$, and let P be a prime divisor over $V_{\bar{k}_i}$. For any $y \in \Pi_P$, the inequality for g_+ and $\lambda_P(y) \geq \mathbf{D}_i(P)$ give

$$q'\langle y, \mu\rangle + m_0\lambda_P(y) = q'\big(\langle y, \mu\rangle + m_1\lambda_P(y)\big) + (m_0 - q'm_1)\lambda_P(y) \geq q' \operatorname{mult}_P \ell_i + (m_0 - q'm_1)\mathbf{D}_i(P),$$

so $J_{m_0}(P) \geq q' \operatorname{mult}_P \ell_i + (m_0 - q'm_1)\mathbf{D}_i(P)$. Since $\lambda_P(y) \geq \mathbf{D}_i(P)$ for any $y \in \Pi_P$, we also have $J_m(P) \geq J_{m_0}(P) + (m - m_0)\mathbf{D}_i(P) \geq J_{m_0}(P)$. Take $y \in \Pi_P$ with $\lambda_P(y) = \mathbf{D}_i(P)$. The inequality for g_- gives $\langle y, \mu\rangle \leq m_1\mathbf{D}_i(P) + \operatorname{ord}_P(g_-) = \operatorname{mult}_P \ell_i - m_1\mathbf{D}_i(P) + \operatorname{mult}_P \Theta$, so

$$J_m(P) \leq q'\langle y, \mu\rangle + m\mathbf{D}_i(P) \leq q' \operatorname{mult}_P \ell_i + (m - q'm_1)\mathbf{D}_i(P) + q' \operatorname{mult}_P \Theta.$$

Let P_i be as in the previous paragraph, and let $\hat{p}_i$ be an integer nearest to $q' \operatorname{mult}_{P_i} \ell_i$. For any sufficiently large i, these inequalities give

$$\hat{p}_i \leq J_{m_0}(P_i) \leq J_m(P_i) \leq \hat{p}_i + \big(C_1 L_i' + m + 2q'm_1(1 + \eta_i)\big)\mathbf{D}_i(P_i),$$

where $C_1 L_i' + m + 2q'm_1(1 + \eta_i) = O(L_i')$. Together with $m - m_0 \geq C_1 L_i'$ and the two interior conditions above, these are the hypotheses of [OAI26b, Proposition D.4] at the scale L_i'. Thus, by **Step 2**, there exist a bounded positive integer q_1 and a non-zero $g \in S(q_1 J_m)$. Let $q_+ := q_1 q'$ and $b_+ := q_1 m$. Then q_+ is bounded, $b_+ = O(L_i')$, and by **Step 1**, $Y_+ := gz^{q_+\mu}$ is a non-zero invariant eigenfunction over k_i of weight $q_+\mu$ with $\operatorname{in}_v(h_i)^{b_+} Y_+$ regular.

The width condition and the integer m_1 do not change when μ is replaced with $-\mu$, and this replacement exchanges ϕ_+ and ϕ_-, hence g_+ and g_-, and does not change Θ. Replacing μ with $-\mu$, the argument above gives a non-zero invariant eigenfunction Y_- over k_i of weight $-q_-\mu$, with q_- bounded and $\operatorname{in}_v(h_i)^{b_-} Y_-$ regular for a positive integer $b_- = O(L_i')$. Let $b := \max\{q_- b_+, q_+ b_-\}$, so that $b = O(L_i')$. Since $\operatorname{in}_v(h_i)$ is regular, $Y_+^{q_-}$ and $Y_-^{q_+}$ are non-zero invariant eigenfunctions over k_i of weights $q_+q_-\mu$ and $-q_+q_-\mu$, and $\operatorname{in}_v(h_i)^b Y_+^{q_-}$ and $\operatorname{in}_v(h_i)^b Y_-^{q_+}$ are regular.

Finally, the multiplier and the cost can be bounded independently of L_i' and μ, that is, there exist a positive integer q_0 and a real number $C_2 > 0$ such that, for any real numbers $L_i' \geq L_i$, any sufficiently large i, and any $\mu \in M_i$ satisfying the width condition at the scale L_i', there exist non-zero invariant eigenfunctions $Y_\pm$ over k_i of weights $\pm q_0\mu$ and a positive integer $b \leq C_2 L_i'$ such that $\operatorname{in}_v(h_i)^b Y_+$ and $\operatorname{in}_v(h_i)^b Y_-$ are regular. Suppose not. Then, for any positive integer j, this statement fails for $q_0 = j!$ and $C_2 = j \cdot j!$, that is, there exist real numbers $L_i' \geq L_i$ such that, for infinitely many i, some $\mu \in M_i$ satisfying the width condition at the scale L_i' admits no such $Y_\pm$ and b. Choosing such indices inductively, we obtain indices $i_1 < i_2 < \cdots$, real numbers $L_{i_j}' \geq L_{i_j}$, and weights $\mu_{i_j} \in M_{i_j}$ satisfying the width condition at the scale L_{i_j}' which admit no such $Y_\pm$ and b with $q_0 = j!$ and $C_2 = j \cdot j!$. The conditions of Set-up 10.6 and the hypotheses of the lemma are preserved under passing to a subsequence. Replacing the sequence with $(i_j)_j$,

L'_i with L'_{i_j}, and μ with μ_{i_j}, the argument above gives a positive integer Q and a real number $C > 0$ such that, for any sufficiently large j, there exist non-zero invariant eigenfunctions $Y'_\pm$ over k_{i_j} of weights $\pm q\mu_{i_j}$ for some positive integer $q \le Q$, and a positive integer $b' \le CL'_{i_j}$ such that $\mathrm{in}_v(h_{i_j})^{b'}Y'_+$ and $\mathrm{in}_v(h_{i_j})^{b'}Y'_-$ are regular. If moreover $j \ge \max\{Q, C\}$, then q divides $j!$, and the eigenfunctions $(Y'_\pm)^{j!/q}$ of weights $\pm j!\mu_{i_j}$, with the cost $b'j!/q \le j \cdot j!L'_{i_j}$, contradict the choice of μ_{i_j}.

Step 4. In this step we show the conclusion of [OAI26b, Corollary D.6] over k_i.

Fix a real number $C_0 > 0$ as in [OAI26b, Corollary D.6]. Let N be a positive integer, let $0 \le t \le C_0NL_i$ be an integer, and let μ satisfy the width condition of [OAI26b, Corollary D.6],

$$|\langle y, \mu\rangle| \le C_0NL_i\langle y, u_i\rangle$$

for any y in the dual cone of the cone of **Step 1**. Let $W := W_{t,\mu}$ be the k_i-vector space of homogeneous elements of $\mathrm{gr}_v \mathcal{O}_{S_i,x_i}$ of degree $tu_i + \mu$ on which G_i acts by the character χ_i^t, and assume that $W \ne 0$. Dividing by $\mathrm{in}_v(h_i)^t$ a non-zero element of W, we obtain a G_i-invariant homogeneous element of degree μ, and hence $\mu \in M_i$. By **Step 3** applied to $-\mu$ at the scale NL_i, there exist a positive integer q, an eigenfunction Y over k_i of weight $-q\mu$, and a positive integer $b \le C_2NL_i$ such that $\mathrm{in}_v(h_i)^bY$ is regular, where q and the real number C_2 are independent of N and i. For $f \in W$, let $\Phi(f) := f^q Y/\mathrm{in}_v(h_i)^{qt}$, which is the map of [OAI26b, Corollary D.6]. Since $\mathrm{in}_v(h_i)^{qt+b}\Phi(f) = f^q\,\mathrm{in}_v(h_i)^bY$ is regular, $\Phi(f) \in S((qt+b)D_i)$ by [OAI26b, Lemma D.3, (D.13)]. Since $qt + b \le (qC_0 + C_2)NL_i$ and $D_i \ge 0$, $\Phi(f) \in S(C'NL_iD_i)$ for $C' := qC_0 + C_2$, which is independent of N and i. Thus Φ is a homogeneous polynomial map of degree q from W to $S(C'NL_iD_i)$, and Φ is defined over k_i. The map Φ is non-zero away from 0. If $\Phi(f)$ and $\Phi(f')$ are proportional over $\bar{k}_i$, then $(f/f')^q \in \bar{k}_i$, and hence $f/f' \in \bar{k}_i$, since f/f' is contained in a field containing the algebraically closed field $\bar{k}_i$. Thus Φ induces an injective morphism from the projective space of $W \otimes_{k_i} \bar{k}_i$ to the projective space of $S(C'NL_iD_i) \otimes_{k_i} \bar{k}_i$, and hence $\dim_{k_i} W \le \dim_{k_i} S(C'NL_iD_i)$. Thus Lemma 10.19(3) implies $\dim_{k_i} W \le C_3v_iN^{e_*}$ for a real number C_3 fixed before N.

Step 5. In this step we show the conclusion of [OAI26b, Corollary D.7] over k_i.

[OAI26b, Corollary D.7] does not assume [OAI26b, (C.5)], and the inequalities in the hypothesis concern symbols defined over k_i and Gauss tests over $\bar{k}_i$. By [OAI26b, Corollary D.7] over $\bar{k}_i$, the space of regular symbols over $\bar{k}_i$ with the weight $q(mu_i - \sum_j r_jk_j)$ and the character χ_i^{qm} of [OAI26b, (D.21)] contains a non-zero element, where q, m, the integers r_j, and the weights $\pm k_j \in M_i$ of the eigenfunctions $Y_j^\pm$ are those of [OAI26b, Corollary D.7]. Let $\mu := -q\sum_j r_jk_j$, and let $W_{qm,\mu}$ be the k_i-vector space of **Step 4** with $t = qm$, namely the space of homogeneous elements of $\mathrm{gr}_v \mathcal{O}_{S_i,x_i}$ of degree $q(mu_i - \sum_j r_jk_j)$ on which G_i acts by the character χ_i^{qm}. Since $\mathrm{gr}_v \mathcal{O}_{S_i,x_i} \otimes_{k_i} \bar{k}_i \cong \mathrm{gr}_{w'_i} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ by Lemma 10.13 and the projection $|G_i|^{-1}\sum_{g\in G_i} \chi_i(g)^{-qm}g^*$ is defined over k_i, the base change of $W_{qm,\mu}$ to $\bar{k}_i$ is the space of regular symbols over $\bar{k}_i$ above, as in **Step 1**. Thus $W_{qm,\mu}$ contains a non-zero element over k_i, with the multiplier and cost of [OAI26b, Corollary D.7]. □

In the proof of Lemma 10.22, **Step 5** uses only **Step 1** and Lemma 10.13, and does not use Proposition 10.16(2). Thus the conclusion of [OAI26b, Corollary D.7] over k_i also holds for the angular data of the exact minimizers of the tilted heights [OAI26b, (G.16)] of [OAI26b, Subsection G.6].

Lemma 10.23 replaces the ϵ-lc condition in the final case of [OAI26b, Subsection G.2] (see Remark 10.17(4)).

Lemma 10.23. *Notation and conditions as in Set-up 10.1, and fix i. Let w be a G_i-invariant valuation on S_i of the form $w = \delta\,\mathrm{ord}_{F_S}$, where $\delta > 0$ is a real number and F_S is a prime divisor over S_i, such that $\mathrm{center}_{S_i} w = x_i$, $a(w, S_i, 0) = w(h_i) = 1$, $\mathrm{gr}_w \mathcal{O}_{S_i,x_i}$ is finitely generated, geometrically normal, and klt, and the induced action of G_i on $\mathrm{Spec}\,\mathrm{gr}_w \mathcal{O}_{S_i,x_i}$ is quasi-étale. Let*

$\bar{w}$ be the restriction of w to $k_i(X_i)$. Then $\bar{w}(k_i(X_i)^*) = \gamma\mathbb{Z}$ for a real number $\gamma > 0$, $\bar{w} = \gamma \operatorname{ord}_F$ for a Kollár component F of $(X_i \ni o_i, 0)$ over k_i, and $a(F, X_i, 0) = 1/\gamma$.

Proof. Write $k := k_i$, $S := S_i$, $X := X_i$, $G := G_i$, $x := x_i$, $o := o_i$, and $p := p_i$. Let $\bar{o}$ be the point of $X_{\bar{k}}$ over o, and let $A := \operatorname{gr}_w \mathcal{O}_{S,x}$. Since p is quasi-étale, $K_S = p^*K_X$ by Lemma 2.5. Thus X is klt by [KM98, Proposition 5.20], and $X_{\bar{k}}$ is klt by Lemma 3.3(3). The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\operatorname{gr}_{\bar{w}} \mathcal{O}_{X,o} = A^G$, that A^G is finitely generated over k, and that $A^G \otimes_k \bar{k}$ is a normal domain.

Since the fiber of p over o is the orbit $\{x\}$, $\mathcal{O}_{X,o} = \mathcal{O}_{S,x}^G$, and $\mathfrak{a}_t(\bar{w}) = \mathfrak{a}_t(w)^G$ for any $t \geq 0$. Let $\pi := |G|^{-1} \sum_{g \in G} g^*$ be the Reynolds operator. Since w is G-invariant, $\pi(\mathfrak{a}_t(w)) \subset \mathfrak{a}_t(w)^G$. If the class of $f \in \mathfrak{a}_t(w)$ in $\mathfrak{a}_t(w)/\mathfrak{a}_{>t}(w)$ is G-invariant, then $\pi(f) - f \in \mathfrak{a}_{>t}(w)$. Thus the natural injection

$$\mathfrak{a}_t(w)^G/\mathfrak{a}_{>t}(w)^G \to (\mathfrak{a}_t(w)/\mathfrak{a}_{>t}(w))^G$$

is surjective, and $\operatorname{gr}_{\bar{w}} \mathcal{O}_{X,o} = A^G$. The ring A^G is finitely generated over k by Noether's finiteness theorem for invariants. Since $\pi \otimes \mathrm{id}_{\bar{k}}$ is the Reynolds operator on $A \otimes_k \bar{k}$, $A^G \otimes_k \bar{k} = (A \otimes_k \bar{k})^G$. The ring $A \otimes_k \bar{k}$ is normal and graded by $\mathbb{R}_{\geq 0}$, and the degree-zero part is $\kappa(x) \otimes_k \bar{k} = \bar{k}$, since $\operatorname{center}_S w = x$. Since $A \otimes_k \bar{k}$ is Noetherian and normal, it is a finite product of normal domains. Any idempotent of a reduced ring graded by $\mathbb{R}_{\geq 0}$ is homogeneous of degree zero, and the degree-zero part $\bar{k}$ has no non-trivial idempotents. Thus $A \otimes_k \bar{k}$ is a normal domain. Thus $(A \otimes_k \bar{k})^G$ is a normal domain.

Step 2. In this step we show that $\bar{w}(k(X)^*) = \gamma\mathbb{Z}$ and $\bar{w} = \gamma \operatorname{ord}_F$ for a real number $\gamma > 0$ and a prime divisor F over X with $a(F, X, 0) = 1/\gamma$.

By Lemma 2.5, $\operatorname{ord}_{F_S}|_{k(X)} = e \operatorname{ord}_F$ for a prime divisor F over X and a positive integer e, and $a(F_S, S, 0) = ea(F, X, 0)$. Thus $\bar{w} = \delta e \operatorname{ord}_F$, so $\bar{w}(k(X)^*) = \gamma\mathbb{Z}$ with $\gamma := \delta e$. Moreover,

$$1 = a(w, S, 0) = \delta a(F_S, S, 0) = \delta e a(F, X, 0) = \gamma a(F, X, 0).$$

Step 3. In this step we construct a valuation ψ on $X_{\bar{k}}$ which is invariant under $\mathrm{Gal}(\bar{k}/k)$, restricts to $\bar{w}$ on $k(X)$, and satisfies $\mathfrak{a}_t(\psi) = \mathfrak{a}_t(\bar{w}) \otimes_k \bar{k}$ for any $t \geq 0$.

Since o is k-rational, $\mathcal{O}_{X_{\bar{k}},\bar{o}} = \mathcal{O}_{X,o} \otimes_k \bar{k}$. For $t \geq 0$, let $\mathcal{F}_t := \mathfrak{a}_t(\bar{w}) \otimes_k \bar{k}$, and for a non-zero $f \in \mathcal{O}_{X_{\bar{k}},\bar{o}}$, let $\psi(f) := \sup\{t \mid f \in \mathcal{F}_t\}$. Since $\bar{k}$ is a free k-module, $\bigcap_t \mathcal{F}_t = (\bigcap_t \mathfrak{a}_t(\bar{w})) \otimes_k \bar{k} = 0$ and $\mathcal{F}_t \cap \mathcal{O}_{X,o} = \mathfrak{a}_t(\bar{w})$. Since $\mathfrak{a}_t(\bar{w}) = \mathfrak{a}_{\gamma\lceil t/\gamma \rceil}(\bar{w})$, we have $\psi(f) \in \gamma\mathbb{Z}_{\geq 0}$ and $f \in \mathcal{F}_{\psi(f)}$. The ideals $\mathcal{F}_t$ decrease in t and satisfy $\mathcal{F}_s\mathcal{F}_t \subset \mathcal{F}_{s+t}$, so $\psi(f+g) \geq \min\{\psi(f), \psi(g)\}$ and $\psi(fg) \geq \psi(f)+\psi(g)$. Since the base change to $\bar{k}$ is exact,

$$\bigoplus_t \mathcal{F}_t/\mathcal{F}_{>t} = \operatorname{gr}_{\bar{w}} \mathcal{O}_{X,o} \otimes_k \bar{k} = A^G \otimes_k \bar{k},$$

which is a domain by **Step 1**. Thus the product of the classes of f and g in degrees $\psi(f)$ and $\psi(g)$ is non-zero, and $\psi(fg) = \psi(f) + \psi(g)$. Hence ψ extends to a valuation of $\bar{k}(X)$ with $\mathfrak{a}_t(\psi) = \mathcal{F}_t$. Since $\mathcal{F}_t \cap \mathcal{O}_{X,o} = \mathfrak{a}_t(\bar{w})$, the restriction of ψ to $k(X)$ is $\bar{w}$. Since $\mathrm{Gal}(\bar{k}/k)$ acts on the second factor of $\mathcal{O}_{X,o} \otimes_k \bar{k}$ and preserves $\mathcal{F}_t$, $\psi \circ \sigma = \psi$ for any $\sigma \in \mathrm{Gal}(\bar{k}/k)$. Moreover, $\psi > 0$ on $\mathfrak{m}_{\bar{o}} = \mathfrak{m}_o \otimes_k \bar{k} = \mathcal{F}_\gamma$, since $\operatorname{center}_X(\bar{w}) = o$.

Step 4. In this step we show that $\psi = \gamma \operatorname{ord}_{E'}$ for a prime divisor E' over $X_{\bar{k}}$ with $\operatorname{center}_{X_{\bar{k}}}(E') = \bar{o}$ and $\operatorname{ord}_{E'} \circ \sigma = \operatorname{ord}_{E'}$ for any $\sigma \in \mathrm{Gal}(\bar{k}/k)$.

The extension $\bar{k}(X)/k(X)$ is algebraic. By [Bou89, Chapter VI, §10, no. 3, Corollary 1], the residue field of ψ is algebraic over the residue field of $\bar{w} = \gamma \operatorname{ord}_F$, which has transcendence degree $d-1$ over k. Thus the residue field of ψ has transcendence degree $d-1$ over $\bar{k}$. By **Step 3**, $\gamma\mathbb{Z} \subset \psi(\bar{k}(X)^*) \subset \gamma\mathbb{Z}$. Thus ψ/γ is a discrete valuation of $\bar{k}(X)$ with value group $\mathbb{Z}$, residue field of transcendence degree $d-1$ over $\bar{k}$, and center $\bar{o}$ on $X_{\bar{k}}$. By [KM98, Lemma 2.45], $\psi/\gamma = \operatorname{ord}_{E'}$ for a prime divisor E' over $X_{\bar{k}}$ with $\operatorname{center}_{X_{\bar{k}}}(E') = \bar{o}$. The Galois invariance of $\operatorname{ord}_{E'}$ follows from **Step 3**.

Step 5. In this step we show that E' is a Kollár component of $(X_{\bar{k}} \ni \bar{o}, 0)$.

Since $\psi(\bar{k}(X)^*) = \gamma\mathbb{Z}$, the graded ring $\mathrm{gr}_{\mathrm{ord}_{E'}} \mathcal{O}_{X_{\bar{k}},\bar{o}}$ is $\mathrm{gr}_\psi \mathcal{O}_{X_{\bar{k}},\bar{o}} = A^G \otimes_k \bar{k}$ with the degrees divided by γ, so it is finitely generated by **Step 1**. The morphism $\operatorname{Spec} A \to \operatorname{Spec} A^G$ is the quotient by G, and it is quasi-étale by assumption. Since $\operatorname{Spec} A$ is klt, $\operatorname{Spec} A^G$ is klt by Lemma 2.5 and [KM98, Proposition 5.20], and $\operatorname{Spec}(A^G \otimes_k \bar{k})$ is klt by Lemma 3.3(3). By Definition 2.9, since the boundary is zero, the marked graded pair of $\mathrm{ord}_{E'}$ is $(\operatorname{Spec}(A^G \otimes_k \bar{k}), 0)$, which is klt. Thus E' is a Kollár component of $(X_{\bar{k}} \ni \bar{o}, 0)$ by Theorem 2.10.

Step 6. In this step we conclude the proof.

Since o is k-rational and $\mathrm{ord}_{E'}$ is invariant under $\mathrm{Gal}(\bar{k}/k)$, Lemma 3.6(2) implies that $\mathrm{ord}_{E'}|_{k(X)} = \mathrm{ord}_{F'}$ for a Kollár component F' of $(X \ni o, 0)$ over k. By **Steps 2–4**, $\gamma \,\mathrm{ord}_{E'}|_{k(X)} = \psi|_{k(X)} = \bar{w} = \gamma \,\mathrm{ord}_F$. Thus $\mathrm{ord}_{F'} = \mathrm{ord}_F$, and $F = F'$ as prime divisors over X. Hence $\bar{w} = \gamma \,\mathrm{ord}_F$ for the Kollár component F, and $a(F, X, 0) = 1/\gamma$ by **Step 2**. □

10.4. **Proof of Theorem 10.2.** In this subsection, we prove Theorem 10.2.

Proof of Theorem 10.2. Suppose that the conclusion fails. Then for any positive integer m and any sufficiently large i, any non-zero $\sigma \in \mathcal{O}_{S_i,x_i}$ with $g^*\sigma = \chi_i(g)^m\sigma$ for any $g \in G_i$ satisfies $T_i(\sigma) \geq m$. By Remark 10.17(5), it suffices to verify the three conditions in Remark 10.17(5) and to exclude the final case of [OAI26b, Subsection G.2]. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the hypotheses of [OAI26b, Theorem G.1] other than ϵ-lc hold for $S_{i,\bar{k}_i}$, and that for any positive integer m and any sufficiently large i, any non-zero $\sigma \in \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ with $g^*\sigma = \chi_i(g)^m\sigma$ for any $g \in G_i$ satisfies $T_i(\sigma) \geq m$.

By Lemma 3.3(3), $S_{i,\bar{k}_i}$ is klt and $(S_{i,\bar{k}_i}, \mathrm{div}(h_i))$ is lc. The base change of p_i to $\bar{k}_i$ is quasi-étale, since any prime divisor on $X_{i,\bar{k}_i}$ is a component of the pullback of a prime divisor on X_i, and étaleness can be checked after the faithfully flat base change to $\bar{k}_i$. The remaining hypotheses of [OAI26b, Theorem G.1] other than ϵ-lc are Set-up 10.1(2)(3)(5)–(7) after the base change to $\bar{k}_i$ (for (5), $\omega_i \otimes 1$ generates the canonical sheaf of $S_{i,\bar{k}_i}$ at $\bar{x}_i$ [Har77, Propositions II.8.10 and III.9.3]).

Since $k_i \supset \mathbb{C}$ contains the values of χ_i, the projection $\pi_m := |G_i|^{-1} \sum_{g \in G_i} \chi_i(g)^{-m} g^*$ is defined over k_i. Since x_i is k_i-rational, $\mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i} = \mathcal{O}_{S_i,x_i} \otimes_{k_i} \bar{k}_i$. Let $\sigma \in \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ be non-zero with $g^*\sigma = \chi_i(g)^m\sigma$ for any $g \in G_i$, and write $\sigma = \sum_l c_l\sigma_l$ with $c_l \in \bar{k}_i$ and $\sigma_l \in \mathcal{O}_{S_i,x_i}$. Then $\sigma = \pi_m(\sigma) = \sum_l c_l\pi_m(\sigma_l)$, and any non-zero $\pi_m(\sigma_l)$ is an element of $\mathcal{O}_{S_i,x_i}$ with $g^*\pi_m(\sigma_l) = \chi_i(g)^m\pi_m(\sigma_l)$ for any $g \in G_i$. Since $T_i(c_l) = 0$ for $c_l \neq 0$,

$$T_i(\sigma) \geq \min_l\{T_i(\pi_m(\sigma_l)) \mid \pi_m(\sigma_l) \neq 0\} \geq m$$

for any sufficiently large i.

Step 2. In this step we show that each exact minimizer w of the induction is invariant under G_i and $\mathrm{Gal}(\bar{k}_i/k_i)$, that the restriction of w to $k_i(S_i)$ has center x_i, and that $\mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is finitely generated, normal, and klt.

Fix i and a stage of the induction. Let f_{ij} be the tests accumulated at the stage, and m_{ij} the costs of f_{ij}. By (I4) of [OAI26b, Subsection G.1], the height of the stage is either $v(h_i)$ or, at a stage with an anchor of anchor scale Λ_i', the height

$$v(h_i) - \lambda v(Y)/\Lambda_i'$$

of [OAI26b, (G.5)], where $\lambda \geq 0$, denoted μ in [OAI26b, (G.5)], is bounded and Y is a member of the anchor pair, a G_i-invariant ratio over k_i of cost $m_Y = O(\Lambda_i')$. Thus the height is the height b of Set-up 6.1 with $b_0 = 1 + \lambda m_Y/\Lambda_i'$, the single function $g_1 := h_i^{m_Y}Y$, and $\beta_1 = \lambda/\Lambda_i' \geq 0$. At the first stage, $b_0 = 1$, and the height involves no function g_l. Since Y is G_i-invariant and h_i is an eigenvector with character χ_i, the numerator $h_i^{m_Y}Y$ is an eigenvector with character $\chi_i^{m_Y}$.

The intersection of the G_i-translates of an affine open neighborhood of x_i is an affine open neighborhood of x_i, since S_i is separated and G_i fixes x_i. Possibly replacing S_i with such a G_i-stable affine open neighborhood, we may assume that h_i, the numerators $h_i^{m_{ij}} f_{ij}$, and the functions g_l are regular on S_i. We apply Set-up 6.1 with $X = S_i$, $x = x_i$, $B = 0$, $c = 1$, $h = h_i$, the numerators of f_{ij} as the functions f_j there with m_{ij} as the integers m_j there, and b_0, β_l, and g_l from the height. The pair $(S_i, \operatorname{div}(h_i))$ is lc, and $h_i(x_i) = 0$ since $T_i(h_i) = 1$ and T_i is centered at $\bar{x}_i$. The set $\mathcal{Q}_{\bar{x}}$ of Set-up 6.1 contains the working valuation of the stage, which is T_i at the first stage, and $b > 0$ on $\mathcal{Q}$ by [OAI26b, (G.4)]. For $v \in \mathcal{Q}$, the conditions $a(v, S_{i,\bar{k}_i}, 0) = v(h_i) = 1$ and $v(h_i^{m_{ij}} f_{ij}) \geq m_{ij}$ are the normalization of Set-up 10.1 and the inequalities $v(f_{ij}) \geq 0$. Thus Definition-Proposition 5.2 implies that $b^d \cdot \mathrm{vol}$ has a unique minimizer w on $\mathcal{Q}_{\bar{x}}$.

For $g \in G_i$, the pullbacks by g of h_i, of the numerators of f_{ij}, and of the functions g_l are non-zero constant multiples of the functions themselves, and valuations vanish on non-zero constants. Thus $\mathcal{Q}_{\bar{x}}$ and b are G_i-invariant. Any $\sigma \in \mathrm{Gal}(\bar{k}_i/k_i)$ fixes $\bar{x}_i$, since x_i is k_i-rational, and the data are defined over k_i. By the uniqueness of w, w is invariant under G_i and $\mathrm{Gal}(\bar{k}_i/k_i)$. By Definition-Proposition 6.3, the restriction of w to $k_i(S_i)$ has center x_i, $\mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is finitely generated, and the marked graded pair of w is klt. Since $B = 0$, the marked graded pair of w is $(\operatorname{Spec} \mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}, 0)$ by Definition 2.9, so $\operatorname{Spec} \mathrm{gr}_w \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$ is klt, and in particular normal. After absorbing the h-denominators, the tilted heights [OAI26b, (G.16)] of [OAI26b, Subsection G.6] have the form of Set-up 6.1 with rational coefficients and G_i-eigenfunctions over k_i. Since the pullbacks of the G_i-eigenfunctions by $g \in G_i$ are non-zero constant multiples of them and the coefficients are defined over k_i, the uniqueness in Definition-Proposition 5.2 implies that the exact minimizers of the tilted heights are invariant under G_i and $\mathrm{Gal}(\bar{k}_i/k_i)$. By Definition-Proposition 6.3, their restrictions to $k_i(S_i)$ have center x_i, their graded rings over $\bar{k}_i$ are finitely generated, and their marked graded pairs are klt, so, as for w (since $B = 0$), the spectra of their graded rings over $\bar{k}_i$ are klt, and in particular normal. By [OAI26b, Proposition F.2], whose hypotheses hold for $S_{i,\bar{k}_i}$ by **Step 1**, the induced action of G_i on the spectra of their graded rings over $\bar{k}_i$ is quasi-étale, and the generators of the canonical sheaves there have character χ_i. Thus the exact minimizers of the tilted heights satisfy the hypotheses of Definition-Proposition 10.14. The height [OAI26b, (G.27)] of [OAI26b, Subsection G.7] is adopted there as a stage height of the form [OAI26b, (G.5)].

Step 3. In this step we show that after each optimization, the conditions of Set-up 10.12 hold with $d_0 = d$, the tests f_{ij}, and the minimizer $w_i := w$ of **Step 2**, so that Proposition 10.16 can be applied.

Conditions (1)–(5) of Set-up 10.12 follow from Set-up 10.1. Since f_{ij} is G_i-invariant and h_i is an eigenvector with character χ_i, the numerator $h_i^{m_{ij}} f_{ij}$ is an eigenvector with character $\chi_i^{m_{ij}}$, which is condition (6). Condition (8) is the exactness of the tests maintained by the induction of [OAI26b, Subsection G.1], and condition (9) follows from **Step 2**. For condition (11), identify $\bar{k}_i$ with $\mathbb{C}$ by Lemma 3.2, and let b^0 be the height induced on $\mathcal{Q}_i^0$. By Proposition 5.8, applied to the germ of $S_{i,\bar{k}_i}$ at $\bar{x}_i$, w_i^0 is the unique minimizer of $(b^0)^d \cdot \mathrm{vol}$ on the elements of $\mathcal{Q}_i^0$ centered at the vertex. Since the heights of the stages are rational by (I4) of [OAI26b, Subsection G.1], the central slice case of [OAI26b, Corollary E.10] implies

$$\varsigma^0(\sigma) \leq 2^d \frac{b^0(\varsigma^0)}{b^0(w_i^0)} w_i^0(\sigma)$$

for any $\varsigma^0 \in \mathcal{Q}_i^0$ and any non-zero homogeneous σ. Since $w_i^0(\mathrm{in}_{w_i} f) = w_i(f)$ for any non-zero regular function f, [OAI26b, (G.4)] implies $b^0(w_i^0) = b(w_i) \geq \rho$ for a real number $\rho > 0$ independent of i. Since $\beta_1 \geq 0$ and ς^0 is non-negative on regular functions,

$$b^0(\varsigma^0) \leq b_0 \varsigma^0(\mathrm{in}_{w_i} h_i) = b_0,$$

and $b_0 = 1 + \lambda m_Y/\Lambda_i'$ is bounded independently of i, since λ is bounded and $m_Y = O(\Lambda_i')$. Thus condition (11) holds with a constant independent of i. At the later stages, condition (11) for the refined valuations follows from Proposition 10.16(1).

Step 4. In this step we show that at each stage, Lemmas 10.19 and 10.22 can be applied to the angular data $(R_i, M_i, u_i, V_{u_i}, D_i, a_i)$ of the restriction v_i' of the valuation w_i' of Proposition 10.16, with the elements $\bar{f}_{ij}$ of Proposition 10.16(2) as test functions.

Let L_i be the scale of the stage, and let L_i^- be the scale L^- of (I1) of [OAI26b, Subsection G.1] at the stage, so that $L_i^- = 1$ at the first stage. By (I1) of [OAI26b, Subsection G.1], there exists a real number $C_1 > 0$ such that $m_{ij} \le C_1 L_i^-$ for any i and j. By the first paragraph of [OAI26b, Subsection G.3], there exists a real number $C' > 0$ such that $L_i^- \le C' L_i$ for any i. Thus $m_{ij} \le C_1 C' L_i$, and by Lemma 10.21(2)(3), the conditions of Set-up 10.6 hold for V_{u_i}, a_i, D_i, L_i, and η_i. Let $\Lambda_i := L_i^-$. By Lemma 10.21(1), $0 \le J \le C_1 \Lambda_i \mathbf{D}_i$ for the functions $\bar{f}_{ij}$, and $1 \le \Lambda_i \le C' L_i$, which is [OAI26b, (C.36)]. The test set $\mathcal{L}$ consists of the prime divisors P over $V_{u_i,\bar{k}_i}$ with $a_i(P) = 0$ and $\mathrm{ord}_P(\bar{f}_{ij}) \ge 0$ for any j. Thus (MO_k) holds at any element of $\mathcal{L}$ by Proposition 10.16(2). The field hypothesis of [OAI26b, Lemma C.3] is (H2) of [OAI26b, Subsection G.1] by the first paragraph of [OAI26b, Subsection G.3]. It holds over k_i if and only if it holds over $\bar{k}_i$: the field generated over $\bar{k}_i$ by $S(\Lambda_i D_i) \otimes_{k_i} \bar{k}_i$ is the compositum $K'\bar{k}_i$ of $\bar{k}_i$ and the field K' generated by $S(\Lambda_i D_i)$ over k_i, and since $K'\bar{k}_i$ is algebraic over K', a prime divisor over $V_{u_i,\bar{k}_i}$ is trivial on K' if and only if it is trivial on $K'\bar{k}_i$ [Bou89, Chapter VI, §10, no. 3, Corollary 1]. The remaining hypotheses of [OAI26b, Lemma C.3] and the hypotheses of [OAI26b, Proposition D.4 and Corollaries D.5, D.6, D.7] other than [OAI26b, (C.5)] are inequalities maintained by the induction of [OAI26b, Section G], and they do not depend on the ground field. Thus Lemma 10.19 can be applied, and Lemma 10.22 can be applied by Proposition 10.16(2) and Lemma 10.21(3).

Step 5. In this step we show that the final case of [OAI26b, Subsection G.2] does not occur for any sufficiently large i.

Suppose that the final case occurs for infinitely many i. Possibly passing to a subsequence, we may assume that it occurs for any i. Let w_i' be the working valuation at the final stage, v_i' the restriction of w_i' to $k_i(S_i)$, and $\bar{v}_i$ the restriction of w_i' to $k_i(X_i)$. At the final stage, M_i has rank one. By [OAI26b, (G.6)], there exist a G_i-invariant ratio Y_i over k_i and a real number $c_0 > 0$ independent of i such that $|v_i'(Y_i)| \ge c_0 \Lambda_i'$ for the anchor scale Λ_i' of **Step 2**, and $\lim_i \Lambda_i' = +\infty$ by the first paragraph of [OAI26b, Subsection G.2]. By (H1) of [OAI26b, Subsection G.1], as applied in the first paragraph of [OAI26b, Subsection G.2], $v_i'(Y_i) = \nu_i \mu_{0,i}$ for a primitive element $\mu_{0,i}$ of M_i and a positive integer $\nu_i \le \nu_0$, where ν_0 is independent of i.

We verify the hypotheses of Lemma 10.23 for $w := v_i'$. By Lemma 10.13, $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i} \otimes_{k_i} \bar{k}_i \cong \mathrm{gr}_{w_i'} \mathcal{O}_{S_{i,\bar{k}_i},\bar{x}_i}$, which is finitely generated, normal, and klt by Proposition 10.16(1). Since v_i' is G_i-invariant by Proposition 10.16, G_i preserves each homogeneous component of $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$. Since G_i is cyclic and k_i contains the values of the characters of G_i, each homogeneous component is spanned by eigenvectors of G_i. For any homogeneous $s \in \mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$ which is an eigenvector of G_i, $s^{|G_i|} \in R_i$. Thus the degrees of the homogeneous elements of $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$ generate a group containing M_i with finite index, and v_i' and w_i' have rational rank one. Since w_i' is quasi-monomial of rational rank one, $w_i' = \delta_i \,\mathrm{ord}_{E_i}$ for a real number $\delta_i > 0$ and a prime divisor E_i over $S_{i,\bar{k}_i}$. By Lemma 3.3(2), $v_i' = \delta_i \,\mathrm{ord}_{E_{i,0}}$ for a prime divisor $E_{i,0}$ over S_i with $a(E_{i,0}, S_i, 0) = a(E_i, S_{i,\bar{k}_i}, 0)$. By Proposition 10.16, v_i' is G_i-invariant with $\mathrm{center}_{S_i} v_i' = x_i$. Since $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i} \otimes_{k_i} \bar{k}_i$ is normal and klt, $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$ is geometrically normal, and klt by Lemma 3.3(3) (its canonical divisor is $\mathbb{Q}$-Cartier over k_i since it is so after the base change to $\bar{k}_i$ [Sta, Tag 05B2]). It is finitely generated, since finitely many homogeneous elements of $\mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$ generate a subalgebra A_i' with $A_i' \otimes_{k_i} \bar{k}_i = \mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i} \otimes_{k_i} \bar{k}_i$, and $\bar{k}_i$ is faithfully flat over k_i. The induced action of G_i on $\mathrm{Spec}\, \mathrm{gr}_{v_i'} \mathcal{O}_{S_i,x_i}$ is quasi-étale by Proposition 10.16(1), since étaleness can be checked after the

base change to $\bar{k}_i$. Since $v'_i = \delta_i \operatorname{ord}_{E_{i,0}}$, $w'_i = \delta_i \operatorname{ord}_{E_i}$, $a(E_{i,0}, S_i, 0) = a(E_i, S_{i,\bar{k}_i}, 0)$, and $w'_i \in \mathcal{Q}_i$,

$$a(v'_i, S_i, 0) = a(w'_i, S_{i,\bar{k}_i}, 0) = w'_i(h_i) = v'_i(h_i) = 1.$$

For any non-zero $f \in \mathcal{O}_{X_i,o_i}$, $\mathrm{in}_{v'_i}(f)$ is a non-zero homogeneous element of R_i of degree $\bar{v}_i(f)$. Thus $\bar{v}_i(k_i(X_i)^*) \subset M_i$. Write $M_i = \gamma'_i\mathbb{Z}$ with $\gamma'_i > 0$. Since $\bar{v}_i(h_i^{|G_i|}) = |G_i| > 0$, $\bar{v}_i(k_i(X_i)^*) = \gamma_i\mathbb{Z}$ for a real number $\gamma_i \geq \gamma'_i$. Since $\mu_{0,i} = \pm\gamma'_i$,

$$\gamma_i \geq \gamma'_i = \frac{|v'_i(Y_i)|}{\nu_i} \geq \frac{c_0\Lambda'_i}{\nu_0}.$$

By Lemma 10.23, there exists a Kollár component F_i of $(X_i \ni o_i, 0)$ over k_i with

$$a(F_i, X_i, 0) = \frac{1}{\gamma_i} \leq \frac{\nu_0}{c_0\Lambda'_i}.$$

Since $\lim_i \Lambda'_i = +\infty$, $a(F_i, X_i, 0) < \epsilon$ for any sufficiently large i, which contradicts the assumption of Theorem 10.2.

Step 6. In this step we conclude the proof.

By **Steps 1–4**, the three conditions of Remark 10.17(5) hold. Thus, by Remark 10.17(5) and [OAI26b, Subsection G.9], at each non-final stage the induction adds an element of $k_i(S_i)$ which is algebraically independent over the field generated by the previous ones, and the rank of M_i is never zero, since $k_i(S_i)$ is a finite extension of $k_i(X_i)$ and a non-trivial valuation of $k_i(S_i)$ restricts to a non-trivial valuation of $k_i(X_i)$. Since $\mathrm{trdeg}_{k_i} k_i(S_i) = d$, the induction has at most d non-final stages, so the final case occurs for any sufficiently large i. This contradicts **Step 5**. □

11. Reduced boundaries

In this section, we prove the following result.

Definition-Theorem 11.1. Let $d \geq 2$ be an integer. Then there exists a positive integer N_0 depending only on d satisfying the following. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension d, and $x \in X$ a closed point such that X is $\mathbb{Q}$-Gorenstein and klt near x. Let $P \geq 0$ be a reduced $\mathbb{Q}$-Cartier Weil divisor on X such that (X, P) is lc and x is an lc center of (X, P). Then there exists a Kollár component E of $(X \ni x, 0)$ over k such that $a(E, X, P) = 0$, $a(E, X, 0) \leq N_0$, and E is geometrically integral over $\kappa(x)$. We let $N_0(d)$ denote the smallest such N_0.

The constant $N_0(d)$ is the only new bound in this paper. We prove it by contradiction in Subsection 11.2: the covers of Subsection 11.1 turn a sequence of germs violating the bound into a sequence of cyclic quotients which Theorem 10.2 excludes.

11.1. Root covers and products.

In this subsection, we construct the covers and the valuations used in the proof of Definition-Theorem 11.1.

Theorem 10.2 needs a generator of the canonical sheaf and an equation of the boundary which are eigenvectors with a common character. On the index cover of K_X and P, the ratio τ/ρ of their characters only satisfies $(\tau/\rho)^{n_0} = 1$, where n_0 can be taken to depend only on d by [Bir19, Corollary 1.9]. On the product of n_0 root covers of degree m with $\mathbb{A}^1_u$ in Construction-Proposition 11.2, both have the character λ^{n_0+1}, and the factor $\mathbb{A}^1_u$ keeps the quotient morphisms quasi-étale.

Construction-Proposition 11.2. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension $d \geq 2$, and $x \in X$ a k-rational point such that X is $\mathbb{Q}$-Gorenstein and klt near x. Let $P \geq 0$ be a reduced $\mathbb{Q}$-Cartier Weil divisor on X such that (X, P) is lc and x is an lc center of (X, P). Let $f : X' \to X$ be the index cover of K_X and P in Construction-Proposition 9.1, $x' \in X'$ the point over x, and $H^\vee$ the Galois group of f. Let $\varphi \in \mathcal{O}_{X',x'}$ be an eigenvector of $H^\vee$ with character ρ such that $f^*P = \mathrm{div}(\varphi)$ near x'. Let ω' be a generator of $\omega_{X'}$ at x' which is an eigenvector of $H^\vee$ with character τ. Let n_0 be a positive integer such that $n_0(K_X + P)$ is Cartier

near x. Let $m \geq 2$ be an integer. Shrinking X around x, we assume that X is affine and klt, $\varphi \in \mathcal{O}(X')$, $f^*P = \operatorname{div}(\varphi)$, and ω' generates $\omega_{X'}$. Let $V := (z^m = \varphi) \subset X' \times \mathbb{A}^1$, $Z := \operatorname{div}_V(z)$, $p : V \to X$ the induced morphism, and

$$\mathcal{G} := \{(g, \lambda) \in H^\vee \times k^* \mid \lambda^m = \rho(g)\},$$

acting on V by g on X' and by $(g, \lambda)^* z = \lambda z$. Let $S := V^{n_0} \times \mathbb{A}^1_u$ over k, with $\mathcal{G}$ acting diagonally on the factors V and by $(g, \lambda)^* u = \lambda u$, let $z_1, \dots, z_{n_0}$ be the functions z on the factors V, $h := z_1 \cdots z_{n_0} u$, $\mathcal{G}_0 := \ker \lambda$, where $\lambda : \mathcal{G} \to k^*$ denotes the character $(g, \lambda) \to \lambda$, $S_0 := S/\mathcal{G}_0$, and $Q := S/\mathcal{G}$. Then the following hold.

(1) V is normal, Gorenstein, and klt, (V, Z) is lc, $p^{-1}(x)$ consists of one point v, v is k-rational, $K_V = p^*(K_X + (1 - \frac{1}{m})P)$, and $K_V + Z = p^*(K_X + P)$.
(2) The projection $\mathcal{G} \to H^\vee$ is surjective with kernel μ_m, and p is the quotient morphism of V by $\mathcal{G}$.
(3) Let $s := (v, \dots, v, 0) \in S$, and let $s_0 \in S_0$ and $o \in Q$ be the images of s. Then the following hold.
  (i) S and S_0 are geometrically integral over k, and $\dim S = dn_0 + 1$.
  (ii) $(S, \operatorname{div}(h))$ is lc.
  (iii) The point s is fixed by $\mathcal{G}$, and s is the only point of S over o.
  (iv) The morphisms $S \to S_0 \to Q$ are quasi-étale.
  (v) $\mathcal{G}/\mathcal{G}_0$ is cyclic and acts faithfully on S_0.
  (vi) S_0 is Gorenstein and klt.
(4) The following hold.
  (i) $(\tau/\rho)^{n_0} = 1$.
  (ii) The generator of ω_S induced by ω' and the element h descend to a generator Ω_0 of ω_{S_0} and an element $h_0 \in \mathcal{O}(S_0)$, which are eigenvectors of $\mathcal{G}/\mathcal{G}_0$ with the character $\chi := \lambda^{n_0+1}$.
  (iii) $(S_0, \operatorname{div}(h_0))$ is lc.

Proof. The proof is long, so we divide it into several steps. An element $(g, \lambda) \in \mathcal{G}$ acts on X' by g, and $(g, \lambda)^* z = \lambda z$. Since φ and ω' are eigenvectors of $H^\vee$ with characters ρ and τ, $g^*\varphi = \rho(g)\varphi$ and $g^*\omega' = \tau(g)\omega'$ for any $g \in H^\vee$. Let $p' : V \to X'$ be the projection, so that $p = f \circ p'$.

Step 1. In this step we prove (1). Since f is quasi-étale, $K_{X'} = f^*K_X$ and X' is klt (Lemma 2.5), so X' is Cohen–Macaulay [KM98, Theorem 5.22, Theorem 5.10]. Since ω' generates $\omega_{X'}$, X' is Gorenstein. The ring $\mathcal{O}(V) = \mathcal{O}(X')[z]/(z^m - \varphi)$ is a free $\mathcal{O}(X')$-module with basis $1, z, \dots, z^{m-1}$, and V is a Cartier divisor in the Gorenstein scheme $X' \times \mathbb{A}^1$. Thus V is Gorenstein, and in particular V satisfies (S_2).

Since X is klt and x is an lc center of (X, P), $x \in \operatorname{Supp} P$, so $\varphi(x') = 0$. Since f is quasi-étale and P is reduced, $\operatorname{div}(\varphi) = f^*P$ is reduced. Let $\xi \in V$ be a point of codimension one and $\xi' := p'(\xi)$. Since p' is finite and flat, ξ' has codimension one [Mat89, Theorem 15.1(ii)]. If φ is a unit at ξ', then z is a unit over ξ', and p' is étale over ξ' as $\partial(z^m - \varphi)/\partial z = mz^{m-1}$, so $\mathcal{O}_{V,\xi}$ is regular. Otherwise ξ' is the generic point of a component of f^*P, φ is a uniformizer of $\mathcal{O}_{X',\xi'}$, and $\mathcal{O}_{X',\xi'}[z]/(z^m - \varphi)$ is a discrete valuation ring with uniformizer z by the Eisenstein criterion. Thus V satisfies (R_1), and V is normal by Serre's criterion [Mat89, Theorem 23.8]. The Eisenstein criterion also shows that $z^m - \varphi$ is irreducible over $k(X')$. Since $\mathcal{O}(V)$ is free over $\mathcal{O}(X')$, $\mathcal{O}(V)$ is contained in the field $k(X')[z]/(z^m - \varphi)$, so V is integral. The fiber of p' over x' is $\operatorname{Spec} \kappa(x')[z]/(z^m)$, and x' is the only point of X' over x, so $p^{-1}(x)$ consists of one point v with $\kappa(v) = \kappa(x') = \kappa(x) = k$ (Construction-Proposition 9.1).

The morphism p' is étale over $X' \setminus \operatorname{Supp} f^*P$, $p'^* f^*P = \operatorname{div}(z^m) = mZ$, and f is quasi-étale. Thus the components of P are the only prime divisors on X over which p is not étale, and p has ramification index m along any component of Z. By the Hurwitz formula,

$$K_V = p^*K_X + (m-1)Z = p^*\Big(K_X + \Big(1 - \frac{1}{m}\Big)P\Big), \qquad K_V + Z = p^*(K_X + P).$$

For any prime divisor E over X,

$$a\Big(E, X, \Big(1-\frac{1}{m}\Big)P\Big) = \frac{1}{m}a(E,X,0) + \Big(1-\frac{1}{m}\Big)a(E,X,P) > 0,$$

since X is klt and (X,P) is lc. Thus $(X,(1-\frac{1}{m})P)$ is klt, and Lemma 2.5 implies that V is klt and (V,Z) is lc. This proves (1).

Step 2. In this step we prove (2). For $(g,\lambda)\in\mathcal{G}$,

$$(g,\lambda)^*(z^m-\varphi) = \lambda^m z^m - \rho(g)\varphi = \rho(g)(z^m-\varphi),$$

so $\mathcal{G}$ acts on V. Since $H^\vee$ is finite, $\rho(g)$ is a root of unity, and $k\supset\mathbb{C}$ contains the m-th roots of $\rho(g)$. Thus $\mathcal{G}\to H^\vee$ is surjective, and the kernel is $\{(1,\zeta)\mid \zeta^m=1\}=\mu_m$. The element $(1,\zeta)$ multiplies z^j by ζ^j, so $\mathcal{O}(V)^{\mu_m}=\mathcal{O}(X')$, on which $\mathcal{G}/\mu_m\cong H^\vee$ acts as the Galois group of f. Hence $\mathcal{O}(V)^{\mathcal{G}}=\mathcal{O}(X')^{H^\vee}=\mathcal{O}(X)$, and p is the quotient morphism of V by $\mathcal{G}$. Moreover, $\mathcal{G}$ acts faithfully on V: if (g,λ) acts trivially, then $\lambda=1$ by the action on z, and $g=1$ as $H^\vee$ acts faithfully on X'.

Step 3. In this step we prove $(\tau/\rho)^{n_0}=1$ and construct generators of ω_V and ω_S. Let σ_X be a generator of the invertible sheaf $\mathcal{O}_X(n_0(K_X+P))$ at x. Let $\mathcal{L}:=\mathcal{O}_{X'}(n_0(K_{X'}+f^*P))$. Since $K_{X'}=f^*K_X$, $f^*\sigma_X$ is an $H^\vee$-invariant generator of $\mathcal{L}$ at x'. Since $f^*P=\operatorname{div}(\varphi)$, $(\omega')^{\otimes n_0}\varphi^{-n_0}$ is also a generator of $\mathcal{L}$ at x', so

$$(\omega')^{\otimes n_0}\varphi^{-n_0} = \epsilon\cdot f^*\sigma_X$$

for a unit $\epsilon\in\mathcal{O}_{X',x'}$ with $g^*\epsilon=(\tau/\rho)^{n_0}(g)\epsilon$ for any $g\in H^\vee$. Since $H^\vee$ acts trivially on $\kappa(x')$ (Construction-Proposition 9.1), evaluating at x' implies $\epsilon(x')=(\tau/\rho)^{n_0}(g)\epsilon(x')$ with $\epsilon(x')\neq 0$. Thus $(\tau/\rho)^{n_0}=1$.

Let $\Omega_V:=p'^*\omega'/z^{m-1}$. The Hurwitz formula for p' is

$$\operatorname{div}(p'^*\omega') = p'^*\operatorname{div}(\omega') + (m-1)Z = (m-1)Z,$$

so $\operatorname{div}(\Omega_V)=0$, and Ω_V is a generator of ω_V as V is normal. Since $\lambda^m=\rho(g)$,

$$(g,\lambda)^*z^{m-1} = \lambda^{m-1}z^{m-1} = \rho(g)\lambda^{-1}z^{m-1},$$

$$(g,\lambda)^*\Omega_V = \frac{\tau(g)\lambda}{\rho(g)}\Omega_V.$$

Let

$$\Omega_S := \operatorname{pr}_1^*\Omega_V\wedge\cdots\wedge\operatorname{pr}_{n_0}^*\Omega_V\wedge du.$$

Since V is normal over a field of characteristic zero, hence geometrically normal, S and $S_{\bar{k}}$ are normal [Sta, Tag 06DG]. On the smooth locus of S, which is the product of the smooth loci of the factors, Ω_S is a generator of ω_S, and the complement has codimension at least two, so Ω_S is a generator of ω_S. In particular, K_S is Cartier. By $(\tau/\rho)^{n_0}=1$,

$$(g,\lambda)^*\Omega_S = \Big(\frac{\tau(g)}{\rho(g)}\Big)^{n_0}\lambda^{n_0+1}\Omega_S = \lambda^{n_0+1}\Omega_S, \qquad (g,\lambda)^*h = \lambda^{n_0+1}h.$$

Step 4. In this step we prove (3) and the remaining part of (4). Since V has the k-rational point v, V is geometrically integral over k (Lemma 3.5(2)), so $S_{\bar{k}}=V_{\bar{k}}^{n_0}\times\mathbb{A}^1_u$ is integral and $\dim S = dn_0+1$. Since taking invariants under the k-linear action of $\mathcal{G}_0$ commutes with the flat base change $k\to\bar{k}$, $S_{0,\bar{k}}=S_{\bar{k}}/\mathcal{G}_0$ is integral.

By Lemma 3.3(3), $V_{\bar{k}}$ is klt and $(V_{\bar{k}},Z_{\bar{k}})$ is lc, where $Z_{\bar{k}}$ is the pullback of Z. Let $\mu_V: Y_V\to V_{\bar{k}}$ be a log resolution of $(V_{\bar{k}},Z_{\bar{k}})$, and let $K_{Y_V}+B_{Y_V}:=\mu_V^*(K_{V_{\bar{k}}}+Z_{\bar{k}})$ and $K_{Y_V}+B'_{Y_V}:=\mu_V^*K_{V_{\bar{k}}}$. The product $\mu: Y:=Y_V^{n_0}\times\mathbb{A}^1_u\to S_{\bar{k}}$ is a log resolution, and since the canonical divisor of a product is the sum of the pullbacks of the canonical divisors of the factors,

$$K_Y + \sum_{a=1}^{n_0}\operatorname{pr}_a^*B_{Y_V} + (u=0) = \mu^*(K_{S_{\bar{k}}}+\operatorname{div}(h)), \qquad K_Y+\sum_{a=1}^{n_0}\operatorname{pr}_a^*B'_{Y_V} = \mu^*K_{S_{\bar{k}}}.$$

Both boundaries have snc support, the coefficients of the first are at most 1, and those of the second are less than 1. Thus $(S_{\bar{k}}, \operatorname{div}(h))$ is lc and $S_{\bar{k}}$ is klt [KM98, Lemma 2.30, Corollary 2.31(3)], so $(S, \operatorname{div}(h))$ is lc and S is klt (Lemma 3.3(3)).

Since $p \circ (g, \lambda) = p$ and $p^{-1}(x) = \{v\}$, v is fixed by $\mathcal{G}$, and hence so is s. The fibers of $S \to Q$ are $\mathcal{G}$-orbits, so s is the only point of S over o. Let $(g, \lambda) \in \mathcal{G}$ be a non-trivial element. If $\lambda \neq 1$, then $(g, \lambda)^* z_a - z_a = (\lambda - 1) z_a$ and $(g, \lambda)^* u - u = (\lambda - 1) u$, so the fixed locus of (g, λ) on S is contained in $(z_1 = \cdots = z_{n_0} = u = 0)$, which is of codimension $n_0 + 1 \geq 2$ in S. If $\lambda = 1$, then $g \neq 1$ by (2). Since f is quasi-étale, the inertia group of the generic point of any prime divisor on X' is trivial, so g fixes no prime divisor on X' pointwise. Thus the fixed locus Σ_g of g on X' is of codimension at least two in X'. Hence $p'^{-1}(\Sigma_g)$ is of codimension at least two in V, and so is the fixed locus of $(g, 1)$ on S, which is contained in $\operatorname{pr}_1^{-1}(p'^{-1}(\Sigma_g))$. Hence the inertia group in $\mathcal{G}$ of the generic point of any prime divisor on S is trivial, and $S \to Q$ is étale over the generic point of any prime divisor on Q. Since ramification indices are multiplicative in $S \to S_0 \to Q$, both $S \to S_0$ and $S_0 \to Q$ are quasi-étale.

The character λ induces an injective homomorphism from $\mathcal{G}/\mathcal{G}_0$ to k^*, so $\mathcal{G}/\mathcal{G}_0$ is a finite subgroup of k^*, which is cyclic. Since $\mathcal{G}$ acts faithfully on V (**Step 2**), it acts faithfully on the integral scheme S, so $k(S)/k(Q)$ is a Galois extension with group $\mathcal{G}$ and $k(S_0) = k(S)^{\mathcal{G}_0}$. By Galois theory, the subgroup of $\mathcal{G}$ acting trivially on $k(S_0)$ is $\mathcal{G}_0$, so $\mathcal{G}/\mathcal{G}_0$ acts faithfully on S_0.

Let $\pi_0 : S \to S_0$ be the quotient morphism. By **Step 3**, Ω_S and h are $\mathcal{G}_0$-invariant. Thus $h = \pi_0^* h_0$ for some $h_0 \in \mathcal{O}(S_0) = \mathcal{O}(S)^{\mathcal{G}_0}$. By Galois descent for the extension $k(S)/k(S_0)$ [Sta, Tag 0CDR], the $\mathcal{G}_0$-invariant rational section Ω_S of ω_S is $\pi_0^* \Omega_0$ for a rational section Ω_0 of ω_{S_0}. Since π_0 is quasi-étale, $0 = \operatorname{div}(\Omega_S) = \pi_0^* \operatorname{div}(\Omega_0)$, so $\operatorname{div}(\Omega_0) = 0$, and Ω_0 is a generator of ω_{S_0}. By **Step 3**, Ω_0 and h_0 are eigenvectors of $\mathcal{G}/\mathcal{G}_0$ with the character $\chi = \lambda^{n_0+1}$. Since π_0 is quasi-étale, $K_S = \pi_0^* K_{S_0}$ and $K_S + \operatorname{div}(h) = \pi_0^*(K_{S_0} + \operatorname{div}(h_0))$. Thus S_0 is klt and $(S_0, \operatorname{div}(h_0))$ is lc (Lemma 2.5). Since S_0 is klt, S_0 is Cohen–Macaulay [KM98, Theorem 5.22, Theorem 5.10], and since K_{S_0} is Cartier, S_0 is Gorenstein. This proves (3) and (4). □

The following valuation on a product is used in Lemma 11.4 and Lemma 11.5.

Definition-Proposition 11.3. Let k be a field finitely generated over $\mathbb{C}$, $\bar{k}$ an algebraic closure of k, $V_1, \ldots, V_l$ normal varieties over $\bar{k}$, $v_j \in V_j$ closed points, w_j quasi-monomial valuations on V_j with $\operatorname{center}_{V_j} w_j = v_j$, and $\theta > 0$ a real number. Let $S := V_1 \times \cdots \times V_l \times \mathbb{A}^1_u$ and $s := (v_1, \ldots, v_l, 0)$. Then there exists a unique valuation w on S with $\operatorname{center}_S w = s$ such that, for any real number $t > 0$, the ideal $\mathfrak{a}_t(w)\mathcal{O}_{S,s}$ is generated by the products $\phi_1 \cdots \phi_l u^e$ of the pullbacks of $\phi_j \in \mathfrak{a}_{t_j}(w_j)$ and u^e, where $t_1, \ldots, t_l \geq 0$ are real numbers and $e \geq 0$ is an integer with $t_1 + \cdots + t_l + \theta e \geq t$. Moreover, w is quasi-monomial, the restriction of w to the function field of V_j is w_j for any j, $\operatorname{gr}_w \mathcal{O}_{S,s}$ is the polynomial ring in $\operatorname{in}_w(u)$ over $\bigotimes_j \operatorname{gr}_{w_j} \mathcal{O}_{V_j, v_j}$, where the tensor product is over $\bar{k}$ and $\deg \operatorname{in}_w(u) = \theta$. If K_{V_j} is $\mathbb{Q}$-Cartier for any j, then

$$a(w, S, 0) = \sum_{j=1}^{l} a(w_j, V_j, 0) + \theta.$$

We call w the *Gauss product* of $w_1, \ldots, w_l$ with u-weight θ.

Proof. Let $R_j := \mathcal{O}_{V_j, v_j}$ and $R := \mathcal{O}_{S,s}$. A valuation w with $\operatorname{center}_S w = s$ is non-negative on R, and $w(\phi) = \max\{t \mid \phi \in \mathfrak{a}_t(w)R\}$ for $\phi \in R \backslash \{0\}$. Since $k(S)$ is the fraction field of R, w is determined by the ideals $\mathfrak{a}_t(w)R$. This proves the uniqueness.

We recall three properties of a monomial valuation ν of a regular local ring $\mathcal{O}$ with residue field κ, in a regular system of parameters $y_1, \ldots, y_r$ with weights $\gamma \in \mathbb{R}^r_{>0}$. They follow from the definition by passing to the completion $\widehat{\mathcal{O}} \cong \kappa[[y_1, \ldots, y_r]]$, which is faithfully flat over $\mathcal{O}$: the ideal $\mathfrak{a}_t(\nu)$ is generated by the monomials y^β with $\langle \gamma, \beta \rangle \geq t$, the ring $\operatorname{gr}_\nu \mathcal{O}$ is the polynomial

ring over κ in $\mathrm{in}_\nu(y_1),\dots,\mathrm{in}_\nu(y_r)$, and

$$\nu\Big(\sum_\beta c_\beta y^\beta\Big) = \min\{\langle\gamma,\beta\rangle \mid c_\beta \neq 0\}$$

for any finite sum with $c_\beta \in \mathcal{O}$ units or zero.

For any j, since w_j is quasi-monomial, there exist a proper birational morphism $\mu_j : Y_j \to V_j$ from a smooth variety, a point $\eta_j \in Y_j$, a regular system of parameters $y_{j,1},\dots,y_{j,r_j}$ of $\mathcal{O}_{Y_j,\eta_j}$ defining an snc divisor D_j near η_j, and $\alpha_j \in \mathbb{R}^{r_j}_{>0}$ such that w_j is the monomial valuation in $y_{j,1},\dots,y_{j,r_j}$ with weights α_j. Since $\mathrm{center}_{V_j} w_j = v_j$, $\mu_j(\eta_j) = v_j$, so $R_j \subset \mathcal{O}_{Y_j,\eta_j}$. Let $\mu : Y := Y_1 \times\cdots\times Y_l \times \mathbb{A}^1_u \to S$ be the product morphism, and $\eta \in Y$ the generic point of $\overline{\{\eta_1\}}\times\cdots\times\overline{\{\eta_l\}}\times\{0\}$. The elements $y_{j,i}$ and u form a regular system of parameters of $\mathcal{O}_{Y,\eta}$. Let w be the monomial valuation in $(y_{1,1},\dots,y_{l,r_l},u)$ with weights $(\alpha_1,\dots,\alpha_l,\theta)$. Then w is quasi-monomial, and $\mathrm{center}_S w = \mu(\eta) = s$.

We show that $w(\phi) = w_j(\phi)$ for any $\phi \in \mathcal{O}_{Y_j,\eta_j}$. Let $\delta := w_j(\phi)$. By the properties above, $\phi = \sum_\beta c_\beta y_j^\beta + \phi'$, where β runs over the finitely many exponents with $\langle\alpha_j,\beta\rangle = \delta$, the $c_\beta \in \mathcal{O}_{Y_j,\eta_j}$ are units or zero and not all zero, and ϕ' belongs to the ideal generated by the monomials y_j^β with $\langle\alpha_j,\beta\rangle > \delta$. Thus $w(\phi') > \delta$, and $w(\sum_\beta c_\beta y_j^\beta) = \delta$ as the c_β are units in $\mathcal{O}_{Y,\eta}$. Thus $w(\phi) = \delta$. In particular, the restriction of w to $k(V_j)$ is w_j, $\mathfrak{a}_\delta(w_j) = \mathfrak{a}_\delta(w)\cap R_j$, and $w(u) = \theta$.

Hence the inclusions $R_j \subset R$ and $u \in R$ induce a homomorphism of graded $\bar{k}$-algebras

$$\iota : \Big(\bigotimes_j \mathrm{gr}_{w_j} R_j\Big)[U] \to \mathrm{gr}_w R, \qquad U \to \mathrm{in}_w(u),$$

where $\deg U = \theta$. We show that ι is injective. By the properties above,

$$\mathrm{gr}_{w_j}\mathcal{O}_{Y_j,\eta_j} = \kappa(\eta_j)[Y_{j,1},\dots,Y_{j,r_j}], \qquad \mathrm{gr}_w \mathcal{O}_{Y,\eta} = \kappa(\eta)[Y_{1,1},\dots,Y_{l,r_l},U],$$

where $Y_{j,i}$ is the image of $y_{j,i}$, and the composite of ι with $\mathrm{gr}_w R \to \mathrm{gr}_w \mathcal{O}_{Y,\eta}$ factors as

$$\Big(\bigotimes_j \mathrm{gr}_{w_j} R_j\Big)[U] \to \Big(\bigotimes_j \kappa(\eta_j)\Big)[Y_{1,1},\dots,Y_{l,r_l},U] \to \kappa(\eta)[Y_{1,1},\dots,Y_{l,r_l},U].$$

The first map is a tensor product over $\bar{k}$ of the injective maps $\mathrm{gr}_{w_j} R_j \to \mathrm{gr}_{w_j}\mathcal{O}_{Y_j,\eta_j}$, hence injective. Since $\bar{k}$ is algebraically closed, $\bigotimes_j \kappa(\eta_j)$ is a domain with fraction field $\kappa(\eta)$, so the second map is injective. Thus ι is injective. In particular, if $b_1,\dots,b_N \in R$ are products $\phi_1\cdots\phi_l u^e$ with $\phi_j \in R_j$ such that, for any δ, the elements $\otimes_j \mathrm{in}_{w_j}(\phi_j)\otimes U^e$ of degree δ are linearly independent over $\bar{k}$, then

$$w\Big(\sum_i c_i b_i\Big) = \min\{w(b_i) \mid c_i \neq 0\} \quad \text{for any } c_1,\dots,c_N \in \bar{k}.$$

We prove the description of $\mathfrak{a}_t(w)R$. Fix $t > 0$, and let $I_t \subset R$ be the ideal generated by the products in the statement. Since $w(\phi_1\cdots\phi_l u^e) = \sum_j w_j(\phi_j) + \theta e$, $I_t \subset \mathfrak{a}_t(w)R$. Conversely, since $\mathrm{center}_{V_j} w_j = v_j$ is a closed point, $\mathfrak{a}_t(w_j)$ is primary to the maximal ideal of R_j, so $\dim_{\bar{k}} R_j/\mathfrak{a}_t(w_j) < +\infty$, and w_j takes finitely many values less than t on R_j. Thus there exist $b_{j,1},\dots,b_{j,N_j} \in R_j$ whose images form a basis of $R_j/\mathfrak{a}_t(w_j)$, such that for any $\delta < t$, the elements $\mathrm{in}_{w_j}(b_{j,i})$ with $w_j(b_{j,i}) = \delta$ form a basis of the degree δ part of $\mathrm{gr}_{w_j} R_j$. Let e_0 be an integer with $\theta e_0 \geq t$. Shrinking V_j around v_j, we may assume that V_j is affine. Then $\bigotimes_j R_j/\mathfrak{a}_t(w_j)\otimes\bar{k}[u]/(u^{e_0})$ is an Artinian local quotient of $\bigotimes_j \mathcal{O}(V_j)\otimes\bar{k}[u]$ supported at s, so

$$R\Big/\Big(\sum_j \mathfrak{a}_t(w_j)R + u^{e_0}R\Big) \cong \bigotimes_j R_j/\mathfrak{a}_t(w_j)\otimes\bar{k}[u]/(u^{e_0}).$$

Thus any $\phi \in R$ can be written as

$$\phi = \sum c_{i_1,\dots,i_l,e} b_{1,i_1}\cdots b_{l,i_l}u^e + \psi,$$

where $c_{i_1,\dots,i_l,e} \in \bar{k}$, $e < e_0$, and $\psi \in \sum_j \mathfrak{a}_t(w_j)R + u^{e_0}R \subset I_t$. Assume that $w(\phi) \geq t$. Then $w(\phi - \psi) \geq t$, since $I_t \subset \mathfrak{a}_t(w)R$. The products $b_{1,i_1} \cdots b_{l,i_l} u^e$ satisfy the independence condition above, so any product with a non-zero coefficient has value $\sum_j w_j(b_{j,i_j}) + \theta e \geq t$, and hence belongs to I_t. Thus $\phi \in I_t$, and $\mathfrak{a}_t(w)R = I_t$.

The expansion of ϕ above, taken with $t > w(\phi)$, shows that $\mathrm{in}_w(\phi)$ is the sum of the images of the products of value $w(\phi)$, so ι is surjective. Therefore $\mathrm{gr}_w R$ is the polynomial ring in $\mathrm{in}_w(u)$ over $\bigotimes_j \mathrm{gr}_{w_j} R_j$, with $\deg \mathrm{in}_w(u) = \theta$.

Finally, assume that K_{V_j} is $\mathbb{Q}$-Cartier for any j. Then $K_S = \sum_j \mathrm{pr}_j^* K_{V_j}$ is $\mathbb{Q}$-Cartier, and $K_{Y/S} = \sum_j \mathrm{pr}_j^* K_{Y_j/V_j}$. Since the log discrepancy of the divisor of any coordinate in a regular system of parameters is 1, [JM12, Proposition 5.1, Remark 5.6] implies

$$a(w, S, 0) = a(w, Y, 0) + w(K_{Y/S}) = \sum_{j,i} \alpha_{j,i} + \theta + \sum_j w_j(K_{Y_j/V_j}) = \sum_j a(w_j, V_j, 0) + \theta,$$

where we used $w(\mathrm{pr}_j^* K_{Y_j/V_j}) = w_j(K_{Y_j/V_j})$ and $a(w_j, V_j, 0) = \sum_i \alpha_{j,i} + w_j(K_{Y_j/V_j})$. □

Lemma 11.4. *Notation as in Construction-Proposition 11.2, and let $\bar{s} \in S_{\bar{k}}$ and $\bar{s}_0 \in S_{0,\bar{k}}$ be the points over s and s_0. Then there exists a quasi-monomial valuation $\bar{T}$ on $S_{\bar{k}}$ with $\mathrm{center}_{S_{\bar{k}}}(\bar{T}) = \bar{s}$, invariant under $\mathcal{G}$ and $\mathrm{Gal}(\bar{k}/k)$, such that $a(\bar{T}, S_{\bar{k}}, 0) = \bar{T}(h) = 1$ and $\bar{T}(z_j) = \bar{T}(u) = \frac{1}{n_0+1}$ for any j. Let $\bar{T}_0$ be the restriction of $\bar{T}$ to the function field of $S_{0,\bar{k}}$. Then $\mathrm{center}_{S_{0,\bar{k}}}(\bar{T}_0) = \bar{s}_0$, $\bar{T}_0$ is quasi-monomial, and $a(\bar{T}_0, S_{0,\bar{k}}, 0) = \bar{T}_0(h_0) = 1$. If b is a positive integer with $m > (n_0+1)b$, then $\bar{T}_0(\sigma) \geq b$ for any non-zero $\sigma \in \mathcal{O}_{S_0,s_0}$ which is an eigenvector of $\mathcal{G}/\mathcal{G}_0$ with character χ^b.*

Proof. Let $\bar{v} \in V_{\bar{k}}$ be the point over v. By Construction-Proposition 11.2(1), V is affine and klt, and (V, Z) is lc. We apply Set-up 6.1 with $(X \ni x, B) := (V \ni v, 0)$, with z in place of h, Z in place of H, $c = 1$, no functions f_j and g_l, $b_0 = 1$, and $\bar{x} := \bar{v}$. The sets $\mathcal{Q}$, $\mathcal{Q}_{\bar{v}}$ and the height are then defined by Set-up 5.1. Then the height is the function $w' \to w'(z)$, and $\mathcal{Q}_{\bar{v}}$ is the set of quasi-monomial valuations w' on $V_{\bar{k}}$ with $\mathrm{center}_{V_{\bar{k}}}(w') = \bar{v}$ and $a(w', V_{\bar{k}}, 0) = w'(z) = 1$. In particular, the height is identically 1 on $\mathcal{Q}$.

We show that $\mathcal{Q}_{\bar{v}} \neq \emptyset$. Since x is an lc center of (X, P), there exists a prime divisor E over X with $\mathrm{center}_X E = x$ and $a(E, X, P) = 0$. Since $k(V)/k(X)$ is finite, ord_E extends to a discrete valuation of $k(V)$ whose residue field is finite over that of ord_E, which is a positive multiple of ord_F for a prime divisor F over V [KM98, Lemma 2.45]. Since $K_V + Z = p^*(K_X + P)$, Lemma 2.5 implies $a(F, V, Z) = 0$, and $\mathrm{center}_V F = v$ as $p(\mathrm{center}_V F) = x$. Let F' be a component of the pullback of F to $\bar{k}$. By Lemma 3.3, $\mathrm{ord}_{F'}|_{k(V)} = \mathrm{ord}_F$ and $a(F', V_{\bar{k}}, Z_{\bar{k}}) = 0$, so $\mathrm{center}_{V_{\bar{k}}}(F') = \bar{v}$ and $a(F', V_{\bar{k}}, 0) = \mathrm{ord}_{F'}(z) > 0$. Thus $\mathrm{ord}_{F'} / \mathrm{ord}_{F'}(z) \in \mathcal{Q}_{\bar{v}}$.

Let w be the valuation in Definition-Proposition 5.2(1). Since the height is identically 1 on $\mathcal{Q}$, w is the unique minimizer of vol on $\mathcal{Q}_{\bar{v}}$, w is quasi-monomial, and $a(w, V_{\bar{k}}, 0) = w(z) = 1$. For $(g, \lambda) \in \mathcal{G}$, the automorphism (g, λ) of $V_{\bar{k}}$ fixes $\bar{v}$ and satisfies $(g, \lambda)^* z = \lambda z$, so the map $w' \to w' \circ (g, \lambda)^*$ preserves $\mathcal{Q}_{\bar{v}}$ and vol. By the uniqueness of w, $w \circ (g, \lambda)^* = w$. Any $\gamma \in \mathrm{Gal}(\bar{k}/k)$ acts on $V_{\bar{k}}$ by an automorphism of schemes fixing $\bar{v}$ and z, which preserves the pullback of K_V and the colengths of ideals. Thus $w' \to w' \circ \gamma^*$ preserves $\mathcal{Q}_{\bar{v}}$ and vol, and $w \circ \gamma^* = w$ by the uniqueness of w.

Let $\bar{T}$ be the Gauss product (Definition-Proposition 11.3) of n_0 copies of $\frac{1}{n_0+1}w$ on the factors $V_{\bar{k}}$ of $S_{\bar{k}} = V_{\bar{k}}^{n_0} \times \mathbb{A}^1_u$, with u-weight $\frac{1}{n_0+1}$. Then $\bar{T}$ is quasi-monomial, $\mathrm{center}_{S_{\bar{k}}}(\bar{T}) = \bar{s}$, $\bar{T}(z_j) = \frac{1}{n_0+1} w(z) = \frac{1}{n_0+1}$, and $\bar{T}(u) = \frac{1}{n_0+1}$ since $\mathrm{in}_{\bar{T}}(u)$ is a non-zero element of degree $\frac{1}{n_0+1}$. Hence $\bar{T}(h) = \sum_j \bar{T}(z_j) + \bar{T}(u) = 1$, and since $K_{V_{\bar{k}}}$ is Cartier,

$$a(\bar{T}, S_{\bar{k}}, 0) = n_0 \cdot \frac{a(w, V_{\bar{k}}, 0)}{n_0 + 1} + \frac{1}{n_0 + 1} = 1.$$

Let γ be an element of $\mathcal{G}$ or of $\mathrm{Gal}(\bar{k}/k)$. Then γ^* acts diagonally on the factors $V_{\bar{k}}$, maps any $\mathfrak{a}_{t_j}(\frac{1}{n_0+1}w)$ onto itself, and maps u to a non-zero constant multiple of u. Thus γ^* maps $\mathfrak{a}_t(\bar{T})\mathcal{O}_{S_{\bar{k}},\bar{s}}$ onto itself for any $t > 0$, and $\bar{T} \circ \gamma^* = \bar{T}$ by the uniqueness in Definition-Proposition 11.3.

Let $\pi_0 : S_{\bar{k}} \to S_{0,\bar{k}} = S_{\bar{k}}/\mathcal{G}_0$ be the quotient morphism. Since π_0 is finite, $\operatorname{center}_{S_{0,\bar{k}}}(\bar{T}_0) = \pi_0(\bar{s}) = \bar{s}_0$. Since $h = \pi_0^* h_0$, $\bar{T}_0(h_0) = \bar{T}(h) = 1$. We show that $\bar{T}_0$ is quasi-monomial. For $\phi \in k(S_{\bar{k}})^*$, the element $\prod_{g\in\mathcal{G}_0} g^*\phi$ belongs to $k(S_{0,\bar{k}})$, and has $\bar{T}$-value $|\mathcal{G}_0|\bar{T}(\phi)$ by the $\mathcal{G}$-invariance of $\bar{T}$. Thus the value group of $\bar{T}_0$ has finite index in that of $\bar{T}$. If $\bar{T}(\phi) \geq 0$, then ϕ is a root of the monic polynomial $\prod_{g\in\mathcal{G}_0}(\xi - g^*\phi)$ in a variable ξ, whose coefficients belong to $k(S_{0,\bar{k}})$ and have non-negative value. Thus the residue field of $\bar{T}$ is algebraic over that of $\bar{T}_0$. Hence the rational rank of $\bar{T}_0$ equals that of $\bar{T}$, and the transcendence degree over $\bar{k}$ of the residue field of $\bar{T}_0$ equals that of $\bar{T}$. A valuation v on a variety W over $\bar{k}$ is quasi-monomial if and only if the rational rank of v plus the transcendence degree over $\bar{k}$ of the residue field of v equals $\dim W$ [JM12, Proposition 3.7], applied on a resolution, so $\bar{T}_0$ is quasi-monomial.

We show that $a(\bar{T}_0, S_{0,\bar{k}}, 0) = 1$. By Construction-Proposition 11.2(4), $(S_{0,\bar{k}}, \operatorname{div}(h_0))$ is lc (Lemma 3.3(3)), so $a(\bar{T}_0, S_{0,\bar{k}}, 0) \geq \bar{T}_0(h_0) = 1$. Conversely, π_0 remains quasi-étale and $K_{S_{0,\bar{k}}} = \operatorname{div}(\Omega_0)$ after the base change to $\bar{k}$, so $K_{S_{\bar{k}}} = \pi_0^* K_{S_{0,\bar{k}}}$, and for any prime divisor F over $S_{\bar{k}}$, Lemma 2.5 implies $\operatorname{ord}_F|_{k(S_{0,\bar{k}})} = e \operatorname{ord}_E$ and $a(F, S_{\bar{k}}, 0) = e a(E, S_{0,\bar{k}}, 0)$ for a prime divisor E over $S_{0,\bar{k}}$ and a positive integer e. Thus $a(v'|_{k(S_{0,\bar{k}})}, S_{0,\bar{k}}, 0) = a(v', S_{\bar{k}}, 0)$ for any positive real multiple v' of a divisorial valuation. Let (Y, D) be a log smooth model of $S_{\bar{k}}$ with $\bar{T} \in \mathrm{QM}_\eta(Y, D)$ for a point $\eta \in Y$, and $v'_1, v'_2, \dots$ divisorial valuations in $\mathrm{QM}_\eta(Y, D)$ converging to $\bar{T}$. By the linearity in [JM12, Proposition 5.1], $a(v'_n, S_{\bar{k}}, 0) \to a(\bar{T}, S_{\bar{k}}, 0) = 1$. Since $v|_{k(S_{0,\bar{k}})}(\mathfrak{b}) = v(\mathfrak{b}\mathcal{O}_{S_{\bar{k}}})$ for any valuation v on $S_{\bar{k}}$ and any ideal $\mathfrak{b}$ on $S_{0,\bar{k}}$, the restriction map is continuous, so $v'_n|_{k(S_{0,\bar{k}})} \to \bar{T}_0$. Let $\mu_0 : Y_0 \to S_{0,\bar{k}}$ be a resolution. Since $S_{0,\bar{k}}$ is Gorenstein and klt, $K_{Y_0/S_{0,\bar{k}}} \geq 0$ is a Cartier divisor, and $a(v, S_{0,\bar{k}}, 0) = A_{Y_0}(v) + v(K_{Y_0/S_{0,\bar{k}}})$ for any valuation v on Y_0 (Definition 2.6 and [JM12, Proposition 5.1, Remark 5.6]). Since $v \to v(K_{Y_0/S_{0,\bar{k}}})$ is continuous, $a(\cdot, S_{0,\bar{k}}, 0)$ is lower semicontinuous [JM12, Lemma 5.7], so $a(\bar{T}_0, S_{0,\bar{k}}, 0) \leq 1$.

Finally, let b be a positive integer with $m > (n_0+1)b$, and $\sigma \in \mathcal{O}_{S_0,s_0}$ a non-zero eigenvector of $\mathcal{G}/\mathcal{G}_0$ with character χ^b. Write $\sigma = \sigma_1/\beta$ with $\sigma_1, \beta \in \mathcal{O}(S_0)$ and $\beta(s_0) \neq 0$. Since s_0 is fixed by $\mathcal{G}/\mathcal{G}_0$, $\beta' := \prod_{g\in\mathcal{G}/\mathcal{G}_0} g^*\beta$ is an invariant element with $\beta'(s_0) \neq 0$, and $\sigma_2 := \sigma\beta' \in \mathcal{O}(S_0)$ is an eigenvector with character χ^b. Since $\operatorname{center}_{S_{0,\bar{k}}}(\bar{T}_0) = \bar{s}_0$, $\bar{T}_0(\beta') = 0$, so it suffices to show $\bar{T}(\pi_0^*\sigma_2) \geq b$. The ring $\mathcal{O}(S)$ is a free module over $\mathcal{O}(X'^{n_0}) \otimes_k k[u]$ with basis $z_1^{j_1} \cdots z_{n_0}^{j_{n_0}}$, $0 \leq j_a < m$, so

$$\pi_0^*\sigma_2 = \sum_{j,l} c_{j,l} z_1^{j_1} \cdots z_{n_0}^{j_{n_0}} u^l$$

for unique $c_{j,l} \in \mathcal{O}(X'^{n_0})$, where $j = (j_1, \dots, j_{n_0})$ with $0 \leq j_a < m$ and $l \geq 0$. Let $\zeta \in k$ be a primitive m-th root of unity. The element $(1, \zeta) \in \mathcal{G}$ fixes $c_{j,l}$, multiplies $z_1^{j_1} \cdots z_{n_0}^{j_{n_0}} u^l$ by $\zeta^{|j|+l}$, where $|j| := j_1 + \cdots + j_{n_0}$, and multiplies $\pi_0^*\sigma_2$ by $\chi^b(1,\zeta) = \zeta^{(n_0+1)b}$. Thus $c_{j,l} \neq 0$ implies

$$|j| + l \equiv (n_0+1)b \pmod{m}.$$

Since $0 < (n_0+1)b < m$ and $|j| + l \geq 0$, the congruence implies $|j| + l \geq (n_0+1)b$. Since $c_{j,l}$ is regular at $\bar{s}$, $\bar{T}(c_{j,l}) \geq 0$, so

$$\bar{T}\big(c_{j,l} z_1^{j_1} \cdots z_{n_0}^{j_{n_0}} u^l\big) \geq \frac{|j| + l}{n_0 + 1} \geq b$$

for any (j, l) with $c_{j,l} \neq 0$. Thus $\bar{T}_0(\sigma) = \bar{T}(\pi_0^*\sigma_2) \geq b$. □

Lemma 11.5. *Let k be a field finitely generated over $\mathbb{C}$, $V_1, \dots, V_l$ normal varieties over k of positive dimension which are $\mathbb{Q}$-Gorenstein and klt, $v_j \in V_j$ k-rational points, and $Z_j \geq 0$ $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on V_j such that (V_j, Z_j) is lc. Let $S := V_1 \times \cdots \times V_l \times \mathbb{A}^1_u$ over k, $s := (v_1, \dots, v_l, 0)$, and $\Theta := \sum_j \mathrm{pr}_j^* Z_j + (u = 0)$. Let F be a Kollár component of $(S \ni s, 0)$ over k such that $a(F, S, \Theta) = 0$. Then for any j, the restriction of ord_F to $k(V_j)$ is $c_j \operatorname{ord}_{F_j}$ for a positive integer c_j and a Kollár component F_j of $(V_j \ni v_j, 0)$ over k with $a(F_j, V_j, Z_j) = 0$. If*

moreover a finite group acts on each V_j fixing v_j, acts on S diagonally and on u by a character, and preserves ord_F*, then it preserves* ord_{F_j} *for any j.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we pass to $\bar{k}$ and show that $\mathrm{ord}_{F_{\bar{k}}}$ is a monomial valuation on a product of log resolutions. Let $\bar{s} \in S_{\bar{k}}$ and $\bar{v}_j \in V_{j,\bar{k}}$ be the points over s and v_j. By Lemma 3.5(2), V_j and S are geometrically integral. Since each V_j is normal over a field of characteristic zero, hence geometrically normal, S and $S_{\bar{k}}$ are normal [Sta, Tag 06DG]. By Lemma 3.3(3), $V_{j,\bar{k}}$ is klt and $(V_{j,\bar{k}}, Z_{j,\bar{k}})$ is lc for any j.

For any j, let $\mu_j : Y_j \to V_{j,\bar{k}}$ be a log resolution of $(V_{j,\bar{k}}, Z_{j,\bar{k}})$, D_j the reduced snc divisor with support $\mathrm{Exc}(\mu_j) \cup \operatorname{Supp} \mu_{j*}^{-1} Z_{j,\bar{k}}$, $K_{Y_j} + B_{Y_j} := \mu_j^*(K_{V_{j,\bar{k}}} + Z_{j,\bar{k}})$, and $K_{Y_j} + B'_{Y_j} := \mu_j^* K_{V_{j,\bar{k}}}$. The coefficients of B_{Y_j} are at most 1, and those of B'_{Y_j} are less than 1.

Let $\mu : Y := Y_1 \times \cdots \times Y_l \times \mathbb{A}^1_u \to S_{\bar{k}}$ be the product, $D := \sum_j \mathrm{pr}_j^* D_j + (u = 0)$, and $B_Y := \sum_j \mathrm{pr}_j^* B_{Y_j} + (u = 0)$. Then μ is a log resolution, D is an snc divisor, $\operatorname{Supp} B_Y \subset D$, and since the canonical divisor of a product is the sum of the pullbacks of the canonical divisors of the factors,

$$K_Y + B_Y = \mu^*(K_{S_{\bar{k}}} + \Theta_{\bar{k}}), \qquad K_Y + \sum_j \mathrm{pr}_j^* B'_{Y_j} = \mu^* K_{S_{\bar{k}}}.$$

Thus $S_{\bar{k}}$ is klt and $(S_{\bar{k}}, \Theta_{\bar{k}})$ is lc [KM98, Lemma 2.30, Corollary 2.31(3)], so S is klt (Lemma 3.3(3)). By Lemma 3.6(1), $F' := F_{\bar{k}}$ is a Kollár component of $(S_{\bar{k}} \ni \bar{s}, 0)$, and $\mathrm{ord}_{F'}|_{k(S)} = \mathrm{ord}_F$. By Lemma 3.3(2), $a(F', S_{\bar{k}}, \Theta_{\bar{k}}) = a(F, S, \Theta) = 0$. Let $w := \mathrm{ord}_{F'}$. Then $a(w, Y, B_Y) = 0$ [KM98, Lemma 2.30]. Write $B_Y = \sum_i b_i D_i$, where the D_i are the components of D and $b_i \le 1$. Let $r := r_{Y,D}$ be the retraction onto $\mathrm{QM}(Y, D)$. Then $r(w)(D_i) = w(D_i)$ for any i [JM12, Section 4.3], and $a(r(w), Y, 0) \le a(w, Y, 0)$ with equality if and only if $w \in \mathrm{QM}(Y, D)$ [JM12, Corollary 5.4]. Since $a(r(w), Y, 0) = \sum_i r(w)(D_i)$ [JM12, Proposition 5.1],

$$0 = a(w, Y, B_Y) = a(w, Y, 0) - \sum_i b_i w(D_i) \ge \sum_i (1 - b_i) w(D_i) \ge 0.$$

Thus $w \in \mathrm{QM}(Y, D)$, and $b_i = 1$ for any D_i with $w(D_i) > 0$. Let $\eta := \mathrm{center}_Y w$. Then w is the monomial valuation at η in local equations of the components of D through η, with positive weights, and any component of D through η has coefficient 1 in B_Y. Since $\mu(\eta) = \bar{s}$, the closure of η is $W_1 \times \cdots \times W_l \times \{0\}$, where W_j is an irreducible component of an intersection of components of D_j with $\mu_j(W_j) = \bar{v}_j$. Let η_j be the generic point of W_j, $y_{j,1}, \ldots, y_{j,r_j}$ local equations at η_j of the components of D_j through η_j, $\alpha_{j,i} := w(\mathrm{pr}_j^* y_{j,i}) > 0$, and $\theta := w(u) > 0$. Then w is the monomial valuation at η in the $\mathrm{pr}_j^* y_{j,i}$ and u with weights $(\alpha_{1,1}, \ldots, \alpha_{l,r_l}, \theta)$.

Step 2. In this step we show that w is a Gauss product. For any j, let w_j be the monomial valuation of $\mathcal{O}_{Y_j,\eta_j}$ in $y_{j,1}, \ldots, y_{j,r_j}$ with weights $\alpha_{j,1}, \ldots, \alpha_{j,r_j}$. Then w_j is a quasi-monomial valuation on $V_{j,\bar{k}}$ with $\mathrm{center}_{V_{j,\bar{k}}}(w_j) = \mu_j(\eta_j) = \bar{v}_j$. Since the components of D_j through η_j have coefficient 1 in B_{Y_j}, [JM12, Proposition 5.1, Remark 5.6] implies

$$a(w_j, V_{j,\bar{k}}, Z_{j,\bar{k}}) = a(w_j, Y_j, B_{Y_j}) = \sum_i \alpha_{j,i} - \sum_i \alpha_{j,i} = 0.$$

Let G be the Gauss product of $w_1, \ldots, w_l$ with u-weight θ (Definition-Proposition 11.3). Since the restriction of G to $k(Y_j) = k(V_{j,\bar{k}})$ is w_j, whose center on Y_j is η_j, we have $\mathrm{pr}_j(\mathrm{center}_Y G) = \eta_j$ for any j. Since $G(u) = \theta > 0$, $\mathrm{center}_Y G \in (u = 0)$. Thus $\mathrm{center}_Y G$ belongs to the closure $W_1 \times \cdots \times W_l \times \{0\}$ of η, and the components of D through $\mathrm{center}_Y G$ are exactly the components of D through η, namely the $\mathrm{pr}_j^*(y_{j,i} = 0)$ and $(u = 0)$. Hence $r(G) \in \mathrm{QM}_\eta(Y, D)$, and $r(G)$ is determined by the values $r(G)(\mathrm{pr}_j^* y_{j,i})$ and $r(G)(u)$ [JM12, Lemma 4.5]. By [JM12, Section 4.3],

$$r(G)(\mathrm{pr}_j^* y_{j,i}) = G(\mathrm{pr}_j^* y_{j,i}) = w_j(y_{j,i}) = \alpha_{j,i}, \qquad r(G)(u) = G(u) = \theta,$$

which are the weights of w. Thus $r(G) = w$. Since $K_{Y/S_{\bar{k}}} = \sum_j \mathrm{pr}_j^* K_{Y_j/V_{j,\bar{k}}}$, Definition-Proposition 11.3 and [JM12, Proposition 5.1, Remark 5.6] imply

$$a(G, Y, 0) = a(G, S_{\bar{k}}, 0) - \sum_j w_j(K_{Y_j/V_{j,\bar{k}}}) = \sum_j a(w_j, Y_j, 0) + \theta = \sum_{j,i} \alpha_{j,i} + \theta = a(w, Y, 0).$$

Thus $a(r(G), Y, 0) = a(G, Y, 0)$, so $G \in \mathrm{QM}(Y, D)$ [JM12, Corollary 5.4], and $G = r(G) = w$. In particular, the restriction of w to $k(V_{j,\bar{k}})$ is w_j, and

$$\mathrm{gr}_w R = \Big(\bigotimes_j \mathrm{gr}_{w_j} R_j \Big)[\mathrm{in}_w(u)],$$

where $R := \mathcal{O}_{S_{\bar{k}},\bar{s}}$ and $R_j := \mathcal{O}_{V_{j,\bar{k}},\bar{v}_j}$.

Step 3. In this step we show that $w_j = c_j \operatorname{ord}_{F_j'}$ for a positive integer c_j and a Kollár component F_j' of $(V_{j,\bar{k}} \ni \bar{v}_j, 0)$. Since w takes values in $\mathbb{Z}$, so does w_j, and the weights $\alpha_{j,i}$ are positive integers. Here $r_j \geq 1$, since $\mu_j(W_j) = \bar{v}_j$ is a point and $\dim V_j > 0$. By the description of the graded ring of a monomial valuation, the residue field of w_j contains $\kappa(\eta_j)$ and the $r_j - 1$ algebraically independent residues of ratios of two monomials in $y_{j,1}, \dots, y_{j,r_j}$ of equal weight, so the transcendence degree of the residue field of w_j is $\dim W_j + r_j - 1 = \dim V_j - 1$. Thus $w_j = c_j \operatorname{ord}_{F_j'}$ for a positive integer c_j and a prime divisor F_j' over $V_{j,\bar{k}}$ [KM98, Lemma 2.45], and $\mathrm{gr}_{w_j} R_j$ is $\mathrm{gr}_{\operatorname{ord}_{F_j'}} R_j$ with degrees multiplied by c_j.

By Theorem 2.10, $A := \mathrm{gr}_w R$ is finitely generated over $\bar{k}$, and the marked graded pair of w, whose boundary is zero, is klt. Thus $X_0 := \operatorname{Spec} A$ is a klt variety. Let $A_j := \mathrm{gr}_{w_j} R_j$ and $X_{0,j} := \operatorname{Spec} A_j$. Since the degree zero part of $A_{j'}$ is $\bar{k}$ for any j', A_j is the quotient of A by the ideal generated by $\mathrm{in}_w(u)$ and the positive degree parts of the $A_{j'}$ with $j' \neq j$, so A_j is finitely generated. Since $A = A_j \otimes_{\bar{k}} A_j'$ for $A_j' := (\bigotimes_{j' \neq j} A_{j'})[\mathrm{in}_w(u)]$, comparing coefficients in a $\bar{k}$-basis of A_j' containing 1 implies $A \cap \operatorname{Frac}(A_j) = A_j$ in $\operatorname{Frac}(A)$. Since A is normal, A_j is normal.

Let $W_j' \subset \operatorname{Spec} A_j'$ be the smooth locus, which is a non-empty open subset, so that $X_{0,j} \times W_j'$ is an open subset of X_0. Let $q : X_{0,j} \times W_j' \to X_{0,j}$ be the projection, which is smooth and surjective. Let n be a positive integer such that nK_{X_0} is Cartier. Since $K_{X_{0,j} \times W_j'} = q^* K_{X_{0,j}} + \mathrm{pr}^* K_{W_j'}$ and $K_{W_j'}$ is Cartier, $q^* \mathcal{O}_{X_{0,j}}(nK_{X_{0,j}})$ is invertible, so $\mathcal{O}_{X_{0,j}}(nK_{X_{0,j}})$ is invertible [Sta, Tag 05B2]. Let $f : X_{0,j}' \to X_{0,j}$ be a log resolution. Then $f \times \mathrm{id}_{W_j'} : X_{0,j}' \times W_j' \to X_{0,j} \times W_j'$ is a log resolution with $K_{X_{0,j}' \times W_j'/X_{0,j} \times W_j'} = q'^* K_{X_{0,j}'/X_{0,j}}$ for the projection q', so $a(E \times W_j', X_0, 0) = a(E, X_{0,j}, 0)$ for any prime divisor E on $X_{0,j}'$. Since X_0 is klt, $X_{0,j}$ is klt [KM98, Lemma 2.30, Corollary 2.31(3)]. Thus $\mathrm{gr}_{\operatorname{ord}_{F_j'}} R_j$ is finitely generated and the marked graded pair of $\operatorname{ord}_{F_j'}$ is klt, so F_j' is a Kollár component of $(V_{j,\bar{k}} \ni \bar{v}_j, 0)$ by Theorem 2.10. Moreover, $a(F_j', V_{j,\bar{k}}, Z_{j,\bar{k}}) = a(w_j, V_{j,\bar{k}}, Z_{j,\bar{k}})/c_j = 0$ by **Step 2**.

Step 4. In this step we descend F_j' to k and conclude. For $\gamma \in \mathrm{Gal}(\bar{k}/k)$, γ maps $F' = F_{\bar{k}}$ to itself, so $w \circ \gamma^* = w$. Since the projections $S_{\bar{k}} \to V_{j,\bar{k}}$ are defined over k, γ^* preserves $k(V_{j,\bar{k}})$, and $w_j \circ \gamma^* = w_j$. Thus $\operatorname{ord}_{F_j'} \circ \gamma^* = \operatorname{ord}_{F_j'}$. By Lemma 3.6(2), $\operatorname{ord}_{F_j'}|_{k(V_j)} = \operatorname{ord}_{F_j}$ for a Kollár component F_j of $(V_j \ni v_j, 0)$ over k with $F_{j,\bar{k}} = F_j'$. Then

$$\operatorname{ord}_F|_{k(V_j)} = w|_{k(V_j)} = w_j|_{k(V_j)} = c_j \operatorname{ord}_{F_j},$$

and $a(F_j, V_j, Z_j) = a(F_j', V_{j,\bar{k}}, Z_{j,\bar{k}}) = 0$ by Lemma 3.3(2).

Assume that a finite group acts as in the statement and preserves ord_F. For any element g of the group, g^* preserves the subfield $k(V_j)$ of $k(S)$ as the action is diagonal. Thus $c_j \operatorname{ord}_{F_j} \circ g^* = (\operatorname{ord}_F \circ g^*)|_{k(V_j)} = c_j \operatorname{ord}_{F_j}$, and g preserves ord_{F_j}. □

The following lemma shows that a Kollár component of $(Q \ni o, 0)$ which is an lc place of (Q, H_Q) restricts to a multiple of a Kollár component of $(X \ni x, 0)$ which is an lc place of (X, P), and that the log discrepancy grows by a factor smaller than m. The order of $H^\vee$, which is the

degree of the index cover $f : X' \to X$, does not enter the estimate. Only the degree m of $V \to X'$ does.

Lemma 11.6. *Notation as in Construction-Proposition 11.2, and let H_Q be the reduced divisor on Q whose pullback to S is $\operatorname{div}(h)$. Then $K_Q + H_Q$ is Cartier near o and (Q, H_Q) is lc. Moreover, for any Kollár component L of $(Q \ni o, 0)$ over k with $a(L, Q, H_Q) = 0$, there exists a Kollár component E of $(X \ni x, 0)$ over k such that $a(E, X, P) = 0$ and $a(E, X, 0) < ma(L, Q, 0)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that H_Q exists, that $K_Q + H_Q$ is Cartier near o, and that (Q, H_Q) is lc.

Let $q_0 : S_0 \to Q$ and $q : S \to Q$ be the quotient morphisms. By Construction-Proposition 11.2(4), Ω_0 and h_0 are eigenvectors of $\mathcal{G}/\mathcal{G}_0$, both with character χ, so the rational section Ω_0/h_0 of ω_{S_0} is invariant under $\mathcal{G}/\mathcal{G}_0$. Since $k(S_0)/k(Q)$ is a Galois extension with group $\mathcal{G}/\mathcal{G}_0$, Galois descent for vector spaces over $k(Q)$ [Sta, Tag 0CDR] implies that $\Omega_0/h_0 = q_0^*\theta$ for a rational section θ of ω_Q. Since q_0 is quasi-étale, it is étale over the generic point of any prime divisor D on Q, so $\operatorname{ord}_{D_0}(q_0^*\theta) = \operatorname{ord}_D(\theta)$ for any prime divisor D_0 on S_0 over D. Since Ω_0 generates ω_{S_0}, we obtain

$$q_0^* \operatorname{div}(\theta) = \operatorname{div}(\Omega_0/h_0) = -\operatorname{div}(h_0).$$

Let $H_Q := -\operatorname{div}(\theta)$. Since h_0 pulls back to h, the pullback of H_Q to S is $\operatorname{div}(h)$. Since $(S, \operatorname{div}(h))$ is lc (Construction-Proposition 11.2(3)) and $\operatorname{div}(h) \geq 0$, $\operatorname{div}(h)$ is reduced. H_Q is reduced and $H_Q \geq 0$, as q is étale over the generic point of any prime divisor on Q. Moreover, θ generates $\mathcal{O}_Q(K_Q + H_Q)$ near o, so $K_Q + H_Q$ is Cartier near o.

By Construction-Proposition 11.2(3), S_0 is Gorenstein and klt, and q_0 is quasi-étale. By the Hurwitz formula, $K_{S_0} = q_0^* K_Q$, so K_Q is $\mathbb{Q}$-Cartier and Q is klt [KM98, Proposition 5.20]. In particular, H_Q is $\mathbb{Q}$-Cartier. Since q is quasi-étale, $K_S + \operatorname{div}(h) = q^*(K_Q + H_Q)$ (Lemma 2.5). Since $(S, \operatorname{div}(h))$ is lc, (Q, H_Q) is lc (Lemma 2.5).

Step 2. In this step we lift L to a Kollár component F of $(S \ni s, 0)$ and restrict F to the first factor V of S.

By Construction-Proposition 11.2(3), q is quasi-étale, s is a closed point, and $q^{-1}(o) = \{s\}$. By **Step 1**, $(Q, 0)$ is klt, and K_Q and $K_Q + H_Q$ are $\mathbb{Q}$-Cartier. By Lemma 9.3 with S, $\mathcal{G}$, q, s, and 0 in place of V, G, q, v, and Δ_W, there exists a unique prime divisor F over S such that $\operatorname{ord}_F|_{k(Q)} = e \operatorname{ord}_L$ for a positive integer e, F is $\mathcal{G}$-invariant, F is a Kollár component of $(S \ni s, 0)$ over k, and

$$a(F, S, 0) = ea(L, Q, 0), \qquad a(F, S, \operatorname{div}(h)) = ea(L, Q, H_Q) = 0. \tag{11.1}$$

By Construction-Proposition 11.2(1), V is Gorenstein and klt, v is k-rational, $Z = \operatorname{div}_V(z) \geq 0$ is a Cartier divisor, and (V, Z) is lc. Moreover, $\operatorname{div}(h) = \sum_{j=1}^{n_0} \operatorname{pr}_j^* Z + (u = 0)$, where $\operatorname{pr}_j : S \to V$ is the projection to the j-th factor. By Lemma 11.5 with $l = n_0$, $V_j = V$, $v_j = v$, and $Z_j = Z$ for any j, the restriction of ord_F to $k(V)$ via pr_1 is $c_1 \operatorname{ord}_{F_1}$ for a positive integer c_1 and a Kollár component F_1 of $(V \ni v, 0)$ over k with $a(F_1, V, Z) = 0$. Since $\mathcal{G}$ fixes s, $\mathcal{G}$ fixes v. Since $\mathcal{G}$ acts on S diagonally and on u by the character λ and preserves ord_F, $\mathcal{G}$ preserves ord_{F_1} (Lemma 11.5).

Step 3. In this step we push F_1 down to a Kollár component E of $(X \ni x, 0)$ with $a(E, X, P) = 0$.

Since F_1 is a Kollár component of $(V \ni v, 0)$, there exists an open neighborhood U_V of v such that F_1 is primitive over U_V and (Y_V, F_1) is plt near F_1, where $f_V : Y_V \to U_V$ is the prime extraction of F_1. Since p is finite and $p^{-1}(x) = \{v\}$ (Construction-Proposition 11.2(1)), $U := X \backslash p(V \backslash U_V)$ is an open neighborhood of x with $p^{-1}(U) \subset U_V$. Possibly replacing X with U and V with $p^{-1}(U)$, we may assume that $U_V = V$.

Let E be the prime divisor over X such that $\operatorname{ord}_{F_1}|_{k(X)} = e_1 \operatorname{ord}_E$ for a positive integer e_1 (Lemma 2.5). By Construction-Proposition 11.2(1)(2), p is the quotient morphism of V by $\mathcal{G}$, v

is fixed by $\mathcal{G}$, and

$$K_V = p^*\Big(K_X + \big(1 - \tfrac{1}{m}\big)P\Big),$$

where $K_X + (1-\frac{1}{m})P$ is $\mathbb{Q}$-Cartier as X is $\mathbb{Q}$-Gorenstein and P is $\mathbb{Q}$-Cartier. By Lemma 9.2 with $\mathcal{G}$, p, X, 0, $(1-\frac{1}{m})P$, F_1, E, and e_1 in place of G, q, W, Δ_V, Δ_W, F, L, and e, $f_X : Y_X := Y_V/\mathcal{G} \to X$ is the prime extraction of E, and

$$\Big(Y_X, \big(1 - \tfrac{1}{m}\big)P_{Y_X} + E\Big)$$

is plt near E, where P_{Y_X} is the strict transform of P. Since P and E are $\mathbb{Q}$-Cartier, $P_{Y_X} = f_X^*P - (\operatorname{mult}_E P)E \geq 0$ is a $\mathbb{Q}$-Cartier divisor, so (Y_X, E) is plt near E [KM98, Corollary 2.35(1)]. Moreover, $\operatorname{center}_X E = p(v) = x$ and X is klt near x. Thus E is a Kollár component of $(X \ni x, 0)$ over k.

By Construction-Proposition 11.2(1), $K_V + Z = p^*(K_X + P)$. By Lemma 2.5,

$$0 = a(F_1, V, Z) = e_1 a(E, X, P),$$

so $a(E, X, P) = 0$.

Step 4. In this step we prove that $a(E, X, 0) < ma(L, Q, 0)$.

Since pr_1 is $\mathcal{G}$-equivariant and p is the quotient morphism by $\mathcal{G}$, the morphism $p \circ \operatorname{pr}_1$ is $\mathcal{G}$-invariant. Thus $p \circ \operatorname{pr}_1 = r \circ q$ for a morphism $r : Q \to X$, and we obtain the commutative diagram

$$\begin{array}{ccc} S & \xrightarrow{\operatorname{pr}_1} & V \\ \downarrow & & \downarrow \\ S_0 & & X' \\ {\scriptstyle q_0}\downarrow & & \downarrow{\scriptstyle f} \\ Q & \xrightarrow{r} & X, \end{array}$$

where $S \to S_0$ is the quotient morphism by $\mathcal{G}_0$, $V \to X'$ is the projection, the left column composes to q, and the right column composes to p. In particular, $k(X) \subset k(Q) \subset k(S)$, and the inclusion $k(X) \subset k(S)$ is also the composition of $k(X) \subset k(V)$ and $k(V) \subset k(S)$ induced by p and pr_1.

Let $c := c_1 e_1$. By **Steps 2 and 3**, $\operatorname{ord}_F|_{k(X)} = c \operatorname{ord}_E$ and $\operatorname{ord}_F|_{k(Q)} = e \operatorname{ord}_L$. By Lemma 9.4, c/e is a positive integer, so

$$e \leq c.$$

Let $z_j := \operatorname{pr}_j^* z$. Since $K_V = p^*(K_X + (1 - \frac{1}{m})P)$ and $K_V + Z = p^*(K_X + P)$, we have $p^*P = mZ$. Since $a(E, X, P) = 0$, $\operatorname{mult}_E P = a(E, X, 0)$. Thus

$$m \operatorname{ord}_F(z_1) = mc_1 \operatorname{ord}_{F_1}(z) = c_1 \operatorname{mult}_{F_1}(p^*P) = c_1 e_1 \operatorname{mult}_E P = ca(E, X, 0).$$

On the other hand, $a(F, S, \operatorname{div}(h)) = 0$ by (11.1), so

$$a(F, S, 0) = \operatorname{ord}_F(h) = \sum_{j=1}^{n_0} \operatorname{ord}_F(z_j) + \operatorname{ord}_F(u).$$

Since $z_2, \ldots, z_{n_0}$ are regular at $s = \operatorname{center}_S F$ and u vanishes at s, $\operatorname{ord}_F(z_j) \geq 0$ for any $j \geq 2$ and $\operatorname{ord}_F(u) > 0$. By (11.1),

$$\frac{c}{m} a(E, X, 0) = \operatorname{ord}_F(z_1) < a(F, S, 0) = ea(L, Q, 0).$$

Since Q is klt, $a(L, Q, 0) > 0$. Since $e \leq c$,

$$a(E, X, 0) < \frac{me}{c} a(L, Q, 0) \leq ma(L, Q, 0).$$

□

The next example shows that restricting a Kollár component of $(Q \ni o, 0)$ to $k(X)$ may multiply the log discrepancy by a factor of order m. In Subsection 11.2, the factor m is absorbed by the quadratic lower bound (11.2).

Example 11.7. Let $k = \mathbb{C}$, $X = \mathbb{A}^2$ with coordinates t_1, t_2, x the origin, $P = (t_1 t_2 = 0)$, and $m \geq 2$ an integer. The index cover of K_X and P is the identity, and in Construction-Proposition 11.2 we may take $\varphi = t_1 t_2$, $\omega' = dt_1 \wedge dt_2$, and $n_0 = 1$. Thus $V = (z^m = t_1 t_2) \subset \mathbb{A}^3$ is an A_{m-1} singularity, $\mathcal{G} \cong \mu_m$ acts by $z \to \lambda z$, $S = V \times \mathbb{A}^1_u$, $h = zu$, $\mathcal{G}_0$ is trivial, $S_0 = S$, and $Q = S/\mu_m$, where μ_m acts by $(z, u) \to (\lambda z, \lambda u)$.

Let E_1 be the exceptional curve of the minimal resolution of V with

$$\operatorname{ord}_{E_1}(t_1) = 1, \qquad \operatorname{ord}_{E_1}(t_2) = m - 1, \qquad \operatorname{ord}_{E_1}(z) = 1.$$

Since the minimal resolution of V is crepant, $a(E_1, V, 0) = 1$. Let F be the prime divisor over S such that ord_F is the Gauss product of ord_{E_1} with u-weight 1 (Definition-Proposition 11.3). Such an F exists since the Gauss product is quasi-monomial, takes values in $\mathbb{Z}$, and takes the value 1 at t_1. Since there is only one factor, the description of $\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{S,s}$ in Definition-Proposition 11.3 gives $\operatorname{ord}_F(\sum_j g_j u^j) = \min_j \left(\operatorname{ord}_{E_1}(g_j) + j\right)$ for $g_j \in \mathcal{O}_{V,v}$, so ord_F is the Gauss extension of ord_{E_1} with weight 1 (Definition 2.8). By Definition-Proposition 11.3,

$$a(F, S, 0) = a(E_1, V, 0) + 1 = 2, \qquad \operatorname{ord}_F(h) = \operatorname{ord}_{E_1}(z) + \operatorname{ord}_F(u) = 2,$$

so $a(F, S, \operatorname{div}(h)) = 0$.

The variety S is an affine toric variety whose cone is simplicial, so S is $\mathbb{Q}$-factorial [CLS11, Proposition 4.2.7] and klt [CLS11, Proposition 11.4.24(b)]. The minimal resolution of the toric surface V is the toric resolution [CLS11, Corollary 10.4.9], so E_1 is a torus-invariant curve on it, and ord_{E_1} is the toric valuation of a lattice vector β_{E_1} of V, with values 1, $m-1$, 1 on t_1, t_2, z. In particular, the valuation ideals of ord_{E_1} are generated by monomials. By Definition-Proposition 11.3 and the formula $\operatorname{ord}_F(\sum_j g_j u^j) = \min_j(\operatorname{ord}_{E_1}(g_j) + j)$, the valuation ideals of ord_F in $\mathcal{O}_{S,s}$ are generated by monomials μu^j with μ a monomial of V, so ord_F is the toric valuation of the lattice vector $\beta = (\beta_{E_1}, 1)$ of $S = V \times \mathbb{A}^1_u$. The vector β is in the interior of the cone of S since ord_F is positive on the monomials t_1, t_2, z, and u, which generate the coordinate ring of S. Thus $\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{S,s}$ is the semigroup ring of S graded by the pairing with β. In particular, $\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{S,s}$ is finitely generated, $\operatorname{Spec} \operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{S,s} \cong S$, and the marked graded pair of ord_F is klt. Thus F is a Kollár component of $(S \ni s, 0)$ (Theorem 2.10). Since μ_m acts through the torus of S, F is μ_m-invariant. Possibly shrinking Q around o and replacing S with the preimage, F is primitive over S (Definition 2.3). By Lemma 9.2 with $\Delta_V = \Delta_W = 0$, the prime divisor L over Q with $\operatorname{ord}_F|_{k(Q)} = e \operatorname{ord}_L$ is a Kollár component of $(Q \ni o, 0)$, and $a(F, S, 0) = e a(L, Q, 0)$. Since $t_1 \in k(Q)$ and $\operatorname{ord}_F(t_1) = 1$, $e = 1$. By Lemma 2.5,

$$a(L, Q, 0) = 2, \qquad a(L, Q, H_Q) = a(F, S, \operatorname{div}(h)) = 0.$$

The restriction of ord_F to $k(V)$ is ord_{E_1}, and the restriction of ord_{E_1} to $k(X) = \mathbb{C}(t_1, t_2)$ is the monomial valuation with weights $(1, m-1)$ on (t_1, t_2). Thus $\operatorname{ord}_L|_{k(X)} = \operatorname{ord}_E$, where E is the exceptional divisor of the weighted blow-up of $\mathbb{A}^2$ with weights $(1, m-1)$, and

$$a(E, X, 0) = m = \frac{m}{2} a(L, Q, 0), \qquad a(E, X, P) = m - \operatorname{ord}_E(t_1 t_2) = 0.$$

11.2. **Proof of Definition-Theorem 11.1.** In this subsection, we prove Definition-Theorem 11.1. The proof is by contradiction: Lemma 11.6 turns a sequence of germs without bounded Kollár components into a sequence of cyclic quotients satisfying the conditions of Theorem 10.2, and Lemma 11.4 contradicts the conclusion of Theorem 10.2.

Proof of Definition-Theorem 11.1. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case when x is a k-rational point.

Let $\kappa := \kappa(x)$, $x_\kappa \in X_\kappa$ the κ-rational point induced by x, X' the irreducible component of X_κ through x_κ, P' the pullback of P to X', and $g : X' \to X$ the induced morphism.

- Since κ/k is a finite separable extension, $X_\kappa \to X$ is finite étale. Since X_κ is normal, X' is a connected component of X_κ, so g is finite étale, and g induces an isomorphism $\kappa(x) \to \kappa(x_\kappa)$.
- The field κ is finitely generated over $\mathbb{C}$, X' is geometrically integral over κ, and $(X' \ni x_\kappa, 0)$ is klt (Corollary 3.10).
- Since g is étale, $P' = g^*P$ is a reduced $\mathbb{Q}$-Cartier divisor and $K_{X'} = g^*K_X$, so X' is $\mathbb{Q}$-Gorenstein near x_κ.
- Since g is quasi-étale and $K_{X'} + P' = g^*(K_X + P)$, (X', P') is lc (Lemma 2.5).
- Let E_0 be a prime divisor over X with $\operatorname{center}_X E_0 = x$ and $a(E_0, X, P) = 0$, and let $Y_0 \to X$ be a log resolution of (X, P) on which E_0 is a divisor. Since g is étale, $Y_0 \times_X X' \to X'$ is a log resolution of (X', P'), and since $E_0 \to \{x\}$ is surjective, the pullback of E_0 to $Y_0 \times_X X'$ has a component E_0' with $\operatorname{center}_{X'} E_0' = x_\kappa$ and $a(E_0', X', P') = a(E_0, X, P) = 0$. Thus x_κ is an lc center of (X', P').

Restriction of valuations induces a bijection between the Kollár components E' of $(X' \ni x_\kappa, 0)$ over κ and the Kollár components E of $(X \ni x, 0)$ over k, with $a(E', X', 0) = a(E, X, 0)$ and $a(E', X', P') = a(E, X, P)$ (Corollary 3.10 with $\Theta = P$). For corresponding E' and E, possibly shrinking X around x and X' around x_κ, the prime extraction of E' is the base change of the prime extraction of E (Proposition 3.9). Since E is reduced and maps to x, E is a $\kappa(x)$-scheme. Since the fiber of g over x near x_κ is $\operatorname{Spec} \kappa(x_\kappa)$, $E' \cong E \times_{\operatorname{Spec} \kappa(x)} \operatorname{Spec} \kappa(x_\kappa)$. Thus E is geometrically integral over $\kappa(x)$ if and only if E' is geometrically integral over $\kappa(x_\kappa)$.

Thus if the conclusion holds for (X', P') at x_κ over κ with a constant N_0, then it holds for (X, P) at x over k with constant N_0. Replacing (k, X, x, P) with $(\kappa, X', x_\kappa, P')$, we may assume that x is a k-rational point.

Then X is geometrically integral over k (Lemma 3.5(2)). For any Kollár component E of $(X \ni x, 0)$ over k, $E_{\bar{k}}$ is a prime divisor (Lemma 3.6(1)), so E is geometrically integral over $k = \kappa(x)$. Thus it suffices to find a positive integer N_0 depending only on d such that, for any k, X, x, and P as in Definition-Theorem 11.1 with x a k-rational point, there exists a Kollár component E of $(X \ni x, 0)$ over k with $a(E, X, P) = 0$ and $a(E, X, 0) \le N_0$.

Step 2. In this step we find a positive integer $n_0 = n_0(d)$ depending only on d such that $n_0(K_X + P)$ is Cartier near x for any k, X, x, and P as in **Step 1**.

Let $\bar{x} \in X_{\bar{k}}$ be the point over x. Since x is k-rational, $\bar{x}$ is the only point of $X_{\bar{k}}$ over x.

- The field $\bar{k}$ is isomorphic to $\mathbb{C}$ (Lemma 3.2).
- The pair $(X_{\bar{k}}, P_{\bar{k}})$ is lc, $X_{\bar{k}}$ is klt near $\bar{x}$ (Lemma 3.3(3)), and $P_{\bar{k}}$ is reduced (Lemma 3.3(1)).
- Let E_0 be a prime divisor over X with $\operatorname{center}_X E_0 = x$ and $a(E_0, X, P) = 0$, let Y_0 be a normal variety birational to X on which E_0 is a prime divisor, and let E_0' be a component of the pullback of E_0 to $Y_{0,\bar{k}}$. By Lemma 3.3(1)(2), the restriction of $\operatorname{ord}_{E_0'}$ to $k(X)$ is ord_{E_0} and $a(E_0', X_{\bar{k}}, P_{\bar{k}}) = 0$. The center of E_0' on $X_{\bar{k}}$ maps to x, so it is $\bar{x}$. Thus $\bar{x}$ is an lc center of $(X_{\bar{k}}, P_{\bar{k}})$.

By [Bir19, Corollary 1.9], applied to $(X_{\bar{k}}, P_{\bar{k}})$ with $R = \{0, 1\}$ (so that $\Phi(R)$ contains the coefficient 1 of $P_{\bar{k}}$), the non-klt center $\bar{x}$, and $\Delta = 0$, there exists a positive integer $n_0 = n_0(d)$ depending only on d such that $n_0(K_{X_{\bar{k}}} + P_{\bar{k}})$ is Cartier near $\bar{x}$. Since the canonical sheaf commutes with base change of the ground field and reflexive hulls commute with flat base change [Mat89, Theorem 7.11],

$$\mathcal{O}_X\big(n_0(K_X + P)\big) \otimes_k \bar{k} \cong \mathcal{O}_{X_{\bar{k}}}\big(n_0(K_{X_{\bar{k}}} + P_{\bar{k}})\big).$$

Since $X_{\bar{k}} \to X$ is integral, hence closed, and $\bar{x}$ is the only point over x, possibly shrinking X around x, we may assume that $\mathcal{O}_{X_{\bar{k}}}(n_0(K_{X_{\bar{k}}} + P_{\bar{k}}))$ is invertible. Since $X_{\bar{k}} \to X$ is faithfully flat, $\mathcal{O}_X(n_0(K_X + P))$ is invertible [Sta, Tag 05B2]. Thus $n_0(K_X + P)$ is Cartier near x.

Step 3. In this step we assume that Definition-Theorem 11.1 fails and construct the corresponding sequence of root products.

Suppose that no positive integer N_0 satisfies the condition at the end of **Step 1**. For any positive integer i, let $m_i := i+1$. Since $N_0 = m_i^2$ does not satisfy the condition, there exist a field k_i finitely generated over $\mathbb{C}$, a normal variety X_i over k_i of dimension d, a k_i-rational point $x_i \in X_i$ such that X_i is $\mathbb{Q}$-Gorenstein and klt near x_i, and a reduced $\mathbb{Q}$-Cartier divisor $P_i \geq 0$ on X_i such that (X_i, P_i) is lc, x_i is an lc center of (X_i, P_i), and

$$a(E, X_i, 0) > m_i^2 \tag{11.2}$$

for any Kollár component E of $(X_i \ni x_i, 0)$ over k_i with $a(E, X_i, P_i) = 0$. The condition is vacuous if no such E exists.

By **Step 2**, $n_0(K_{X_i} + P_i)$ is Cartier near x_i for $n_0 = n_0(d)$. We apply Construction-Proposition 11.2 to k_i, X_i, x_i, and P_i with $n_0 = n_0(d)$ and $m = m_i \geq 2$ (such φ and ω' exist by the last statement of Construction-Proposition 9.1, since x_i is k_i-rational), and we denote by V_i, $\mathcal{G}_i$, $\mathcal{G}_{0,i}$, S_i, $S_{0,i}$, Q_i, h_i, s_i, $s_{0,i}$, o_i, $\Omega_{0,i}$, $h_{0,i}$, and χ_i the objects V, $\mathcal{G}$, $\mathcal{G}_0$, S, S_0, Q, h, s, s_0, o, Ω_0, h_0, and χ constructed there. We write H_{Q_i} for H_Q in Lemma 11.6 applied to Q_i. By Lemma 11.4, there exists a quasi-monomial valuation $\bar{T}_i$ on $S_{i,\bar{k}_i}$ with center the point over s_i, invariant under $\mathcal{G}_i$ and $\mathrm{Gal}(\bar{k}_i/k_i)$, such that $a(\bar{T}_i, S_{i,\bar{k}_i}, 0) = \bar{T}_i(h_i) = 1$ and $\bar{T}_i(z_j) = \bar{T}_i(u) = \frac{1}{n_0+1}$ for any j. Let $\bar{T}_{0,i}$ be the restriction of $\bar{T}_i$ to the function field of $S_{0,i,\bar{k}_i}$.

Step 4. In this step we verify the conditions of Set-up 10.1 in dimension $dn_0 + 1$ for $(S_{0,i}, \mathcal{G}_i/\mathcal{G}_{0,i}, s_{0,i}, \Omega_{0,i}, h_{0,i}, \bar{T}_{0,i})$, which play the roles of $(S_i, G_i, x_i, \omega_i, h_i, T_i)$ there, while Q_i and $o_i \in Q_i$ play the roles of X_i and o_i there.

- By Construction-Proposition 11.2(3), $S_{0,i}$ is geometrically integral over k_i, normal, and klt, and $\dim S_{0,i} = \dim S_i = dn_0 + 1 \geq 2$. Since S_i is affine, so is $S_{0,i}$.
- The point of V_i over x_i is k_i-rational (Construction-Proposition 11.2(1)), so s_i and $s_{0,i}$ are k_i-rational.
- The group $\mathcal{G}_i/\mathcal{G}_{0,i}$ is cyclic, acts faithfully on $S_{0,i}$ over k_i, and fixes $s_{0,i}$ since $\mathcal{G}_i$ fixes s_i. The quotient morphism $S_{0,i} \to Q_i$ is quasi-étale (Construction-Proposition 11.2(3)).
- The generator $\Omega_{0,i}$ of $\omega_{S_{0,i}}$ and the element $h_{0,i} \neq 0$ are eigenvectors of $\mathcal{G}_i/\mathcal{G}_{0,i}$ with the character χ_i, and $(S_{0,i}, \mathrm{div}(h_{0,i}))$ is lc (Construction-Proposition 11.2(4)).
- The valuation $\bar{T}_{0,i}$ is quasi-monomial with center the point of $S_{0,i,\bar{k}_i}$ over $s_{0,i}$, and $a(\bar{T}_{0,i}, S_{0,i,\bar{k}_i}, 0) = \bar{T}_{0,i}(h_{0,i}) = 1$ (Lemma 11.4).

Step 5. In this step we prove that $a(L, Q_i, 0) \geq 1$ for any i and any Kollár component L of $(Q_i \ni o_i, 0)$ over k_i.

By Lemma 11.6, $K_{Q_i} + H_{Q_i}$ is Cartier near o_i and (Q_i, H_{Q_i}) is lc. Thus $a(L, Q_i, H_{Q_i})$ is a non-negative integer. If $a(L, Q_i, H_{Q_i}) \geq 1$, then since $H_{Q_i} \geq 0$ is $\mathbb{Q}$-Cartier,

$$a(L, Q_i, 0) = a(L, Q_i, H_{Q_i}) + \mathrm{mult}_L H_{Q_i} \geq a(L, Q_i, H_{Q_i}) \geq 1.$$

If $a(L, Q_i, H_{Q_i}) = 0$, then by Lemma 11.6 there exists a Kollár component E of $(X_i \ni x_i, 0)$ over k_i such that $a(E, X_i, P_i) = 0$ and $a(E, X_i, 0) < m_i a(L, Q_i, 0)$. By (11.2),

$$m_i^2 < a(E, X_i, 0) < m_i a(L, Q_i, 0),$$

so $a(L, Q_i, 0) > m_i > 1$.

Step 6. In this step we derive a contradiction.

By **Steps 4 and 5**, the conditions of Theorem 10.2 hold in dimension $dn_0 + 1$ with $\epsilon = 1$. Thus there exist a positive integer b and infinitely many i such that $\bar{T}_{0,i}(\sigma_i) < b$ for some non-zero $\sigma_i \in \mathcal{O}_{S_{0,i}, s_{0,i}}$ satisfying $\gamma^* \sigma_i = \chi_i(\gamma)^b \sigma_i$ for any $\gamma \in \mathcal{G}_i/\mathcal{G}_{0,i}$. Choose such an i with $i + 1 > (n_0 + 1)b$. Then $m_i > (n_0 + 1)b$, and σ_i is an eigenvector of $\mathcal{G}_i/\mathcal{G}_{0,i}$ with character χ_i^b, so $\bar{T}_{0,i}(\sigma_i) \geq b$ by Lemma 11.4. This is a contradiction.

Thus some positive integer N_0 depending only on d satisfies the condition at the end of **Step 1**, and Definition-Theorem 11.1 follows from **Step 1**. □

12. Fermat suspensions

In this section, we construct two Fermat suspensions over $\mathbb{C}$ and prove that the Kollár components on them that are compatible with the boundary return to Kollár components of the base. Both suspensions reduce the boundaries with a fixed denominator in Section 13 to the setting of Definition-Theorem 11.1, a variety with a reduced boundary: the marked suspension absorbs a klt boundary with denominator N into the variety and keeps log discrepancies, while the additive suspension turns an lc boundary with denominator N into a reduced boundary, and log discrepancies are divided by N on return. Section 13 uses the results of this section over k by Lemma 3.2 and Lemma 3.6.

We begin with three lemmas used in both subsections.

Lemma 12.1. *Let $\mathbb{K}$ be a field of characteristic zero, $a \in \mathbb{K}\backslash\{0\}$, and $N \geq 1$ and $s \geq 2$ integers. Then the affine hypersurface $(a + \sum_{j=1}^{s} U_j^N = 0) \subset \mathbb{A}^s_{\mathbb{K}}$ is geometrically integral. In particular, $a + \sum_{j=1}^{s} U_j^N$ is irreducible in $\mathbb{K}[U_1, \dots, U_s]$.*

Proof. Let $\bar{\mathbb{K}}$ be an algebraic closure of $\mathbb{K}$. By definition [Har77, Exercise II.3.15(c)], it suffices to prove that the hypersurface $(a + \sum_j U_j^N = 0) \subset \mathbb{A}^s_{\bar{\mathbb{K}}}$ is integral. The partial derivatives NaU_0^{N-1} and NU_j^{N-1} of $aU_0^N + \sum_{j=1}^{s} U_j^N$ have no common zero in $\mathbb{P}^s_{\bar{\mathbb{K}}}$. Assume that $aU_0^N + \sum_j U_j^N = f_1 f_2$ with non-constant homogeneous polynomials $f_1, f_2 \in \bar{\mathbb{K}}[U_0, \dots, U_s]$. Since $s \geq 2$, f_1 and f_2 have a common zero in $\mathbb{P}^s_{\bar{\mathbb{K}}}$ by [Har77, Theorem I.7.2], and $d(f_1 f_2) = f_1 df_2 + f_2 df_1$ vanishes there, a contradiction. Thus the projective hypersurface $(aU_0^N + \sum_j U_j^N = 0) \subset \mathbb{P}^s_{\bar{\mathbb{K}}}$ is integral, and so is its non-empty open subset $(U_0 \neq 0)$, which is $(a + \sum_j U_j^N = 0) \subset \mathbb{A}^s_{\bar{\mathbb{K}}}$. In particular, $\mathbb{K}[U_1, \dots, U_s]/(a + \sum_j U_j^N)$ is an integral domain, so $a + \sum_j U_j^N$ is irreducible. □

Lemma 12.2. *Let $(X \ni x, B)$ and v be as in Definition 2.9, and let $c > 0$ be a real number. Then $\mathrm{gr}_{cv} \mathcal{O}_{X,x}$ is the ring $\mathrm{gr}_v \mathcal{O}_{X,x}$ with each degree multiplied by c, and $\mathrm{in}_{cv}(f) = \mathrm{in}_v(f)$ for any non-zero $f \in \mathcal{O}_{X,x}$. In particular, $\mathrm{gr}_{cv} \mathcal{O}_{X,x}$ is finitely generated if and only if $\mathrm{gr}_v \mathcal{O}_{X,x}$ is, and in this case the marked graded pairs of cv and v coincide.*

Proof. For a real number $\rho \geq 0$, $\mathfrak{a}_\rho(cv) = \mathfrak{a}_{\rho/c}(v)$ and $\mathfrak{a}_{>\rho}(cv) = \mathfrak{a}_{>\rho/c}(v)$. Thus the degree ρ part of $\mathrm{gr}_{cv} \mathcal{O}_{X,x}$ is the degree ρ/c part of $\mathrm{gr}_v \mathcal{O}_{X,x}$, the multiplications are compatible, and $\mathrm{in}_{cv}(f) = \mathrm{in}_v(f)$ for any non-zero $f \in \mathcal{O}_{X,x}$. In Definition 2.9, X_0 and D_0 are determined by the graded ring $\mathrm{gr}_v \mathcal{O}_{X,x}$ and the initial forms $\mathrm{in}_v(f)$, so the marked graded pairs of cv and v coincide. □

Lemma 12.3. *Let A be a $\mathbb{C}$-algebra which is an integral domain, n a positive integer, and $P := A[T_1, \dots, T_n]$ the polynomial ring in n variables over A. Assume that P is finitely generated over $\mathbb{C}$ and normal, and that $\operatorname{Spec} P$ is $\mathbb{Q}$-Gorenstein. Then A is finitely generated over $\mathbb{C}$ and normal, and $\operatorname{Spec} A$ is $\mathbb{Q}$-Gorenstein. If moreover $\operatorname{Spec} P$ is klt, then $\operatorname{Spec} A$ is klt.*

Proof. Since $A = P/(T_1, \dots, T_n)$, A is finitely generated over $\mathbb{C}$. If an element of $\mathrm{Frac}(A)$ is integral over A, then it is integral over P, hence it belongs to $P \cap \mathrm{Frac}(A) = A$. Thus A is normal. The projection $\operatorname{Spec} P \to \operatorname{Spec} A$ is smooth and $\Omega_{P/A}$ is free, so for any integer l, the reflexive sheaf $\mathcal{O}(lK_{\operatorname{Spec} P})$ is the pullback of $\mathcal{O}(lK_{\operatorname{Spec} A})$. If $lK_{\operatorname{Spec} P}$ is Cartier for a positive integer l, then the restriction along the zero section $T_1 = \cdots = T_n = 0$ shows that $lK_{\operatorname{Spec} A}$ is Cartier. Thus $\operatorname{Spec} A$ is $\mathbb{Q}$-Gorenstein.

Assume moreover that $\operatorname{Spec} P$ is klt. For a prime divisor E' over $\operatorname{Spec} A$, realized on a higher model $W' \to \operatorname{Spec} A$, the prime divisor $E' \times \mathbb{A}^n$ on $W' \times \mathbb{A}^n$ satisfies

$$a(E', \operatorname{Spec} A, 0) = a(E' \times \mathbb{A}^n, \operatorname{Spec} P, 0) > 0,$$

since $K_{W' \times \mathbb{A}^n}$ is the pullback of $K_{W'}$. Thus $\operatorname{Spec} A$ is klt. □

12.1. **The additive suspension.** In this subsection, we construct the additive suspension of an lc pair whose boundary has denominator N and prove Theorem 12.7.

Construction 12.4. Let X be a $\mathbb{Q}$-Gorenstein klt variety over $\mathbb{C}$ of dimension $d \geq 2$, $x \in X$ a closed point, $N \geq 3$ an integer, and f a non-zero regular function on X with $f(x) = 0$ such that $(X, \frac{1}{N}\operatorname{div}(f))$ is lc and x is an lc center of $(X, \frac{1}{N}\operatorname{div}(f))$. Let $s := N - 1$, $Y := X \times \mathbb{A}^s$ with coordinates $u_1, \dots, u_s$ on $\mathbb{A}^s$, $o := (x, 0) \in Y$, $h := f + \sum_{j=1}^{s} u_j^N$, and $H := \operatorname{div}(h)$. Let $G := \mu_N^s$ act on Y trivially on X and by multiplying the coordinates $u_1, \dots, u_s$ separately, and let $Q := Y/G \cong X \times \mathbb{A}^s$ with coordinates $z_j := u_j^N$. We call (Y, H) the *additive suspension* of $(X \ni x, \frac{1}{N}\operatorname{div}(f))$.

Definition-Proposition 12.5. Notation as in Construction 12.4. Then Y is a $\mathbb{Q}$-Gorenstein klt variety, H is reduced, (Y, H) is lc, and o is an lc center of (Y, H). Let F be a prime divisor over Y with $\operatorname{center}_Y F = o$ and $a(F, Y, H) = 0$. Then the restriction of ord_F to $\mathbb{C}(X)$ is $c \operatorname{ord}_E$ for a positive integer c and a prime divisor E over X with $\operatorname{center}_X E = x$, and the following hold.

(1) $m := \operatorname{ord}_E(f) > 0$, $a(E, X, \frac{1}{N}\operatorname{div}(f)) = 0$, and $Nca(E, X, 0) \leq a(F, Y, 0)$.
(2) Let $p := N/\gcd(m, N)$ and $q := m/\gcd(m, N)$, let R_E be the valuation ring of ord_E, t a uniformizer of R_E, and $\beta := t^{-m} f$, a unit of R_E. Let Y_Γ be the weighted blow-up of $\operatorname{Spec} R_E[u_1, \dots, u_s]$ along $(t, u_1, \dots, u_s)$ with weights $(p, q, \dots, q)$, Γ the exceptional divisor of $Y_\Gamma \to \operatorname{Spec} R_E[u_1, \dots, u_s]$, and H' the strict transform of H on Y_Γ. Then the crepant pullback of (Y, H) to Y_Γ has coefficient one along Γ and along H', and $\Gamma \cap H'$ is irreducible. Moreover, let $R' := R_E[v_0]/(v_0^p - t)$, and consider the morphism from $\operatorname{Spec} R'[v_1, \dots, v_s]$ to Y_Γ given by $t = v_0^p$ and $u_j = v_0^q v_j$. Then $h = v_0^{mp} r$ with $r := \beta + \sum_{j=1}^{s} v_j^N$, the preimages of Γ, H', and $\Gamma \cap H'$ are $(v_0 = 0)$, $(r = 0)$, and $(v_0 = r = 0)$, and $(v_0 = r = 0)$ is integral.
(3) There exist a prime divisor $\tilde{F}$ over $\operatorname{Spec} R'[v_1, \dots, v_s]$ and integers $e_0, \lambda > 0$ and $\mu \geq 0$ such that the restriction of $\operatorname{ord}_{\tilde{F}}$ to the function field of Y is $e_0 \operatorname{ord}_F$ and $\operatorname{ord}_{\tilde{F}}$ is the monomial valuation in v_0 and r with $\operatorname{ord}_{\tilde{F}}(v_0) = \lambda$ and $\operatorname{ord}_{\tilde{F}}(r) = \mu$ at the generic point of $(v_0 = r = 0)$. In particular, $c = \lambda p / e_0$, $\operatorname{ord}_F(u_j^N) = cm$ for any j, and $\operatorname{ord}_F(h) = a(F, Y, 0) = cm + \mu/e_0$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that Y is a $\mathbb{Q}$-Gorenstein klt variety, that H is reduced, and that $a(F', Y, H) \geq 0$ for any prime divisor F' over Y such that $\operatorname{ord}_{F'}$ is trivial on $\mathbb{C}(X)$.

Let $\operatorname{pr} : Y \to X$ be the projection. Since $K_Y = \operatorname{pr}^* K_X$, Y is $\mathbb{Q}$-Gorenstein. Let $\tilde{X} \to X$ be a log resolution. Then $\tilde{X} \times \mathbb{A}^s \to Y$ is a log resolution, the exceptional prime divisors of $\tilde{X} \times \mathbb{A}^s \to Y$ are the divisors $D \times \mathbb{A}^s$ for the exceptional prime divisors D of $\tilde{X} \to X$, and $a(D \times \mathbb{A}^s, Y, 0) = a(D, X, 0) > 0$. Thus Y is klt [KM98, Lemma 2.30 and Corollary 2.31(3)].

For any prime divisor P on X, the restriction of h to $P \times \mathbb{A}^s$ is $f|_P + \sum_j u_j^N \neq 0$. Thus any component of H dominates X, and the coefficient of a component of H is the multiplicity of the corresponding prime factor of h in $\mathbb{C}(X)[u_1, \dots, u_s]$. If ϖ^2 divides h for a non-constant $\varpi \in \mathbb{C}(X)[u_1, \dots, u_s]$, then ϖ divides $\partial h / \partial u_j = N u_j^{N-1}$ for any j, which is impossible since $s \geq 2$. Thus H is reduced.

Let F' be a prime divisor over Y such that $\operatorname{ord}_{F'}$ is trivial on $\mathbb{C}(X)$. Then $\operatorname{center}_Y F'$ dominates X, and $a(F', Y, H)$ can be computed on the generic fiber $\mathbb{A}^s_{\mathbb{C}(X)}$ of pr, with boundary the hypersurface $(f + \sum_j u_j^N = 0)$. Since $f \in \mathbb{C}(X)^*$, the partial derivatives $N u_j^{N-1}$ vanish simultaneously only at $u = 0$, where $h = f \neq 0$. Thus $(f + \sum_j u_j^N = 0)$ is smooth, and $a(F', Y, H) \geq 0$ by [KM98, Corollary 2.31(3)].

Step 2. In this step we fix a prime divisor E over X, write the crepant pullback of (Y, H) near the generic point of $E \times \mathbb{A}^s$, and prove that $a(F', Y, H) > 0$ for the prime divisors F' over Y in two cases.

Let $\alpha := a(E, X, 0)$ and $m := \operatorname{ord}_E(f) \geq 0$. Since X is klt, $\alpha > 0$. Let $g : X' \to X$ be a proper birational morphism from a normal variety X' such that E is a prime divisor on X', let R_E be the valuation ring of ord_E, t a uniformizer of R_E, and $\beta := t^{-m} f$. Choose an affine open subset $X'' \subset X'$ such that X'' is smooth, $E'' := E \cap X''$ is non-empty, t and β are regular on X'', β is a unit on X'', $E'' = (t = 0)$, the morphism $t : X'' \to \mathbb{A}^1$ is smooth, and E is the only g-exceptional prime divisor intersecting X''. Let $Y'' := X'' \times \mathbb{A}^s$, $S := E'' \times \mathbb{A}^s = (t = 0)$, $V_0 := E'' \times \{0\}$, and $H_{Y''} := \operatorname{div}(t^m \beta + \sum_j u_j^N)$ on Y''. Since $K_{X''} = g^* K_X|_{X''} + (\alpha - 1)E''$,

$$K_{Y''} + H_{Y''} + (1 - \alpha)S = (g \times \mathrm{id}_{\mathbb{A}^s})^*(K_Y + H)|_{Y''}. \tag{12.1}$$

Let F' be a prime divisor over Y such that $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = c' \operatorname{ord}_E$ for a positive integer c'. Then the center of F' on $X' \times \mathbb{A}^s$ dominates E, so F' is a prime divisor over Y'', and $Z' := \operatorname{center}_{Y''} F'$ dominates E'' and is contained in S as $\operatorname{ord}_{F'}(t) = c' > 0$. By (12.1), $a(F', Y, H) = a(F', Y'', H_{Y''} + (1 - \alpha)S)$.

Assume that $m = 0$ or that $Z' \not\subset V_0$. If $m = 0$, then $h|_S = \beta|_{E''} + \sum_j u_j^N$ with $\beta|_{E''}$ a unit, so some u_j does not vanish at any zero of $h|_S$. If $m > 0$, then $h|_S = \sum_j u_j^N$, and some u_j does not vanish at the generic point of Z'. In both cases $\partial(h|_S)/\partial u_j = N u_j^{N-1} \neq 0$ there. Thus near the generic point of Z', the divisor $H_{Y''}$ is smooth and transversal to S, and $(Y'', H_{Y''} + (1 - \alpha)S)$ is snc with coefficient-one part $H_{Y''}$, as $1 - \alpha < 1$. Replacing the coefficient $1 - \alpha$ with $\max\{1 - \alpha, 0\}$ does not increase log discrepancies. Since the lc centers of an snc pair with coefficients at most one are the strata of the coefficient-one part [Kol13, Section 4.2, the paragraph after Definition 4.15], and $Z' \subset S$ is not a stratum of $H_{Y''}$, we obtain $a(F', Y, H) > 0$.

Step 3. In this step we assume that $m > 0$, describe the weighted blow-up of Y'' along V_0 by charts, and prove that (Y, H) is lc.

Let $p := N/\gcd(m, N)$, $q := m/\gcd(m, N)$, and $\delta := p\alpha - q$. Since $(X, \frac{1}{N}\operatorname{div}(f))$ is lc and $q = pm/N$,

$$\delta = p\Big(\alpha - \frac{m}{N}\Big) = pa\Big(E, X, \frac{1}{N}\operatorname{div}(f)\Big) \geq 0.$$

Let $Y''_\Gamma \to Y''$ be the weighted blow-up along $(t, u_1, \dots, u_s)$ with weights $(p, q, \dots, q)$. Since weighted blow-ups commute with flat base change, the weighted blow-up Y_Γ of Definition-Proposition 12.5(2) is the base change of Y''_Γ along $\operatorname{Spec} R_E \to X''$, and we also denote by Γ and H' the exceptional divisor of $Y''_\Gamma \to Y''$ and the strict transform of $H_{Y''}$. Let S' be the strict transform of S, and let $K_{Y''_\Gamma} + \Delta_\Gamma$ be the pullback of $K_Y + H$ to Y''_Γ.

Let $X''_0 := \operatorname{Spec} \mathcal{O}(X'')[v_0]/(v_0^p - t)$ and $\tilde{U}_0 := X''_0 \times \mathbb{A}^s$ with coordinates $v_1, \dots, v_s$, and let $\varphi_0 : \tilde{U}_0 \to Y''_\Gamma$ be given by $t = v_0^p$ and $u_j = v_0^q v_j$. For $1 \leq k \leq s$, let $X''_k := \operatorname{Spec} \mathcal{O}(X'')[v_0, \tau]/(v_0^p \tau - t)$ and $\tilde{U}_k := X''_k \times \mathbb{A}^{s-1}$ with coordinates v_j, $j \neq k$, and let $\varphi_k : \tilde{U}_k \to Y''_\Gamma$ be given by $t = v_0^p \tau$, $u_k = v_0^q$, and $u_j = v_0^q v_j$ for $j \neq k$. A direct calculation of the degree zero parts of the localized Rees algebra of Y''_Γ shows that the images $U_0, U_1, \dots, U_s$ of $\varphi_0, \varphi_1, \dots, \varphi_s$ form an open cover of Y''_Γ, that φ_0 is the quotient by μ_p acting by $v_0 \to \zeta v_0$ and $v_j \to \zeta^{-q} v_j$, and that φ_k is the quotient by μ_q acting by $v_0 \to \zeta v_0$ and $\tau \to \zeta^{-p} \tau$. In particular, φ_i is finite and surjective onto U_i, $\varphi_i^{-1}(\Gamma) = (v_0 = 0)$, $\varphi_0^{-1}(S') = \emptyset$, $\varphi_k^{-1}(S') = (\tau = 0)$, and $\varphi_k^{-1}(U_0) = (\tau \neq 0)$.

Since $t : X'' \to \mathbb{A}^1$ is smooth, X''_0 and X''_k are smooth, as base changes of t along $v_0 \to v_0^p$ and $(v_0, \tau) \to v_0^p \tau$. Thus $\tilde{U}_i$ is smooth, $(v_0 = 0) \subset \tilde{U}_0$ is $E'' \times \mathbb{A}^s$, and $(v_0 = 0) \subset \tilde{U}_k$ is $E'' \times \mathbb{A}^1_\tau \times \mathbb{A}^{s-1}$. Since $qN = mp$,

$$h = v_0^{mp} r \text{ on } \tilde{U}_0, \qquad h = v_0^{mp} r_k \text{ on } \tilde{U}_k, \qquad r := \beta + \sum_{j=1}^{s} v_j^N, \quad r_k := \tau^m \beta + 1 + \sum_{j \neq k} v_j^N. \tag{12.2}$$

Let $\psi_i : \tilde{U}_i \to Y''$ be the composition of φ_i and $Y''_\Gamma \to Y''$. A direct calculation of the Jacobian of ψ_i shows that $K_{\tilde{U}_i} = \psi_i^* K_{Y''} + (p + sq - 1)(v_0 = 0)$. Moreover, $\psi_0^* S = p(v_0 = 0)$ and $\psi_k^* S = p(v_0 = 0) + (\tau = 0)$. Since $mp - sq = Nq - (N - 1)q = q$, (12.1) and (12.2) imply that

$K_{\tilde{U}_i} + \Delta_i = \varphi_i^*(K_{U_i} + \Delta_\Gamma|_{U_i})$, where

$$\Delta_0 := (r = 0) + (1 - \delta)(v_0 = 0), \qquad \Delta_k := (r_k = 0) + (1 - \delta)(v_0 = 0) + (1 - \alpha)(\tau = 0).$$

We show that Δ_i is snc near $(v_0 = 0)$. Let $\beta_0 := \beta|_{E''}$, a unit on E''. On $(v_0 = 0) \subset \tilde{U}_0$, r restricts to $\beta_0 + \sum_j v_j^N$, so some v_j does not vanish at any zero, and $\partial/\partial v_j$ of the restriction is $Nv_j^{N-1} \neq 0$ there. On $(v_0 = 0) \subset \tilde{U}_k$, r_k restricts to $\tau^m\beta_0 + 1 + \sum_{j\neq k} v_j^N$. At a zero, either some v_j with $j \neq k$ does not vanish, and the derivative in v_j is $Nv_j^{N-1} \neq 0$, or all such v_j vanish, so $\tau \neq 0$ and the derivative in τ is $m\tau^{m-1}\beta_0 \neq 0$. On $(v_0 = \tau = 0) \subset \tilde{U}_k$, r_k restricts to $1 + \sum_{j\neq k} v_j^N$, and since $s \geq 2$, some v_j with $j \neq k$ does not vanish at any zero. Thus Δ_i is snc near $(v_0 = 0)$, and the coefficients of Δ_i are at most one since $\delta \geq 0$ and $\alpha > 0$.

Let F' be a prime divisor over Y as in **Step 2** with $Z' = V_0$. Then the center of F' on Y''_Γ is contained in Γ. Let i be an index such that the center of F' on Y''_Γ intersects U_i. By [Kol13, 2.42], there exists a prime divisor $\tilde{F}'$ over $\tilde{U}_i$ such that $\operatorname{ord}_{\tilde{F}'}|_{\mathbb{C}(Y)} = e' \operatorname{ord}_{F'}$ for a positive integer e', and by Lemma 2.5, $a(\tilde{F}', \tilde{U}_i, \Delta_i) = e'a(F', Y, H)$. Since $\operatorname{center}_{\tilde{U}_i}(\tilde{F}') \subset (v_0 = 0)$, [KM98, Corollary 2.31(3)] and [Kol13, Section 4.2, the paragraph after Definition 4.15] imply that $a(F', Y, H) \geq 0$, and that

$$\begin{aligned} &\text{if } a(F', Y, H) = 0, \text{ then } \operatorname{center}_{\tilde{U}_i}(\tilde{F}') \text{ is a stratum of the} \\ &\qquad \text{coefficient-one part of } \Delta_i \text{ contained in } (v_0 = 0). \end{aligned} \tag{12.3}$$

By Lemma 9.5, for any prime divisor F' over Y, either $\operatorname{ord}_{F'}$ is trivial on $\mathbb{C}(X)$, or $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = c' \operatorname{ord}_E$ for a positive integer c' and a prime divisor E over X. By **Steps 1 and 2** and the previous paragraph (if $m > 0$ and $Z' \subset V_0$, then $Z' = V_0$, since Z' dominates E'' and $\dim V_0 = \dim E''$), $a(F', Y, H) \geq 0$ in both cases. Since $K_Y + H$ is $\mathbb{Q}$-Cartier, (Y, H) is lc.

Step 4. In this step we keep the assumption $m > 0$ and prove that $\Gamma \cap H'$ is irreducible.

Let $\kappa := \mathbb{C}(E)$ be the residue field of R_E, and regard β_0 as an element of κ^*. Since $s \geq 2$, Lemma 12.1 with $\mathbb{K} = \kappa$ and $a = \beta_0$ shows that $\kappa[v_1, \dots, v_s]/(\beta_0 + \sum_j v_j^N)$ is geometrically integral over κ.

The subset $(v_0 = r = 0) \subset \tilde{U}_0$ is the hypersurface $(\beta_0 + \sum_j v_j^N = 0)$ in $E'' \times \mathbb{A}^s$, which is smooth of dimension $d + s - 2$ by **Step 3**. The fiber over any closed point of E'' is a hypersurface in $\mathbb{A}^s$, of dimension $s - 1$, so any irreducible component of $(v_0 = r = 0)$ dominates E'', as $\dim E'' = d - 1$. Since the generic fiber is integral, $(v_0 = r = 0) \subset \tilde{U}_0$ is integral, and $\Gamma \cap H' \cap U_0 = \varphi_0((v_0 = r = 0))$ is irreducible. On $\tilde{U}_k$, the preimage of $\Gamma \cap H'$ is $(v_0 = r_k = 0)$, which is smooth of dimension $d + s - 2$, while $\dim(v_0 = \tau = r_k = 0) = d + s - 3$ by **Step 3**. Thus no irreducible component of $(v_0 = r_k = 0)$ is contained in $(\tau = 0) = \varphi_k^{-1}(U_k \backslash U_0)$. Therefore any irreducible component of $\Gamma \cap H'$ intersects U_0, and $\Gamma \cap H'$ is irreducible.

Step 5. In this step we prove that o is an lc center of (Y, H).

Since x is an lc center of $(X, \frac{1}{N}\operatorname{div}(f))$, there exists a prime divisor E_1 over X with $\operatorname{center}_X E_1 = x$ and $a(E_1, X, \frac{1}{N}\operatorname{div}(f)) = 0$. Let $\alpha_1 := a(E_1, X, 0)$. Then $\alpha_1 = \frac{1}{N}\operatorname{ord}_{E_1}(f)$, and $\alpha_1 > 0$ as X is klt. Let v be the Gauss extension of ord_{E_1} with weights $(\alpha_1, \dots, \alpha_1)$ (Definition 2.8), a valuation of $\mathbb{C}(Y) = \mathbb{C}(X)(u_1, \dots, u_s)$. Let $X_1 \to X$ be a proper birational morphism from a normal variety X_1 such that E_1 is a prime divisor on X_1, and let t_1 be a local equation of E_1 at the generic point of E_1. Then v is the monomial valuation in $(t_1, u_1, \dots, u_s)$ with weights $(1, \alpha_1, \dots, \alpha_1)$ at the generic point of $E_1 \times \{0\} \subset X_1 \times \mathbb{A}^s$. Since $\alpha_1 \in \mathbb{Q}$, v is a positive rational multiple of ord_{F_1} for a prime divisor F_1 over Y. By Definition 2.6, since $a(E_1 \times \mathbb{A}^s, Y, 0) = \alpha_1$ and $a((u_j = 0), Y, 0) = 1$,

$$a(v, Y, 0) = \alpha_1 + s\alpha_1 = N\alpha_1,$$

and by Definition 2.8, $v(h) = \min\{\operatorname{ord}_{E_1}(f), N\alpha_1\} = N\alpha_1$. Thus $a(F_1, Y, H) = 0$. Since $v > 0$ on the maximal ideal of x and $v(u_j) = \alpha_1 > 0$, $\operatorname{center}_Y F_1 = o$. Since (Y, H) is lc by **Step 3**, o is an lc center of (Y, H).

Step 6. In this step we prove (1), (2), and (3).

Let F be a prime divisor over Y with $\operatorname{center}_Y F = o$ and $a(F,Y,H) = 0$. By Lemma 9.5, $\operatorname{ord}_F|_{\mathbb{C}(X)} = c \operatorname{ord}_E$ for a positive integer c and a prime divisor E over X with $\operatorname{center}_X E = x$. We apply **Steps 2–4** to E, with t the uniformizer of (2). By **Step 2**, $m > 0$ and $Z' \subset V_0$, where $Z' := \operatorname{center}_{Y''} F$. Since Z' dominates E'' and $\dim V_0 = \dim E''$, $Z' = V_0$. Let i and $\tilde{F}'$ be as in **Step 3** for $F' = F$. If $\delta > 0$, then the coefficient-one part of Δ_i near $(v_0 = 0)$ is $(r = 0)$ or $(r_k = 0)$, which has no stratum contained in $(v_0 = 0)$, contradicting (12.3). Thus $\delta = 0$, that is, $a(E, X, \frac{1}{N}\operatorname{div}(f)) = 0$ and $a(E,X,0) = \frac{m}{N}$.

We prove (2). By Lemma 2.5 applied to the prime divisor $(v_0 = 0)$ on $\tilde{U}_0$, which maps onto $\Gamma \cap U_0$, the log discrepancy of Γ with respect to (Y,H) is a positive rational multiple of $a((v_0 = 0), \tilde{U}_0, \Delta_0) = \delta = 0$. Thus Δ_Γ has coefficient one along Γ, and along H' since H is reduced. By **Step 4**, $\Gamma \cap H'$ is irreducible. Since $R' = R_E \otimes_{\mathcal{O}(X'')} \mathcal{O}(X_0'')$, the morphism $\operatorname{Spec} R'[v_1, \dots, v_s] \to Y_\Gamma$ is the base change of φ_0 along $\operatorname{Spec} R_E \to X''$. Thus (12.2) and **Step 3** imply that $h = v_0^{mp} r$ and that the preimages of Γ, H', and $\Gamma \cap H'$ are $(v_0 = 0)$, $(r = 0)$, and $(v_0 = r = 0)$. Since $R'/(v_0) = R_E/(t) = \kappa$, $(v_0 = r = 0) = \operatorname{Spec} \kappa[v_1, \dots, v_s]/(\beta_0 + \sum_j v_j^N)$ is integral by **Step 4**.

We prove (3). By (12.3) with $\delta = 0$, $\operatorname{center}_{\tilde{U}_i}(\tilde{F}')$ is $(v_0 = 0)$ or an irreducible component of $(v_0 = r = 0)$ or of $(v_0 = r_k = 0)$, and the image of $\operatorname{center}_{\tilde{U}_i}(\tilde{F}')$ in Y_Γ'' is Γ or $\Gamma \cap H'$. Both intersect U_0 by **Step 4**, so we may assume that $i = 0$, and we let $\tilde{F} := \tilde{F}'$ and $e_0 := e'$. Let ξ be the generic point of $(v_0 = r = 0) \subset \tilde{U}_0$. By **Step 3**, v_0 and r form a regular system of parameters of $\mathcal{O}_{\tilde{U}_0,\xi}$, and $\Delta_0 = (v_0 = 0) + (r = 0)$ near ξ. If $\operatorname{center}_{\tilde{U}_0}(\tilde{F}) = (v_0 = 0)$, then $\tilde{F} = (v_0 = 0)$, and $\operatorname{ord}_{\tilde{F}}$ is the monomial valuation in v_0 and r at ξ with weights $(1,0)$. Otherwise $\operatorname{center}_{\tilde{U}_0}(\tilde{F}) = \overline{\{\xi\}}$. By [Kol13, 2.10], $\tilde{F}$ is obtained by a sequence of blow-ups along regular centers over ξ, and by the blow-up formula in [Kol13, 2.10] applied inductively (see also [Kol13, Section 4.2, the paragraph after Definition 4.15]), the log discrepancy of $\tilde{F}$ with respect to $(\tilde{U}_0, (v_0 = 0) + (r = 0))$ vanishes only if every center in the sequence is a stratum of the coefficient-one part of the crepant pullback. Such blow-ups are toric with respect to v_0 and r, so $\operatorname{ord}_{\tilde{F}}$ is the monomial valuation in v_0 and r at ξ with weights $\lambda := \operatorname{ord}_{\tilde{F}}(v_0) > 0$ and $\mu := \operatorname{ord}_{\tilde{F}}(r) > 0$. In both cases, $\operatorname{center}_{\tilde{U}_0}(\tilde{F})$ dominates $(v_0 = 0) \subset X_0''$, so $\tilde{F}$ is a prime divisor over $\operatorname{Spec} R'[v_1, \dots, v_s]$.

Since $\operatorname{ord}_E(t) = 1$ and $t = v_0^p$, $ce_0 = \operatorname{ord}_{\tilde{F}}(t) = \lambda p$, so $c = \lambda p / e_0$. Since $(v_0 = r = 0)$ is integral and contains points with $v_j \neq 0$, v_j is a unit at ξ. Thus $\operatorname{ord}_{\tilde{F}}(u_j^N) = Nq\lambda = mp\lambda$ and $\operatorname{ord}_{\tilde{F}}(h) = mp\lambda + \mu$, so $\operatorname{ord}_F(u_j^N) = cm$ and $\operatorname{ord}_F(h) = cm + \mu/e_0$. Since $a(F,Y,H) = a(F,Y,0) - \operatorname{ord}_F(h) = 0$, $a(F,Y,0) = \operatorname{ord}_F(h)$. This proves (3). Finally,

$$a(F,Y,0) = cm + \frac{\mu}{e_0} \geq cm = Nca(E,X,0),$$

which proves (1). □

Remark 12.6. The condition $N \geq 3$ in Construction 12.4 is used only through $s = N - 1 \geq 2$, in **Steps 1, 3, and 4** of the proof of Definition-Proposition 12.5. In **Step 4** it is the assumption $s \geq 2$ of Lemma 12.1, whereas for $s = 1$ the polynomial $\beta_0 + v_1^N$ splits into N linear factors over an algebraic closure of κ.

The difficulty in Theorem 12.7 is a valuation whose value on h is larger than the value on f and on the powers u_j^N, so that the filtration on Y is not a Gauss extension in the coordinates $u_1, \dots, u_s$. Since $N \geq 3$, the irreducibility in Definition-Proposition 12.5(2) forces such a valuation to be invariant under G, and on Q the filtration becomes a Gauss extension after replacing z_s with h, so that finite generation and the klt property descend to the associated graded ring of ord_E.

Theorem 12.7. *Notation as in Construction 12.4. Let F be a Kollár component of $(Y \ni o, 0)$ with $a(F,Y,H) = 0$, and let c and E be as in Definition-Proposition 12.5. Then F is G-invariant, E is a Kollár component of $(X \ni x, 0)$, $a(E, X, \frac{1}{N}\operatorname{div}(f)) = 0$, and $Nca(E,X,0) \leq a(F,Y,0)$.*

Moreover, let w be the restriction of ord_F to $\mathbb{C}(Q)$, $m := \operatorname{ord}_E(f)$, and $b := \operatorname{ord}_F(h)$. Then $b \geq cm$. Regard the G-invariant function h as the function $f + \sum_{j=1}^{s} z_j$ on Q, so that $z_1, \dots, z_{s-1}, h$ are coordinates of $Q \cong X \times \mathbb{A}^s$ over X. With respect to the coordinates $z_1, \dots, z_{s-1}, h$,

$$\operatorname{gr}_w \mathcal{O}_{Q,(x,0)} \cong \big(\operatorname{gr}_{c\operatorname{ord}_E} \mathcal{O}_{X,x}\big)[Z_1, \dots, Z_{s-1}, T], \quad \deg Z_j = cm, \ \deg T = b.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix notation and prove that F is G-invariant.

Since F is a Kollár component of $(Y \ni o, 0)$, $\operatorname{center}_Y F = o$ (Definition 2.3). Let p, q, R_E, t, β, R', r, $\tilde{F}$, e_0, λ, and μ be as in Definition-Proposition 12.5(2)(3), let κ be the residue field of R_E, $\beta_0 \in \kappa^*$ the residue of β, and ξ the generic point of $(v_0 = r = 0) \subset \operatorname{Spec} R'[v_1, \dots, v_s]$. By Definition-Proposition 12.5(1), $a(E, X, \frac{1}{N}\operatorname{div}(f)) = 0$ and $Nca(E, X, 0) \leq a(F, Y, 0)$, and by Definition-Proposition 12.5(3), $b = cm + \mu/e_0 \geq cm$.

Let $\gamma \in G$ act by $u_j \to \zeta_j u_j$ with $\zeta_j \in \mu_N$, and let $\tilde{\gamma}$ be the automorphism of $R'[v_1, \dots, v_s]$ over R' with $\tilde{\gamma}(v_j) = \zeta_j v_j$. Since $u_j = v_0^q v_j$, the morphism from $\operatorname{Spec} R'[v_1, \dots, v_s]$ to Y is equivariant with respect to $\tilde{\gamma}$ and γ. Since $\zeta_j^N = 1$, we have $\tilde{\gamma}(v_0) = v_0$ and $\tilde{\gamma}(r) = r$, so $\tilde{\gamma}$ fixes ξ. Since $\operatorname{ord}_{\tilde{F}}$ is the monomial valuation in v_0 and r at ξ with weights (λ, μ), $\operatorname{ord}_{\tilde{F}} \circ \tilde{\gamma}$ is the monomial valuation in $\tilde{\gamma}(v_0) = v_0$ and $\tilde{\gamma}(r) = r$ at $\tilde{\gamma}^{-1}(\xi) = \xi$ with weights (λ, μ), that is, $\operatorname{ord}_{\tilde{F}} \circ \tilde{\gamma} = \operatorname{ord}_{\tilde{F}}$. Restricting to $\mathbb{C}(Y)$, $\operatorname{ord}_F \circ \gamma^* = \operatorname{ord}_F$. Since $\gamma \in G$ is arbitrary, F is G-invariant.

Step 2. In this step we prove that for any non-zero polynomial $P = \sum_{(\alpha,k)} a_{\alpha,k} z^\alpha h^k$ with $a_{\alpha,k} \in \mathbb{C}(X)$, where $\alpha = (\alpha_1, \dots, \alpha_{s-1}) \in \mathbb{N}^{s-1}$, $k \in \mathbb{N}$, and $z^\alpha := \prod_{j<s} z_j^{\alpha_j}$,

$$w(P) = \min_{(\alpha,k)} \big(c\operatorname{ord}_E(a_{\alpha,k}) + cm|\alpha| + bk\big), \tag{12.4}$$

where $|\alpha| := \sum_j \alpha_j$ and the minimum runs over the pairs with $a_{\alpha,k} \neq 0$.

Replacing P with $t^l P$ for an integer l changes both sides of (12.4) by cl, so we may assume that $a_{\alpha,k} \in R_E$. Write $a_{\alpha,k} = t^{n_{\alpha,k}} \epsilon_{\alpha,k}$ with $\epsilon_{\alpha,k}$ a unit of R_E, and let $\bar{\epsilon}_{\alpha,k} \in \kappa^*$ be the residue of $\epsilon_{\alpha,k}$. Since $t = v_0^p$, $z_j = u_j^N = v_0^{mp} v_j^N$, and $h = v_0^{mp} r$ by Definition-Proposition 12.5(2),

$$P = \sum_{(\alpha,k)} \epsilon_{\alpha,k} v_0^{pn_{\alpha,k} + mp(|\alpha|+k)} r^k \prod_{j<s} v_j^{N\alpha_j}.$$

First assume that $\mu > 0$. By Definition-Proposition 12.5(2),

$$(v_0 = r = 0) = \operatorname{Spec} \kappa[v_1, \dots, v_s] \big/ \big(\beta_0 + \textstyle\sum_j v_j^N\big),$$

so the residue field $\kappa(\xi)$ is algebraic over $\kappa(\bar{v}_1, \dots, \bar{v}_{s-1})$ as $\bar{v}_s^N = -\beta_0 - \sum_{j<s} \bar{v}_j^N$, and has transcendence degree $s-1$ over κ. Thus $\bar{v}_1^N, \dots, \bar{v}_{s-1}^N$ are algebraically independent over κ. For a pair (i,k) of integers, let $\Phi_{i,k}$ be the sum of $\epsilon_{\alpha,k} \prod_{j<s} v_j^{N\alpha_j}$ over the α with $a_{\alpha,k} \neq 0$ and $pn_{\alpha,k} + mp(|\alpha| + k) = i$. Distinct α correspond to distinct monomials in $\bar{v}_1^N, \dots, \bar{v}_{s-1}^N$, so the residue of $\Phi_{i,k}$ in $\kappa(\xi)$ is non-zero if the sum defining $\Phi_{i,k}$ is non-empty, and then $\Phi_{i,k}$ is a unit at ξ. Since $P = \sum_{(i,k)} \Phi_{i,k} v_0^i r^k$, where the sum runs over the pairs (i,k) for which the sum defining $\Phi_{i,k}$ is non-empty, and $\operatorname{ord}_{\tilde{F}}$ is the monomial valuation in v_0 and r at ξ,

$$\operatorname{ord}_{\tilde{F}}(P) = \min_{(\alpha,k)} \big(\lambda pn_{\alpha,k} + \lambda mp(|\alpha| + k) + \mu k\big).$$

Dividing by e_0, and using $c = \lambda p/e_0$ and $b = cm + \mu/e_0$, we obtain (12.4).

Next assume that $\mu = 0$. Then $\operatorname{ord}_{\tilde{F}} = \lambda \operatorname{ord}_{(v_0=0)}$, and the residue field of $\operatorname{ord}_{(v_0=0)}$ is $\kappa(\bar{v}_1, \dots, \bar{v}_s)$, which has transcendence degree s over κ. Since $\bar{v}_s^N = \bar{r} - \beta_0 - \sum_{j<s} \bar{v}_j^N$ with $\bar{r} = \beta_0 + \sum_j \bar{v}_j^N$, $\kappa(\bar{v}_1, \dots, \bar{v}_s)$ is algebraic over $\kappa(\bar{v}_1^N, \dots, \bar{v}_{s-1}^N, \bar{r})$, so $\bar{v}_1^N, \dots, \bar{v}_{s-1}^N, \bar{r}$ are algebraically independent over κ. For an integer i, the residue of the sum of $\epsilon_{\alpha,k} r^k \prod_{j<s} v_j^{N\alpha_j}$ over the (α,k)

with $a_{\alpha,k} \neq 0$ and $pn_{\alpha,k} + mp(|\alpha| + k) = i$ is $\sum \bar{\epsilon}_{\alpha,k} \bar{r}^k \prod_{j<s} \bar{v}_j^{N\alpha_j}$, which is non-zero if the sum is non-empty. Thus

$$\operatorname{ord}_{\tilde{F}}(P) = \min_{(\alpha,k)} \lambda\big(pn_{\alpha,k} + mp(|\alpha| + k)\big).$$

Dividing by e_0, and using $c = \lambda p/e_0$ and $b = cm$, we obtain (12.4).

Step 3. In this step we prove the isomorphism of graded rings.

Let $\mathcal{P} := \mathcal{O}_{X,x}[z_1, \dots, z_{s-1}, h]$ and $\mathfrak{n} := \mathfrak{m}_x\mathcal{P} + (z_1, \dots, z_{s-1}, h)$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. Since $z_s = h - f - \sum_{j<s} z_j$ and $f(x) = 0$, $\mathcal{O}_{Q,(x,0)} = \mathcal{P}_{\mathfrak{n}}$. By (12.4), for any real number $\rho \geq 0$,

$$\mathfrak{a}_\rho(w) \cap \mathcal{P} = \bigoplus_{(\alpha,k)} \mathfrak{a}_{\rho - cm|\alpha| - bk}(c \operatorname{ord}_E)\mathcal{O}_{X,x} \cdot z^\alpha h^k,$$

where $\mathfrak{a}_\sigma(c \operatorname{ord}_E)\mathcal{O}_{X,x} := \mathcal{O}_{X,x}$ for $\sigma \leq 0$. The decomposition of $\mathfrak{a}_\rho(w) \cap \mathcal{P}$ also holds with $\mathfrak{a}_\rho(w)$ and $\mathfrak{a}_{\rho - cm|\alpha| - bk}(c \operatorname{ord}_E)$ replaced by $\mathfrak{a}_{>\rho}(w)$ and $\mathfrak{a}_{>\rho - cm|\alpha| - bk}(c \operatorname{ord}_E)$, where $\mathfrak{a}_{>\sigma}(c \operatorname{ord}_E)\mathcal{O}_{X,x} := \mathcal{O}_{X,x}$ for $\sigma < 0$. Thus there exists an isomorphism of graded rings

$$\operatorname{gr}_w \mathcal{P} \cong \big(\operatorname{gr}_{c \operatorname{ord}_E} \mathcal{O}_{X,x}\big)[Z_1, \dots, Z_{s-1}, T]$$

sending $\operatorname{in}_w(z_j)$ to Z_j and $\operatorname{in}_w(h)$ to T, with $\deg Z_j = w(z_j) = cm$ and $\deg T = w(h) = b$. Since $\operatorname{center}_X E = x$, the degree zero part of $\operatorname{gr}_{c \operatorname{ord}_E} \mathcal{O}_{X,x}$ is $\mathcal{O}_{X,x}/\mathfrak{m}_x = \mathbb{C}$. Let $D \in \mathcal{P} \backslash \mathfrak{n}$, and let $D_0 \in \mathcal{O}_{X,x} \backslash \mathfrak{m}_x$ be the constant term of D as a polynomial in $z_1, \dots, z_{s-1}, h$. By (12.4), $w(D) = 0$ and $\operatorname{in}_w(D) = D_0(x) \in \mathbb{C}^*$. Thus for $P \in \mathcal{P}$, $\operatorname{in}_w(P/D) = \operatorname{in}_w(D_0(x)^{-1}P)$, so the natural injection $\operatorname{gr}_w \mathcal{P} \to \operatorname{gr}_w \mathcal{O}_{Q,(x,0)}$ is surjective. This proves the isomorphism in the theorem.

Step 4. In this step we prove that the image of F on Q is a Kollár component of $(Q \ni (x,0), 0)$.

Let $\varrho : Y \to Q$ be the quotient morphism, and let $\Delta_Q := (1 - \frac{1}{N}) \sum_{j=1}^s \operatorname{div}(z_j)$. Since ϱ is ramified with index N along $(u_j = 0)$ and étale over the complement of $\bigcup_j (z_j = 0)$, $K_Y = \varrho^*(K_Q + \Delta_Q)$ by the Hurwitz formula. Since $Q \cong X \times \mathbb{A}^s$, Q is a $\mathbb{Q}$-Gorenstein klt variety by Definition-Proposition 12.5, so $K_Q + \Delta_Q$ is $\mathbb{Q}$-Cartier. Replacing Y with its G-invariant open neighborhood $Y \backslash \varrho^{-1}(\varrho(Y \backslash U))$ of o (open since ϱ is finite, contained in U, and containing o since $\varrho^{-1}(\varrho(o)) = \{o\} \subset U$), where U is an open neighborhood of o over which F is primitive (Definition 2.3), and Q with $Q \backslash \varrho(Y \backslash U)$, let $f_Y : Y' \to Y$ be the prime extraction of F (Definition 2.3), let J be the prime divisor over Q with $w = e \operatorname{ord}_J$ for a positive integer e, and let $f_Q : Y'_Q \to Q$ be the induced morphism. We apply Lemma 9.2 with $k = \mathbb{C}$, $V = Y$, the quotient morphism $\varrho : Y \to Q$, the closed point o, $\Delta_W = \Delta_Q$, $\Delta_V = 0$, the prime divisor F, and $L = J$. Its hypotheses hold: Y is normal of dimension $d + s \geq 2$, $\Delta_Q \geq 0$, o is fixed by G and $\varrho^{-1}(\varrho(o)) = \{o\}$, $K_Q + \Delta_Q$ is $\mathbb{Q}$-Cartier and $K_Y = \varrho^*(K_Q + \Delta_Q)$, and F is G-invariant by **Step 1**, primitive over Y, and satisfies $\operatorname{center}_Y F = o$. Thus f_Q is the prime extraction of J, and $(Y'_Q, (f_Q)_*^{-1}\Delta_Q + J)$ is plt near J since (Y', F) is plt. Since $-J$ is $\mathbb{Q}$-Cartier and Δ_Q is $\mathbb{Q}$-Cartier, $(f_Q)_*^{-1}\Delta_Q = f_Q^*\Delta_Q - (\operatorname{mult}_J \Delta_Q)J$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor and $(f_Q)_*^{-1}\Delta_Q \geq 0$, so (Y'_Q, J) is plt near J [KM98, Corollary 2.35(1)]. Since the exceptional locus of f_Q is J and Q is klt, (Y'_Q, J) is plt. Thus J is a Kollár component of $(Q \ni (x,0), 0)$.

Step 5. In this step we prove that E is a Kollár component of $(X \ni x, 0)$.

By Theorem 2.10 applied to J and the klt pair $(Q \ni (x,0), 0)$, $\operatorname{gr}_{\operatorname{ord}_J} \mathcal{O}_{Q,(x,0)}$ is finitely generated and the marked graded pair $(\operatorname{Spec} \operatorname{gr}_{\operatorname{ord}_J} \mathcal{O}_{Q,(x,0)}, 0)$ is klt. Since $w = e \operatorname{ord}_J$, Lemma 12.2 implies that $\operatorname{gr}_w \mathcal{O}_{Q,(x,0)}$ is finitely generated and that the marked graded pair $(\operatorname{Spec} \operatorname{gr}_w \mathcal{O}_{Q,(x,0)}, 0)$ is klt. Let $A := \operatorname{gr}_{c \operatorname{ord}_E} \mathcal{O}_{X,x}$. By **Step 3**, $A[Z_1, \dots, Z_{s-1}, T]$ is a finitely generated normal domain, $K_{\operatorname{Spec} A[Z_1, \dots, Z_{s-1}, T]}$ is $\mathbb{R}$-Cartier, hence $\mathbb{Q}$-Cartier as it has integer coefficients, and $\operatorname{Spec} A[Z_1, \dots, Z_{s-1}, T]$ is klt.

By Lemma 12.3, A is finitely generated and normal, and $\operatorname{Spec} A$ is $\mathbb{Q}$-Gorenstein and klt.

By Lemma 12.2, $\operatorname{gr}_{\operatorname{ord}_E} \mathcal{O}_{X,x}$ is finitely generated, and the marked graded pair of ord_E with respect to $(X \ni x, 0)$ is $(\operatorname{Spec} A, 0)$, which is klt. Since $\operatorname{center}_X E = x$, Theorem 2.10 implies that E is a Kollár component of $(X \ni x, 0)$. □

12.2. **The marked suspension.** In this subsection, we construct the marked suspension of a klt pair whose boundary is a rational multiple of a principal divisor, and we prove that Kollár components of the marked suspension over the closed point return to Kollár components of the pair (Theorem 12.12).

Construction 12.8. Let $(X \ni x, B)$ be a klt pair over $\mathbb{C}$ of dimension $d \geq 2$ such that x is a closed point, X is $\mathbb{Q}$-Gorenstein, and B is a $\mathbb{Q}$-divisor. Let $N \geq 3$ be an integer and $g \in \mathcal{O}_{X,x}$ an element with $g(x) = 0$ such that $NB = \operatorname{div}(g)$ near x. Shrinking X around x, we assume that X is affine, $g \in \mathcal{O}(X)$, and $NB = \operatorname{div}(g)$ on X. Let $s := N - 1$,

$$Y := \Big(g + \sum_{j=1}^{s} y_j^N = 0\Big) \subset X \times \mathbb{A}^s,$$

$o := (x, 0) \in Y$, and $p : Y \to X$ the projection. Let $G := \mu_N^s$ act on Y by multiplying the coordinates $y_1, \dots, y_s$ separately, let $q : Y \to W := Y/G \cong X \times \mathbb{A}^{s-1}$ be the quotient morphism, where W has coordinates $T_j := y_j^N$ for $1 \leq j \leq s-1$, and let

$$\Delta_W := \Big(1 - \frac{1}{N}\Big)\Big(\sum_{j=1}^{s-1} \operatorname{div}(T_j) + \operatorname{div}\Big(g + \sum_{j=1}^{s-1} T_j\Big)\Big).$$

The isomorphism $Y/G \cong X \times \mathbb{A}^{s-1}$ holds because $\mathcal{O}(X)[y_1, \dots, y_s]^G = \mathcal{O}(X)[y_1^N, \dots, y_s^N]$, the polynomial $g + \sum_j y_j^N$ is G-invariant, and taking G-invariants is exact since $|G|$ is invertible in $\mathbb{C}$. Thus

$$\mathcal{O}(Y)^G = \mathcal{O}(X)[T_1, \dots, T_s] \Big/ \Big(g + \sum_{j=1}^{s} T_j\Big) \cong \mathcal{O}(X)[T_1, \dots, T_{s-1}], \qquad T_j := y_j^N.$$

We call Y the *marked suspension* of $(X \ni x, B)$ with respect to g and N.

Remark 12.9. The condition $N \geq 3$ in Construction 12.8 is used only through $s = N - 1 \geq 2$: it is the assumption $s \geq 2$ of Lemma 12.1, which is used to prove that Y is normal (together with $\operatorname{codim}_Y(y_1 = \cdots = y_s = 0) = s \geq 2$), that a prime divisor F over Y with $\operatorname{center}_Y F = o$ and $a(F, Y, p^*D) = 0$, for D as in Definition-Proposition 12.11, is determined by the restriction of ord_F to $\mathbb{C}(W)$ and hence is G-invariant (Definition-Proposition 12.11), and that the ring $R_0[z_1, \dots, z_s]/(g_0 + \sum_j z_j^N)$ in Theorem 12.12 is an integral domain.

The key point is a toric computation on the quotient: the branch boundary consists of hyperplanes with coefficient $1 - \frac{1}{N}$, and the coefficient of the smallest weight cancels because the number of Fermat variables is one less than N (Lemma 12.10). The cancellation forces the restriction to $\mathbb{C}(W)$ of an lc place over o of (Y, p^*D), with D as in Definition-Proposition 12.11, to be a Gauss extension, which determines the lc place (Definition-Proposition 12.11).

Let X be a smooth variety, $x \in X$ a point, and $t_1, \dots, t_r \in \mathcal{O}_{X,x}$ part of a regular system of parameters of $\mathcal{O}_{X,x}$. Near x, the functions $t_1, \dots, t_r$ define a morphism $\varphi : X \to \mathbb{A}^r$, which is smooth near x since $dt_1, \dots, dt_r$ are linearly independent in $\Omega_X \otimes \kappa(x)$, and étale near x if $r = \dim X$. A log resolution of a pair on X near x is *toric with respect to* $t_1, \dots, t_r$ if it is the base change along φ of a toric log resolution $X_\Sigma \to \mathbb{A}^r$.

Lemma 12.10. *Let $N \geq 3$ and $n \geq 1$ be integers, $s := N - 1$, X a smooth variety over $\mathbb{C}$, $x \in X$ a point, and $t_1, \dots, t_n, T_1, \dots, T_{s-1} \in \mathcal{O}_{X,x}$ a regular system of parameters of $\mathcal{O}_{X,x}$. Let $m_1, \dots, m_n$ be non-negative integers and $\delta_1, \dots, \delta_n$ real numbers such that $\delta_i \geq \frac{m_i}{N}$ for any i, and let*

$$\Delta := \sum_{i=1}^{n} (1 - \delta_i) \operatorname{div}(t_i) + \Big(1 - \frac{1}{N}\Big)\Big(\sum_{j=1}^{s-1} \operatorname{div}(T_j) + \operatorname{div}\Big(\prod_{i=1}^{n} t_i^{m_i} + \sum_{j=1}^{s-1} T_j\Big)\Big)$$

near x. For real numbers $\alpha_1,\dots,\alpha_n,\beta_1,\dots,\beta_{s-1}\ge 0$, not all zero, let w be the monomial valuation at x with $w(t_i)=\alpha_i$ and $w(T_j)=\beta_j$, $M:=\sum_i m_i\alpha_i$, and $\gamma:=\min\{M,\beta_1,\dots,\beta_{s-1}\}$. Then

$$a(w,X,\Delta)=\sum_{i=1}^{n}\Big(\delta_i-\frac{m_i}{N}\Big)\alpha_i+\frac{M-\gamma}{N}+\sum_{j=1}^{s-1}\frac{\beta_j-\gamma}{N}. \tag{12.5}$$

Moreover, near x there exists a log resolution of (X,Δ) which is toric with respect to $t_1,\dots,t_n,T_1,\dots,T_{s-1}$, (X,Δ) is sub-lc near x, and (X,Δ) is sub-klt near x if $\delta_i>\frac{m_i}{N}$ for any i.

Proof. Let $h:=\prod_i t_i^{m_i}+\sum_j T_j$ and

$$\Delta':=\sum_{i=1}^{n}(1-\delta_i)\operatorname{div}(t_i)+\Big(1-\frac{1}{N}\Big)\sum_{j=1}^{s-1}\operatorname{div}(T_j),$$

so that $\Delta=\Delta'+(1-\frac{1}{N})\operatorname{div}(h)$. Near x, the pair $(X,\sum_i\operatorname{div}(t_i)+\sum_j\operatorname{div}(T_j))$ is log smooth, and w is a monomial valuation on it. By Definition 2.6 and [JM12, Proposition 5.1], $a(w,X,0)=\sum_i\alpha_i+\sum_j\beta_j$. Since

$$w(\Delta')=\sum_{i=1}^{n}(1-\delta_i)\alpha_i+\Big(1-\frac{1}{N}\Big)\sum_{j=1}^{s-1}\beta_j,$$

we obtain

$$a(w,X,\Delta')=a(w,X,0)-w(\Delta')=\sum_{i=1}^{n}\delta_i\alpha_i+\sum_{j=1}^{s-1}\frac{\beta_j}{N}.$$

Here Δ' need not be effective, since $1-\delta_i$ may be negative. The value of a monomial valuation on a polynomial in the coordinates is the minimum of the weights of the monomials of the polynomial, so $w(h)=\gamma$. Since $a(w,X,\Delta)=a(w,X,\Delta')-(1-\frac{1}{N})w(h)$, $\sum_i m_i\alpha_i=M$, and the number of the T_j is $s-1=N-2$, we obtain

$$\begin{aligned}a(w,X,\Delta)&=\sum_{i=1}^{n}\delta_i\alpha_i+\sum_{j=1}^{s-1}\frac{\beta_j}{N}-\frac{N-1}{N}\gamma\\&=\sum_{i=1}^{n}\Big(\delta_i-\frac{m_i}{N}\Big)\alpha_i+\frac{M-\gamma}{N}+\sum_{j=1}^{s-1}\frac{\beta_j-\gamma}{N},\end{aligned}$$

where the coefficient of γ cancels because $1+(s-1)=N-1$. This proves (12.5).

Near x, the functions $t_1,\dots,t_n,T_1,\dots,T_{s-1}$ define a morphism $\varphi:X\to\mathbb{A}^{n+s-1}$ which is smooth near x, with $\varphi(x)=0$ and $\Delta=\varphi^*\Delta_{\mathbb{A}}$, where $\Delta_{\mathbb{A}}$ is given by the formula for Δ with $t_1,\dots,t_n,T_1,\dots,T_{s-1}$ replaced by the coordinates of $\mathbb{A}^{n+s-1}$. Since φ is smooth, the base change along φ of a log resolution $X_\Sigma\to\mathbb{A}^{n+s-1}$ of $(\mathbb{A}^{n+s-1},\Delta_{\mathbb{A}})$ near 0 is a log resolution of (X,Δ) near x, and $K_{X\times_{\mathbb{A}^{n+s-1}}X_\Sigma/X}$ is the pullback of $K_{X_\Sigma/\mathbb{A}^{n+s-1}}$, so the coefficients of the boundary are pulled back. The base change of a toric log resolution is toric with respect to $t_1,\dots,t_n,T_1,\dots,T_{s-1}$. Thus a toric log resolution and the sub-lc and sub-klt properties of $(\mathbb{A}^{n+s-1},\Delta_{\mathbb{A}})$ near 0 imply those of (X,Δ) near x, and it suffices to treat $X=\mathbb{A}^{n+s-1}$ near 0.

For $u\in\mathbb{R}^{n+s-1}_{\ge 0}$, let $\operatorname{in}_u(h)$ be the sum of the monomials of h of minimal u-weight. If $\operatorname{in}_u(h)$ contains some T_j, then $\partial\operatorname{in}_u(h)/\partial T_j=1$. Otherwise $\operatorname{in}_u(h)=\prod_i t_i^{m_i}$, which has no zero on the torus. Thus $(\operatorname{in}_u(h)=0)$ is smooth in the torus for any u, that is, h is non-degenerate with respect to the Newton polyhedron of h. Let Σ_0 be the fan of cones in $\mathbb{R}^{n+s-1}_{\ge 0}$ on which the function $u\to\min\{\sum_i m_iu_i,u_{n+1},\dots,u_{n+s-1}\}$ is linear. Since Σ_0 is the normal fan of the Newton polyhedron of h, $X_{\Sigma_0}\to\mathbb{A}^{n+s-1}$ is projective [CLS11, Theorem 7.1.10(b)]. Let Σ be a smooth refinement of Σ_0 such that $X_\Sigma\to X_{\Sigma_0}$ is projective, as in [CLS11, Theorem 11.1.9], and let X_Σ be the corresponding toric projective resolution, with structure morphism $\pi:X_\Sigma\to\mathbb{A}^{n+s-1}$. On

the chart of a cone $\sigma \in \Sigma$, π^*h is a monomial times a function h_σ which is not divisible by any coordinate, and for any face τ of σ the restriction of h_σ to the orbit of τ is, up to a monomial unit, $\mathrm{in}_u(h)$ for u in the relative interior of τ. Hence $(h_\sigma = 0)$ is smooth and transversal to any torus orbit, so π is a log resolution of $(\mathbb{A}^{n+s-1}, \Delta_{\mathbb{A}})$.

Write $K_{X_\Sigma} + \Gamma = \pi^*(K_{\mathbb{A}^{n+s-1}} + \Delta_{\mathbb{A}})$. The coefficient of the toric divisor with primitive generator (α, β) in Γ is one minus the right hand side of (12.5), and the coefficient of the strict transform of $(h = 0)$ is $1 - \frac{1}{N}$. Since $\delta_i \geq \frac{m_i}{N}$, $M \geq \gamma$, and $\beta_j \geq \gamma$,

$$\sum_{i=1}^{n} \Big(\delta_i - \frac{m_i}{N}\Big)\alpha_i \geq 0, \qquad \frac{M-\gamma}{N} \geq 0, \qquad \frac{\beta_j - \gamma}{N} \geq 0,$$

so all coefficients of Γ are at most one, and $(\mathbb{A}^{n+s-1}, \Delta_{\mathbb{A}})$ is sub-lc by [Kol13, Corollary 2.13(1)], applied to the log resolution π. Here sub-lc and sub-klt in the sense of Definition 2.1 are lc and klt in the sense of [Kol13, Definition 2.8], which allows negative coefficients, and [Kol13, Corollary 2.13] holds for the $\mathbb{R}$-divisor $\Delta_{\mathbb{A}}$ by [Kol13, 2.20], since $K_{\mathbb{A}^{n+s-1}} + \Delta_{\mathbb{A}}$ is a linear combination of Cartier divisors with real coefficients. Assume that $\delta_i > \frac{m_i}{N}$ for any i. If $\alpha_i > 0$ for some i, then the right hand side of (12.5) is at least $(\delta_i - \frac{m_i}{N})\alpha_i > 0$. If $\alpha = 0$, then $M = \gamma = 0$, and the right hand side of (12.5) is $\sum_j \frac{\beta_j}{N} > 0$. Thus all coefficients of Γ are less than one, and $(\mathbb{A}^{n+s-1}, \Delta_{\mathbb{A}})$ is sub-klt by [Kol13, Corollary 2.13(2)]. □

Definition-Proposition 12.11. Notation as in Construction 12.8. Then Y is normal, $\mathbb{Q}$-Gorenstein, and klt near o, and $K_Y = q^*(K_W + \Delta_W)$. Let $D \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $(X, B + D)$ is lc and x is an lc center of $(X, B + D)$. Then (Y, p^*D) is lc near o, and o is an lc center of (Y, p^*D). Moreover, for any prime divisor F over Y with $\mathrm{center}_Y F = o$ and $a(F, Y, p^*D) = 0$, F is G-invariant, and the restriction of ord_F to $\mathbb{C}(X)$ is $c\,\mathrm{ord}_E$ for a positive integer c and a prime divisor E over X with $\mathrm{center}_X E = x$ such that $a(E, X, B + D) = 0$ and $a(F, Y, 0) = ca(E, X, B)$. Furthermore, let $m := \mathrm{ord}_E(g)$. Then $\mathrm{ord}_F(y_j) = \frac{cm}{N}$ for any j, the restriction of ord_F to $\mathbb{C}(W) = \mathbb{C}(X)(T_1, \dots, T_{s-1})$ is the Gauss extension of $c\,\mathrm{ord}_E$ with weights $(cm, \dots, cm)$ (Definition 2.8), and ord_F is the only valuation of $\mathbb{C}(Y)$ whose restriction to $\mathbb{C}(W)$ equals $\mathrm{ord}_F|_{\mathbb{C}(W)}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that Y is normal. Let $h := g + \sum_{j=1}^{s} y_j^N$. Since h is monic of degree N in y_s, $\mathcal{O}(Y) = \mathcal{O}(X)[y_1, \dots, y_s]/(h)$ is a free $\mathcal{O}(X)[y_1, \dots, y_{s-1}]$-module of rank N. Thus p is flat with fibers of pure dimension $s - 1$, and $\mathcal{O}(Y)$ is a subring of $\mathcal{O}(Y) \otimes_{\mathcal{O}(X)} \mathbb{C}(X) = \mathbb{C}(X)[y_1, \dots, y_s]/(h)$, which is an integral domain by Lemma 12.1 with $\mathbb{K} = \mathbb{C}(X)$ and $a = g$. Hence Y is integral. Since X is klt, X is Cohen–Macaulay [KM98, Theorem 5.22 with Theorem 5.10], so $\mathcal{O}(X)[y_1, \dots, y_s]$ and $\mathcal{O}(Y) = \mathcal{O}(X)[y_1, \dots, y_s]/(h)$ are Cohen–Macaulay, as $h \neq 0$, and Y satisfies S_2. At a point of Y over the smooth locus of X with $y_j \neq 0$ for some j, $\partial h/\partial y_j = N y_j^{N-1} \neq 0$, so Y is smooth there. Since p is flat with equidimensional fibers and X is normal, $\mathrm{codim}_Y p^{-1}(\mathrm{Sing}\, X) \geq 2$. The locus $Y \cap (y_1 = \cdots = y_s = 0) = (g = 0) \times \{0\}$ has codimension $(d + s - 1) - (d - 1) = s \geq 2$ in Y. Hence Y is regular in codimension one, and Y is normal by [Har77, Theorem II.8.22A].

Step 2. In this step we prove that $K_Y = q^*(K_W + \Delta_W)$ and that Y is $\mathbb{Q}$-Gorenstein. Let $T_s := -g - \sum_{j=1}^{s-1} T_j \in \mathcal{O}(W)$. Then

$$\mathcal{O}(Y) = \mathcal{O}(W)[y_1, \dots, y_s]/(y_1^N - T_1, \dots, y_s^N - T_s).$$

The divisors $(T_j = 0)$, $1 \leq j \leq s$, on W are prime and pairwise distinct: $(T_s = 0) = (T_{s-1} = -g - \sum_{j=1}^{s-2} T_j)$ is the graph of a function of the coordinates of X and $T_1, \dots, T_{s-2}$, and T_s is not divisible by T_j for $j \leq s - 1$. Thus q is étale over the complement of $\bigcup_{j=1}^{s}(T_j = 0)$, and the ramification index of q along $(T_j = 0)$ is N, since T_j is a local equation of $(T_j = 0)$

and T_k is a unit at the generic point of $(T_j = 0)$ for $k \neq j$. By the Hurwitz formula and $\operatorname{div}(T_s) = \operatorname{div}(g + \sum_{j=1}^{s-1} T_j)$,

$$K_Y = q^*\Big(K_W + \Big(1 - \frac{1}{N}\Big)\sum_{j=1}^{s} \operatorname{div}(T_j)\Big) = q^*(K_W + \Delta_W). \tag{12.6}$$

Since $W = X \times \mathbb{A}^{s-1}$ is $\mathbb{Q}$-Gorenstein and Δ_W is a rational combination of principal divisors, $K_W + \Delta_W$ is $\mathbb{Q}$-Cartier. Hence K_Y is $\mathbb{Q}$-Cartier by [KM98, Proposition 5.20(1)].

Step 3. In this step we prove that (Y, p^*D) is lc near o and that Y is klt near o. The points of Y over $(x, 0)$ satisfy $y_j^N = T_j = 0$ for $j \leq s-1$ and $y_s^N = -g(x) = 0$, so $q^{-1}((x,0)) = \{o\}$. Let $f : X' \to X$ be a log resolution of $(X, \operatorname{div}(g) + D)$, and write $K_{X'} + \Gamma_D = f^*(K_X + D)$. Let $W' := X' \times \mathbb{A}^{s-1}$ and $f_W := f \times \mathrm{id}_{\mathbb{A}^{s-1}} : W' \to W$. Then $K_{W'} + \Gamma_W = f_W^*(K_W + \Delta_W + \operatorname{pr}_X^* D)$ for

$$\Gamma_W := \operatorname{pr}_{X'}^* \Gamma_D + \Big(1 - \frac{1}{N}\Big)\Big(\sum_{j=1}^{s-1} \operatorname{div}(T_j) + \operatorname{div}\Big(f^*g + \sum_{j=1}^{s-1} T_j\Big)\Big).$$

Let $z \in f^{-1}(x)$ be a closed point. Choose a regular system of parameters $t_1, \dots, t_d$ of $\mathcal{O}_{X',z}$ such that, near z, $f^*g = u\prod_i t_i^{m_i}$ with u a unit and $\Gamma_D = \sum_i (1 - \delta_i)\operatorname{div}(t_i)$. Since $f^*B = \frac{1}{N}\operatorname{div}(f^*g)$,

$$\delta_i - \frac{m_i}{N} = a\big((t_i = 0), X, B + D\big) \geq 0.$$

Near $(z, 0)$, the functions $t_1, \dots, t_d, T_1/u, \dots, T_{s-1}/u$ form a regular system of parameters, $\operatorname{div}(T_j/u) = \operatorname{div}(T_j)$, and $\operatorname{div}(f^*g + \sum_j T_j) = \operatorname{div}(\prod_i t_i^{m_i} + \sum_j T_j/u)$. Thus (W', Γ_W) is sub-lc near $(z, 0)$ by Lemma 12.10. Since $f_W^{-1}((x,0)) = f^{-1}(x) \times \{0\}$ and f_W is proper, $(W, \Delta_W + \operatorname{pr}_X^* D)$ is lc near $(x, 0)$ by [KM98, Lemma 2.30]. Since $p^*D = q^* \operatorname{pr}_X^* D$, (12.6) implies $K_Y + p^*D = q^*(K_W + \Delta_W + \operatorname{pr}_X^* D)$, and (Y, p^*D) is lc near o by Lemma 2.5. Replacing D with 0, we have $\delta_i - \frac{m_i}{N} = a((t_i = 0), X, B) > 0$ since (X, B) is klt. Thus (W', Γ_W) is sub-klt near $(z, 0)$ by Lemma 12.10, (W, Δ_W) is klt near $(x, 0)$ by [KM98, Lemma 2.30], and Y is klt near o by (12.6) and Lemma 2.5.

Step 4. In this step we prove that o is an lc center of (Y, p^*D). Since x is an lc center of $(X, B + D)$, there exists a prime divisor E' over X with $\operatorname{center}_X E' = x$ and $a(E', X, B + D) = 0$. Let $m_{E'} := \operatorname{ord}_{E'}(g)$, which is positive since $g(x) = 0$. Let $f : X' \to X$ be a projective birational morphism from a normal variety X' on which E' is a divisor, write $K_{X'} + \Gamma_D = f^*(K_X + D)$, and define W', f_W, and Γ_W as in **Step 3**. Let $z \in E'$ be a general closed point. Then X' is smooth near z, and there exist a regular system of parameters $t_1, \dots, t_d$ of $\mathcal{O}_{X',z}$ and a unit u near z such that $E' = (t_1 = 0)$, $f^*g = ut_1^{m_{E'}}$, and $\Gamma_D = (1 - \delta_1)\operatorname{div}(t_1)$ near z, where $\delta_1 = a(E', X, D)$. Thus $\delta_1 - \frac{m_{E'}}{N} = a(E', X, B + D) = 0$. Let w' be the monomial valuation at $(z, 0)$ in the coordinates $t_1, \dots, t_d, T_1/u, \dots, T_{s-1}/u$ with weights $(1, 0, \dots, 0, m_{E'}, \dots, m_{E'})$. Then $M = \gamma = m_{E'}$ in Lemma 12.10, and (12.5) together with [KM98, Lemma 2.30] implies

$$a(w', W, \Delta_W + \operatorname{pr}_X^* D) = a(w', W', \Gamma_W) = \Big(\delta_1 - \frac{m_{E'}}{N}\Big) + \frac{m_{E'} - m_{E'}}{N} + \sum_{j=1}^{s-1} \frac{m_{E'} - m_{E'}}{N} = 0.$$

Since the weights are integers and the first weight is one, $w' = \operatorname{ord}_{L'}$ for the toric divisor L' over W' corresponding to the primitive vector $(1, 0, \dots, 0, m_{E'}, \dots, m_{E'})$. Moreover, $\operatorname{center}_W L' = (x, 0)$, since $w'|_{\mathbb{C}(X)} = \operatorname{ord}_{E'}$ and $w'(T_j) = m_{E'} > 0$. Let $W'' \to W$ be a projective birational morphism such that L' is a divisor on W''. Let Y'' be the normalization of W'' in $\mathbb{C}(Y)$, and F' a prime divisor on Y'' over L'. Then the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(W)$ is a positive integral multiple of $\operatorname{ord}_{L'}$, so $a(F', Y, p^*D) = 0$ by Lemma 2.5, and $\operatorname{center}_Y F' \in q^{-1}((x,0)) = \{o\}$. Thus o is an lc center of (Y, p^*D).

Step 5. In this step we define c and E. Let F be a prime divisor over Y with $\operatorname{center}_Y F = o$ and $a(F, Y, p^*D) = 0$. By (12.6) and Lemma 2.5 applied to q, the restriction of ord_F to $\mathbb{C}(W)$ is $e \operatorname{ord}_L$ for a positive integer e and a prime divisor L over W with $\operatorname{center}_W L = q(o) = (x, 0)$, and

$$0 = a(F, Y, p^*D) = ea(L, W, \Delta_W + \operatorname{pr}_X^* D), \qquad a(F, Y, 0) = ea(L, W, \Delta_W). \tag{12.7}$$

Since $W = X \times \mathbb{A}^{s-1}$, $\dim X = d \geq 1$, and $s - 1 = N - 2 \geq 1$, Lemma 9.5 implies that the restriction of ord_L to $\mathbb{C}(X)$ is $c' \operatorname{ord}_E$ for a positive integer c' and a prime divisor E over X with $\operatorname{center}_X E = x$. Let $c := ec'$ and $m := \operatorname{ord}_E(g)$, which is positive since $g(x) = 0$.

Step 6. In this step we prove that $c' = 1$, that ord_L is the Gauss extension of ord_E with weights $(m, \dots, m)$, and the equalities for log discrepancies. Let $f : X' \to X$ be a projective birational morphism from a normal variety X' on which E is a divisor, write $K_{X'} + \Gamma_D = f^*(K_X + D)$, define W', f_W, and Γ_W as in **Step 3**, and let $z \in E$ be a general closed point. As in **Step 4**, there exist a regular system of parameters $t_1, \dots, t_d$ of $\mathcal{O}_{X',z}$ and a unit u near z such that $E = (t_1 = 0)$, $f^*g = ut_1^m$, $\Gamma_D = (1 - \delta_1) \operatorname{div}(t_1)$, and $\Gamma_0 = (1 - \delta_1^0) \operatorname{div}(t_1)$ near z, where $K_{X'} + \Gamma_0 = f^*K_X$, $\delta_1 := a(E, X, D)$, and $\delta_1^0 := a(E, X, 0)$. Since $(X, B + D)$ is lc and (X, B) is klt,

$$\delta_1 - \frac{m}{N} = a(E, X, B + D) \geq 0, \qquad \delta_1^0 - \frac{m}{N} = a(E, X, B) > 0.$$

The image in X' of the center of ord_L on W' is the center of $c' \operatorname{ord}_E$, which is the generic point of E, and $\operatorname{ord}_L(T_j) > 0$ for any j. Thus the center of ord_L on W' is the generic point ξ of $E \times \{0\}$, and ξ is contained in any neighborhood of $(z, 0)$.

Let $\pi : Z \to U$ be a log resolution of (W', Γ_W) over a neighborhood U of $(z, 0)$ which is toric with respect to $t_1, \dots, t_d, T_1/u, \dots, T_{s-1}/u$ (Lemma 12.10), and write $K_Z + \Gamma_Z = \pi^*(K_U + \Gamma_W|_U)$. By (12.7) and [KM98, Lemma 2.30], $a(L, Z, \Gamma_Z) = 0$. In Lemma 12.10 we have $m_1 = m$, $m_i = 0$, and $\delta_i = 1$ for $i \geq 2$. By (12.5), a toric divisor over U with primitive generator (α, β) has log discrepancy zero if and only if $\alpha_i = 0$ for $i \geq 2$, $(\delta_1 - \frac{m}{N})\alpha_1 = 0$, and $\beta_j = \gamma = m\alpha_1$ for any j. If $\alpha_1 = 0$, then $(\alpha, \beta) = 0$. Thus such a toric divisor exists only if $\delta_1 = \frac{m}{N}$, and then it is the toric divisor L_0 corresponding to

$$\rho_0 := (1, 0, \dots, 0, m, \dots, m).$$

The strict transforms of $(T_j = 0)$ and of the branch divisor have coefficient $1 - \frac{1}{N} < 1$ in Γ_Z. The strict transforms of $(t_1 = 0)$ and of $(t_i = 0)$, $i \geq 2$, have coefficients $1 - \delta_1 < 1$ and 0, since $\delta_1 \geq \frac{m}{N} > 0$. Hence Γ_Z has at most one component with coefficient one, namely L_0. If L were exceptional over Z, then $a(L, Z, \Gamma_Z) > 0$ by [KM98, Corollary 2.31(3)]. So L is a divisor on Z with coefficient one in Γ_Z. Since $\operatorname{codim}_{W'} \bar{\xi} = s \geq 2$, L is exceptional over W', so L is a toric divisor and $L = L_0$. Therefore $\delta_1 = \frac{m}{N}$, that is,

$$a(E, X, B + D) = 0,$$

and ord_L is the monomial valuation with weights ρ_0. Since u is a unit in $\mathcal{O}_{X',z}$, ord_L sends $\sum_\alpha b_\alpha T^\alpha$, with $b_\alpha \in \mathcal{O}_{X',z}$, to $\min_\alpha\{\operatorname{ord}_E(b_\alpha) + m|\alpha|\}$. Hence ord_L is the Gauss extension of ord_E with weights $(m, \dots, m)$ (Definition 2.8). In particular $c' = 1$, $c = e$, and the restriction $e \operatorname{ord}_L$ of ord_F to $\mathbb{C}(W)$ is the Gauss extension of $c \operatorname{ord}_E$ with weights $(cm, \dots, cm)$. Applying (12.5) to the pullback of (W, Δ_W), with δ_1^0 in place of δ_1, and the weights ρ_0, we obtain $a(L, W, \Delta_W) = \delta_1^0 - \frac{m}{N} = a(E, X, B)$. Thus $a(F, Y, 0) = ca(E, X, B)$ by (12.7).

Step 7. In this step we prove that ord_F is the only valuation of $\mathbb{C}(Y)$ extending $e \operatorname{ord}_L$, that F is G-invariant, and that $\operatorname{ord}_F(y_j) = \frac{cm}{N}$. Let $t := t_1$, so that $\operatorname{ord}_L(t) = \operatorname{ord}_E(t) = 1$, and let $\mathcal{O}_L \subset \mathbb{C}(W)$ be the valuation ring of ord_L. Let $\tau_j := T_j/t^m$ for $1 \leq j \leq s$, where $T_s = -g - \sum_{j=1}^{s-1} T_j$ as in **Step 2**, so that $\tau_s = -u - \sum_{j=1}^{s-1} \tau_j$. Since ord_L is the Gauss extension of ord_E with weights $(m, \dots, m)$, the τ_j are units of $\mathcal{O}_L$. Moreover, $\mathbb{C}(W) = \mathbb{C}(X)(\tau_1, \dots, \tau_{s-1})$, and

$$\sum_\alpha a_\alpha T^\alpha = \sum_\alpha \left(a_\alpha t^{m|\alpha|}\right)\tau^\alpha$$

for $a_\alpha \in \mathbb{C}(X)$, so ord_L is the Gauss extension of ord_E with weights $(0,\dots,0)$ with respect to $\tau_1,\dots,\tau_{s-1}$ (Definition 2.8). By [Bou89, Chapter VI, §10, no. 1, Proposition 2] for $\tau_1,\dots,\tau_{s-1}$ successively, the residue field of ord_L is $\kappa(E)(\bar\tau_1,\dots,\bar\tau_{s-1})$ with $\bar\tau_1,\dots,\bar\tau_{s-1}$ algebraically independent over $\kappa(E)$, where $\bar\tau_j$ is the residue of τ_j. Let $\bar u \in \kappa(E)\backslash\{0\}$ be the residue of u. Let $m' := m/\gcd(m,N)$ and $N' := N/\gcd(m,N)$, and let

$$A'' := \mathcal{O}_L[\ell, U_1,\dots,U_s]/\big(\ell^{N'} - t,\ U_1^N - \tau_1,\dots,U_s^N - \tau_s\big).$$

Then A'' is a free $\mathcal{O}_L$-module of finite rank, and

$$A''/\ell A'' \cong \kappa(E)(U_1,\dots,U_{s-1})[U_s]\Big/\Big(\bar u + \sum_{j=1}^s U_j^N\Big).$$

By Lemma 12.1 with $\mathbb{K} = \kappa(E)$ and $a = \bar u$, the polynomial $\bar u + \sum_j U_j^N$ is irreducible in $\kappa(E)[U_1,\dots,U_s]$. Since it is monic in U_s, it is irreducible in $\kappa(E)(U_1,\dots,U_{s-1})[U_s]$ by the Gauss lemma, so $A''/\ell A''$ is a field. Any maximal ideal of A'' contains $t = \ell^{N'}$, hence ℓ. Thus A'' is a Noetherian local ring of dimension one with principal maximal ideal $\ell A''$, that is, a discrete valuation ring. Let ν be the valuation of $\operatorname{Frac}(A'')$ with $\nu(\ell) = 1$. Since $\ell^{m'N} = t^m$, the assignment $y_j \to \ell^{m'} U_j$ defines an embedding of $\mathbb{C}(Y) = \mathbb{C}(W)[y_1,\dots,y_s]/(y_1^N - T_1,\dots,y_s^N - T_s)$ into $\operatorname{Frac}(A'')$ over $\mathbb{C}(W)$. The ring A'' is the integral closure of $\mathcal{O}_L$ in $\operatorname{Frac}(A'')$, so any maximal ideal of the integral closure of $\mathcal{O}_L$ in $\mathbb{C}(Y)$ is the contraction of the maximal ideal of A''. By [Ser79, Chapter I, §4, Proposition 11], any valuation of $\mathbb{C}(Y)$ extending ord_L up to a positive multiple is a positive multiple of $\nu|_{\mathbb{C}(Y)}$. Since a positive multiple is determined by the restriction to $\mathbb{C}(W)$, ord_F is the only valuation of $\mathbb{C}(Y)$ whose restriction to $\mathbb{C}(W)$ is $e\operatorname{ord}_L$. For $\sigma \in G$, $(\operatorname{ord}_F \circ\sigma)|_{\mathbb{C}(W)} = \operatorname{ord}_F|_{\mathbb{C}(W)}$ since $\mathbb{C}(W) = \mathbb{C}(Y)^G$, so $\operatorname{ord}_F \circ\sigma = \operatorname{ord}_F$, and F is G-invariant. Finally, write $\operatorname{ord}_F = \lambda\nu|_{\mathbb{C}(Y)}$ with $\lambda > 0$. Then $e = \operatorname{ord}_F(t) = \lambda N'$, and since U_j is a unit of A'',

$$\operatorname{ord}_F(y_j) = \lambda\nu(\ell^{m'}U_j) = \lambda m' = \frac{em'}{N'} = \frac{cm}{N}.$$

□

To return to the pair (X,B), and not only to the variety X, we keep the branch boundary in the marked graded pair, from which the boundary of (X,B) is recovered (Theorem 12.12).

Theorem 12.12. *Notation as in Construction 12.8, and let D, F, c, and E be as in Definition-Proposition 12.11. Assume that F is a Kollár component of $(Y \ni o, 0)$. Then E is a Kollár component of $(X \ni x, B)$. Moreover, let $R_0 := \operatorname{gr}_{c\operatorname{ord}_E} \mathcal{O}_{X,x}$ and $g_0 := \operatorname{in}_{c\operatorname{ord}_E}(g) \in R_0$. Then*

$$\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{Y,o} \cong R_0[z_1,\dots,z_s]\Big/\Big(g_0 + \sum_{j=1}^s z_j^N\Big), \quad \deg z_j = \frac{c\operatorname{ord}_E(g)}{N},$$

and the pair $(\operatorname{Spec} R_0, \frac{1}{N}\operatorname{div}(g_0))$ is klt.

Proof. The proof is long, so we divide it into several steps. Let $R := \mathcal{O}_{X,x}$, $v := c\operatorname{ord}_E$, $m := \operatorname{ord}_E(g)$, and $\mu := \frac{cm}{N}$, which is an integer since $\mu = \operatorname{ord}_F(y_j)$ (Definition-Proposition 12.11). Since $\operatorname{center}_X v = x$, $\mathfrak{a}_{>0}(v)\mathcal{O}_{X,x}$ is the maximal ideal of R, so the degree zero part of R_0 is $\mathbb{C}$.

Step 1. In this step we compute $\operatorname{gr}_{\operatorname{ord}_F} \mathcal{O}_{Y,o}$. Let $P := R[y_1,\dots,y_s]$, $h := g + \sum_j y_j^N$, and let ν_P be the Gauss extension of v with weights $(\mu,\dots,\mu)$ (Definition 2.8). By the definition of ν_P,

$$\mathfrak{a}_\lambda(\nu_P) \cap P = \bigoplus_\alpha \mathfrak{a}_{\lambda-\mu|\alpha|}(v)y^\alpha,$$

so $\operatorname{gr}_{\nu_P} P = R_0[z_1,\dots,z_s]$ with $z_j := \operatorname{in}_{\nu_P}(y_j)$ of degree μ, and

$$\nu_P(h) = cm, \qquad \operatorname{in}_{\nu_P}(h) = g_0 + \sum_{j=1}^s z_j^N.$$

By Lemma 12.1 with $\mathbb{K} = \mathrm{Frac}(R_0)$ and $a = g_0$, the polynomial $g_0 + \sum_j z_j^N$ is irreducible in $\mathrm{Frac}(R_0)[z_1, \dots, z_s]$. Since $g_0 + \sum_j z_j^N$ is monic in z_s, $R_0[z_1, \dots, z_s]/(g_0 + \sum_j z_j^N)$ is free over $R_0[z_1, \dots, z_{s-1}]$ and is a subring of $\mathrm{Frac}(R_0)[z_1, \dots, z_s]/(g_0 + \sum_j z_j^N)$. Hence $R_0[z_1, \dots, z_s]/(g_0 + \sum_j z_j^N)$ is an integral domain.

Let $A_Y := P/(h)$. Since h is monic in y_s, any element of A_Y has a unique representative $r \in P$ with $\deg_{y_s} r < N$, and we set $\bar{\nu}(r + (h)) := \nu_P(r)$. For any other representative $r + hb$, $\nu_P(r + hb) \le \nu_P(r)$: otherwise $\mathrm{in}_{\nu_P}(r) = -\mathrm{in}_{\nu_P}(h)\,\mathrm{in}_{\nu_P}(b)$, which is impossible since a non-zero multiple of the monic polynomial $\mathrm{in}_{\nu_P}(h)$ has z_s-degree at least N. Thus $\bar{\nu}$ is the maximum of ν_P over the representatives, and $\mathrm{in}_{\nu_P}(r) \notin (g_0 + \sum_j z_j^N)$ for $r \neq 0$. Let r_1, r_2 be such representatives and $r_1 r_2 = hb + r$ with $\deg_{y_s} r < N$. If $\nu_P(r) > \nu_P(r_1 r_2)$, then $\mathrm{in}_{\nu_P}(r_1)\,\mathrm{in}_{\nu_P}(r_2) \in (g_0 + \sum_j z_j^N)$, which is impossible since $R_0[z_1, \dots, z_s]/(g_0 + \sum_j z_j^N)$ is an integral domain. Since $\bar{\nu}$ is the maximum over the representatives, $\bar{\nu}(r_1 r_2 + (h)) = \nu_P(r_1 r_2) = \nu_P(r_1) + \nu_P(r_2)$. Hence $\bar{\nu}$ is multiplicative, $\bar{\nu}$ is a valuation of $\mathbb{C}(Y)$, and the map $r + (h) \to \mathrm{in}_{\nu_P}(r)$ induces

$$\mathrm{gr}_{\bar{\nu}} A_Y \cong R_0[z_1, \dots, z_s] \Big/ \Big(g_0 + \sum_{j=1}^{s} z_j^N\Big).$$

An element $\sum_\alpha b_\alpha T^\alpha$ of $R[T_1, \dots, T_{s-1}]$ is the element of A_Y represented by $\sum_\alpha b_\alpha y^{N\alpha}$, and $\deg_{y_s}\big(\sum_\alpha b_\alpha y^{N\alpha}\big) = 0 < N$, so $\bar{\nu}(\sum_\alpha b_\alpha T^\alpha) = \min_\alpha\{v(b_\alpha) + cm|\alpha|\}$. Thus the restriction of $\bar{\nu}$ to $\mathbb{C}(W)$ is the Gauss extension of $c\,\mathrm{ord}_E$ with weights $(cm, \dots, cm)$, and $\bar{\nu} = \mathrm{ord}_F$ by Definition-Proposition 12.11. The ring $\mathcal{O}_{Y,o}$ is a localization of A_Y, and for $b \in A_Y$ with $b(o) \neq 0$, $\mathrm{in}_{\bar{\nu}}(b) = b(o) \in \mathbb{C}\backslash\{0\}$. Hence $\mathrm{gr}_{\mathrm{ord}_F} \mathcal{O}_{Y,o} = \mathrm{gr}_{\bar{\nu}} A_Y$, which is the isomorphism in the theorem with $\deg z_j = \mu = \frac{c\,\mathrm{ord}_E(g)}{N}$.

Step 2. In this step we compute the graded ring of $w := \mathrm{ord}_F|_{\mathbb{C}(W)}$. By Definition-Proposition 12.11, w is the Gauss extension of v with weights $(cm, \dots, cm)$, so

$$\mathfrak{a}_\lambda(w) \cap R[T_1, \dots, T_{s-1}] = \bigoplus_\alpha \mathfrak{a}_{\lambda - cm|\alpha|}(v) T^\alpha.$$

For $b \in R[T_1, \dots, T_{s-1}]$ with $b((x,0)) \neq 0$, $\mathrm{in}_w(b) = b((x,0)) \in \mathbb{C}\backslash\{0\}$. Hence

$$\begin{aligned} &\mathrm{gr}_w \mathcal{O}_{W,(x,0)} = R_0[T_1, \dots, T_{s-1}], \qquad \deg T_j = cm, \\ &\mathrm{in}_w(T_j) = T_j, \qquad \mathrm{in}_w\Big(g + \sum_{j=1}^{s-1} T_j\Big) = g_0 + \sum_{j=1}^{s-1} T_j. \end{aligned} \tag{12.8}$$

Step 3. In this step we prove that the prime divisor L over W with $w = e\,\mathrm{ord}_L$ for a positive integer e is a Kollár component of $(W \ni (x,0), \Delta_W)$. The hypotheses of Lemma 9.2 hold for $k = \mathbb{C}$, $V = Y$, the quotient morphism $q : Y \to W$, the closed point o, the $\mathbb{Q}$-divisors Δ_W and $\Delta_V = 0$, the prime divisor F, and L, as follows. The variety Y is normal of dimension $d + s - 1 \ge 2$. By its definition in Construction 12.8, $\Delta_W \ge 0$. The group G fixes o, and $q^{-1}(q(o)) = \{o\}$ since $y_j^N = 0$ for any j on the fiber of q over $(x,0)$. The $\mathbb{Q}$-divisor $K_W + \Delta_W$ is $\mathbb{Q}$-Cartier since X is $\mathbb{Q}$-Gorenstein and Δ_W is a rational combination of principal divisors, and $K_Y = q^*(K_W + \Delta_W)$. The prime divisor F is G-invariant with $\mathrm{center}_Y F = o$ (Definition-Proposition 12.11). Since F is a Kollár component of $(Y \ni o, 0)$, F is primitive over a neighborhood of o (Definition 2.3), and since q is finite and $q^{-1}(q(o)) = \{o\}$, possibly shrinking W around $(x,0)$ and replacing Y with the preimage, F is primitive over Y. Let $f_Y : Y' \to Y$ be the prime extraction of F. By Lemma 9.2, the induced morphism $f_W : Y'_W \to W$ is the prime extraction of L, and $(Y'_W, (f_W)_*^{-1}\Delta_W + L)$ is plt near L since (Y', F) is plt near F. Moreover, $\mathrm{center}_W L = q(o) = (x,0)$, and (W, Δ_W) is klt near $(x,0)$ by Lemma 2.5, since Y is klt near o. Thus L is a Kollár component of $(W \ni (x,0), \Delta_W)$ by Definition 2.3.

Step 4. In this step we prove that R_0 is a finitely generated normal $\mathbb{Q}$-Gorenstein domain and that the marked graded pair of (W, Δ_W) along w is klt. By **Step 3**, Theorem 2.10, and Lemma 12.2 (as $w = e\,\mathrm{ord}_L$), $\mathrm{gr}_w \mathcal{O}_{W,(x,0)}$ is finitely generated and the marked graded pair of (W, Δ_W) along w is klt. The components of Δ_W are the prime divisors $(T_j = 0)$, $1 \le j \le s-1$, and $(g + \sum_j T_j = 0)$, whose ideals are generated by T_j and $g + \sum_j T_j$. For such a generator φ and any b, $\mathrm{in}_w(\varphi b) = \mathrm{in}_w(\varphi)\,\mathrm{in}_w(b)$ since $\mathrm{gr}_w \mathcal{O}_{W,(x,0)}$ is an integral domain, so the ideal generated by the initial forms is $(\mathrm{in}_w(\varphi))$. By (12.8), the marked graded pair of (W, Δ_W) along w is

$$\Big(\operatorname{Spec} R_0[T_1, \dots, T_{s-1}],\ \Delta_{W,0} := \Big(1 - \frac{1}{N}\Big)\Big(\sum_{j=1}^{s-1} \operatorname{div}(T_j) + \operatorname{div}\Big(g_0 + \sum_{j=1}^{s-1} T_j\Big)\Big)\Big). \tag{12.9}$$

Thus $R_0[T_1, \dots, T_{s-1}]$ is a finitely generated normal domain, and since $K_{\operatorname{Spec} R_0[T_1,\dots,T_{s-1}]} + \Delta_{W,0}$ is $\mathbb{Q}$-Cartier (it is $\mathbb{R}$-Cartier by Definition 2.9, hence $\mathbb{Q}$-Cartier since its coefficients are rational) and $\Delta_{W,0}$ is a rational combination of principal divisors, $\operatorname{Spec} R_0[T_1, \dots, T_{s-1}]$ is $\mathbb{Q}$-Gorenstein. By Lemma 12.3, R_0 is a finitely generated normal domain and $\operatorname{Spec} R_0$ is $\mathbb{Q}$-Gorenstein.

Step 5. In this step we prove that $(\operatorname{Spec} R_0, \frac{1}{N}\operatorname{div}(g_0))$ is klt. By **Step 4**, R_0 is normal and $K_{\operatorname{Spec} R_0} + \frac{1}{N}\operatorname{div}(g_0)$ is $\mathbb{Q}$-Cartier. Let S be a prime divisor over $\operatorname{Spec} R_0$, $l := \mathrm{ord}_S(g_0) \ge 0$, and $\delta := a(S, \operatorname{Spec} R_0, 0)$. Let $\tilde{S}$ be the prime divisor over $\operatorname{Spec} R_0[T_1, \dots, T_{s-1}]$ whose valuation is the Gauss extension of ord_S with weights $(l, \dots, l)$ (Definition 2.8). Let $Z \to \operatorname{Spec} R_0$ be a projective birational morphism from a normal variety Z on which S is a divisor. Near $(z, 0)$ for a general closed point $z \in S$, $\mathrm{ord}_{\tilde{S}}$ is the monomial valuation on $Z \times \mathbb{A}^{s-1}$ with weight one on a local equation of S, weight zero on the remaining coordinates, and weight l on the T_j. Since z is a general point of S and $\delta = a(S, \operatorname{Spec} R_0, 0)$, near $(z, 0)$ we have

$$K_{Z \times \mathbb{A}^{s-1}} + (1 - \delta)S \times \mathbb{A}^{s-1} = \phi^* K_{\operatorname{Spec} R_0[T_1,\dots,T_{s-1}]}$$

for the induced morphism $\phi : Z \times \mathbb{A}^{s-1} \to \operatorname{Spec} R_0[T_1, \dots, T_{s-1}]$, so log discrepancies with respect to $(\operatorname{Spec} R_0[T_1, \dots, T_{s-1}], \Delta_{W,0})$ are computed on $Z \times \mathbb{A}^{s-1}$ with the boundary $(1-\delta)S \times \mathbb{A}^{s-1} + \phi^*\Delta_{W,0}$. By Definition 2.6 and [JM12, Proposition 5.1], $a(\mathrm{ord}_{\tilde{S}}, Z \times \mathbb{A}^{s-1}, 0) = 1 + (s-1)l$ on a smooth neighborhood of $(z, 0)$. Thus, by Definition 2.1, the log discrepancy of $\mathrm{ord}_{\tilde{S}}$ with respect to $(1-\delta)S \times \mathbb{A}^{s-1} + (1 - \frac{1}{N})\sum_j \operatorname{div}(T_j)$ is

$$1 + (s-1)l - (1-\delta) - (s-1)\Big(1 - \frac{1}{N}\Big)l = \delta + (s-1)\frac{l}{N},$$

and $\mathrm{ord}_{\tilde{S}}(g_0 + \sum_j T_j) = \min\{l, l\} = l$. Since $s - 1 = N - 2$,

$$\begin{aligned} a(\tilde{S}, \operatorname{Spec} R_0[T_1, \dots, T_{s-1}], \Delta_{W,0}) &= \delta + (N-2)\frac{l}{N} - \Big(1 - \frac{1}{N}\Big)l \\ &= \delta - \frac{l}{N} = a\Big(S, \operatorname{Spec} R_0, \frac{1}{N}\operatorname{div}(g_0)\Big). \end{aligned} \tag{12.10}$$

The left hand side is positive by **Step 4**. Hence $(\operatorname{Spec} R_0, \frac{1}{N}\operatorname{div}(g_0))$ is klt.

Step 6. In this step we prove that the marked graded pair of (X, B) along v is $(\operatorname{Spec} R_0, \frac{1}{N}\operatorname{div}(g_0))$. Write $B = \sum_i b_i B_i$ with prime divisors B_i. Since v is integer valued, let $\mathfrak{a}_\lambda := \mathfrak{a}_\lambda(v) \subset \mathcal{O}(X)$ for $\lambda \in \mathbb{Z}$, so that $\mathfrak{a}_\lambda = \mathcal{O}(X)$ for $\lambda \le 0$. Since $\mathfrak{a}_\lambda$ is cosupported at the closed point x for $\lambda > 0$, $\mathfrak{a}_\lambda/\mathfrak{a}_{\lambda+1} = \mathfrak{a}_\lambda R/\mathfrak{a}_{\lambda+1}R$. Let

$$\mathcal{R} := \bigoplus_{\lambda \in \mathbb{Z}} t^{-\lambda}\mathfrak{a}_\lambda \subset \mathcal{O}(X)[t, t^{-1}], \qquad \mathcal{X} := \operatorname{Spec} \mathcal{R}.$$

Then $\mathcal{R}/t\mathcal{R} = \bigoplus_{\lambda \ge 0} \mathfrak{a}_\lambda/\mathfrak{a}_{\lambda+1} = R_0$, and $\mathcal{R}[t^{-1}] = \mathcal{O}(X)[t, t^{-1}]$.

We show that $\mathcal{R}$ is finitely generated over $\mathbb{C}$. By **Step 4** and Lemma 12.2, $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x}$ is finitely generated. The ring $\mathrm{gr}_{\mathrm{ord}_E}$ in [XZ25] is $\bigoplus_{q \ge 0} \mathfrak{a}_q(\mathrm{ord}_E)/\mathfrak{a}_{q+1}(\mathrm{ord}_E)$ with $\mathfrak{a}_q(\mathrm{ord}_E) \subset \mathcal{O}(X)$ for the affine variety X, and $\mathfrak{a}_q(\mathrm{ord}_E)$ is cosupported at x for $q > 0$, so it is $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x}$. Thus E is primitive over X by [XZ25, Lemma 2.14]. Let $f_E : X_E \to X$ be the prime extraction

of E (Definition 2.3), and r_E a positive integer such that $r_E E$ is Cartier. By Lemma 2.7, $\mathfrak{a}_q(\mathrm{ord}_E) = (f_E)_*\mathcal{O}_{X_E}(-qE)$ for $q \geq 0$, and $\bigoplus_{n\geq 0}\mathfrak{a}_{nr_E}(\mathrm{ord}_E)$ is a finitely generated $\mathcal{O}(X)$-algebra. For $0 \leq i' \leq r_E - 1$,

$$\bigoplus_{n\geq 0}\mathfrak{a}_{i'+nr_E}(\mathrm{ord}_E) = \bigoplus_{n\geq 0}(f_E)_*\big(\mathcal{O}_{X_E}(-i'E)\otimes\mathcal{O}_{X_E}(-r_E E)^{\otimes n}\big)$$

is a finitely generated $\bigoplus_{n\geq 0}\mathfrak{a}_{nr_E}(\mathrm{ord}_E)$-module, since $-r_E E$ is an f_E-ample Cartier divisor [Har77, Proposition II.5.17, Theorem III.5.2, and Exercise II.5.9]. Thus $\bigoplus_{q\geq 0}\mathfrak{a}_q(\mathrm{ord}_E)$ is a finitely generated $\mathcal{O}(X)$-algebra. Since $\mathfrak{a}_q(\mathrm{ord}_E) = \mathcal{O}(X)$ for $q \leq 0$, the ring

$$\mathcal{R}_E := \bigoplus_{q\in\mathbb{Z}} t^{-cq}\mathfrak{a}_q(\mathrm{ord}_E)$$

is generated by t^c and $\bigoplus_{q\geq 0} t^{-cq}\mathfrak{a}_q(\mathrm{ord}_E)$, so $\mathcal{R}_E$ is finitely generated. Since $\mathfrak{a}_\lambda = \mathfrak{a}_{\lceil\lambda/c\rceil}(\mathrm{ord}_E)$, for $\lambda = cq - r$ with $q \in \mathbb{Z}$ and $0 \leq r \leq c-1$ we have

$$t^{-\lambda}\mathfrak{a}_\lambda = t^r \cdot t^{-cq}\mathfrak{a}_q(\mathrm{ord}_E).$$

Hence $\mathcal{R} = \sum_{r=0}^{c-1} t^r\mathcal{R}_E$ is a finite $\mathcal{R}_E$-module, and $\mathcal{R}$ is a finitely generated $\mathbb{C}$-algebra. As a subring of $\mathcal{O}(X)[t,t^{-1}]$, $\mathcal{R}$ is an integral domain, and $\mathcal{R}$ is flat over $\mathbb{C}[t]$ since it is torsion free. Thus $\mathcal{X}$ is a variety of dimension $d+1$, $X_0 := (t=0) \cong \operatorname{Spec} R_0$ is a Cartier divisor on $\mathcal{X}$, and $\mathcal{X}\backslash X_0 \cong X \times \mathbb{G}_m$.

(a) Let $\mathcal{B}_i$ be the closure of $B_i \times \mathbb{G}_m$ in $\mathcal{X}$. The ideal of $\mathcal{B}_i$ is

$$\mathcal{I}_i := \bigoplus_{\lambda} t^{-\lambda}(\mathfrak{a}_\lambda \cap I_{B_i}),$$

and t is a nonzerodivisor on $\mathcal{R}/\mathcal{I}_i$, which is a subring of $(\mathcal{O}(X)/I_{B_i})[t,t^{-1}]$. The image of $\mathcal{I}_i$ in R_0 is generated by $\mathrm{in}_v(f)$, $f \in I_{B_i}\backslash\{0\}$. Hence $\mathcal{B}_i \not\subset X_0$, and by [Ful98, Definition 2.3] and Definition 2.9,

$$X_0 \cdot [\mathcal{B}_i] = (B_i)_0.$$

(b) Let $\tilde{g} := t^{-cm}g \in \mathcal{R}$. The restriction of $\tilde{g}$ to X_0 is $g_0 \neq 0$, and X_0 is integral, so $\mathrm{div}(\tilde{g})$ has no component contained in X_0. On $\mathcal{X}\backslash X_0$, $\mathrm{div}(\tilde{g}) = \mathrm{div}(g) \times \mathbb{G}_m = \sum_i Nb_i B_i \times \mathbb{G}_m$. Hence $[\mathrm{div}(\tilde{g})] = \sum_i Nb_i[\mathcal{B}_i]$.

(c) By [Ful98, Definition 2.3 and Theorem 2.4],

$$\mathrm{div}(g_0) = \mathrm{div}(\tilde{g}) \cdot [X_0] = X_0 \cdot [\mathrm{div}(\tilde{g})] = \sum_i Nb_i(B_i)_0$$

in $A_{d-1}(|X_0| \cap |\mathrm{div}(\tilde{g})|)$. Since $\dim(|X_0| \cap |\mathrm{div}(\tilde{g})|) = d-1$, the equality holds as cycles. Dividing by N, the marked boundary B_0 of (X,B) along v is $\frac{1}{N}\mathrm{div}(g_0)$.

Step 7. In this step we prove that E is a Kollár component of $(X \ni x, B)$. By **Steps 4, 5, and 6** and Lemma 12.2, $\mathrm{gr}_{\mathrm{ord}_E}\mathcal{O}_{X,x}$ is finitely generated and the marked graded pair of ord_E is klt. Since (X,B) is klt and B is a $\mathbb{Q}$-divisor, E is a Kollár component of $(X \ni x, B)$ by Theorem 2.10. □

13. Boundaries with fixed denominators

In this section, we prove Theorem 13.1, which extends Definition-Theorem 11.1 to boundaries with fixed denominators. The degrees of the index covers and of the Fermat suspensions cancel, so the constant depends only on d, N_1, and N_2.

Theorem 13.1. *Let $d \geq 2$ and $N_1, N_2 \geq 3$ be integers, let $N_0(\cdot)$ be the function defined in Definition-Theorem 11.1, and let*

$$s(d,N_1,N_2) := \frac{\max\{N_0(d+N_2-1), N_0(d+N_1+N_2-3)\}}{N_2}.$$

Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension d, $x \in X$ a closed point, and $B \geq 0$ and $D \geq 0$ two $\mathbb{Q}$-divisors on X such that $K_X + B$ and D are $\mathbb{Q}$-Cartier, $N_1 B$ and $N_2 D$ are Weil divisors, X is $\mathbb{Q}$-Gorenstein near x, (X, B) is klt, $(X, B + D)$ is lc, and x is an lc center of $(X, B + D)$. Then there exists a Kollár component E of $(X \ni x, B)$ over k such that $a(E, X, B + D) = 0$ and

$$a(E, X, B) \leq s(d, N_1, N_2).$$

13.1. **Suspensions over the ground field.** In this subsection, we prove the versions over k of Definition-Propositions 12.11 and 12.5 and of Theorems 12.12 and 12.7.

Definition-Propositions 12.11 and 12.5 and Theorems 12.12 and 12.7 concern varieties over $\mathbb{C}$, whereas the Kollár component F in Proposition 13.2 is defined over k. Theorems 12.12 and 12.7 attach to $F_{\bar{k}}$ a divisor over $X_{\bar{k}}$ without any choice, so the divisor is Galois invariant and descends to k by Lemma 3.6.

Proposition 13.2. *Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension $d \geq 2$ which is $\mathbb{Q}$-Gorenstein and klt, and $x \in X$ a closed point.*

(1) Let $B \geq 0$ be a $\mathbb{Q}$-divisor on X such that (X, B) is klt. Let $N \geq 3$ be an integer such that $NB = \operatorname{div}(g)$ near x for some $g \in \mathcal{O}_{X,x}$ with $g(x) = 0$. Let Y, o, and p be as in Construction 12.8 with $\mathbb{C}$ replaced by k, and let $D \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $(X, B+D)$ is lc and x is an lc center of $(X, B+D)$. Then Y is normal, $\mathbb{Q}$-Gorenstein, and klt near o, (Y, p^*D) is lc near o, o is an lc center of (Y, p^*D), p is flat, and the flat pullback p^*S of any prime divisor S on X is reduced near o. In particular, $N'p^*\Theta$ is a Weil divisor near o for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor Θ on X and any positive integer N' such that $N'\Theta$ is a Weil divisor. Moreover, for any Kollár component F of $(Y \ni o, 0)$ over k with $a(F, Y, p^*D) = 0$, the restriction of ord_F to $k(X)$ is $c \operatorname{ord}_E$ for a positive integer c and a Kollár component E of $(X \ni x, B)$ over k such that $a(E, X, B + D) = 0$ and $a(F, Y, 0) = ca(E, X, B)$.

(2) Let $N \geq 3$ be an integer and f a non-zero regular function on X with $f(x) = 0$ such that $(X, \frac{1}{N}\operatorname{div}(f))$ is lc and x is an lc center of $(X, \frac{1}{N}\operatorname{div}(f))$, and let Y, o, and H be as in Construction 12.4 with $\mathbb{C}$ replaced by k. Then (Y, H) is lc and o is an lc center of (Y, H). Moreover, for any Kollár component F of $(Y \ni o, 0)$ over k with $a(F, Y, H) = 0$, the restriction of ord_F to $k(X)$ is $c \operatorname{ord}_E$ for a positive integer c and a Kollár component E of $(X \ni x, 0)$ over k such that $a(E, X, \frac{1}{N}\operatorname{div}(f)) = 0$ and $Nca(E, X, 0) \leq a(F, Y, 0)$.

Proof. The proof is long, so we divide it into several steps. Let $\Theta_X := B + D$ in (1) and $\Theta_X := \frac{1}{N}\operatorname{div}(f)$ in (2).

Step 1. In this step we show that the conditions on X and x are preserved by algebraic extensions of k. Let L/k be an algebraic extension, $x_L \in X_L$ a point over x, and X° the connected component of X_L containing x_L. By Lemma 3.3(1), X_L is normal, so X° is a normal variety over L of dimension d. Since $\Omega_{X_L/L}$ is the pullback of $\Omega_{X/k}$, K_{X° is the pullback of K_X. Thus X° is $\mathbb{Q}$-Gorenstein, and the pullbacks of D and Θ_X are $\mathbb{Q}$-Cartier. By Lemma 3.3(3), X° is klt, the pullback of (X, B) to X° is klt, and the pullback of (X, Θ_X) to X° is lc. The pullbacks of g and f vanish at x_L, and the pullback of NB is the divisor of the pullback of g near x_L.

It remains to show that x_L is an lc center of the pullback of (X, Θ_X) to X°. Let $h : Z \to X$ be a projective birational morphism from a normal variety and E_0 a prime divisor on Z with $h(E_0) = x$ and $a(E_0, X, \Theta_X) = 0$. Since $E_0 \to x$ is surjective, $(E_0)_L \to x \times_k L$ is surjective. Since $x \times_k L$ is a finite discrete set containing x_L, some irreducible component E_0° of $(E_0)_L$ maps to x_L. By Lemma 3.3(1), Z_L is normal and a local equation of E_0 has order one along E_0°, so E_0° is a prime divisor over X° with $\operatorname{center}_{X^\circ}(E_0^\circ) = x_L$ and $\operatorname{ord}_{E_0^\circ}|_{k(X)} = \operatorname{ord}_{E_0}$. By Lemma 3.3(2), the log discrepancy of E_0° with respect to the pullback of (X, Θ_X) is $a(E_0, X, \Theta_X) = 0$.

Step 2. In this step we prove the statements of (1) and (2) that do not concern F. By Lemma 3.2, there exists a field isomorphism $\bar{k} \cong \mathbb{C}$. By transport of structure along it, Constructions 12.8

and 12.4, Definition-Propositions 12.11 and 12.5, and Theorems 12.12 and 12.7 hold with $\mathbb{C}$ replaced by $\bar{k}$.

Let $\bar{o} \in Y_{\bar{k}}$ be a point over o. Then $\bar{o} = (\bar{x}, 0)$ for a point $\bar{x} \in X_{\bar{k}}$ over x. Let X° be the connected component of $X_{\bar{k}}$ containing $\bar{x}$. By **Step 1** with $L = \bar{k}$, the variety X°, the point $\bar{x}$, and the pullbacks of B, g, and D (resp. of f) satisfy the conditions of Construction 12.8 and Definition-Proposition 12.11 (resp. of Construction 12.4) over $\bar{k}$. The suspensions commute with base change, so $Y_{\bar{k}} \times_{X_{\bar{k}}} X^\circ$ is the suspension of Construction 12.8 (resp. Construction 12.4) for X°, $\bar{x}$, and the pullbacks of B and g (resp. of f). Hence by Definition-Proposition 12.11 (resp. Definition-Proposition 12.5), $\bar{o}$ has an open neighborhood in $Y_{\bar{k}}$ which is normal, $\mathbb{Q}$-Gorenstein, and klt, on which the pullback of (Y, p^*D) (resp. (Y, H)) is lc, and $\bar{o}$ is an lc center of the pullback of (Y, p^*D) (resp. (Y, H)).

We first treat (1). Since $\bar{o}$ is an arbitrary point over o, there exists an open subset $V \subset Y_{\bar{k}}$ containing the fiber over o with the properties above. Since $Y_{\bar{k}} \to Y$ is integral, hence closed, the complement U of the image of $Y_{\bar{k}} \backslash V$ is an open neighborhood of o in Y with $U_{\bar{k}} \subset V$. For any $y \in U$ and any point $\bar{y} \in U_{\bar{k}}$ over y, the local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{Y_{\bar{k}},\bar{y}}$ is flat and $\mathcal{O}_{Y_{\bar{k}},\bar{y}}$ is normal, so $\mathcal{O}_{Y,y}$ is normal [Mat89, Theorem 23.9, Corollary (i)]. Since U is normal and Noetherian, the connected component of U through o is open and irreducible. Replacing U with it, we may assume that U is a normal variety.

Since $U_{\bar{k}}$ is quasi-compact, $mK_{U_{\bar{k}}}$ is Cartier for some positive integer m. Choose finitely many open subsets $V_1, \dots, V_r$ covering $U_{\bar{k}}$ and rational functions ϕ_i on $U_{\bar{k}}$ with $mK_{U_{\bar{k}}}|_{V_i} = \operatorname{div}(\phi_i)|_{V_i}$, where $K_{U_{\bar{k}}}$ is the pullback of a canonical divisor K_U of U. By a standard limit argument [Gro66, §8], there exists a finite extension k'/k in $\bar{k}$ such that each V_i is the pullback of an open subset V_i' of $U_{k'}$ and each ϕ_i is the pullback of a rational function ϕ_i' on $U_{k'}$. Since $U_{\bar{k}} \to U_{k'}$ is surjective, the V_i' cover $U_{k'}$, and since the pullback of Weil divisors from $U_{k'}$ to $U_{\bar{k}}$ is injective by Lemma 3.3(1), $mK_{U_{k'}}|_{V_i'} = \operatorname{div}(\phi_i')|_{V_i'}$, so $mK_{U_{k'}}$ is Cartier. Since $U_{k'} \to U$ is finite and étale, $K_{U_{k'}}$ is the pullback of K_U, so K_U is $\mathbb{Q}$-Cartier by [KM98, Proposition 5.20(1)] (applied to a connected component of $U_{k'}$, which is a normal variety mapping finitely and surjectively onto U). By Lemma 3.3(3), U is klt and $(U, p^*D|_U)$ is lc. Let $\bar{F}_0$ be a prime divisor over $Y_{\bar{k}}$ with $\operatorname{center}_{Y_{\bar{k}}}(\bar{F}_0) = \bar{o}$ and log discrepancy zero with respect to the pullback of (Y, p^*D). By Lemma 3.3(2), $\operatorname{ord}_{\bar{F}_0}|_{k(Y)} = \operatorname{ord}_{F_0}$ for a prime divisor F_0 over Y with $a(F_0, Y, p^*D) = 0$, and $\operatorname{center}_Y F_0$ is the image of $\bar{o}$, which is o. Thus o is an lc center of (Y, p^*D).

Next we show the flatness statements in (1). The coordinate ring of Y is

$$\mathcal{O}(X)[y_1, \dots, y_{N-1}] \Big/ \Big(g + \sum_{j=1}^{N-1} y_j^N\Big).$$

The equation is monic of degree N in y_{N-1}, so the coordinate ring of Y is a free module over $\mathcal{O}(X)[y_1, \dots, y_{N-2}]$ with basis $1, y_{N-1}, \dots, y_{N-1}^{N-1}$. Thus p is flat. Let $\xi \in X$ be a point and $\overline{\kappa(\xi)}$ an algebraic closure of $\kappa(\xi)$. If r^2 divides $g(\xi) + \sum_j y_j^N$ in $\overline{\kappa(\xi)}[y_1, \dots, y_{N-1}]$, then r divides $\partial(g(\xi) + \sum_j y_j^N)/\partial y_j = N y_j^{N-1}$ for any j. Since $N - 1 \geq 2$, the polynomials $N y_1^{N-1}$ and $N y_2^{N-1}$ have no common factor, so r is a constant. Hence any fiber of p is geometrically reduced. Let S be a prime divisor on X with generic point η_S, t a uniformizer of $\mathcal{O}_{X,\eta_S}$, and Γ an irreducible component of $p^{-1}(S)$ through o. Since p is flat, generizations lift along p, so Γ is the closure of an irreducible component of $p^{-1}(\eta_S)$ and Γ is a prime divisor on Y. The homomorphism $\mathcal{O}_{X,\eta_S} \to \mathcal{O}_{Y,\Gamma}$ is flat and local, and $\mathcal{O}_{Y,\Gamma}/t\mathcal{O}_{Y,\Gamma}$ is the local ring of the reduced fiber $p^{-1}(\eta_S)$ at a generic point, which is a field. Thus $\operatorname{ord}_\Gamma(t) = 1$, and p^*S is reduced near o. If Θ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, then, since the pullback of a Cartier divisor under the flat morphism p is the flat pullback, $p^*\Theta = \sum_S (\operatorname{mult}_S \Theta) p^*S$. Hence $N' p^*\Theta$ is a Weil divisor near o if $N'\Theta$ is a Weil divisor.

Now we treat (2). The variety $Y = X \times \mathbb{A}^{N-1}$ is normal, $\mathbb{Q}$-Gorenstein, and klt, since X is normal, $\mathbb{Q}$-Gorenstein, and klt and K_Y is the pullback of K_X. By Lemma 3.5(1), X is

geometrically integral over the algebraic closure k_0 of k in $k(X)$, and $k_0 \subset \Gamma(X, \mathcal{O}_X)$. Hence $X_{\bar{k}}$ is the disjoint union of the integral schemes $X \times_{k_0,\tau} \bar{k}$ over the k-embeddings $\tau : k_0 \to \bar{k}$, and each of them maps onto X. Thus every connected component X° of $X_{\bar{k}}$ contains a point $\bar{x}$ over x. By **Step 1** with $L = \bar{k}$ and Definition-Proposition 12.5 over $\bar{k}$, applied to X°, $\bar{x}$, and the pullback of f, the pullback of (Y, H) to $Y_{\bar{k}} \times_{X_{\bar{k}}} X^\circ = X^\circ \times \mathbb{A}^{N-1}$ is lc for each X°, so the pullback of (Y, H) to $Y_{\bar{k}}$ is lc. By Lemma 3.3(3), (Y, H) is lc. Let $\bar{F}_0$ be a prime divisor over $Y_{\bar{k}}$ with $\operatorname{center}_{Y_{\bar{k}}}(\bar{F}_0) = \bar{o}$ and log discrepancy zero with respect to $(Y_{\bar{k}}, H_{\bar{k}})$. By Lemma 3.3(2), $\operatorname{ord}_{\bar{F}_0}|_{k(Y)} = \operatorname{ord}_{F_0}$ for a prime divisor F_0 over Y with $a(F_0, Y, H) = 0$ and $\operatorname{center}_Y F_0 = o$. Thus o is an lc center of (Y, H).

Step 3. In this step we prove the statements on F when x is a k-rational point. Then o is a k-rational point. By **Step 2**, possibly replacing Y with an open neighborhood of o, we may assume that Y is a normal klt variety. By Lemma 3.5(2), X and Y are geometrically integral over k. Let $\bar{x} \in X_{\bar{k}}$ and $\bar{o} \in Y_{\bar{k}}$ be the points over x and o.

Let F be as in (1). By Lemma 3.6(1), $\bar{F} := F_{\bar{k}}$ is a Kollár component of $(Y_{\bar{k}} \ni \bar{o}, 0)$, $\operatorname{ord}_{\bar{F}}|_{k(Y)} = \operatorname{ord}_F$, and $a(\bar{F}, Y_{\bar{k}}, 0) = a(F, Y, 0)$. By Lemma 3.3(2), the log discrepancy of $\bar{F}$ with respect to $(Y_{\bar{k}}, (p^*D)_{\bar{k}})$ is $a(F, Y, p^*D) = 0$. By Definition-Proposition 12.11 and Theorem 12.12 over $\bar{k}$, the restriction of $\operatorname{ord}_{\bar{F}}$ to $\bar{k}(X_{\bar{k}})$ is $c \operatorname{ord}_{\bar{E}}$ for a positive integer c and a Kollár component $\bar{E}$ of $(X_{\bar{k}} \ni \bar{x}, B_{\bar{k}})$ such that

$$a(\bar{E}, X_{\bar{k}}, B_{\bar{k}} + D_{\bar{k}}) = 0 \quad \text{and} \quad a(\bar{F}, Y_{\bar{k}}, 0) = ca(\bar{E}, X_{\bar{k}}, B_{\bar{k}}).$$

Let $\sigma \in \operatorname{Gal}(\bar{k}/k)$. Then σ acts on $Y_{\bar{k}}$ and $X_{\bar{k}}$ compatibly with $p_{\bar{k}}$, and $\operatorname{ord}_{\bar{F}} \circ \sigma = \operatorname{ord}_{\bar{F}}$ since $\bar{F}$ is the base change of F. Restricting to $\bar{k}(X_{\bar{k}})$, we obtain $c \operatorname{ord}_{\bar{E}} \circ \sigma = c \operatorname{ord}_{\bar{E}}$. Since σ is arbitrary, Lemma 3.6(2) implies that $\operatorname{ord}_{\bar{E}}|_{k(X)} = \operatorname{ord}_E$ for a Kollár component E of $(X \ni x, B)$ over k with $E_{\bar{k}} = \bar{E}$. Thus

$$\operatorname{ord}_F|_{k(X)} = \operatorname{ord}_{\bar{F}}|_{k(X)} = c \operatorname{ord}_{\bar{E}}|_{k(X)} = c \operatorname{ord}_E.$$

By Lemma 3.6(1), $a(E, X, B) = a(\bar{E}, X_{\bar{k}}, B_{\bar{k}})$, and by Lemma 3.3(2),

$$a(E, X, B + D) = a(\bar{E}, X_{\bar{k}}, B_{\bar{k}} + D_{\bar{k}}) = 0.$$

Hence $a(F, Y, 0) = ca(E, X, B)$.

Let F be as in (2). Replacing Definition-Proposition 12.11 and Theorem 12.12 with Definition-Proposition 12.5 and Theorem 12.7, p^*D with H, and B with 0, the argument above for (1) shows that $\operatorname{ord}_F|_{k(X)} = c \operatorname{ord}_E$ for a positive integer c and a Kollár component E of $(X \ni x, 0)$ over k with $a(E, X, \frac{1}{N} \operatorname{div}(f)) = 0$ and

$$Nca(E, X, 0) = Nca(\bar{E}, X_{\bar{k}}, 0) \le a(\bar{F}, Y_{\bar{k}}, 0) = a(F, Y, 0).$$

Step 4. In this step we prove the statements on F for an arbitrary closed point x. Let $\kappa := \kappa(x)$, $x_1 \in X_\kappa$ the κ-rational point induced by x, X_1 the irreducible component of X_κ through x_1, and B_1, D_1, g_1, and f_1 the pullbacks of B, D, g, and f to X_1. By Lemma 3.3(1), X_κ is normal, so X_1 is also the connected component of X_κ containing x_1. Thus we may take $X^\circ = X_1$ in **Step 1** with $L = \kappa$. Since $\kappa(o) = \kappa$, the κ-rational point of Y_κ induced by o is $o_1 := (x_1, 0)$. Let Y_1 be the irreducible component of Y_κ through o_1.

The field κ is finitely generated over $\mathbb{C}$. By **Step 1** with $L = \kappa$, the data X_1, x_1, B_1, g_1, and D_1 (resp. X_1, x_1, and f_1) over κ satisfy the conditions of (1) (resp. (2)). The suspensions commute with base change, so $Y_\kappa \times_{X_\kappa} X_1$ is the suspension of Construction 12.8 (resp. Construction 12.4) for X_1, x_1, B_1, and g_1 (resp. X_1, x_1, and f_1). By **Step 2** over κ, $Y_\kappa \times_{X_\kappa} X_1$ is normal near o_1, hence it coincides with Y_1 near o_1. Let $p_1 : Y_1 \to X_1$ be the projection. Then the pullbacks of p^*D and H to Y_1 are $p_1^*D_1$ and the divisor H_1 of Construction 12.4 for X_1 and f_1.

Let F be as in (1) (resp. (2)). By **Step 2**, Y is klt near o. By Corollary 3.10 applied to $(Y \ni o, 0)$, there exists a Kollár component F_1 of $(Y_1 \ni o_1, 0)$ over κ such that $\operatorname{ord}_{F_1}|_{k(Y)} = \operatorname{ord}_F$, $a(F_1, Y_1, 0) = a(F, Y, 0)$, and $a(F_1, Y_1, p_1^*D_1) = 0$ (resp. $a(F_1, Y_1, H_1) = 0$). By **Step 3** over κ, $\operatorname{ord}_{F_1}|_{\kappa(X_1)} = c \operatorname{ord}_{E_1}$ for a positive integer c and a Kollár component E_1 of $(X_1 \ni x_1, B_1)$

(resp. $(X_1 \ni x_1, 0)$) over κ with $a(E_1, X_1, B_1 + D_1) = 0$ and $a(F_1, Y_1, 0) = ca(E_1, X_1, B_1)$ (resp. $a(E_1, X_1, \frac{1}{N}\operatorname{div}(f_1)) = 0$ and $Nca(E_1, X_1, 0) \le a(F_1, Y_1, 0)$). By Corollary 3.10 applied to $(X \ni x, B)$ (resp. $(X \ni x, 0)$), $\operatorname{ord}_{E_1}|_{k(X)} = \operatorname{ord}_E$ for a Kollár component E of $(X \ni x, B)$ (resp. $(X \ni x, 0)$) over k, and the log discrepancies of E_1 with respect to B_1, $B_1 + D_1$, 0, and $\frac{1}{N}\operatorname{div}(f_1)$ are those of E with respect to B, $B + D$, 0, and $\frac{1}{N}\operatorname{div}(f)$. Since $k(X) \subset \kappa(X_1) \subset \kappa(Y_1)$ and $k(X) \subset k(Y) \subset \kappa(Y_1)$,

$$\operatorname{ord}_F|_{k(X)} = \operatorname{ord}_{F_1}|_{k(X)} = c\operatorname{ord}_{E_1}|_{k(X)} = c\operatorname{ord}_E.$$

Hence E and c satisfy the conclusions of (1) (resp. (2)). □

13.2. **Selection.** In this subsection, we prove Lemma 13.3 and Theorem 13.1.

The difficulty is that the Cartier index of D is not bounded. On the index cover X' of N_2D, the pullback of N_2D is principal, but a Kollár component taken on X' need not descend to X. We therefore take a Kollár component on a quotient of the additive suspension of X' by Definition-Theorem 11.1, lift it uniquely by Lemma 9.3, restrict it to $k(X')$ by Proposition 13.2(2), take the quotient by Lemma 9.2, and cancel the degree of the cover by Lemma 9.4.

Lemma 13.3. *Let $d \ge 2$ and $N_2 \ge 3$ be integers, and let $N_0(\cdot)$ be the function defined in Definition-Theorem 11.1. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension d, and $x \in X$ a k-rational point such that X is $\mathbb{Q}$-Gorenstein and klt near x. Let $D \ge 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that N_2D is a Weil divisor, (X, D) is lc, and x is an lc center of (X, D). Then there exists a Kollár component E of $(X \ni x, 0)$ over k such that $a(E, X, D) = 0$ and*

$$a(E, X, 0) \le \frac{N_0(d + N_2 - 1)}{N_2}.$$

Proof. Possibly shrinking X around x, we may assume that X is affine, $\mathbb{Q}$-Gorenstein, and klt. Let $f : X' \to X$ be the index cover of N_2D (Construction-Proposition 9.1), H_0 the Galois group of f, and $x' \in X'$ the unique point over x. Then $\kappa(x') = k$, and H_0 fixes x'. Since $\mathcal{O}_{X',x'}$ is a domain, x' is the only point over x, and f is finite, possibly shrinking X around x, we may assume that any irreducible component of X' contains x'. Since X' is normal, X' is irreducible. Let $D' := f^*D$. Since f is quasi-étale, $K_{X'} = f^*K_X$ and $K_{X'} + D' = f^*(K_X + D)$, so X' is $\mathbb{Q}$-Gorenstein and klt, and (X', D') is lc (Lemma 2.5).

We show that x' is an lc center of (X', D'). Let $h : Z \to X$ be a projective birational morphism from a normal variety and E_0 a prime divisor on Z with $h(E_0) = x$ and $a(E_0, X, D) = 0$. Let Z' be the normalization of Z in $k(X')$ and E_0' a prime divisor on Z' over E_0. Then $\operatorname{ord}_{E_0'}|_{k(X)} = e_0 \operatorname{ord}_{E_0}$ for a positive integer e_0, and the center of E_0' on X' maps to x, so it is x'. By Lemma 2.5, $a(E_0', X', D') = e_0 a(E_0, X, D) = 0$.

By Construction-Proposition 9.1, applied to the H_0-linearized invertible ideal sheaf $\mathcal{O}_{X'}(-f^*(N_2D))$ on an H_0-invariant neighborhood of x', there exists $\varphi \in \mathcal{O}_{X',x'}$ with $f^*(N_2D) = \operatorname{div}(\varphi)$ near x' which is an eigenvector of H_0 with character χ, that is, $g^*\varphi = \chi(g)\varphi$ for any $g \in H_0$. Since f is finite and x' is the only point over x, possibly shrinking X around x, we may assume that φ is a regular function on X' and $f^*(N_2D) = \operatorname{div}(\varphi)$ on X'. Since X is klt and x is an lc center of (X, D), $x \in \operatorname{Supp} D$, so $\varphi(x') = 0$. Moreover $\frac{1}{N_2}\operatorname{div}(\varphi) = D'$.

Let Y, o, h, and H be as in Construction 12.4 over k, with X, x, N, and f replaced by X', x', N_2, and φ. Thus $Y = X' \times \mathbb{A}^{N_2-1}$ with coordinates $u_1, \dots, u_{N_2-1}$, $o = (x', 0)$, $h = \varphi + \sum_{j=1}^{N_2-1} u_j^{N_2}$, $H = \operatorname{div}(h)$, and $\dim Y = d + N_2 - 1$. By Proposition 13.2(2), (Y, H) is lc and o is an lc center of (Y, H). We show that H is reduced. A prime divisor on Y which does not dominate X' is of the form $P \times \mathbb{A}^{N_2-1}$ for a prime divisor P on X', and h does not vanish on $P \times \mathbb{A}^{N_2-1}$. A prime divisor Γ on Y which dominates X' has local ring a localization of $k(X')[u_1, \dots, u_{N_2-1}]$, and $\varphi + \sum_j u_j^{N_2}$ is square-free in $\overline{k(X')}[u_1, \dots, u_{N_2-1}]$ because a square factor would divide $\partial h/\partial u_j = N_2 u_j^{N_2-1}$ for every j, hence would be a common factor of $u_1^{N_2-1}$ and $u_2^{N_2-1}$ (the coordinate u_2 exists since $N_2 - 1 \ge 2$), so it is a constant. Thus $\operatorname{ord}_\Gamma(h) \le 1$, and H is reduced.

Let

$$\mathcal{G} := \{(g, \lambda) \in H_0 \times k^* \mid \lambda^{N_2} = \chi(g)\},$$

a finite group acting on Y by $(g, \lambda) \cdot (x'', u_1, \dots, u_{N_2-1}) = (g \cdot x'', \lambda u_1, \dots, \lambda u_{N_2-1})$. Then $(g, \lambda)^* h = \chi(g)h$, so H is $\mathcal{G}$-invariant. Let $q : Y \to W' := Y/\mathcal{G}$ be the quotient morphism and $w := q(o)$. We show that q is quasi-étale. Let $(g, \lambda) \in \mathcal{G}$ be a non-identity element. Since the residue fields have characteristic zero, the order of the inertia group of a prime divisor equals the ramification index along it [Ser79, Chapter I, Proposition 21 and its Corollary]. If $g \neq 1$, then, since f is quasi-étale, the inertia group of any prime divisor on X' is trivial, so g fixes no prime divisor on X' pointwise and $\mathrm{Fix}(g)$ has codimension at least two in X'. Thus the fixed locus of (g, λ) is contained in $\mathrm{Fix}(g) \times \mathbb{A}^{N_2-1}$, which has codimension at least two in Y. If $g = 1$, then $\lambda \neq 1$, and the fixed locus of (g, λ) is $X' \times \{0\}$, which has codimension $N_2 - 1 \geq 2$ in Y. Hence the inertia group of any prime divisor on Y is trivial, and q is quasi-étale.

Since o is fixed by $\mathcal{G}$, we have $q^{-1}(w) = \{o\}$. Moreover, w is a k-rational point. By Lemma 3.5(2), W' is geometrically integral over k, and $\dim W' = d + N_2 - 1$. Let $H_{W'}$ be the reduced divisor $q(H)$. Since H is reduced and $\mathcal{G}$-invariant and q is quasi-étale, $q^* H_{W'} = H$. Thus $K_Y = q^* K_{W'}$ and $K_Y + H = q^*(K_{W'} + H_{W'})$, and K_Y and $K_Y + H$ are $\mathbb{Q}$-Cartier. By [KM98, Proposition 5.20(1)], applied with $\Delta = 0$ and with $\Delta = H_{W'}$, both $K_{W'}$ and $K_{W'} + H_{W'}$ are $\mathbb{Q}$-Cartier, so $H_{W'}$ is $\mathbb{Q}$-Cartier. By Lemma 2.5, W' is klt and $(W', H_{W'})$ is lc. Let F_0 be a prime divisor over Y with $\mathrm{center}_Y F_0 = o$ and $a(F_0, Y, H) = 0$. By Lemma 2.5, $\mathrm{ord}_{F_0}|_{k(W')}$ is a positive multiple of ord_{L_0} for a prime divisor L_0 over W' with $a(L_0, W', H_{W'}) = 0$, and $\mathrm{center}_{W'} L_0 = q(o) = w$. Thus w is an lc center of $(W', H_{W'})$.

By Definition-Theorem 11.1 in dimension $d + N_2 - 1$, applied to $W' \ni w$ and $P = H_{W'}$, there exists a Kollár component L of $(W' \ni w, 0)$ over k such that

$$a(L, W', H_{W'}) = 0 \quad \text{and} \quad a(L, W', 0) \leq N_0(d + N_2 - 1).$$

By Lemma 9.3, applied to $V = Y$, $G = \mathcal{G}$, $v = o$, and $\Delta_W = 0$, there exists a unique prime divisor F over Y such that $\mathrm{ord}_F|_{k(W')} = e\,\mathrm{ord}_L$ for a positive integer e. Moreover, F is $\mathcal{G}$-invariant, F is a Kollár component of $(Y \ni o, 0)$ over k, and, since $q^* H_{W'} = H$,

$$a(F, Y, 0) = e a(L, W', 0) \quad \text{and} \quad a(F, Y, H) = e a(L, W', H_{W'}) = 0.$$

By Proposition 13.2(2), applied to X', x', N_2, and φ, the restriction of ord_F to $k(X')$ is $c\,\mathrm{ord}_{E'}$ for a positive integer c and a Kollár component E' of $(X' \ni x', 0)$ over k such that

$$a(E', X', D') = 0 \quad \text{and} \quad N_2 c a(E', X', 0) \leq a(F, Y, 0).$$

For any $g \in H_0$, there exists $\lambda \in k^*$ with $(g, \lambda) \in \mathcal{G}$, since $\chi(g)$ is a root of unity and $\mathbb{C} \subset k$ is algebraically closed, and (g, λ) acts on $k(X') \subset k(Y)$ as g. Since F is $\mathcal{G}$-invariant, $c\,\mathrm{ord}_{E'} \circ g = c\,\mathrm{ord}_{E'}$, so E' is H_0-invariant. Let E be the prime divisor over X with $\mathrm{ord}_{E'}|_{k(X)} = b\,\mathrm{ord}_E$ for a positive integer b. Since $X = X'/H_0$, x' is fixed by H_0, $f^{-1}(x) = \{x'\}$, and $K_{X'} = f^* K_X$, possibly shrinking X around x so that E' is primitive over X', Lemma 9.2 with $\Delta_W = 0$ and $\Delta_V = 0$ implies that E is a Kollár component of $(X \ni x, 0)$ over k. By Lemma 2.5, $a(E', X', 0) = b a(E, X, 0)$ and $a(E', X', D') = b a(E, X, D)$, so $a(E, X, D) = 0$.

Since $\mathcal{G}$ acts on $k(X')$ through H_0, $k(X) = k(X')^{H_0} \subset k(Y)^{\mathcal{G}} = k(W')$. Thus $k(X) \subset k(W') \subset k(Y)$, $\mathrm{ord}_F|_{k(W')} = e\,\mathrm{ord}_L$, and $\mathrm{ord}_F|_{k(X)} = cb\,\mathrm{ord}_E$. By Lemma 9.4, cb/e is a positive integer. Hence

$$N_2 cba(E, X, 0) = N_2 c a(E', X', 0) \leq a(F, Y, 0) = e a(L, W', 0) \leq e N_0(d + N_2 - 1),$$

and

$$a(E, X, 0) \leq \frac{e}{cb} \cdot \frac{N_0(d + N_2 - 1)}{N_2} \leq \frac{N_0(d + N_2 - 1)}{N_2}.$$

□

Proof of Theorem 13.1. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case when x is a k-rational point. Let $\kappa := \kappa(x)$, $x_1 \in X_\kappa$ the κ-rational point induced by x, X_1 the irreducible component of X_κ through x_1, and B_1 and D_1 the pullbacks of B and D to X_1. By Corollary 3.10, X_1 is geometrically integral over κ and $(X_1 \ni x_1, B_1)$ is klt. By Lemma 3.3(1) applied to the finite extension κ/k, X_κ is normal, so X_1 is a connected component of X_κ. Since $\Omega_{X_\kappa/\kappa}$ is the pullback of $\Omega_{X/k}$, K_{X_1} is the pullback of K_X. Thus $K_{X_1} + B_1$ and D_1 are $\mathbb{Q}$-Cartier, and X_1 is $\mathbb{Q}$-Gorenstein near x_1 since X is $\mathbb{Q}$-Gorenstein near x. By Lemma 3.3(1), the pullback of a prime divisor on X to X_κ is reduced, so N_1B_1 and N_2D_1 are Weil divisors. By Lemma 3.3(3), $(X_1, B_1 + D_1)$ is lc. Let $h : Z \to X$ be a projective birational morphism from a normal variety and E_0 a prime divisor on Z with $h(E_0) = x$ and $a(E_0, X, B + D) = 0$. Then $(E_0)_\kappa$ maps onto $x \times_k \kappa$, so some irreducible component E_0° of $(E_0)_\kappa$ maps to x_1, and $\operatorname{center}_{X_\kappa}(E_0^\circ) = x_1$. By Lemma 3.3(1), $\operatorname{ord}_{E_0^\circ}|_{k(X)} = \operatorname{ord}_{E_0}$, and by Lemma 3.3(2), $a(E_0^\circ, X_1, B_1 + D_1) = a(E_0, X, B + D) = 0$. Thus x_1 is an lc center of $(X_1, B_1 + D_1)$.

The field κ is finitely generated over $\mathbb{C}$, and $s(\cdot, N_1, N_2)$ does not depend on the ground field. By Corollary 3.10, applied with $\Theta = B + D$, any Kollár component E_1 of $(X_1 \ni x_1, B_1)$ over κ with $a(E_1, X_1, B_1 + D_1) = 0$ restricts to a Kollár component E of $(X \ni x, B)$ over k with $a(E, X, B + D) = 0$ and $a(E, X, B) = a(E_1, X_1, B_1)$. Hence, possibly replacing k, X, x, B, and D with κ, X_1, x_1, B_1, and D_1, we may assume that x is a k-rational point. By Lemma 3.5(2), X is geometrically integral over k. Possibly shrinking X around x, we may assume that X is affine.

Step 2. In this step we treat the case when $x \notin \operatorname{Supp} B$. Since K_X and $K_X + B$ are $\mathbb{Q}$-Cartier near x, B is $\mathbb{Q}$-Cartier near x. Since (X, B) is klt and $B \geq 0$, X is klt near x [KM98, Corollary 2.35(1)]. Possibly shrinking X around x, we may assume that $B = 0$. Then (X, D) is lc and x is an lc center of (X, D). By Lemma 13.3, there exists a Kollár component E of $(X \ni x, 0)$ over k such that $a(E, X, D) = 0$ and

$$a(E, X, 0) \leq \frac{N_0(d + N_2 - 1)}{N_2} \leq s(d, N_1, N_2).$$

Step 3. In this step we treat the case when $x \in \operatorname{Supp} B$. Since K_X and $K_X + B$ are $\mathbb{Q}$-Cartier near x, B is $\mathbb{Q}$-Cartier near x, and X is klt near x [KM98, Corollary 2.35(1)]. Possibly shrinking X around x, we may assume that X is affine, $\mathbb{Q}$-Gorenstein, and klt. Let $f : X' \to X$ be the index cover of N_1B (Construction-Proposition 9.1), H_0 the Galois group of f, and $x' \in X'$ the unique point over x. Then $\kappa(x') = k$ and H_0 fixes x'. Since $\mathcal{O}_{X',x'}$ is a domain, x' is the only point over x, f is finite, and X' is normal, possibly shrinking X around x, we may assume that X' is irreducible. Let $B' := f^*B$ and $D' := f^*D$. Since f is quasi-étale, $K_{X'} = f^*K_X$, $K_{X'} + B' = f^*(K_X + B)$, and $K_{X'} + B' + D' = f^*(K_X + B + D)$. By Lemma 2.5, X' is $\mathbb{Q}$-Gorenstein and klt, (X', B') is klt, and $(X', B' + D')$ is lc. Since f is étale over the generic point of any prime divisor on X, the pullback of a prime divisor on X is reduced, so N_1B' and N_2D' are Weil divisors. Let $h : Z \to X$ be a projective birational morphism from a normal variety and E_0 a prime divisor on Z with $h(E_0) = x$ and $a(E_0, X, B + D) = 0$, and let E_0' be a prime divisor over E_0 on the normalization of Z in $k(X')$ with $\operatorname{ord}_{E_0'}|_{k(X)} = e_0 \operatorname{ord}_{E_0}$ for a positive integer e_0. The center of E_0' on X' is x', and by Lemma 2.5, $a(E_0', X', B' + D') = e_0 a(E_0, X, B + D) = 0$. Thus x' is an lc center of $(X', B' + D')$.

By Construction-Proposition 9.1, applied to the H_0-linearized invertible ideal sheaf $\mathcal{O}_{X'}(-f^*(N_1B))$ on an H_0-invariant neighborhood of x', there exists $\psi \in \mathcal{O}_{X',x'}$ with $f^*(N_1B) = \operatorname{div}(\psi)$ near x' which is an eigenvector of H_0 with character χ. Since f is finite and x' is the only point over x, possibly shrinking X around x, we may assume that ψ is a regular function on X' and $f^*(N_1B) = \operatorname{div}(\psi)$ on X'. Since $x \in \operatorname{Supp} B$, $\psi(x') = 0$.

Let Y, o, and p be as in Construction 12.8 over k, with X, x, B, N, and g replaced by X', x', B', N_1, and ψ. Thus

$$Y = \Big(\psi + \sum_{j=1}^{N_1-1} y_j^{N_1} = 0\Big) \subset X' \times \mathbb{A}^{N_1-1},$$

$o = (x', 0)$, $p : Y \to X'$ is the projection, and $\dim Y = d + N_1 - 2$. By Proposition 13.2(1), applied to X', x', B', N_1, ψ, and D', the variety Y is normal, $\mathbb{Q}$-Gorenstein, and klt near o, (Y, p^*D') is lc near o, o is an lc center of (Y, p^*D'), p is flat, and $N_2 p^*D'$ is a Weil divisor near o.

Let

$$\mathcal{G} := \{(g, \lambda) \in H_0 \times k^* \mid \lambda^{N_1} = \chi(g)\},$$

a finite group acting on Y by $(g, \lambda) \cdot (x'', y_1, \dots, y_{N_1-1}) = (g \cdot x'', \lambda y_1, \dots, \lambda y_{N_1-1})$. Since $(g, \lambda)^*(\psi + \sum_j y_j^{N_1}) = \chi(g)(\psi + \sum_j y_j^{N_1})$, the action preserves Y. Since o is fixed by $\mathcal{G}$, possibly replacing Y with the connected component through o of a $\mathcal{G}$-invariant open neighborhood of o, we may assume that Y is a normal, $\mathbb{Q}$-Gorenstein, and klt variety, (Y, p^*D') is lc, and $N_2 p^*D'$ is a Weil divisor. Let $q : Y \to W' := Y/\mathcal{G}$ be the quotient morphism and $w := q(o)$. We show that q is quasi-étale. Let $(g, \lambda) \in \mathcal{G}$ be a non-identity element. Since the residue fields have characteristic zero, the order of the inertia group of a prime divisor equals the ramification index along it [Ser79, Chapter I, Proposition 21 and its Corollary]. If $g \neq 1$, then the fixed locus of (g, λ) is contained in $p^{-1}(\mathrm{Fix}(g))$. Since the inertia group of any prime divisor on X' is trivial as f is quasi-étale, $\mathrm{Fix}(g)$ has codimension at least two in X', and since p is flat, $p^{-1}(\mathrm{Fix}(g))$ has codimension at least two in Y [Mat89, Theorem 15.1(ii)]. If $g = 1$, then $\lambda \neq 1$, and the fixed locus of (g, λ) is $Y \cap (X' \times \{0\}) = (\psi = 0) \times \{0\}$. Since $\dim((\psi = 0) \times \{0\}) = d - 1$ and $\dim Y = d + N_1 - 2$, the fixed locus has codimension $N_1 - 1 \geq 2$ in Y. Hence the inertia group of any prime divisor on Y is trivial, and q is quasi-étale.

Since o is fixed by $\mathcal{G}$, we have $q^{-1}(w) = \{o\}$. Moreover, w is a k-rational point. By Lemma 3.5(2), W' is geometrically integral over k, and $\dim W' = d + N_1 - 2$. The $\mathbb{Q}$-divisor p^*D' is $\mathcal{G}$-invariant since $p \circ (g, \lambda) = g \circ p$ and D' is H_0-invariant. Since q is quasi-étale, there exists a $\mathbb{Q}$-divisor $D_{W'}$ on W' with $q^*D_{W'} = p^*D'$, and for any prime divisor P on Y the coefficient of $D_{W'}$ along $q(P)$ equals the coefficient of p^*D' along P. Thus $N_2 D_{W'}$ is a Weil divisor, $K_Y = q^*K_{W'}$, and $K_Y + p^*D' = q^*(K_{W'} + D_{W'})$. By [KM98, Proposition 5.20(1)], applied with $\Delta = 0$ and with $\Delta = D_{W'}$, both $K_{W'}$ and $K_{W'} + D_{W'}$ are $\mathbb{Q}$-Cartier, so $D_{W'}$ is $\mathbb{Q}$-Cartier. By Lemma 2.5, W' is klt and $(W', D_{W'})$ is lc. Let F_0 be a prime divisor over Y with $\mathrm{center}_Y F_0 = o$ and $a(F_0, Y, p^*D') = 0$. By Lemma 2.5, $\mathrm{ord}_{F_0}|_{k(W')}$ is a positive multiple of ord_{L_0} for a prime divisor L_0 over W' with $a(L_0, W', D_{W'}) = 0$ and $\mathrm{center}_{W'} L_0 = w$. Thus w is an lc center of $(W', D_{W'})$.

By Lemma 13.3 in dimension $d + N_1 - 2 \geq 2$, applied to $W' \ni w$ and $D_{W'}$, there exists a Kollár component L of $(W' \ni w, 0)$ over k such that

$$a(L, W', D_{W'}) = 0 \quad \text{and} \quad a(L, W', 0) \leq \frac{N_0(d + N_1 + N_2 - 3)}{N_2}.$$

By Lemma 9.3, applied to $V = Y$, $G = \mathcal{G}$, $v = o$, and $\Delta_W = 0$, there exists a unique prime divisor F over Y such that $\mathrm{ord}_F|_{k(W')} = e\,\mathrm{ord}_L$ for a positive integer e. Moreover, F is $\mathcal{G}$-invariant, F is a Kollár component of $(Y \ni o, 0)$ over k, and, since $q^*D_{W'} = p^*D'$,

$$a(F, Y, 0) = e a(L, W', 0) \quad \text{and} \quad a(F, Y, p^*D') = e a(L, W', D_{W'}) = 0.$$

By Proposition 13.2(1), applied to X', x', B', N_1, ψ, and D', the restriction of ord_F to $k(X')$ is $c\,\mathrm{ord}_{E'}$ for a positive integer c and a Kollár component E' of $(X' \ni x', B')$ over k such that

$$a(E', X', B' + D') = 0 \quad \text{and} \quad a(F, Y, 0) = c a(E', X', B').$$

For any $g \in H_0$, there exists $\lambda \in k^*$ with $(g, \lambda) \in \mathcal{G}$, since $\chi(g)$ is a root of unity and $\mathbb{C} \subset k$ is algebraically closed, and (g, λ) acts on $k(X') \subset k(Y)$ as g. Since F is $\mathcal{G}$-invariant, $c\,\mathrm{ord}_{E'} \circ g = c\,\mathrm{ord}_{E'}$, so E' is H_0-invariant. Let E be the prime divisor over X with $\mathrm{ord}_{E'}|_{k(X)} =$

$b\operatorname{ord}_E$ for a positive integer b. Since $X = X'/H_0$, x' is fixed by H_0, $f^{-1}(x) = \{x'\}$, and $K_{X'} + B' = f^*(K_X + B)$, possibly shrinking X around x so that E' is primitive over X', Lemma 9.2 with $\Delta_W = B$ and $\Delta_V = B'$ implies that E is a Kollár component of $(X \ni x, B)$ over k and $a(E', X', B') = ba(E, X, B)$. By Lemma 2.5, $a(E', X', B' + D') = ba(E, X, B + D)$, so $a(E, X, B + D) = 0$.

Since $\mathcal{G}$ acts on $k(X')$ through H_0, $k(X) = k(X')^{H_0} \subset k(Y)^{\mathcal{G}} = k(W')$. Thus $k(X) \subset k(W') \subset k(Y)$, $\operatorname{ord}_F|_{k(W')} = e\operatorname{ord}_L$, and $\operatorname{ord}_F|_{k(X)} = cb\operatorname{ord}_E$. By Lemma 9.4, cb/e is a positive integer. Hence

$$cba(E, X, B) = ca(E', X', B') = a(F, Y, 0) = ea(L, W', 0) \leq e\frac{N_0(d + N_1 + N_2 - 3)}{N_2},$$

and

$$a(E, X, B) \leq \frac{e}{cb} \cdot \frac{N_0(d + N_1 + N_2 - 3)}{N_2} \leq s(d, N_1, N_2).$$

□

14. Closed points over finitely generated fields

In this section, we prove Theorem 14.1: at any closed point of a klt pair over a field finitely generated over $\mathbb{C}$ with coefficients in a DCC set, there exists a Kollár component whose log discrepancy is bounded in terms of the dimension and the coefficient set. In the bound, the only multiplicative loss is the factor 2^{c_J} of Theorem 4.5. The torus tower of Construction-Theorem 8.5 only raises the dimension, and the degrees of the covers behind Theorem 13.1 cancel out.

Theorem 14.1. *Let $d \geq 2$ be an integer and $I \subset [0, 1]$ a DCC set. Then there exists a positive integer $N = N(d, I)$ depending only on d and I satisfying the following. Let k be a field finitely generated over $\mathbb{C}$, X a normal variety over k of dimension d, $x \in X$ a closed point, and (X, B) a klt pair over k such that $B \in I$. Then there exists a Kollár component E of $(X \ni x, B)$ over k such that $a(E, X, B) \leq N$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we define the constant $N(d, I)$.

Let $J = J(d, I) \subset [0, 1]$ be a finite set satisfying Corollary 7.5, and let $J^+ = J \cup \{0, 1\}$ and c_J be defined by Definition-Theorem 4.1 for the set J. The integer c_J depends only on J. For any integer e such that

$$d \leq e \leq d + 1 + |J\backslash\{0\}|,$$

let ρ_e be the map of Definition-Theorem 4.1 for the dimension e and the set J, and let q_e be the common denominator of the finite set $\rho_e(J^+) \subset \mathbb{Q}$, that is, the smallest positive integer such that $q_e\rho_e(J^+) \subset \mathbb{Z}$. The map ρ_e and the integer q_e vary with e. Let

$$N(d, I) := \left\lceil \max_{d \leq e \leq d+1+|J\backslash\{0\}|} 2^{c_J} s(e, 3q_e, 3q_e) \right\rceil,$$

where $s(e, 3q_e, 3q_e)$ is the constant of Theorem 13.1, namely

$$s(e, 3q_e, 3q_e) = \frac{\max\{N_0(e + 3q_e - 1), N_0(e + 6q_e - 3)\}}{3q_e},$$

and $N_0(\cdot)$ is the function of Definition-Theorem 11.1, whose values are positive integers depending only on the dimension. Thus $N(d, I)$ is a positive integer. (The factor 3 ensures $3q_e \geq 3$, which is the assumption $N_1, N_2 \geq 3$ of Theorem 13.1.) Thus $N(d, I)$ depends only on d and I, through $J = J(d, I)$, c_J, and the integers q_e. In particular, $N(d, I)$ does not depend on the field k, the degree of the residue field of x over k, the number of components of B, or the Cartier indices of divisors on X. The constant $N(d, I)$ is not explicit.

Let k, X, x, and B satisfy the assumptions of Theorem 14.1. We show that E exists with $a(E, X, B) \leq N(d, I)$.

Step 2. In this step we reduce to the case when X is geometrically integral over k.

Let k_0 be the algebraic closure of k in the function field of X. By Lemma 3.5(1), k_0 is a finite extension of k, X is a geometrically integral normal variety over k_0, (X, B) is a klt pair over k_0, and the Kollár components of $(X \ni x, B)$ over k and over k_0 coincide. Let E be a Kollár component of $(X \ni x, B)$ over k (equivalently, over k_0), and, possibly shrinking X around x (shrinking X around x changes neither the Kollár components of $(X \ni x, B)$ nor their log discrepancies), let $f : X' \to X$ be the prime extraction of E. Since the function field of X' is the function field of X, Lemma 3.5(1) for X and for X' implies that the canonical divisors of X and of X' over k coincide with those over k_0. Thus $a(E, X, B)$ computed over k equals $a(E, X, B)$ computed over k_0. Since k_0 is finite over k, k_0 is finitely generated over $\mathbb{C}$, $\dim X = d$ as a variety over k_0, and x is a closed point of X. Possibly replacing k with k_0, we may assume that X is geometrically integral over k.

Step 3. In this step we construct an $\mathbb{R}$-divisor $\Delta \geq B$ with coefficients in J such that (X, Δ) is lc and x is an lc center of (X, Δ).

By Corollary 7.5, possibly shrinking X around x, there exists an $\mathbb{R}$-divisor $\Delta \geq B$ on X such that $\Delta \in J$, (X, Δ) is lc, and x is an lc center of (X, Δ). Thus X, x, B, and Δ satisfy (i) and (ii) of Construction-Theorem 8.5 for the set J. Let B^{eq} be defined by (iii) of Construction-Theorem 8.5.

Step 4. In this step we pass to the top of the torus tower.

By Construction-Theorem 8.5, possibly shrinking X around x, there exist an integer $0 \leq l \leq |J\backslash\{0\}| + 1$, rank one torus quasi-torsors $Y_i \to Y_{i-1}$ over k for $1 \leq i \leq l$ with $Y_0 = X$, and closed points $y_i \in Y_i$ with $y_0 = x$ satisfying (1)–(3) of Construction-Theorem 8.5, where B_i, Δ_i, and B_i^{eq} are the pullbacks of B, Δ, and B^{eq} to Y_i. Let $e := d + l$. Then

$$d \leq e \leq d + 1 + |J\backslash\{0\}|,$$

and by Construction-Theorem 8.5(1), Y_l is a geometrically integral normal variety over k of dimension e, (Y_l, Δ_l) is lc, and y_l is an lc center of (Y_l, Δ_l). By Construction-Theorem 8.5(2), after shrinking Y_l around y_l, K_{Y_l} is $\mathbb{Q}$-Cartier and (Y_l, B_l^{eq}) is klt. We fix such a shrinking, so that we are in the situation of Construction-Theorem 8.5(3).

For any prime divisor P on X, $\operatorname{mult}_P B^{\mathrm{eq}} \in \{0, \operatorname{mult}_P B\}$ and $\operatorname{mult}_P B^{\mathrm{eq}} \in \{0, \operatorname{mult}_P \Delta\}$. Since $B \geq 0$ and $\Delta \in J$, $B^{\mathrm{eq}} \leq B$ and the coefficients of B^{eq} belong to J. By Construction-Theorem 8.5(1), the coefficients of B_l^{eq} and Δ_l belong to J, so B_l^{eq} and Δ_l have coefficients in J^+. By Proposition 8.2(1), the pullback of an $\mathbb{R}$-divisor under a rank one torus quasi-torsor is taken coefficientwise, so $B^{\mathrm{eq}} \leq B$ implies $B_l^{\mathrm{eq}} \leq B_l$. Together with $B_l \leq \Delta_l$ (Construction-Theorem 8.5(1)), we obtain

$$B_l^{\mathrm{eq}} \leq B_l \leq \Delta_l.$$

Step 5. In this step we replace the coefficients of B_l^{eq} and Δ_l by rational numbers.

Let ρ be the map ρ_e of **Step 1**, that is, the map of Definition-Theorem 4.1 for the dimension e and the set J, and let $(B_l^{\mathrm{eq}})^\rho$ and Δ_l^ρ be defined with respect to ρ. By **Step 4**, Y_l, B_l^{eq}, and Δ_l satisfy the assumptions of Theorem 4.5 in dimension e. By Theorem 4.5(1),

$$(B_l^{\mathrm{eq}})^\rho \leq \Delta_l^\rho,$$

both $K_{Y_l} + (B_l^{\mathrm{eq}})^\rho$ and $K_{Y_l} + \Delta_l^\rho$ are $\mathbb{Q}$-Cartier, $(Y_l, (B_l^{\mathrm{eq}})^\rho)$ is klt, and (Y_l, Δ_l^ρ) is lc. By Theorem 4.5(2), the lc places of (Y_l, Δ_l) and (Y_l, Δ_l^ρ) coincide. Since y_l is an lc center of (Y_l, Δ_l), y_l is an lc center of (Y_l, Δ_l^ρ).

Let

$$D := \Delta_l^\rho - (B_l^{\mathrm{eq}})^\rho.$$

Then $D \geq 0$ (as $(B_l^{\mathrm{eq}})^\rho \leq \Delta_l^\rho$), and D is $\mathbb{Q}$-Cartier as

$$D = \left(K_{Y_l} + \Delta_l^\rho\right) - \left(K_{Y_l} + (B_l^{\mathrm{eq}})^\rho\right).$$

Since the coefficients of $(B_l^{\mathrm{eq}})^\rho$ and Δ_l^ρ belong to $\rho(J^+)$ and $q_e\rho(J^+) \subset \mathbb{Z}$, the $\mathbb{Q}$-divisors $q_e(B_l^{\mathrm{eq}})^\rho$ and $q_e\Delta_l^\rho$ have integer coefficients. Hence $3q_e(B_l^{\mathrm{eq}})^\rho$ and $3q_eD$ are Weil divisors.

Step 6. In this step we construct a Kollár component F of $(Y_l \ni y_l, (B_l^{\mathrm{eq}})^\rho)$ which is an lc place of (Y_l, Δ_l^ρ).

By **Steps 4 and 5**, Y_l is a normal variety over k of dimension $e \geq 2$ which is $\mathbb{Q}$-Gorenstein near y_l, $(B_l^{\mathrm{eq}})^\rho \geq 0$ and $D \geq 0$ are $\mathbb{Q}$-divisors such that $K_{Y_l} + (B_l^{\mathrm{eq}})^\rho$ and D are $\mathbb{Q}$-Cartier, $3q_e(B_l^{\mathrm{eq}})^\rho$ and $3q_eD$ are Weil divisors, $(Y_l, (B_l^{\mathrm{eq}})^\rho)$ is klt, $(Y_l, (B_l^{\mathrm{eq}})^\rho + D) = (Y_l, \Delta_l^\rho)$ is lc, and y_l is an lc center of (Y_l, Δ_l^ρ). By Theorem 13.1 for $N_1 = N_2 = 3q_e$, there exists a Kollár component F of $(Y_l \ni y_l, (B_l^{\mathrm{eq}})^\rho)$ over k such that $a(F, Y_l, \Delta_l^\rho) = 0$ and

$$a(F, Y_l, (B_l^{\mathrm{eq}})^\rho) \leq s(e, 3q_e, 3q_e).$$

Step 7. In this step we show that F is a Kollár component of $(Y_l \ni y_l, B_l^{\mathrm{eq}})$ over k with $a(F, Y_l, \Delta_l) = 0$ and $a(F, Y_l, B_l^{\mathrm{eq}}) \leq N(d, I)$.

Possibly shrinking Y_l around y_l, let $f : Y' \to Y_l$ be the prime extraction of F over k. Then f is a projective birational morphism from a normal variety Y' over k with exactly one exceptional prime divisor F (Lemma 2.4(1), as $\mathrm{center}_{Y_l} F = y_l$ is a closed point), $\mathrm{center}_{Y_l} F = y_l$, $-F$ is $\mathbb{Q}$-Cartier and f-ample, and $(Y', f_*^{-1}(B_l^{\mathrm{eq}})^\rho + F)$ is plt near F. By Theorem 4.5(3) for the closed point y_l, $(Y', f_*^{-1}B_l^{\mathrm{eq}} + F)$ is plt near F. Thus F is a Kollár component of $(Y_l \ni y_l, B_l^{\mathrm{eq}})$ over k. Since the lc places of (Y_l, Δ_l) and (Y_l, Δ_l^ρ) coincide (**Step 5**), $a(F, Y_l, \Delta_l) = 0$. By Theorem 4.5(2) for $D = B_l^{\mathrm{eq}}$, and by **Step 6**,

$$a(F, Y_l, B_l^{\mathrm{eq}}) \leq 2^{c_J} a(F, Y_l, (B_l^{\mathrm{eq}})^\rho) \leq 2^{c_J} s(e, 3q_e, 3q_e) \leq N(d, I),$$

where the last inequality follows from the definition of $N(d, I)$ in **Step 1**, as $d \leq e \leq d+1+|J\backslash\{0\}|$ by **Step 4**.

Step 8. In this step we return to (X, B).

Recall that in **Step 4**, Y_l is shrunk around y_l so that (Y_l, B_l^{eq}) is klt. By **Step 7**, F is a Kollár component of $(Y_l \ni y_l, B_l^{\mathrm{eq}})$ over k with $a(F, Y_l, \Delta_l) = 0$. By Construction-Theorem 8.5(3), there exists a Kollár component E of $(X \ni x, B)$ over k such that $a(E, X, \Delta) = 0$ and

$$a(E, X, B) \leq a(F, Y_l, B_l^{\mathrm{eq}}) \leq N(d, I).$$

This proves Theorem 14.1. □

15. Weil indices

In this section, we prove the results on orbifold cones used in the proof of Corollary 1.3, and we prove the assertions of Example 1.4. The Weil index of a Fano variety X becomes the log discrepancy of the zero section over the vertex of an orbifold cone over X, and weak exceptionality of X implies that the zero section is the unique Kollár component over the vertex.

Definition 15.1. A *Fano variety* is a normal projective variety X such that K_X is $\mathbb{Q}$-Cartier and $-K_X$ is ample. A Fano variety X is *weakly exceptional* if (X, D) is lc for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$. The *Weil index* $q_W(X)$ of a Fano variety X is the supremum of the positive integers q such that $-K_X \sim_{\mathbb{Q}} qA$ for some Weil divisor A on X.

Remark 15.2. Weakly exceptional Fano varieties are called *weakly special* in [Zhu24b, Section 7], and we follow the terminology of [CPS10, Definition 1.1.10]. The Weil index is defined in [Zhu24b, Section 7] as the largest such integer q. We define it as a supremum, and Corollary 1.3 implies that $q_W(X) < +\infty$ when X is weakly exceptional, so the supremum is a maximum in this case.

Lemma 15.3. *Let X be a weakly exceptional Fano variety. Then X is klt.*

Proof. Suppose that X is not klt. Then there exists a prime divisor F over X such that $a(F, X, 0) \leq 0$. Let V be the closure of $\mathrm{center}_X F$ and $\mathcal{I}_V$ the ideal sheaf of V. Then $V \neq X$, so $\mathcal{I}_V \neq 0$. Since $-K_X$ is an ample $\mathbb{Q}$-Cartier divisor, there exists a positive integer m_0 such that $-m_0K_X$ is a very ample Cartier divisor [Har77, Theorem II.7.6], and $\mathcal{I}_V \otimes \mathcal{O}_X(-km_0K_X)$ is globally generated for $k \gg 0$ [Har77, Theorem II.5.17]. Let $m := km_0$ for such a k. Thus

$\mathcal{I}_V \otimes \mathcal{O}_X(-mK_X)$ has a non-zero global section. Let $H \geq 0$ be the divisor of a non-zero global section of $\mathcal{I}_V \otimes \mathcal{O}_X(-mK_X)$, so $H \sim -mK_X$. A local equation of H at $\operatorname{center}_X F$ is contained in the maximal ideal of $\mathcal{O}_{X,\operatorname{center}_X F}$, hence $\operatorname{mult}_F H > 0$. Let $D := \frac{1}{m}H$. Then $D \geq 0$, $D \sim_{\mathbb{Q}} -K_X$, and

$$a(F, X, D) = a(F, X, 0) - \frac{1}{m}\operatorname{mult}_F H < 0.$$

Thus (X, D) is not lc, which contradicts the assumption that X is weakly exceptional. □

15.1. **Orbifold cones.** In this subsection, we construct the orbifold cone over a klt variety with an ample $\mathbb{Q}$-Cartier Weil divisor (Construction-Proposition 15.4), and we prove Lemma 15.6.

Construction-Proposition 15.4. Let X be a projective klt variety of dimension $d \geq 1$, A an ample $\mathbb{Q}$-Cartier Weil divisor on X, and q a positive integer such that $-K_X \sim_{\mathbb{Q}} qA$. Let $R(X, A) := \bigoplus_{m\geq 0} H^0(X, \mathcal{O}_X(mA))$, $C(X, A) := \operatorname{Spec} R(X, A)$, $o \in C(X, A)$ the vertex defined by the elements of positive degree, $Y := \operatorname{Spec}_X \bigoplus_{m\geq 0} \mathcal{O}_X(mA)$, and $E \subset Y$ the zero section. Then $R(X, A)$ is a finitely generated normal domain, $\dim C(X, A) = d + 1$, o is a closed point of $C(X, A)$ of codimension $d + 1$, $C(X, A)$ is klt, and the natural morphism $\pi : Y \to C(X, A)$ is projective, birational, and an isomorphism over $C(X, A)\backslash\{o\}$. Moreover, $E \cong X$, the different of (Y, E) on E is zero, $-E$ is $\mathbb{Q}$-Cartier and π-ample, E is a Kollár component of $(C(X, A) \ni o, 0)$, $a(E, C(X, A), 0) = q$, and $\operatorname{ord}_E(\sigma) = m$ for any non-zero $\sigma \in H^0(X, \mathcal{O}_X(mA))$.

Proof. Let $K(X)$ be the function field of X and t an indeterminate. For any integer $m \geq 0$,

$$R(X, A)_m = H^0(X, \mathcal{O}_X(mA)) = \{ft^m \mid f \in K(X)^*,\ \operatorname{div}(f) + mA \geq 0\} \cup \{0\} \subset K(X)[t].$$

Thus any $\sigma \in H^0(X, \mathcal{O}_X(mA))$ is a homogeneous regular function of degree m on $C(X, A)$, and $\operatorname{ord}_E(\sigma)$ is the order along E of the pullback of σ to Y. For any open subset $V \subset X$ and any $m \geq 0$,

$$\Gamma(V, \mathcal{O}_X(mA)) = \{ft^m \mid f \in K(X)^*,\ (\operatorname{div}(f) + mA)|_V \geq 0\} \cup \{0\}.$$

Thus $\mathcal{S} := \bigoplus_{m\geq 0} \mathcal{O}_X(mA)t^m$ is a sheaf of subalgebras of the constant sheaf $K(X)[t]$, and $Y = \operatorname{Spec}_X \mathcal{S}$. Let $p : Y \to X$ be the natural morphism, r a positive integer such that rA is a very ample Cartier divisor, and $\mathcal{S}^{(r)} := \bigoplus_{k\geq 0} \mathcal{O}_X(krA)t^{kr}$.

Since rA is Cartier, $\mathcal{O}_X(mA) \otimes \mathcal{O}_X(rA) \cong \mathcal{O}_X((m + r)A)$ for any $m \geq 0$. Thus $\mathcal{S}$ is generated by $\bigoplus_{0\leq i<r} \mathcal{O}_X(iA)t^i$ as an $\mathcal{S}^{(r)}$-module, and $R(X, A)$ is the direct sum of the modules $\bigoplus_{k\geq 0} H^0(X, \mathcal{O}_X(iA) \otimes \mathcal{O}_X(krA))$, $0 \leq i < r$, over $R^{(r)} := \bigoplus_{k\geq 0} H^0(X, \mathcal{O}_X(krA))$. The algebra $R^{(r)}$ is finitely generated, and $\bigoplus_{k\geq 0} H^0(X, \mathcal{O}_X(iA) \otimes \mathcal{O}_X(krA))$ is a finitely generated $R^{(r)}$-module for any $0 \leq i < r$ [Har77, Theorem II.5.17, Theorem III.5.2, and Exercise II.5.9]. Thus $R(X, A)$ is a finitely generated algebra.

For any prime divisor P on X, let w_P be the Gauss extension of ord_P with weight $\operatorname{mult}_P A$ (Definition 2.8), that is, the valuation of $K(X)(t)$ such that

$$w_P\Big(\sum_m f_m t^m\Big) = \min_m\big(\operatorname{ord}_P(f_m) + m \operatorname{mult}_P A\big),$$

and let $\mathcal{O}_{w_P}$ be the valuation ring of w_P. Since $\sum_m f_m t^m \in K(X)[t]$ belongs to $R(X, A)$ if and only if $\operatorname{ord}_P(f_m) + m \operatorname{mult}_P A \geq 0$ for any P and any m with $f_m \neq 0$, we have

$$R(X, A) = K(X)[t] \cap \bigcap_P \mathcal{O}_{w_P}.$$

Since $K(X)[t]$ and the valuation rings $\mathcal{O}_{w_P}$ are integrally closed in $K(X)(t)$, $R(X, A)$ is integrally closed in $K(X)(t)$. Hence $R(X, A)$ is a normal domain. For any affine open subset $V \subset X$,

$$\Gamma(V, \mathcal{S}) = K(X)[t] \cap \bigcap_{P\cap V\neq\emptyset} \mathcal{O}_{w_P},$$

where P runs over the prime divisors on X such that $P \cap V \neq \emptyset$, so $\Gamma(V, \mathcal{S})$ is integrally closed in $K(X)(t)$. Thus Y is normal.

Let $T := \operatorname{Spec}_X \mathcal{S}^{(r)}$, $E_T \subset T$ the zero section, and $g : Y \to T$ the morphism induced by $\mathcal{S}^{(r)} \subset \mathcal{S}$. Since $\mathcal{S}$ is a finite $\mathcal{S}^{(r)}$-module, g is finite. Since rA is Cartier, T is the total space of the line bundle $\mathcal{O}_X(-rA)$ over X, and $\mathcal{O}_T(-E_T)$ is the pullback of $\mathcal{O}_X(rA)$. Let X° be the smooth locus of X. Then $\operatorname{codim}_X(X\backslash X^\circ) \geq 2$. Locally over X°, $\mathcal{O}_X(A)$ is invertible, Y is the total space of $\mathcal{O}_X(-A)$ with a fiber coordinate u, $E = (u = 0)$, and g is $u \to u^r$. Since g is finite and the fibers of $T \to X$ are lines, $\operatorname{codim}_Y p^{-1}(X\backslash X^\circ) \geq 2$. Thus $g^*E_T = rE$, and since the ground field has characteristic zero, the Hurwitz formula implies $K_Y = g^*K_T + (r-1)E$. Hence

$$K_Y + E = g^*(K_T + E_T) \quad \text{and} \quad \mathcal{O}_Y(-rE) \cong g^*\mathcal{O}_T(-E_T) \cong p^*\mathcal{O}_X(rA).$$

In particular, rE is Cartier and $K_Y + E$ is $\mathbb{Q}$-Cartier.

Locally over X, $(T, E_T) \cong (U \times \mathbb{A}^1, U \times \{0\})$ for an open subset $U \subset X$. Let $h : W \to U$ be a log resolution and write $K_W + B_W := h^*K_U$. Since U is klt, the coefficients of B_W are less than 1. Thus $B_W \times \mathbb{A}^1 + W \times \{0\}$ has snc support, $W \times \{0\}$ has coefficient 1 in $B_W \times \mathbb{A}^1 + W \times \{0\}$, and the other coefficients of $B_W \times \mathbb{A}^1 + W \times \{0\}$ are less than 1. By [KM98, Lemma 2.30 and Corollary 2.31(3)], $(U \times \mathbb{A}^1, U \times \{0\})$ is plt. Thus (T, E_T) is plt, and (Y, E) is plt by Lemma 2.5.

Since E is defined by $\bigoplus_{m>0} \mathcal{O}_X(mA)t^m$ and $\mathcal{S}/\bigoplus_{m>0} \mathcal{O}_X(mA)t^m = \mathcal{O}_X$, p restricts to an isomorphism $E \to X$. Since $E \cap p^{-1}(X^\circ)$ is the zero section of a line bundle over the smooth variety X° and $\operatorname{codim}_E(E\backslash p^{-1}(X^\circ)) \geq 2$, Y and E are smooth at the generic point of any prime divisor on E. Thus $\operatorname{Diff}_E(0) = 0$ [Kol13, Definition 4.2].

Over the generic point of X, Y is $\operatorname{Spec} K(X)[t]$ and E is $(t = 0)$. Thus for any non-zero $\sigma = ft^m \in H^0(X, \mathcal{O}_X(mA))$, $\operatorname{ord}_E(\sigma) = m$, since $f \in K(X)^*$ is a unit at the generic point of E. In particular, $\operatorname{ord}_E(t) = 1$, and the value group of ord_E on $K(X)(t)$ is $\mathbb{Z}$. Since $R(X, A)_0 = H^0(X, \mathcal{O}_X) = \mathbb{C}$ [Har77, Theorem I.3.4(a)], the ideal of elements of positive degree is a maximal ideal, so o is a closed point of $C(X, A)$. Since $\operatorname{ord}_E(\sigma) > 0$ for any non-zero homogeneous $\sigma \in R(X, A)$ of positive degree, $\pi(E) = \{o\}$.

Let $\mathcal{S}' \subset \mathcal{S}$ be the image of $R(X, A) \otimes \mathcal{O}_X \to \mathcal{S}$. Since rA is a very ample Cartier divisor, for any $0 \leq i < r$, $\mathcal{O}_X(iA) \otimes \mathcal{O}_X(krA)$ is globally generated for $k \gg 0$ [Har77, Theorem II.5.17]. Thus there exists a positive integer m_0 such that $\mathcal{S}'_m = \mathcal{S}_m$ for any $m \geq m_0$. Hence $\mathcal{S}$ is generated by the coherent sheaf $\bigoplus_{m<m_0} \mathcal{S}_m$ as an $\mathcal{S}'$-module, so $\mathcal{S}$ is a finite $\mathcal{S}'$-module and $Y \to \operatorname{Spec}_X \mathcal{S}'$ is finite. Since $\operatorname{Spec}_X \mathcal{S}'$ is a closed subscheme of $X \times C(X, A)$ and X is projective, π is proper.

Since X is proper and A is ample, π contracts the zero section E to the point o of the affine variety $C(X, A)$ [Kol04, Paragraph 14]. Thus π is birational, and π restricts to an isomorphism $Y\backslash E \to C(X, A)\backslash\{o\}$. Thus $\pi^{-1}(o)_{\mathrm{red}} = E$. Let $Z \subset Y$ be a closed integral subscheme of positive dimension such that $\pi(Z)$ is a point. Since π is an isomorphism over $C(X, A)\backslash\{o\}$, $Z \subset E$. Since $\mathcal{O}_Y(-rE)|_E \cong \mathcal{O}_X(rA)$ and A is ample, $((-rE)^{\dim Z} \cdot Z) = ((rA)^{\dim Z} \cdot Z) > 0$ under $E \cong X$. Thus the Cartier divisor $-rE$ is π-ample [KM98, Theorem 1.42], and hence $-E$ is π-ample. Since $C(X, A)$ is affine and π is proper, a power of $\mathcal{O}_Y(-rE)$ is very ample over $C(X, A)$ [Har77, Theorem II.7.6], so there exists an immersion $Y \to \mathbb{P}^n_{R(X,A)}$ over $C(X, A)$, which is a closed immersion since π is proper. Thus π is projective. Since the function field of Y is $K(X)(t)$, $\dim C(X, A) = d + 1$, so the closed point o has codimension $d + 1$ [Har77, Exercise II.3.20(a)(b)].

By [Zhu24c, Definition 3.2], $C(X, A)$ is the orbifold cone over (X, A), and since the coefficients of A are integers, the different attached to A in [Zhu24c, Section 3.1] is zero. Since X is klt, A is ample, and $-K_X \sim_{\mathbb{Q}} qA$, $C(X, A)$ is klt by [Zhu24c, Lemma 3.3], and $C(X, A)\backslash\{o\} \cong Y\backslash E$ is klt, with $\mathbb{Q}$-Cartier canonical divisor, since (Y, E) is plt. In particular, $K_{C(X,A)}$ is $\mathbb{Q}$-Cartier. Since π is an isomorphism over $C(X, A)\backslash\{o\}$ and $\pi^{-1}(o)_{\mathrm{red}} = E$, there exists $c \in \mathbb{Q}$ such that

$$K_Y + E = \pi^*K_{C(X,A)} + cE,$$

and $a(E, C(X, A), 0) = c$ by Definition 2.1. Since $\pi(E) = \{o\}$, $(\pi^*K_{C(X,A)})|_E \sim_{\mathbb{Q}} 0$. Under $E \cong X$, the adjunction formula with $\operatorname{Diff}_E(0) = 0$ and $\mathcal{O}_Y(-rE)|_E \cong \mathcal{O}_X(rA)$ imply

$$K_X \sim_{\mathbb{Q}} (K_Y + E)|_E \sim_{\mathbb{Q}} (\pi^*K_{C(X,A)} + cE)|_E \sim_{\mathbb{Q}} -cA.$$

Since $-K_X \sim_{\mathbb{Q}} qA$, we have $(c-q)A \sim_{\mathbb{Q}} 0$, so $(c-q)(A^d) = 0$. Since A is ample and $d \geq 1$, $(A^d) > 0$. Thus $a(E, C(X,A), 0) = c = q$, which is also the value of [Zhu24c, Lemma 3.3].

Since Y is normal and $-E$ is $\mathbb{Q}$-Cartier and π-ample, π is the prime extraction of E. Since $C(X,A)$ is klt, $\pi(E) = \{o\}$, and (Y,E) is plt, E is a Kollár component of $(C(X,A) \ni o, 0)$. □

Remark 15.5. We use the section ring of A instead of the section ring of rA, where r is the Cartier index of A: over the vertex of $\operatorname{Spec} \bigoplus_{k\geq 0} H^0(X, \mathcal{O}_X(krA))$, the zero section has log discrepancy q/r [Zhu24c, Lemma 3.3], so Theorem 1.2 only implies $q \leq rN$, and the Cartier index r of A is not bounded (for instance, for the hypersurface X_{8n+4} of Section 1, $A = -K_{X_{8n+4}}$, which corresponds to $\mathcal{O}_{X_{8n+4}}(1)$, has Cartier index divisible by $4n+1$).

In the proof of Corollary 1.3, the following lemma shows that the Kollár component over the vertex provided by Theorem 1.2 is the zero section E. For K-semistable Fano varieties, the Kollár component provided by Theorem 1.2 may differ from E, see Example 1.4.

Lemma 15.6. *Notation and conditions as in Construction-Proposition 15.4. Assume that X is weakly exceptional. Then E is the unique Kollár component of $(C(X,A) \ni o, 0)$.*

Proof. By Construction-Proposition 15.4, o is a closed point. Let F be a Kollár component of $(C(X,A) \ni o, 0)$. Then there exist an open neighborhood $U \subset C(X,A)$ of o and the prime extraction $f : Y_F \to U$ of F such that (Y_F, F) is plt near F. Since $f(F) = \{o\}$ (Definition 2.3), F is f-exceptional as $\dim U = d+1 \geq 2$, and $-F$ is $\mathbb{Q}$-Cartier and f-ample, Lemma 2.4(1) implies that f is an isomorphism over $U\backslash\{o\}$ and $\operatorname{Supp} f^{-1}(o) = F$. Since $Y_F \backslash F \cong U\backslash\{o\}$ is klt, (Y_F, F) is plt. Thus f is a plt blow up of $o \in U$ in the sense of [CS14, Section 1].

By Construction-Proposition 15.4, $C(X,A)$ is klt, E is a Kollár component of $(C(X,A) \ni o, 0)$ with prime extraction π, $E \cong X$, and $\operatorname{Diff}_E(0) = 0$. Hence π is a plt blow up of $o \in C(X,A)$ in the sense of [CS14, Section 1]. Since X is weakly exceptional, $(E, \operatorname{Diff}_E(0) + D_E)$ is lc for any $\mathbb{Q}$-divisor $D_E \geq 0$ with $D_E \sim_{\mathbb{Q}} -(K_E + \operatorname{Diff}_E(0))$ (Definition 15.1). By [CS14, Theorem 1.5] (due to [Pro00, Theorem 4] for $\mathbb{Q}$-factorial singularities of dimension at most three and to [Kud01, Theorem 2.1] in general), $o \in C(X,A)$ has a unique plt blow up (also see [CPS18, Corollary 12], which follows from [CPS18, Theorem 11]). Thus f coincides with π over U, and $F = E$. □

15.2. **A K-semistable toric surface.** In this subsection, we prove the claims in Example 1.4.

Lemma 15.7. *Let $\Lambda \subset \mathbb{Q}^2$ be a lattice of rank two generated by $w_1, \dots, w_n$. The* covolume *of Λ is $|\det(b_1, b_2)|$ for a basis b_1, b_2 of Λ. Then the covolume of Λ is the positive generator of the subgroup of $\mathbb{Q}$ generated by $\det(w_i, w_j)$, $1 \leq i < j \leq n$. Moreover, for any non-zero $w \in \Lambda$, the largest positive integer e with $w/e \in \Lambda$ is the positive generator of the subgroup of $\mathbb{Q}$ generated by $\det(w, w_i)$, $1 \leq i \leq n$, divided by the covolume of Λ.*

Proof. Two bases of Λ differ by a matrix in $\mathrm{GL}_2(\mathbb{Z})$, so the covolume does not depend on the basis. Let b_1, b_2 be a basis of Λ, let B be the 2×2 matrix with columns b_1, b_2, and write $w_i = Bx_i$ with $x_i \in \mathbb{Z}^2$. Then

$$\det(w_i, w_j) = \det(B)\det(x_i, x_j) \quad \text{for any } i, j.$$

Since $w_1, \dots, w_n$ generate Λ, the vectors $x_1, \dots, x_n$ generate $\mathbb{Z}^2$. Thus the Smith normal form of the $2 \times n$ integer matrix with columns $x_1, \dots, x_n$ has diagonal entries $1, 1$. Since the greatest common divisor of the 2×2 minors of an integer matrix is invariant under left and right multiplication by invertible integer matrices, $\gcd_{i<j} \det(x_i, x_j) = 1$. Hence the subgroup of $\mathbb{Q}$ generated by the $\det(w_i, w_j)$ is $|\det B|\mathbb{Z}$, which proves the first part of the lemma.

Let $w = Bx$ with $x = (a,b) \in \mathbb{Z}^2\backslash\{0\}$. Since $x_1, \dots, x_n$ generate $\mathbb{Z}^2$, the subgroup of $\mathbb{Z}$ generated by $\det(x, x_i)$, $1 \leq i \leq n$, is

$$\{\det(x,y) \mid y \in \mathbb{Z}^2\} = \{ay_2 - by_1 \mid y_1, y_2 \in \mathbb{Z}\} = \gcd(a,b)\mathbb{Z}.$$

Thus the subgroup of $\mathbb{Q}$ generated by $\det(w, w_i)$, $1 \leq i \leq n$, is $\gcd(a,b)|\det B|\mathbb{Z}$. On the other hand, for a positive integer e, $w/e \in \Lambda$ if and only if $x/e \in \mathbb{Z}^2$, that is, $e \mid \gcd(a,b)$. So the largest such e is $\gcd(a,b)$, which proves the second part of the lemma. □

Lemma 15.8. *Let Λ be a lattice of rank two, Σ a complete fan in $\Lambda\otimes\mathbb{R}$, and X the toric surface associated to Σ. Let $v_0,\dots,v_r$ be the primitive generators of the rays of Σ, and assume that $v_0,\dots,v_r$ generate Λ and that the subgroup $\Lambda'\subset\Lambda$ generated by $v_1-v_0,\dots,v_r-v_0$ has rank two. Then the largest positive integer q such that $-K_X\sim_{\mathbb{Q}} qA$ for some Weil divisor A on X exists and is equal to the index $[\Lambda:\Lambda']$.*

Proof. Let $M:=\mathrm{Hom}(\Lambda,\mathbb{Z})$ with pairing $\langle\cdot,\cdot\rangle$, let χ^u be the character of $u\in M$, and let D_i be the torus-invariant prime divisor corresponding to the ray $\mathbb{R}_{\geq 0}v_i$. By [CLS11, Theorem 8.2.3] and [CLS11, Proposition 4.1.2],

$$K_X=-\sum_{i=0}^{r}D_i,\qquad \mathrm{div}(\chi^u)=\sum_{i=0}^{r}\langle u,v_i\rangle D_i\quad\text{for any } u\in M. \tag{15.1}$$

Since Λ' has rank two, $q_0:=[\Lambda:\Lambda']$ is finite. Since $v_i-v_0\in\Lambda'$ for any i and $v_0,\dots,v_r$ generate Λ, the group Λ/Λ' is generated by the class $[v_0]$, and $[v_i]=[v_0]$ for any i. Thus Λ/Λ' is cyclic of order q_0.

Let q be a positive integer and A a Weil divisor on X such that $-K_X\sim_{\mathbb{Q}} qA$. We show that $q\mid q_0$. By [CLS11, Theorem 4.1.3], A is linearly equivalent to $\sum_{i=0}^{r}a_iD_i$ for some integers $a_0,\dots,a_r$. The condition $-K_X\sim_{\mathbb{Q}} qA$ depends only on the linear equivalence class of A. Possibly replacing A with $\sum_{i=0}^{r}a_iD_i$, we may assume that $A=\sum_{i=0}^{r}a_iD_i$. Then there exists a positive integer l such that $\sum_{i=0}^{r}l(1-qa_i)D_i$ is principal. By the exactness of the sequence $M\to\mathrm{Div}_T(X)\to\mathrm{Cl}(X)\to 0$ of [CLS11, Theorem 4.1.3] and by (15.1), there exists $u'\in M$ with $l(1-qa_i)=\langle u',v_i\rangle$ for any i. Let $u:=u'/l\in M\otimes\mathbb{Q}$. Then

$$1-qa_i=\langle u,v_i\rangle\quad\text{for } 0\leq i\leq r. \tag{15.2}$$

Thus $\langle u,v_i\rangle\in\mathbb{Z}$ for any i, and since $v_0,\dots,v_r$ generate Λ, $u\in M$. By (15.2),

$$\langle u,v_i-v_0\rangle=q(a_0-a_i)\in q\mathbb{Z},\qquad \langle u,v_0\rangle\equiv 1\pmod q.$$

Hence the homomorphism $\Lambda\to\mathbb{Z}/q\mathbb{Z}$, $v\to\langle u,v\rangle \bmod q$, vanishes on Λ' and is surjective. So it induces a surjection $\Lambda/\Lambda'\to\mathbb{Z}/q\mathbb{Z}$, and $q\mid q_0$.

Conversely, let $\varphi:\Lambda/\Lambda'\to\mathbb{Z}/q_0\mathbb{Z}$ be the isomorphism with $\varphi([v_0])=1$. Since Λ is free, the composite $\Lambda\to\Lambda/\Lambda'\to\mathbb{Z}/q_0\mathbb{Z}$ lifts to a homomorphism $u:\Lambda\to\mathbb{Z}$, that is, $u\in M$. Since $[v_i]=[v_0]$, $\langle u,v_i\rangle\equiv 1\pmod{q_0}$ for any i. Let $a_i:=(1-\langle u,v_i\rangle)/q_0\in\mathbb{Z}$ and $A:=\sum_{i=0}^{r}a_iD_i$. By (15.1),

$$-K_X=\sum_{i=0}^{r}D_i=q_0A+\mathrm{div}(\chi^u).$$

So $-K_X\sim q_0A$, which is stronger than $-K_X\sim_{\mathbb{Q}} q_0A$. Therefore $q=q_0$ satisfies $-K_X\sim_{\mathbb{Q}} qA$ for some Weil divisor A, and any positive integer q such that $-K_X\sim_{\mathbb{Q}} qA$ for some Weil divisor A divides q_0. Thus the largest such q exists and is equal to $q_0=[\Lambda:\Lambda']$. □

Lemma 15.9. *Let m be a positive integer divisible by $2\cdot 3\cdot 5^2\cdot 7\cdot 11\cdot 31$, and let*

$$\begin{aligned} g_{01}&:=3(m-1)(7m+5)(11m+1), & g_{02}&:=(2m+3)(7m-5)(11m+1),\\ g_{03}&:=-5m(7m-5)(7m+5), & g_{12}&:=m(11m-1)(11m+1),\\ g_{13}&:=(2m-3)(7m+5)(11m-1), & g_{23}&:=3(m+1)(7m-5)(11m-1). \end{aligned}$$

Then

$$\gcd(g_{01},g_{02},g_{03})=\gcd(g_{01},g_{12},g_{13})=\gcd(g_{02},g_{12},g_{23})=\gcd(g_{03},g_{13},g_{23})=15.$$

Proof. We call (g_{01},g_{02},g_{03}), (g_{01},g_{12},g_{13}), (g_{02},g_{12},g_{23}), and (g_{03},g_{13},g_{23}) the *rows* 0, 1, 2, and 3. Let $S:=\{2,3,5,7,11,31\}$.

We first show that no prime $p\notin S$ divides the greatest common divisor of a row. Rows 0 and 1 contain g_{01}, and rows 2 and 3 contain g_{23}. If $p\mid m$, then

$$g_{01}\equiv 3\cdot(-1)\cdot 5\cdot 1=-15\pmod p,\qquad g_{23}\equiv 3\cdot 1\cdot(-5)\cdot(-1)=15\pmod p,$$

and $p \nmid 15$. Assume that $p \nmid m$. If p divides the greatest common divisor of row 0 or row 3, then $p \mid g_{03} = -5m(7m-5)(7m+5)$ and $p \nmid 5m$, so $p \mid 7m-5$ or $p \mid 7m+5$. Similarly, if p divides the greatest common divisor of row 1 or row 2, then $p \mid g_{12} = m(11m-1)(11m+1)$ and $p \nmid m$, so $p \mid 11m-1$ or $p \mid 11m+1$. For $0 \le i < j \le 3$, let $g_{ij}(s) \in \mathbb{Z}[s]$ be the polynomial obtained by replacing m with a variable s in the definition of g_{ij}, so that $g_{ij} = g_{ij}(m)$. Since 7 and 11 are invertible modulo p, if $p \mid 7m \mp 5$ (resp. $p \mid 11m \mp 1$), then for $g(s) \in \mathbb{Z}[s]$, the image of $g(m)$ in $\mathbb{Z}/p\mathbb{Z}$ is the image of the rational number $g(\pm 5/7)$ (resp. $g(\pm 1/11)$), whose denominator is prime to p. A direct calculation shows the values in Table 1. Since

$$3720 = 2^3 \cdot 3 \cdot 5 \cdot 31, \qquad 5280 = 2^5 \cdot 3 \cdot 5 \cdot 11, \qquad 3360 = 2^5 \cdot 3 \cdot 5 \cdot 7,$$

p divides none of the numerators in Table 1. Thus in any case one entry of the row is non-zero modulo p.

TABLE 1. For each row and each prime factor of g_{03} or g_{12} outside S and prime to m, an entry of the row and the value of the entry at the corresponding root.

row	condition	entry	value
0	$p \mid 7m-5$	g_{01}	$g_{01}(5/7) = -3720/49$
0	$p \mid 7m+5$	g_{02}	$g_{02}(-5/7) = 5280/49$
1	$p \mid 11m-1$	g_{01}	$g_{01}(1/11) = -3720/121$
1	$p \mid 11m+1$	g_{13}	$g_{13}(-1/11) = 3360/121$
2	$p \mid 11m-1$	g_{02}	$g_{02}(1/11) = -3360/121$
2	$p \mid 11m+1$	g_{23}	$g_{23}(-1/11) = 3720/121$
3	$p \mid 7m-5$	g_{13}	$g_{13}(5/7) = -5280/49$
3	$p \mid 7m+5$	g_{23}	$g_{23}(-5/7) = 3720/49$

Next we treat the primes in S. Since $15 \mid m$, we have $5 \mid 7m+5$, $5 \mid 7m-5$, $3 \mid 2m+3$, and $3 \mid 2m-3$. Hence 15 divides each of $g_{01}, \dots, g_{23}$. Since m is even, the factors $m \mp 1$, $7m \pm 5$, $11m \pm 1$ are odd, so g_{01} and g_{23} are odd. Since 7, 11, and 31 divide m, the congruences $g_{01} \equiv -15$ and $g_{23} \equiv 15$ modulo $p \mid m$ above show that 7, 11, and 31 divide neither g_{01} nor g_{23}. As g_{01} belongs to rows 0 and 1, and g_{23} belongs to rows 2 and 3, the primes 2, 7, 11, 31 do not divide the greatest common divisor of any row.

It remains to show that the 3-adic and 5-adic valuations of g_{01} and g_{23} are 1. Since $3 \mid m$,

$$m-1 \equiv 2, \quad 7m+5 \equiv 2, \quad 11m+1 \equiv 1, \quad m+1 \equiv 1, \quad 7m-5 \equiv 1, \quad 11m-1 \equiv 2 \pmod 3,$$

so $v_3(g_{01}) = v_3(g_{23}) = 1$. Since $25 \mid m$,

$$7m+5 \equiv 5 \pmod{25}, \qquad 7m-5 \equiv -5 \pmod{25},$$

so $v_5(7m+5) = v_5(7m-5) = 1$, while $m \mp 1 \equiv \mp 1$ and $11m \pm 1 \equiv \pm 1$ modulo 5. Hence $v_5(g_{01}) = v_5(g_{23}) = 1$. Therefore the greatest common divisor of each row is $3 \cdot 5 = 15$. □

The choice of Q_t was suggested by the parametrization in [Hal24, Remark 5.7]. In the proof below, the K-semistability of X_t is reduced to the vanishing of the barycenter of the polygon of $-K_{X_t}$ [BJ20, Corollary 7.17], and $q_W(X_t)$ is computed as a lattice index by Lemma 15.8, which requires the vertices of Q_t to be primitive in Λ_t. The divisibility of m by 358050 is used to obtain the primitivity of the vertices of Q_t through Lemma 15.9, and the divisibility of m by 2 and 5 is also used in the computation of $q_W(X_t)$.

Proof of the claims in Example 1.4. The proof is long, so we divide it into several steps. Recall from Example 1.4 that $m = 358050t = 2 \cdot 3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 31 \cdot t$, that Λ_t is the group generated by $v_0, \dots, v_3$, that Q_t is the convex hull of $v_0, \dots, v_3$, that Σ_t consists of the zero cone and the cones over the proper faces of Q_t, and that X_t is the toric surface associated to Σ_t. In particular $m \ge 358050$, and m satisfies the hypothesis of Lemma 15.9. We pair $\mathbb{Q}^2$ with itself by the dot product $\langle \cdot, \cdot \rangle$.

Step 1. In this step we construct a quadrilateral P with 0 in the interior and with barycenter 0. Let

$$u_1 := \Big(-\frac{11m+1}{8m}, -\frac{11m-1}{8m}\Big), \quad u_2 := (1,0), \quad u_3 := \Big(\frac{7m+5}{8m}, \frac{7m-5}{8m}\Big), \quad u_4 := (0,1),$$

and $P := \mathrm{Conv}(u_1,u_2,u_3,u_4) \subset \mathbb{R}^2$. With indices modulo 4, a direct calculation shows

$$\big(\det(u_j - u_{j-1}, u_{j+1} - u_j)\big)_{j=1,2,3,4} = \Big(\frac{15}{4}, \frac{9(m-1)}{4m}, \frac{3}{4}, \frac{9(m+1)}{4m}\Big).$$

All four determinants are positive since $m \geq 358050$, so P is a convex quadrilateral with vertices u_1,u_2,u_3,u_4 in counterclockwise order. Both coordinates of u_1 are negative, so 0 is a combination of u_1, u_2, u_4 with positive coefficients, and 0 is in the interior of $T := \mathrm{Conv}(u_1,u_2,u_4) \subset P$.

Let $T' := \mathrm{Conv}(u_2,u_3,u_4)$. The sum of the coordinates is $-11/4$ at u_1, 1 at u_2 and u_4, and $7/4$ at u_3. So T and T' are on opposite sides of the line through u_2 and u_4, $P = T \cup T'$, and $T \cap T' = \mathrm{Conv}(u_2,u_4)$. A direct calculation shows

$$\mathrm{area}(T) = \frac{15}{8}, \qquad \mathrm{area}(T') = \frac{3}{8}, \qquad 5u_1 + 6u_2 + u_3 + 6u_4 = 0.$$

Since the barycenter of a triangle is the mean of the vertices, and the barycenter of P is the average of the barycenters of T and T' weighted by their areas, the barycenter of P is

$$\frac{1}{3\,\mathrm{area}(P)}\Big(\frac{15}{8}(u_1+u_2+u_4) + \frac{3}{8}(u_2+u_3+u_4)\Big) = \frac{1}{8\,\mathrm{area}(P)}(5u_1+6u_2+u_3+6u_4) = 0. \tag{15.3}$$

Step 2. In this step we show that Q_t is the polar of P, that 0 is in the interior of Q_t, and that Λ_t is a lattice of rank two. We use the polar

$$P^\vee := \{v \in \mathbb{R}^2 \mid \langle u, v\rangle \geq -1 \text{ for any } u \in P\}.$$

A direct calculation shows that, with indices of u modulo 4,

$$\langle u_{j+1}, v_j\rangle = \langle u_{j+2}, v_j\rangle = -1 \quad \text{for } 0 \leq j \leq 3. \tag{15.4}$$

So v_0, v_1, v_2, v_3 correspond to the edges $\mathrm{Conv}(u_1,u_2)$, $\mathrm{Conv}(u_2,u_3)$, $\mathrm{Conv}(u_3,u_4)$, $\mathrm{Conv}(u_4,u_1)$ of P. Since 0 is in the interior of P, the line $\{\langle u, v_j\rangle = -1\}$ through the edge $\mathrm{Conv}(u_{j+1},u_{j+2})$ has P on the side containing 0. Thus

$$P = \{u \in \mathbb{R}^2 \mid \langle u, v_j\rangle \geq -1 \text{ for } 0 \leq j \leq 3\},$$

and v_j is an inner normal of the edge $\mathrm{Conv}(u_{j+1},u_{j+2})$, normalized so that $\langle u, v_j\rangle = -1$ for $u \in \mathrm{Conv}(u_{j+1},u_{j+2})$ by (15.4). By [CLS11, Section 2.2 and Exercise 2.2.1],

$$P^\vee = \mathrm{Conv}(v_0,v_1,v_2,v_3) = Q_t, \qquad Q_t^\vee = (P^\vee)^\vee = P. \tag{15.5}$$

Since P is bounded, there exists $\epsilon > 0$ such that $|\langle u, v\rangle| \leq 1$ for any $u \in P$ and any v with $|v| \leq \epsilon$. So the ball of radius ϵ is contained in $P^\vee = Q_t$, and 0 is in the interior of Q_t.

Let h be a common denominator of the coordinates of $v_0,\ldots,v_3$. Then $\Lambda_t \subset \frac{1}{h}\mathbb{Z}^2$, so Λ_t is a free abelian group of rank at most two. The first coordinates of v_0 and v_1 are both -1, while their second coordinates satisfy

$$\frac{19m+1}{11m-1} > 0 > \frac{5-m}{7m-5}$$

since $m \geq 358050$. So v_0 and v_1 are linearly independent, and Λ_t is a lattice of rank two.

Step 3. In this step we show that $v_0,\ldots,v_3$ are primitive in Λ_t and compute the covolume of Λ_t. Let

$$\Pi := (7m-5)(7m+5)(11m-1)(11m+1), \qquad c_0 := \frac{48m}{\Pi} > 0,$$

and let g_{ij}, $0 \leq i < j \leq 3$, be the integers in Lemma 15.9. A direct calculation shows

$$\det(v_i, v_j) = c_0 g_{ij} \quad \text{for } 0 \leq i < j \leq 3. \tag{15.6}$$

By Lemma 15.9, $\gcd(g_{01}, g_{02}, g_{03}, g_{12}, g_{13}, g_{23}) = 15$. Thus by Lemma 15.7 and (15.6), the covolume of Λ_t is

$$\operatorname{covol}(\Lambda_t) = 15c_0. \tag{15.7}$$

For any i, the numbers $\det(v_i, v_j)$, $j \neq i$, are $\pm c_0$ times the entries of row i in Lemma 15.9. So by Lemma 15.9 they generate $15c_0\mathbb{Z}$. By Lemma 15.7 and (15.7), the largest positive integer e with $v_i/e \in \Lambda_t$ is $15c_0/(15c_0) = 1$. Thus v_i is primitive in Λ_t.

Step 4. In this step we show that Σ_t is a complete fan, that X_t is a $\mathbb{Q}$-factorial klt Fano surface, and that $\delta(X_t) = 1$. Let $M_t := \operatorname{Hom}(\Lambda_t, \mathbb{Z})$, identified by $\langle \cdot, \cdot \rangle$ with a lattice in $\mathbb{Q}^2$. Since $u_1, \dots, u_4 \in \mathbb{Q}^2 = M_t \otimes \mathbb{Q}$, there exists a positive integer r such that rP is a two-dimensional lattice polygon with respect to M_t. Since $v_0, \dots, v_3$ are primitive in Λ_t (**Step 3**),

$$rP = \{u \in \mathbb{R}^2 \mid \langle u, v_j \rangle \geq -r \text{ for } 0 \leq j \leq 3\}$$

is the presentation of rP by primitive inner normals of the edges. By (15.4), the vertex ru_{j+1} of rP is on the edges with inner normals v_{j-1} and v_j (indices of v modulo 4), so the maximal cone of the inner normal fan of rP at ru_{j+1} is the cone over $\operatorname{Conv}(v_{j-1}, v_j)$, which is an edge of Q_t by (15.5). Thus Σ_t is the inner normal fan of rP. Hence Σ_t is a complete fan [CLS11, Theorem 2.3.2 and Proposition 2.3.8(b)].

For $0 \leq j \leq 3$, let D_j be the torus-invariant prime divisor of X_t corresponding to the ray $\mathbb{R}_{\geq 0} v_j$. Since $v_0, \dots, v_3$ are the primitive inner normals in the presentation of rP above, the divisor of rP defined in [CLS11, (4.2.6) and (4.2.7)] is

$$D_{rP} = r \sum_{j=0}^{3} D_j,$$

and it is an ample Cartier divisor on X_t [CLS11, Proposition 6.1.10(a)]. In particular, X_t is projective, and X_t is normal [CLS11, Theorem 3.1.5]. By [CLS11, Theorem 8.2.3], $K_{X_t} = -\sum_{j=0}^{3} D_j$, so $D_{rP} = -rK_{X_t}$. So K_{X_t} is $\mathbb{Q}$-Cartier and $-K_{X_t}$ is ample. The maximal cones of Σ_t are generated by pairs of linearly independent vectors, so Σ_t is simplicial, and X_t is $\mathbb{Q}$-factorial [CLS11, Proposition 4.2.7]. Since K_{X_t} is $\mathbb{Q}$-Cartier, X_t is klt [CLS11, Proposition 11.4.24(b)]. Thus X_t is a $\mathbb{Q}$-factorial klt Fano surface.

The polygon of $-K_{X_t} = \sum_{j=0}^{3} D_j$ is $\{u \mid \langle u, v_j \rangle \geq -1 \text{ for } 0 \leq j \leq 3\} = P$, and the barycenter $\bar{u}$ of P is 0 by (15.3). By [BJ20, Corollary 7.17], X_t is K-semistable (compare [BB13, Theorem 1.2]), and by [BJ20, Corollary 7.16],

$$\delta(X_t) = \min_{0 \leq j \leq 3} \frac{1}{\langle \bar{u}, v_j \rangle + 1} = 1.$$

Step 5. In this step we show that $q_W(X_t) = 429660t$. Let $w_i := v_i - v_0$ for $i = 1, 2, 3$, and let $\Lambda'_t := \mathbb{Z}w_1 + \mathbb{Z}w_2 + \mathbb{Z}w_3 \subset \Lambda_t$. Recall that $c_0 = 48m/\Pi$ is the positive number of **Step 3**. A direct calculation shows

$$\det(w_1, w_2) = 18mc_0 g'_1, \qquad \det(w_1, w_3) = 18mc_0 g'_2, \qquad \det(w_2, w_3) = 18mc_0 g'_3, \tag{15.8}$$

where

$$g'_1 := (m-1)(11m+1), \qquad g'_2 := 5(m-1)(7m+5), \qquad g'_3 := 5(m+1)(7m-5).$$

In particular, $\det(w_1, w_2) \neq 0$, so Λ'_t has rank two.

We show that $\gcd(g'_1, g'_2, g'_3) = 1$. A direct calculation shows $g'_3 - g'_2 = 20m$, and $g'_1 \equiv -1 \pmod{m}$. So a common divisor of g'_1, g'_2, g'_3 divides $20m$ and is prime to m, hence divides 20. Since m is even and $5 \mid m$, g'_1 is odd and $g'_1 \equiv -1 \pmod{5}$. Thus $\gcd(g'_1, g'_2, g'_3) = 1$, and by Lemma 15.7 and (15.8),

$$\operatorname{covol}(\Lambda'_t) = 18mc_0.$$

If C is the integer matrix of the coordinates of a basis of Λ'_t with respect to a basis of Λ_t, then $\operatorname{covol}(\Lambda'_t) = |\det C| \operatorname{covol}(\Lambda_t)$, and $[\Lambda_t : \Lambda'_t] = |\det C|$ by the Smith normal form of C. By (15.7),

$$[\Lambda_t : \Lambda'_t] = \frac{\operatorname{covol}(\Lambda'_t)}{\operatorname{covol}(\Lambda_t)} = \frac{18mc_0}{15c_0} = \frac{6m}{5} = 429660t.$$

By **Steps 3 and 4**, $v_0, \dots, v_3$ are the primitive generators of the rays of the complete fan Σ_t, and they generate Λ_t by definition. By Definition 15.1, $q_W(X_t)$ is the supremum of the positive integers q such that $-K_{X_t} \sim_{\mathbb{Q}} qA$ for some Weil divisor A on X_t. By Lemma 15.8, the supremum is a maximum and

$$q_W(X_t) = [\Lambda_t : \Lambda'_t] = 429660t.$$

Since t is an arbitrary positive integer and X_t is a K-semistable Fano surface by **Step 4**, the Weil indices of K-semistable Fano surfaces are not bounded. □

16. Proofs of the main theorems

In this section, we prove Theorem 1.2 and Corollary 1.3.

Proof of Theorem 1.2. We show that Theorem 1.2 holds with $N = N(d, I)$, where $N(1, I) := 1$ and, for $d \geq 2$, $N(d, I)$ is the integer given by Theorem 14.1. Let (X, B) be a klt pair such that $B \in I$, and let $\eta \in X$ be a point of codimension d. We show that there exists a Kollár component E of $(X \ni \eta, B)$ such that $a(E, X, B) \leq N(d, I)$. The inequality $\operatorname{mld}^K(X \ni \eta, B) \leq N(d, I)$ then follows from Definition 2.3.

Case 1. $d = 1$. Let b be the coefficient of $\overline{\{\eta\}}$ in B, where $b := 0$ if $\overline{\{\eta\}}$ is not a component of B. Since X is normal, $\mathcal{O}_{X,\eta}$ is a DVR [Har77, Theorem I.6.2A], so η is a smooth point of X. Since the set of points of a variety with regular local ring is open [Har77, Exercise II.8.1(d)], applied to X and to $\overline{\{\eta\}}$, there exists an open neighborhood U of η such that U and $E := \overline{\{\eta\}} \cap U$ are smooth, and the components of B other than $\overline{\{\eta\}}$ do not intersect U. Then E is a Cartier divisor on U with $\operatorname{center}_X E = \eta$, and the identity morphism of U is the prime extraction of E. Moreover, since U and E are smooth, (U, E) is plt [KM98, Corollary 2.31(3)]. Thus E is a Kollár component of $(X \ni \eta, B)$, and since $b \geq 0$,

$$a(E, X, B) = 1 - b \leq 1 = N(1, I).$$

Case 2. $d \geq 2$. Let U, p, k, V, B_V, and x be as in Proposition 3.8, applied to (X, B) and η. By Proposition 3.8(1)(2), V is a normal variety over k of dimension d, $x \in V$ is a closed point, and (V, B_V) is a klt pair over k such that $B_V \in I$. Since k is finitely generated over $\mathbb{C}$, by Theorem 14.1 there exists a Kollár component E_V of $(V \ni x, B_V)$ over k such that

$$a(E_V, V, B_V) \leq N(d, I).$$

By Proposition 3.8(3), E_V corresponds to a Kollár component E of $(X \ni \eta, B)$ such that

$$a(E, X, B) = a(E_V, V, B_V) \leq N(d, I).$$

Since $N(d, I)$ depends only on d and I, Theorem 1.2 follows. □

Proof of Corollary 1.3. Let X be a weakly exceptional Fano variety with $\dim X = d$, and let $N(d+1, \{0\})$ be the constant of Theorem 1.2 for the dimension $d+1$ and the DCC set $\{0\}$. We show that $q_W(X) \leq N(d+1, \{0\})$.

Let q be a positive integer and A a Weil divisor on X such that $-K_X \sim_{\mathbb{Q}} qA$. Since K_X is $\mathbb{Q}$-Cartier and $-K_X$ is ample, A is $\mathbb{Q}$-Cartier and ample. By Lemma 15.3, X is klt. By Construction-Proposition 15.4, $C(X, A)$ is a klt variety with $\dim C(X, A) = d + 1$, o is a closed point of $C(X, A)$ of codimension $d + 1$, and the zero section E is a Kollár component of $(C(X, A) \ni o, 0)$ such that

$$a(E, C(X, A), 0) = q.$$

By Theorem 1.2, applied to the klt pair $(C(X,A),0)$, the point o of codimension $d+1$, and $I=\{0\}$, there exists a Kollár component F of $(C(X,A)\ni o,0)$ such that

$$a(F,C(X,A),0)\le N(d+1,\{0\}).$$

By Lemma 15.6, $F=E$. Therefore

$$q=a(E,C(X,A),0)=a(F,C(X,A),0)\le N(d+1,\{0\}).$$

Since q is an arbitrary positive integer such that $-K_X\sim_{\mathbb{Q}} qA$ for some Weil divisor A on X, we have $q_W(X)\le N(d+1,\{0\})$. Since $N(d+1,\{0\})$ depends only on d, Corollary 1.3 follows. □

References

[Abh56] S. Abhyankar, *On the valuations centered in a local domain*, Amer. J. Math. **78** (1956), no. 2, 321–348.

[ADHL10] I. Arzhantsev, U. Derenthal, J. Hausen, and A. Laface, *Cox rings*, arXiv:1003.4229.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley, Reading, MA (1969).

[BP24] O. Benoist and A. Pirutka, *On the rationality of some real threefolds*, arXiv:2412.13624.

[BB13] R. J. Berman and B. Berndtsson, *Real Monge–Ampère equations and Kähler–Ricci solitons on toric log Fano varieties*, Ann. Fac. Sci. Toulouse Math. (6) **22** (2013), no. 4, 649–711.

[BM08] E. Bierstone and P. D. Milman, *Functoriality in resolution of singularities*, Publ. Res. Inst. Math. Sci. **44** (2008), no. 2, 609–639.

[Bir19] C. Birkar, *Anti-pluricanonical systems on Fano varieties*, Ann. of Math. (2) **190** (2019), no. 2, 345–463.

[Bir21] C. Birkar, *Singularities of linear systems and boundedness of Fano varieties*, Ann. of Math. (2) **193** (2021), no. 2, 347–405.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[BJ20] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. **365** (2020), 107062.

[BL21] H. Blum and Y. Liu, *The normalized volume of a singularity is lower semicontinuous*, J. Eur. Math. Soc. **23** (2021), no. 4, 1225–1256.

[Bou89] N. Bourbaki, *Commutative algebra. Chapters 1–7*, Elements of Mathematics, Springer-Verlag, Berlin (1989).

[CPS10] I. Cheltsov, J. Park, and C. Shramov, *Exceptional del Pezzo hypersurfaces*, J. Geom. Anal. **20** (2010), no. 4, 787–816.

[CPS18] I. Cheltsov, J. Park, and C. Shramov, *Alpha-invariants and purely log terminal blow-ups*, Eur. J. Math. **4** (2018), no. 3, 845–858.

[CS14] I. Cheltsov and C. Shramov, *Weakly-exceptional singularities in higher dimensions*, J. Reine Angew. Math. **689** (2014), 201–241.

[CLO15] D. A. Cox, J. Little, and D. O'Shea, *Ideals, varieties, and algorithms*, 4th ed., Undergraduate Texts in Mathematics, Springer, Cham (2015).

[CLS11] D. A. Cox, J. B. Little, and H. K. Schenck, *Toric varieties*, Graduate Studies in Mathematics **124**, Amer. Math. Soc., Providence, RI (2011).

[Cut14] S. D. Cutkosky, *Asymptotic multiplicities of graded families of ideals and linear series*, Adv. Math. **264** (2014), 55–113.

[ELS03] L. Ein, R. Lazarsfeld, and K. E. Smith, *Uniform approximation of Abhyankar valuation ideals in smooth function fields*, Amer. J. Math. **125** (2003), no. 2, 409–440.

[Ful93] W. Fulton, *Introduction to toric varieties*, Ann. of Math. Stud. **131**, Princeton University Press, Princeton, NJ (1993).

[Ful98] W. Fulton, *Intersection theory*, 2nd ed., Ergeb. Math. Grenzgeb. (3) **2**, Springer-Verlag, Berlin (1998).

[Gro65] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. II*, Inst. Hautes Études Sci. Publ. Math. (1965), no. 24, 5–231.

[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. (1966), no. 28, 5–255.

[Hal24] T. Hall, *On the uniqueness of Kähler–Einstein polygons in mutation-equivalence classes*, arXiv:2402.02832.

[HLQ24] J. Han, J. Liu, and L. Qi, *ACC for local volumes*, arXiv:2408.15090.

[HLS26] J. Han, J. Liu, and V. V. Shokurov, *ACC for minimal log discrepancies of exceptional singularities*, Peking Math. J. **9** (2026), no. 2, 225–257.

[HLQ20] J. Han, Y. Liu, and L. Qi, *ACC for local volumes and boundedness of singularities*, arXiv:2011.06509.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York (1977).

[Hir64] H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II*, Ann. of Math. (2) **79** (1964), 109–203; ibid. (2) **79** (1964), 205–326.
[HK00] Y. Hu and S. Keel, *Mori dream spaces and GIT*, Michigan Math. J. **48** (2000), 331–348.
[Jia20] C. Jiang, *Boundedness of $\mathbb{Q}$-Fano varieties with degrees and alpha-invariants bounded from below*, Ann. Sci. Éc. Norm. Supér. (4) **53** (2020), no. 5, 1235–1248.
[JM12] M. Jonsson and M. Mustaţă, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.
[KKMS73] G. Kempf, F. Knudsen, D. Mumford, and B. Saint-Donat, *Toroidal embeddings I*, Lecture Notes in Mathematics **339**, Springer-Verlag, Berlin (1973).
[Kol04] J. Kollár, *Seifert G_m-bundles*, arXiv:math/0404386.
[Kol13] J. Kollár, *Singularities of the minimal model program*, Cambridge Tracts in Math. **200**, Cambridge University Press, Cambridge (2013).
[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).
[Kud01] S. A. Kudryavtsev, *Pure log terminal blow-ups*, Math. Notes **69** (2001), 814–819.
[Laz04a] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergeb. Math. Grenzgeb. (3) **48**, Springer-Verlag, Berlin (2004).
[Laz04b] R. Lazarsfeld, *Positivity in algebraic geometry. II. Positivity for vector bundles, and multiplier ideals*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge **49**, Springer-Verlag, Berlin (2004).
[Li18] C. Li, *Minimizing normalized volumes of valuations*, Math. Z. **289** (2018), no. 1-2, 491–513.
[LX18] C. Li and C. Xu, *Stability of valuations: higher rational rank*, Peking Math. J. **1** (2018), no. 1, 1–79.
[LX20] C. Li and C. Xu, *Stability of valuations and Kollár components*, J. Eur. Math. Soc. **22** (2020), no. 8, 2573–2627.
[Liu26] J. Liu, *Upper bound of minimal log discrepancies*, preprint (2026), https://jihaoliu.org/notes/upper-bound-of-minimal-log-discrepancies-2026-10-09.pdf.
[LX23] J. Liu and L. Xie, *Divisors computing minimal log discrepancies on lc surfaces*, Math. Proc. Cambridge Philos. Soc. **175** (2023), no. 1, 107–128.
[Liu02] Q. Liu, *Algebraic geometry and arithmetic curves*, Oxford Graduate Texts in Mathematics **6**, Oxford University Press, Oxford (2002).
[LMS26] Y. Liu, J. Moraga, and H. Süß, *On the boundedness of singularities via normalized volume*, J. Inst. Math. Jussieu **25** (2026), no. 6, 3033–3081.
[Mat89] H. Matsumura, *Commutative ring theory*, translated from the Japanese by M. Reid, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1989).
[Mor21] J. Moraga, *On minimal log discrepancies and Kollár components*, Proc. Edinb. Math. Soc. (2) **64** (2021), no. 4, 982–1001.
[Nak16] Y. Nakamura, *On minimal log discrepancies on varieties with fixed Gorenstein index*, Michigan Math. J. **65** (2016), no. 1, 165–187.
[OAI26a] OpenAI, *Bounded klt complements for Fano contractions*, Paper No. 066 of https://github.com/openai/math.
[OAI26b] OpenAI, *Uniform Cartier sections for Fano type contractions*, Paper No. 066 of https://github.com/openai/math.
[Pro00] Yu. G. Prokhorov, *Blow-ups of canonical singularities*, Algebra (Moscow, 1998), de Gruyter, Berlin, 2000, pp. 301–317.
[PS09] Yu. G. Prokhorov and V. V. Shokurov, *Towards the second main theorem on complements*, J. Algebraic Geom. **18** (2009), no. 1, 151–199.
[RT11] J. Ross and R. Thomas, *Weighted projective embeddings, stability of orbifolds, and constant scalar curvature Kähler metrics*, J. Differential Geom. **88** (2011), 109–159.
[Ser79] J.-P. Serre, *Local fields*, Graduate Texts in Mathematics **67**, Springer-Verlag, New York–Berlin (1979).
[Sho88] V. V. Shokurov, *Problems about Fano varieties*, in: *Birational geometry of algebraic varieties: open problems*, The XXIIIrd International Symposium of the Taniguchi Foundation, Katata (1988), 30–32.
[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.
[Sho00] V. V. Shokurov, *Complements on surfaces*, J. Math. Sci. (New York) **102** (2000), no. 2, 3876–3932.
[Sta] The Stacks project authors, *The Stacks project*, https://stacks.math.columbia.edu, 2026.
[Ste10] E. Steinitz, *Algebraische Theorie der Körper*, J. Reine Angew. Math. **137** (1910), 167–309.
[Tia87] G. Tian, *On Kähler–Einstein metrics on certain Kähler manifolds with $C_1(M) > 0$*, Invent. Math. **89** (1987), 225–246.
[Xu14] C. Xu, *Finiteness of algebraic fundamental groups*, Compos. Math. **150** (2014), no. 3, 409–414.
[XZ21] C. Xu and Z. Zhuang, *Uniqueness of the minimizer of the normalized volume function*, Camb. J. Math. **9** (2021), no. 1, 149–176.

[XZ24] C. Xu and Z. Zhuang, *Boundedness of log Fano cone singularities and discreteness of local volumes*, arXiv:2404.17134.

[XZ25] C. Xu and Z. Zhuang, *Stable degenerations of singularities*, J. Amer. Math. Soc. **38** (2025), no. 3, 585–626.

[Zhu24a] Z. Zhuang, *Direct summands of klt singularities*, Invent. Math. **237** (2024), no. 3, 1683–1695.

[Zhu24b] Z. Zhuang, *On boundedness of singularities and minimal log discrepancies of Kollár components*, J. Algebraic Geom. **33** (2024), no. 3, 521–565.

[Zhu24c] Z. Zhuang, *On boundedness of singularities and minimal log discrepancies of Kollár components, II*, Geom. Topol. **28** (2024), no. 8, 3909–3933.

[Zhu25] Z. Zhuang, *Stability of klt singularities*, in: *Handbook of geometry and topology of singularities VII*, Springer, Cham (2025), 501–551.

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